

# Book I of Euclid's Elements

Formal Reconstruction — Technical Documentation

Jurek Róžański

Military University of Technology, Warsaw, Poland

CGJTEAMLAB

August 24, 2026

# Document Status

This volume is the technical documentation on formal reconstruction of Book I of Euclid's *Elements*. It records the mathematical architecture of the reconstruction, the Lean formalization, the dependency structure of individual propositions, the reusable Hilbert and scissors-congruence infrastructure, and the relation between the formal development and the classical sources.

The documentation accompanies the formal development. The Lean source is authoritative for theorem statements, hypotheses, dependencies, and proof status.

# Contents

|                                                            |              |
|------------------------------------------------------------|--------------|
| <b>Document Status</b>                                     | <b>ii</b>    |
| <b>Introduction</b>                                        | <b>xxvii</b> |
| <b>List of Figures</b>                                     | <b>xxxix</b> |
| <b>1 Proposition I.1</b>                                   | <b>1</b>     |
| 1.1 The Classical Configuration and Construction . . . . . | 1            |
| 1.2 Separating the Two Problems . . . . .                  | 2            |
| 1.3 Theorem . . . . .                                      | 2            |
| 1.4 Proof . . . . .                                        | 3            |
| 1.5 Proof Architecture of the Reconstruction . . . . .     | 3            |
| 1.6 Significance of the Reconstruction . . . . .           | 4            |
| 1.7 Relationship with the Foundations . . . . .            | 4            |
| 1.8 Hilbert and Book Zero Components Used . . . . .        | 5            |
| 1.8.1 Segment Congruence . . . . .                         | 5            |
| 1.8.2 Betweenness Case Analysis . . . . .                  | 5            |
| 1.8.3 Book Zero Order Lemmas . . . . .                     | 5            |
| 1.8.4 Noncollinearity . . . . .                            | 5            |
| 1.9 Formalization Notes . . . . .                          | 6            |
| 1.10 Dependency Summary . . . . .                          | 6            |
| 1.11 Conclusion . . . . .                                  | 6            |
| <b>2 Proposition I.2</b>                                   | <b>8</b>     |
| 2.1 Statement . . . . .                                    | 8            |
| 2.2 Euclid’s Classical Construction . . . . .              | 8            |
| 2.3 Hilbert Reconstruction . . . . .                       | 9            |
| 2.4 Formalization . . . . .                                | 10           |
| 2.5 Proof Architecture of the Reconstruction . . . . .     | 10           |
| 2.6 Discussion . . . . .                                   | 11           |
| 2.7 Relationship with Earlier Propositions . . . . .       | 11           |
| 2.7.1 Proposition I.1 . . . . .                            | 11           |
| 2.7.2 Transition to Proposition I.3 . . . . .              | 11           |
| 2.8 Hilbert and Book Zero Components Used . . . . .        | 11           |
| 2.8.1 Book Zero #49: Layoff . . . . .                      | 11           |
| 2.8.2 Same-Ray Structure . . . . .                         | 11           |
| 2.8.3 Segment Congruence . . . . .                         | 12           |
| 2.9 Formalization Notes . . . . .                          | 12           |
| 2.10 Dependency Summary . . . . .                          | 12           |
| 2.11 Conclusion . . . . .                                  | 13           |
| <b>3 Proposition I.3</b>                                   | <b>14</b>    |
| 3.1 Introduction . . . . .                                 | 14           |

|          |                                          |           |
|----------|------------------------------------------|-----------|
| 3.2      | The Classical Configuration              | 14        |
| 3.2.1    | Final Configuration                      | 14        |
| 3.2.2    | Construction Algorithm                   | 15        |
| 3.3      | Classical Proof                          | 16        |
| 3.4      | Proof Architecture of the Reconstruction | 17        |
| 3.5      | Hilbert Reconstruction                   | 17        |
| 3.6      | Lean Formalization                       | 18        |
| 3.7      | Relationship with Earlier Propositions   | 18        |
| 3.7.1    | Proposition I.2                          | 18        |
| 3.7.2    | Proposition I.1                          | 18        |
| 3.7.3    | Transition to Proposition I.4            | 18        |
| 3.8      | Hilbert and Book Zero Components Used    | 19        |
| 3.8.1    | Strict Segment Comparison                | 19        |
| 3.8.2    | Betweenness                              | 19        |
| 3.8.3    | Congruence Symmetry                      | 19        |
| 3.9      | Formalization Notes                      | 19        |
| 3.10     | Dependency Summary                       | 20        |
| 3.11     | Conclusion                               | 20        |
| <b>4</b> | <b>Proposition I.4</b>                   | <b>21</b> |
| 4.1      | Introduction                             | 21        |
| 4.2      | The Classical Configuration              | 21        |
| 4.2.1    | Final Configuration                      | 21        |
| 4.3      | Statement                                | 22        |
| 4.4      | Classical Proof                          | 23        |
| 4.5      | Proof Architecture of the Reconstruction | 24        |
| 4.6      | Hilbert Reconstruction                   | 24        |
| 4.7      | Lean Formalization                       | 25        |
| 4.8      | Relationship with Earlier Propositions   | 25        |
| 4.8.1    | Proposition I.2                          | 25        |
| 4.8.2    | Proposition I.3                          | 25        |
| 4.8.3    | Transition to Proposition I.5            | 25        |
| 4.9      | Hilbert and Book Zero Components Used    | 26        |
| 4.9.1    | Hilbert SAS                              | 26        |
| 4.9.2    | Nondegenerate Triangles                  | 26        |
| 4.9.3    | Angle Orientation                        | 26        |
| 4.9.4    | Segment Congruence                       | 26        |
| 4.10     | Formalization Notes                      | 26        |
| 4.11     | Dependency Summary                       | 27        |
| 4.12     | Conclusion                               | 27        |
| <b>5</b> | <b>Proposition I.5</b>                   | <b>28</b> |
| 5.1      | Introduction                             | 28        |
| 5.2      | The Classical Configuration              | 28        |
| 5.2.1    | Initial Isosceles Triangle               | 29        |
| 5.2.2    | Extension of the Equal Sides             | 29        |
| 5.3      | Statement                                | 29        |
| 5.4      | Classical Proof                          | 30        |
| 5.5      | Proof Architecture of the Reconstruction | 31        |
| 5.6      | Hilbert Reconstruction                   | 32        |
| 5.7      | Lean Formalization                       | 32        |
| 5.8      | Relationship with Earlier Propositions   | 33        |

|          |                                          |           |
|----------|------------------------------------------|-----------|
| 5.8.1    | Proposition I.2                          | 33        |
| 5.8.2    | Proposition I.3                          | 33        |
| 5.8.3    | Proposition I.4                          | 33        |
| 5.8.4    | Transition to Proposition I.6            | 34        |
| 5.9      | Hilbert and Book Zero Components Used    | 34        |
| 5.9.1    | Segment Extension and Layoff             | 34        |
| 5.9.2    | SAS                                      | 34        |
| 5.9.3    | Angle Congruence Transport               | 34        |
| 5.9.4    | Betweenness and Collinearity             | 34        |
| 5.10     | Formalization Notes                      | 34        |
| 5.11     | Dependency Summary                       | 35        |
| 5.12     | Conclusion                               | 35        |
| <b>6</b> | <b>Proposition I.6</b>                   | <b>36</b> |
| 6.1      | Introduction                             | 36        |
| 6.2      | The Classical Configuration              | 36        |
| 6.2.1    | Initial Triangle                         | 36        |
| 6.2.2    | Assuming Unequal Sides                   | 37        |
| 6.3      | Statement                                | 37        |
| 6.4      | Classical Proof                          | 38        |
| 6.5      | Proof Architecture of the Reconstruction | 38        |
| 6.6      | Hilbert Reconstruction                   | 39        |
| 6.7      | Lean Formalization                       | 40        |
| 6.8      | Relationship with Earlier Propositions   | 40        |
| 6.8.1    | Proposition I.3                          | 40        |
| 6.8.2    | Proposition I.4                          | 41        |
| 6.8.3    | Proposition I.5                          | 41        |
| 6.8.4    | Transition to Proposition I.7            | 41        |
| 6.9      | Hilbert and Book Zero Components Used    | 41        |
| 6.9.1    | Segment Trichotomy                       | 41        |
| 6.9.2    | Segment Layoff                           | 41        |
| 6.9.3    | Isosceles Base Angles                    | 42        |
| 6.9.4    | SAS and Angle Transport                  | 42        |
| 6.10     | Formalization Notes                      | 42        |
| 6.11     | Dependency Summary                       | 42        |
| 6.12     | Conclusion                               | 42        |
| <b>7</b> | <b>Proposition I.7</b>                   | <b>44</b> |
| 7.1      | Introduction                             | 44        |
| 7.2      | The Classical Configuration              | 44        |
| 7.2.1    | The Contradictory Configuration          | 45        |
| 7.3      | Statement                                | 45        |
| 7.4      | Classical Proof                          | 46        |
| 7.5      | Proof Architecture of the Reconstruction | 47        |
| 7.6      | Hilbert Reconstruction                   | 48        |
| 7.7      | Lean Formalization                       | 48        |
| 7.8      | Relationship with Earlier Propositions   | 49        |
| 7.8.1    | Proposition I.5                          | 49        |
| 7.8.2    | Proposition I.6                          | 50        |
| 7.8.3    | Transition to Proposition I.8            | 50        |
| 7.9      | Hilbert and Book Zero Components Used    | 50        |
| 7.9.1    | Same-Side Geometry                       | 50        |

|          |                                                    |           |
|----------|----------------------------------------------------|-----------|
| 7.9.2    | Segment Congruence . . . . .                       | 50        |
| 7.9.3    | Isosceles Angle Theorems . . . . .                 | 50        |
| 7.9.4    | Order and Collinearity . . . . .                   | 50        |
| 7.10     | Formalization Notes . . . . .                      | 51        |
| 7.11     | Dependency Summary . . . . .                       | 51        |
| 7.12     | Conclusion . . . . .                               | 51        |
| <b>8</b> | <b>Proposition I.8</b>                             | <b>53</b> |
| 8.1      | Introduction . . . . .                             | 53        |
| 8.2      | The Classical Configuration . . . . .              | 53        |
| 8.2.1    | Superposition . . . . .                            | 54        |
| 8.2.2    | Application of Proposition I.7 . . . . .           | 54        |
| 8.3      | Statement . . . . .                                | 55        |
| 8.4      | Classical Proof . . . . .                          | 56        |
| 8.5      | Proof Architecture of the Reconstruction . . . . . | 56        |
| 8.6      | Hilbert Reconstruction . . . . .                   | 57        |
| 8.7      | Lean Formalization . . . . .                       | 57        |
| 8.8      | Relationship with Earlier Propositions . . . . .   | 58        |
| 8.8.1    | Proposition I.4 . . . . .                          | 58        |
| 8.8.2    | Proposition I.5 . . . . .                          | 58        |
| 8.8.3    | Proposition I.7 . . . . .                          | 58        |
| 8.8.4    | Transition to Proposition I.9 . . . . .            | 58        |
| 8.9      | Hilbert and Book Zero Components Used . . . . .    | 59        |
| 8.9.1    | Hilbert SSS . . . . .                              | 59        |
| 8.9.2    | Same-Side and Opposite-Side Geometry . . . . .     | 59        |
| 8.9.3    | Angle Congruence . . . . .                         | 59        |
| 8.9.4    | Nondegeneracy . . . . .                            | 59        |
| 8.10     | Formalization Notes . . . . .                      | 59        |
| 8.11     | Dependency Summary . . . . .                       | 60        |
| 8.12     | Conclusion . . . . .                               | 60        |
| <b>9</b> | <b>Proposition I.9</b>                             | <b>61</b> |
| 9.1      | Introduction . . . . .                             | 61        |
| 9.2      | The Classical Configuration . . . . .              | 61        |
| 9.2.1    | Construction . . . . .                             | 62        |
| 9.2.2    | Application of Proposition I.8 . . . . .           | 62        |
| 9.3      | Statement . . . . .                                | 63        |
| 9.4      | Classical Proof . . . . .                          | 63        |
| 9.5      | Proof Architecture of the Reconstruction . . . . . | 64        |
| 9.6      | Hilbert Reconstruction . . . . .                   | 65        |
| 9.7      | Lean Formalization . . . . .                       | 65        |
| 9.8      | Relationship with Earlier Propositions . . . . .   | 66        |
| 9.8.1    | Proposition I.1 . . . . .                          | 66        |
| 9.8.2    | Proposition I.3 . . . . .                          | 67        |
| 9.8.3    | Proposition I.8 . . . . .                          | 67        |
| 9.8.4    | Transition to Proposition I.10 . . . . .           | 67        |
| 9.9      | Hilbert and Book Zero Components Used . . . . .    | 67        |
| 9.9.1    | Angle-Bisector Representation . . . . .            | 67        |
| 9.9.2    | Segment Layoff . . . . .                           | 67        |
| 9.9.3    | SSS . . . . .                                      | 67        |
| 9.9.4    | Interior and Ray Information . . . . .             | 67        |
| 9.10     | Formalization Notes . . . . .                      | 68        |

|           |                                                    |           |
|-----------|----------------------------------------------------|-----------|
| 9.11      | Dependency Summary . . . . .                       | 68        |
| 9.12      | Conclusion . . . . .                               | 68        |
| <b>10</b> | <b>Proposition I.10</b>                            | <b>70</b> |
| 10.1      | Introduction . . . . .                             | 70        |
| 10.2      | The Classical Configuration . . . . .              | 70        |
| 10.2.1    | Construction . . . . .                             | 70        |
| 10.2.2    | Application of Proposition I.4 . . . . .           | 71        |
| 10.3      | Statement . . . . .                                | 71        |
| 10.4      | Classical Proof . . . . .                          | 72        |
| 10.5      | Proof Architecture of the Reconstruction . . . . . | 73        |
| 10.6      | Hilbert Reconstruction . . . . .                   | 73        |
| 10.7      | Lean Formalization . . . . .                       | 74        |
| 10.8      | Relationship with Earlier Propositions . . . . .   | 74        |
| 10.8.1    | Proposition I.1 . . . . .                          | 74        |
| 10.8.2    | Proposition I.4 . . . . .                          | 74        |
| 10.8.3    | Proposition I.9 . . . . .                          | 74        |
| 10.8.4    | Transition to Proposition I.11 . . . . .           | 75        |
| 10.9      | Hilbert and Book Zero Components Used . . . . .    | 75        |
| 10.9.1    | Midpoint Representation . . . . .                  | 75        |
| 10.9.2    | Interface Theorem . . . . .                        | 75        |
| 10.10     | Formalization Notes . . . . .                      | 75        |
| 10.11     | Dependency Summary . . . . .                       | 76        |
| 10.12     | Conclusion . . . . .                               | 76        |
| <b>11</b> | <b>Proposition I.11</b>                            | <b>77</b> |
| 11.1      | Introduction . . . . .                             | 77        |
| 11.2      | The Classical Configuration . . . . .              | 77        |
| 11.3      | Statement . . . . .                                | 78        |
| 11.4      | Classical Proof . . . . .                          | 79        |
| 11.5      | Proof Architecture of the Reconstruction . . . . . | 80        |
| 11.6      | Hilbert Reconstruction . . . . .                   | 81        |
| 11.7      | Lean Formalization . . . . .                       | 83        |
| 11.8      | Relationship with Earlier Propositions . . . . .   | 85        |
| 11.8.1    | Proposition I.1 . . . . .                          | 85        |
| 11.8.2    | Proposition I.3 . . . . .                          | 86        |
| 11.8.3    | Proposition I.8 . . . . .                          | 86        |
| 11.8.4    | Transition to Proposition I.12 . . . . .           | 86        |
| 11.9      | Hilbert and Book Zero Components Used . . . . .    | 86        |
| 11.9.1    | Segment Layoff . . . . .                           | 86        |
| 11.9.2    | Hilbert III.4 . . . . .                            | 86        |
| 11.9.3    | HilbertRightAngle . . . . .                        | 86        |
| 11.9.4    | Betweenness and Collinearity . . . . .             | 86        |
| 11.10     | Formalization Notes . . . . .                      | 87        |
| 11.11     | Dependency Summary . . . . .                       | 88        |
| 11.12     | Conclusion . . . . .                               | 88        |
| <b>12</b> | <b>Proposition I.12</b>                            | <b>89</b> |
| 12.1      | Introduction . . . . .                             | 89        |
| 12.2      | The Classical Configuration . . . . .              | 90        |
| 12.2.1    | Final Configuration . . . . .                      | 90        |
| 12.2.2    | Construction Algorithm . . . . .                   | 90        |

|           |                                                       |            |
|-----------|-------------------------------------------------------|------------|
| 12.3      | Classical Proof                                       | 92         |
| 12.4      | Proof Architecture of the Reconstruction              | 93         |
| 12.4.1    | Dependency Diagram                                    | 93         |
| 12.5      | Hilbert Reconstruction                                | 94         |
| 12.6      | Lean Formalization                                    | 96         |
| 12.7      | Relationship with Earlier Propositions                | 98         |
| 12.7.1    | Proposition I.8                                       | 98         |
| 12.7.2    | Proposition I.10                                      | 98         |
| 12.7.3    | Proposition I.11                                      | 98         |
| 12.8      | Hilbert and Book Zero Components Used                 | 99         |
| 12.8.1    | <code>hilbert_collinear_angle_transport</code>        | 99         |
| 12.8.2    | <code>hilbert_equidistant_line_bisects_segment</code> | 99         |
| 12.8.3    | <code>hilbert_perpendicular_from_point_exists</code>  | 99         |
| 12.8.4    | Hilbert SSS and Vertical Angles                       | 99         |
| 12.9      | Formalization Notes                                   | 99         |
| 12.10     | Dependency Summary                                    | 101        |
| 12.11     | Conclusion                                            | 101        |
| <b>13</b> | <b>Proposition I.13</b>                               | <b>102</b> |
| 13.1      | Introduction                                          | 102        |
| 13.2      | The Classical Configuration                           | 102        |
| 13.3      | Statement                                             | 103        |
| 13.4      | Classical Proof                                       | 103        |
| 13.4.1    | Case 1. The Adjacent Angles Are Equal                 | 104        |
| 13.4.2    | Case 2. The Adjacent Angles Are Not Equal             | 104        |
| 13.5      | Proof Architecture of the Reconstruction              | 105        |
| 13.6      | Hilbert Reconstruction                                | 106        |
| 13.7      | Lean Formalization                                    | 107        |
| 13.8      | Discussion                                            | 108        |
| 13.9      | Relationship with Earlier Propositions                | 109        |
| 13.9.1    | Proposition I.11                                      | 109        |
| 13.9.2    | Proposition I.12                                      | 109        |
| 13.9.3    | Transition to Proposition I.14                        | 109        |
| 13.10     | Hilbert and Book Zero Components Used                 | 110        |
| 13.11     | Formalization Notes                                   | 111        |
| 13.12     | Dependency Summary                                    | 111        |
| 13.13     | Conclusion                                            | 112        |
| <b>14</b> | <b>Proposition I.14</b>                               | <b>113</b> |
| 14.1      | Introduction                                          | 113        |
| 14.2      | The Classical Configuration                           | 113        |
| 14.3      | Statement                                             | 114        |
| 14.4      | Classical Proof                                       | 114        |
| 14.5      | Proof Architecture of the Reconstruction              | 116        |
| 14.6      | Hilbert Reconstruction                                | 117        |
| 14.6.1    | Encoding the Angle Hypothesis                         | 117        |
| 14.6.2    | The Collinear Case                                    | 118        |
| 14.6.3    | The Noncollinear Case                                 | 119        |
| 14.7      | Lean Formalization                                    | 121        |
| 14.8      | Relationship with Earlier Propositions                | 122        |
| 14.8.1    | Proposition I.13                                      | 122        |
| 14.8.2    | Proposition I.4                                       | 123        |



|           |                                                    |            |
|-----------|----------------------------------------------------|------------|
| 14.8.3    | Transition to Proposition I.15                     | 123        |
| 14.9      | Hilbert and Book Zero Components Used              | 123        |
| 14.9.1    | <code>hilbert_third_side_endpoints_sameSide</code> | 123        |
| 14.9.2    | Hilbert III.4: Uniqueness of Angle Construction    | 124        |
| 14.9.3    | <code>bookZero_36_ray3</code>                      | 124        |
| 14.9.4    | <code>hilbert_between_transport_sameRays</code>    | 125        |
| 14.10     | Formalization Notes                                | 125        |
| 14.11     | Dependency Summary                                 | 126        |
| 14.12     | Conclusion                                         | 126        |
| <b>15</b> | <b>Proposition I.15</b>                            | <b>128</b> |
| 15.1      | Introduction                                       | 128        |
| 15.2      | The Classical Configuration                        | 128        |
| 15.3      | Statement                                          | 129        |
| 15.4      | Classical Proof                                    | 130        |
| 15.5      | Proof Architecture of the Reconstruction           | 131        |
| 15.6      | Hilbert Reconstruction                             | 133        |
| 15.6.1    | The First Pair of Vertical Angles                  | 133        |
| 15.6.2    | The Second Pair of Vertical Angles                 | 133        |
| 15.7      | Lean Formalization                                 | 134        |
| 15.8      | Relationship with Earlier Propositions             | 136        |
| 15.8.1    | Proposition I.13                                   | 136        |
| 15.8.2    | Proposition I.14                                   | 137        |
| 15.8.3    | Transition to Proposition I.16                     | 137        |
| 15.9      | Hilbert and Book Zero Components Used              | 137        |
| 15.9.1    | <code>hilbert_vertical_angles</code>               | 137        |
| 15.9.2    | <code>HilbertOrder.between_incidence</code>        | 138        |
| 15.9.3    | <code>PrimCollinearCycle</code>                    | 139        |
| 15.9.4    | <code>hilbert_primCollinear_trans</code>           | 139        |
| 15.10     | Formalization Notes                                | 140        |
| 15.11     | Dependency Summary                                 | 140        |
| 15.12     | Conclusion                                         | 141        |
| <b>16</b> | <b>Proposition I.16</b>                            | <b>142</b> |
| 16.1      | Introduction                                       | 142        |
| 16.2      | The Classical Configuration                        | 142        |
| 16.3      | Statement                                          | 143        |
| 16.4      | Classical Proof                                    | 143        |
| 16.5      | Proof Architecture of the Reconstruction           | 144        |
| 16.6      | Hilbert Reconstruction                             | 146        |
| 16.6.1    | The First Opposite Interior Angle                  | 146        |
| 16.6.2    | The Second Opposite Interior Angle                 | 147        |
| 16.7      | Lean Formalization                                 | 147        |
| 16.8      | Relationship with Earlier Propositions             | 149        |
| 16.8.1    | Proposition I.10                                   | 149        |
| 16.8.2    | Proposition I.15                                   | 149        |
| 16.8.3    | Transition to Proposition I.17                     | 150        |
| 16.9      | Hilbert and Book Zero Components Used              | 150        |
| 16.9.1    | <code>HilbertAngleLess</code>                      | 150        |
| 16.9.2    | <code>hilbert_exterior_angle_aux</code>            | 150        |
| 16.9.3    | <code>hilbert_exterior_angle_aux_parallel</code>   | 150        |
| 16.9.4    | <code>hilbert_exterior_angle_less</code>           | 150        |

|           |                                                          |            |
|-----------|----------------------------------------------------------|------------|
| 16.9.5    | Interior-subangle transport                              | 150        |
| 16.9.6    | VerticalAngles                                           | 151        |
| 16.9.7    | hilbert_angleLess_transport_right                        | 151        |
| 16.10     | Formalization Notes                                      | 151        |
| 16.11     | Dependency Summary                                       | 152        |
| 16.12     | Conclusion                                               | 152        |
| <b>17</b> | <b>Proposition I.17</b>                                  | <b>153</b> |
| 17.1      | Introduction                                             | 153        |
| 17.2      | The Classical Configuration                              | 153        |
| 17.3      | Classical Proof                                          | 155        |
| 17.4      | Synthetic Representation of “Less Than Two Right Angles” | 155        |
| 17.5      | Proof Architecture of the Reconstruction                 | 156        |
| 17.6      | Lean Formalization                                       | 157        |
| 17.6.1    | The Pair $\angle BAC$ and $\angle ACB$                   | 157        |
| 17.6.2    | The Pair $\angle ABC$ and $\angle ACB$                   | 158        |
| 17.6.3    | The Pair $\angle BAC$ and $\angle ABC$                   | 159        |
| 17.6.4    | The Final Wrapper                                        | 160        |
| 17.7      | Relationship with Earlier Propositions                   | 161        |
| 17.7.1    | Proposition I.13                                         | 161        |
| 17.7.2    | Proposition I.14                                         | 161        |
| 17.7.3    | Proposition I.16                                         | 161        |
| 17.8      | Hilbert and Book Zero Components Used                    | 162        |
| 17.8.1    | HilbertAngleLess                                         | 162        |
| 17.8.2    | HilbertOrder.between_extension                           | 162        |
| 17.8.3    | Collinearity Symmetry and Rotation                       | 162        |
| 17.8.4    | Angle Reversal                                           | 162        |
| 17.9      | Formalization Notes                                      | 162        |
| 17.10     | Dependency Summary                                       | 163        |
| 17.11     | Conclusion                                               | 163        |
| <b>18</b> | <b>Proposition I.18</b>                                  | <b>164</b> |
| 18.1      | Introduction                                             | 164        |
| 18.2      | The Classical Configuration                              | 164        |
| 18.3      | Classical Proof                                          | 166        |
| 18.4      | Strict Angle Comparison                                  | 167        |
| 18.5      | Proof Architecture of the Reconstruction                 | 167        |
| 18.5.1    | Recovering the Interior Point                            | 168        |
| 18.5.2    | The Isosceles Triangle                                   | 168        |
| 18.5.3    | The Exterior-Angle Comparison                            | 168        |
| 18.5.4    | The Interior Subangle                                    | 169        |
| 18.5.5    | Transitivity                                             | 169        |
| 18.6      | Lean Formalization                                       | 169        |
| 18.7      | Relationship with Earlier Propositions                   | 171        |
| 18.7.1    | Proposition I.5                                          | 171        |
| 18.7.2    | Proposition I.16                                         | 172        |
| 18.7.3    | Proposition I.17                                         | 172        |
| 18.8      | Hilbert and Book Zero Components Used                    | 172        |
| 18.8.1    | HilbertSegmentLess                                       | 172        |
| 18.8.2    | hilbert_isosceles_base_angles                            | 172        |
| 18.8.3    | HilbertSameRay                                           | 172        |
| 18.8.4    | hilbert_interior_angle_less                              | 172        |

|           |                                                |            |
|-----------|------------------------------------------------|------------|
| 18.8.5    | <code>hilbert_angleLess_trans</code>           | 173        |
| 18.9      | Formalization Notes                            | 173        |
| 18.10     | Dependency Summary                             | 173        |
| 18.11     | Conclusion                                     | 174        |
| <b>19</b> | <b>Proposition I.19</b>                        | <b>175</b> |
| 19.1      | Introduction                                   | 175        |
| 19.2      | The Classical Configuration                    | 176        |
| 19.3      | Classical Proof                                | 176        |
| 19.4      | Segment Trichotomy                             | 177        |
| 19.5      | Irreflexivity of Strict Angle Comparison       | 178        |
| 19.6      | Proof Architecture of the Reconstruction       | 178        |
| 19.6.1    | Nondegeneracy of the Compared Side             | 179        |
| 19.6.2    | Apply Segment Trichotomy                       | 179        |
| 19.6.3    | Eliminate the Two Impossible Branches          | 179        |
| 19.7      | Lean Formalization                             | 180        |
| 19.8      | Relationship with Earlier Propositions         | 182        |
| 19.8.1    | Proposition I.5                                | 182        |
| 19.8.2    | Proposition I.18                               | 182        |
| 19.8.3    | Logical Relation between I.18 and I.19         | 182        |
| 19.9      | Hilbert and Book Zero Components Used          | 183        |
| 19.9.1    | <code>hilbert_segment_trichotomy</code>        | 183        |
| 19.9.2    | <code>hilbert_angleLess_irrefl</code>          | 183        |
| 19.9.3    | <code>hilbert_isosceles_base_angles</code>     | 183        |
| 19.9.4    | <code>hilbert_angleLess_transport_right</code> | 183        |
| 19.9.5    | <code>hilbert_angleLess_trans</code>           | 183        |
| 19.10     | Formalization Notes                            | 183        |
| 19.11     | Dependency Summary                             | 184        |
| 19.12     | Conclusion                                     | 184        |
| <b>20</b> | <b>Proposition I.20</b>                        | <b>185</b> |
| 20.1      | Introduction                                   | 185        |
| 20.2      | The Classical Configuration                    | 185        |
| 20.3      | Classical Proof                                | 186        |
| 20.4      | Synthetic Form of the Triangle Inequality      | 187        |
| 20.5      | Proof Architecture of the Reconstruction       | 187        |
| 20.5.1    | Construction of the Extension Point            | 188        |
| 20.5.2    | The Interior-Angle Step                        | 189        |
| 20.5.3    | Transport Through the Isosceles Triangle       | 189        |
| 20.5.4    | Application of Proposition I.19                | 190        |
| 20.6      | Lean Formalization                             | 190        |
| 20.7      | Relationship with Earlier Propositions         | 194        |
| 20.7.1    | Proposition I.5                                | 194        |
| 20.7.2    | Proposition I.16                               | 194        |
| 20.7.3    | Proposition I.19                               | 194        |
| 20.7.4    | Transition to Proposition I.21                 | 194        |
| 20.8      | Hilbert and Book Zero Components Used          | 195        |
| 20.9      | Dependency Summary                             | 195        |
| 20.10     | Conclusion                                     | 195        |
| <b>21</b> | <b>Proposition I.21</b>                        | <b>197</b> |
| 21.1      | Introduction                                   | 197        |

|           |                                                  |            |
|-----------|--------------------------------------------------|------------|
| 21.2      | The Classical Configuration                      | 197        |
| 21.3      | Classical Proof                                  | 198        |
| 21.4      | From Euclid’s Diagram to the Formal Argument     | 199        |
| 21.5      | Synthetic Segment Arithmetic                     | 200        |
| 21.6      | Proof Architecture of the Reconstruction         | 200        |
| 21.6.1    | The Segment Helper                               | 201        |
| 21.6.2    | The Angle Branch                                 | 202        |
| 21.6.3    | Final Segment Comparison                         | 203        |
| 21.7      | Lean Formalization                               | 203        |
| 21.8      | Relationship with Earlier Propositions           | 204        |
| 21.8.1    | Proposition I.16                                 | 204        |
| 21.8.2    | Proposition I.20                                 | 204        |
| 21.8.3    | The Cumulative Character of I.21                 | 205        |
| 21.9      | Hilbert and Book Zero Components Used            | 205        |
| 21.9.1    | HilbertSegmentLess                               | 205        |
| 21.9.2    | Congruence transport                             | 205        |
| 21.9.3    | Difference of parts                              | 205        |
| 21.9.4    | Addition of inequalities                         | 206        |
| 21.9.5    | Segment-Inequality Transitivity                  | 206        |
| 21.9.6    | Sum of parts                                     | 206        |
| 21.10     | What I.21 Reveals About Book Zero                | 206        |
| 21.11     | Dependency Summary                               | 207        |
| 21.12     | Formalization Notes                              | 208        |
| 21.13     | Conclusion                                       | 208        |
| <b>22</b> | <b>Proposition I.22</b>                          | <b>209</b> |
| 22.1      | Introduction                                     | 209        |
| 22.2      | The Classical Configuration                      | 210        |
| 22.3      | Classical Proof                                  | 211        |
| 22.4      | Statement                                        | 212        |
| 22.5      | A Synthetic Predicate for “Two Together Greater” | 213        |
| 22.6      | Proof Architecture of the Reconstruction         | 214        |
| 22.6.1    | Transporting the outside inequality              | 214        |
| 22.6.2    | Same-ray trichotomy                              | 215        |
| 22.6.3    | Cancellation for strict segment inequalities     | 215        |
| 22.6.4    | The inside point                                 | 216        |
| 22.6.5    | Constructing explicit sum representatives        | 216        |
| 22.6.6    | Circle witnesses                                 | 216        |
| 22.7      | Circle-Circle Continuity                         | 217        |
| 22.8      | The Core and the Public Theorem                  | 218        |
| 22.9      | Lean Formalization                               | 219        |
| 22.10     | Relationship with Earlier Propositions           | 220        |
| 22.10.1   | Proposition I.20                                 | 220        |
| 22.10.2   | Proposition I.21                                 | 220        |
| 22.10.3   | Proposition I.23                                 | 221        |
| 22.11     | Hilbert and Book Zero Components Used            | 221        |
| 22.11.1   | HilbertSegmentLess                               | 221        |
| 22.11.2   | bookZero_49_layoff                               | 221        |
| 22.11.3   | bookZero_sumOfParts                              | 221        |
| 22.11.4   | Congruence transport for inequalities            | 222        |
| 22.11.5   | bookZero_31_trichotomy1                          | 222        |
| 22.11.6   | bookZero_51_lessThanBetween                      | 222        |

|           |                                                      |            |
|-----------|------------------------------------------------------|------------|
| 22.11.7   | Circle-circle intersection                           | 222        |
| 22.12     | Comparison with Euclid's Original Organization       | 223        |
| 22.13     | What Proposition I.22 Adds to Book Zero              | 223        |
| 22.14     | Dependency Summary                                   | 224        |
| 22.15     | Formalization Notes                                  | 224        |
| 22.16     | Conclusion                                           | 225        |
| <b>23</b> | <b>Proposition I.23</b>                              | <b>226</b> |
| 23.1      | Introduction                                         | 226        |
| 23.2      | The Classical Configuration                          | 226        |
| 23.3      | Classical Proof                                      | 227        |
| 23.4      | Hilbert III.4                                        | 228        |
| 23.5      | Proof Architecture of the Reconstruction             | 228        |
| 23.5.1    | The Source Angle                                     | 229        |
| 23.5.2    | The Prescribed Ray                                   | 229        |
| 23.5.3    | Selecting the Side                                   | 229        |
| 23.5.4    | Applying Angle Construction                          | 229        |
| 23.6      | Lean Formalization                                   | 230        |
| 23.7      | The Euclidean Route and the Hilbert Route            | 231        |
| 23.7.1    | The Euclidean Route                                  | 231        |
| 23.7.2    | The Hilbert Route                                    | 231        |
| 23.7.3    | Foundational Interpretation                          | 232        |
| 23.8      | Relationship with Earlier Propositions               | 232        |
| 23.8.1    | Proposition I.8                                      | 232        |
| 23.8.2    | Proposition I.22                                     | 232        |
| 23.8.3    | Book I Order versus Formal Dependency                | 232        |
| 23.9      | Hilbert and Book Zero Components Used                | 233        |
| 23.9.1    | HilbertCongruence.angle_construction                 | 233        |
| 23.9.2    | HilbertSameSide                                      | 233        |
| 23.9.3    | HilbertSameRay                                       | 233        |
| 23.9.4    | No New Book Zero Lemma                               | 233        |
| 23.10     | Uniqueness and the Strength of the Hilbert Statement | 233        |
| 23.11     | Dependency Summary                                   | 234        |
| 23.12     | Formalization Notes                                  | 234        |
| 23.13     | Conclusion                                           | 234        |
| <b>24</b> | <b>Proposition I.24</b>                              | <b>236</b> |
| 24.1      | Introduction                                         | 236        |
| 24.2      | The Classical Configuration                          | 236        |
| 24.3      | Classical Proof                                      | 237        |
| 24.4      | The Classical Configuration Gap                      | 238        |
| 24.5      | Strict Angle and Segment Comparison                  | 239        |
| 24.6      | Proof Architecture of the Reconstruction             | 239        |
| 24.6.1    | The SAS Construction                                 | 239        |
| 24.6.2    | The Hinge Configuration                              | 240        |
| 24.6.3    | Case 1: $A - H - Y$                                  | 240        |
| 24.6.4    | Case 2: $H = Y$                                      | 241        |
| 24.6.5    | Case 3: $A - Y - H$                                  | 241        |
| 24.6.6    | The Hinge Helper                                     | 241        |
| 24.6.7    | Completion of I.24                                   | 242        |
| 24.7      | Lean Formalization                                   | 242        |
| 24.8      | Relationship with Earlier Propositions               | 243        |

|           |                                                       |            |
|-----------|-------------------------------------------------------|------------|
| 24.8.1    | Proposition I.5                                       | 243        |
| 24.8.2    | Proposition I.16                                      | 243        |
| 24.8.3    | Proposition I.19                                      | 243        |
| 24.8.4    | Proposition I.23                                      | 244        |
| 24.9      | Hilbert and Book Zero Components Used                 | 244        |
| 24.9.1    | HilbertAngleLess                                      | 244        |
| 24.9.2    | HilbertSegmentLess                                    | 244        |
| 24.9.3    | HilbertSameRay                                        | 244        |
| 24.9.4    | Same-ray trichotomy                                   | 244        |
| 24.9.5    | hilbert_interior_angle_less                           | 244        |
| 24.9.6    | Angle-congruence transport                            | 244        |
| 24.10     | Historical Note: Euclid, Proclus, Simson, and Hilbert | 245        |
| 24.11     | Dependency Summary                                    | 245        |
| 24.12     | Conclusion                                            | 245        |
| <b>25</b> | <b>Proposition I.25</b>                               | <b>247</b> |
| 25.1      | Introduction                                          | 247        |
| 25.2      | The Classical Configuration                           | 247        |
| 25.3      | Classical Proof                                       | 248        |
| 25.4      | Angle Trichotomy                                      | 248        |
| 25.5      | Proof Architecture of the Reconstruction              | 249        |
| 25.6      | Hilbert Reconstruction                                | 249        |
| 25.6.1    | The Congruent-Angle Branch                            | 249        |
| 25.6.2    | The Wrong Strict-Inequality Branch                    | 249        |
| 25.6.3    | The Desired Branch                                    | 250        |
| 25.7      | Lean Formalization                                    | 250        |
| 25.8      | Relationship with Earlier Propositions                | 250        |
| 25.8.1    | Proposition I.4                                       | 250        |
| 25.8.2    | Proposition I.24                                      | 251        |
| 25.9      | Hilbert and Book Zero Components Used                 | 251        |
| 25.9.1    | angle_trichotomy                                      | 251        |
| 25.9.2    | SAS                                                   | 251        |
| 25.9.3    | hilbert_segmentLess_not_congruent                     | 251        |
| 25.9.4    | hilbert_segmentLess_asymm                             | 251        |
| 25.10     | Formalization Notes                                   | 251        |
| 25.11     | Dependency Summary                                    | 252        |
| 25.12     | Conclusion                                            | 253        |
| <b>26</b> | <b>Proposition I.26</b>                               | <b>254</b> |
| 26.1      | Introduction                                          | 254        |
| 26.2      | Classical Configurations                              | 254        |
| 26.2.1    | ASA auxiliary configuration                           | 255        |
| 26.2.2    | SAA auxiliary configuration                           | 256        |
| 26.3      | Diagrammatic Audit                                    | 257        |
| 26.4      | Hilbert Reconstruction                                | 257        |
| 26.4.1    | ASA                                                   | 257        |
| 26.4.2    | SAA                                                   | 258        |
| 26.5      | Proof Architecture of the Reconstruction              | 258        |
| 26.6      | Lean Formalization                                    | 260        |
| 26.7      | Relationship with Earlier Propositions                | 260        |
| 26.7.1    | Proposition I.4                                       | 260        |
| 26.7.2    | Proposition I.16                                      | 261        |

|           |                                                |            |
|-----------|------------------------------------------------|------------|
| 26.7.3    | Segment trichotomy                             | 261        |
| 26.8      | Hilbert and Book Zero Components Used          | 261        |
| 26.8.1    | hilbert_asa_sides                              | 261        |
| 26.8.2    | HilbertSSS                                     | 261        |
| 26.8.3    | euclid_proposition_26_saa_not_less             | 261        |
| 26.8.4    | bookZero_31_trichotomy1                        | 262        |
| 26.9      | Formalization Notes                            | 262        |
| 26.10     | Dependency Summary                             | 262        |
| 26.11     | Conclusion                                     | 264        |
| <b>27</b> | <b>Proposition I.27</b>                        | <b>265</b> |
| 27.1      | Introduction                                   | 265        |
| 27.2      | Classical Statement                            | 265        |
| 27.3      | Euclid's Proof                                 | 266        |
| 27.4      | Diagrammatic Audit                             | 267        |
| 27.5      | Hilbert Reconstruction                         | 267        |
| 27.6      | Two Formal Formulations                        | 268        |
| 27.6.1    | The interface-level formulation                | 268        |
| 27.6.2    | The explicit transversal formulation           | 268        |
| 27.7      | Normalization of the Transversal Configuration | 269        |
| 27.8      | Proof Architecture                             | 270        |
| 27.9      | Proof-Theoretic Status                         | 271        |
| 27.10     | Relationship with Proposition I.16             | 271        |
| 27.11     | Neutral Geometry and the Fifth Postulate       | 272        |
| 27.12     | Lean Formalization                             | 272        |
| 27.13     | Conclusion                                     | 272        |
| <b>28</b> | <b>Proposition I.28</b>                        | <b>274</b> |
| 28.1      | Introduction                                   | 274        |
| 28.2      | Classical Configuration                        | 274        |
| 28.2.1    | Corresponding-angle criterion                  | 274        |
| 28.2.2    | Same-side interior-angle criterion             | 275        |
| 28.3      | Hilbert Reconstruction                         | 276        |
| 28.3.1    | First branch                                   | 276        |
| 28.3.2    | Second branch                                  | 277        |
| 28.4      | Neutral Geometry                               | 278        |
| 28.5      | Dependency Summary                             | 279        |
| 28.6      | Conclusion                                     | 279        |
| <b>29</b> | <b>Proposition I.29</b>                        | <b>280</b> |
| 29.1      | Introduction                                   | 280        |
| 29.2      | Classical Configuration                        | 281        |
| 29.3      | First Conclusion: Alternate Interior Angles    | 281        |
| 29.4      | Second Conclusion: Corresponding Angles        | 282        |
| 29.5      | Third Conclusion: Same-Side Interior Angles    | 284        |
| 29.6      | Lean Formalization                             | 284        |
| 29.7      | Proof Architecture                             | 286        |
| 29.8      | The Logical Boundary at I.29                   | 286        |
| 29.9      | Existence and Uniqueness Behind I.29           | 287        |
| 29.10     | Conclusion                                     | 287        |
| <b>30</b> | <b>Proposition I.30</b>                        | <b>289</b> |

|           |                                                                   |            |
|-----------|-------------------------------------------------------------------|------------|
| 30.1      | Introduction . . . . .                                            | 289        |
| 30.2      | Classical Configuration . . . . .                                 | 289        |
| 30.3      | Euclid’s Proof Architecture . . . . .                             | 290        |
| 30.4      | Hilbert Reconstruction . . . . .                                  | 291        |
| 30.5      | Lean Formalization . . . . .                                      | 291        |
| 30.6      | Logical Position of Proposition I.30 . . . . .                    | 293        |
| 30.7      | Conclusion . . . . .                                              | 293        |
| <b>31</b> | <b>Proposition I.31</b>                                           | <b>295</b> |
| 31.1      | Introduction . . . . .                                            | 295        |
| 31.2      | Classical Configuration . . . . .                                 | 295        |
| 31.3      | Why No Right Angle Is Needed . . . . .                            | 296        |
| 31.4      | The Hidden Configuration . . . . .                                | 297        |
| 31.5      | Hilbert Reconstruction . . . . .                                  | 297        |
| 31.6      | Lean Formalization . . . . .                                      | 298        |
| 31.7      | Neutral Existence Versus Euclidean Uniqueness . . . . .           | 299        |
| 31.8      | Final Configuration . . . . .                                     | 300        |
| 31.9      | Conclusion . . . . .                                              | 300        |
| <b>32</b> | <b>Proposition I.32</b>                                           | <b>301</b> |
| 32.1      | Introduction . . . . .                                            | 301        |
| 32.2      | Classical Statement . . . . .                                     | 301        |
| 32.3      | The Formal Configuration . . . . .                                | 302        |
| 32.4      | Synthetic Representation of the Exterior-Angle Equality . . . . . | 302        |
| 32.5      | Construction of the Correctly Oriented Parallel Ray . . . . .     | 303        |
| 32.6      | First Remote Interior Angle . . . . .                             | 304        |
| 32.7      | Preparing Proposition I.29 . . . . .                              | 305        |
| 32.8      | Second Remote Interior Angle . . . . .                            | 306        |
| 32.9      | Proof Architecture of the Exterior-Angle Part . . . . .           | 307        |
| 32.10     | Where the Euclidean Parallel Axiom Enters . . . . .               | 307        |
| 32.11     | The Triangle-Angle-Sum Statement . . . . .                        | 308        |
| 32.12     | What “Equal to Two Right Angles” Means Here . . . . .             | 309        |
| 32.13     | Lean Formalization . . . . .                                      | 309        |
| 32.14     | Relation to the Classical Proof . . . . .                         | 310        |
| 32.15     | Conclusion . . . . .                                              | 311        |
| <b>33</b> | <b>Proposition I.33</b>                                           | <b>312</b> |
| 33.1      | Introduction . . . . .                                            | 312        |
| 33.2      | The Classical Configuration . . . . .                             | 312        |
| 33.3      | Classical Proof . . . . .                                         | 313        |
| 33.4      | The Hidden Orientation Data . . . . .                             | 313        |
| 33.5      | Proof Architecture of the Reconstruction . . . . .                | 314        |
| 33.6      | Lean Formalization . . . . .                                      | 315        |
| 33.7      | Relationship with Earlier Propositions . . . . .                  | 316        |
| 33.8      | Hilbert and Book Zero Components Used . . . . .                   | 316        |
| 33.9      | Formalization Notes . . . . .                                     | 317        |
| 33.10     | Dependency Summary . . . . .                                      | 318        |
| 33.11     | Conclusion . . . . .                                              | 318        |
| <b>34</b> | <b>Proposition I.34</b>                                           | <b>319</b> |
| 34.1      | Introduction . . . . .                                            | 319        |
| 34.2      | The Classical Configuration . . . . .                             | 319        |



|           |                                                             |            |
|-----------|-------------------------------------------------------------|------------|
| 34.3      | Classical Proof                                             | 320        |
| 34.4      | The Main Formalization Issue: Reusing Parallelogram Theory  | 320        |
| 34.5      | Proof Architecture of the Reconstruction                    | 321        |
| 34.6      | Lean Formalization                                          | 321        |
| 34.7      | Relationship with Earlier Propositions                      | 322        |
| 34.8      | Hilbert and Book Zero Components Used                       | 323        |
| 34.9      | Formalization Notes                                         | 323        |
| 34.9.1    | Axiomatic status                                            | 323        |
| 34.9.2    | Hidden configuration data                                   | 323        |
| 34.9.3    | Reusable helpers introduced                                 | 324        |
| 34.10     | Dependency Summary                                          | 324        |
| 34.11     | Conclusion                                                  | 325        |
| <b>35</b> | <b>Proposition I.35</b>                                     | <b>326</b> |
| 35.1      | Introduction                                                | 326        |
| 35.2      | The Classical Configuration                                 | 327        |
| 35.3      | Classical Proof                                             | 327        |
| 35.4      | Equidecomposition and Equicomplementability                 | 328        |
| 35.4.1    | Why numerical area is deliberately avoided                  | 329        |
| 35.5      | The Hilbert Scissors Calculus                               | 329        |
| 35.5.1    | Triangulation lemmas                                        | 331        |
| 35.6      | Proof Architecture of the Reconstruction                    | 332        |
| 35.6.1    | Stage 1: recover the upper-line order                       | 332        |
| 35.6.2    | Stage 2: solve the three scissors cases                     | 333        |
| 35.6.3    | The complete Lean DAG                                       | 335        |
| 35.7      | Lean Formalization                                          | 335        |
| 35.8      | Relationship with Earlier Propositions                      | 336        |
| 35.8.1    | Proposition I.29                                            | 336        |
| 35.8.2    | Proposition I.34                                            | 337        |
| 35.8.3    | Transition to Proposition I.36                              | 337        |
| 35.9      | Hilbert and Book Zero Components Used                       | 337        |
| 35.10     | Formalization Notes                                         | 338        |
| 35.10.1   | Axiomatic status                                            | 338        |
| 35.10.2   | Hidden configuration data                                   | 338        |
| 35.10.3   | The scissors language is formal, not yet semantic           | 338        |
| 35.10.4   | Reusable helpers introduced                                 | 339        |
| 35.11     | Dependency Summary                                          | 339        |
| 35.12     | Conclusion                                                  | 340        |
| <b>36</b> | <b>Proposition I.36</b>                                     | <b>341</b> |
| 36.1      | Introduction                                                | 341        |
| 36.2      | The Classical Configuration                                 | 341        |
| 36.3      | Classical Proof                                             | 341        |
| 36.4      | The Main Issue: Constructing the Intermediate Parallelogram | 343        |
| 36.4.1    | The upper and lower parallel sides                          | 343        |
| 36.4.2    | The shifted bottom segments                                 | 343        |
| 36.4.3    | The corresponding angle                                     | 343        |
| 36.4.4    | SAS across the two bases                                    | 344        |
| 36.4.5    | Recovering the second pair of parallel sides                | 344        |
| 36.5      | Proof Architecture of the Reconstruction                    | 345        |
| 36.6      | Lean Formalization                                          | 346        |
| 36.7      | Relationship with Earlier Propositions                      | 347        |

|           |                                                              |            |
|-----------|--------------------------------------------------------------|------------|
| 36.7.1    | Proposition I.33                                             | 347        |
| 36.7.2    | Proposition I.34                                             | 347        |
| 36.7.3    | Proposition I.35                                             | 347        |
| 36.8      | Hilbert and Book Zero Components Used                        | 347        |
| 36.9      | Formalization Notes                                          | 348        |
| 36.9.1    | Geometry, content, and representation                        | 348        |
| 36.9.2    | Hidden configuration data                                    | 348        |
| 36.9.3    | Axiomatic status                                             | 349        |
| 36.9.4    | Axiom audit                                                  | 349        |
| 36.9.5    | Source audit                                                 | 350        |
| 36.10     | Dependency Summary                                           | 351        |
| 36.11     | Conclusion                                                   | 351        |
| <b>37</b> | <b>Proposition I.37</b>                                      | <b>352</b> |
| 37.1      | Introduction                                                 | 352        |
| 37.2      | The Classical Configuration                                  | 352        |
| 37.3      | Classical Proof                                              | 353        |
| 37.4      | Hilbert's Theorems 44–46 and the Halving Issue               | 354        |
| 37.5      | Proof Architecture of the Reconstruction                     | 355        |
| 37.5.1    | Step 1: nondegeneracy from the parallel hypothesis           | 355        |
| 37.5.2    | Step 2: four midpoint witnesses and Hilbert 45 twice         | 355        |
| 37.5.3    | Step 3: the common upper carrier                             | 355        |
| 37.5.4    | Step 4: equal upper sides                                    | 356        |
| 37.5.5    | Step 5: representation normalization                         | 356        |
| 37.5.6    | Step 6: order normalization on the common upper line         | 357        |
| 37.5.7    | Step 7: the orientation lemma for the strict cases           | 358        |
| 37.5.8    | Step 8: Proposition I.35 and transport back to the triangles | 358        |
| 37.6      | Lean Formalization                                           | 359        |
| 37.7      | Relationship with Earlier Propositions                       | 360        |
| 37.7.1    | Proposition I.31                                             | 360        |
| 37.7.2    | Proposition I.34                                             | 360        |
| 37.7.3    | Proposition I.35                                             | 360        |
| 37.7.4    | Proposition I.36                                             | 360        |
| 37.8      | Hilbert and Book Zero Components Used                        | 360        |
| 37.8.1    | Midpoint and midsegment infrastructure                       | 361        |
| 37.8.2    | Hilbert Theorem 45 in the scissors layer                     | 361        |
| 37.8.3    | Parallel and parallelogram infrastructure                    | 361        |
| 37.8.4    | Order, Pasch, and segment comparison                         | 362        |
| 37.9      | Formalization Notes                                          | 362        |
| 37.9.1    | Geometry, representation, and content                        | 362        |
| 37.9.2    | Hidden configuration data                                    | 362        |
| 37.9.3    | Axiomatic status                                             | 363        |
| 37.9.4    | Why the conclusion is equicomplementability                  | 363        |
| 37.9.5    | Reusable helpers introduced around I.37                      | 363        |
| 37.9.6    | Source audit                                                 | 364        |
| 37.10     | Dependency Summary                                           | 365        |
| 37.11     | Conclusion                                                   | 365        |
| <b>38</b> | <b>Proposition I.38</b>                                      | <b>366</b> |
| 38.1      | Introduction                                                 | 366        |
| 38.2      | The Classical Configuration                                  | 366        |
| 38.3      | Classical Proof                                              | 368        |

|           |                                                                     |            |
|-----------|---------------------------------------------------------------------|------------|
| 38.4      | Equal Bases by Congruent Copy and Transitivity . . . . .            | 368        |
| 38.5      | Proof Architecture of the Reconstruction . . . . .                  | 369        |
| 38.5.1    | Step 1: recovering one common base carrier . . . . .                | 369        |
| 38.5.2    | Step 2: nondegeneracy is derived, not assumed . . . . .             | 369        |
| 38.5.3    | Step 3: constructing the congruent copy on the equal base . . . . . | 370        |
| 38.5.4    | Step 4: proving that the copy is a parallelogram . . . . .          | 370        |
| 38.5.5    | Step 5: identifying the upper carrier . . . . .                     | 371        |
| 38.5.6    | Step 6: the content proof is now short . . . . .                    | 372        |
| 38.6      | Lean Formalization . . . . .                                        | 373        |
| 38.7      | Relationship with Earlier Propositions . . . . .                    | 373        |
| 38.7.1    | Proposition I.28 . . . . .                                          | 373        |
| 38.7.2    | Proposition I.33 . . . . .                                          | 374        |
| 38.7.3    | Proposition I.34 . . . . .                                          | 374        |
| 38.7.4    | Proposition I.36 . . . . .                                          | 374        |
| 38.7.5    | Proposition I.37 . . . . .                                          | 374        |
| 38.8      | Hilbert and Book Zero Components Used . . . . .                     | 374        |
| 38.8.1    | Incidence, order, and side-of-line infrastructure . . . . .         | 374        |
| 38.8.2    | Congruence construction and SAS . . . . .                           | 375        |
| 38.8.3    | Parallel and parallelogram infrastructure . . . . .                 | 375        |
| 38.8.4    | Scissors and content infrastructure . . . . .                       | 375        |
| 38.9      | Formalization Notes . . . . .                                       | 376        |
| 38.9.1    | Geometry, representation, and content . . . . .                     | 376        |
| 38.9.2    | Hidden configuration data . . . . .                                 | 376        |
| 38.9.3    | Axiomatic status . . . . .                                          | 377        |
| 38.9.4    | Why the conclusion remains equicomplementability . . . . .          | 377        |
| 38.9.5    | Reusable helpers and proposition-local helpers . . . . .            | 377        |
| 38.9.6    | Source audit . . . . .                                              | 378        |
| 38.10     | Dependency Summary . . . . .                                        | 379        |
| 38.11     | Conclusion . . . . .                                                | 379        |
| <b>39</b> | <b>Proposition I.39</b> . . . . .                                   | <b>381</b> |
| 39.1      | Introduction . . . . .                                              | 381        |
| 39.2      | The Classical Configuration . . . . .                               | 382        |
| 39.3      | Classical Proof . . . . .                                           | 382        |
| 39.4      | Property (Z), De Zolt, and the Converse Content Problem . . . . .   | 383        |
| 39.4.1    | Why I.39 is different from I.38 . . . . .                           | 383        |
| 39.4.2    | Hilbert's Theorem 48 . . . . .                                      | 384        |
| 39.4.3    | Positivity and the derived De Zolt principle . . . . .              | 384        |
| 39.5      | Proof Architecture of the Reconstruction . . . . .                  | 385        |
| 39.5.1    | Step 1: the coincident-vertex boundary case . . . . .               | 385        |
| 39.5.2    | Step 2: the already-parallel branch . . . . .                       | 386        |
| 39.5.3    | Step 3: construction of the auxiliary point . . . . .               | 386        |
| 39.5.4    | Step 4: normalization of the order on BD . . . . .                  | 386        |
| 39.5.5    | Step 5: the I.37 content comparison . . . . .                       | 387        |
| 39.5.6    | Step 6: the proper-part contradiction . . . . .                     | 387        |
| 39.6      | Lean Formalization . . . . .                                        | 388        |
| 39.7      | Relationship with Earlier Propositions . . . . .                    | 389        |
| 39.7.1    | Proposition I.31 . . . . .                                          | 389        |
| 39.7.2    | Proposition I.37 . . . . .                                          | 390        |
| 39.7.3    | Proposition I.38 . . . . .                                          | 390        |
| 39.7.4    | Proposition I.40 . . . . .                                          | 390        |
| 39.8      | Hilbert and Book Zero Components Used . . . . .                     | 390        |

|           |                                                                     |            |
|-----------|---------------------------------------------------------------------|------------|
| 39.8.1    | Incidence and order . . . . .                                       | 390        |
| 39.8.2    | SameSide and OppositeSide . . . . .                                 | 391        |
| 39.8.3    | Euclidean parallelism . . . . .                                     | 391        |
| 39.8.4    | Scissors content . . . . .                                          | 391        |
| 39.8.5    | Book Zero . . . . .                                                 | 392        |
| 39.9      | Formalization Notes . . . . .                                       | 392        |
| 39.9.1    | Geometry, representation, and content . . . . .                     | 392        |
| 39.9.2    | Hidden configuration data . . . . .                                 | 393        |
| 39.9.3    | Axiomatic status . . . . .                                          | 393        |
| 39.9.4    | Why Hilbert Theorem 48 is not an imported axiom . . . . .           | 394        |
| 39.9.5    | Source audit . . . . .                                              | 394        |
| 39.10     | Dependency Summary . . . . .                                        | 395        |
| 39.11     | Conclusion . . . . .                                                | 396        |
| <b>40</b> | <b>Proposition I.40</b>                                             | <b>397</b> |
| 40.1      | Introduction . . . . .                                              | 397        |
| 40.2      | The Classical Configuration . . . . .                               | 398        |
| 40.3      | Classical Proof . . . . .                                           | 398        |
| 40.4      | Equal Bases by Reduction to the Same-Base Converse . . . . .        | 400        |
| 40.5      | Proof Architecture of the Reconstruction . . . . .                  | 401        |
| 40.5.1    | Step 1: putting the second base on the common carrier . . . . .     | 401        |
| 40.5.2    | Step 2: reusing the I.38 congruent-copy construction . . . . .      | 402        |
| 40.5.3    | Step 3: transporting equal content to the copied triangle . . . . . | 402        |
| 40.5.4    | Step 4: reducing to I.39 on the common base $EF$ . . . . .          | 403        |
| 40.5.5    | Step 5: identifying the final top line . . . . .                    | 403        |
| 40.6      | Lean Formalization . . . . .                                        | 403        |
| 40.7      | Relationship with Earlier Propositions . . . . .                    | 405        |
| 40.7.1    | Proposition I.38 . . . . .                                          | 405        |
| 40.7.2    | Proposition I.39 . . . . .                                          | 405        |
| 40.7.3    | Proposition I.37 . . . . .                                          | 405        |
| 40.8      | Hilbert and Book Zero Components Used . . . . .                     | 405        |
| 40.8.1    | Incidence and order . . . . .                                       | 405        |
| 40.8.2    | Side-of-line geometry . . . . .                                     | 406        |
| 40.8.3    | Parallelism and parallelograms . . . . .                            | 406        |
| 40.8.4    | Scissors content . . . . .                                          | 406        |
| 40.9      | Source Audit . . . . .                                              | 406        |
| 40.9.1    | Euclid and Joyce . . . . .                                          | 406        |
| 40.9.2    | Wilkins . . . . .                                                   | 407        |
| 40.9.3    | Hilbert and Hartshorne . . . . .                                    | 407        |
| 40.9.4    | Borsuk–Szmielew and Greenberg . . . . .                             | 407        |
| 40.9.5    | Beeson . . . . .                                                    | 407        |
| 40.10     | Formalization Notes . . . . .                                       | 407        |
| 40.10.1   | Geometry . . . . .                                                  | 407        |
| 40.10.2   | Content . . . . .                                                   | 408        |
| 40.10.3   | Representation . . . . .                                            | 408        |
| 40.10.4   | Axiomatic status . . . . .                                          | 408        |
| 40.11     | Dependency Summary . . . . .                                        | 409        |
| 40.12     | Conclusion . . . . .                                                | 409        |
| <b>41</b> | <b>Proposition I.41</b>                                             | <b>411</b> |
| 41.1      | Introduction . . . . .                                              | 411        |
| 41.2      | The Classical Configuration . . . . .                               | 412        |

|           |                                                                    |            |
|-----------|--------------------------------------------------------------------|------------|
| 41.3      | Classical Proof                                                    | 412        |
| 41.4      | Doubling in the Scissors Calculus                                  | 413        |
| 41.5      | Proof Architecture of the Reconstruction                           | 414        |
| 41.5.1    | Step 1: replace the second diagonal triangle by the first          | 414        |
| 41.5.2    | Step 2: apply Proposition I.37 on the common base                  | 414        |
| 41.5.3    | Step 3: add the equal-content relation to itself                   | 415        |
| 41.5.4    | Step 4: transport from the doubled triangle to the parallelogram   | 415        |
| 41.6      | Lean Formalization                                                 | 416        |
| 41.7      | Relationship with Earlier Propositions                             | 416        |
| 41.7.1    | Proposition I.34                                                   | 416        |
| 41.7.2    | Proposition I.37                                                   | 417        |
| 41.7.3    | Propositions I.35–I.40                                             | 417        |
| 41.8      | Hilbert and Book Zero Components Used                              | 417        |
| 41.8.1    | Triangle congruence and parallelogram infrastructure               | 417        |
| 41.8.2    | Same-base triangle content                                         | 417        |
| 41.8.3    | Scissors representation                                            | 418        |
| 41.9      | Source Audit                                                       | 418        |
| 41.9.1    | Euclid and Joyce                                                   | 418        |
| 41.9.2    | Hartshorne                                                         | 418        |
| 41.9.3    | Hilbert, Borsuk–Szmielew, Greenberg, and Beeson                    | 418        |
| 41.10     | Formalization Notes                                                | 419        |
| 41.10.1   | Geometry                                                           | 419        |
| 41.10.2   | Content                                                            | 419        |
| 41.10.3   | Representation                                                     | 419        |
| 41.10.4   | Axiomatic status                                                   | 419        |
| 41.11     | Dependency Summary                                                 | 420        |
| 41.12     | Conclusion                                                         | 420        |
| <b>42</b> | <b>Proposition I.42</b>                                            | <b>422</b> |
| 42.1      | Introduction                                                       | 422        |
| 42.2      | The Classical Configuration                                        | 423        |
| 42.3      | Classical Proof                                                    | 423        |
| 42.4      | Application of Areas and the Hidden Intersection                   | 424        |
| 42.5      | Proof Architecture of the Reconstruction                           | 425        |
| 42.5.1    | Step 1: midpoint and equal-content halves                          | 425        |
| 42.5.2    | Step 2: copy the angle on the correct side and draw the parallel   | 425        |
| 42.5.3    | Step 3: prove that the two carriers meet                           | 426        |
| 42.5.4    | Step 4: orient the intersection onto the correct ray               | 426        |
| 42.5.5    | Step 5: complete the parallelogram and identify its upper carrier  | 427        |
| 42.5.6    | Step 6: use I.41 on a genuine side and transport along the carrier | 427        |
| 42.5.7    | Step 7: assemble the content equality                              | 427        |
| 42.6      | Lean Formalization                                                 | 428        |
| 42.6.1    | The midpoint-content helper                                        | 429        |
| 42.6.2    | The construction helper                                            | 429        |
| 42.6.3    | The carrier form of I.41                                           | 430        |
| 42.7      | Relationship with Earlier Propositions                             | 430        |
| 42.7.1    | Proposition I.10                                                   | 430        |
| 42.7.2    | Propositions I.23 and I.31                                         | 430        |
| 42.7.3    | Proposition I.37                                                   | 431        |
| 42.7.4    | Proposition I.38                                                   | 431        |
| 42.7.5    | Proposition I.41                                                   | 431        |
| 42.7.6    | Propositions I.44 and I.45                                         | 431        |

|           |                                                                |            |
|-----------|----------------------------------------------------------------|------------|
| 42.8      | Hilbert and Book Zero Components Used . . . . .                | 431        |
| 42.8.1    | Midpoint, betweenness, and segment construction . . . . .      | 431        |
| 42.8.2    | Angle construction, SameSide, and SameRay . . . . .            | 432        |
| 42.8.3    | Euclidean parallel and intersection infrastructure . . . . .   | 432        |
| 42.8.4    | Parallelogram infrastructure . . . . .                         | 432        |
| 42.8.5    | Scissors content infrastructure . . . . .                      | 432        |
| 42.9      | Source Audit . . . . .                                         | 433        |
| 42.9.1    | Euclid, Heath, and Joyce . . . . .                             | 433        |
| 42.9.2    | Wilkins . . . . .                                              | 433        |
| 42.9.3    | Beeson . . . . .                                               | 433        |
| 42.9.4    | Hilbert . . . . .                                              | 433        |
| 42.9.5    | Hartshorne . . . . .                                           | 433        |
| 42.9.6    | Borsuk–Szmielew and Greenberg . . . . .                        | 434        |
| 42.10     | Formalization Notes . . . . .                                  | 434        |
| 42.10.1   | Geometry . . . . .                                             | 434        |
| 42.10.2   | Content . . . . .                                              | 435        |
| 42.10.3   | Representation . . . . .                                       | 435        |
| 42.10.4   | Axiomatic status . . . . .                                     | 435        |
| 42.11     | Dependency Summary . . . . .                                   | 436        |
| 42.12     | Conclusion . . . . .                                           | 436        |
| <b>43</b> | <b>Proposition I.43</b> . . . . .                              | <b>438</b> |
| 43.1      | Introduction . . . . .                                         | 438        |
| 43.2      | The Classical Configuration . . . . .                          | 438        |
| 43.3      | Classical Proof . . . . .                                      | 439        |
| 43.4      | Complements and Equicomplementability . . . . .                | 440        |
| 43.5      | Proof Architecture of the Reconstruction . . . . .             | 440        |
| 43.5.1    | Step 1: decompose the first large diagonal triangle . . . . .  | 440        |
| 43.5.2    | Step 2: decompose the second large diagonal triangle . . . . . | 440        |
| 43.5.3    | Step 3: obtain the three I.34 equalities . . . . .             | 441        |
| 43.5.4    | Step 4: assemble and regroup the full decompositions . . . . . | 441        |
| 43.5.5    | Step 5: package the equicomplementability witness . . . . .    | 441        |
| 43.6      | Lean Formalization . . . . .                                   | 442        |
| 43.7      | Relationship with Earlier Propositions . . . . .               | 443        |
| 43.7.1    | Proposition I.34 . . . . .                                     | 443        |
| 43.7.2    | Propositions I.35–I.41 . . . . .                               | 443        |
| 43.7.3    | Proposition I.42 . . . . .                                     | 443        |
| 43.7.4    | Proposition I.44 . . . . .                                     | 444        |
| 43.8      | Hilbert and Book Zero Components Used . . . . .                | 444        |
| 43.8.1    | Betweenness and incidence . . . . .                            | 444        |
| 43.8.2    | Parallelogram diagonal congruence . . . . .                    | 444        |
| 43.8.3    | Scissors content infrastructure . . . . .                      | 444        |
| 43.9      | Source Audit . . . . .                                         | 444        |
| 43.9.1    | Euclid, Heath, and Joyce . . . . .                             | 444        |
| 43.9.2    | Wilkins . . . . .                                              | 445        |
| 43.9.3    | Hilbert . . . . .                                              | 445        |
| 43.9.4    | Hartshorne . . . . .                                           | 445        |
| 43.9.5    | Beeson, Borsuk–Szmielew, and Greenberg . . . . .               | 445        |
| 43.10     | Formalization Notes . . . . .                                  | 445        |
| 43.10.1   | Geometry . . . . .                                             | 445        |
| 43.10.2   | Content . . . . .                                              | 446        |
| 43.10.3   | Representation . . . . .                                       | 446        |

|           |                                                            |            |
|-----------|------------------------------------------------------------|------------|
| 43.10.4   | Axiomatic status                                           | 446        |
| 43.11     | Dependency Summary                                         | 447        |
| 43.12     | Conclusion                                                 | 447        |
| <b>44</b> | <b>Proposition I.44</b>                                    | <b>448</b> |
| 44.1      | Introduction                                               | 448        |
| 44.2      | The Classical Configuration                                | 449        |
| 44.3      | Classical Proof                                            | 450        |
| 44.4      | Application of Areas and Rigid Placement                   | 451        |
| 44.5      | Proof Architecture of the Reconstruction                   | 452        |
| 44.5.1    | Step 1: obtain a content-and-angle parallelogram from I.42 | 452        |
| 44.5.2    | Step 2: rigidly place the I.42 parallelogram beyond B      | 453        |
| 44.5.3    | Step 3: certify the content of the rigid copy              | 453        |
| 44.5.4    | Step 4: construct the positioned Euclid configuration      | 453        |
| 44.5.5    | Step 5: recognize the four parallelograms needed by I.43   | 454        |
| 44.5.6    | Step 6: use I.43 and transfer the angle                    | 454        |
| 44.5.7    | Step 7: chain the two content transports                   | 455        |
| 44.6      | Lean Formalization                                         | 456        |
| 44.7      | Relationship with Earlier Propositions                     | 458        |
| 44.7.1    | Proposition I.15                                           | 458        |
| 44.7.2    | Proposition I.31 and the Euclidean parallel layer          | 458        |
| 44.7.3    | Proposition I.42                                           | 458        |
| 44.7.4    | Proposition I.43                                           | 458        |
| 44.7.5    | Propositions I.34–I.41                                     | 459        |
| 44.7.6    | Proposition I.45                                           | 459        |
| 44.8      | Hilbert and Book Zero Components Used                      | 459        |
| 44.8.1    | Betweenness, extensions, and carrier incidence             | 459        |
| 44.8.2    | SameSide and OppositeSide                                  | 459        |
| 44.8.3    | Segment construction, SameRay, and angle construction      | 460        |
| 44.8.4    | Euclidean parallelism                                      | 460        |
| 44.8.5    | Congruence and parallelogram infrastructure                | 460        |
| 44.8.6    | Scissors content infrastructure                            | 460        |
| 44.9      | Source Audit                                               | 461        |
| 44.9.1    | Euclid, Heath, and Joyce                                   | 461        |
| 44.9.2    | Wilkins                                                    | 461        |
| 44.9.3    | Hilbert and Borsuk–Szmielew                                | 461        |
| 44.9.4    | Hartshorne                                                 | 462        |
| 44.9.5    | Greenberg                                                  | 462        |
| 44.9.6    | Beeson proof corpus                                        | 462        |
| 44.10     | Formalization Notes                                        | 462        |
| 44.10.1   | Geometry                                                   | 462        |
| 44.10.2   | Content                                                    | 463        |
| 44.10.3   | Representation                                             | 463        |
| 44.10.4   | Axiomatic status                                           | 463        |
| 44.11     | Dependency Summary                                         | 464        |
| 44.12     | Conclusion                                                 | 465        |
| <b>45</b> | <b>Proposition I.45</b>                                    | <b>466</b> |
| 45.1      | Introduction                                               | 466        |
| 45.2      | The Classical Configuration                                | 467        |
| 45.3      | Classical Proof                                            | 468        |
| 45.4      | Triangulated Figures and the Representation Boundary       | 469        |

|          |                                                                          |            |
|----------|--------------------------------------------------------------------------|------------|
| 45.5     | Proof Architecture of the Reconstruction . . . . .                       | 470        |
| 45.5.1   | Step 1: turn a triangle list into a scissors term . . . . .              | 470        |
| 45.5.2   | Step 2: prove the geometry of pasting adjacent parallelograms . . . . .  | 471        |
| 45.5.3   | Step 3: refine the paste into a scissors identity . . . . .              | 471        |
| 45.5.4   | Step 4: preserve the prescribed angle . . . . .                          | 472        |
| 45.5.5   | Step 5: reduce the orientation problem to one strict order . . . . .     | 472        |
| 45.5.6   | Step 6: construct the new triangle piece on the correct side . . . . .   | 473        |
| 45.5.7   | Step 7: package one induction extension . . . . .                        | 474        |
| 45.5.8   | Step 8: induction on the triangle list . . . . .                         | 475        |
| 45.6     | Lean Formalization . . . . .                                             | 475        |
| 45.6.1   | The rectilineal term . . . . .                                           | 475        |
| 45.6.2   | The adjacent-piece theorem . . . . .                                     | 476        |
| 45.6.3   | The induction-step theorem . . . . .                                     | 476        |
| 45.6.4   | The public proposition . . . . .                                         | 477        |
| 45.7     | Relationship with Earlier Propositions . . . . .                         | 477        |
| 45.7.1   | Proposition I.42 . . . . .                                               | 477        |
| 45.7.2   | Proposition I.44 . . . . .                                               | 477        |
| 45.7.3   | Proposition I.43 . . . . .                                               | 478        |
| 45.7.4   | The Euclidean parallel block . . . . .                                   | 478        |
| 45.7.5   | The content block I.35–I.41 . . . . .                                    | 478        |
| 45.8     | Hilbert and Book Zero Components Used . . . . .                          | 478        |
| 45.8.1   | Betweenness and SameRay . . . . .                                        | 478        |
| 45.8.2   | SameSide, OppositeSide, and plane separation . . . . .                   | 479        |
| 45.8.3   | Pasch and crossing . . . . .                                             | 479        |
| 45.8.4   | Euclidean uniqueness of parallels . . . . .                              | 479        |
| 45.8.5   | Congruence and angle construction . . . . .                              | 479        |
| 45.8.6   | Scissors operations . . . . .                                            | 479        |
| 45.9     | Formalization Notes . . . . .                                            | 480        |
| 45.9.1   | Geometry . . . . .                                                       | 480        |
| 45.9.2   | Content . . . . .                                                        | 480        |
| 45.9.3   | Representation . . . . .                                                 | 481        |
| 45.9.4   | Axiomatic status . . . . .                                               | 481        |
| 45.10    | Dependency Summary . . . . .                                             | 482        |
| 45.11    | Conclusion . . . . .                                                     | 483        |
| <b>A</b> | <b>Greenberg: A Hilbert-Style Reconstruction</b> . . . . .               | <b>484</b> |
| A.1      | Purpose and scope . . . . .                                              | 484        |
| A.2      | Why the fourth edition matters . . . . .                                 | 485        |
| A.3      | Comparison table . . . . .                                               | 485        |
| A.4      | Structural observations from the audit . . . . .                         | 495        |
| A.5      | Notes that should be retained for future work . . . . .                  | 496        |
| A.6      | I.43–I.45: the same construction/content split at larger scale . . . . . | 497        |
| A.7      | Result of the fourth-edition query . . . . .                             | 498        |
| A.8      | Current checkpoint . . . . .                                             | 499        |
| <b>B</b> | <b>Hartshorne: Euclid in a Hilbert Plane</b> . . . . .                   | <b>501</b> |
| B.1      | Purpose and scope . . . . .                                              | 501        |
| B.2      | Hartshorne’s organization of the problem . . . . .                       | 501        |
| B.3      | Comparison table . . . . .                                               | 502        |
| B.4      | Foundational observations retained from Hartshorne . . . . .             | 509        |
| B.5      | I.43–I.45: completion of the application-of-areas block . . . . .        | 511        |
| B.6      | Result of the Hartshorne audit . . . . .                                 | 511        |



|          |                                                                                      |            |
|----------|--------------------------------------------------------------------------------------|------------|
| B.7      | Current checkpoint . . . . .                                                         | 512        |
| <b>C</b> | <b>Wilkins: The Euclidean Dependency Graph</b>                                       | <b>514</b> |
| C.1      | Purpose and scope . . . . .                                                          | 514        |
| C.2      | Proposition-by-proposition audit . . . . .                                           | 514        |
| C.3      | Major dependency blocks . . . . .                                                    | 525        |
| C.3.1    | Construction and continuity: I.1–I.3, I.9–I.12, I.22–I.23 . . . . .                  | 525        |
| C.3.2    | Congruence and triangle comparison: I.4–I.8, I.18–I.26 . . . . .                     | 526        |
| C.3.3    | The neutral-to-Euclidean parallel boundary: I.27–I.34 . . . . .                      | 526        |
| C.3.4    | The area and application block: I.35–I.45 . . . . .                                  | 527        |
| C.4      | Configuration completeness . . . . .                                                 | 528        |
| C.5      | Converses, equivalences, and alternative proof paths . . . . .                       | 528        |
| C.6      | Common Notions as proof operations . . . . .                                         | 529        |
| C.7      | Source-audit observations retained for the formal project . . . . .                  | 530        |
| C.8      | I.43–I.45: complements and finite application of areas . . . . .                     | 531        |
| C.9      | Conclusion . . . . .                                                                 | 531        |
| <b>D</b> | <b>Hilbert: The <i>Grundlagen</i> Architecture</b>                                   | <b>533</b> |
| D.1      | Purpose and scope . . . . .                                                          | 533        |
| D.2      | Hilbert’s organization of the elementary theory . . . . .                            | 534        |
| D.3      | Comparison table . . . . .                                                           | 534        |
| D.4      | Hilbert counterparts and structural reductions in the I.1–I.45 block . . . . .       | 543        |
| D.5      | The neutral theory before Group IV . . . . .                                         | 544        |
| D.6      | The exact role of Group IV . . . . .                                                 | 544        |
| D.7      | Congruence architecture and the status of SSS . . . . .                              | 545        |
| D.8      | Continuity and the I.1–I.45 checkpoint . . . . .                                     | 546        |
| D.9      | The transition at I.35: from congruence to plane area . . . . .                      | 546        |
| D.10     | The I.36–I.42 extension: forward content, converse, and construction . . . . .       | 547        |
| D.10.1   | I.36–I.38: the forward content theorems . . . . .                                    | 547        |
| D.10.2   | I.39–I.40: the converse boundary . . . . .                                           | 547        |
| D.10.3   | I.41–I.42: composition and application of areas . . . . .                            | 548        |
| D.10.4   | I.43–I.45: from complement calculus to finite polygon construction . . . . .         | 549        |
| D.11     | Result of the Grundlagen audit . . . . .                                             | 549        |
| D.12     | Current checkpoint . . . . .                                                         | 550        |
| <b>E</b> | <b>Borsuk–Szmielw: The Axiomatic Layers</b>                                          | <b>552</b> |
| E.1      | Why this source matters . . . . .                                                    | 552        |
| E.2      | Layer notation . . . . .                                                             | 553        |
| E.3      | Euclid in the Borsuk–Szmielw development . . . . .                                   | 553        |
| E.4      | The strongest architectural correspondences . . . . .                                | 559        |
| E.4.1    | Transport is primitive-strength infrastructure . . . . .                             | 560        |
| E.4.2    | Pairs of Euclidean propositions become biconditionals . . . . .                      | 560        |
| E.4.3    | Midpoints and bisectors include positional information . . . . .                     | 560        |
| E.5      | The neutral parallel theory is already large . . . . .                               | 560        |
| E.6      | Theorem 12 of Chapter V as an interface theorem . . . . .                            | 561        |
| E.7      | What the book contributes at I.35–I.45 . . . . .                                     | 562        |
| E.8      | I.42 as a geometry/content separation test . . . . .                                 | 563        |
| E.9      | I.43–I.45: representation and construction without equal-content semantics . . . . . | 563        |
| E.10     | Consequences for the formal library . . . . .                                        | 564        |
| E.10.1   | Hilbert interface candidates . . . . .                                               | 564        |
| E.10.2   | Book Zero candidates . . . . .                                                       | 564        |
| E.10.3   | Euclidean interface candidates . . . . .                                             | 565        |

|                           |            |
|---------------------------|------------|
| E.11 Conclusion . . . . . | 565        |
| <b>Bibliography</b>       | <b>566</b> |

# Introduction

## From Hilbert’s Foundations to Synthetic Practice

This volume documents a formal reconstruction of the first part of Euclid’s *Elements*, Book I, over a Hilbert-style foundation in Lean 4.

The original objective was comparatively modest: to determine whether classical Euclidean proofs could be reconstructed faithfully in a modern proof assistant without replacing synthetic geometry by coordinates, real-number measurement, or algebraic computation.

The reconstruction of Propositions I.1–I.36 has revealed a more specific problem.

Hilbert’s axioms provide a foundation for geometry. They separate incidence, order, congruence, and continuity, and they make explicit many assumptions that remain implicit in Euclid. But there is still a substantial distance between such a foundational system and the language in which ordinary synthetic proofs are actually carried out.

A mathematician moves almost invisibly between statements such as:

- three points are collinear in a different order;
- one point lies between two others;
- two descriptions determine the same ray;
- an angle may be transported along congruent rays;
- a proper part of a segment is smaller than the whole;
- equal parts may be added to or removed from segment inequalities;
- a configuration suggested by a diagram really has the required order and separation properties.

None of these operations is mysterious. But neither are they merely syntactic details. In a formal proof they must be represented by definitions, lemmas, and explicit chains of inference.

This is the layer that the present reconstruction has progressively made visible.

## Book Zero

Beeson, Narboux, and Wiedijk use the informal name *Book Zero* for fundamental preliminary theorems that are used in Euclid’s Book I but do not themselves depend on propositions of Book

I. Their Book Zero includes elementary results about congruence, betweenness, collinearity and noncollinearity, segment order, rays, laying off segments, angle comparison, and plane separation.<sup>1</sup>

The CGJteamLab development adopts this terminology. In the present Hilbert-style reconstruction, Book Zero plays the same structural role, but now specifically as an intermediate theory between Hilbert’s foundational axioms and the level at which ordinary synthetic arguments are naturally conducted. The formalization of Book I suggests the following interpretation.

Book Zero functions as an intermediate language between Hilbert’s axioms and classical synthetic reasoning:

$$\boxed{\text{Hilbert axioms} \longrightarrow \text{Book Zero} \longrightarrow \text{classical synthetic geometry}}$$

Its purpose is not to replace Hilbert’s foundation and not to introduce a second axiomatic system. Rather, it packages the elementary consequences of that foundation at the level at which geometric arguments naturally operate.

This role becomes particularly clear in Proposition I.21. There the proof requires what looks informally like arithmetic with segments: parts are added, removed, compared, and inequalities are transported through congruence. No numerical length function is necessary. Betweenness, congruence, and strict segment comparison already carry the required arithmetic through geometric witnesses.

Thus Book Zero is not merely a library for shortening Lean proofs. It is an attempt to identify the missing operational layer between foundational geometry and working synthetic mathematics.

## Euclid and Hilbert Do Not Have the Same Dependency Graph

The reconstruction also makes clear that the order of Euclid’s propositions cannot simply be identified with the logical dependency order of a Hilbert-style theory.

Several different situations occur.

First, some Euclidean propositions correspond to principles that are already foundational in Hilbert geometry. Euclid’s Proposition I.4, for example, proves SAS by superposition, whereas SAS belongs to the Hilbert congruence structure. Similarly, the angle-copying construction of Proposition I.23 is already contained in Hilbert’s angle-construction principle.

Second, some Euclidean propositions genuinely require more than neutral Hilbert geometry. Propositions I.1 and I.22 expose intersection or continuity assumptions that cannot be obtained from incidence, order, and congruence alone. A different boundary appears at Proposition I.29. The criteria I.27 and I.28 derive parallelism from angle relations in neutral geometry, whereas I.29 reverses the direction and requires the Euclidean uniqueness of parallels.

The distinction is sharper than the numerical order of the *Elements* suggests. Proposition I.31 comes after I.29, but its existence statement is already neutral: an angle can be transported and I.27 then proves that the constructed line is parallel. Only uniqueness of that parallel requires the Euclidean parallel principle. Thus the sequence of propositions and the logical layers of the theory cross one another.

A second structural boundary appears at Proposition I.35. Here the subject changes from

---

<sup>1</sup>Michael Beeson, Julien Narboux, and Freek Wiedijk, “Proof-checking Euclid,” *Annals of Mathematics and Artificial Intelligence* 85 (2019), 213–257, doi:10.1007/s10472-018-9606-x.

congruence and parallelism to equality of rectilinear figures. The reconstruction does not introduce numerical area. Instead it follows Hilbert’s Chapter IV and represents the relevant equality by a synthetic theory of equidecomposition and equicomplementability. Proposition I.36 immediately confirms that this is not an isolated device: its content argument is obtained by applying I.35 twice and using transitivity, while the new work lies in constructing the intermediate parallelogram.

Third, and most importantly for the present project, many propositions require no stronger axiom but depend on a considerable intermediate theory of order, rays, segment comparison, angle comparison, noncollinearity, separation, and transport.

It is this third class that explains much of the apparent complexity of formal synthetic geometry.

## What Formalization Reveals

A classical diagram compresses information.

It may show immediately that a point lies inside an angle, that two points occur on the same ray, that one point is between two others, or that an auxiliary construction has a particular orientation. A proof assistant cannot obtain these facts by looking at the figure.

Consequently formalization repeatedly exposes obligations that are easy to overlook in textbook prose:

$$\text{diagrammatic configuration} \not\Rightarrow \text{formal order information.}$$

Proposition I.24 provides a particularly clear example. Euclid’s construction suggests one relative position of the auxiliary points, but that position is not forced merely by the stated construction. The formal proof must analyze the possible configurations explicitly.

The same phenomenon reappears in the theory of parallels. In I.27 the informal argument must exclude an intersection on either side of the transversal. In I.29 the decisive step is not another angle calculation but uniqueness: a neutrally constructed candidate parallel must be identified with the given parallel. Proposition I.31 then shows that existence and uniqueness, often compressed into one familiar statement, belong to different logical layers.

Such phenomena should not be regarded simply as inconveniences caused by Lean. They reveal logical structure already present in the mathematics.

## The Purpose of This Documentation

The chapters that follow are therefore not intended merely as line-by-line explanations of Lean source files.

For each proposition we record several distinct objects:

- Euclid’s classical construction and proof;
- the corresponding reconstruction over the Hilbert foundation;
- the actual dependency architecture of the Lean theorem;

- the Hilbert and Book Zero components used in the proof;
- formal obligations hidden by the traditional diagram;
- the relationship of the proposition to earlier and later propositions of Book I.

The appendices compare this reconstruction with the Hilbert-style developments of Greenberg and Hartshorne and with Wilkins's detailed study of the internal dependency structure of Euclid's Book I. These comparisons are not included merely as historical commentary. They help distinguish genuine foundational dependencies from intermediate lemmas that a formal development must make explicit, and they expose alternative proof paths, converse relations, and configuration cases hidden by the linear order of the *Elements*.

## Current Scope

The present volume covers Euclid Book I through Proposition I.36.

Thirty-six propositions are still too small a sample from which to claim a final architecture for synthetic geometry. They do, however, cross two genuine structural boundaries. Proposition I.29 marks the transition from neutral geometry to the Euclidean theory of parallels. Proposition I.35 marks a second transition, from congruence and parallelism to the synthetic theory of equality of rectilinear figures.

The treatment of I.35–I.36 suggests that this second transition can also be handled without numerical measurement. Hilbert's equicomplementability theory and the formal scissors calculus provide a synthetic content layer which can be reused by subsequent propositions.

Nevertheless, the evidence is already substantial enough to identify a recurring structure.

Hilbert supplied the foundational theory. Euclid supplied the classical sequence of geometric problems and proofs. Between them lies a layer of elementary synthetic operations that is normally left implicit because mathematicians use it automatically.

Formal reconstruction forces that layer into view.

The principal question of the project is therefore no longer only

Can Euclid's proofs be formalized over Hilbert geometry?

It has become the more structural question

What intermediate theory is required to turn Hilbert's axioms into a practical language for classical synthetic geometry?

The reconstruction of Book I is one way of investigating that question.

# List of Figures

|      |                                                                                    |     |
|------|------------------------------------------------------------------------------------|-----|
| 1.2  | Constructing an equilateral triangle by intersecting circles . . . . .             | 2   |
| 2.2  | Transporting a segment with two circles and an equilateral triangle . . . . .      | 9   |
| 3.2  | Cutting off a prescribed segment by a circle . . . . .                             | 16  |
| 4.1  | SAS data for triangle congruence . . . . .                                         | 22  |
| 5.2  | Extension construction for the isosceles base-angle theorem . . . . .              | 30  |
| 6.2  | Reductio for the converse isosceles theorem by cutting off an equal side . . . . . | 37  |
| 7.1  | The forbidden double-isosceles configuration . . . . .                             | 45  |
| 7.2  | Same-side contradiction after joining the rival vertices . . . . .                 | 46  |
| 8.2  | Superposition forces coincidence of the third vertex . . . . .                     | 54  |
| 9.2  | Angle bisection by an equilateral triangle and SSS . . . . .                       | 62  |
| 10.2 | Segment bisection by angle bisection and SAS . . . . .                             | 71  |
| 11.1 | Perpendicular construction from equal adjacent angles . . . . .                    | 78  |
| 12.2 | Opposite-side choice in the external perpendicular construction . . . . .          | 91  |
| 12.3 | Equal radii from two circle-line intersections . . . . .                           | 91  |
| 12.4 | Euclid's external perpendicular construction completed by SSS . . . . .            | 92  |
| 12.5 | Hilbert reflection by angle construction and segment layoff . . . . .              | 95  |
| 12.6 | Perpendicular-bisector argument in the Hilbert reconstruction . . . . .            | 96  |
| 13.1 | Decomposing a straight angle into adjacent angles . . . . .                        | 103 |
| 13.2 | Perpendicular decomposition of the unequal-angle case . . . . .                    | 104 |

|      |                                                                                |     |
|------|--------------------------------------------------------------------------------|-----|
| 14.1 | Supplementary adjacent angles forcing a straight line . . . . .                | 114 |
| 14.2 | Comparing the given ray with the constructed opposite ray . . . . .            | 115 |
| 14.3 | Angle-construction uniqueness in the noncollinear branch . . . . .             | 120 |
| 15.1 | Vertical angles formed by two intersecting straight lines . . . . .            | 129 |
| 16.1 | Exterior angle formed by extending a side of a triangle . . . . .              | 143 |
| 16.2 | Midpoint-and-extension proof of the exterior-angle theorem . . . . .           | 144 |
| 18.1 | An isosceles subtriangle converts side order into angle order . . . . .        | 165 |
| 19.1 | A larger angle forces the opposite side to be larger . . . . .                 | 176 |
| 20.1 | Triangle inequality by side extension and an isosceles triangle . . . . .      | 186 |
| 21.1 | An interior broken path compared with two sides of a triangle . . . . .        | 198 |
| 21.2 | Synthetic encoding of segment sums without numerical lengths . . . . .         | 200 |
| 22.1 | Constructing a triangle from three prescribed side lengths . . . . .           | 210 |
| 22.2 | Boundary and failure cases for the triangle inequalities . . . . .             | 211 |
| 23.2 | Euclid's triangle construction for copying an angle . . . . .                  | 227 |
| 23.3 | Hilbert angle construction with explicit side choice . . . . .                 | 230 |
| 24.1 | Three positional cases hidden in Euclid's proof . . . . .                      | 237 |
| 24.2 | Hinge comparison through nested angles in Euclid's treated case . . . . .      | 239 |
| 24.3 | Order trichotomy in the formal hinge proof . . . . .                           | 240 |
| 25.1 | A larger third side forces a larger included angle . . . . .                   | 248 |
| 26.1 | Two-angle congruence data for ASA and SAA . . . . .                            | 255 |
| 26.2 | ASA reductio by cutting the included side . . . . .                            | 255 |
| 26.3 | SAA reductio by cutting the opposite side . . . . .                            | 256 |
| 27.1 | Equal alternate interior angles as a criterion for parallelism . . . . .       | 266 |
| 27.2 | One direction of intersection contradicts the exterior-angle theorem . . . . . | 266 |
| 27.3 | The opposite direction gives the reverse contradiction . . . . .               | 267 |
| 28.2 | Corresponding-angle criterion reduced to alternate angles . . . . .            | 276 |
| 28.3 | Same-side supplementary-angle criterion reduced to alternate angles . . . . .  | 277 |



|      |                                                                                   |     |
|------|-----------------------------------------------------------------------------------|-----|
| 29.1 | Parallel lines cut by a transversal . . . . .                                     | 281 |
| 29.2 | Parallelism implies congruent alternate interior angles . . . . .                 | 282 |
| 29.3 | Corresponding-angle congruence from vertical angles . . . . .                     | 283 |
| 30.1 | Transitivity of parallelism through a common parallel . . . . .                   | 290 |
| 30.2 | Uniqueness of parallels as the Hilbert contradiction . . . . .                    | 292 |
| 31.2 | Constructing a parallel by copying an alternate angle . . . . .                   | 296 |
| 31.3 | Neutral existence of a parallel through an external point . . . . .               | 300 |
| 32.1 | Synthetic decomposition of an exterior angle by an auxiliary parallel . . . . .   | 303 |
| 33.1 | Equal parallel segments joined toward the same parts . . . . .                    | 313 |
| 34.1 | A diagonal divides a parallelogram into congruent triangles . . . . .             | 319 |
| 35.1 | Three relative positions of parallelograms on the same base . . . . .             | 327 |
| 35.2 | Triangulation before the diagonal flip . . . . .                                  | 331 |
| 35.3 | Common refinement at the diagonal crossing . . . . .                              | 332 |
| 35.4 | Triangulation after the diagonal flip . . . . .                                   | 332 |
| 35.5 | Equicomplementability by adding the same triangular complement . . . . .          | 334 |
| 35.6 | Scissors decomposition of the first augmented parallelogram . . . . .             | 334 |
| 35.7 | Congruent-triangle replacement in the scissors decomposition . . . . .            | 335 |
| 35.8 | Scissors reconstruction of the second augmented parallelogram . . . . .           | 335 |
| 36.1 | An intermediate parallelogram links equal bases in the same parallels . . . . .   | 342 |
| 37.1 | Midpoint reduction of same-base triangles to parallelograms . . . . .             | 353 |
| 38.1 | A congruent copy reduces equal bases to a common base . . . . .                   | 367 |
| 39.1 | The two proper-part alternatives in the I.39 reductio . . . . .                   | 382 |
| 40.1 | Equal bases reduce I.40 to the same-base converse I.39 . . . . .                  | 399 |
| 41.1 | Diagonal bisection and same-base content comparison in I.41 . . . . .             | 412 |
| 42.1 | Midpoint reduction, ray intersection, and the constructed parallelogram . . . . . | 423 |
| 43.1 | Complementary regions around a divided diagonal . . . . .                         | 439 |

|                                                        |     |
|--------------------------------------------------------|-----|
| 44.1 Complement transfer about the diameter . . . . .  | 450 |
| 44.2 Rigid placement on the prescribed base . . . . .  | 452 |
| 45.1 Pasting adjacent parallelograms . . . . .         | 468 |
| 45.2 Oriented placement on the extension ray . . . . . | 474 |

# Chapter 1

## Proposition I.1

Euclid's first proposition in Book I is usually presented as the straightforward construction obtained from the intersection of two circles. A formal reconstruction, however, reveals two logically independent problems: the existence of a common point of the corresponding circles and the proof that the resulting point does not lie on the line determined by the given base.

The proof of non-collinearity relies on the classical idea that a proper part of a segment cannot be congruent to the entire segment. This argument is well known from modern axiomatic reconstructions of geometry, including the work of Michael Beeson and his collaborators.

The purpose of this note is not to present a new geometric proof. Rather, it is to demonstrate that, once Book Zero has been developed, this argument becomes a direct application of the previously established theory. Proposition I.1 therefore serves as the first experimental test of Book Zero as an intermediate language for synthetic proofs.

### 1.1 The Classical Configuration and Construction

Given a nondegenerate segment  $AB$ , Euclid constructs an equilateral triangle on it.

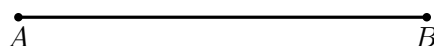


Figure 1.1: The data of Proposition I.1: a nondegenerate segment  $AB$ .

Draw the circle centered at  $A$  through  $B$ , and the circle centered at  $B$  through  $A$ . Choose an intersection point  $C$ , and join  $C$  to  $A$  and  $B$ .

The metric conclusion is immediate from the definitions of the two circles:

$$AC \cong AB, \quad BC \cong AB.$$

Therefore the three sides are equal and  $ABC$  is equilateral.

**Diagrammatic audit.** The four successive classical snapshots have been reduced to the initial segment and the completed two-circle construction. The intermediate states merely add the first circle and then the second; they are not distinct geometric cases.

Wilkins draws attention to the genuinely nontrivial point hidden by the familiar picture: Euclid tacitly assumes that the two circles intersect. Thus the diagram should not be read as a proof of

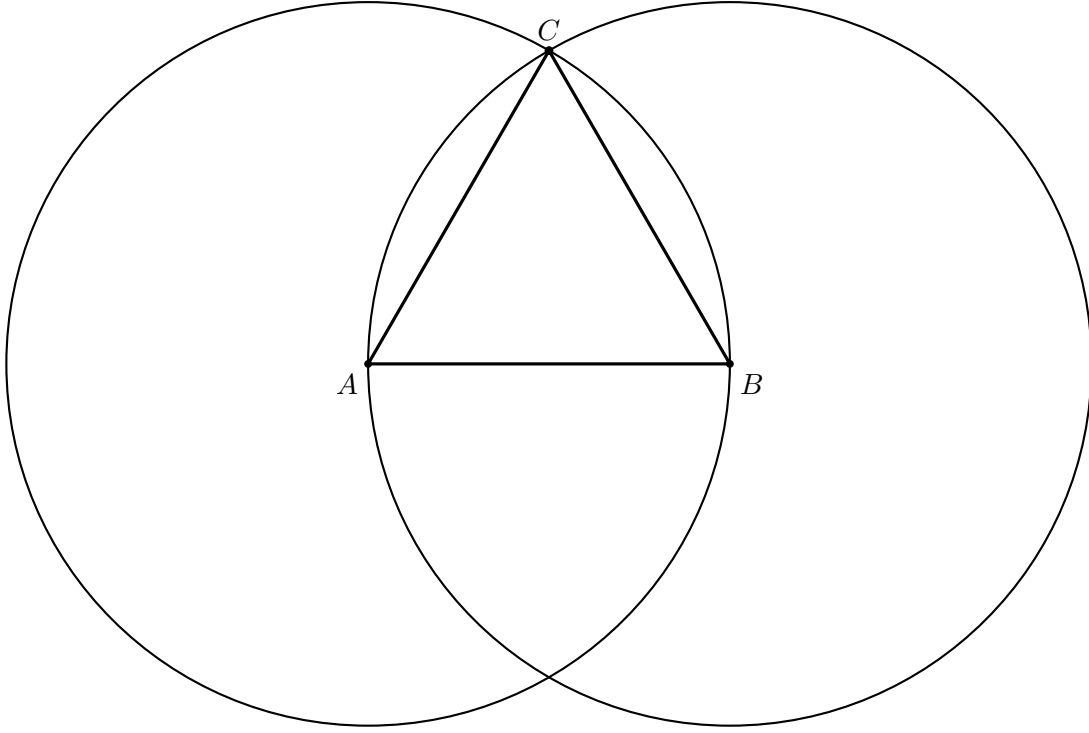


Figure 1.2: The complete classical construction. Since  $C$  lies on both circles,  $AC = AB$  and  $BC = AB$ ; hence  $AC = BC$  by Common Notion 1.

intersection. The existence of  $C$  requires an additional circle-intersection or continuity principle beyond the mere permission to draw the two circles.

## 1.2 Separating the Two Problems

The reconstruction reveals two logically independent stages.

1. **Existence stage.** One must obtain a point  $C$  such that

$$AC \cong AB, \quad BC \cong AB.$$

In the current version of the project, this stage is provided by the temporary principle `hilbert_equidistant_point_exists`. Ultimately, it is intended to be derived from an appropriate theory of continuity in Hilbert geometry.

2. **Synthetic stage.** From the congruences above one must derive

$$\neg \text{Collinear}(A, B, C).$$

This stage no longer requires any theory of circles.

## 1.3 Theorem

**Theorem.** Let  $A \neq B$ . Assume that there exists a point  $C$  such that

$$AC \cong AB, \quad BC \cong AB.$$

Then the points  $A, B, C$  are not collinear.

## 1.4 Proof

The formal proof separates existence of an equidistant point from proof that the point is genuinely off the base line.

First `hilbert_equidistant_point_exists` supplies a point  $C$  with

$$AC \cong AB, \quad BC \cong AB.$$

The proof then excludes the degenerate possibilities  $C = A$  and  $C = B$  using the null-segment theorem.

It remains to prove noncollinearity. Assume that  $A, B, C$  are collinear. Since the three points are pairwise distinct, betweenness trichotomy gives exactly three possibilities:

$$A - B - C, \quad B - A - C, \quad A - C - B.$$

In every case one of the congruent segments would be a proper part of the other. Book Zero #45, `bookZero_45_partNotEqualWhole`, rules this out. Hence  $A, B, C$  are not collinear.

This is a significant foundational difference from Euclid. The formal proof does not construct  $C$  as an intersection of two circles; existence is provided by the Hilbert interface, and Book Zero order theory is then used to prove that the witness cannot be collinear with  $A, B$ .

□

## 1.5 Proof Architecture of the Reconstruction

Euclid

-----

segment AB

|

circle center A through B

+ circle center B through A

|

intersection C

|

AC = AB and BC = AB

|

Common Notion 1

|

equilateral ABC

Hilbert / Lean

-----

A != B

|

`hilbert_equidistant_point_exists`

|

C with AC ~= AB and BC ~= AB

|

exclude C=A and C=B

```

|
assume A,B,C collinear
|
betweenness trichotomy
|
bookZero_45_partNotEqualWhole
in each of the three cases
|
not collinear A B C
|
euclid_proposition_1

```

## 1.6 Significance of the Reconstruction

The geometric result itself is not new. Constructing an equilateral triangle on a given segment is one of the classical theorems of elementary geometry.

Likewise, the idea underlying the proof of non-collinearity is not new. Modern axiomatic reconstructions, in particular the work of Michael Beeson and his collaborators, employ an analogous argument based on the theorem that a proper part of a segment cannot be congruent to the entire segment.

The novelty of the present reconstruction lies elsewhere.

The objective of the CGJteamLab project was not to discover a new proof of Proposition I.1, but to investigate whether the theory developed in Book Zero is sufficiently rich to serve as an intermediate language for synthetic proofs.

The experiment shows that, once the existence problem for an equidistant point has been isolated, the entire argument can be expressed almost exclusively in terms of the theorems previously established in Book Zero.

From the perspective of the architecture of the theory, this is more significant than Proposition I.1 itself. It demonstrates that Book Zero is not merely a collection of auxiliary lemmas, but rather forms a new layer of abstraction between Hilbert's axioms and classical synthetic reasoning.

Schematically, the situation may be represented as

$$\text{Hilbert's axioms} \longrightarrow \text{Book Zero} \longrightarrow \text{Proposition I.1.}$$

In this sense, Proposition I.1 constitutes the first experimental test confirming that Book Zero can serve as a language of synthetic proofs for the further reconstruction of Euclidean geometry.

## 1.7 Relationship with the Foundations

Proposition I.1 has no earlier proposition of Book I on which it can depend. It therefore exposes the foundational boundary of the reconstruction more directly than any later proposition.

Euclid begins Book I with a circle-based construction: two circles with equal radii are drawn from the endpoints of the given segment, and an intersection point supplies the third vertex of the equilateral triangle.

In the Hilbert reconstruction, however, the essential issue is not segment congruence. Constructing points at prescribed distances is already supported by the congruence layer. The difficult step is to obtain a point realizing both distance conditions simultaneously and in the required nondegenerate configuration.

Thus I.1 immediately tests the interaction of

$$\text{incidence} \quad + \quad \text{order} \quad + \quad \text{congruence}$$

at the very beginning of Book I.

## 1.8 Hilbert and Book Zero Components Used

### 1.8.1 Segment Congruence

The equilateral conclusion is expressed synthetically:

$$AB \cong AC \quad \text{and} \quad AB \cong BC.$$

No numerical length function is used.

### 1.8.2 Betweenness Case Analysis

The reconstruction separates the possible collinear orders of three points. The three cases

$$A - B - C, \quad B - A - C, \quad A - C - B$$

are handled explicitly.

This is one of the earliest examples of a recurring formalization phenomenon: a configuration that is visually excluded in Euclid's diagram must be ruled out logically from order and congruence.

### 1.8.3 Book Zero Order Lemmas

The contradiction in each collinear branch depends on elementary results expressing that a proper part of a segment cannot be congruent to the whole segment.

These low-level theorems are precisely the kind of facts collected in Book Zero so that Book I proofs do not repeatedly reconstruct elementary betweenness arithmetic.

### 1.8.4 Noncollinearity

An equilateral triangle must be a genuine triangle. The formal construction therefore requires not only three pairwise congruent side relations but also a proof that the third point does not lie on the given line.

This nondegeneracy obligation is explicit in Lean and becomes an important pattern throughout the rest of Book I.

## 1.9 Formalization Notes

The complete implementation is available in the repository. Two excerpts show the mathematical structure.

Existence and the two equal-distance facts arrive together:

```
obtain <C, hAC, hBC> :=
  hilbert_equidistant_point_exists Geo A B hAB
```

After degeneracies are excluded, collinearity is split into the three betweenness cases:

```
rcases
  hilbert_between_trichotomy
    Geo A B C hAB hBCne hACne hCol with
  hABC | hBAC | hACB
```

Each branch is closed by `bookZero_45_partNotEqualWhole`, with the necessary congruence orientation supplied by symmetry or endpoint reversal. Those branch details are verification bookkeeping; the common geometric idea is simply that a proper part cannot equal the whole.

## 1.10 Dependency Summary

Classical:

Postulate III + circle intersection + Common Notion 1 -> I.1

Formal:

```
hilbert_equidistant_point_exists
+
null-segment exclusion
+
betweenness trichotomy
+
bookZero_45_partNotEqualWhole
|
v
euclid_proposition_1
```

The formal theorem therefore replaces Euclid’s circle-intersection mechanism by an abstract equidistant-point existence theorem and an explicit proof of noncollinearity.

## 1.11 Conclusion

Proposition I.1 is the natural starting point of Euclid’s Book I, but its formal reconstruction immediately reveals that “starting point” does not mean “foundationally primitive.”

Euclid’s two-circle diagram suppresses elementary facts about point order, segment comparison, and nondegeneracy. Once these are made explicit, the proof reaches below Book I into the Hilbert



and Book Zero layers.

The reconstruction therefore establishes an important theme for the entire project at the very first proposition: Book I is the visible classical development, while Book Zero supplies the elementary synthetic language needed to make its diagrammatic reasoning precise.

From this point onward, later propositions can reuse that language instead of rebuilding it. Proposition I.1 is consequently both the first Euclidean construction and the first demonstration of why the formal reconstruction requires a substantial interface beneath Euclid's text.

## Chapter 2

# Proposition I.2

This note compares Euclid's original construction for Proposition I.2 with its reconstruction over the Hilbert foundation. In the CGJteamLab development the proposition follows immediately from Book Zero #49 (Layoff), illustrating the difference between the architectural roles of segment construction in Euclid's geometry and the Hilbert axiomatic framework.

### 2.1 Statement

Given a point  $A$  and a segment  $BC$ , construct a point  $D$  such that

$$AD \cong BC.$$

### 2.2 Euclid's Classical Construction

Let  $A$  be the prescribed endpoint and  $BC$  the given segment. Euclid constructs a segment beginning at  $A$  and equal to  $BC$ .



Figure 2.1: The data of Proposition I.2: a prescribed point  $A$  and a given segment  $BC$ .

Join  $A$  to  $B$ , and construct the equilateral triangle  $ABD$  on  $AB$  by Proposition I.1. Draw the circle centered at  $B$  through  $C$ , and produce  $DB$  until it meets that circle at  $G$ . Next draw the circle centered at  $D$  through  $G$ , and produce  $DA$  until it meets this second circle at  $L$ .

The proof can be summarized by

$$BC = BG, \quad DG = DB + BG, \quad DL = DA + AL,$$

together with

$$DB = DA, \quad DG = DL.$$

Subtracting the equal sides  $DB$  and  $DA$  from the equal radii  $DG$  and  $DL$  gives  $BG = AL$ . Since  $BG = BC$ , it follows that

$$AL = BC.$$

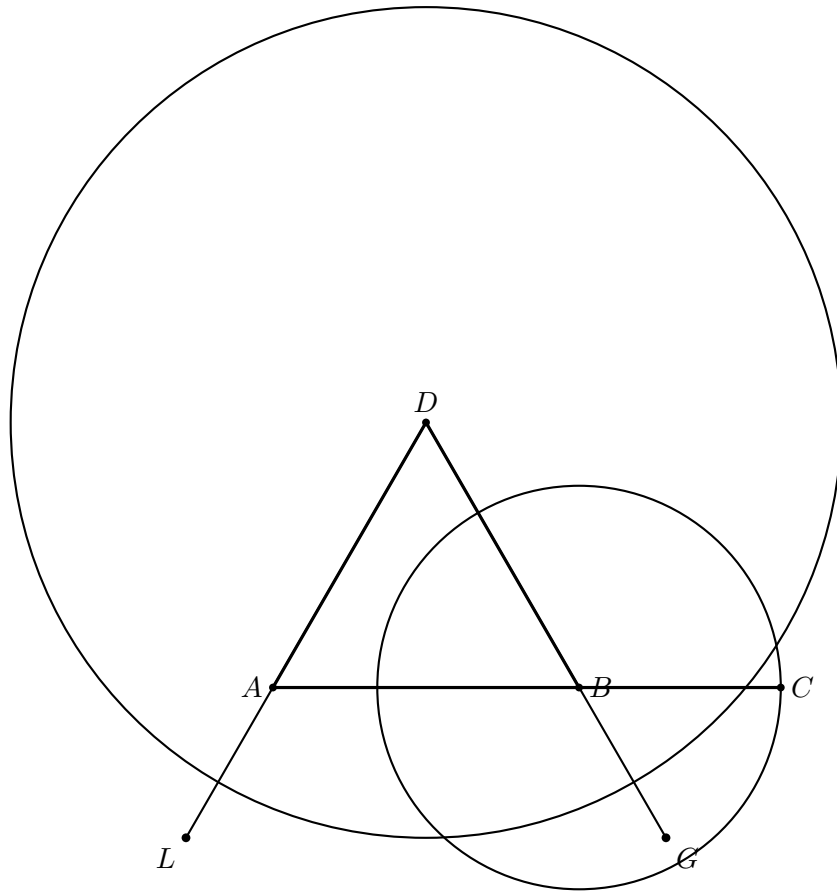


Figure 2.2: The complete classical construction. The first circle gives  $BG = BC$ ; the second gives  $DG = DL$ . Since  $ABD$  is equilateral,  $DB = DA$ , and subtraction of equals yields  $AL = BG = BC$ .

**Diagrammatic audit.** The six successive construction snapshots in the previous version have been reduced to two figures: the initial data and the completed construction. The intermediate pictures added points and circles one at a time but did not represent distinct geometric cases.

Wilkins's account also exposes a foundational subtlety that deserves mention. The sides  $DB$  and  $DA$  must be produced far enough to meet the relevant circles. This uses a strong reading of Euclid's Second Postulate together with a tacit continuity or intersection principle. That issue belongs to the feasibility of Euclid's construction, not to the Hilbert/Book Zero proof used below.

## 2.3 Hilbert Reconstruction

The formal theorem is not a literal transcription of Euclid's construction. Its statement includes a chosen nondegenerate ray  $AE$ . The task is to place a copy of  $BC$  on that ray.

This is exactly Book Zero #49, `bookZero_49_layoff`. Given

$$A \neq E, \quad B \neq C,$$

the theorem produces a point  $D$  such that

$$\text{HilbertSameRay Geo A E D}$$

and

$$AD \cong BC.$$

Thus the equilateral triangle, both circles, the two produced sides, and the subtraction argument of Euclid disappear from Proposition I.2 itself. Their role is replaced by the already established abstract operation of laying off a segment on a prescribed ray.

## 2.4 Formalization

The complete implementation is available in the repository. In this case the mathematical content of the wrapper is literally one theorem call:

```
exact
  bookZero_49_layoff
    Geo A E B C hAE hBC
```

There is no additional incidence or congruence bookkeeping in `Proposition02.lean`. The two nondegeneracy assumptions are precisely the hypotheses required by the Book Zero layoff theorem.

## 2.5 Proof Architecture of the Reconstruction

```
Euclid
-----
point A + segment BC
      |
join AB
      |
I.1: equilateral ABD
      |
circle centered B through C
+ produce DB to G
      |
circle centered D through G
+ produce DA to L
      |
subtract equal segments
      |
AL = BC
```

```
Hilbert / Lean
-----
chosen ray AE + segment BC
      |
bookZero_49_layoff
      |
D on ray AE and AD ~= BC
      |
euclid_proposition_2
```

## 2.6 Discussion

Proposition I.2 is an especially clear example of compression by the Book Zero interface. Euclid devotes a nontrivial construction to transporting a length to a prescribed endpoint. The formal development treats that operation as an already established synthetic primitive.

This does not mean that the classical construction is dispensable mathematically. Rather, the proof burden has moved to the theorem `bookZero_49_layoff`. The Book I wrapper can therefore express the operation at the level at which later geometric proofs actually use it.

## 2.7 Relationship with Earlier Propositions

### 2.7.1 Proposition I.1

Proposition I.2 is the first substantial reuse of the equilateral triangle construction of I.1.

Euclid constructs equilateral triangles as auxiliary devices for transporting a given length to a prescribed point. The equilateral property creates rigid pairs of congruent segments from which the target segment congruence can be obtained.

Thus I.1 is not an isolated construction: its symmetry becomes working infrastructure immediately in I.2.

### 2.7.2 Transition to Proposition I.3

Proposition I.2 establishes free segment transport: given a segment and a prescribed point, a congruent segment can be constructed from that point.

Proposition I.3 adds an order constraint. When a longer segment is given, the transported length must be cut from inside it.

Hence the progression is

$$\text{I.2 : copy a length} \quad \longrightarrow \quad \text{I.3 : cut that length from a longer segment.}$$

## 2.8 Hilbert and Book Zero Components Used

### 2.8.1 Book Zero #49: Layoff

The sole substantive dependency of the formal proposition is `bookZero_49_layoff`. It packages existence of a point on a chosen ray together with congruence to a prescribed nondegenerate segment.

### 2.8.2 Same-Ray Structure

The conclusion records position by

`HilbertSameRay Geo A E D.`

This is stronger and more useful than merely asserting collinearity: it fixes the direction from  $A$  in which the copied segment is laid off.

### 2.8.3 Segment Congruence

The metric part of the conclusion remains purely synthetic:

$$AD \cong BC.$$

No numerical length function is introduced.

## 2.9 Formalization Notes

Proposition I.2 reveals an early difference between Euclid's constructional development and the Hilbert foundation.

Euclid has not yet postulated a general segment-layoff operation. Consequently he constructs the transported length indirectly using equilateral triangles and joins.

In Hilbert geometry, segment construction is already part of the congruence structure. The central mathematical content of I.2 is therefore available at a lower foundational level.

The dependency comparison is

Euclid: I.1 + auxiliary construction      versus      Hilbert: segment layoff.

This is analogous to what occurs much later in Proposition I.23, where Euclid derives angle transport through earlier triangle constructions while Hilbert III.4 provides angle construction directly.

Thus I.2 is the segment-level version of a recurring phenomenon in the reconstruction: a Euclidean construction theorem may become a thin interface theorem when the corresponding construction has been made foundational.

## 2.10 Dependency Summary

Classical:

I.1 + Postulates II, III  
+ circle/line intersection  
+ Common Notions

|  
v

I.2

Formal:

bookZero\_49\_layoff

|  
v

euclid\_proposition\_2

The formal theorem is therefore a thin Book I interface over a construction already established in Book Zero.

## 2.11 Conclusion

Proposition I.2 establishes the transport of segment length to a prescribed point.

For Euclid this is a genuine derived construction built from the equilateral-triangle machinery of I.1. In the Hilbert reconstruction, the corresponding segment-layoff principle belongs to the foundational congruence interface.

The contrast is mathematically informative. It shows, already at the second proposition of Book I, that the historical order of Euclid's constructions need not coincide with the dependency order of a modern axiomatic reconstruction.

I.2 immediately becomes infrastructure for I.3, where free transport is combined with order to cut a prescribed length from a longer segment. It is therefore the first general segment-construction interface of the Book I development.

## Chapter 3

# Proposition I.3

### 3.1 Introduction

Euclid's third proposition asks for the construction of a segment equal to a given shorter segment on a given greater segment.

Classically, this construction is obtained by combining Proposition I.2 with an additional circle construction. A segment equal to the shorter one is first laid off from an endpoint of the greater segment, and a circle is then used to determine the required point on the greater segment.

In the present reconstruction over Hilbert geometry, the logical structure is substantially different. The theory developed in Book Zero already contains an abstract notion of one segment being less than another. This notion is defined by the existence of a point between the endpoints of the greater segment such that the corresponding subsegment is congruent to the smaller one.

Consequently, the existential content of Euclid's Proposition I.3 is already embodied in the definition of segment inequality. The reconstruction therefore consists not in repeating Euclid's geometric construction, but in exposing the witness contained in the definition of `HilbertSegmentLess`.

This proposition provides the first clear example where a classical Euclidean construction becomes an immediate consequence of the abstract language developed in Book Zero.

### 3.2 The Classical Configuration

Euclid's third proposition asks for the construction of a segment equal to a given shorter segment on a given greater segment.

Unlike Proposition I.2, the present construction does not introduce a new geometric technique. Instead, it combines Proposition I.2 with one additional circle construction.

The construction proceeds as follows.

#### 3.2.1 Final Configuration

The objective is to determine a point  $E$  on the greater segment  $AB$  such that

$$AE \cong CD.$$



The final geometric configuration is shown in Figure 3.1.

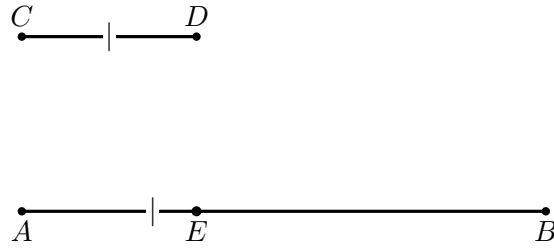


Figure 3.1: The final configuration of Proposition I.3:  $A - E - B$  and  $AE \cong CD$ .

From the construction we obtain

$$AE \cong CD.$$

The remaining task is to justify that the point  $E$  lies between the endpoints of the greater segment.

### 3.2.2 Construction Algorithm

The construction uses Proposition I.2 as an already established construction primitive.

#### Step 1. Given Segments

Let  $AB$  be the greater segment and  $CD$  the given shorter segment.

#### Step 2. Apply Proposition I.2

By Proposition I.2, construct a segment  $AF$  congruent to the given segment  $CD$ .

Thus

$$AF \cong CD.$$

#### Step 3. Draw the Circle Centered at $A$

Draw the circle centered at  $A$  passing through  $F$ .

#### Step 4. Determine the Required Point

Let  $E$  be the intersection of the circle with the greater segment.

Since

$$AE \cong AF$$

and

$$AF \cong CD,$$

it follows that

$$AE \cong CD.$$

The required segment has therefore been obtained.

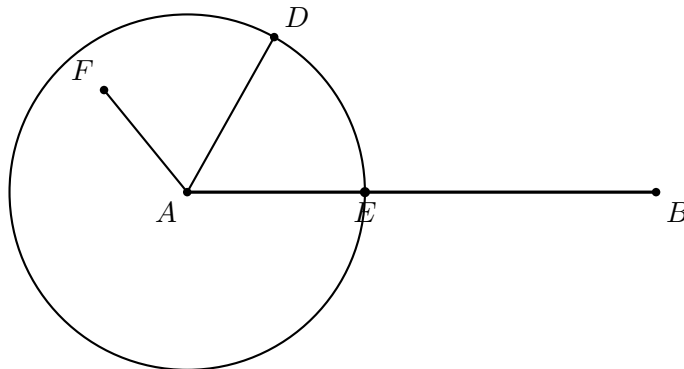


Figure 3.2: The classical cutting step. Proposition I.2 first provides a point  $D$  with  $AD$  equal to the given shorter segment. The circle centered at  $A$  through  $D$  meets the greater segment  $AB$  at  $E$ , so  $AE = AD$ . The auxiliary point  $F$  belongs to the I.2 construction and need not carry further metric markings here.

### 3.3 Classical Proof

Let  $AB$  be the greater segment and  $CD$  the shorter one. By Proposition I.2 construct a point  $D'$  such that

$$AD' \cong CD.$$

Draw the circle with center  $A$  through  $D'$ . Because  $CD < AB$ , this circle meets the segment  $AB$  at a point  $E$  between  $A$  and  $B$ . Every radius of the circle is congruent, so

$$AE \cong AD'.$$

Together with  $AD' \cong CD$ , the First Common Notion gives

$$AE \cong CD.$$

Hence the required part  $AE$  has been cut from the greater segment.

□

**Diagrammatic audit.** Two figures are useful. The first shows only the target relation  $A-E-B$  and  $AE \cong CD$ . The second shows the actual classical mechanism: the transferred length from I.2 and the circle centered at  $A$  whose intersection with  $AB$  determines  $E$ . Separate snapshots for each construction instruction would add no geometric information.

Wilkins emphasizes that I.3 completes a strategy distributed over the first three propositions: I.1 supplies the equilateral-triangle construction used inside I.2, I.2 transports a length, and I.3 uses the resulting radius to cut the required part from the longer segment.

### 3.4 Proof Architecture of the Reconstruction

```

Euclid
-----
CD < AB
  |
I.2: construct D with AD = CD
  |
circle centered at A through D
  |
intersection E with AB
  |
AE = AD = CD
  |
I.3

```

```

Hilbert / Lean
-----
HilbertSegmentLess Geo C D A B
  |
unpack witness E
  |
A-E-B and CD ~= AE
  |
congruence symmetry
  |
A-E-B and AE ~= CD
  |
euclid_proposition_3

```

### 3.5 Hilbert Reconstruction

The Book Zero formulation changes the logical status of the proposition. The hypothesis

$$\text{HilbertSegmentLess Geo C D A B}$$

already contains a witness  $E$  satisfying

$$A - E - B \quad \text{and} \quad CD \cong AE.$$

Consequently Proposition I.3 does not invoke a segment-layoff theorem at all. It merely extracts that witness and reverses the orientation of the congruence to obtain

$$AE \cong CD.$$

This is stronger than saying that Book Zero supplies convenient infrastructure for Euclid's construction. With the present definition of strict segment comparison, the existential conclusion of I.3 is literally part of the hypothesis.

## 3.6 Lean Formalization

The complete implementation is available in the repository. The entire formal argument is worth showing because it exposes the definitional collapse of the classical construction:

```
rcases hCDAB with <E, hAEB, hCDAE>

exact <E, hAEB,
  hilbert_congruent_symmetry
  Geo C D A E hCDAE>
```

There is no hidden construction step in `Proposition03.lean`. Unpacking `hCDAB` produces exactly the required point and betweenness statement; only congruence symmetry is needed to match the orientation of the Euclidean conclusion.

## 3.7 Relationship with Earlier Propositions

### 3.7.1 Proposition I.2

Proposition I.3 is the first direct application of the segment-transfer construction established in I.2.

I.2 says that a segment congruent to a given segment can be placed at a prescribed point. I.3 adds an order constraint: when the target segment is longer than the given one, the transferred length can be realized as an initial part of that longer segment.

Thus I.3 converts free segment transport into controlled segment cutting.

### 3.7.2 Proposition I.1

Proposition I.1 enters indirectly through I.2. Euclid's segment-transfer construction uses equilateral triangles as part of the mechanism for moving a length to a new position.

By the time I.3 is reached, this construction has already been packaged as Proposition I.2 and can be used as a reusable operation.

### 3.7.3 Transition to Proposition I.4

Propositions I.1–I.3 form the initial construction block of Book I: equilateral triangle construction, segment transport, and segment cutting.

Proposition I.4 changes the character of the development. It is no longer primarily a construction theorem but a rigidity theorem for triangles: SAS.

Thus I.3 closes the first elementary segment-construction sequence and prepares the congruence theory beginning with I.4.

## 3.8 Hilbert and Book Zero Components Used

### 3.8.1 Strict Segment Comparison

The decisive object is `HilbertSegmentLess`. In the present Book Zero language,  $CD < AB$  is witnessed by an interior point  $E$  of  $AB$  whose initial subsegment is congruent to  $CD$ . Thus order and congruence are already coupled in the definition.

### 3.8.2 Betweenness

The witness contains the positional fact

$$A - E - B.$$

This is the formal counterpart of the classical requirement that the circle meet the greater segment itself, not merely its supporting line.

### 3.8.3 Congruence Symmetry

The witness supplies  $CD \cong AE$ , while the proposition is stated with  $AE \cong CD$ . The only theorem used after unpacking the witness is segment-congruence symmetry.

## 3.9 Formalization Notes

Proposition I.3 is mathematically simple but formally instructive.

In modern numerical language one might describe the theorem as follows: if

$$|AB| > |CD|,$$

then there is a point  $E$  on  $AB$  such that

$$|AE| = |CD|.$$

The reconstruction deliberately avoids this numerical formulation. “Greater than” is represented by a synthetic segment-order relation, and the conclusion consists of two separate geometric facts:

$$A - E - B \quad \text{and} \quad AE \cong CD.$$

This distinction is important. Congruence alone would allow a copy of  $CD$  to be placed anywhere; Proposition I.3 requires the copy to occur inside the prescribed longer segment.

The formal theorem therefore combines the two foundational layers that will recur throughout Book I:

$$\text{congruence} \quad + \quad \text{order}.$$

In the Hilbert reconstruction the actual segment-layoff operation is already available at the interface level. Consequently the public Euclidean theorem is relatively short: its main role

is to package layoff together with the betweenness condition dictated by the strict segment comparison.

### 3.10 Dependency Summary

Classical:

I.1 → I.2 → circle construction → I.3

Formal:

```
HilbertSegmentLess
  |
  +-- witness E
  +-- A-E-B
  +-- CD ~= AE
  |
congruence symmetry
  |
euclid_proposition_3
```

The contrast is especially sharp: the classical proposition performs a construction, whereas the formal theorem exposes existential data already encoded by the Book Zero segment-order relation.

### 3.11 Conclusion

Proposition I.3 completes the first elementary segment-construction block of Book I.

Proposition I.2 established free transport of a segment length to a prescribed point. Proposition I.3 adds order: the copied segment is cut from a longer prescribed segment and its endpoint must lie between the original endpoints.

The formalization makes this distinction particularly clear. The theorem is not simply about equality of lengths; it is about the interaction of segment congruence with betweenness.

This combination of congruence and order becomes fundamental throughout the later development. I.3 therefore serves as a compact early example of the two Hilbert layers working together before Book I turns in Proposition I.4 to the rigidity theory of triangles.

## Chapter 4

# Proposition I.4

### 4.1 Introduction

Euclid's fourth proposition is the first theorem on triangle congruence.

Unlike the preceding propositions, it does not describe a geometric construction. Instead, it establishes that two triangles are congruent whenever two corresponding sides and the included angle are equal.

Euclid's original proof relies on the method of superposition, one of the most discussed arguments in the *Elements*. Modern axiomatic systems generally avoid superposition as a primitive method of proof.

In the present reconstruction over Hilbert geometry, the proposition is obtained from the Side–Angle–Side (SAS) congruence theorem, which has already been established within the Hilbert theory.

Thus, while the mathematical content coincides with Euclid's Proposition I.4, the logical structure of the proof is fundamentally different.

### 4.2 The Classical Configuration

Unlike the first three propositions of Book I, Proposition I.4 is not a geometric construction. Instead, it establishes a criterion for the congruence of two triangles.

The proposition considers two triangles satisfying three geometric hypotheses:

- two pairs of corresponding sides are congruent;
- the included angles are congruent.

The objective is to prove that the triangles are congruent. As a consequence, the remaining corresponding side and the remaining corresponding angles are also congruent.

#### 4.2.1 Final Configuration

The geometric configuration is shown in Figure [4.1](#).

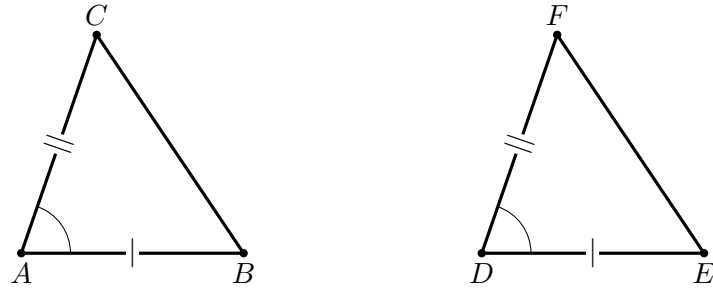


Figure 4.1: The SAS data of Proposition I.4:  $AB \cong DE$ ,  $AC \cong DF$ , and  $\angle BAC \cong \angle EDF$ .

The hypotheses are

$$AB \cong DE,$$

$$AC \cong DF,$$

and

$$\angle BAC \cong \angle EDF.$$

The desired conclusions are

$$BC \cong EF,$$

together with

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

### 4.3 Statement

#### Proposition I.4.

If two triangles have two corresponding sides congruent and the included angles congruent, then the triangles are congruent.

More precisely, let  $ABC$  and  $DEF$  be two non degenerate triangles:

$$AB \cong DE,$$

$$AC \cong DF,$$



and

$$\angle BAC \cong \angle EDF.$$

Then

$$BC \cong EF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

## 4.4 Classical Proof

Euclid's proof is a superposition argument. Conceive the triangle  $ABC$  as being applied to  $DEF$  so that  $A$  coincides with  $D$  and the ray  $AB$  coincides in direction with the ray  $DE$ . Since

$$AB \cong DE,$$

the point  $B$  must then coincide with  $E$ .

There is a further positional condition that is easy to miss in the bare classical diagram. The transported point  $C$  must be chosen on the same side of the line  $DE$  as  $F$ . With that choice, the equality

$$\angle BAC \cong \angle EDF$$

makes the ray  $AC$  coincide with the ray  $DF$ . Since also

$$AC \cong DF,$$

the point  $C$  coincides with  $F$ .

The endpoints of the remaining sides therefore coincide pairwise, so  $BC$  coincides with  $EF$ . The two triangles coincide under the superposition, and hence

$$BC \cong EF, \quad \angle ABC \cong \angle DEF, \quad \angle ACB \cong \angle DFE.$$

Euclid also concludes that the two triangles are equal in area. That additional magnitude statement belongs to the classical proposition but is not part of the present `TriangleCongruenceResult` interface.

□

**Diagrammatic audit.** Only one geometric figure is needed. Proposition I.4 has no auxiliary construction: the proof acts on the two triangles already present in the hypotheses. The figure therefore records exactly the SAS data and nothing more. Wilkins's discussion is useful here not because it supplies additional construction diagrams, but because it makes explicit the same-side condition on the transported third vertex that Euclid leaves implicit in words.

## 4.5 Proof Architecture of the Reconstruction

```

Euclid
-----
AB ~= DE
AC ~= DF
angle BAC ~= angle EDF
      |
apply triangle ABC to triangle DEF
A -> D, ray AB -> ray DE
      |
B -> E
      |
choose the same side of DE for C and F
      |
equal included angle + AC ~= DF
      |
C -> F
      |
triangles coincide
      |
BC ~= EF and corresponding angles equal

```

```

Hilbert / Lean
-----
two noncollinear triangles
+ two side congruences
+ included-angle congruence
      |
TriangleCongruentFromSAS
      |
TriangleCongruenceResult
      |
euclid_proposition_4

```

## 4.6 Hilbert Reconstruction

The foundational organization is different. Superposition is not replayed inside Proposition I.4. The required rigidity is already available through the Hilbert SAS interface.

For two noncollinear triangles  $ABC$  and  $DEF$ , the hypotheses

$$AB \cong DE, \quad AC \cong DF, \quad \angle BAC \cong \angle EDF$$

are precisely the inputs of `TriangleCongruentFromSAS`. Its result is a `TriangleCongruenceResult`, which packages the corresponding side and angle congruences.

Thus the difference from Euclid is foundational rather than mathematical: Euclid proves SAS by superposition; the Hilbert development has already established SAS before the Book I wrapper is reached.

## 4.7 Lean Formalization

The complete implementation is available in the repository. Here the entire mathematical step is a single interface call, so only that call is worth displaying:

```
exact
  TriangleCongruentFromSAS
    Geo
    A B C
    D E F
    hABC
    hDEF
    hAB
    hAngle
    hAC
```

The explicit hypotheses  $hABC$  and  $hDEF$  record nondegeneracy, which the classical picture leaves implicit. No auxiliary points or intermediate geometric constructions occur in `Proposition04.lean`.

## 4.8 Relationship with Earlier Propositions

### 4.8.1 Proposition I.2

The classical proof of SAS is expressed by Euclid through superposition rather than by a sequence of earlier construction propositions. Nevertheless, the preceding segment-transport theorem already establishes the general idea that congruent lengths can be reproduced independently of their original position.

In the Hilbert reconstruction, segment congruence is part of the foundational congruence structure, so no separate segment-copying construction is needed inside I.4.

### 4.8.2 Proposition I.3

Proposition I.3 develops the controlled cutting of one segment by a length congruent to another. It belongs to the same early congruence infrastructure, although it is not the decisive logical ingredient in SAS.

The contrast is useful: I.2 and I.3 are construction theorems about segments, whereas I.4 is the first major rigidity theorem for complete triangles.

### 4.8.3 Transition to Proposition I.5

Proposition I.5 uses SAS repeatedly. The auxiliary construction in the isosceles-triangle proof creates pairs of triangles with two corresponding sides and the included angle congruent, and I.4 propagates that local symmetry into further side and angle congruences.

Thus I.4 becomes one of the principal reusable congruence principles of Book I immediately after it is established.

## 4.9 Hilbert and Book Zero Components Used

### 4.9.1 Hilbert SAS

The decisive formal ingredient is the Hilbert SAS congruence axiom or its established interface theorem. Given two corresponding side congruences and congruence of the included angles, it supplies the remaining corresponding side and angle congruences.

The public Euclidean theorem therefore packages a foundational Hilbert congruence principle into the form of Proposition I.4.

### 4.9.2 Nondegenerate Triangles

The formal hypotheses explicitly require genuine triangles. The noncollinearity assumptions prevent the angle data from degenerating and make precise a condition that the classical diagram leaves implicit.

### 4.9.3 Angle Orientation

Angles are represented by ordered triples of points. Consequently the formal proof must ensure that the included angles supplied to SAS and the corresponding angles returned from it have exactly the orientations required by the Euclidean statement.

Symmetry and transport lemmas handle these changes explicitly.

### 4.9.4 Segment Congruence

The side hypotheses are expressed directly by the Hilbert segment congruence relation. No numerical length function or coordinate representation is introduced.

## 4.10 Formalization Notes

Proposition I.4 occupies a special position in the reconstruction.

For Euclid, SAS is proved by superposition: one triangle is imagined as being placed on another so that a side and the included angle coincide. The remaining equal side then forces the third vertex and hence the whole triangle to coincide.

In a Hilbert-style axiomatization, the same rigidity is foundational. SAS belongs to the congruence axioms. The Lean proof of the Euclidean proposition is therefore much shorter than Euclid's argument because the geometric content has already been placed in the Hilbert interface.

This is not a loss of information. It records a genuine difference in foundational organization:

Euclid: superposition proof      versus      Hilbert: SAS congruence principle.

The formal statement also makes nondegeneracy explicit. A diagram of two triangles silently suggests that each triple of vertices is noncollinear; Lean requires this information as part of the hypotheses needed to apply the angle-congruence machinery.

Proposition I.4 is therefore an early example where the Book I order and the logical order of the Hilbert foundation diverge sharply. What is a theorem for Euclid is already foundational infrastructure for the reconstruction.

## 4.11 Dependency Summary

Classical:

SAS data  $\rightarrow$  superposition  $\rightarrow$  coincidence  $\rightarrow$  I.4

Formal:

TriangleCongruentFromSAS  $\rightarrow$  euclid\_proposition\_4

No new local axiom is introduced by the Book I theorem. The proof burden has already been discharged in the Hilbert congruence layer.

## 4.12 Conclusion

Proposition I.4 is the first fundamental triangle-congruence theorem of Book I.

Its historical proof uses superposition, one of the most discussed features of Euclid’s early geometry. In the Hilbert reconstruction the same mathematical rigidity is represented directly by the SAS congruence principle.

The resulting Lean theorem is therefore short for a substantive reason: the proof burden has moved from Book I into the axiomatic foundation.

This distinction becomes increasingly important in the propositions that follow. I.5 uses SAS as working infrastructure, and later constructions repeatedly rely on the ability to convert local side and angle congruences into congruence information about an entire triangle.

Proposition I.4 thus marks the point where the early segment constructions of I.1–I.3 give way to the systematic theory of triangle rigidity.

## Chapter 5

# Proposition I.5

### 5.1 Introduction

Proposition I.5 is the first theorem of Book I devoted specifically to the geometry of isosceles triangles. It establishes that the angles at the base of an isosceles triangle are congruent. Moreover, if the equal sides are extended beyond the base, the exterior angles formed by these extensions are also congruent.

In the Elements this proposition is one of the best known results of classical geometry and later became known under the traditional name *pons asinorum*. Although the theorem itself is elementary, it marks the beginning of a systematic study of congruence properties of triangles.

Euclid proves the proposition by combining the congruence theorem of Proposition I.4 with auxiliary constructions on the extensions of the equal sides.

In the present reconstruction the logical organization is different. The theorem is obtained by composing two previously established results of the Hilbert interface. The congruence of the base angles is provided by the theorem on isosceles triangles, while the congruence of the exterior angles follows from Hilbert's theorem on adjacent angles.

Consequently, Proposition I.5 is the first example in which a Euclidean proposition is reconstructed by combining several independent interface theorems rather than by introducing new geometric techniques.

### 5.2 The Classical Configuration

Euclid's proof of Proposition I.5 does not use only the original isosceles triangle and the extensions of its equal sides. It introduces additional auxiliary points and joins, producing a larger configuration to which Proposition I.4 can be applied.

Let  $ABC$  be an isosceles triangle satisfying

$$AB \cong AC.$$

Produce the equal sides  $AB$  and  $AC$  beyond the base vertices  $B$  and  $C$ . Euclid then chooses an auxiliary point on one extension, transfers a corresponding segment to the other extension, and joins the resulting points to the opposite vertices of the base.

The following sequence shows the successive stages of Euclid's configuration.

### 5.2.1 Initial Isosceles Triangle

The construction begins with an isosceles triangle  $ABC$  satisfying

$$AB \cong AC.$$

At this stage no auxiliary points or segments have yet been introduced.

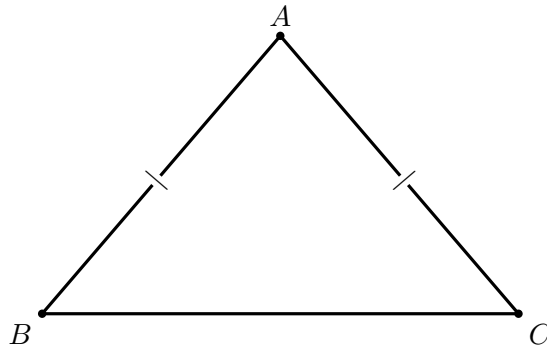


Figure 5.1: The initial isosceles triangle, with  $AB \cong AC$ .

### 5.2.2 Extension of the Equal Sides

Euclid next produces the equal sides  $AB$  and  $AC$  beyond the base vertices. Let  $D$  lie beyond  $B$  on the line  $AB$ , and let  $E$  lie beyond  $C$  on the line  $AC$ .

Thus

$$A - B - D, \quad A - C - E.$$

## 5.3 Statement

Let  $ABC$  be a nondegenerate isosceles triangle satisfying

$$AB \cong AC.$$

Let the equal sides be extended beyond the base vertices so that

$$A - B - D, \quad A - C - E.$$

Then

$$\angle ABC \cong \angle ACB,$$

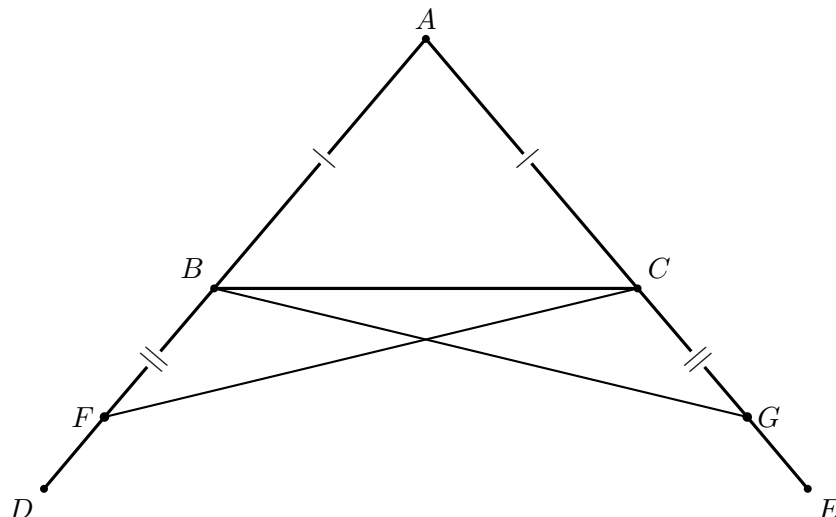


Figure 5.2: The complete classical configuration for Proposition I.5. The sides are produced beyond  $B, C$ ;  $F$  is chosen on the left extension,  $G$  on the right so that  $BF \cong CG$ ; finally  $FC$  and  $GB$  are joined. The outer points  $D, E$  record the rays used in the statement.

and

$$\angle CBD \cong \angle BCE.$$

Thus both the base angles of the isosceles triangle and the exterior angles formed after extending the equal sides are congruent.

**Diagrammatic audit.** The four successive construction snapshots have been replaced by one complete classical figure. For I.5 the successive states do not represent genuinely different geometric cases: they merely add  $F$ , then  $G$ , then the two joins. A single well-spaced configuration makes the proof easier to follow.

Wilkins's note confirms that the two genuinely important stages are the two SAS comparisons. First  $FAC$  is compared with  $GAB$ ; this gives, among other consequences,  $FC \cong GB$ . Then  $FBC$  is compared with  $GCB$ , which yields the equal angles below the base. Subtracting equal angles then gives the equality of the base angles. These logical stages are described in the proof rather than represented by repeated copies of the same diagram.

## 5.4 Classical Proof

Euclid's proof is considerably more elaborate than the modern reconstruction presented later in this chapter. Rather than reasoning directly from the properties of an isosceles triangle, he introduces an auxiliary construction that allows repeated applications of the Side–Angle–Side congruence theorem established in Proposition I.4.

After extending the equal sides of the triangle, an arbitrary point is chosen on one extension. Proposition I.3 is then used to construct an equal segment on the opposite extension. The newly obtained points are joined to the opposite vertices of the base, completing the auxiliary configuration.



The proof proceeds by applying Proposition I.4 twice. The first application establishes congruence relations between the auxiliary triangles and yields equal corresponding sides and angles. These newly obtained equalities provide the hypotheses required for a second application of Proposition I.4.

Finally, Euclid combines the resulting congruence relations with the elementary properties of equal segments and equal angles to conclude that the interior base angles of the original isosceles triangle are congruent. The same argument immediately establishes the congruence of the corresponding exterior angles formed after extending the equal sides.

The auxiliary construction is therefore not an end in itself. Its sole purpose is to transform Proposition I.5 into two successive applications of Proposition I.4, allowing the theorem to be derived using only previously established geometric results.

## 5.5 Proof Architecture of the Reconstruction

The formal proof of Proposition I.5 is obtained by composing two previously established theorems of the Hilbert interface.

The first theorem establishes the congruence of the base angles of an isosceles triangle. The second theorem transfers this congruence to the adjacent exterior angles after the equal sides have been extended.

The resulting logical dependency is shown below.

Euclid

-----

```

AB ~= AC
  |
produce AB, AC beyond B, C
  |
choose F; I.3 gives G with BF ~= CG
  |
join FC and GB
  |
I.4: SAS on FAC and GAB
  |
I.4: SAS on FBC and GCB
  |
subtract equal angles
  |
base angles equal
and angles below base equal

```

Hilbert / Lean

-----

```

AB ~= AC
  |
hilbert_isosceles_base_angles
  |
angle ABC ~= angle ACB

```

```

|
A-B-D and A-C-E
|
hilbert_adjacent_angles_congruent
|
angle CBD ~= angle BCE

```

Unlike the classical proof of Euclid, the formal development does not repeat the geometric argument establishing the base-angle theorem. Instead, the proposition is reconstructed by composing two independent results already available in the Hilbert interface. Consequently, Proposition I.5 illustrates the modular nature of the library: new Euclidean propositions are obtained by combining previously verified synthetic theorems.

## 5.6 Hilbert Reconstruction

At the level of Proposition I.5, the Hilbert proof is much shorter than Euclid’s classical construction.

The first conclusion,

$$\angle ABC \cong \angle ACB,$$

is already the theorem

```
hilbert_isosceles_base_angles
```

applied to the noncollinear isosceles triangle  $ABC$ .

For the second conclusion, the hypotheses

$$A - B - D, \quad A - C - E$$

say that the equal sides have been produced beyond the base vertices. The established theorem

```
hilbert_adjacent_angles_congruent
```

transports the congruence of the base angles to their adjacent exterior angles, giving

$$\angle CBD \cong \angle BCE.$$

Thus the formal proposition does not construct Euclid’s auxiliary points  $F, G$ , and it does not replay the two SAS arguments. Those facts have already been packaged into the Hilbert interface theorems used here.

## 5.7 Lean Formalization

The complete implementation is available in the repository. The two interface calls contain the mathematical content of the formal proof.

The base-angle conclusion is immediate:

```

have hBase :=
  hilbert_isosceles_base_angles
    Geo A B C hABC hABAC

```

The exterior-angle conclusion is then obtained from the two extensions and the already established base-angle congruence:

```

have hExterior :=
  hilbert_adjacent_angles_congruent
    Geo
    A B C D
    A C B E
    hABD
    hACE
    hABC
    hACB
    hBase

```

The theorem returns the pair  $\langle hBase, hExterior \rangle$ . The only additional bookkeeping is the rotated noncollinearity hypothesis required for the second call.

## 5.8 Relationship with Earlier Propositions

### 5.8.1 Proposition I.2

Euclid's proof requires the transfer of a prescribed segment length to a new position in the auxiliary construction. Proposition I.2 supplies the underlying segment-transport mechanism.

In the Hilbert reconstruction this role is absorbed by the established segment-layoff infrastructure.

### 5.8.2 Proposition I.3

After extending the equal sides of the original isosceles triangle, Euclid cuts off an auxiliary segment equal to a prescribed segment. Proposition I.3 provides exactly this operation.

This construction creates the equal-side data needed for the subsequent triangle-congruence arguments.

### 5.8.3 Proposition I.4

SAS is the decisive congruence theorem in the classical proof. It is applied to auxiliary triangles in order to propagate the original equality of the two sides into the angle congruences required at the base.

The formal reconstruction uses the corresponding Hilbert SAS theorem and angle-congruence transport.

### 5.8.4 Transition to Proposition I.6

Proposition I.5 proves

$$AB \cong AC \implies \angle ABC \cong \angle BCA.$$

Proposition I.6 proves the converse implication. Together they establish the first full equivalence between side symmetry and angle symmetry in a triangle.

## 5.9 Hilbert and Book Zero Components Used

### 5.9.1 Segment Extension and Layoff

The proof extends the equal sides beyond the base vertices and chooses auxiliary points at prescribed distances. These operations are supplied by the established Hilbert order and congruence interfaces.

### 5.9.2 SAS

The Hilbert SAS theorem is the central rigidity principle in the reconstruction. It converts the constructed side congruences and the appropriate included-angle congruence into further corresponding side and angle congruences.

### 5.9.3 Angle Congruence Transport

The auxiliary triangles naturally produce angles in orientations that do not always coincide syntactically with the final base angles. Symmetry, transitivity, and transport of angle congruence are therefore part of the formal proof rather than silent diagrammatic steps.

### 5.9.4 Betweenness and Collinearity

The extensions of the two equal sides are represented by explicit betweenness hypotheses. These yield the collinearity and ray information needed to identify the auxiliary angles with the angles occurring in the Euclidean statement.

## 5.10 Formalization Notes

Proposition I.5 marks an important transition in the reconstruction of Book I.

The earlier propositions are centred on elementary constructions and a single application of the SAS theorem. Beginning with Proposition I.5, most subsequent propositions of Book I are expected to be reconstructed by composing previously established interface theorems.

This modular organization differs substantially from the presentation of the *Elements*. Euclid develops each proposition as an independent geometric argument, whereas the present reconstruction emphasizes reusable formal components. The resulting proofs are shorter, their logical dependencies are explicit, and each proposition becomes a thin verification layer built upon the synthetic theory already developed in the Hilbert interface.

## 5.11 Dependency Summary

The classical and formal architectures are sharply different:

Classical:

I.3 + two uses of I.4 + angle subtraction  $\rightarrow$  I.5

Formal:

hilbert\_isosceles\_base\_angles

+

hilbert\_adjacent\_angles\_congruent  $\rightarrow$  euclid\_proposition\_5

The Book I theorem is therefore a thin interface layer over results already proved in the Hilbert development.

## 5.12 Conclusion

Proposition I.5 is the first substantial theorem about the internal symmetry of a triangle.

Starting from two congruent sides, Euclid constructs an extended auxiliary configuration and uses SAS twice to prove congruence of the base angles. The theorem also establishes the corresponding congruence of the exterior angles below the base.

The formal reconstruction shows that the geometric idea remains elementary, but its successful reuse depends on a well-developed interface for segment extension, layoff, betweenness, collinearity, and angle-congruence transport.

I.5 is structurally important far beyond the immediate statement. Its converse is Proposition I.6; the pair then supplies the isosceles machinery used in the uniqueness argument of I.7 and ultimately in the SSS theorem of I.8.

Thus Proposition I.5 begins a coherent four-proposition development

$$I.5 \longrightarrow I.6 \longrightarrow I.7 \longrightarrow I.8,$$

moving from isosceles symmetry to general triangle rigidity.

## Chapter 6

# Proposition I.6

### 6.1 Introduction

Proposition I.6 establishes the converse of the preceding proposition. Where Proposition I.5 proves that equal sides of a triangle imply equal base angles, Proposition I.6 shows that equal base angles imply the equality of the opposite sides.

This result completes the fundamental correspondence between the sides and angles of an isosceles triangle. Together, Propositions I.5 and I.6 establish the equivalence between the two defining properties of an isosceles triangle.

Euclid's original proof is considerably more involved than the modern reconstruction. It proceeds by contradiction, introducing an auxiliary construction obtained from Proposition I.3 and ultimately deriving an impossible configuration by repeated use of Proposition I.4.

In the Hilbert reconstruction developed in this project, the theorem admits a substantially shorter proof. The previously established Angle–Side–Angle theorem is sufficient to derive the equality of the opposite sides directly, eliminating the auxiliary construction and the *reductio ad absurdum* argument required by Euclid.

### 6.2 The Classical Configuration

Euclid's proof begins with a triangle whose base angles are assumed to be equal. The desired conclusion is that the opposite sides are also equal.

Instead of arguing directly, Euclid assumes that the two sides are unequal and derives a contradiction. The proof therefore combines an auxiliary construction with a *reductio ad absurdum* argument.

The following sequence illustrates the successive stages of Euclid's construction.

#### 6.2.1 Initial Triangle

Let  $ABC$  be a triangle satisfying

$$\angle ABC \cong \angle ACB.$$

At this stage no auxiliary construction has yet been introduced.

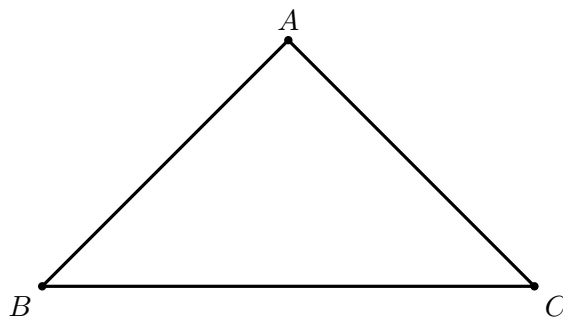


Figure 6.1: The data of Proposition I.6: the base angles satisfy  $\angle ABC \cong \angle ACB$ .

### 6.2.2 Assuming Unequal Sides

To obtain a contradiction, Euclid assumes that the two sides opposite the equal angles are not equal. Without loss of generality, suppose

$$AB > AC.$$

Using Proposition I.3, a point  $D$  is chosen on the segment  $AB$  so that

$$DB \cong AC.$$

Consequently,

$$A - D - B.$$

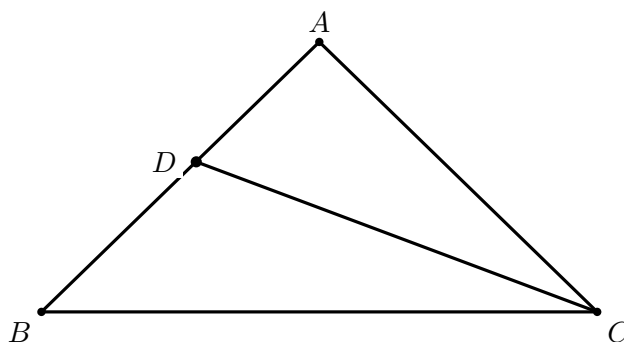


Figure 6.2: Euclid's contradiction configuration for the case  $AB > AC$ . Proposition I.3 chooses  $D$  on  $AB$  with  $DB \cong AC$ , and  $D$  is joined to  $C$ . The congruence is stated in the text rather than marked again on the diagram.

## 6.3 Statement

**Theorem 1** (Euclid I.6). *Let  $ABC$  be a nondegenerate triangle.*

If

$$\angle ABC \cong \angle ACB,$$

then

$$AB \cong AC.$$

*That is, equal angles in a triangle subtend equal opposite sides.*

**Diagrammatic audit.** The three former auxiliary pictures were successive snapshots of the same construction and have been merged into one. Two figures are sufficient: the initial triangle and the contradiction configuration with the point  $D$  and the segment  $DC$ .

Wilkins’s discussion confirms that the essential classical move is exactly this one: assuming, say,  $AB > AC$ , choose  $D \in AB$  with  $DB = AC$ , then compare  $DBC$  with  $ABC$  by SAS. His note also records an alternative angle-based contradiction, but that is another proof rather than another necessary diagram for the reconstruction.

## 6.4 Classical Proof

Euclid proves the theorem by contradiction.

Assume that the sides opposite the equal angles are unequal. Without loss of generality, suppose that  $AB$  is greater than  $AC$ . By Proposition I.3, a point is constructed on the longer side so that the remaining segment is equal to the shorter side.

Joining the auxiliary point to the remaining vertex produces a second triangle. Proposition I.4 is then applied to show that the auxiliary triangle is congruent to the original one.

The resulting congruence forces equalities that are incompatible with the assumption that one side is strictly longer than the other. Since the assumption leads to a contradiction, the two sides must be equal.

Thus equal angles in a triangle necessarily subtend equal opposite sides.

## 6.5 Proof Architecture of the Reconstruction

The formal proof follows a substantially different route from Euclid’s classical argument. Rather than introducing an auxiliary point, constructing additional triangles, and reasoning by contradiction, the implementation exploits a higher-level theorem already available in the Hilbert interface.

The proof compares the given triangle with itself after exchanging the roles of the two equal angles. Since the common side is trivially congruent to itself, the derived Angle–Side–Angle theorem immediately implies the equality of the remaining sides.

Consequently, the entire classical construction disappears from the formal proof and is replaced by a single application of a previously established congruence theorem.



```

Euclid
-----
ABC with angle ABC ~= angle ACB
|
assume AB != AC
|
choose one strict inequality, e.g. AB > AC
|
I.3: choose D on AB with DB ~= AC
|
join DC
|
I.4: SAS
|
auxiliary triangle equals whole triangle
|
contradiction
|
AB ~= AC

```

```

Hilbert / Lean
-----
angle ABC ~= angle ACB
|
reorient the two angle congruences
|
BC ~= CB
|
hilbert_asa_sides
on triangles BCA and CBA
|
AB ~= AC

```

## 6.6 Hilbert Reconstruction

The formal proof uses a substantially shorter route than Euclid's contradiction argument. It applies an established ASA theorem directly.

From the hypothesis

$$\angle ABC \cong \angle ACB$$

the proof first obtains the corresponding reversed angle congruences needed for the triangles

$$\triangle BCA \quad \text{and} \quad \triangle CBA.$$

The side  $BC$  is common to both triangles, so

$$BC \cong CB.$$

The theorem `hilbert_asa_sides` then yields the equality of the remaining corresponding sides. After reorienting the segment congruence, this is exactly

$$AB \cong AC.$$

Thus no segment trichotomy, point  $D$ , Proposition I.3 construction, or contradiction appears in `Proposition06.lean`. Those belong to the classical Euclidean proof, not to the formal wrapper used here.

## 6.7 Lean Formalization

The complete implementation is available in the repository. Only the two steps that explain the formal proof are reproduced here.

First the common side and the two angle orientations are prepared. The decisive theorem call is then:

```
have hASA :=
  hilbert_asa_sides
    Geo
    B C A
    C B A
    hBCA
    hCBA
    hBC
    hAngleB
    hAngleC
```

The returned side congruence has the orientation dictated by those two triangles, so the final line merely reverses its endpoints:

```
exact
  (Geo.congruent_reverse_second
    A B C A).mp
  ((Geo.congruent_reverse_first
    B A C A).mp hASA.1)
```

The remaining code establishes the two noncollinearity variants and reorients the given angle congruence. These are formal interface obligations, not additional geometric constructions.

## 6.8 Relationship with Earlier Propositions

### 6.8.1 Proposition I.3

The contradiction argument assumes that one of the two sides is longer than the other and then cuts from the longer side a segment congruent to the shorter one.

This is precisely the segment-cutting operation established in Proposition I.3. It creates the auxiliary point from which the isosceles configuration used in the proof is obtained.

### 6.8.2 Proposition I.4

Once the auxiliary point has been constructed, SAS is used to compare the relevant triangles. Proposition I.4 transports the available side and angle congruences into the additional congruence information needed for the contradiction.

### 6.8.3 Proposition I.5

Proposition I.6 is the converse of Proposition I.5.

I.5 states that equal sides imply equal opposite angles:

$$AB \cong AC \implies \angle ABC \cong \angle BCA.$$

I.6 reverses the implication:

$$\angle ABC \cong \angle BCA \implies AB \cong AC.$$

The two propositions therefore establish the first clean equivalence between side symmetry and angle symmetry in a triangle.

### 6.8.4 Transition to Proposition I.7

Proposition I.7 uses this isosceles infrastructure inside a uniqueness argument. Equal distances to the endpoints of a base generate isosceles triangles, and the resulting angle congruences help exclude a second possible vertex on the same side of the base.

Thus I.6 is part of the rigidity machinery leading directly to I.7 and then to SSS in I.8.

## 6.9 Hilbert and Book Zero Components Used

### 6.9.1 Segment Trichotomy

The formal proof must turn the assumption

$$AB \not\cong AC$$

into a usable strict comparison. Segment trichotomy separates the possibilities

$$AB < AC, \quad AB \cong AC, \quad AC < AB.$$

The congruent branch contradicts the hypothesis, while the two strict branches are treated symmetrically.

### 6.9.2 Segment Layoff

In each strict branch a point is constructed on the longer side so that the cut-off segment is congruent to the shorter side. This is the formal counterpart of Euclid's application of Proposition I.3.

### 6.9.3 Isosceles Base Angles

The auxiliary equal-side configuration is converted into angle congruence by the established Hilbert isosceles theorem.

### 6.9.4 SAS and Angle Transport

SAS and angle-congruence transitivity transport the local auxiliary equalities into the orientation required to contradict the original angle hypothesis.

## 6.10 Formalization Notes

Proposition I.6 provides one of the clearest examples of the difference between a classical geometric proof and its modern formal reconstruction.

Euclid establishes the theorem by introducing an auxiliary point, constructing an additional triangle, and arguing by contradiction. The proof occupies a central place in Book I because it demonstrates how previous propositions can be combined to derive a converse statement.

Within the Hilbert framework, the same mathematical result becomes an almost immediate consequence of the previously established Angle–Side–Angle theorem. Once ASA has been derived from Hilbert’s axioms, the converse property of isosceles triangles follows directly, without auxiliary constructions or *reductio ad absurdum*.

This proposition therefore illustrates an important principle of the CGJteamLab reconstruction: as the interface theory grows, increasingly complex classical arguments collapse into short applications of high-level synthetic theorems.

## 6.11 Dependency Summary

The two proof paths are genuinely different:

Classical:

I.3 + I.4 + contradiction -> I.6

Formal:

hilbert\_asa\_sides -> euclid\_proposition\_6

The formal proof is therefore not a line-by-line rendering of Euclid’s argument. It reuses a stronger theorem already available in the Hilbert interface.

## 6.12 Conclusion

Proposition I.6 completes the converse half of Euclid’s elementary isosceles-triangle theory.

Together, I.5 and I.6 establish the equivalence between equal sides and equal opposite angles. The classical proof of I.6 is by contradiction: if the sides were unequal, the longer side could

be cut down to create an auxiliary isosceles configuration incompatible with the assumed angle congruence.

The formal reconstruction exposes one additional logical ingredient that is almost invisible on paper: from noncongruence of the two sides one must obtain a definite strict order. Segment trichotomy supplies that step and forces the proof to account for both possible orientations of the inequality.

This side-angle equivalence becomes immediate infrastructure for Proposition I.7 and, through I.7, for the SSS theorem of Proposition I.8.

## Chapter 7

# Proposition I.7

### 7.1 Introduction

Proposition I.7 occupies a special position in Book I of Euclid's *Elements*. Unlike the preceding propositions, its principal purpose is not to establish a new geometric construction or a direct property of triangles, but to prove a uniqueness principle.

Euclid shows that, on the same side of a given base, there cannot exist two distinct vertices determining triangles whose corresponding sides are equal. This uniqueness result immediately precedes Proposition I.8, where it is used as a crucial ingredient in the classical proof of the Side–Side–Side congruence theorem.

The Hilbert reconstruction follows a different logical order. The general Side–Side–Side theorem has already been established within the Hilbert interface. Consequently, Proposition I.7 becomes a concise uniqueness argument derived from the developed congruence theory, without reproducing Euclid's original chain of auxiliary arguments.

This proposition therefore provides one of the clearest examples of the difference between the historical organization of Euclid's *Elements* and the logical organization of a modern axiomatic development.

### 7.2 The Classical Configuration

Let  $AB$  be a given finite straight line and let  $C$  and  $D$  be two points lying on the same side of the line  $AB$ .

Assume that

$$AC \cong AD$$

and

$$BC \cong BD.$$

The proposition asks whether the two vertices  $C$  and  $D$  can be distinct under these assumptions. Euclid's objective is to prove that such a configuration is impossible.

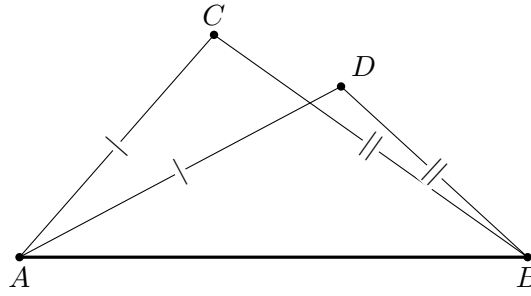


Figure 7.1: The forbidden configuration of Proposition I.7: distinct points  $C, D$  lie on the same side of the base  $AB$ , while  $AC \cong AD$  and  $BC \cong BD$ .

### 7.2.1 The Contradictory Configuration

Euclid joins the two assumed vertices by the segment  $CD$ .

Since

$$AC \cong AD,$$

the triangle  $ACD$  is isosceles. By Proposition I.5,

$$\angle ACD \cong \angle CDA.$$

From the geometric position of the points, one of these angles is strictly greater than the corresponding angle formed with the base vertex  $B$ .

On the other hand, since

$$BC \cong BD,$$

the triangle  $BCD$  is also isosceles. Proposition I.5 therefore gives

$$\angle BCD \cong \angle CDB.$$

The same pair of angles is thus forced to be both equal and unequal, which is impossible. Hence the two assumed vertices cannot be distinct.

## 7.3 Statement

**Theorem 2** (Euclid I.7). *Let  $AB$  be a given base, and let  $C$  and  $D$  lie on the same side of the line  $AB$ .*

*Assume*

$$AC \cong AD$$

*and*

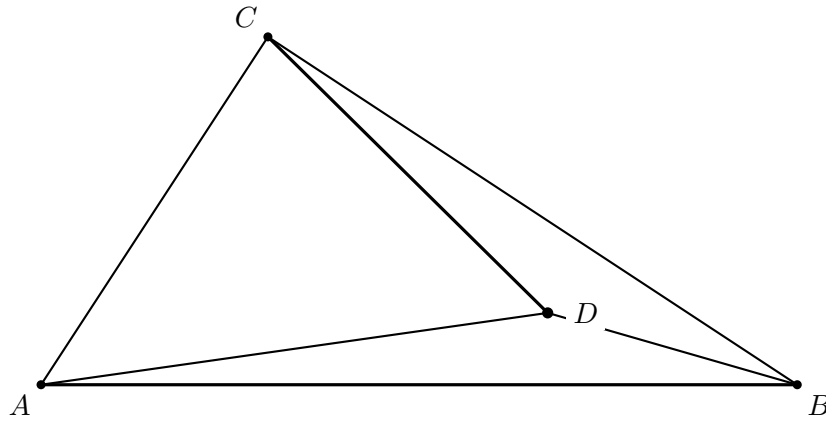


Figure 7.2: Euclid’s representative configuration after joining  $C$  to  $D$ . The point  $D$  is placed deep inside the configuration so that the four side segments and the auxiliary segment  $CD$  are visually separated. The equalities  $AC \cong AD$  and  $BC \cong BD$  are hypotheses of the proposition and are stated in the text rather than marked again on this diagram. This is one positional case, not a diagram of every same-side configuration.

$$BC \cong BD.$$

Then

$$C = D.$$

*Equivalently, on a fixed side of a given base there is at most one point having prescribed distances from the two endpoints of that base.*

**Diagrammatic audit.** The two original introductory figures were duplicates and have been merged. One figure now records the actual hypotheses, with the two pairs of equal segments marked; a second records the configuration obtained after joining  $C$  and  $D$ .

There is an important limitation to the second picture. Wilkins emphasizes that a complete proof of I.7 requires several positional configurations, whereas Euclid explicitly treats only one. In particular, after excluding the cases in which  $D$  lies on one of the rays through  $C$ , the relevant half-plane is divided into four regions  $K, L, M, N$ . The classical diagram below should therefore be read as Euclid’s representative case, not as a picture of every possible same-side configuration.

## 7.4 Classical Proof

Assume, for contradiction, that  $C$  and  $D$  are distinct.

Join  $C$  to  $D$ . Since

$$AC \cong AD,$$

the triangle  $ACD$  is isosceles. By Proposition I.5, its base angles are congruent:



$$\angle ACD \cong \angle CDA.$$

Similarly, since

$$BC \cong BD,$$

the triangle  $BCD$  is isosceles, and Proposition I.5 gives

$$\angle BCD \cong \angle CDB.$$

The position of the points on the same side of the base forces one of these angles to contain the other as a proper part. Hence an angle would have to be congruent to one of its proper parts, which is impossible.

Therefore the assumption that  $C$  and  $D$  are distinct leads to a contradiction. Hence

$$C = D.$$

Thus, on a fixed side of a given base, a point is uniquely determined by its distances from the two endpoints of the base.

## 7.5 Proof Architecture of the Reconstruction

The formal proof bears little resemblance to Euclid's original argument.

Instead of constructing auxiliary triangles and reasoning by contradiction, the implementation first applies the already established Side-Side-Side theorem of the Hilbert interface. This immediately shows that the two candidate triangles are congruent.

From the resulting congruence, the corresponding angle at the common base vertex is obtained. The uniqueness of angle construction then forces both candidate vertices to lie on the same ray issued from the base vertex.

Finally, since the two points lie on the same ray and are at the same distance from the common endpoint, the uniqueness of segment construction implies that the two points coincide.

```

Euclid
-----
same-side distinct C,D
AC ~= AD, BC ~= BD
    |
join CD
    |
I.5 + angle-order argument
    |
contradiction

```

Hilbert / Lean

```

-----
same-side C,D off base AB
  |
noncollinearity of ABC and ABD
  |
HilbertSSS on ABC and ABD
  |
equal angles at B
  |
hilbert_angle_unique_common_ray
  |
C and D lie on the same ray from B
  |
hilbert_segment_construction_unique
  |
C = D

```

## 7.6 Hilbert Reconstruction

The formal proof avoids the positional case analysis highlighted by the classical discussion.

Assume first that  $C \neq D$ . Since  $C$  and  $D$  lie on the same side of the line through  $A, B$ , both are off that line. The interface theorem `hilbert_not_collinear_of_off_line` therefore gives the noncollinearity of  $ABC$  and  $ABD$ .

The common side  $AB$ , together with

$$AC \cong AD, \quad BC \cong BD,$$

allows Hilbert SSS to be applied to the triangles  $ABC$  and  $ABD$ . In particular,

$$\angle ABC \cong \angle ABD.$$

The same-side hypothesis now matters a second time. Angle uniqueness on the common ray at  $B$  shows that  $C$  and  $D$  determine the same ray from  $B$ . Finally  $BC \cong BD$ , together with uniqueness of segment construction on that ray, forces

$$C = D,$$

contradicting the assumption  $C \neq D$ .

Thus the Hilbert proof replaces Euclid's region-sensitive angle-order argument by two uniqueness principles: uniqueness of the ray determined by an angle on a fixed side, followed by uniqueness of a point at a prescribed distance on that ray.

## 7.7 Lean Formalization

The complete implementation is available in the repository. Three short excerpts expose the structure of the formal argument.

First SSS is applied to the two triangles sharing the base  $AB$ :

```
have hSSS :=
  HilbertSSS
    Geo
    A B C
    A B D
    hABC
    hABAB
    hBCBD
    hACAD
```

The angle at  $B$  is then fed into the same-ray uniqueness theorem:

```
rcases
  hilbert_angle_unique_common_ray
    Geo
    A B C D
    base
    hAB
    hAbase
    hBbase
    hCbase
    hSame
    hAngleB with
  <X, hRayXC, hRayXD>
```

Finally, the equal distances from  $B$  and the common ray identify the two points:

```
exact
  hilbert_segment_construction_unique
    Geo
    B C
    B X
    C D
    hRayXC
    hRayXD
    (hilbert_congruent_reflexive Geo B C)
    hBD_BC
```

The remaining code establishes the off-line and noncollinearity conditions required by these interface theorems. Those bookkeeping steps are important for verification but do not add a new geometric idea to the proof.

## 7.8 Relationship with Earlier Propositions

### 7.8.1 Proposition I.5

The classical proof of I.7 creates isosceles triangles from the assumed equal distances to the endpoints of the common base. Proposition I.5 then converts those segment congruences into base-angle congruences.

This is the first essential step in the contradiction: metric symmetry at the level of sides becomes angular symmetry.

### 7.8.2 Proposition I.6

Proposition I.6 is the converse of the isosceles-triangle theorem. It allows angle congruence to be converted back into segment congruence when the contradiction argument reaches the appropriate auxiliary triangle.

Thus I.5 and I.6 together provide the two-way bridge between side congruence and angle congruence used in the uniqueness argument.

### 7.8.3 Transition to Proposition I.8

The importance of I.7 becomes fully visible in Proposition I.8.

Euclid's SSS proof superposes one triangle on another. After two vertices and the common base have been matched, I.7 excludes the possibility that the third vertex could occupy a different position on the same side of the base while preserving both distances to the base endpoints.

Thus I.7 supplies the uniqueness principle hidden inside Euclid's superposition proof of SSS.

## 7.9 Hilbert and Book Zero Components Used

### 7.9.1 Same-Side Geometry

The statement of I.7 is fundamentally a same-side uniqueness theorem. The two candidate vertices are assumed to lie on the same side of the common base. This positional information is explicit in the Hilbert reconstruction rather than inferred from the drawing.

### 7.9.2 Segment Congruence

The hypotheses prescribe equal distances from each candidate vertex to the two endpoints of the base. The formal proof repeatedly transports and composes these congruences in order to create the isosceles configurations used in the contradiction.

### 7.9.3 Isosceles Angle Theorems

The established Hilbert isosceles infrastructure converts the equal-side hypotheses into angle congruences. These are the formal counterparts of Euclid's applications of I.5.

### 7.9.4 Order and Collinearity

The contradictory configuration depends on the relative positions of the candidate vertices and the base. Betweenness, collinearity, and same-side information replace the diagrammatic assumptions in the classical proof.

## 7.10 Formalization Notes

Proposition I.7 illustrates a fundamental difference between the historical development of Euclid's geometry and its modern axiomatic reconstruction.

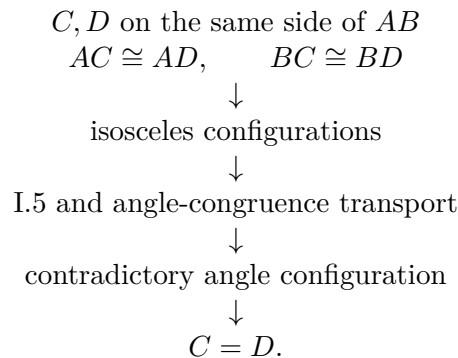
In the *Elements*, Proposition I.7 serves as an essential lemma for the proof of the Side–Side–Side congruence theorem in Proposition I.8. Euclid therefore establishes the uniqueness result before proving the general congruence theorem.

The Hilbert reconstruction follows the opposite logical direction. Once the general Side–Side–Side theorem has been derived from the Hilbert axioms, Proposition I.7 becomes an immediate uniqueness corollary obtained by combining triangle congruence with the uniqueness of angle and segment constructions.

This proposition therefore demonstrates that the historical order of the *Elements* need not coincide with the logical organization of a modern formal theory. As the interface develops, results that were once foundational become concise consequences of more general synthetic theorems.

## 7.11 Dependency Summary

The geometric content of Proposition I.7 can be summarized as



Equivalently, on a prescribed side of a base, the two distances to the base endpoints determine the vertex uniquely.

The formal reconstruction uses only established incidence, order, and congruence infrastructure. No coordinate representation or numerical distance is introduced.

## 7.12 Conclusion

Proposition I.7 is best understood as a rigidity or uniqueness theorem.

Its classical formulation is somewhat indirect: two different points on the same side of a base cannot have the same prescribed distances to both endpoints of that base. The formal reconstruction exposes the structural content more clearly: the intersection of two distance conditions is unique once the side of the base has been fixed.

The proof depends on the isosceles theory developed in I.5 and I.6 and turns segment congruence into a contradiction among angles and point positions.

Its main importance is prospective. Proposition I.8 uses exactly this uniqueness principle to justify the final step of the SSS superposition argument. I.7 therefore forms the bridge between the elementary isosceles results and the general triangle-rigidity theorem of I.8.

## Chapter 8

# Proposition I.8

### 8.1 Introduction

Proposition I.8 is one of the central results of Book I of Euclid's *Elements*. It establishes the Side–Side–Side criterion for triangle congruence, proving that triangles having three corresponding sides equal must also have their corresponding angles equal.

Within Euclid's development, this proposition concludes a carefully prepared sequence of results. Proposition I.7 provides the essential uniqueness argument that allows the final congruence theorem to be established.

The Hilbert reconstruction follows a different logical organization. The general Side–Side–Side theorem has already been derived from the Hilbert axioms before the Euclidean propositions are reconstructed. Consequently, Proposition I.8 becomes a direct application of this general theorem rather than the culmination of a long geometric argument.

This proposition therefore marks the point where the historical proof of Euclid and the modern Hilbert reconstruction differ most significantly while establishing exactly the same mathematical result.

### 8.2 The Classical Configuration

Consider two triangles,

$$\triangle ABC \quad \text{and} \quad \triangle DEF,$$

whose corresponding sides satisfy

$$AB \cong DE,$$

$$BC \cong EF,$$

and

$$AC \cong DF.$$

The objective is to prove that the corresponding angles of the two triangles are also congruent.

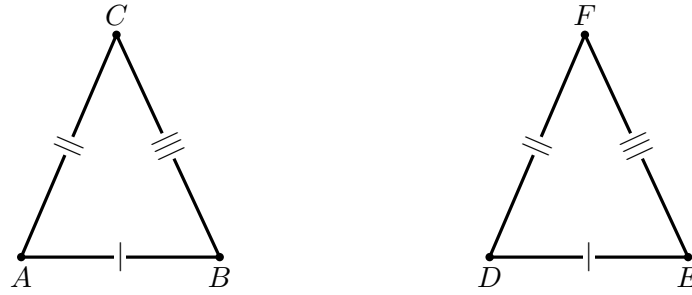


Figure 8.1: The data of Proposition I.8:  $AB \cong DE$ ,  $AC \cong DF$ , and  $BC \cong EF$ .

### 8.2.1 Superposition

Euclid conceptually places one triangle upon the other so that the vertex  $D$  coincides with  $A$  and the side  $DE$  lies along the side  $AB$ .

Since

$$AB \cong DE,$$

the endpoint  $E$  coincides with  $B$ .

The remaining vertex  $F$  must then lie at the intersection of two circles centered at  $A$  and  $B$  with radii equal to the corresponding sides of the first triangle.

Proposition I.7 shows that, on a fixed side of the base, such a point is unique. Consequently, the vertex  $F$  must coincide with  $C$ .

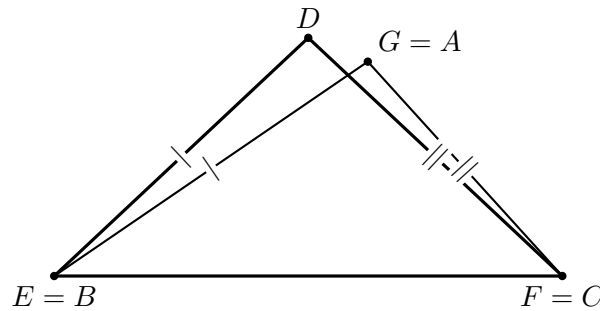


Figure 8.2: Euclid's superposition argument. After  $BC$  is placed on  $EF$ , the transported vertex  $G$  and the vertex  $D$  lie on the same side of  $EF$ . If  $G \neq D$ , then  $EG = ED$  and  $FG = FD$ , contradicting Proposition I.7. Hence the vertices coincide.

### 8.2.2 Application of Proposition I.7

Once the triangles have been superposed, the only remaining possibility is that the third vertices do not coincide.

In that case there would exist two distinct points on the same side of the common base satisfying the same two distance conditions.



Proposition I.7 excludes precisely this situation.

Therefore the remaining vertices must also coincide, and the two triangles agree completely. Consequently, all corresponding angles are congruent.

### 8.3 Statement

**Theorem 3** (Euclid I.8). *Let*

$$\triangle ABC \quad \text{and} \quad \triangle DEF$$

*be nondegenerate triangles.*

*Assume*

$$AB \cong DE,$$

$$BC \cong EF,$$

*and*

$$AC \cong DF.$$

*Then*

$$\angle BAC \cong \angle EDF,$$

$$\angle ABC \cong \angle DEF,$$

*and*

$$\angle ACB \cong \angle DFE.$$

*That is, triangles having three corresponding sides congruent also have their corresponding angles congruent.*

**Diagrammatic audit.** Two geometric figures are useful here. The first records the three side correspondences. The second isolates the genuinely nontrivial classical move: superposition followed by Proposition I.7. Wilkins's study note makes this same logical point explicit: after the first triangle is moved so that  $B, C$  occupy  $E, F$  on the chosen side, Proposition I.7 forces the remaining vertex to coincide with  $D$ . The much longer alternative proof attributed to Philo of Byzantium introduces five positional cases; it is historically interesting, but it is not the proof formalized here and is not duplicated diagrammatically.

## 8.4 Classical Proof

Superpose the triangle  $ABC$  onto the triangle  $DEF$  so that the vertex  $B$  coincides with  $E$  and the side  $BC$  lies along the side  $EF$ .

Since

$$BC \cong EF,$$

the point  $C$  coincides with  $F$ .

Suppose that the remaining vertex  $A$  does not coincide with  $D$ . Then there would exist two distinct points on the same side of the base  $EF$  satisfying

$$EA \cong ED$$

and

$$FA \cong FD.$$

This contradicts Proposition I.7, which asserts the uniqueness of such a point.

Therefore the vertex  $A$  must coincide with  $D$ , and the two triangles coincide completely.

Hence all corresponding angles are congruent:

$$\angle BAC \cong \angle EDF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

This establishes the Side–Side–Side congruence theorem.

## 8.5 Proof Architecture of the Reconstruction

The formal reconstruction is dramatically simpler than Euclid’s classical proof.

No superposition argument is required. Once the three corresponding side congruences are available, the previously established Side–Side–Side theorem of the Hilbert interface immediately provides the congruence of the two triangles together with the congruence of all corresponding angles.

The entire classical argument is therefore replaced by a single application of a high-level interface theorem.

```

Euclid
-----
three corresponding side equalities
      |
superpose BC on EF
      |
I.7 excludes a distinct remaining vertex
      |
triangles coincide
      |
corresponding angles are equal

```

```

Hilbert / Lean
-----
three corresponding segment congruences
      |
      v
HilbertSSS
      |
      v
angleA, angleB, angleC
      |
      v
euclid_proposition_8

```

## 8.6 Hilbert Reconstruction

The Hilbert reconstruction is structurally much shorter than Euclid’s proof at the level of Proposition I.8. The hypotheses are exactly the three side congruences required by the established theorem `HilbertSSS`. Applying that theorem to  $ABC$  and  $DEF$  yields the complete triangle congruence data, including all three corresponding angle congruences.

Consequently no superposition, rigid motion, or separate appeal to Proposition I.7 appears in `Proposition08.lean`. Those issues belong to the foundational proof of SSS rather than to this Book I wrapper.

## 8.7 Lean Formalization

The complete implementation is available in the repository. Here the code is short enough that only its decisive interface call needs to be displayed.

```

have hSSS :=
  HilbertSSS
    Geo
    A B C
    D E F
    hABC
    hAB
    hBC

```

hAC

The result returned by `HilbertSSS` already contains the three corresponding angle congruences. Proposition I.8 merely exposes them in the order required by the Euclidean statement:

```
exact
  <hSSS.2.angleA,
    hSSS.2.angleB,
    hSSS.2.angleC>
```

Thus the formal proposition does not reproduce Euclid's superposition argument. That foundational work has already been absorbed into the Hilbert SSS theorem.

## 8.8 Relationship with Earlier Propositions

### 8.8.1 Proposition I.4

Proposition I.4 establishes SAS: two sides and the included angle determine the remaining side and angles of a triangle.

Proposition I.8 provides the complementary rigidity theorem. If all three corresponding sides are congruent, then the corresponding angles are congruent as well.

Together I.4 and I.8 provide the two fundamental triangle-congruence principles used repeatedly in the rest of Book I.

### 8.8.2 Proposition I.5

The isosceles-triangle theorem supplies an early example of recovering angle information from segment congruence. Proposition I.8 generalizes that phenomenon from the special isosceles configuration to arbitrary triangles with three corresponding sides congruent.

### 8.8.3 Proposition I.7

Euclid uses Proposition I.7 to exclude the possibility that two different vertices on the same side of a base could realize the same two prescribed distances from the endpoints of that base.

This uniqueness result is the critical classical ingredient in I.8: after superposition, I.7 forces the remaining vertex to coincide with its counterpart.

### 8.8.4 Transition to Proposition I.9

Proposition I.9 uses SSS immediately. Its auxiliary construction creates two triangles with three corresponding sides congruent, and I.8 then yields equality of the two angles forming the desired angle bisector.

Thus I.8 becomes a central reusable congruence theorem as soon as it is established.

## 8.9 Hilbert and Book Zero Components Used

### 8.9.1 Hilbert SSS

The formal reconstruction packages the SSS rigidity statement as a Hilbert theorem rather than reproducing Euclid’s superposition argument inside every later application.

Its conclusion supplies the corresponding angle congruences required by the public Euclidean proposition.

### 8.9.2 Same-Side and Opposite-Side Geometry

The classical proof relies on the phrase “apply one triangle to the other” and on the location of the transported vertex. In the Hilbert setting this positional information is expressed explicitly through side-of-line relations.

### 8.9.3 Angle Congruence

The target of I.8 is not merely coincidence of triangles but congruence of corresponding angles. The formal theorem therefore returns the angle-congruence information in the orientation required by subsequent Book I propositions.

### 8.9.4 Nondegeneracy

The triangles must be genuine triangles. The formal hypotheses record noncollinearity explicitly, preventing degenerate configurations that are silently excluded by the classical diagram.

## 8.10 Formalization Notes

Proposition I.8 concludes the first major block of triangle congruence theory developed in Book I.

Historically, Euclid establishes the Side–Side–Side criterion only after proving the uniqueness result of Proposition I.7 and employing the method of superposition. The classical proof therefore depends on a carefully ordered sequence of earlier propositions.

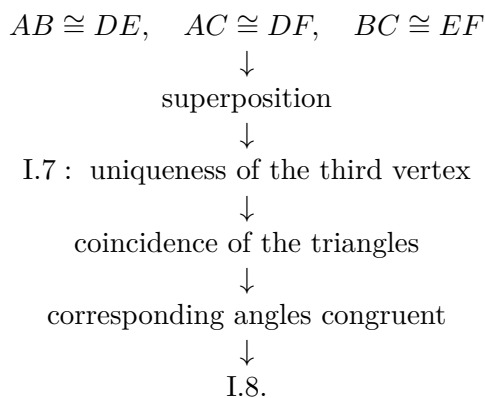
The Hilbert reconstruction follows a different logical organization. The general Side–Side–Side theorem has already been established as a theorem of the Hilbert interface. Consequently, Proposition I.8 becomes a direct application of this previously developed result, requiring no additional geometric construction.

Together, Propositions I.4 through I.8 illustrate the progressive development of triangle congruence theory. The classical arguments of Euclid are systematically replaced by applications of increasingly powerful synthetic theorems derived from the Hilbert axioms.

This transition marks the completion of the first coherent layer of the CGJteamLab Hilbert interface and demonstrates how a modern axiomatic framework can substantially simplify the formal verification of classical geometry.

## 8.11 Dependency Summary

The classical dependency structure is



In the Hilbert reconstruction, the same rigidity is isolated as the SSS theorem and used through the congruence interface.

No numerical lengths, coordinates, or new geometric axioms are required by the public Proposition I.8.

## 8.12 Conclusion

Proposition I.8 establishes Side-Side-Side congruence for triangles.

Historically, Euclid proves the result by superposition and uses Proposition I.7 to rule out a second possible position of the remaining vertex. The reconstruction makes the underlying rigidity theorem explicit in the Hilbert layer.

This is a major infrastructural point in Book I. From I.8 onward, three segment congruences can be converted into angle congruences without reconstructing the superposition argument each time.

The theorem is used immediately in Proposition I.9 and continues to support midpoint, perpendicular, and later triangle constructions. Thus I.8 is not only a classical proposition but one of the principal congruence interfaces of the formal development.

## Chapter 9

# Proposition I.9

### 9.1 Introduction

Proposition I.9 presents Euclid's construction of the bisector of a given rectilinear angle.

The classical proof combines several earlier propositions. A segment of one side of the angle is first copied onto the other side, after which an auxiliary equilateral triangle is constructed. The Side–Side–Side criterion established in Proposition I.8 is then used to prove that the joining segment bisects the angle.

The Hilbert reconstruction follows a different route. Rather than constructing an auxiliary equilateral triangle, it first lays off a segment congruent to one side of the angle, constructs the midpoint of the resulting segment, and finally applies the general Side–Side–Side theorem.

Although the geometric constructions differ substantially, both approaches establish the existence of an angle bisector within neutral geometry.

### 9.2 The Classical Configuration

Let  $\angle BAC$  be the given rectilinear angle; see Figure 9.1.

The objective is to construct a ray issuing from the vertex  $A$  that divides the angle into two congruent angles.

No assumption is made on the lengths of the sides forming the angle; only the angle itself is given.

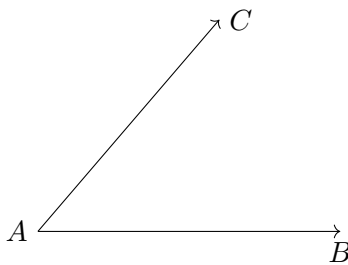


Figure 9.1: The given angle  $\angle BAC$ .

### 9.2.1 Construction

Choose an arbitrary point  $D$  on the side  $AB$ .

On the side  $AC$ , construct a point  $E$  such that

$$AE \cong AD.$$

Join the points  $D$  and  $E$ , and construct the equilateral triangle  $DEF$  on the segment  $DE$ .

Finally, join the vertex  $A$  to the vertex  $F$ ; the completed construction is shown in Figure 9.2.

The claim is that the segment  $AF$  bisects the given angle.

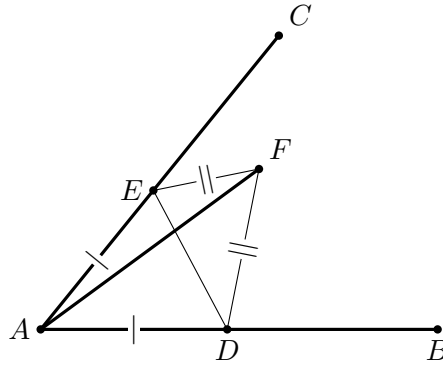


Figure 9.2: Euclid's construction for Proposition I.9. The points  $D, E$  satisfy  $AD \cong AE$ , and  $DEF$  is equilateral. With  $AF$  common, SSS for  $ADF$  and  $AEF$  proves that  $AF$  bisects the given angle.

### 9.2.2 Application of Proposition I.8

Since the triangle  $DEF$  is equilateral,

$$DF \cong EF.$$

Moreover,

$$AD \cong AE$$

by construction, while

$$AF \cong AF$$

is common to the two triangles.

Therefore the triangles

$$\triangle ADF \quad \text{and} \quad \triangle AEF$$

have three corresponding sides congruent.



By Proposition I.8,

$$\angle DAF \cong \angle FAE.$$

Since  $D$  lies on the ray  $AB$  and  $E$  lies on the ray  $AC$ , these are precisely the two parts of the original angle  $\angle BAC$ .

Hence  $AF$  bisects the given angle.

### 9.3 Statement

**Theorem 4** (Euclid I.9). *Let  $\angle BAC$  be a nondegenerate angle.*

*Then there exists a point  $M$  such that*

$$\angle BAM \cong \angle MAC.$$

*Equivalently, the ray  $AM$  bisects the angle  $\angle BAC$ .*

**Diagrammatic audit.** Two classical figures are sufficient: the given angle and the completed auxiliary construction. Wilkins's essential diagram is the latter: points  $D, E$  are equidistant from  $A$ ,  $DEF$  is equilateral, and  $AF$  is the candidate bisector. These are exactly the data used by SSS. No additional classical snapshots are needed. The Hilbert proof, however, uses a genuinely different configuration based on a midpoint of  $DC$ ; that difference is recorded in the architecture diagram rather than hidden behind a copy of Euclid's picture.

### 9.4 Classical Proof

Choose a point  $D$  on the side  $AB$  and construct a point  $E$  on the side  $AC$  such that

$$AD \cong AE.$$

Join the points  $D$  and  $E$ , and construct the equilateral triangle  $DEF$ .

Finally, join the vertex  $A$  to the vertex  $F$ .

Since

$$AD \cong AE,$$

$$DF \cong EF,$$

and

$$AF \cong AF,$$

the triangles

$$\triangle ADF \quad \text{and} \quad \triangle AEF$$

have three corresponding sides congruent.

By Proposition I.8,

$$\angle DAF \cong \angle FAE.$$

Because  $D$  lies on the ray  $AB$  and  $E$  lies on the ray  $AC$ , these angles are exactly

$$\angle BAF \quad \text{and} \quad \angle FAC.$$

Hence the ray  $AF$  bisects the given angle.

This completes the construction.

## 9.5 Proof Architecture of the Reconstruction

The Hilbert reconstruction replaces Euclid's auxiliary equilateral triangle by two fundamental constructions already available in the Hilbert interface.

First, a segment congruent to one side of the angle is laid off on the other side of the angle. The midpoint of the resulting segment is then constructed, and the Side-Side-Side theorem is applied to two auxiliary triangles.

Thus the existence of the angle bisector follows entirely from previously established neutral-geometric results.

Euclid

-----

angle BAC

|

I.3: choose D,E with  $AD \cong AE$

|

I.1: construct equilateral DEF

|

I.8: SSS on ADF and AEF

|

AF bisects angle BAC

Hilbert / Lean

-----

noncollinear A,B,C

|

segment\_construction: D on ray AB,  $AD \cong AC$

|

HilbertMidpointExists: M midpoint of DC

```

|
HilbertSSS: ADM ~= ACM
|
same-ray angle transport: DAM = BAM
|
AM bisects angle BAC

```

## 9.6 Hilbert Reconstruction

The formal reconstruction uses a different symmetric configuration from Euclid's. From the noncollinear points  $A, B, C$ , segment construction produces a point  $D$  on the ray  $AB$  such that

$$AD \cong AC.$$

The midpoint theorem is then applied to  $DC$ , producing  $M$  with

$$D - M - C, \quad DM \cong MC.$$

The triangles  $ADM$  and  $ACM$  have the three corresponding sides

$$AD \cong AC, \quad DM \cong CM, \quad AM \cong AM.$$

Hilbert SSS therefore yields

$$\angle DAM \cong \angle CAM.$$

Finally, because  $D$  lies on the same ray from  $A$  as  $B$ , angle transport identifies  $\angle DAM$  with  $\angle BAM$ . Hence

$$\angle BAM \cong \angle CAM,$$

which is the required angle bisection.

This route uses neither Euclid's equilateral triangle  $DEF$  nor a direct invocation of Proposition I.8. Its symmetry is created instead by segment layoff and midpoint construction.

## 9.7 Lean Formalization

The complete implementation is available in the repository. The purpose of the present section is therefore not to reproduce the whole source file, but to isolate the steps that explain the formal geometry.

The first characteristic move is the replacement of Euclid's pair of equal segments  $AD = AE$  by a single laid-off segment on the ray  $AB$ :

```

rcases
  HilbertCongruence.segment_construction

```

```

(Geo := Geo)
A C
A B
hAB with
<D, hRayBD, hAD_AC>

```

Thus  $D$  lies on the ray  $AB$  and satisfies

$$AD \cong AC.$$

This is the first point at which the Hilbert reconstruction visibly departs from Euclid's construction.

The second decisive step is the use of a midpoint of  $DC$ :

```

rcases
  HilbertMidpointExists
    Geo D C hDC with
  <M, hMid>

```

The midpoint provides

$$D - M - C, \quad DM \cong MC.$$

Together with  $AD \cong AC$  and the common side  $AM$ , this creates the two triangles used for SSS.

The proof then applies

```

have hSSS :=
  HilbertSSS
    Geo
    A D M
    A C M
    hADM
    hACM
    hAD_AC
    hDM_CM
    hAM_AM

```

and extracts the equality of the angles at  $A$ . A final same-ray transport replaces the ray  $AD$  by the original ray  $AB$ , yielding

$$\angle BAM \cong \angle CAM.$$

The full theorem contains the incidence and noncollinearity bookkeeping needed to justify these steps. Those details remain in the source code; the three excerpts above are the parts that explain the mathematical structure of the Lean proof.

## 9.8 Relationship with Earlier Propositions

### 9.8.1 Proposition I.1

Euclid begins the construction by choosing points on the two arms of the given angle and constructing an equilateral triangle on the segment joining them. Proposition I.1 supplies the

symmetric auxiliary configuration from which the angle bisector will be obtained.

### 9.8.2 Proposition I.3

The classical proof uses segment cutting to arrange equal auxiliary lengths on the two sides of the given angle. Proposition I.3 supplies this segment-layoff step.

### 9.8.3 Proposition I.8

Proposition I.8 is the decisive congruence theorem in Euclid's proof. The two triangles adjacent to the candidate bisector have three corresponding sides congruent, so SSS yields congruence of the two angles at the original vertex.

Thus the classical dependency pattern is

$$\text{I.3} \longrightarrow \text{I.1} \longrightarrow \text{I.8} \longrightarrow \text{I.9}.$$

### 9.8.4 Transition to Proposition I.10

Proposition I.10 uses the angle-bisection construction immediately: an equilateral triangle is erected on the given segment and its apex angle is bisected. SAS then shows that the point where the bisector meets the base is the midpoint.

Thus I.9 becomes constructional infrastructure as soon as it is proved.

## 9.9 Hilbert and Book Zero Components Used

### 9.9.1 Angle-Bisector Representation

The conclusion is expressed synthetically by a ray from the vertex whose two adjacent angles are congruent. No numerical angle measure is used.

### 9.9.2 Segment Layoff

The Hilbert segment-construction machinery supplies the equal auxiliary segments needed to create the symmetric configuration around the given angle.

### 9.9.3 SSS

The congruence of the two auxiliary triangles is obtained from the established Hilbert SSS theorem. The desired angle congruence is then a corresponding-angle consequence.

### 9.9.4 Interior and Ray Information

The constructed bisector must lie in the intended interior region of the given angle. The formal proof therefore carries ray and side-of-line information that is implicit in Euclid's diagram.

## 9.10 Formalization Notes

Proposition I.9 is the first construction in Book I whose Hilbert reconstruction differs substantially from Euclid’s original proof.

Euclid constructs an auxiliary equilateral triangle and applies the Side–Side–Side criterion established in Proposition I.8. The proof is therefore closely tied to the historical sequence of propositions in the *Elements*.

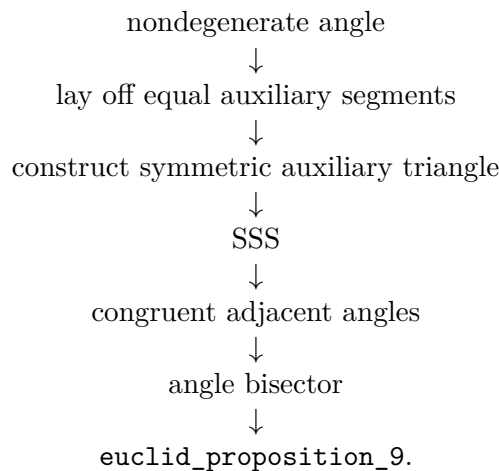
The Hilbert reconstruction adopts a different strategy. A congruent segment is first laid off on one side of the angle, the midpoint of the resulting segment is constructed, and the general Side–Side–Side theorem is applied directly to two auxiliary triangles.

This approach illustrates a general principle of the CGJteamLab library: classical geometric constructions are not reproduced verbatim when a simpler synthetic argument is available within the existing Hilbert theory.

Proposition I.9 therefore demonstrates how a modern axiomatic development can replace a classical construction by a shorter proof while preserving the same mathematical content.

## 9.11 Dependency Summary

The formal reconstruction follows the same basic synthetic geometry as the classical proof:



The reconstruction uses only the established Hilbert and Book Zero infrastructure. It introduces neither numerical angle measure nor a new geometric axiom.

## 9.12 Conclusion

Proposition I.9 constructs the bisector of a given rectilinear angle.

The classical proof is an elegant combination of the early construction theorems and SSS: equal auxiliary segments create a symmetric configuration, and triangle congruence converts that symmetry into equality of the two angles at the vertex.

The formal reconstruction preserves this synthetic character while making the positional content explicit. A bisector is not merely an abstract equality of two angle magnitudes; it is represented

by a specific ray lying in the appropriate geometric configuration.

I.9 is also an important infrastructural milestone. The construction is used immediately in Proposition I.10 and becomes part of the reusable angle-construction language needed throughout the later development.

## Chapter 10

# Proposition I.10

### 10.1 Introduction

Proposition I.10 constructs the midpoint of a given finite straight line.

In Euclid's development this construction depends on several earlier results. An equilateral triangle is first erected on the given segment using Proposition I.1. The vertex angle of this triangle is then bisected by Proposition I.9, and the Side–Angle–Side congruence criterion of Proposition I.4 is used to prove that the intersection point is the midpoint of the original segment.

The Hilbert reconstruction follows a fundamentally different approach. The existence of a midpoint has already been established as an independent theorem of the Hilbert interface. Consequently, Proposition I.10 becomes a direct application of this previously derived result.

This proposition therefore provides one of the clearest examples of the difference between Euclid's historical development and a modern axiomatic reconstruction.

### 10.2 The Classical Configuration

Let  $AB$  be the given finite straight line; see Figure 10.1.

The objective is to construct a point lying on the segment  $AB$  that divides it into two congruent parts.

Such a point is called the midpoint of the segment.

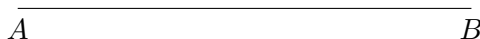


Figure 10.1: The given finite straight line  $AB$ .

#### 10.2.1 Construction

Construct the equilateral triangle  $ABC$  on the segment  $AB$  by Proposition I.1.

Next, bisect the angle  $\angle ACB$  by Proposition I.9.

Let the angle bisector meet the segment  $AB$  at the point  $D$ ; the completed construction is Figure 10.2.



The claim is that  $D$  is the midpoint of the segment  $AB$ .

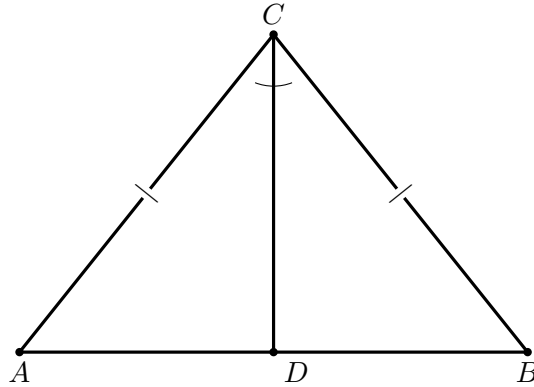


Figure 10.2: Euclid's construction for Proposition I.10. The equilateral triangle gives  $AC \cong BC$ ; the ray  $CD$  bisects the angle at  $C$ . SAS for  $ACD$  and  $BCD$  yields  $AD \cong DB$ .

### 10.2.2 Application of Proposition I.4

Consider the triangles

$$\triangle ACD \quad \text{and} \quad \triangle BCD.$$

Since the triangle  $ABC$  is equilateral,

$$AC \cong BC.$$

The segment

$$CD$$

is common to both triangles, and by construction,

$$\angle ACD \cong \angle DCB.$$

Therefore the hypotheses of Proposition I.4 are satisfied.

It follows that

$$AD \cong DB.$$

Since the point  $D$  lies on the segment  $AB$ , it is the midpoint of the given finite straight line.

## 10.3 Statement

**Theorem 5** (Euclid I.10). *Let  $AB$  be a finite straight line with*

$$A \neq B.$$

*Then there exists a midpoint of the segment  $AB$ . That is, there exists a point  $M$  such that*

$$A, M, B$$

*are collinear,*

$$M$$

*lies between  $A$  and  $B$ , and*

$$AM \cong MB.$$

*Equivalently, every nondegenerate segment admits a midpoint.*

**Diagrammatic audit.** Two geometric figures are sufficient. The first records the input segment; the second contains the entire Euclidean construction and the data used by SAS. Wilkins likewise presents the equilateral-triangle/angle-bisector configuration as the essential picture. His note also records a distinct historical construction attributed by Proclus to Apollonius, using two circles and SSS; that alternative is mathematically interesting but is not part of Euclid's proof adopted here and is therefore not duplicated as a second construction.

## 10.4 Classical Proof

Construct the equilateral triangle  $ABC$  on the segment  $AB$  by Proposition I.1.

Bisect the angle  $\angle ACB$  by Proposition I.9, and let the bisecting ray meet the segment  $AB$  at the point  $D$ .

Since

$$AC \cong BC,$$

$$CD \cong CD,$$

and

$$\angle ACD \cong \angle DCB,$$

the triangles

$$\triangle ACD \quad \text{and} \quad \triangle BCD$$

satisfy the hypotheses of Proposition I.4.

Hence

$$AD \cong DB.$$

Because the point  $D$  lies on the segment  $AB$ , it is the midpoint of the given finite straight line.

This completes the construction.

## 10.5 Proof Architecture of the Reconstruction

The Hilbert reconstruction replaces the entire classical construction by a previously established theorem of the Hilbert interface.

Rather than constructing an auxiliary equilateral triangle, bisecting an angle, and applying the Side–Angle–Side criterion, the existence of a midpoint is obtained directly from the general midpoint existence theorem.

Consequently, Proposition I.10 becomes a simple wrapper around an existing interface theorem.

Euclid

-----

given segment AB

|

I.1: equilateral triangle ABC

|

I.9: bisect angle ACB

|

I.4: SAS

|

AD = DB

|

D is midpoint of AB

Hilbert / Lean

-----

A ≠ B

|

∨

HilbertMidpointExists

|

∨

exists M, HilbertIsMidpoint Geo M A B

|

∨

euclid\_proposition\_10

## 10.6 Hilbert Reconstruction

The reconstruction consists of a direct application of the general midpoint existence theorem.

Given a nondegenerate segment, the theorem `HilbertMidpointExists` immediately produces a point lying between the endpoints and equidistant from them.

No auxiliary geometric construction is required, since the existence of midpoints has already been established independently within the Hilbert development.

This represents one of the largest simplifications obtained by the layered architecture of the Hilbert interface.

## 10.7 Lean Formalization

The Lean theorem is correspondingly short:

```
theorem euclid_proposition_10
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B : Geo.Point)
  (hAB : A != B) :
  exists M : Geo.Point,
    HilbertIsMidpoint Geo M A B := by
exact
  HilbertMidpointExists
    Geo A B hAB
```

Thus the proposition itself introduces no auxiliary construction. Its mathematical content has already been packaged in `HilbertMidpointExists`; I.10 is the Euclidean-facing wrapper exposing that result in the Book I development.

## 10.8 Relationship with Earlier Propositions

### 10.8.1 Proposition I.1

Euclid's construction of the midpoint begins by constructing an equilateral triangle on the given segment. Proposition I.1 therefore supplies the first auxiliary point and the equal sides needed later in the congruence argument.

### 10.8.2 Proposition I.4

The decisive classical step is SAS. After the angle at the apex has been bisected, the two resulting triangles share one side, have a pair of equal sides from the equilateral construction, and have congruent included angles. Proposition I.4 then yields congruence of the two triangles and hence equality of the two halves of the original segment.

### 10.8.3 Proposition I.9

Proposition I.9 is the immediate constructional input: it bisects the angle at the apex of the equilateral triangle.

Thus Euclid’s dependency chain is particularly transparent:

$$\text{I.1} \longrightarrow \text{I.9} \longrightarrow \text{I.4} \longrightarrow \text{I.10}.$$

#### 10.8.4 Transition to Proposition I.11

The midpoint construction becomes useful immediately in the classical proof of Proposition I.11, where a symmetric configuration about a point on a line is used to construct a perpendicular.

More generally, midpoint and segment-bisection infrastructure becomes a recurring component of later Book I constructions.

### 10.9 Hilbert and Book Zero Components Used

#### 10.9.1 Midpoint Representation

The formal conclusion is expressed by the project midpoint predicate: the constructed point lies between the endpoints and the two resulting segments are congruent. Thus midpoint remains entirely synthetic:

$$A - M - B \quad \text{and} \quad AM \cong MB.$$

#### 10.9.2 Interface Theorem

At the level of Proposition I.10 itself, the only substantive geometric dependency is

```
HilbertMidpointExists
```

together with the incidence and congruence structures required by that theorem. Angle bisection and SAS belong to Euclid’s historical proof; they are not invoked by `euclid_proposition_10`.

### 10.10 Formalization Notes

Proposition I.10 provides one of the clearest illustrations of the difference between Euclid’s historical development and a layered axiomatic reconstruction.

In the *Elements*, the midpoint construction depends on three earlier propositions. Euclid first constructs an equilateral triangle, then bisects one of its angles, and finally applies the Side–Angle–Side criterion to prove that the resulting point divides the original segment into two equal parts.

The Hilbert reconstruction is fundamentally different. The existence of midpoints has already been established as an independent theorem of the Hilbert interface. Consequently, Proposition I.10 becomes a direct application of this theorem, requiring no additional geometric construction.

This proposition demonstrates the principal advantage of the layered architecture developed in CGJteamLab. Once sufficiently rich interface theorems have been established, many classical

propositions cease to be independent proofs and instead become concise applications of the existing theory.

Proposition I.10 is perhaps the simplest example of this philosophy, reducing an entire classical construction to a single theorem call.

## 10.11 Dependency Summary

The two proof architectures should be kept distinct:

Classical:

I.1 -> I.9 -> I.4 -> I.10

Formal:

HilbertMidpointExists -> euclid\_proposition\_10

The formal theorem does not replay Euclid's equilateral-triangle, angle-bisection, and SAS construction. It reuses a midpoint theorem already proved in the Hilbert interface. No coordinate system, numerical length, or new geometric axiom is introduced.

## 10.12 Conclusion

Proposition I.10 completes the first ten propositions of Book I with the construction of a midpoint.

The classical proof is a compact synthesis of three earlier results: I.1 supplies an equilateral triangle, I.9 bisects its apex angle, and I.4 converts the resulting SAS configuration into equality of the two halves of the given segment.

The formal reconstruction preserves this synthetic character while making one feature especially explicit: a midpoint is not merely a point with two equal distances. Its position between the endpoints is part of the theorem and must be carried by the Hilbert order structure.

I.10 therefore consolidates the early construction and congruence machinery into a reusable segment-bisection result. This midpoint infrastructure becomes immediately relevant to the perpendicular constructions beginning with Proposition I.11.

# Chapter 11

## Proposition I.11

### 11.1 Introduction

Proposition I.11 states:

To draw a straight line at right angles to a given straight line from a given point on it.

Let  $AB$  be the given straight line and let  $C$  be the given point on  $AB$ .

Euclid's construction proceeds as follows. Choose a point  $D$  on  $AC$ . By Proposition I.3, construct a point  $E$  on the opposite side of  $C$  such that

$$CD \cong CE.$$

On the segment  $DE$  construct the equilateral triangle  $DFE$  by Proposition I.1, and join  $C$  to  $F$ .

Since

$$CD \cong CE, \quad DF \cong EF,$$

and  $CF$  is common to the two triangles  $DCF$  and  $ECF$ , Proposition I.8 gives

$$\angle DCF \cong \angle ECF.$$

The points  $D$ ,  $C$ , and  $E$  lie on one straight line. Hence the two congruent angles  $DCF$  and  $FCE$  are adjacent angles formed when the straight line  $CF$  stands on the straight line  $DE$ .

By Definition 10, each of these angles is therefore a right angle. Consequently,  $CF$  is drawn at right angles to the given straight line  $AB$  at the given point  $C$ .

Thus the required perpendicular has been constructed.

### 11.2 The Classical Configuration

Let  $AB$  be the given straight line and let  $C$  be the prescribed point on it. The task is to construct a straight line through  $C$  perpendicular to  $AB$ .

Euclid's construction chooses a point  $D$  on one side of  $C$  on the given line, then chooses  $E$  on the opposite side so that

$$CD \cong CE.$$

An equilateral triangle  $DFE$  is constructed on the base  $DE$ .

The relevant configuration is therefore

$$D - C - E, \quad CD \cong CE, \quad DF \cong FE.$$

The segment  $CF$  is the candidate perpendicular. The proof compares the two triangles

$$\triangle DCF \quad \text{and} \quad \triangle ECF.$$

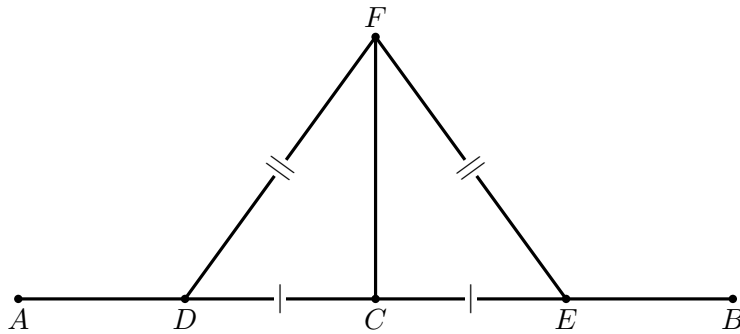


Figure 11.1: Classical construction for Proposition I.11. The points  $D, C, E$  are collinear with  $CD \cong CE$ , and  $DFE$  is equilateral. SSS applied to  $DCF$  and  $ECF$  makes the adjacent angles at  $C$  equal.

### 11.3 Statement

The synthetic content of Proposition I.11 is the existence of a perpendicular through a prescribed point of a given straight line.

**Theorem 6** (Euclid I.11). *Let  $A, C, B$  be three points such that*

$$A * C * B.$$

*Then there exists a point  $X$  such that the angle  $ACX$  is a right angle.*

In the Hilbert development, a right angle is represented directly by the Euclidean definition: an angle is right when it is congruent to its adjacent angle obtained by extending one of its sides through the vertex.

More precisely, we define

$$\text{RightAngle}(A, O, B)$$

to mean that there exists a point  $C$  such that

$$A * O * C$$

and

$$\angle AOB \cong \angle BOC.$$



Thus Proposition I.11 is represented formally by

$$A * C * B \implies \exists X \text{ RightAngle}(A, C, X).$$

The corresponding Lean definition is

```
def HilbertRightAngle
  (Geo : Geometry.Geo)
  (A O B : Geo.Point) : Prop :=
  exists C : Geo.Point,
  Geo.Between A O C /\
  Geo.AngleCongruent A O B B O C
```

and the proposition itself has the form

```
theorem euclid_proposition_11
  [HilbertCongruence Geo]
  (A C B : Geo.Point)
  (hACB : Geo.Between A C B) :
  exists X : Geo.Point,
  HilbertRightAngle Geo A C X
```

Notice that the formal statement does not attempt to encode Euclid's particular construction of the perpendicular. It expresses the geometric result of that construction: through the prescribed point  $C$  on the given line there exists a ray  $CX$  forming a right angle with the line.

**Diagrammatic audit.** Only one classical geometric figure is needed here. Wilkins's diagram contains exactly the construction that carries the proof: the symmetric points  $D, E$ , the equilateral triangle  $DFE$ , and the segment  $CF$ . There is no hidden sequence of geometrically different states worth duplicating. In contrast, the Hilbert proof below uses a substantially different auxiliary construction; its complexity is primarily order-theoretic and is better exposed by the dependency architecture than by copying the classical picture repeatedly.

## 11.4 Classical Proof

Let  $AB$  be the given straight line and let  $C$  be the prescribed point on it.

Choose a point  $D$  on  $AC$ . By Proposition I.3, cut off from the ray starting at  $C$  in the direction of  $B$  a segment  $CE$  congruent to  $CD$ . Thus

$$CD \cong CE,$$

and the points  $D, C, E$  lie on one straight line, with  $C$  between  $D$  and  $E$ .

Construct the equilateral triangle  $DFE$  on the segment  $DE$  by Proposition I.1, and join  $C$  to  $F$ ; see Figure 11.1. Since the triangle is equilateral,

$$DF \cong EF.$$

Moreover,

$$CF \cong CF$$

by reflexivity.

Consider the triangles

$$\triangle DCF \quad \text{and} \quad \triangle ECF.$$

Their three corresponding sides satisfy

$$CD \cong CE, \quad DF \cong EF, \quad CF \cong CF.$$

Therefore Proposition I.8 yields

$$\angle DCF \cong \angle ECF.$$

Since  $D, C, E$  are collinear and  $C$  lies between  $D$  and  $E$ , the rays  $CD$  and  $CE$  are opposite rays. Hence

$$\angle DCF \quad \text{and} \quad \angle FCE$$

are adjacent angles formed by the straight line  $CF$  standing on the straight line  $DE$ .

The two adjacent angles are congruent. By Euclid's Definition 10, each of them is therefore a right angle.

Consequently,

$$\angle DCF$$

is a right angle, and the straight line through  $C$  and  $F$  is perpendicular to the given straight line  $AB$  at the prescribed point  $C$ .

This proves Proposition I.11.

## 11.5 Proof Architecture of the Reconstruction

The formal reconstruction separates the notion of a right angle from the particular statement of Proposition I.11.

At the foundational level, the predicate `HilbertRightAngle` expresses Euclid's Definition 10. An angle  $AOB$  is right when there exists a point  $C$  on the extension of  $AO$  through  $O$  such that

$$\angle AOB \cong \angle BOC.$$

The substantial geometric result is then established independently as

```
hilbert_right_angle_exists
```

which proves that, whenever

$$A * C * B,$$

there exists a point  $X$  such that

$$\text{HilbertRightAngle}(A, C, X).$$

The proof of this existence theorem is considerably more elaborate than the final formalization of Proposition I.11. It uses the developed Hilbert theory of angle construction, segment construction, midpoints, triangle congruence, ray order, plane separation, and angle addition.

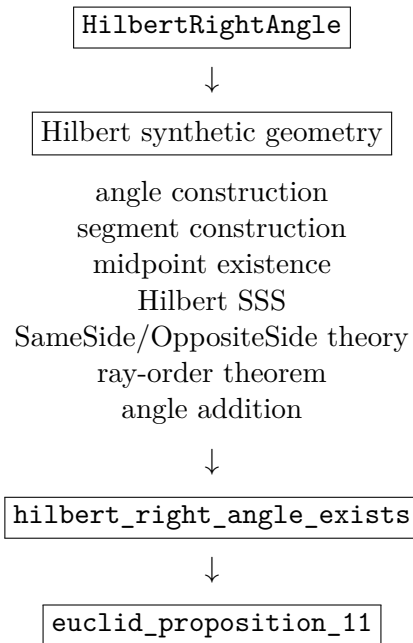
Once this general result has been established, Proposition I.11 itself becomes a direct interface theorem:

```

hilbert_right_angle_exists
|
v
euclid_proposition_11

```

More explicitly, the architecture is



Thus the formal dependency structure differs sharply from Euclid's historical construction. Euclid obtains the perpendicular by constructing an equilateral triangle and applying Proposition I.8. The Hilbert development instead proves a general right-angle existence theorem from the already developed synthetic theory and then obtains Proposition I.11 as an immediate consequence.

This distinction is intentional. The proposition file records the mathematical content of Euclid I.11, while the underlying Hilbert layer is free to establish that content by a more general synthetic argument.

## 11.6 Hilbert Reconstruction

The Hilbert proof of the existence theorem does not reproduce Euclid's equilateral-triangle construction. Instead, it derives a right angle from the general machinery already available in the Hilbert layer.

Assume

$$A * C * B.$$

Thus  $A, C, B$  lie on a common line, which we denote by  $b$ .

Choose a point  $D$  outside  $b$ . The angle-construction theorem is then applied at  $C$  to construct a point  $E$ , on the same side of  $b$  as  $D$ , such that

$$\angle ACD \cong \angle BCE.$$

The uniqueness part of angle construction determines the ray  $CE$ .

Next choose a point  $F$  on the ray  $CD$  such that

$$CF \cong CE.$$

Let  $X$  be the midpoint of  $FE$ . Hence

$$F * X * E$$

and

$$FX \cong XE.$$

Consider the triangles

$$\triangle CFX \quad \text{and} \quad \triangle CEX.$$

They satisfy

$$CF \cong CE, \quad FX \cong XE, \quad CX \cong CX.$$

Hilbert SSS therefore gives

$$\angle FCX \cong \angle ECX.$$

Since  $F$  and  $D$  lie on the same ray from  $C$ , replacing the ray  $CF$  by the ray  $CD$  does not change the corresponding angle. Consequently,

$$\angle DCX \cong \angle ECX.$$

We have therefore obtained the two component congruences

$$\angle ACD \cong \angle BCE$$

and

$$\angle DCX \cong \angle ECX.$$

The remaining problem is to justify that these component angles may be added in the required configuration.

This is the technically most substantial part of the reconstruction. The ray-order theorem applied to the noncollinear configuration determines two possible cases:

$$\text{RayMeetsSegment}(CD, EA)$$

or

$$\text{RayMeetsSegment}(CE, DA).$$

In the first case, the ray  $CD$  meets the open segment  $EA$  at a point  $Y$ . The SameSide/OppositeSide calculus shows that

$$A \text{ and } X$$

lie on opposite sides of the line  $CD$ , while

$$B \text{ and } X$$

lie on opposite sides of the line  $CE$ .

In the second case, the ray  $CE$  meets the open segment  $DA$  at a point  $Y$ . Plane-separation arguments show instead that

$$A \text{ and } X$$

lie on the same side of  $CD$ , while

$$B \text{ and } X$$

lie on the same side of  $CE$ .

Thus in both cases the two configurations have the same side pattern:

$$\text{SameSide}(A, X; CD) \iff \text{SameSide}(B, X; CE).$$

The proof also establishes that  $X$  does not lie on the original line  $AB$ . Hence both

$$\angle ACX \quad \text{and} \quad \angle BCX$$

are nondegenerate angles.

The angle-addition theorem can now be applied to

$$\angle ACD \cong \angle BCE$$

and

$$\angle DCX \cong \angle ECX.$$

It yields

$$\angle ACX \cong \angle BCX.$$

Since  $A * C * B$ , the rays  $CA$  and  $CB$  are opposite. Reversing the order of the sides in the second angle gives

$$\angle ACX \cong \angle XCB.$$

Therefore the two adjacent angles formed by  $CX$  with the straight line  $AB$  are congruent.

By the definition of `HilbertRightAngle`,

$$\text{HilbertRightAngle}(A, C, X).$$

Hence

$$\exists X \text{ HilbertRightAngle}(A, C, X),$$

which is the required Hilbert form of Proposition I.11.

## 11.7 Lean Formalization

The formalization is divided into three levels: the definition of a right angle, the general Hilbert existence theorem, and the final Euclid I.11 interface theorem.

The notion of a right angle is represented by the following definition:

```
def HilbertRightAngle
  (Geo : Geometry.Geo)
  (A O B : Geo.Point) : Prop :=
  exists C : Geo.Point,
  Geo.Between A O C /\
  Geo.AngleCongruent A O B B O C
```

This is a direct formalization of the adjacent-angle characterization of a right angle. The witness  $C$  extends the ray  $OA$  through  $O$ , and the congruence

$$\angle AOB \cong \angle BOC$$

states that the two adjacent angles are equal.

The main work is contained in the general existence theorem:

```
theorem hilbert_right_angle_exists
[HilbertCongruence Geo]
(A C B : Geo.Point)
(hACB : Geo.Between A C B) :
exists X : Geo.Point,
HilbertRightAngle Geo A C X := by
...
```

The proof first extracts the incidence and nondegeneracy information contained in `hACB`. It then chooses an auxiliary point off the line  $AB$  and constructs a congruent angle on the opposite ray at  $C$ .

A point  $F$  is subsequently constructed on the auxiliary ray with

$$CF \cong CE,$$

and the midpoint  $X$  of  $FE$  is introduced. The central triangle congruence step is encoded by an application of Hilbert SSS:

```
have hSSS :=
HilbertSSS ...
```

This produces the congruence of the two angles at  $C$  required for the second component of the final angle-addition argument.

The principal technical issue is not the congruence calculation but the relative position of the rays. The proof therefore invokes the ray-order theorem and splits into its two possible cases:

```
rcases hRayOrder with hCaseD | hCaseE
```

Each branch establishes the same logical condition required by the angle-addition theorem:

```
have hSideConfiguration :
HilbertSameSide Geo A X lineCD <->
HilbertSameSide Geo B X lineCE := by
...
```

In one branch both propositions are false; in the other both are true. This case distinction isolates the plane-separation reasoning from the final angle calculation.

After proving the necessary noncollinearity conditions, the two component angle congruences are combined by

```
have hACX_BCX :
Geo.AngleCongruent A C X B C X :=
hilbert_angle_addition
...
hSideConfiguration
```

```
...
hAngleE
hDCX_ECX
```

The second angle is then reversed:

```
have hACX_XCB :
  Geo.AngleCongruent A C X X C B :=
  (Geo.angle_congruent_reverse_second
   A C X B C X).mp hACX_BCX
```

Since the original hypothesis already supplies

$$A * C * B,$$

the point  $B$  itself is the witness required by the definition of `HilbertRightAngle`. The existence theorem is therefore completed by

```
refine <X, ?_>
refine <B, hACB, ?_>
exact hACX_XCB
```

Once this theorem is available, the formal statement of Proposition I.11 contains no additional geometric argument:

```
theorem euclid_proposition_11
  [HilbertCongruence Geo]
  (A C B : Geo.Point)
  (hACB : Geo.Between A C B) :
  exists X : Geo.Point,
  HilbertRightAngle Geo A C X := by
  exact
  hilbert_right_angle_exists
  Geo A C B hACB
```

Thus the Lean theorem corresponding to Euclid I.11 is deliberately small. The substantial synthetic argument has been factored into `hilbert_right_angle_exists`, making right-angle existence available independently of Proposition I.11 and reusable in the subsequent development.

## 11.8 Relationship with Earlier Propositions

### 11.8.1 Proposition I.1

Euclid's classical proof invokes Proposition I.1 to construct the equilateral triangle  $DFE$  on the auxiliary segment  $DE$ .

In the Hilbert reconstruction the perpendicular construction is obtained more directly from angle construction and uniqueness, so the public proof need not reproduce the equilateral-triangle construction.

### 11.8.2 Proposition I.3

The classical construction requires a point  $E$  on the opposite side of  $C$  satisfying

$$CE \cong CD.$$

This is a segment-layoff step of the kind developed in the early propositions.

The Hilbert layer provides the corresponding layoff operation directly.

### 11.8.3 Proposition I.8

In Euclid's proof, the two triangles  $DCF$  and  $ECF$  have three corresponding sides congruent. Proposition I.8 therefore gives the congruence of the adjacent angles at  $C$ , from which their right-angle character follows.

The Hilbert reconstruction packages the same rigidity more directly through angle construction and the definition of a right angle.

### 11.8.4 Transition to Proposition I.12

Proposition I.11 solves the perpendicular problem when the prescribed point lies on the line. Proposition I.12 treats the complementary case of a point outside the line.

Together they form the basic perpendicular-construction block used in the subsequent straight-angle theory.

## 11.9 Hilbert and Book Zero Components Used

### 11.9.1 Segment Layoff

The construction needs a point on the opposite ray at a prescribed distance from the given point. This is supplied by the established Hilbert segment-construction machinery.

### 11.9.2 Hilbert III.4

Angle construction provides a ray forming an angle congruent to a prescribed angle on a selected side of the supporting line. Its uniqueness clause is particularly important in identifying the constructed perpendicular direction.

### 11.9.3 HilbertRightAngle

The final statement does not use a numerical value of  $90^\circ$ . Perpendicularity is expressed through the synthetic right-angle predicate already defined in the Hilbert interface.

### 11.9.4 Betweenness and Collinearity

The fact that the auxiliary points lie on opposite sides of  $C$  along the given line must be carried by explicit betweenness and primitive collinearity data. These replace the informal reading of



point order from the classical diagram.

## 11.10 Formalization Notes

Proposition I.11 provides an instructive example of the distinction between the organization of Euclid's *Elements* and the organization of the Hilbert formalization.

In Euclid's development, Proposition I.11 is itself a construction. Its proof depends explicitly on earlier propositions:

I.1,      I.3,      I.8,

together with the definition of a right angle. The perpendicular is obtained by constructing an equilateral triangle and comparing two triangles by SSS.

The formal Hilbert development reaches the same result through a different dependency structure. Proposition I.11 does not carry the construction itself. Instead, the substantial argument is isolated as the general theorem

```
hilbert_right_angle_exists
```

and Proposition I.11 merely exposes this theorem at the Euclidean interface.

This is similar in form to Proposition I.10. In both cases the final Euclid theorem is short because the required existence result has already been established in the underlying Hilbert theory. The amount of work hidden behind the interface, however, is very different.

For Proposition I.10, midpoint existence was already a natural primitive result of the developed theory. Proposition I.11 required the introduction of a new general notion,

```
HilbertRightAngle
```

and a substantial synthetic existence proof.

The proof also reveals where the formal difficulty of the result lies. The elementary congruence argument is comparatively short: after constructing  $F$  and the midpoint  $X$  of  $FE$ , SSS gives

$$\angle FCX \cong \angle ECX.$$

The difficult part is proving that the component angles occur in the correct relative configuration so that angle addition is legitimate.

This requires the developed theory of

SameSide,      OppositeSide,      rays,      betweenness,

together with the ray-order theorem. The resulting two-case analysis is substantially longer than the final congruence calculation.

Thus Proposition I.11 illustrates an important feature of the formalization. Statements that are elementary in an informal diagrammatic proof may require a significant amount of explicit order and separation theory once all geometric configurations must be justified formally.

At the same time, this complexity is not local overhead that must be repeated in later propositions. The result has been packaged as `hilbert_right_angle_exists`. Subsequent arguments may use right-angle existence directly without reconstructing the SameSide/OppositeSide analysis.

The resulting architecture is therefore

foundational Hilbert theory  $\longrightarrow$  general synthetic lemmas  $\longrightarrow$  Euclid propositions.

Proposition I.11 is a particularly clear instance of this pattern: its Euclidean interface is almost trivial precisely because the geometric content has been absorbed into the reusable Hilbert layer.

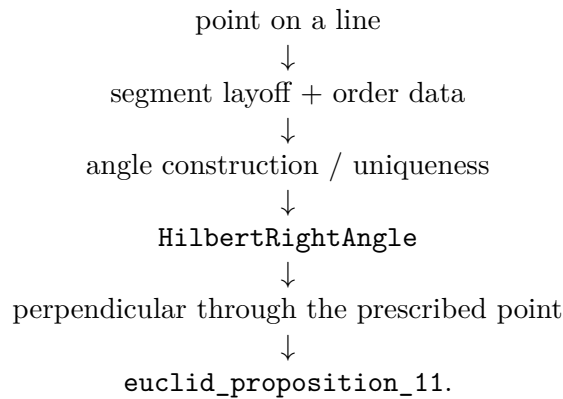
Finally, the comparison with Euclid should not be interpreted as a requirement that the formal proof reproduce Euclid's historical proof step by step. The purpose of the proposition layer is to recover the mathematical statements of Book I over the developed Hilbert theory. When that theory already supplies a stronger or more reusable result, the Euclid proposition can appropriately appear as a thin interface theorem.

## 11.11 Dependency Summary

The classical dependency pattern is

segment layoff  $\longrightarrow$  I.1  $\longrightarrow$  I.8  $\longrightarrow$  equal adjacent angles  $\longrightarrow$  I.11.

The Hilbert reconstruction packages the construction differently:



No numerical angle measure and no new geometric axiom are introduced.

## 11.12 Conclusion

Proposition I.11 begins the explicit perpendicular-construction block of Book I.

Euclid reaches the perpendicular through an auxiliary equilateral triangle and SSS. The Hilbert reconstruction can express the same geometric requirement more directly through segment construction, angle construction, uniqueness, and the synthetic right-angle predicate.

The formalization therefore illustrates the same phenomenon already visible elsewhere in the reconstruction: the Euclidean proposition is preserved as the public milestone, while the dependency graph underneath it reflects the stronger organization of Hilbert geometry.

Together with Proposition I.12, I.11 supplies the perpendicular infrastructure needed for the straight-angle results beginning with Proposition I.13.

# Chapter 12

## Proposition I.12

### 12.1 Introduction

Euclid's Proposition I.12 asks for the construction of a perpendicular to a given straight line from a given point lying outside that line.

Classically, Euclid obtains the perpendicular by choosing a point on the opposite side of the given line, drawing a circle centered at the external point, and using the two intersections of the circle with the given line. The segment joining these intersections is bisected by Proposition I.10, and Proposition I.8 is then applied to prove that the line joining the external point to the midpoint is perpendicular to the given line.

The reconstruction over the Hilbert foundation follows a different synthetic route. No circle-line intersection theorem is used. Instead, a second point is constructed on the opposite side of the given line so that two distinct points of the line are equidistant from the original external point and the constructed point.

This leads naturally to the perpendicular-bisector configuration. The required midpoint property is isolated in the general theorem

```
hilbert_equidistant_line_bisects_segment
```

which states that if two distinct points of a line are equidistant from two points lying on opposite sides of that line, then the intersection of the joining segment with the line bisects that segment.

A second auxiliary result,

```
hilbert_collinear_angle_transport
```

encapsulates the angle-order argument needed in the proof of this perpendicular-bisector theorem.

Consequently, Proposition I.12 provides another example of the role of the Book Zero layer: a comparatively elaborate geometric configuration is separated into reusable synthetic lemmas before the final Euclidean proposition is formulated.

## 12.2 The Classical Configuration

Euclid's twelfth proposition asks for a perpendicular to a given straight line from a point outside it.

Let  $AB$  be the given straight line and let  $C$  be the given point outside it.

### 12.2.1 Final Configuration

The objective is to determine a point  $H$  on the straight line  $AB$  such that

$$CH \perp AB.$$

Euclid constructs two points  $E$  and  $G$  on  $AB$  satisfying

$$CE \cong CG,$$

and then bisects  $EG$  at  $H$ .

The final proof configuration, shown later in Figure 12.4, satisfies

$$E - H - G, \quad EH \cong HG, \quad CE \cong CG.$$

### 12.2.2 Construction Algorithm

The construction uses Proposition I.10 as an already established construction primitive.

#### Step 1. Given Line and External Point

Let  $AB$  be the given straight line and let  $C$  be a point not lying on  $AB$ ; see Figure 12.1.

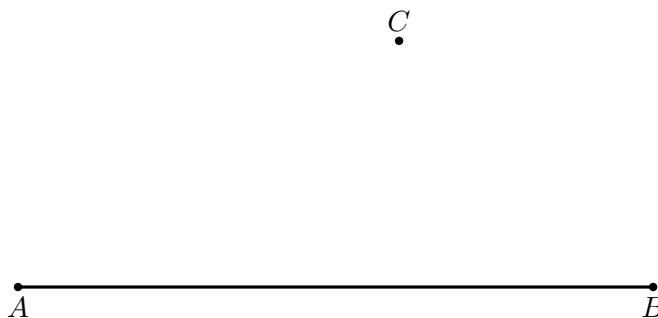


Figure 12.1: Input data for Proposition I.12: a straight line  $AB$  and a point  $C$  outside it.

#### Step 2. Choose a Point on the Opposite Side

Choose a point  $D$  on the opposite side of the straight line  $AB$  from  $C$ , as in Figure 12.2.

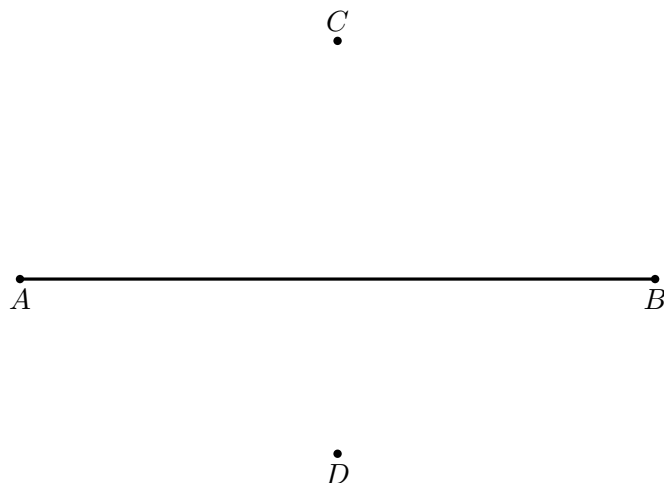


Figure 12.2: Choose  $D$  on the side of the line  $AB$  opposite  $C$ . This side-of-line information is essential for the later intersection argument.

### Step 3. Draw the Circle Centered at $C$

Draw the circle centered at  $C$  and passing through  $D$ ; the resulting intersection configuration is shown in Figure 12.3.

Let the circle meet the given straight line at  $E$  and  $G$ .

Since  $E$  and  $G$  lie on the same circle centered at  $C$ ,

$$CE \cong CG.$$

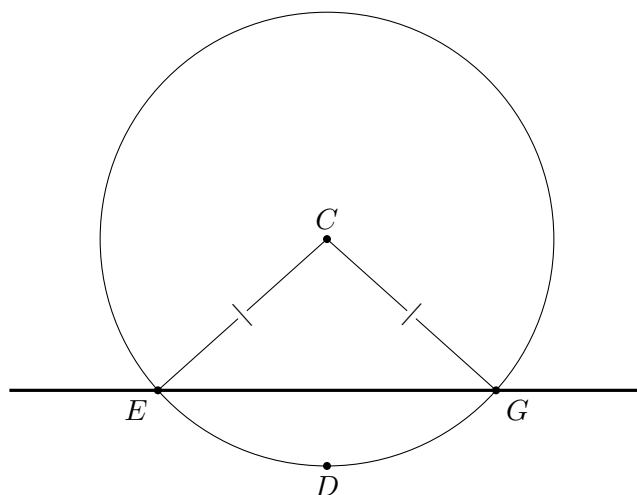


Figure 12.3: The circle centered at  $C$  through  $D$  meets the given line at two distinct points  $E$  and  $G$ , so  $CE \cong CG$ . The existence of two such intersections is a genuine geometric existence issue, not something proved by the picture itself.

### Step 4. Bisect $EG$

Apply Proposition I.10 to bisect the segment  $EG$  at  $H$ .

Thus

$$E - H - G$$

and

$$EH \cong HG.$$

**Step 5. Join  $CH$**

Join the external point  $C$  to the midpoint  $H$ ; the completed proof configuration is Figure 12.4.

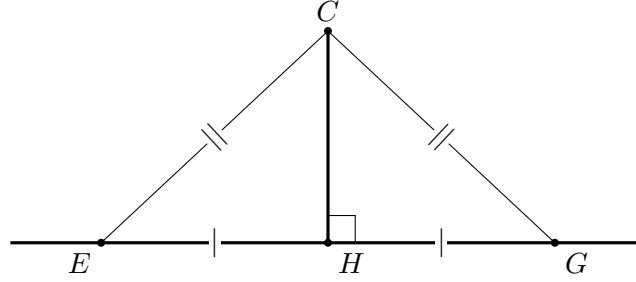


Figure 12.4: Completed Euclidean proof configuration. The midpoint relation  $EH \cong HG$ , the radii  $CE \cong CG$ , and the common side  $CH$  give SSS for triangles  $CEH$  and  $CGH$ .

The two triangles  $EHC$  and  $GHC$  now satisfy

$$EH \cong HG, \quad HC \cong HC, \quad CE \cong CG.$$

By Proposition I.8,

$$\angle EHC \cong \angle GHC.$$

Since  $E, H, G$  are collinear, these are adjacent angles. By Definition 10 they are right angles.

Therefore

$$CH \perp AB.$$

The required perpendicular has been constructed.

**Diagrammatic audit.** The construction sequence is retained only where the geometry genuinely changes: the input data, the opposite-side choice, the circle intersection, and the completed SSS configuration. The former midpoint-only snapshot is omitted because it adds only the label  $H$  to the preceding circle diagram. Conversely, the circle-intersection figure is essential: Wilkins explicitly notes that the existence of two distinct points  $E, G$  requires a tacit topological intersection principle. The picture records those intersections but does not prove their existence.

## 12.3 Classical Proof

Euclid's construction is obtained from the configuration described above.

Choose a point  $D$  on the opposite side of the given line  $AB$  from the external point  $C$ . Draw the circle centered at  $C$  through  $D$ , and let it meet  $AB$  at  $E$  and  $G$ . Then

$$CE \cong CG.$$

Apply Proposition I.10 to bisect  $EG$  at  $H$ . Hence

$$E - H - G \quad \text{and} \quad EH \cong HG.$$

Now compare the triangles

$$\triangle EHC \quad \text{and} \quad \triangle GHC.$$

They satisfy

$$EH \cong HG, \quad HC \cong HC, \quad CE \cong CG.$$

By Proposition I.8, the triangles are congruent by SSS, and therefore

$$\angle EHC \cong \angle GHC.$$

Since  $E, H, G$  are collinear, these congruent adjacent angles are right angles. Consequently

$$CH \perp AB.$$

The classical dependency pattern is therefore

circle construction  $\longrightarrow CE \cong CG \longrightarrow$  I.10  $\longrightarrow EH \cong HG \longrightarrow$  I.8  $\longrightarrow$  equal adjacent angles  $\longrightarrow CH \perp AB$ .

The formal Hilbert reconstruction below proves the same proposition by a different synthetic route and deliberately avoids the circle-line intersection used by Euclid.

## 12.4 Proof Architecture of the Reconstruction

### 12.4.1 Dependency Diagram

Hilbert / Book Zero dependency path

```

-----
hilbert_collinear_angle_transport
      |
      v
hilbert_equidistant_line_bisects_segment
      |
      v
hilbert_perpendicular_from_point_exists
      |
      v
euclid_proposition_12

```

The formal proof separates Proposition I.12 into two reusable Book Zero lemmas and one final construction theorem.

The first auxiliary result is

```

hilbert_collinear_angle_transport

```

It transports an angle congruence when one side of the angle is replaced by another point on the same supporting line. The proof handles the two possible ray configurations: direct transport along the same ray, and transport through supplementary angles when the new ray is opposite.

The second auxiliary result is

```
hilbert_equidistant_line_bisects_segment
```

Suppose two distinct points  $A$  and  $B$  lie on a line  $b$ , while  $P$  and  $Q$  lie on opposite sides of  $b$ , and

$$AP \cong AQ, \quad BP \cong BQ.$$

If the segment  $PQ$  meets  $b$  at  $H$ , then

$$PH \cong HQ.$$

The logical structure is therefore

```
hilbert_collinear_angle_transport
|
hilbert_equidistant_line_bisects_segment
|
hilbert_perpendicular_from_point_exists
|
euclid_proposition_12
```

The main construction theorem

```
hilbert_perpendicular_from_point_exists
```

uses the perpendicular-bisector result to obtain the midpoint  $H$  of  $PQ$ . It then chooses a point  $R$  on the given line distinct from  $H$ , applies Hilbert SSS to the triangles  $RHP$  and  $RHQ$ , and uses vertical angles together with the definition of a right angle.

Finally,

```
euclid_proposition_12
```

is a thin interface theorem exposing this general Hilbert result as Euclid's Proposition I.12.

## 12.5 Hilbert Reconstruction

The Hilbert reconstruction preserves the geometric objective of Proposition I.12 but does not reproduce Euclid's circle construction.

Let  $A$  and  $B$  be distinct points of the given line  $b$ , and let  $P$  be the prescribed point outside  $b$ .

The first task is to construct a point  $Q$  on the opposite side of  $b$  from  $P$  such that

$$AP \cong AQ, \quad BP \cong BQ.$$



To obtain  $Q$ , extend  $PA$  through  $A$  to a point  $S$ . Thus  $P$  and  $S$  lie on opposite sides of  $b$ .

Using Hilbert angle construction at  $A$ , construct a ray on the side of  $b$  containing  $S$  whose angle with  $AB$  is congruent to the angle formed by  $AP$  and  $AB$ . On this ray lay off a segment congruent to  $AP$ . Denote its endpoint by  $Q$ ; see Figure 12.5.

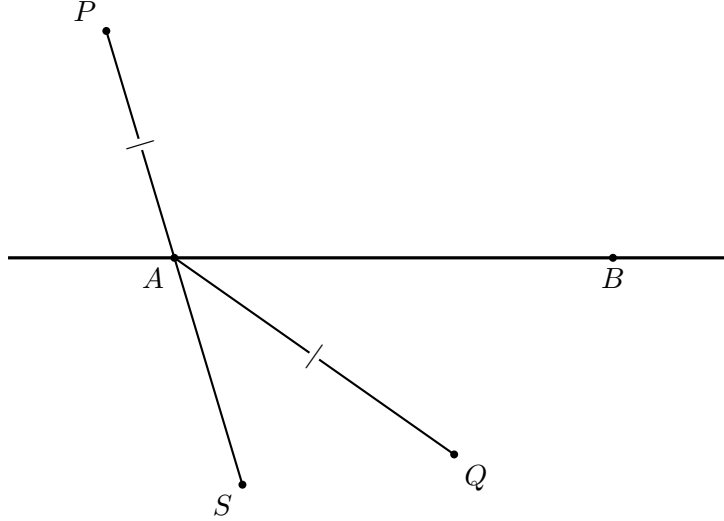


Figure 12.5: Hilbert construction of the reflected point  $Q$ . The line  $PA$  is extended through  $A$  to  $S$ ; angle construction chooses the direction of  $AQ$ , and segment layoff gives  $AQ \cong AP$ .

The construction gives

$$AP \cong AQ.$$

The two triangles

$$\triangle APB \quad \text{and} \quad \triangle AQB$$

now have

$$AP \cong AQ, \quad AB \cong AB,$$

and the included angles at  $A$  are congruent. By SAS,

$$BP \cong BQ.$$

Moreover, the construction places  $Q$  on the opposite side of  $b$  from  $P$ . Hence the segment  $PQ$  intersects  $b$  at a point  $H$ . The resulting configuration is shown in Figure 12.6.

At this point the general Book Zero theorem

```
hilbert_equidistant_line_bisects_segment
```

applies. Since

$$AP \cong AQ, \quad BP \cong BQ,$$

and  $A$  and  $B$  are distinct points of  $b$ , the intersection point  $H$  bisects  $PQ$ :

$$PH \cong HQ.$$

Thus the difficult incidence and angle-order argument has been encapsulated before Proposition I.12 itself is completed.

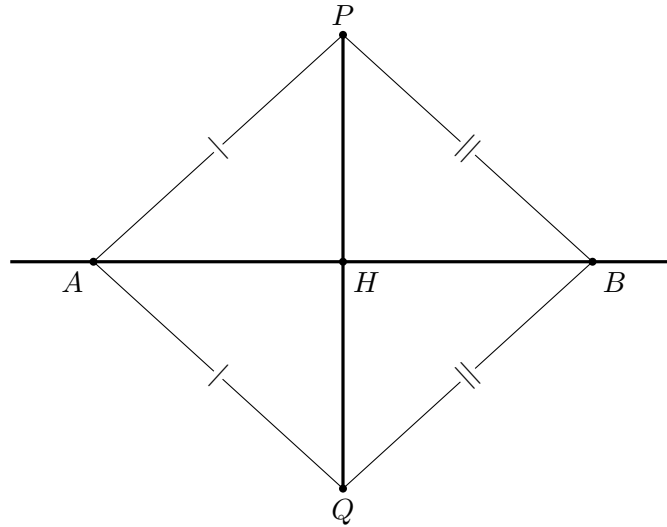


Figure 12.6: Hilbert perpendicular-bisector configuration. The constructed equalities  $AP \cong AQ$  and  $BP \cong BQ$ , together with the intersection  $P - H - Q$ , allow the Book Zero bisector theorem to conclude  $PH \cong HQ$ .

To obtain the right angle, choose one of  $A$  and  $B$  distinct from  $H$  and call it  $R$ . Since  $R$  is one of the two equidistant points,

$$RP \cong RQ.$$

Together with

$$PH \cong HQ \quad \text{and} \quad RH \cong RH,$$

Hilbert SSS applied to

$$\triangle RHP \quad \text{and} \quad \triangle RHQ$$

gives congruent angles at  $H$ .

Since  $P - H - Q$ , these congruent angles determine a perpendicular to  $b$  at  $H$ . In the formal development this last step is expressed using vertical angles and the predicate

`HilbertRightAngle`

The reconstruction therefore replaces Euclid's use of a circle and Propositions I.10 and I.8 by a Hilbert construction based on angle construction, segment layoff, SAS, the perpendicular-bisector theorem, and SSS.

The resulting geometric object is nevertheless the same: a point  $H$  on the given line such that the line through  $P$  and  $H$  is perpendicular to the given line.

## 12.6 Lean Formalization

The Lean formalization separates the reusable geometric content from the final statement of Proposition I.12.

The main construction theorem is

```
theorem hilbert_perpendicular_from_point_exists
```

```

[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B P : Geo.Point)
(base : Geo.Line)
(hAB : A != B)
(hAbase : HilbertIncidence.OnLine A base)
(hBbase : HilbertIncidence.OnLine B base)
(hPbase : not HilbertIncidence.OnLine P base) :
exists H R : Geo.Point,
HilbertIncidence.OnLine H base /\
HilbertIncidence.OnLine R base /\
HilbertRightAngle Geo R H P

```

Here  $H$  is the foot of the perpendicular and  $R$  is a second point of the given line used to determine its direction at  $H$ .

The presence of  $R$  in the conclusion is important. The intersection point  $H$  may coincide with either  $A$  or  $B$ , so neither of the original points can be fixed in advance as the first point in the right-angle predicate. Since

$$A \neq B,$$

at least one of them is distinct from  $H$  and can therefore be chosen as  $R$ .

The formal proof first constructs a point  $Q$  on the opposite side of the given line from  $P$ . The construction establishes

$$AP \cong AQ$$

and, by SAS,

$$BP \cong BQ.$$

The opposite-side relation supplies an intersection point  $H$  with

$$P - H - Q.$$

The reusable Book Zero theorem

```

hilbert_equidistant_line_bisects_segment

```

then gives

$$PH \cong HQ.$$

The remaining argument selects  $R$ , applies Hilbert SSS to the triangles  $RHP$  and  $RHQ$ , and converts the resulting angle congruence into a right angle by means of vertical angles.

The Euclidean proposition itself is exposed by the wrapper

```

theorem euclid_proposition_12
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B P : Geo.Point)
(base : Geo.Line)
(hAB : A != B)
(hAbase : HilbertIncidence.OnLine A base)
(hBbase : HilbertIncidence.OnLine B base)

```

```

(hPbase : not HilbertIncidence.OnLine P base) :
exists H R : Geo.Point,
HilbertIncidence.OnLine H base /\
HilbertIncidence.OnLine R base /\
HilbertRightAngle Geo R H P := by

exact
hilbert_perpendicular_from_point_exists
Geo
A B P
base
hAB
hAbase
hBbase
hPbase

```

Thus the theorem carrying Euclid's name is deliberately only a thin wrapper. The geometric work is performed by the Hilbert construction theorem and by the reusable results already established in the Book Zero layer.

## 12.7 Relationship with Earlier Propositions

### 12.7.1 Proposition I.8

Euclid uses Proposition I.8 at the final stage of the classical proof. Once the two radii from  $C$  and the two half-segments at  $H$  are known to be congruent, SSS gives congruent adjacent angles and hence the perpendicular.

The Hilbert reconstruction also uses SSS, but in a different configuration: the triangles  $RHP$  and  $RHQ$  arise from the perpendicular-bisector construction rather than from Euclid's circle.

### 12.7.2 Proposition I.10

In Euclid's proof, Proposition I.10 is the construction that bisects the chord  $EG$ .

The Hilbert reconstruction does not call I.10. Instead, the midpoint property

$$PH \cong HQ$$

is derived from the more general theorem `hilbert_equidistant_line_bisects_segment`.

Thus the Euclidean midpoint construction is replaced by a reusable perpendicular-bisector principle.

### 12.7.3 Proposition I.11

Proposition I.11 constructs a perpendicular at a point already lying on the given line. Proposition I.12 solves the complementary problem: the prescribed point lies outside the line and the foot of the perpendicular must itself be constructed.

Together I.11 and I.12 complete the two elementary perpendicular construction problems needed later in Book I.

## 12.8 Hilbert and Book Zero Components Used

### 12.8.1 `hilbert_collinear_angle_transport`

This helper transports angle congruence when one endpoint on a supporting line is replaced by another collinear point. It handles both the same-ray and opposite-ray configurations.

### 12.8.2 `hilbert_equidistant_line_bisects_segment`

This is the central reusable theorem extracted during the reconstruction of I.12. Two distinct points of a line, each equidistant from  $P$  and  $Q$ , force the intersection of  $PQ$  with that line to bisect  $PQ$ .

### 12.8.3 `hilbert_perpendicular_from_point_exists`

This theorem assembles the construction: it produces the opposite-side point, obtains the midpoint from the equidistant-line theorem, chooses a second point of the base line, and proves the resulting angle is right.

### 12.8.4 Hilbert SSS and Vertical Angles

SSS converts the three segment congruences in the triangles  $RHP$  and  $RHQ$  into angle congruence at  $H$ . Vertical-angle congruence then puts that equality into the adjacent-angle orientation required by `HilbertRightAngle`.

## 12.9 Formalization Notes

Proposition I.12 reveals a substantial difference between Euclid's constructional language and the Hilbert/Book Zero architecture.

Euclid obtains the perpendicular through a concrete circle construction. The circle centered at the external point produces two points of the given line at equal distance from that point. Proposition I.10 supplies their midpoint, and Proposition I.8 converts the resulting three side equalities into equality of the adjacent angles at the midpoint.

The Hilbert reconstruction does not attempt to reproduce this sequence literally. In particular, no circle is introduced. Instead, the essential geometric configuration behind Euclid's construction is isolated:

$$AP \cong AQ, \quad BP \cong BQ.$$

Thus two distinct points  $A$  and  $B$  of the given line are equidistant from  $P$  and  $Q$ . The line through  $A$  and  $B$  therefore plays the role of the perpendicular-bisector axis of  $PQ$ .

This observation leads to the theorem

```
hilbert_equidistant_line_bisects_segment
```

which extracts the reusable geometric content of the configuration. Once the segment  $PQ$  intersects the line at  $H$ , the theorem gives

$$PH \cong HQ.$$

The proposition is therefore no longer organized around a particular circle construction. It is organized around an abstract property of two equidistant points and the line determined by them.

The proof of the perpendicular-bisector theorem itself exposed another reusable operation. When an angle is known relative to one point of a line, replacing that point by another collinear point requires attention to the order of the points: the relevant ray may remain unchanged or become the opposite ray. This argument is encapsulated in

```
hilbert_collinear_angle_transport
```

rather than being repeated inside Proposition I.12.

This is characteristic of the role of Book Zero. Elementary incidence, order, congruence, ray, and angle manipulations are packaged as synthetic lemmas. Later Euclidean propositions can then operate with larger geometric units instead of repeatedly descending to the Hilbert axioms.

There is also an important difference from the preceding elementary propositions. Proposition I.12 does not collapse to a single previously available Book Zero operation. Its reconstruction required the identification of genuinely reusable intermediate geometry. In this sense, the formalization of I.12 contributed new infrastructure to Book Zero rather than merely consuming existing infrastructure.

The final theorem

```
euclid_proposition_12
```

is consequently a wrapper around

```
hilbert_perpendicular_from_point_exists
```

The wrapper records the Euclidean proposition in the formal library, while the underlying theorem records the more general Hilbert construction on which it depends.

Proposition I.12 therefore illustrates the intended division of labor in the development:

|                              |   |                                                    |
|------------------------------|---|----------------------------------------------------|
| Hilbert axioms               | → | elementary incidence, order, and congruence theory |
| Book Zero                    | → | reusable synthetic geometric mechanisms            |
| Hilbert construction theorem | → | existence of the required perpendicular            |
| Euclid wrapper               | → | Proposition I.12                                   |

The result is not a literal transcription of Euclid's proof. It is a reconstruction of the same geometric theorem over the Hilbert foundation, with the intermediate reasoning reorganized into reusable components.

## 12.10 Dependency Summary

The Euclidean and Hilbert dependency structures are visibly different.

For Euclid,

$$\text{circle} \longrightarrow \text{I.10} \longrightarrow \text{I.8} \longrightarrow \text{I.12}.$$

For the reconstruction,

$$\begin{array}{c}
 \text{Hilbert angle construction + segment layoff + SAS} \\
 \downarrow \\
 AP \cong AQ, \quad BP \cong BQ, \quad P, Q \text{ on opposite sides} \\
 \downarrow \\
 \text{hilbert\_equidistant\_line\_bisects\_segment} \\
 \downarrow \\
 PH \cong HQ \\
 \downarrow \\
 \text{Hilbert SSS + vertical angles} \\
 \downarrow \\
 \text{HilbertRightAngle} \\
 \downarrow \\
 \text{hilbert\_perpendicular\_from\_point\_exists} \\
 \downarrow \\
 \text{euclid\_proposition\_12.}
 \end{array}$$

The reconstruction introduces no new axiom. Its main contribution is the extraction of reusable synthetic lemmas from the geometry hidden inside the classical construction.

## 12.11 Conclusion

Proposition I.12 is an especially clear example of reconstruction rather than literal transcription.

Euclid uses a circle, Proposition I.10, and Proposition I.8. The Hilbert development identifies the underlying perpendicular-bisector geometry and proves the result without reproducing the circle-line construction.

This change of route is productive. The proof yields `hilbert_collinear_angle_transport` and, more importantly, `hilbert_equidistant_line_bisects_segment` as reusable synthetic infrastructure.

The public theorem remains Euclid I.12: from a point outside a given line, a perpendicular to that line exists. Underneath that statement, however, the formal dependency graph records a more general Hilbert construction whose intermediate results can be reused independently.

Together with I.11, Proposition I.12 completes the basic perpendicular construction block and prepares the straight-angle theory beginning with Proposition I.13.

# Chapter 13

## Proposition I.13

### 13.1 Introduction

Euclid's Proposition I.13 concerns the two adjacent angles formed when one straight line stands on another straight line.

Let  $C, B, D$  lie on one straight line, with  $B$  between  $C$  and  $D$ , and let  $A$  be a point outside that line. The two adjacent angles are

$$\angle CBA \quad \text{and} \quad \angle ABD.$$

Euclid states that these angles are either two right angles or are together equal to two right angles.

The classical proof distinguishes two cases. If the adjacent angles are equal, then by Definition 10 they are both right angles. If they are not equal, Euclid erects a perpendicular at  $B$  using Proposition I.11 and compares the resulting angle decompositions.

The reconstruction over the Hilbert/Book Zero layer is much shorter. Book Zero already contains an explicit relation expressing that one angle is supplementary to another:

`BookZeroSupplement`

For the configuration of Proposition I.13, this relation contains exactly the required geometric data. Consequently, the formal theorem does not reconstruct Euclid's case distinction or introduce an arithmetic operation on angles. It identifies the two adjacent angles directly as a supplementary pair.

Thus Proposition I.13 provides a particularly clear example of the compression obtained from the Book Zero language: a classical geometric argument becomes essentially an instance of an already available synthetic relation.

### 13.2 The Classical Configuration

Proposition I.13 is not a construction theorem. The geometric configuration is already given.

Let  $C, B, D$  lie on one straight line with

$$C - B - D,$$



and let  $A$  lie outside the straight line  $CD$ .

The straight line  $BA$  stands on the straight line  $CD$  as shown in Figure 13.1 and forms the adjacent angles

$$\angle CBA \quad \text{and} \quad \angle ABD.$$

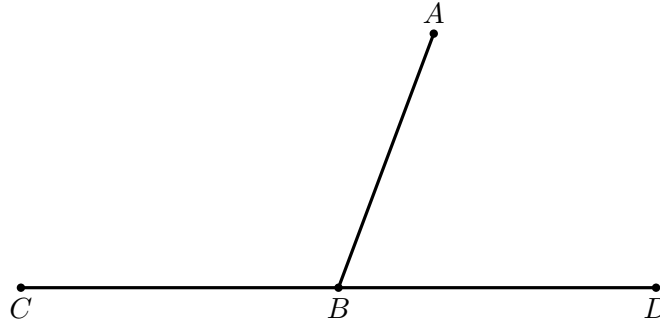


Figure 13.1: Classical configuration for Proposition I.13. The relation  $C - B - D$  makes  $BC$  and  $BD$  opposite rays, while  $BA$  is the standing ray.

### 13.3 Statement

Assume

$$C - B - D$$

and let  $A$  be a point distinct from  $B$ .

Then the adjacent angles

$$\angle CBA \quad \text{and} \quad \angle ABD$$

form a supplementary pair.

In Euclid's formulation, they are therefore either two right angles or together equal to two right angles.

In the Book Zero language this conclusion is represented by

$$\text{BookZeroSupplement}(C, B, A, A, D).$$

By definition,

$$\text{BookZeroSupplement}(C, B, A, A, D)$$

means

$$\text{SameRay}(B, A, A)$$

together with

$$C - B - D.$$

The first condition expresses the trivial fact that the ray  $BA$  coincides with itself, while the second is precisely the given straight-line configuration.

### 13.4 Classical Proof

Euclid proves Proposition I.13 by distinguishing two cases.

### 13.4.1 Case 1. The Adjacent Angles Are Equal

Suppose first that

$$\angle CBA \cong \angle ABD.$$

Since  $C, B, D$  lie on one straight line, the two angles are adjacent angles formed by the straight line  $BA$  standing on  $CD$ .

By Euclid's Definition 10, when a straight line standing on another straight line makes the adjacent angles equal, each of those angles is a right angle.

Therefore

$$\angle CBA \quad \text{and} \quad \angle ABD$$

are two right angles.

This is the first alternative in Proposition I.13.

### 13.4.2 Case 2. The Adjacent Angles Are Not Equal

Suppose now that

$$\angle CBA \not\cong \angle ABD.$$

By Proposition I.11, construct through  $B$  a perpendicular  $BE$  to the straight line  $CD$ ; see Figure 13.2.

Thus

$$\angle CBE \quad \text{and} \quad \angle EBD$$

are right angles.

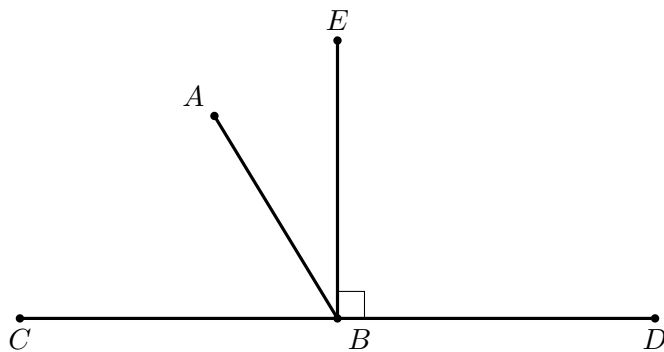


Figure 13.2: Auxiliary perpendicular configuration in the non-equal-angle case. The ray  $BE$  is perpendicular to  $CD$ , and  $BA$  lies between the rays  $BC$  and  $BE$  in the displayed arrangement.

Since  $BE$  is perpendicular to  $CD$ ,

$$\angle CBE + \angle EBD$$

is equal to two right angles.

The original ray  $BA$  divides one of these right-angle configurations. In the arrangement shown in Figure 13.2,

$$\angle CBE = \angle CBA + \angle ABE.$$

Adding  $\angle EBD$  gives

$$\angle CBE + \angle EBD = \angle CBA + \angle ABE + \angle EBD.$$

But

$$\angle ABE + \angle EBD = \angle ABD.$$

Hence

$$\angle CBE + \angle EBD = \angle CBA + \angle ABD.$$

The left-hand side consists of two right angles. Therefore

$$\angle CBA + \angle ABD$$

is equal to two right angles.

This is the second alternative in Proposition I.13.

Only the two numbered figures above are needed. Figure 13.1 states the given geometry; Figure 13.2 introduces the one genuinely new construction. Additional copies of the same rays would not correspond to new geometric information.

Thus, whether the two original adjacent angles are equal or unequal, they are either themselves two right angles or are together equal to two right angles.

## 13.5 Proof Architecture of the Reconstruction

Euclid's proof proceeds through the geometry of right angles and angle addition:

Euclid

-----

C-B-D and standing ray BA

|  
are the adjacent angles equal?

|  
+-----+-----+

|                      |  
yes                      no

|                      |  
Definition 10   erect perpendicular BE

|                      |  
right angles   angle decomposition

|                      |  
+-----+-----+

|  
two right angles

Hilbert / Book Zero

-----

C-B-D

```

+
SameRay(B,A,A)
|
v
BookZeroSupplement(C,B,A,A,D)
|
v
euclid_proposition_13

```

The two architectures should not be represented by additional geometric figures. The difference is logical rather than spatial: Euclid proves a statement about angle sums, whereas the Book Zero theorem recognizes the same straight-angle configuration directly.

The contrast is substantial. Euclid proves the proposition by introducing a perpendicular and reasoning about decompositions and combinations of angles. At the Book Zero level, the same geometric configuration has already been abstracted as the supplementary-angle relation.

No new geometric construction is therefore required, and Proposition I.11 disappears from the formal dependency path.

## 13.6 Hilbert Reconstruction

The Hilbert reconstruction of Proposition I.13 does not reproduce Euclid's proof.

The essential observation is that the conclusion of the proposition has already been isolated in Book Zero as the relation

**BookZeroSupplement**

For five points  $A, B, C, D, F$ , this relation is defined by

$$\text{BookZeroSupplement}(A, B, C, D, F)$$

if and only if

$$\text{SameRay}(B, C, D)$$

and

$$A - B - F.$$

Thus the relation records synthetically the configuration in which one side of an angle is continued through its vertex while the other side is preserved along the same ray.

For Proposition I.13 we instantiate this relation as

$$\text{BookZeroSupplement}(C, B, A, A, D).$$

Expanding the definition gives exactly two conditions:

$$\text{SameRay}(B, A, A)$$

and

$$C - B - D.$$

The second condition is one of the hypotheses of Proposition I.13.

The first condition says only that the ray  $BA$  is the same ray as itself. Since  $A \neq B$ , it follows immediately from reflexivity of the same-ray relation.

Consequently,

$$\text{BookZeroSupplement}(C, B, A, A, D)$$

follows directly.

No perpendicular is constructed. No case distinction between equal and unequal adjacent angles is required. No decomposition or addition of angles occurs in the proof.

This does not mean that Euclid's geometric argument has been discarded. Rather, the geometric notion expressed by that argument has already been incorporated into the language of Book Zero. Proposition I.13 therefore becomes a recognition theorem: the given configuration is an instance of an existing synthetic relation.

## 13.7 Lean Formalization

The Lean theorem is correspondingly short:

```
theorem euclid_proposition_13
  [HilbertIncidence Geo]
  [HilbertOrder Geo]
  (C B D A : Geo.Point)
  (hCBD : Geo.Between C B D)
  (hBA : B != A) :
  BookZeroSupplement Geo C B A A D := by

  constructor

  · exact
    hilbert_sameRay_refl
    Geo B A hBA.symm

  · exact hCBD
```

The proof follows the definition of

```
BookZeroSupplement
```

literally.

The first component of the conclusion is

$$\text{SameRay}(B, A, A).$$

It is obtained from

`hilbert_sameRay_refl`

using  $A \neq B$ , which is supplied by the symmetric form of the hypothesis  $B \neq A$ .

The second component is

$$C - B - D,$$

which is exactly the hypothesis

`hCBD`

Thus the formal proof contains no intermediate geometric construction and requires no theorem corresponding to Euclid's use of Proposition I.11.

## 13.8 Discussion

Proposition I.13 is a particularly clear test of the role assigned to Book Zero in the reconstruction.

In Euclid's original development, the statement requires an actual proof. The argument distinguishes whether the two adjacent angles are equal. In the nontrivial case, Proposition I.11 is invoked to erect a perpendicular, and the conclusion is obtained by decomposing and recombining angles.

At the Hilbert level, one could reconstruct this argument explicitly. The existing theory already contains the necessary machinery for right angles, supplementary angles, and angle addition.

That reconstruction is unnecessary, however, once the Book Zero language is used.

The relation

`BookZeroSupplement`

was introduced independently as a reusable description of a standard synthetic configuration. Proposition I.13 is precisely an instance of that configuration.

The resulting compression is therefore structural rather than tactical. Lean is not discovering a shorter proof of Euclid's argument. The formal language has been organized at a level at which the conclusion of I.13 can already be stated directly.

This produces an unusually short dependency path:

betweenness + same-ray reflexivity  $\longrightarrow$  BookZeroSupplement  $\longrightarrow$  Euclid I.13.

In particular, Proposition I.11, although essential to Euclid's classical proof, is not a dependency of the reconstructed theorem.

This distinction is important. The objective of the reconstruction is not to force every Euclidean proposition to retain the dependency graph of the original text. The objective is to express Euclidean geometry over a stable synthetic interface derived from the Hilbert foundation.

Earlier propositions exhibited several possible relationships with this interface. Some consumed already available Book Zero results, while others exposed missing reusable geometry that had to be added to the interface.

Proposition I.13 exhibits a third and especially simple situation: the proposition is already expressed by the vocabulary of Book Zero.

Schematically,

|                   |   |                                              |
|-------------------|---|----------------------------------------------|
| Euclid            | : | case distinction + I.11 + angle composition  |
| Hilbert/Book Zero | : | recognition of a supplementary configuration |
| Lean              | : | same-ray reflexivity + given betweenness     |

The four-line proof is therefore not merely an accidental shortening of Proposition I.13. It is evidence that the Book Zero layer is functioning as intended: elementary synthetic arguments can be encapsulated once and subsequently used as part of the language in which later geometry is formulated.

## 13.9 Relationship with Earlier Propositions

### 13.9.1 Proposition I.11

Proposition I.11 constructs a perpendicular to a given straight line at a given point on it. In the classical development this supplies the canonical right-angle configuration against which the adjacent angles in I.13 can be compared.

The formal reconstruction can package the same geometry through the Hilbert and Book Zero angle machinery, but the conceptual role remains the same: a perpendicular provides the reference configuration for two right angles.

### 13.9.2 Proposition I.12

Proposition I.12 extends the perpendicular construction to a point outside a line. It is not the principal dependency of I.13, but together with I.11 it completes the elementary perpendicular infrastructure that precedes the straight-angle block.

### 13.9.3 Transition to Proposition I.14

Proposition I.13 is the direct theorem: when a straight line stands on another straight line, the adjacent angles are two right angles or are together equal to two right angles.

Proposition I.14 reverses this logic. From the corresponding supplementary-angle condition it recovers the straight-line configuration.

Thus I.13 and I.14 form a natural theorem/converse pair and should be read as one local unit in the development of straight-angle geometry.

### 13.10 Hilbert and Book Zero Components Used

The reconstruction of Proposition I.13 relies on the Book Zero relation

`BookZeroSupplement`

defined by

$$\text{BookZeroSupplement}(A, B, C, D, F)$$

if and only if

$$\text{SameRay}(B, C, D)$$

and

$$A - B - F.$$

Geometrically, this represents a pair of supplementary angles. The angle determined by the rays  $BA$  and  $BC$  has as its supplement the angle determined by the rays  $BD$  and  $BF$ , where  $BD$  lies on the same ray as  $BC$  and  $BF$  is the opposite extension of  $BA$ .

For Proposition I.13 the relation is instantiated as

$$\text{BookZeroSupplement}(C, B, A, A, D).$$

Thus the required conditions become

$$\text{SameRay}(B, A, A)$$

and

$$C - B - D.$$

The first is immediate from same-ray reflexivity, while the second is the given betweenness hypothesis.

The proof also uses

`hilbert_sameRay_refl`



which expresses reflexivity of a nondegenerate ray:

$$A \neq B \implies \text{SameRay}(A, B, B).$$

## 13.11 Formalization Notes

The formal reconstruction is considerably shorter than Euclid’s classical proof because the supplementary-angle configuration has already been packaged in Book Zero.

The theorem does not reproduce Euclid’s case distinction and does not construct the perpendicular from Proposition I.11. Instead, it proves

$$\text{BookZeroSupplement}(C, B, A, A, D)$$

directly from the two components required by the definition:

$$\text{SameRay}(B, A, A) \quad \text{and} \quad C - B - D.$$

The first component follows from

`\thilbert_sameRay_refl`

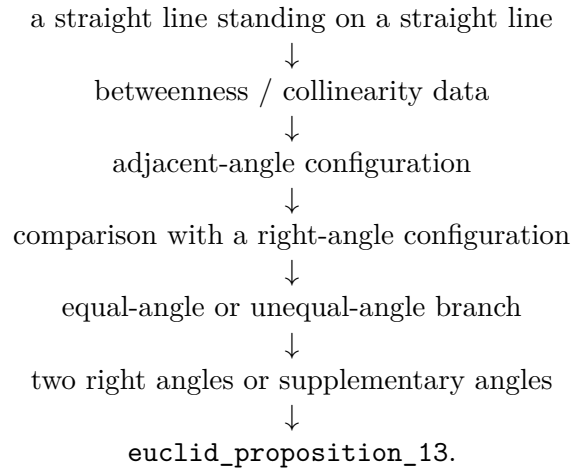
using the nondegeneracy hypothesis  $A \neq B$ . The second component is exactly the given betweenness hypothesis.

Thus no numerical angle measure, angle addition, right-angle case analysis, or auxiliary perpendicular is needed in the Lean proof. The formalization records the same synthetic content at a more abstract level: Proposition I.13 is recognized as an instance of the already established Book Zero notion of supplementary angles.

This is also the form needed by Proposition I.14, which uses the same supplementary-angle structure in the converse direction.

## 13.12 Dependency Summary

The formal dependency structure is



The proof uses only the established Hilbert and Book Zero infrastructure. No new geometric axiom is introduced.

### 13.13 Conclusion

Proposition I.13 begins the straight-angle block of Book I.

Its familiar classical statement looks almost arithmetical: the two adjacent angles made by a line standing on another line amount to two right angles. The formal reconstruction shows that no numerical angle measure is required to express or prove this fact.

Instead, the theorem is recovered from explicit incidence, order, right-angle, and angle-congruence structure. The classical distinction between equal and unequal adjacent angles is preserved, while the diagrammatic assumptions concerning the supporting straight line are made explicit.

The proposition is also structurally important. I.14 supplies its converse, and I.15 completes the local theory by proving the congruence of vertical angles. I.13 therefore marks the beginning of a coherent three-proposition development of the geometry of intersecting straight lines.

## Chapter 14

# Proposition I.14

### 14.1 Introduction

Euclid's Proposition I.14 is the converse in spirit of the preceding Proposition I.13.

Proposition I.13 starts from a straight-line configuration and proves that the adjacent angles are together equal to two right angles. Proposition I.14 reverses this direction: if two adjacent angles are together equal to two right angles, and their outer rays lie on opposite sides of the common straight line, then those outer rays form one straight line.

In Euclid's original proof the argument is indirect. The ray opposite to one of the given rays is constructed, Proposition I.13 is applied, and equality of the resulting angles forces the constructed ray to coincide with the remaining given ray.

The Hilbert reconstruction follows the same geometric idea at a more structural level. The condition "equal to two right angles" is encoded without numerical angle measure, using an opposite ray and angle congruence. The decisive step is then supplied by the uniqueness clause of Hilbert's angle-construction axiom.

Proposition I.14 is therefore an instructive example of the interaction between Euclid's angle language, Hilbert plane separation, angle construction, and the ray calculus developed in Book Zero.

### 14.2 The Classical Configuration

Let  $AB$  be a straight line, and let two further straight lines  $BC$  and  $BD$  meet it at the common point  $B$ , with  $C$  and  $D$  lying on opposite sides of  $AB$ .

The two adjacent angles are therefore

$$\angle ABC \quad \text{and} \quad \angle ABD.$$

The hypothesis of Proposition I.14 is that these two angles are together equal to two right angles. The conclusion is that the two outer rays  $BC$  and  $BD$  are opposite rays and hence form one straight line.

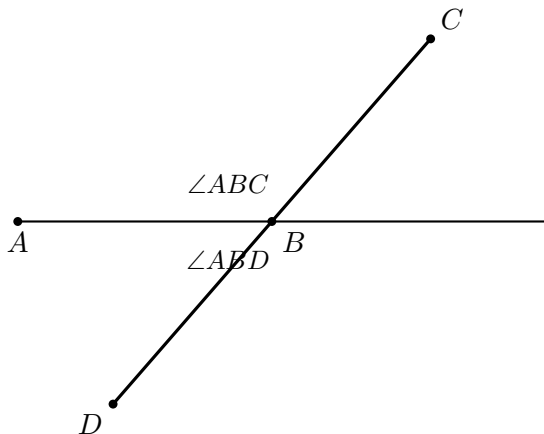


Figure 14.1: The classical configuration of Proposition I.14. The rays  $BC$  and  $BD$  lie on opposite sides of the straight line through  $A$  and  $B$ .

### 14.3 Statement

**Theorem 7** (Euclid I.14). *Let  $AB$  be a straight line, and let the straight lines  $BC$  and  $BD$  meet it at the point  $B$  on opposite sides of  $AB$ .*

*If the adjacent angles*

$$\angle ABC \quad \text{and} \quad \angle ABD$$

*are together equal to two right angles, then  $BC$  and  $BD$  form one straight line.*

*Equivalently,*

$$C - B - D.$$

The proposition may therefore be viewed as a converse to the straight-line angle relation established in Proposition I.13.

The geometric content of the conclusion is stronger than mere collinearity. The point  $B$  lies between  $C$  and  $D$ , so the rays  $BC$  and  $BD$  are opposite rays with common endpoint  $B$ .

### 14.4 Classical Proof

Euclid argues by extending one of the two outer rays through the common vertex; the auxiliary configuration is shown in Figure 14.2.

Extend  $CB$  through  $B$  to a point  $E$ . Then

$$C - B - E,$$

so  $BC$  and  $BE$  form one straight line.

By Proposition I.13, the adjacent angles

$$\angle ABC \quad \text{and} \quad \angle ABE$$

are together equal to two right angles.

But by hypothesis,

$$\angle ABC \quad \text{and} \quad \angle ABD$$

are also together equal to two right angles.

Therefore the remaining angles must be equal:

$$\angle ABE = \angle ABD.$$

But if  $BE$  and  $BD$  were distinct, one of these two angles would be a proper part of the other. A proper part cannot be equal to the whole. Hence the rays  $BE$  and  $BD$  cannot be distinct.

Therefore  $BE$  coincides with  $BD$ . Since

$$C - B - E,$$

it follows that

$$C - B - D.$$

Thus  $BC$  and  $BD$  form one straight line.

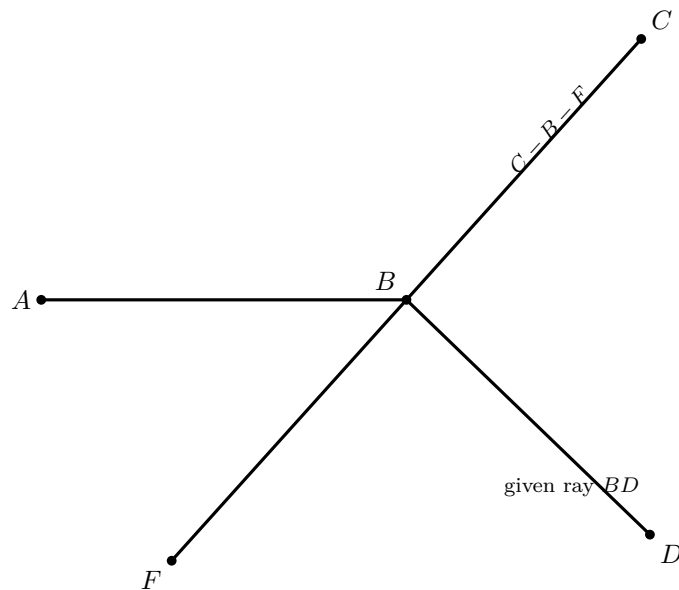


Figure 14.2: Wilkins auxiliary construction before the uniqueness step. The straight line  $CBF$  is the constructed extension through  $B$ ;  $BD$  is the given ray. The proof shows that  $BD$  must coincide with the ray  $BF$ .

The essential structure of Euclid's argument is therefore

$$\text{extension} \longrightarrow \text{I.13} \longrightarrow \text{equality of the remaining angles} \longrightarrow \text{coincidence of the rays.}$$

The second diagram is genuinely useful, but it must be read as a *pre-coincidence* configuration. It would be misleading to draw  $BD$  as a visibly different ray and simultaneously treat  $\angle ABF =$

$\angle ABD$  as already geometrically realized. The equality is precisely what forces the two rays to coincide. Wilkins makes the missing uniqueness step explicit by using the fact that  $D$  and  $F$  lie on the same side of  $AB$ .

## 14.5 Proof Architecture of the Reconstruction

The Hilbert reconstruction preserves the central geometric idea of Euclid's proof but reorganizes its logical dependencies.

Euclid extends  $CB$  through  $B$  to a point  $E$  and uses Proposition I.13 to obtain the supplementary angle relation. He then compares this relation with the hypothesis of Proposition I.14 and concludes that the rays  $BD$  and  $BE$  must coincide.

In the formal reconstruction, the point  $E$  is incorporated directly into the synthetic representation of the hypothesis. We require

$$C - B - E$$

together with

$$\angle ABD \cong \angle ABE.$$

Thus the intermediate appeal to Proposition I.13 is no longer needed: the supplementary configuration required by Euclid's argument is already present in the formal hypothesis.

The remaining problem is to prove that  $BD$  and  $BE$  determine the same ray from  $B$ .

At this point the reconstruction divides into two cases according to whether  $C, D, E$  are collinear.

Hilbert hypothesis

-----

C-B-E

angle ABD ~= angle ABE

C and D opposite sides of base AB

|

v

derive: C and E opposite sides of base AB

|

v

case split: are C,D,E collinear?

|

+-----+-----+

|

yes

|

line

uniqueness

|

|

|

|

no

|

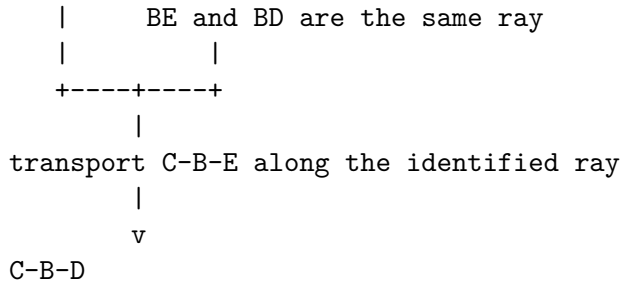
D and E are on the same side of AB

|

v

Hilbert III.4 angle uniqueness

|



The two classical figures have different jobs. The first states the problem; the second introduces the extension used in the proof. Beyond these, another geometric copy of the same crossing is unnecessary. The formal case split is better represented as a dependency diagram, because its branches are logical alternatives rather than two prescribed Euclidean constructions.

In the collinear case, incidence and line uniqueness are sufficient.

In the noncollinear case, plane separation first places  $D$  and  $E$  on the same side of the base line. The uniqueness clause of Hilbert's angle-construction axiom then identifies the rays  $BD$  and  $BE$ . Ray calculus and transport of betweenness finally yield

$$C - B - D.$$

Schematically, the two proof architectures are

Euclid : extension  $\rightarrow$  I.13  $\rightarrow$  angle comparison  $\rightarrow$  ray coincidence,

Hilbert reconstruction : encoded supplement  $\rightarrow$  case split  $\rightarrow$  line/angle uniqueness  $\rightarrow$  betweenness.

## 14.6 Hilbert Reconstruction

The Hilbert reconstruction begins by replacing Euclid's phrase "together equal to two right angles" by a purely synthetic configuration.

No numerical measure of angles and no operation of angle addition are introduced. Instead, the supplementary relation is represented by an opposite ray together with angle congruence.

### 14.6.1 Encoding the Angle Hypothesis

Choose a point  $E$  on the continuation of  $CB$  beyond  $B$ , so that

$$C - B - E.$$

The rays  $BC$  and  $BE$  are therefore opposite. Consequently,  $\angle ABE$  is the angle supplementary to  $\angle ABC$ .

The hypothesis that

$$\angle ABC \quad \text{and} \quad \angle ABD$$

are together equal to two right angles is represented by requiring

$$\angle ABD \cong \angle ABE.$$

The geometry is already visible in Figure 14.2: the witness  $E$  lies on the continuation of  $CB$  beyond  $B$ , and the formal hypothesis identifies the angle made by  $BD$  with the supplementary angle made by  $BE$ . A second drawing of the same rays would add no new information.

In Lean this condition is packaged as

```
def HilbertAnglesEqualTwoRightAngles
  [HilbertIncidence Geo]
  (A B C D : Geo.Point) : Prop :=
  exists E : Geo.Point,
  Geo.Between C B E /\
  Geo.AngleCongruent
  A B D
  A B E
```

Thus the witness  $E$  contains exactly the two pieces of geometric information needed by the reconstruction:

$$C - B - E$$

and

$$\angle ABD \cong \angle ABE.$$

This definition should not be interpreted as a numerical definition of the sum of two angles. It is a synthetic encoding of the particular supplementary configuration occurring in Proposition I.14.

### 14.6.2 The Collinear Case

Assume first that  $C$ ,  $D$ , and  $E$  are collinear.

From the definition of `HilbertAnglesEqualTwoRightAngles` we already have

$$C - B - E.$$

Hence  $B$  lies on the line determined by  $C$  and  $E$ .

Since  $C$  and  $D$  lie on opposite sides of the base line  $AB$ , the segment  $CD$  meets that line. Let  $X$  be a point of intersection, so that

$$C - X - D$$

and  $X$  lies on the base line.



No separate figure is needed for this branch. The point  $X$  is only an incidence witness: if the line through  $C, D, E$  meets the base line at  $X$ , then line uniqueness forces  $X = B$ . Drawing  $X$  as a second visible point near  $B$  is more confusing than informative, because the proof immediately shows that the two points coincide.

Both  $B$  and  $X$  now lie on two lines: the base line  $AB$  and the line containing  $C, D, E$ .

Suppose that

$$X \neq B.$$

By uniqueness of the line through two distinct points, the two lines would then have to coincide. In particular,  $C$  would lie on the base line.

This contradicts the hypothesis that  $C$  and  $D$  lie on opposite sides of the base line, since a point belonging to the line itself cannot be one of the two opposite-side points.

Therefore

$$X = B.$$

Substituting this equality into

$$C - X - D$$

gives

$$C - B - D.$$

Thus the collinear branch requires no angle argument at all. Once  $C, D, E$  are known to be collinear, the conclusion follows entirely from betweenness, plane separation, incidence, and uniqueness of the line through two distinct points.

### 14.6.3 The Noncollinear Case

Assume now that  $C, D$ , and  $E$  are not collinear.

The segment  $CD$  meets the base line because  $C$  and  $D$  lie on opposite sides of it. The segment  $CE$  also meets the base line, namely at  $B$ , since

$$C - B - E.$$

Because  $C, D, E$  form a genuine triangle, the Hilbert plane-separation theory implies that the remaining endpoints  $D$  and  $E$  lie on the same side of the base line; this is the configuration shown in Figure 14.3. Thus

$$\text{SameSide}(D, E; AB).$$

We now use Hilbert's angle-construction axiom.

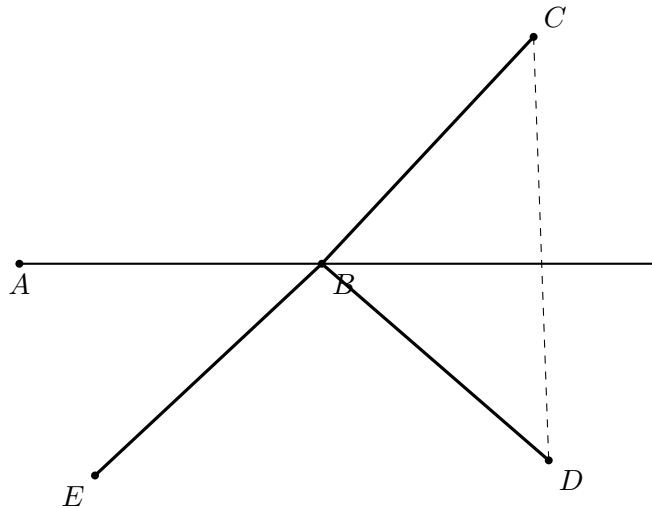


Figure 14.3: Noncollinear branch of the Hilbert reconstruction. The relation  $C - B - E$  puts  $E$  opposite  $C$  across  $B$ ; plane separation places  $D$  and  $E$  on the same side of the base line  $AB$ . The equality of the two angles then invokes the uniqueness clause of Hilbert III.4.

The angle  $\angle ABE$  is nondegenerate. Construct, on the side of the base line containing  $D$ , a ray  $BZ$  satisfying

$$\angle ABZ \cong \angle ABE.$$

Hilbert III.4 includes a uniqueness clause: on the prescribed side of the base ray there is only one ray realizing the prescribed angle.

Since  $E$  lies on the same side of the base line as  $D$ , and

$$\angle ABE \cong \angle ABE,$$

the uniqueness clause gives

$$\text{SameRay}(B, Z, E).$$

On the other hand, the encoded hypothesis of Proposition I.14 gives

$$\angle ABD \cong \angle ABE,$$

and hence, by symmetry,

$$\angle ABE \cong \angle ABD.$$

The point  $D$  lies on the side on which  $BZ$  was constructed. Applying the same uniqueness clause again gives

$$\text{SameRay}(B, Z, D).$$

We therefore have two same-ray relations with the common reference ray  $BZ$ :

$$\text{SameRay}(B, Z, E), \quad \text{SameRay}(B, Z, D).$$

The Book Zero ray calculus now yields

$$\text{SameRay}(B, E, D).$$

Finally, from the encoded supplementary configuration we already know

$$C - B - E.$$

Replacing  $E$  by another point  $D$  on the same ray from  $B$  preserves the betweenness relation. Transport of betweenness along equal rays therefore gives

$$C - B - D.$$

This completes the noncollinear branch and hence the Hilbert reconstruction of Proposition I.14.

## 14.7 Lean Formalization

The reconstructed proposition is formalized by the theorem

```
theorem euclid_proposition_14
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point)
  (base : Geo.Line)
  (hAB : A != B)
  (hAbase :
    HilbertIncidence.OnLine A base)
  (hBbase :
    HilbertIncidence.OnLine B base)
  (hOppCD :
    HilbertOppositeSide Geo C D base)
  (hTwoRight :
    HilbertAnglesEqualTwoRightAngles
    Geo A B C D) :
  Geo.Between C B D := by
```

The formal conclusion

$$\text{Between}(C, B, D)$$

expresses directly that  $B$  lies between  $C$  and  $D$ , and hence that the rays  $BC$  and  $BD$  form one straight line.

The principal dependency structure of the proof is

```

HilbertAnglesEqualTwoRightAngles
|
+--- C-B-E
+--- angle ABD ~= angle ABE
|
HilbertOppositeSide(C,D; base AB)
|
case split on Collinear(C,D,E)
|
+-----+-----+
|                                     |
collinear                           noncollinear
|                                     |
line uniqueness                     SameSide(D,E; base AB)
|                                     |
X = B                               Hilbert III.4 uniqueness
|                                     |
|                                     same ray BE, BD
|                                     |
+-----+-----+
|
hilbert_between_transport_sameRays
|
v
C-B-D

```

The proof introduces no new geometric axiom. The new predicate

```
HilbertAnglesEqualTwoRightAngles
```

serves only as an interface representation of the angle hypothesis. The actual derivation uses the existing incidence, order, plane separation, angle-construction, and ray machinery.

A notable difference from Euclid's dependency graph is that the Lean theorem does not invoke Proposition I.13. The supplementary configuration needed in the classical proof has already been encoded in the hypothesis, and the identification of the relevant rays is obtained directly from Hilbert III.4.

## 14.8 Relationship with Earlier Propositions

### 14.8.1 Proposition I.13

Proposition I.13 gives the direct statement: when one straight line stands on another, the two adjacent angles are either two right angles or together equal to two right angles.

Proposition I.14 is the converse. Starting from the supplementary-angle condition, it proves that the two outer rays form one straight line.

Thus I.13 and I.14 form a natural direct/converse pair.

### 14.8.2 Proposition I.4

The reconstruction ultimately relies on the Hilbert congruence infrastructure whose triangle-level consequences include SAS. More importantly for I.14, Hilbert III.4 supplies uniqueness of angle construction on a prescribed side of a ray.

This uniqueness principle plays the role of a rigidity theorem: once two rays on the same side make congruent angles with the same initial ray, they must coincide.

### 14.8.3 Transition to Proposition I.15

Propositions I.13 and I.14 establish the supplementary and straight-line geometry surrounding an intersection. Proposition I.15 then turns to the opposite angles produced by two intersecting lines and proves their congruence.

The three propositions therefore form a coherent local block: straight-angle structure, its converse, and vertical angles.

## 14.9 Hilbert and Book Zero Components Used

The proof of Proposition I.14 uses several previously established components of the Hilbert and Book Zero layers. Their roles in the reconstruction are recorded here explicitly.

### 14.9.1 hilbert\_third\_side\_endpoints\_sameSide

In the noncollinear branch, the points  $C, D, E$  form a genuine triangle. The base line meets the interior of  $CD$ , because  $C$  and  $D$  lie on opposite sides of it, and it meets the interior of  $CE$  at  $B$ , since

$$C - B - E.$$

The theorem

`hilbert_third_side_endpoints_sameSide`

then shows that the remaining endpoints  $D$  and  $E$  lie on the same side of the base line.

Schematically,

$$\begin{array}{l} C - X - D, \\ C - B - E, \\ X, B \in AB \end{array} \quad \implies \quad \text{SameSide}(D, E; AB).$$

This is the plane-separation step that makes the uniqueness clause of Hilbert's angle-construction axiom applicable to both  $D$  and  $E$ .

### 14.9.2 Hilbert III.4: Uniqueness of Angle Construction

Hilbert's angle-construction axiom provides not only the existence of a copy of a given angle but also uniqueness on a prescribed side of the base ray.

In Proposition I.14 a ray  $BZ$  is constructed so that

$$\angle ABZ \cong \angle ABE.$$

Both  $E$  and  $D$  lie on the prescribed side of the base line. The uniqueness clause can therefore be applied twice, yielding

$$\text{SameRay}(B, Z, E)$$

and

$$\text{SameRay}(B, Z, D).$$

This is the formal replacement for Euclid's final inference that equal angles on the same side determine the same straight line.

### 14.9.3 bookZero\_36\_ray3

The Book Zero theorem

$$\text{bookZero\_36\_ray3}$$

is a ray-composition principle.

In the present proof,

$$\text{SameRay}(B, Z, E)$$

and

$$\text{SameRay}(B, Z, D)$$

imply

$$\text{SameRay}(B, E, D).$$

Thus the auxiliary ray  $BZ$  introduced by Hilbert III.4 can be eliminated, leaving the geometrically relevant statement that  $BE$  and  $BD$  are the same ray.

#### 14.9.4 hilbert\_between\_transport\_sameRays

The final step uses

`hilbert_between_transport_sameRays`

to transport strict betweenness along corresponding rays.

We know

$$C - B - E$$

and have established

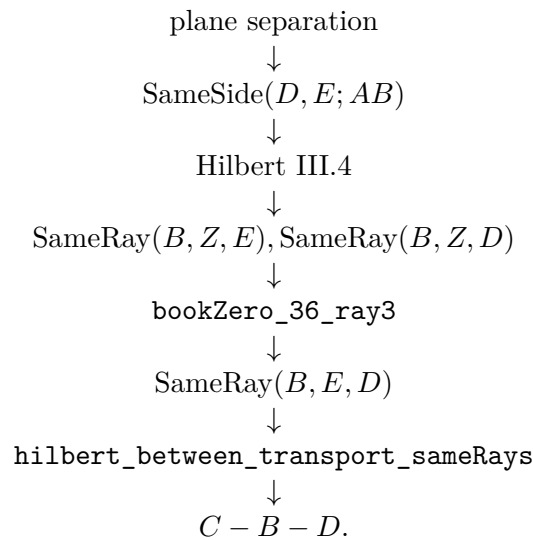
$$\text{SameRay}(B, E, D).$$

The ray from  $B$  toward  $C$  is unchanged, while  $E$  is replaced by the point  $D$  on the same ray from  $B$ . Therefore the theorem gives

$$C - B - D.$$

This is precisely the formal conclusion of Proposition I.14.

The dependency path of the noncollinear branch can therefore be summarized as



## 14.10 Formalization Notes

Proposition I.14 is a good example of a classical converse whose formal proof is governed by uniqueness rather than by numerical angle arithmetic.

The hypothesis says that two adjacent angles have the same total effect as a straight angle. In a traditional proof this is often read directly from the diagram. In the Hilbert reconstruction, the decisive step is to compare rays using the uniqueness clause of angle construction.

The proof naturally separates into two cases.

If the relevant points are already collinear, the desired straight-line conclusion follows from order and collinearity reasoning.

If they are not collinear, the angle hypothesis determines a ray that would have to occupy the same side and form the same angle as another already specified ray. Hilbert III.4 then forces the two rays to coincide. The resulting same-ray information is transported back to betweenness, contradicting the assumed noncollinear configuration.

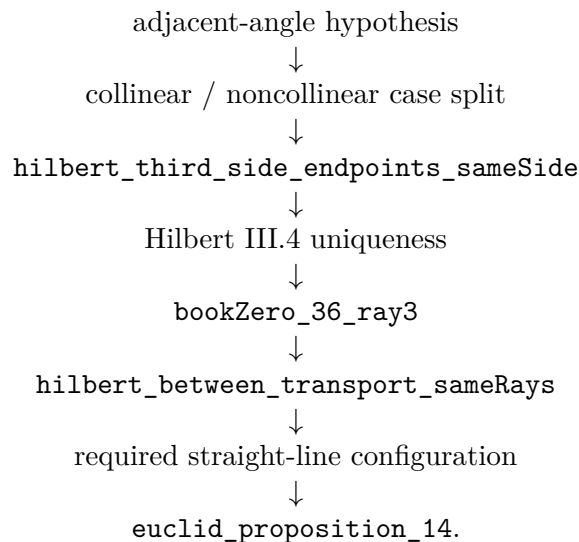
Schematically,

supplementary-angle hypothesis  $\longrightarrow$  same-side information  $\longrightarrow$  angle-construction uniqueness  $\longrightarrow$  same ray  $\longrightarrow$

This is structurally different from simply writing an equation such as  $\alpha + \beta = 180^\circ$ . No numerical angle measure is introduced. The straight-line conclusion is recovered entirely from incidence, order, angle congruence, side-of-line information, and uniqueness.

## 14.11 Dependency Summary

The formal dependency structure is



## 14.12 Conclusion

Proposition I.14 completes the converse half of the supplementary-angle theory begun in I.13.

Its formal reconstruction is especially instructive because it avoids introducing angle addition or numerical angle measures. The proof instead uses Hilbert's side-of-line and angle-construction uniqueness machinery to show that the geometry of the rays is rigid enough to force a straight line.

The proposition therefore illustrates a recurring feature of the project: statements that look algebraic in modern notation can often be recovered synthetically from incidence, order, congruence, and uniqueness.



Together with I.13, it prepares the local theory of intersecting lines used immediately in Proposition I.15.

## Chapter 15

# Proposition I.15

### 15.1 Introduction

Euclid's Proposition I.15 establishes the equality of vertical angles formed by two intersecting straight lines.

If two straight lines intersect at a point, each pair of opposite angles is equal. In modern terminology, these are the vertical-angle relations.

Euclid proves the proposition by applying Proposition I.13 to two adjacent linear pairs and subtracting a common angle from two sums that are each equal to two right angles.

The Hilbert reconstruction is considerably shorter. The corresponding vertical-angle theorem has already been derived in the Hilbert layer as

`hilbert_vertical_angles`

from Hilbert's theory of adjacent angles. Proposition I.15 therefore becomes an application of an existing synthetic theorem rather than a new geometric derivation.

This proposition provides another example in which a classical Euclidean proof is compressed by the intermediate Hilbert interface: the mathematical content required by Euclid has already been isolated as a reusable theorem.

### 15.2 The Classical Configuration

Let the straight lines  $AB$  and  $CD$  intersect at the point  $E$ , with

$$A - E - B$$

and

$$C - E - D.$$

The intersection determines four angles around  $E$ .

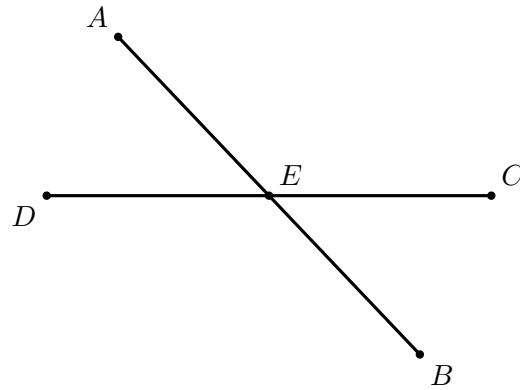


Figure 15.1: Classical configuration for Proposition I.15. The betweenness relations  $A - E - B$  and  $D - E - C$  encode the two pairs of opposite rays.

The opposite pairs are

$$\angle AEC \quad \text{and} \quad \angle BED,$$

and

$$\angle CEB \quad \text{and} \quad \angle DEA.$$

Proposition I.15 states that each pair is equal.

Only one geometric diagram is needed. The two conclusions are not two different configurations: they are the two opposite pairs determined by the same crossing. Repeating the intersection would add no mathematical information.

The betweenness relations

$$A - E - B, \quad C - E - D$$

express synthetically that  $EA$  and  $EB$  are opposite rays and that  $EC$  and  $ED$  are opposite rays. Thus the classical configuration is already naturally adapted to Hilbert's order geometry.

### 15.3 Statement

Euclid's Proposition I.15 states:

If two straight lines cut one another, they make the vertical angles equal to one another.

For the configuration

$$A - E - B, \quad C - E - D,$$

the proposition gives the two vertical-angle congruences

$$\angle AEC \cong \angle BED$$

and

$$\angle CEB \cong \angle DEA.$$

In the Hilbert reconstruction we assume, in addition, that the two lines are genuinely distinct. This is expressed by

$$\neg \text{Collinear}(A, E, C).$$

The formal statement is therefore

$$\begin{array}{l} A - E - B, \\ C - E - D, \\ \neg \text{Collinear}(A, E, C) \end{array} \implies \begin{cases} \angle AEC \cong \angle BED, \\ \angle CEB \cong \angle DEA. \end{cases}$$

The noncollinearity hypothesis is the synthetic expression of a proper intersection of two distinct straight lines. It also guarantees that the angles occurring in the proposition are nondegenerate.

Thus the Lean formulation makes explicit a condition that is implicit in Euclid's phrase "two straight lines cut one another." Strict betweenness specifies the two pairs of opposite rays, while noncollinearity excludes the case in which all five points lie on a single straight line.

## 15.4 Classical Proof

Euclid first proves the equality of one pair of vertical angles.

Since the straight line  $AE$  stands upon the straight line  $CD$ , Proposition I.13 gives

$$\angle CEA + \angle AED$$

equal to two right angles.

Again, since the straight line  $DE$  stands upon the straight line  $AB$ , Proposition I.13 gives

$$\angle AED + \angle DEB$$

equal to two right angles.

Therefore

$$\angle CEA + \angle AED = \angle AED + \angle DEB.$$

Subtracting the common angle  $\angle AED$  from both sides, by Common Notion 3, gives

$$\angle CEA = \angle DEB.$$

Thus one pair of vertical angles is equal.

In the same way,

$$\angle CEB = \angle AED.$$

Therefore, when two straight lines cut one another, they make the vertical angles equal to one another.

Wilkins's diagram is especially effective here because the geometry is entirely local at  $E$ . The proof needs only the four rays and two linear pairs. The colored sectors in Wilkins's note help distinguish the angles, but no additional construction is involved, so the reconstruction keeps the diagram unshaded and records the equalities in the text.

## 15.5 Proof Architecture of the Reconstruction

The formal reconstruction and Euclid's original proof establish the same geometric fact at different levels of the theory.

Euclid

-----

A-E-B and C-E-D

|

$$+-- \text{CEA} + \text{AED} = 2\text{R} \quad (\text{I.13})$$

|

$$+-- \text{DEB} + \text{AED} = 2\text{R} \quad (\text{I.13})$$

|

v

$$\text{CEA} + \text{AED} = \text{DEB} + \text{AED}$$

|

Common Notion 3

|

v

$$\text{CEA} = \text{DEB}$$

same argument

|

v

$$\text{CEB} = \text{AED}$$

Hilbert / Lean

-----

A-E-B, C-E-D, noncollinearity

|

hilbert\_vertical\_angles

|

v

$$\text{AEC} \sim \text{BED}$$

|

```

reorder the same crossing
  |
transport noncollinearity
  |
hilbert_vertical_angles
  |
  v
CEB ~= DEA

```

The formal proof therefore introduces no new geometric point and no new diagrammatic case. Its extra work is purely structural: the second pair is obtained by reorienting the same intersection and proving that the reordered triple is still noncollinear.

Euclid derives the vertical-angle theorem from Proposition I.13. Each of two adjacent angle pairs is equal to two right angles. Since the two sums contain a common angle, Common Notion 3 allows that angle to be subtracted, leaving the two vertical angles equal.

Thus the classical dependency is

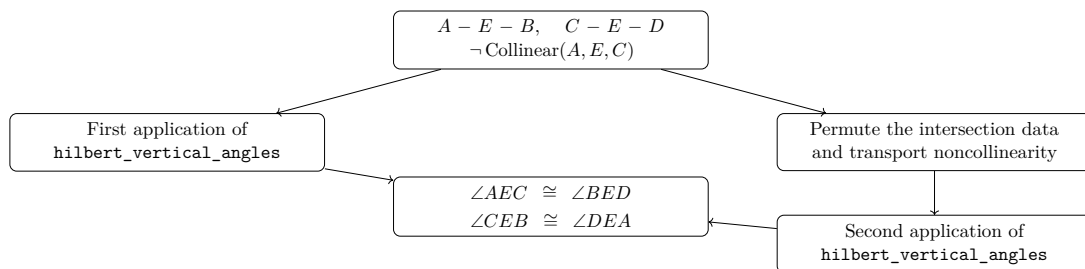
$$\text{I.13} \longrightarrow \text{two linear-pair equalities} \longrightarrow \text{Common Notion 3} \longrightarrow \text{I.15.}$$

The Hilbert development has already isolated the corresponding geometric argument in the theorem

`hilbert_vertical_angles`

Consequently, Proposition I.15 does not require a new reconstruction of angle addition or subtraction. The first pair of vertical angles follows by a direct application of this theorem.

The second pair is obtained by viewing the same intersection with the roles of the two straight lines exchanged. The only additional work is to transport the required noncollinearity hypothesis to the reordered triple of points.



Schematically, the two proof architectures are therefore

Euclid : I.13 → angle sums → C.N.3 → vertical angles,

Hilbert reconstruction : intersection data → `hilbert_vertical_angles` → vertical angles.

The difference is architectural rather than mathematical. The Hilbert layer has already packaged the vertical-angle argument as a reusable synthetic theorem, so the Euclidean proposition becomes a thin interface theorem over previously established geometry.

## 15.6 Hilbert Reconstruction

The Hilbert reconstruction starts directly from the synthetic data of the intersection:

$$A - E - B, \quad C - E - D,$$

together with

$$\neg \text{Collinear}(A, E, C).$$

Unlike Euclid's proof, no explicit notion of a sum of angles equal to two right angles is required. The relevant geometric content has already been established in the Hilbert layer as the vertical-angle theorem.

### 15.6.1 The First Pair of Vertical Angles

The theorem

`hilbert_vertical_angles`

has precisely the form needed for the first pair.

Applied to

$$A - E - B, \quad C - E - D, \quad \neg \text{Collinear}(A, E, C),$$

it gives directly

$$\angle AEC \cong \angle BED.$$

Thus the first conclusion of Proposition I.15 requires no auxiliary construction.

### 15.6.2 The Second Pair of Vertical Angles

For the second pair we view the same intersection with the two straight lines exchanged.

From

$$A - E - B$$

the symmetry of strict betweenness gives

$$B - E - A.$$

We also require the noncollinearity condition

$$\neg \text{Collinear}(C, E, B).$$

Suppose, for contradiction, that

$$\text{Collinear}(C, E, B).$$

Since  $A - E - B$ , the points  $A, E, B$  are collinear. Permuting the first collinearity relation and using transitivity of collinearity through the distinct points  $E$  and  $B$  would then imply

$$\text{Collinear}(A, E, C),$$

contrary to the original hypothesis.

Hence

$$\neg \text{Collinear}(C, E, B).$$

We may therefore apply `hilbert_vertical_angles` a second time, now to

$$C - E - D, \quad B - E - A.$$

This yields

$$\angle CEB \cong \angle DEA.$$

Combining the two applications gives the complete conclusion

$$\angle AEC \cong \angle BED, \quad \angle CEB \cong \angle DEA.$$

The Hilbert reconstruction is therefore substantially shorter than Euclid's original argument. The classical use of Proposition I.13 and Common Notion 3 has already been absorbed into the previously proved vertical-angle theorem; the only additional formal work is the permutation of the intersection data required for its second application.

## 15.7 Lean Formalization

The reconstructed proposition is formalized by the theorem

```
theorem euclid_proposition_15
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E : Geo.Point)
  (hAEB : Geo.Between A E B)
  (hCED : Geo.Between C E D)
  (hAEC : not PrimCollinear Geo A E C) :
  Geo.AngleCongruent A E C B E D /\
  Geo.AngleCongruent C E B D E A := by
```



The conclusion records both pairs of vertical angles:

$$\angle AEC \cong \angle BED$$

and

$$\angle CEB \cong \angle DEA.$$

The first pair is obtained directly:

```
have hFirst :
  Geo.AngleCongruent A E C B E D :=
  hilbert_vertical_angles
  Geo A E C B D
  hAEB
  hCED
  hAEC
```

For the second pair, the first line is read in the opposite direction:

```
have hBEC : Geo.Between B E A :=
  (HilbertOrder.between_incidence
  A E B hAEB).2.2.2.2
```

The required noncollinearity condition is then derived by contradiction. If  $C, E, B$  were collinear, cyclic permutation gives  $E, B, C$  collinear. Since  $A, E, B$  are already collinear and  $E \neq B$ , transitivity would imply that  $A, E, C$  are collinear, contradicting **hAEC**.

In Lean this is expressed by

```
have hCEB :
  not PrimCollinear Geo C E B := by
  intro h

  have hEBC :
    PrimCollinear Geo E B C :=
    PrimCollinearCycle Geo C E B h

  have hAEBcol :
    PrimCollinear Geo A E B :=
    (HilbertOrder.between_incidence
    A E B hAEB).2.2.2.1

  have hEB : E != B :=
    (HilbertOrder.between_incidence
    A E B hAEB).2.1

  have hAECcol :
    PrimCollinear Geo A E C :=
```

```

hilbert_primCollinear_trans
Geo A E B C
hEB
hAEBcol
hEBC

exact hAEC hAECcol

```

The second pair of vertical angles is now another direct application:

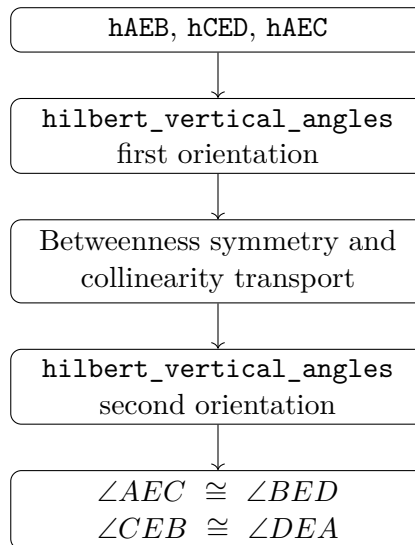
```

have hSecond :
Geo.AngleCongruent C E B D E A :=
hilbert_vertical_angles
Geo C E B D A
hCED
hBEC
hCEB

exact <hFirst, hSecond>

```

Thus the formal dependency structure is



No new geometric axiom or auxiliary geometric construction is introduced in the formalization. Proposition I.15 is obtained entirely from the existing Hilbert incidence, order, collinearity, and vertical-angle theory.

## 15.8 Relationship with Earlier Propositions

### 15.8.1 Proposition I.13

Proposition I.13 establishes the supplementary-angle configuration created when a straight line stands on another straight line. That result belongs to the same local geometry of intersecting lines that underlies I.15.

The logical roles are different, however. I.13 concerns adjacent angles whose sum forms a straight angle; I.15 concerns opposite angles and proves their congruence.

### 15.8.2 Proposition I.14

Proposition I.14 gives a converse criterion for recognizing a straight line from a supplementary configuration. Together, I.13 and I.14 develop the straight-angle infrastructure surrounding an intersection.

Proposition I.15 then extracts a different invariant of the same configuration: opposite, or vertical, angles are congruent.

### 15.8.3 Transition to Proposition I.16

Vertical-angle congruence becomes an explicit transport tool in Proposition I.16. The exterior-angle proof naturally produces angles in orientations that do not always match the public Euclidean statement, and I.15 supplies the congruence needed to move between those orientations.

Thus I.15 is not merely the end of the elementary intersecting-lines block. It becomes part of the infrastructure for the strict angle-comparison theory beginning with I.16.

## 15.9 Hilbert and Book Zero Components Used

Proposition I.15 is formally short because its main geometric content has already been isolated in the Hilbert layer.

The proof uses one principal theorem together with elementary order and collinearity machinery required to apply it in the two orientations of the intersection.

### 15.9.1 hilbert\_vertical\_angles

The central result is

`hilbert_vertical_angles`

which states that if

$$C - E - B, \quad D - E - F,$$

and

$$\neg \text{Collinear}(C, E, D),$$

then

$$\angle CED \cong \angle BEF.$$

This theorem is the immediate vertical-angle corollary developed in the Hilbert layer after the theorem on adjacent congruent angles.

In Proposition I.15 it is used twice.

The first application gives

$$\angle AEC \cong \angle BED.$$

The second application, after reversing the first straight line and transporting noncollinearity, gives

$$\angle CEB \cong \angle DEA.$$

Thus essentially all of the geometric content of Euclid's Proposition I.15 is already contained in `hilbert_vertical_angles`.

### 15.9.2 HilbertOrder.between\_incidence

The theorem

`HilbertOrder.between_incidence`

extracts the elementary order and incidence consequences of strict betweenness.

From

$$A - E - B$$

the proof obtains both

$$B - E - A$$

and the collinearity relation

$$\text{Collinear}(A, E, B).$$

It also supplies the inequality

$$E \neq B,$$

which is needed for the transitivity of collinearity.

Thus a single betweenness hypothesis provides all of the auxiliary order data required for the second application of the vertical-angle theorem.

**15.9.3 PrimCollinearCycle**

The permutation theorem

`PrimCollinearCycle`

cyclically reorders a collinearity relation.

In the contradiction argument,

$$\text{Collinear}(C, E, B)$$

is transformed into

$$\text{Collinear}(E, B, C).$$

This places the points in exactly the orientation required by the transitivity theorem used in the next step.

**15.9.4 hilbert\_primCollinear\_trans**

The theorem

`hilbert_primCollinear_trans`

expresses the transitivity of collinearity when two distinct common points determine the same line.

In Proposition I.15 we combine

$$\text{Collinear}(A, E, B)$$

with

$$\text{Collinear}(E, B, C)$$

and the inequality

$$E \neq B.$$

The result is

$$\text{Collinear}(A, E, C),$$

contradicting the original noncollinearity hypothesis.

Hence

$$\neg \text{Collinear}(C, E, B),$$

which is precisely the hypothesis needed for the second invocation of `hilbert_vertical_angles`.

The complete formal dependency may therefore be summarized as

$$\begin{array}{c}
 A - E - B, \quad C - E - D, \quad \neg \text{Collinear}(A, E, C) \\
 \downarrow \\
 \text{hilbert\_vertical\_angles} \\
 \downarrow \\
 \angle AEC \cong \angle BED \\
 \text{betweenness symmetry + collinearity transport} \\
 \downarrow \\
 \neg \text{Collinear}(C, E, B) \\
 \downarrow \\
 \text{hilbert\_vertical\_angles} \\
 \downarrow \\
 \angle CEB \cong \angle DEA.
 \end{array}$$

No genuinely new Book Zero theorem is required for Proposition I.15. The proposition is therefore primarily an interface theorem over the already established Hilbert vertical-angle theory.

## 15.10 Formalization Notes

Proposition I.15 is classically almost immediate from the supplementary angle theory, but its formal reconstruction highlights a different issue: orientation.

At an intersection there are two pairs of vertical angles. The geometric theorem is symmetric, while the formal predicates are written with ordered point triples. Consequently the proof must explicitly recover the appropriate collinearities from betweenness, rotate or cycle them into the required orientation, and then invoke the reusable vertical-angle theorem.

The essential pattern is

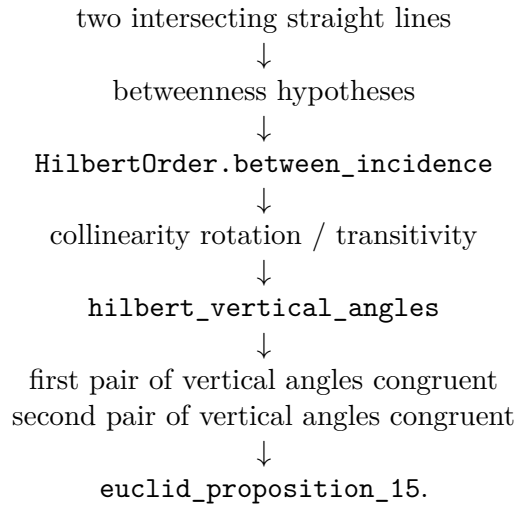
betweenness  $\longrightarrow$  primitive collinearity  $\longrightarrow$  orientation normalization  $\longrightarrow$  `hilbert_vertical_angles`  $\longrightarrow$  angle

This is a good example of why the Book Zero and Hilbert interface layers contain many apparently small symmetry and rotation lemmas. On paper, one simply says that two lines intersect. In Lean, the exact ordering of the points on those lines determines which theorem application type-checks.

No new geometric axiom is introduced by I.15. The proposition packages the already established intersection and angle-congruence machinery into the classical Euclidean statement.

## 15.11 Dependency Summary

The formal dependency structure is



## 15.12 Conclusion

Proposition I.15 completes the elementary theory of angles formed by intersecting straight lines.

Its classical content is the familiar theorem that vertical angles are congruent. The formal reconstruction shows that the principal burden is not angle arithmetic but the management of incidence and orientation: betweenness must be converted into primitive collinearity, and the resulting triples must be placed in exactly the orientation expected by the vertical-angle theorem.

The proposition is also structurally important for what follows. Vertical-angle congruence becomes a reusable transport mechanism in Proposition I.16, where Book I moves from congruence statements to strict angle comparison.

Thus I.15 closes one local geometric block and simultaneously provides infrastructure for the next.

# Chapter 16

## Proposition I.16

### 16.1 Introduction

Euclid's Proposition I.16 is the exterior-angle inequality for a triangle. If a side of a triangle is extended beyond one of its vertices, the exterior angle formed at that vertex is greater than either of the two remote interior angles.

For a nondegenerate triangle  $ABC$ , extend  $BC$  through  $C$  to a point  $D$ , so that

$$B - C - D.$$

The proposition asserts

$$\angle BAC < \angle ACD \quad \text{and} \quad \angle ABC < \angle ACD.$$

Unlike Proposition I.15, this proposition is not merely a thin application of one previously packaged theorem. Its reconstruction requires a substantial piece of Hilbert order-and-congruence geometry: an interior ray must be transported between congruent angles, and the resulting angle comparison must be expressed without numerical angle measure.

The formal development therefore isolates the geometric machinery in the Hilbert interface and leaves the Euclidean proposition itself comparatively short. In particular, the final Lean file contains three theorems:

- `euclid_proposition_16_first`,
- `euclid_proposition_16_second`,
- `euclid_proposition_16`.

### 16.2 The Classical Configuration

Let  $ABC$  be a nondegenerate triangle and let the side  $BC$  be produced through  $C$  to  $D$ :

$$B - C - D.$$

The angle  $\angle ACD$  is exterior to the triangle. The two interior opposite angles are  $\angle BAC$  and  $\angle ABC$ .

The formal hypotheses are

$$\neg \text{Collinear}(A, B, C) \quad \text{and} \quad B - C - D.$$



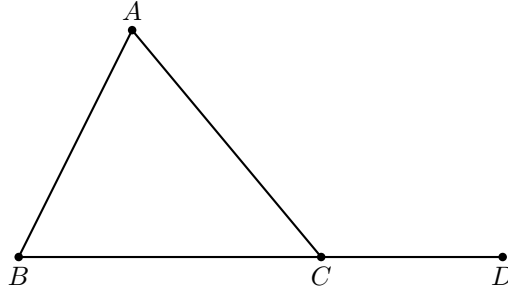


Figure 16.1: Statement configuration for Proposition I.16:  $BC$  is produced beyond  $C$  to  $D$ .

The noncollinearity hypothesis makes the triangle nondegenerate, while strict betweenness records that  $D$  lies on the continuation of  $BC$  beyond  $C$ .

### 16.3 Statement

Euclid states Proposition I.16 as follows:

If one side of a triangle is produced, the exterior angle is greater than either of the two interior opposite angles.

In the Hilbert reconstruction, strict comparison of angles is represented by `HilbertAngleLess`. Thus the formal conclusion is

$$\text{HilbertAngleLess}(BAC, ACD) \quad \wedge \quad \text{HilbertAngleLess}(ABC, ACD).$$

More explicitly,

$$\neg \text{Collinear}(A, B, C), \quad B - C - D \implies \begin{cases} \angle BAC < \angle ACD, \\ \angle ABC < \angle ACD. \end{cases}$$

The relation `HilbertAngleLess` is synthetic. It does not compare numerical measures. An angle is smaller than another when a congruent copy of the first can be realized as a proper interior subangle of the second.

### 16.4 Classical Proof

Euclid proves the first inequality by a midpoint construction.

Let  $E$  be the midpoint of  $AC$ . Join  $BE$  and extend it beyond  $E$  to a point  $F$  such that

$$BE = EF.$$

Join  $CF$ .

Since  $AE = EC$ ,  $BE = EF$ , and the vertical angles  $\angle AEB$  and  $\angle CEF$  are equal by Proposition I.15, the triangles  $AEB$  and  $CEF$  are congruent by Proposition I.4. Hence

$$\angle BAE = \angle ECF.$$

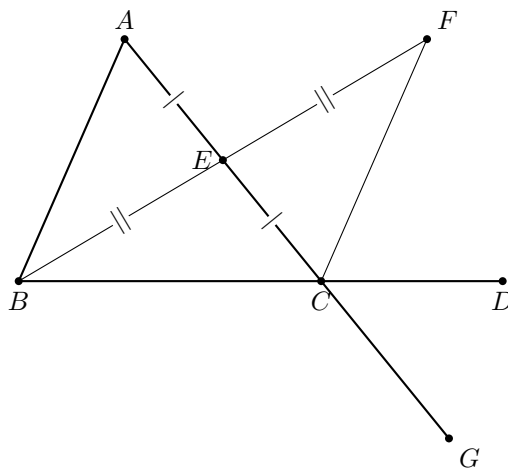


Figure 16.2: Wilkins's auxiliary construction for the first inequality:  $E$  bisects  $AC$ ,  $BE$  is produced to  $F$  with  $BE \cong EF$ , and  $AC$  is produced beyond  $C$  to  $G$ .

Since  $E$  lies on  $AC$ , this gives

$$\angle BAC = \angle ACF.$$

But the ray  $CF$  lies inside the exterior angle  $ACD$ . Therefore

$$\angle BAC < \angle ACD.$$

This last sentence is exactly where the diagram carries hidden order information. Wilkins spells it out:  $F$  must lie on the same side of  $CD$  as  $A$  and on the same side of  $AC$  as  $D$ ; these two half-plane conditions characterize the interior of  $\angle ACD$ . Thus the inclusion of the ray  $CF$  in the exterior angle is a theorem, not a fact that may be read off the picture.

For the second opposite angle Euclid repeats the same argument after extending another side of the triangle. The equality of the corresponding vertical exterior angles then identifies the resulting exterior angle with  $\angle ACD$ .

Thus the classical architecture is

$$\text{midpoint construction} \rightarrow \text{I.15} \rightarrow \text{SAS (I.4)} \rightarrow \text{interior subangle} \rightarrow \text{I.16}.$$

## 16.5 Proof Architecture of the Reconstruction

The Hilbert reconstruction retains the synthetic structure of Euclid's argument but packages the difficult order-theoretic steps into reusable interface theorems.

```
statement: B-C-D
|
+----- first opposite angle -----+
|
v
Hilbert auxiliary construction
A-M-C, B-M-E + congruence data
|
v
```

```

AB || CE
|
hilbert_exterior_angle_less
|
v
BAC < ACD
|
+----- second opposite angle -----+
|
extend AC beyond C: A-C-F
|
apply first half to triangle BAC
|
ABC < BCF
|
vertical angles at C:
BCF ~= ACD
|
transport strict angle order
|
ABC < ACD
|
v
euclid_proposition_16

```

Here the distinction between the classical and formal diagrams matters. Wilkins's  $E, F$  construction explains Euclid's SAS argument and the nontrivial claim that  $F$  lies inside the exterior angle. The Lean helper uses its own auxiliary points  $M, E$  and packages that order geometry into the Hilbert interface. The second Lean inequality is then obtained by a genuinely new extension  $A - C - F$  and vertical-angle transport.

For the first opposite angle the auxiliary construction produces points  $M$  and  $E$  with

$$A - M - C, \quad B - M - E,$$

and the required segment and angle congruences. From these data the interface proves that  $AB$  is parallel to  $CE$ . The theorem `hilbert_exterior_angle_less` then concludes directly that

$$\angle BAC < \angle ACD.$$

Schematically:

$$\begin{array}{c}
\neg \text{Collinear}(A, B, C) \\
\downarrow \\
\text{hilbert\_exterior\_angle\_aux} \\
\downarrow \\
A - M - C, B - M - E, \text{ congruence data} \\
\downarrow \\
\text{hilbert\_exterior\_angle\_aux\_parallel} \\
\downarrow \\
AB \parallel CE \\
\downarrow \\
\text{hilbert\_exterior\_angle\_less} \\
\downarrow \\
\angle BAC < \angle ACD.
\end{array}$$

The second inequality is deliberately reduced to the first. Extend  $AC$  through  $C$  to  $F$ . Apply the first part to triangle  $BAC$  to obtain

$$\angle ABC < \angle BCF.$$

The lines  $ACF$  and  $BCD$  intersect at  $C$ , so Proposition I.15 gives the vertical-angle congruence

$$\angle BCF \cong \angle ACD.$$

Transporting the right-hand side of the strict angle comparison across this congruence yields

$$\angle ABC < \angle ACD.$$

Thus the second half has the compact architecture

$$\text{first half of I.16} \rightarrow \text{I.15 / vertical angles} \rightarrow \text{transport of } < \rightarrow \text{second half of I.16.}$$

## 16.6 Hilbert Reconstruction

### 16.6.1 The First Opposite Interior Angle

The theorem `euclid_proposition_16_first` begins from

$$\neg \text{Collinear}(A, B, C), \quad B - C - D.$$

It invokes

$$\text{hilbert\_exterior\_angle\_aux}$$

to obtain points  $M, E$  and the auxiliary configuration required for the exterior-angle argument.

The same configuration is passed to

$$\text{hilbert\_exterior\_angle\_aux\_parallel},$$

which proves

$$AB \parallel CE.$$

Finally,

$$\text{hilbert\_exterior\_angle\_less}$$

combines the auxiliary construction, the extension  $B - C - D$ , and the parallelism to establish

$$\angle BAC < \angle ACD.$$

The Euclidean theorem itself therefore contains no unresolved construction. The substantial work has been moved into the Hilbert interface, where it can be reused independently of Proposition I.16.

### 16.6.2 The Second Opposite Interior Angle

For the second inequality, the proof first derives  $A \neq C$  from the noncollinearity of  $A, B, C$ . Hilbert's order axiom then extends  $AC$  through  $C$  to a point  $F$ :

$$A - C - F.$$

The triangle is viewed in the order  $BAC$ . Applying `euclid_proposition_16_first` to this reordered triangle gives

$$\angle ABC < \angle BCF.$$

It remains to identify  $\angle BCF$  with the desired exterior angle  $\angle ACD$ . From  $B - C - D$  we obtain the reversed betweenness relation

$$D - C - B.$$

The two straight lines  $ACF$  and  $BCD$  therefore form a pair of vertical angles at  $C$ . The theorem `VerticalAngles` yields, after the required symmetry and reversal operations,

$$\angle BCF \cong \angle ACD.$$

The theorem

`hilbert_angleLess_transport_right`

then transports the comparison across the congruent right-hand angle:

$$\angle ABC < \angle BCF, \quad \angle BCF \cong \angle ACD \implies \angle ABC < \angle ACD.$$

## 16.7 Lean Formalization

The first half is formalized as follows:

```

theorem euclid_proposition_16_first
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point)
  (hABC : not PrimCollinear Geo A B C)
  (hBCD : Geo.Between B C D) :
  HilbertAngleLess Geo B A C A C D := by

rcases
  hilbert_exterior_angle_aux
    Geo A B C hABC with
  <M, E,
  hAMC,
  hAMMC,
  hBME,
  hBMME,
  hAngle>

have hParallel :
  Geo.Parallel A B C E :=
  hilbert_exterior_angle_aux_parallel

```

```

Geo A B C M E
hABC
hAMC
hBME
hAngle

exact
hilbert_exterior_angle_less
  Geo A B C D M E
  hABC
  hAMC
  hBME
  hBCD
  hAngle
  hParallel

```

The second half deliberately reuses the first:

```

have hLessBCF :
  HilbertAngleLess Geo A B C B C F :=
  euclid_proposition_16_first
  Geo
  B A C F
  hBACnc
  hACF

```

The vertical-angle relation is then constructed at  $C$  and normalized to the required orientation:

```

have hVerticalRaw :
  Geo.AngleCongruent A C D F C B :=
  VerticalAngles
  Geo
  A C D F B
  hACF
  hDCA
  hACDnc

have hVerticalSymm :
  Geo.AngleCongruent F C B A C D :=
  Geo.angle_congruent_symmetry
  A C D
  F C B
  hVerticalRaw

have hVertical :
  Geo.AngleCongruent B C F A C D :=
  (Geo.angle_congruent_reverse_first
   F C B A C D).mp
  hVerticalSymm

```

Finally the strict inequality is transported:

```

exact
hilbert_angleLess_transport_right

```

```

Geo
A B C
B C F
A C D
hLessBCF
hACDnc
hVertical

```

The complete proposition simply combines the two parts:

```

theorem euclid_proposition_16
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point)
  (hABC : not PrimCollinear Geo A B C)
  (hBCD : Geo.Between B C D) :
  HilbertAngleLess Geo B A C A C D /\
  HilbertAngleLess Geo A B C A C D := by

  have hFirst :=
    euclid_proposition_16_first
      Geo A B C D hABC hBCD

  have hSecond :=
    euclid_proposition_16_second
      Geo A B C D hABC hBCD

  exact <hFirst, hSecond>

```

## 16.8 Relationship with Earlier Propositions

### 16.8.1 Proposition I.10

Euclid’s classical proof uses a midpoint construction as part of the auxiliary configuration. Proposition I.10 is therefore one of the constructional ingredients behind the traditional argument.

In the Hilbert reconstruction, however, the decisive exterior-angle comparison has already been isolated in reusable infrastructure, so the public I.16 proof does not need to reproduce the entire midpoint construction locally.

### 16.8.2 Proposition I.15

Vertical-angle congruence is used when the auxiliary configuration produces an angle in the opposite orientation from the one required in the final statement.

The theorem `VerticalAngles` therefore acts as an orientation bridge inside the reconstruction.

### 16.8.3 Transition to Proposition I.17

Proposition I.16 supplies the strict exterior-angle comparison that drives Proposition I.17. In I.17, each assertion that two interior angles are together less than two right angles is obtained by extending a side and applying the exterior-angle theorem in an appropriate orientation.

Thus I.16 begins the strict angle-order block of Book I that continues through I.17–I.19.

## 16.9 Hilbert and Book Zero Components Used

### 16.9.1 HilbertAngleLess

The proposition requires a genuinely synthetic notion of strict angle comparison. The relation `HilbertAngleLess` expresses that a congruent copy of one angle occurs as a proper interior subangle of another. This avoids introducing numerical angle measure and keeps Proposition I.16 inside pure incidence, order, and congruence geometry.

### 16.9.2 hilbert\_exterior\_angle\_aux

This theorem supplies the auxiliary points used in the first exterior-angle construction. It packages the midpoint-style configuration that underlies Euclid’s original proof and returns the betweenness and congruence data required by the subsequent argument.

### 16.9.3 hilbert\_exterior\_angle\_aux\_parallel

From the auxiliary configuration, this theorem proves

$$AB \parallel CE.$$

The parallel relation is not an additional Euclidean postulate here; it is a derived relation inside the neutral Hilbert development used to organize the exterior-angle argument.

### 16.9.4 hilbert\_exterior\_angle\_less

This is the principal interface theorem used by the first half of I.16. It converts the auxiliary construction and its parallelism into the strict angle comparison

$$\angle BAC < \angle ACD.$$

Its proof depends on the interior-subangle machinery developed in the Hilbert layer.

### 16.9.5 Interior-subangle transport

A substantial part of the work required for Proposition I.16 lies below the final Euclidean file. The Hilbert interface establishes that an interior ray can be transported between congruent angles while preserving the corresponding subangle.

The completed infrastructure includes the segment and crossbar mechanisms needed for that transport, in particular the ability to transport strict segment comparison through congruence,



transport an interior point between congruent segments, and preserve ray–segment intersection when the endpoints are moved along the same two rays.

This is the part of the reconstruction where the formal proof exposes geometric structure that is largely implicit in the paper proof.

### 16.9.6 VerticalAngles

The second half uses the vertical-angle theorem already exposed by Proposition I.15. After  $AC$  is extended through  $C$  to  $F$ , the angles

$$\angle BCF \quad \text{and} \quad \angle ACD$$

are vertical. Their congruence allows the second inequality to be reduced to the first.

### 16.9.7 hilbert\_angleLess\_transport\_right

This theorem expresses invariance of strict angle comparison under congruence of the right-hand angle. In the final step,

$$\angle ABC < \angle BCF$$

and

$$\angle BCF \cong \angle ACD$$

imply

$$\angle ABC < \angle ACD.$$

## 16.10 Formalization Notes

Proposition I.16 is a major transition from angle congruence to strict angle comparison.

Earlier propositions establish equalities of angles through triangle congruence, isosceles geometry, supplementary configurations, and vertical angles. I.16 requires a genuinely ordered conclusion: an exterior angle is strictly greater than either opposite interior angle.

The reconstruction therefore uses `HilbertAngleLess` rather than a numerical angle measure. The strict comparison is obtained from an explicit geometric subangle configuration and then transported through angle congruences until its endpoints match the Euclidean statement.

A second important feature is that the two opposite interior angles are not handled by pretending that orientation is irrelevant. One branch can be obtained directly from the exterior-angle infrastructure; the other requires additional transport, including vertical-angle congruence. Lean forces these orientations to be recorded explicitly.

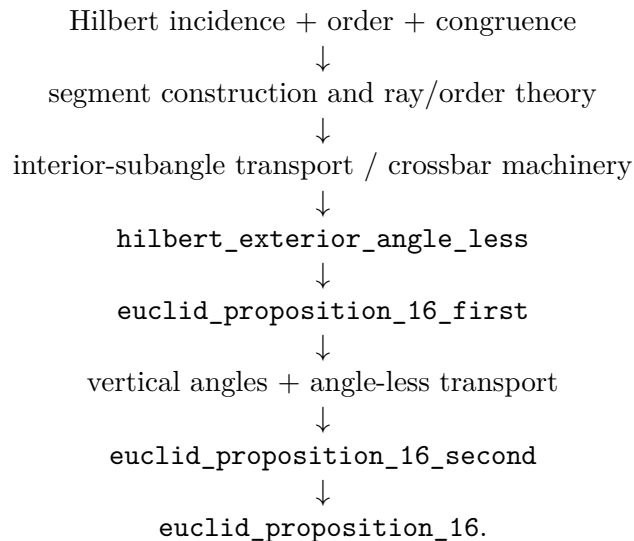
Schematically, the formal mechanism is

auxiliary exterior-angle configuration  $\longrightarrow$  strict subangle comparison  $\longrightarrow$  angle-congruence transport  $\longrightarrow$  I.16.

This infrastructure becomes immediately reusable. Proposition I.17 uses I.16 to control sums of two interior angles without introducing angle addition, while Propositions I.18 and I.19 use the resulting strict angle-order theory to relate side order and angle order in a triangle.

## 16.11 Dependency Summary

The formal dependency structure can be summarized as



The final Euclidean theorem is therefore compact, but its compactness is meaningful: the difficult geometry has been factored into reusable Hilbert-interface results rather than hidden in an axiom or a local `sorry`.

## 16.12 Conclusion

Proposition I.16 marks a significant step beyond the immediately preceding propositions. It introduces a strict comparison of angles and requires a robust synthetic notion of interiority. In Euclid’s presentation this is handled by a midpoint construction, SAS, and the observation that one angle is a proper part of another. In the formal Hilbert reconstruction the same content is decomposed into explicit reusable mechanisms for rays, betweenness, congruence, crossbar behavior, and transport of strict angle comparison.

The final Lean theorem is short precisely because those mechanisms have been established independently. No additional axiom is introduced in `Proposition16.lean`, and the completed development compiles without `sorry`.

## Chapter 17

# Proposition I.17

### 17.1 Introduction

Euclid’s Proposition I.17 states that in every triangle any two interior angles are together less than two right angles.

For a nondegenerate triangle  $ABC$ , the three conclusions are

$$\angle BAC + \angle ABC < 2R,$$

$$\angle BAC + \angle ACB < 2R,$$

and

$$\angle ABC + \angle ACB < 2R.$$

In Euclid’s classical proof, Proposition I.16 supplies an exterior-angle inequality and Proposition I.13 identifies an adjacent pair of angles with two right angles. The informal argument then adds the same angle to both sides of an inequality.

The formal reconstruction takes a different route. No numerical angle measure and no binary operation of angle addition are introduced. Instead, the phrase “two angles are together less than two right angles” is represented synthetically: the first angle is required to be strictly smaller than the supplementary angle of the second. This follows the same design principle already used in the reconstruction of Proposition I.14, where “equal to two right angles” was encoded by comparison with a supplementary angle.

The final Lean development consists of a new interface predicate, three pairwise forms of Proposition I.17, and one wrapper theorem combining them.

### 17.2 The Classical Configuration

Let  $ABC$  be a nondegenerate triangle. Euclid proves a representative case by extending one side of the triangle. In Wilkins’s presentation,  $BC$  is produced beyond  $C$  to a point  $D$ :

$$B - C - D.$$

Proposition I.16 gives

$$\angle ABC < \angle ACD.$$

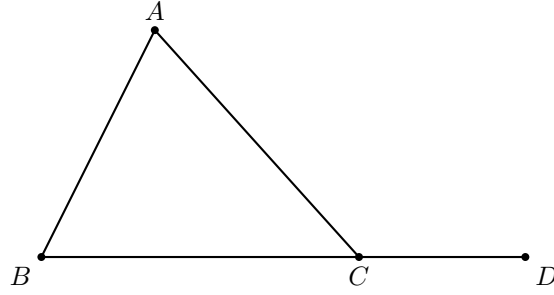


Figure 17.1: The classical extension used for the pair  $\angle ABC, \angle ACB$ :  $BC$  is produced beyond  $C$ .

Since  $CB$  and  $CD$  are opposite rays, the adjacent angles  $\angle ACB$  and  $\angle ACD$  form a supplementary pair. Proposition I.13 therefore gives the classical conclusion

$$\angle ABC + \angle ACB < 2R.$$

The formal reconstruction uses the same extension for two of the three pairwise conclusions. I.16 supplies both

$$\angle BAC < \angle ACD \quad \text{and} \quad \angle ABC < \angle ACD,$$

so the single witness  $D$  proves both

$$\angle BAC + \angle ACB < 2R$$

and

$$\angle ABC + \angle ACB < 2R.$$

For the remaining pair, the formal proof uses the symmetric extension at the other endpoint of the base. Produce  $CB$  beyond  $B$  to a point  $E$ :

$$C - B - E.$$

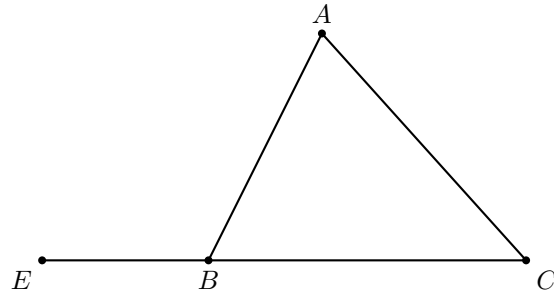


Figure 17.2: The second extension used by the Lean wrapper:  $CB$  is produced beyond  $B$ .

Applying I.16 to the reordered triangle  $ACB$  yields

$$\angle CAB < \angle ABE.$$

After reversing the first angle,

$$\angle BAC < \angle ABE,$$

and  $\angle ABE$  is supplementary to  $\angle ABC$ . Hence

$$\angle BAC + \angle ABC < 2R.$$

The two figures therefore have genuinely different roles. They are not duplicate views of the same proof: the first extension supplies the two pairs containing  $\angle ACB$ , while the second supplies the remaining pair.

## 17.3 Classical Proof

Euclid proves one representative case and then observes that the same argument applies to the other pairs.

Extend  $BC$  beyond  $C$  to  $D$ . By Proposition I.16,

$$\angle ABC < \angle ACD.$$

Add  $\angle ACB$  to both sides:

$$\angle ABC + \angle ACB < \angle ACD + \angle ACB.$$

By Proposition I.13, the adjacent angles  $\angle ACD$  and  $\angle ACB$  are together equal to two right angles. Hence

$$\angle ABC + \angle ACB < 2R.$$

The other two pairs follow in the same way after choosing the appropriate side extension.

The classical dependency pattern is therefore

I.16 exterior-angle inequality  $\longrightarrow$  add the remaining interior angle  $\longrightarrow$  I.13 supplementary pair  $\longrightarrow < 2R$ .

The formal reconstruction deliberately removes the middle arithmetic step.

## 17.4 Synthetic Representation of “Less Than Two Right Angles”

The central design choice is the predicate

`HilbertAnglesLessThanTwoRightAngles.`

For two angles  $\alpha$  and  $\beta$ , instead of introducing a sum  $\alpha + \beta$ , choose a supplementary angle  $\beta'$  to  $\beta$ . Then

$$\alpha + \beta < 2R$$

is represented by

$$\alpha < \beta'.$$

In the formal language, if the second angle is  $\angle CPD$ , a witness  $E$  is chosen with

$$D - P - E.$$

Then  $PE$  is the opposite ray to  $PD$ , so  $\angle CPE$  is supplementary to  $\angle CPD$ . The definition requires

$$\angle AOB < \angle CPE.$$

The Lean definition is:

```
def HilbertAnglesLessThanTwoRightAngles
  [HilbertIncidence Geo]
  (A O B C P D : Geo.Point) : Prop :=
  exists E : Geo.Point,
    Geo.Between D P E /\
    HilbertAngleLess Geo A O B C P E
```

This representation is entirely synthetic. It records only strict betweenness and strict comparison of angles. No real-valued measure, no degree notation, and no algebraic operation on angles is required.

## 17.5 Proof Architecture of the Reconstruction

The diagrammatic structure of the Lean proof is:

```

triangle ABC
|
+-- extend BC beyond C: B-C-D
|   |
|   +-- I.16 first
|   |   BAC < ACD
|   |   -> BAC + ACB < 2R
|   |
|   +-- I.16 second
|   |   ABC < ACD
|   |   -> ABC + ACB < 2R
|   |
+-- extend CB beyond B: C-B-E
|   |
|   +-- I.16 on triangle ACB
|   |   CAB < ABE
|   |   -> reverse CAB to BAC
|   |   -> BAC + ABC < 2R
|
three pairwise results
|
v
euclid_proposition_17

```

This is a case where two geometric diagrams are justified: the public theorem packages three conclusions, and the formal proof actually uses two different side extensions to obtain them.

The reconstruction is especially compact because Proposition I.16 already gives exactly the inequalities needed against exterior angles.

For the two pairs containing  $\angle ACB$ , extend  $BC$  beyond  $C$ :

$$B - C - D.$$

Then Proposition I.16 gives

$$\angle BAC < \angle ACD$$

and

$$\angle ABC < \angle ACD.$$

Since  $\angle ACD$  is the supplement of  $\angle ACB$ , these are precisely the synthetic forms of

$$\angle BAC + \angle ACB < 2R$$

and

$$\angle ABC + \angle ACB < 2R.$$

For the remaining pair, extend  $CB$  beyond  $B$ :

$$C - B - E.$$

Apply Proposition I.16 to the reordered triangle  $ACB$ . This gives

$$\angle CAB < \angle ABE.$$

Reversal of the first angle identifies  $\angle CAB$  with  $\angle BAC$ , while  $\angle ABE$  is supplementary to  $\angle ABC$ . Hence

$$\angle BAC + \angle ABC < 2R.$$

Schematically:

$$\begin{array}{c}
 \neg \text{Collinear}(A, B, C) \\
 \downarrow \\
 \text{extend } BC \text{ beyond } C \\
 \downarrow \\
 B - C - D \\
 \downarrow \\
 \text{I.16} \\
 \swarrow \quad \searrow \\
 \angle BAC < \angle ACD \quad \angle ABC < \angle ACD \\
 \downarrow \quad \downarrow \\
 \angle BAC + \angle ACB < 2R \quad \angle ABC + \angle ACB < 2R
 \end{array}$$

and separately

$$\begin{array}{c}
 \text{extend } CB \text{ beyond } B \\
 \downarrow \\
 C - B - E \\
 \downarrow \\
 \text{I.16 on triangle } ACB \\
 \downarrow \\
 \angle CAB < \angle ABE \\
 \downarrow \\
 \angle BAC + \angle ABC < 2R.
 \end{array}$$

## 17.6 Lean Formalization

### 17.6.1 The Pair $\angle BAC$ and $\angle ACB$

The first theorem extends  $BC$  through  $C$  and uses the first half of Proposition I.16.

```

theorem euclid_proposition_17_BAC_ACB
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not PrimCollinear Geo A B C) :
  HilbertAnglesLessThanTwoRightAngles
    Geo B A C A C B := by

```

```

have hBCA :
  not PrimCollinear Geo B C A := by
intro h

have hACB :
  PrimCollinear Geo A C B :=
  PrimCollinearSymm Geo B C A h

have hABC' :
  PrimCollinear Geo A B C :=
  PrimCollinearRotate Geo A C B hACB

exact hABC hABC'

have hBC : B != C :=
  hilbert_noncollinear_ne_first
  Geo B C A hBCA

rcases
  HilbertOrder.between_extension
  B C hBC with
  <D, hBCD>

refine <D, hBCD, ?_>

exact
  euclid_proposition_16_first
  Geo A B C D hABC hBCD

```

The witness  $D$  simultaneously records the supplementary configuration and serves as the exterior-angle target in I.16.

### 17.6.2 The Pair $\angle ABC$ and $\angle ACB$

The second theorem uses the same extension but invokes the second half of I.16.

```

theorem euclid_proposition_17_ABC_ACB
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not PrimCollinear Geo A B C) :
  HilbertAnglesLessThanTwoRightAngles
  Geo A B C A C B := by

have hBCA :
  not PrimCollinear Geo B C A := by
intro h

have hACB :
  PrimCollinear Geo A C B :=
  PrimCollinearSymm Geo B C A h

have hABC' :
  PrimCollinear Geo A B C :=

```



```

    PrimCollinearRotate Geo A C B hACB

    exact hABC hABC'

    have hBC : B != C :=
      hilbert_noncollinear_ne_first
        Geo B C A hBCA

    rcases
      HilbertOrder.between_extension
        B C hBC with
      <D, hBCD>

    refine <D, hBCD, ?_>

    exact
      euclid_proposition_16_second
        Geo A B C D hABC hBCD

```

Thus the two inequalities associated with the vertex  $C$  differ only in which conclusion of I.16 is selected.

### 17.6.3 The Pair $\angle BAC$ and $\angle ABC$

For the remaining pair, extend  $CB$  through  $B$  to  $E$  and apply I.16 to the triangle in the order  $ACB$ .

```

theorem euclid_proposition_17_BAC_ABC
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not PrimCollinear Geo A B C) :
  HilbertAnglesLessThanTwoRightAngles
    Geo B A C A B C := by

    have hCBA :
      not PrimCollinear Geo C B A := by
    intro h

    have hABC' :
      PrimCollinear Geo A B C :=
      PrimCollinearSymm Geo C B A h

    exact hABC hABC'

    have hCB : C != B :=
      hilbert_noncollinear_ne_first
        Geo C B A hCBA

    rcases
      HilbertOrder.between_extension
        C B hCB with
      <E, hCBE>

    refine <E, hCBE, ?_>

```

```

have hACB :
  not PrimCollinear Geo A C B := by
intro h

have hABC' :
  PrimCollinear Geo A B C :=
  PrimCollinearRotate Geo A C B h

exact hABC hABC'

have hLessCAB :
  HilbertAngleLess Geo C A B A B E :=
  euclid_proposition_16_first
  Geo A C B E hACB hCBE

rcases hLessCAB with
<hCABnc, hABEnc, X, hMeet, hAngle>

have hBACnc :
  not PrimCollinear Geo B A C := by
intro h
exact hABC
  (PrimCollinearSwap Geo B A C h)

have hAngle' :
  Geo.AngleCongruent B A C A B X :=
  (Geometry.Geo.angle_congruent_reverse_first
   Geo C A B A B X).mp hAngle

exact
<hBACnc,
  hABEnc,
  X,
  hMeet,
  hAngle'>

```

The only additional work here is normalization of the orientation of the first angle: I.16 returns  $\angle CAB < \angle ABE$ , whereas the final interface uses  $\angle BAC$ . The unordered-angle representation makes this a reversal operation rather than a new geometric argument.

#### 17.6.4 The Final Wrapper

The final proposition simply combines the three pairwise results.

```

theorem euclid_proposition_17
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not PrimCollinear Geo A B C) :
  HilbertAnglesLessThanTwoRightAngles Geo B A C A B C /\
  HilbertAnglesLessThanTwoRightAngles Geo B A C A C B /\
  HilbertAnglesLessThanTwoRightAngles Geo A B C A C B := by

have h1 :=

```

```

euclid_proposition_17_BAC_ABC
  Geo A B C hABC

have h2 :=
  euclid_proposition_17_BAC_ACB
  Geo A B C hABC

have h3 :=
  euclid_proposition_17_ABC_ACB
  Geo A B C hABC

exact <h1, h2, h3>

```

## 17.7 Relationship with Earlier Propositions

Proposition I.17 sits at an interesting point in the reconstruction because its classical statement appears to require angle arithmetic, while the existing synthetic interface allows that arithmetic to be avoided.

### 17.7.1 Proposition I.13

I.13 identifies two adjacent angles on a straight line as a supplementary pair. In the Book Zero reconstruction this is represented structurally by `BookZeroSupplement`, rather than by an equation involving angle sums.

For I.17 we use the same geometric idea but do not need to invoke the theorem I.13 explicitly. The witness produced by `between_extension` already records the opposite ray required to construct the supplementary angle.

### 17.7.2 Proposition I.14

I.14 introduced the useful design pattern of representing “equal to two right angles” by congruence with a supplementary angle. The I.17 predicate is the strict-inequality analogue of that pattern:

$$\alpha + \beta = 2R \quad \rightsquigarrow \quad \alpha \cong \text{supp}(\beta),$$

while

$$\alpha + \beta < 2R \quad \rightsquigarrow \quad \alpha < \text{supp}(\beta).$$

Thus I.17 extends an already established interface convention rather than introducing a new angle algebra.

### 17.7.3 Proposition I.16

I.16 is the decisive geometric input. Once a side is extended, it provides strict comparison of each remote interior angle with the exterior angle. Since that exterior angle is automatically supplementary to the adjacent interior angle, the conclusions of I.16 are already in the exact form required by `HilbertAnglesLessThanTwoRightAngles`.

This gives the particularly short dependency path

$$\text{betweenness extension} \longrightarrow \text{I.16} \longrightarrow \text{synthetic} < 2R.$$

## 17.8 Hilbert and Book Zero Components Used

### 17.8.1 HilbertAngleLess

The strict comparison relation introduced for I.16 remains the fundamental notion of angle inequality. It is synthetic: one angle is less than another when a congruent copy of the first occurs as a proper interior subangle of the second.

### 17.8.2 HilbertOrder.between\_extension

This construction supplies the points  $D$  and  $E$  on the required opposite rays:

$$B - C - D \quad \text{or} \quad C - B - E.$$

The witness is simultaneously the side extension used by I.16 and the supplementary-ray witness used by the definition of “less than two right angles.”

### 17.8.3 Collinearity Symmetry and Rotation

The three applications of I.16 require the triangle vertices in different orders. The small uses of `PrimCollinearSymm`, `PrimCollinearRotate`, and `PrimCollinearSwap` normalize noncollinearity hypotheses without adding any geometry.

### 17.8.4 Angle Reversal

In the third pair, Proposition I.16 naturally produces the angle  $\angle CAB$ , while the public form of I.17 uses  $\angle BAC$ . The theorem `angle_congruent_reverse_first` transfers the congruence across reversal of the first angle.

## 17.9 Formalization Notes

Proposition I.17 is a useful example of how a familiar numerical-looking statement is represented without introducing angle measures.

The phrase “two angles of a triangle are together less than two right angles” is not formalized by adding numerical angle magnitudes. Instead, the reconstruction uses an extension of one side and compares the relevant interior angle with an exterior angle. Proposition I.16 then provides the strict comparison.

Consequently the three conclusions of I.17 are proved by three geometric instances of the same mechanism rather than by a single algebraic inequality involving an angle-sum operation.

Schematically,

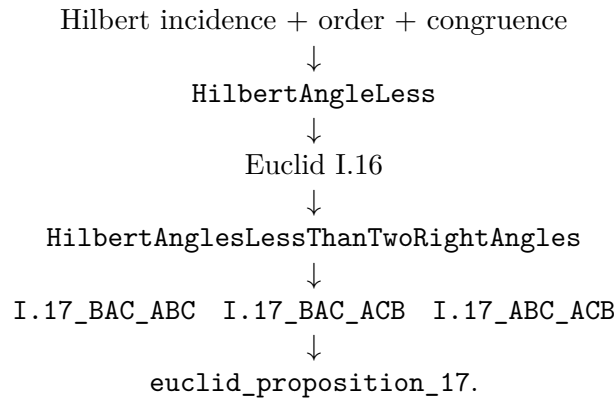
extend a side  $\longrightarrow$  straight-angle configuration  $\longrightarrow$  I.16  $\longrightarrow$  strict comparison  $\longrightarrow$  “two angles  $<$  two right angles”

The repetition in the Lean proof is therefore mathematically meaningful. Each pair of interior angles requires the appropriate extension, orientation of collinearity, and angle reversal. The formalization makes these orientation changes explicit rather than treating them as automatic consequences of a diagram.

This proposition also marks an important design choice in the project: there is still no need for a numerical angle measure or a general addition operation on angles. The classical statement can be recovered entirely from incidence, order, angle congruence, and strict angle comparison.

## 17.10 Dependency Summary

The formal dependency structure is



No additional geometric axiom is introduced. More importantly, no general arithmetic of angles is introduced merely to reproduce Euclid’s phrase “together less than two right angles.” The proposition is expressed entirely in the existing synthetic language of rays, betweenness, supplementary configurations, and strict angle comparison.

## 17.11 Conclusion

Proposition I.17 is a useful example of the difference between reproducing a classical proof literally and reconstructing its mathematical content over a synthetic interface.

Euclid’s paper proof uses an implicit monotonicity principle for angle addition: from  $\alpha < \beta$  one passes to  $\alpha + \gamma < \beta + \gamma$ . A literal formalization would therefore require an angle-sum operation and an order theory for such sums.

The Hilbert/Book Zero reconstruction avoids that additional layer. The statement “two angles are together less than two right angles” is encoded directly by comparison of one angle with the supplement of the other. Once this representation is chosen, Proposition I.16 supplies exactly the needed inequalities, and Proposition I.17 becomes a short structural consequence of side extension and exterior-angle comparison.

The completed Lean development compiles cleanly and contains no `sorry`.

## Chapter 18

# Proposition I.18

### 18.1 Introduction

Euclid's Proposition I.18 states that in a triangle the greater side is opposite the greater angle.

For a nondegenerate triangle  $ABC$ , if

$$AB < AC,$$

then

$$\angle ACB < \angle ABC.$$

The classical proof is short but structurally rich. A copy of the shorter side  $AB$  is laid off on the longer side  $AC$ . This produces a point  $D$  with

$$A - D - C \quad \text{and} \quad AD \cong AB.$$

The auxiliary triangle  $ABD$  is therefore isosceles. Proposition I.5 identifies its base angles, while Proposition I.16 compares one of these angles with an angle of the original triangle. Finally, the fact that an interior subangle is smaller than the whole angle completes the comparison.

The formal reconstruction follows this synthetic argument closely. It does not introduce numerical angle measures. Instead, it uses the strict comparison relation `HilbertAngleLess` already developed for Proposition I.16.

The reconstruction also isolates two general facts about strict angle comparison:

$$\text{hilbert\_interior\_angle\_less}$$

and

$$\text{hilbert\_angleLess\_trans.}$$

These belong naturally to the Hilbert interface rather than to Proposition I.18 itself.

### 18.2 The Classical Configuration

Let  $ABC$  be a nondegenerate triangle and suppose

$$AB < AC.$$

Choose  $D$  on  $AC$  so that

$$A - D - C \quad \text{and} \quad AD \cong AB.$$

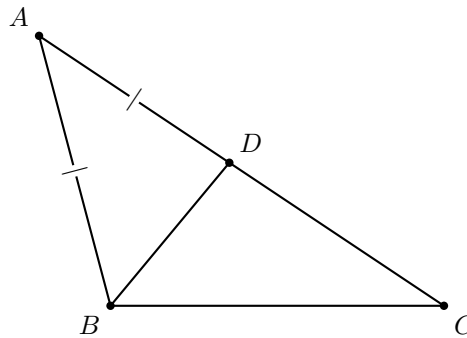


Figure 18.1: Classical configuration for Proposition I.18. The point  $D$  lies on  $AC$ , with  $AD \cong AB$ , and  $BD$  is joined.

The construction creates two auxiliary triangles.

First, triangle  $ABD$  is isosceles because

$$AB \cong AD.$$

Hence Proposition I.5 gives

$$\angle ABD \cong \angle ADB.$$

Second, consider triangle  $BCD$ . Since  $A - D - C$ , the ray  $DA$  is the continuation of  $DC$  through  $D$ . Thus  $\angle BDA$  is an exterior angle of triangle  $BCD$ .

Proposition I.16 therefore gives

$$\angle BCD < \angle BDA.$$

Because  $A, D, C$  are collinear with  $D$  between  $A$  and  $C$ , the rays  $CD$  and  $CA$  coincide. Hence

$$\angle BCD = \angle BCA.$$

Combining these observations gives

$$\angle BCA < \angle BDA \cong \angle ABD.$$

Finally, the ray  $BD$  lies inside  $\angle ABC$ , since it meets the segment  $AC$  at the interior point  $D$ . Therefore

$$\angle ABD < \angle ABC.$$

By transitivity,

$$\angle BCA < \angle ABC,$$

which is the desired conclusion.

### 18.3 Classical Proof

Assume that in triangle  $ABC$ ,

$$AC > AB.$$

Choose a point  $D$  on  $AC$  such that

$$AD = AB.$$

Since  $AC > AB$ , the point  $D$  lies strictly between  $A$  and  $C$ .

The triangle  $ABD$  is isosceles, so by Proposition I.5,

$$\angle ABD = \angle ADB.$$

Now  $\angle ADB$  is an exterior angle of triangle  $BCD$ . Proposition I.16 gives

$$\angle ADB > \angle BCD.$$

Since  $D$  lies on  $AC$ ,

$$\angle BCD = \angle BCA.$$

Consequently,

$$\angle ABD > \angle BCA.$$

But  $BD$  lies inside  $\angle ABC$ , so

$$\angle ABC > \angle ABD.$$

Therefore

$$\angle ABC > \angle BCA.$$

The classical dependency pattern is thus

$$\begin{array}{c}
 AB < AC \\
 \downarrow \\
 A - D - C, \quad AB \cong AD \\
 \downarrow \\
 \text{I.5} \\
 \downarrow \\
 \angle ABD \cong \angle ADB \\
 \downarrow \\
 \text{I.16 on } BCD \\
 \downarrow \\
 \angle BCA < \angle ADB \\
 \downarrow \\
 \angle BCA < \angle ABD \\
 \downarrow \\
 \angle ABD < \angle ABC \\
 \downarrow \\
 \angle BCA < \angle ABC.
 \end{array}$$



## 18.4 Strict Angle Comparison

The formal statement uses the synthetic relation

`HilbertAngleLess.`

No numerical measure of angles is introduced. Informally,

$$\alpha < \beta$$

means that a congruent copy of  $\alpha$  occurs as a proper interior subangle of  $\beta$ .

This relation was already required for Proposition I.16. Proposition I.18 reveals that two further structural properties are needed if strict angle comparison is to behave as a genuine order relation.

The first is that an interior subangle is smaller than the whole angle.

```
hilbert_interior_angle_less
```

Geometrically, if a ray from the vertex of  $\angle AOB$  meets the opposite side  $AB$  internally, then the angle cut off by that ray is strictly smaller than  $\angle AOB$ .

The second property is transitivity.

```
hilbert_angleLess_trans
```

It expresses the expected implication

$$\alpha < \beta \quad \text{and} \quad \beta < \gamma \quad \implies \quad \alpha < \gamma.$$

Both results are independent of Proposition I.18 and were therefore added to the general Hilbert interface.

## 18.5 Proof Architecture of the Reconstruction

The proof is carried by one auxiliary point and one chain of angle comparisons:

```
given: AB < AC
      |
choose D on AC with AD ~= AB
      |
triangle ABD is isosceles
      |
I.5: ABD ~= ADB
      |
I.16 in triangle DBC:
ACB < ADB
      |
D lies inside angle ABC:
```

```

ABD < ABC
  |
combine:
ACB < ADB ~= ABD < ABC
  |
therefore ACB < ABC

```

The single classical diagram is sufficient. The different inequalities are not different configurations; they are successive readings of the same configuration.

The formal proof separates naturally into five geometric stages.

### 18.5.1 Recovering the Interior Point

The hypothesis

$$AB < AC$$

is represented by `HilbertSegmentLess`.

Unpacking it produces a witness  $D$  satisfying

$$A - D - C$$

and

$$AB \cong AD.$$

Thus the first step of Euclid's construction is already contained in the definition of strict segment comparison.

### 18.5.2 The Isosceles Triangle

From

$$AB \cong AD$$

the existing theorem

$$\text{hilbert\_isosceles\_base\_angles}$$

gives

$$\angle ABD \cong \angle ADB.$$

This is precisely Proposition I.5 in the form required by the reconstruction.

### 18.5.3 The Exterior-Angle Comparison

Since

$$A - D - C,$$

we also have

$$C - D - A.$$

Apply Proposition I.16 to triangle  $BCD$  with  $DA$  as the extension of  $DC$ . This yields

$$\angle BCD < \angle BDA.$$

The ray from  $C$  through  $D$  is the same ray as the ray from  $C$  through  $A$ . Hence the first angle can be normalized:

$$\angle BCD = \angle BCA.$$

Thus

$$\angle BCA < \angle BDA.$$

Using the isosceles-angle congruence and the symmetry of the unordered angle representation gives

$$\angle BCA < \angle ABD.$$

### 18.5.4 The Interior Subangle

The point  $D$  lies on the segment  $AC$ . Consequently the ray  $BD$  meets the interior of that segment.

Therefore  $BD$  lies inside  $\angle ABC$ , and `hilbert_interior_angle_less` gives

$$\angle ABD < \angle ABC.$$

### 18.5.5 Transitivity

We now have

$$\angle BCA < \angle ABD$$

and

$$\angle ABD < \angle ABC.$$

Applying `hilbert_angleLess_trans` gives

$$\angle BCA < \angle ABC.$$

The final public statement writes the first angle as  $\angle ACB$ . Since the formal angle representation is insensitive to reversal of its two rays, this is a final orientation normalization rather than an additional geometric argument.

## 18.6 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_18
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not PrimCollinear Geo A B C)
  (hABltAC : HilbertSegmentLess Geo A B A C) :
  HilbertAngleLess Geo A C B A B C := by
```

The hypothesis `hABltAC` is immediately unpacked:

```
rcases hABltAC with
<D, hADC, hAB_AD>
```

Thus `hADC` records

$$A - D - C,$$

while `hAB_AD` records

$$AB \cong AD.$$

The isosceles part of the argument is then:

```
have hIso :
Geo.AngleCongruent A B D A D B :=
hilbert_isosceles_base_angles
Geo
A B D
hABD
hAB_AD
```

The exterior-angle step uses Proposition I.16:

```
have hBCD_BDA :
HilbertAngleLess Geo B C D B D A :=
euclid_proposition_16_second
Geo B C D A hBCD hCDA
```

Here

$$C - D - A$$

is obtained from the original betweenness relation

$$A - D - C.$$

The ray identity  $CD = CA$  is expressed by `HilbertSameRay`:

```
have hRayCDA :
HilbertSameRay Geo C D A :=
hilbert_sameRay_of_between
Geo C D A hCDA
```

This permits normalization of

$$\angle BCD$$

to

$$\angle BCA.$$

After transporting the right-hand angle across the isosceles congruence, the proof obtains

```
have hBCA_ABD :
HilbertAngleLess Geo B C A A B D := ...
```

which represents

$$\angle BCA < \angle ABD.$$

For the second comparison, the ray  $BD$  meets the segment  $AC$  at  $D$ :

```
have hInsideBD :
HilbertRayMeetsSegment Geo B D A C :=
<D, hADC, hRayBDD>
```

Hence

```
have hABD_ABC :
HilbertAngleLess Geo A B D A B C :=
hilbert_interior_angle_less
Geo
B D A C
hABC
hInsideBD
```

which is exactly

$$\angle ABD < \angle ABC.$$

Transitivity now gives

```
have hFinal :
HilbertAngleLess Geo B C A A B C :=
hilbert_angleLess_trans
Geo
B C A
A B D
A B C
hBCA_ABD
hABD_ABC
```

The final lines merely reverse the first angle from

$$\angle BCA$$

to

$$\angle ACB,$$

matching the public orientation of the proposition.

## 18.7 Relationship with Earlier Propositions

### 18.7.1 Proposition I.5

Proposition I.5 supplies the equality of the base angles in the auxiliary isosceles triangle  $ABD$ :

$$AB \cong AD \implies \angle ABD \cong \angle ADB.$$

This is the bridge between the comparison of sides and the comparison of angles.

### 18.7.2 Proposition I.16

Proposition I.16 supplies the strict inequality. Applied to triangle  $BCD$ , it compares the remote interior angle at  $C$  with the exterior angle at  $D$ :

$$\angle BCD < \angle BDA.$$

Thus I.18 depends essentially on the exterior-angle theorem proved two propositions earlier.

### 18.7.3 Proposition I.17

Proposition I.17 is not required by the proof of I.18. The formal dependency follows Euclid's direct argument through I.5 and I.16.

This is useful structurally: I.18 does not rely on an angle-sum theory developed for I.17. Both propositions instead reuse the synthetic strict-angle relation introduced for I.16.

## 18.8 Hilbert and Book Zero Components Used

### 18.8.1 HilbertSegmentLess

The hypothesis that one side is shorter than another already contains the point required by Euclid's construction. Unpacking

$$AB < AC$$

produces

$$A - D - C \quad \text{and} \quad AB \cong AD.$$

Thus no separate segment-layoff construction is required inside Proposition I.18.

### 18.8.2 hilbert\_isosceles\_base\_angles

This theorem supplies the I.5 step for the auxiliary triangle  $ABD$ .

### 18.8.3 HilbertSameRay

Same-ray machinery is used to recognize that, because  $A - D - C$ , the rays  $CD$  and  $CA$  determine the same side of the angle at  $C$ .

This turns the I.16 result about  $\angle BCD$  into the required comparison involving the original angle  $\angle BCA$ .

### 18.8.4 hilbert\_interior\_angle\_less

This new interface theorem expresses the elementary principle that a proper interior part of an angle is smaller than the whole angle.

In I.18 it gives

$$\angle ABD < \angle ABC$$

from the fact that  $BD$  meets  $AC$  internally.

### 18.8.5 hilbert\_angleLess\_trans

This new interface theorem provides transitivity of strict angle comparison.

It combines

$$\angle BCA < \angle ABD$$

and

$$\angle ABD < \angle ABC$$

into the final inequality.

## 18.9 Formalization Notes

Proposition I.18 is an important transition point in the reconstruction. The classical proof is short, but its formal version depends on several different notions of strict geometric order.

The hypothesis that one side is greater than another is represented by `HilbertSegmentLess`. It is not treated as a numerical inequality between lengths. Instead, the relation supplies the geometric data needed to recover an interior point on the longer side.

The proof then changes domains. Segment comparison produces a point configuration; the isosceles construction produces angle congruence; Proposition I.16 supplies a strict exterior-angle comparison; and `hilbert_interior_angle_less` supplies the second strict angle comparison required for transitivity.

Thus the decisive chain is

strict segment comparison  $\longrightarrow$  betweenness  $\longrightarrow$  isosceles angle congruence  $\longrightarrow$  I.16  $\longrightarrow$  interior-angle comparison

This explains why I.18 is more than a local theorem about one triangle. It is the first proposition in this part of Book I where the developed segment-order and angle-order interfaces interact in a sustained way.

It also prepares the formal structure of Proposition I.19. Once I.18 provides the direct implication from side order to angle order, I.19 can be proved as a converse by segment trichotomy and elimination of the two impossible alternatives.

## 18.10 Dependency Summary

The formal dependency structure is

$$\begin{array}{c}
\text{Hilbert incidence + order + congruence} \\
\downarrow \\
\text{HilbertSegmentLess} \quad \text{HilbertAngleLess} \\
\downarrow \\
A - D - C, \quad AB \cong AD \\
\downarrow \\
\text{hilbert\_isosceles\_base\_angles} \quad (\text{I.5}) \\
\downarrow \\
\angle ABD \cong \angle ADB \\
\downarrow \\
\text{Euclid I.16} \\
\downarrow \\
\angle BCA < \angle ABD \\
\downarrow \\
\text{hilbert\_interior\_angle\_less} \\
\downarrow \\
\angle ABD < \angle ABC \\
\downarrow \\
\text{hilbert\_angleLess\_trans} \\
\downarrow \\
\text{euclid\_proposition\_18.}
\end{array}$$

No new geometric axiom is introduced.

The two results added to the Hilbert interface are structural properties of the already existing strict-angle relation rather than assumptions specific to Proposition I.18.

## 18.11 Conclusion

Proposition I.18 is a particularly clear example of how Euclid's construction survives formal reconstruction.

The classical proof introduces a point  $D$  on the longer side, creates an isosceles triangle, applies the exterior-angle theorem, and finally observes that a part of an angle is smaller than the whole. The Lean proof follows exactly this geometry.

What changes is not the mathematical argument but the amount of structure that must be made explicit. The informal steps

$$\angle BCA < \angle ABD < \angle ABC$$

require a formal notion of strict angle comparison, a theorem asserting that an interior subangle is smaller than its containing angle, and a proof of transitivity.

Consequently, Proposition I.18 does more than add another theorem of Book I. It strengthens the reusable language of synthetic angle order. The new results `hilbert_interior_angle_less` and `hilbert_angleLess_trans` are available for subsequent propositions independently of I.18.

The completed Lean development compiles cleanly and contains no `sorry`.



## Chapter 19

# Proposition I.19

### 19.1 Introduction

Euclid's Proposition I.19 is the converse of Proposition I.18. In a nondegenerate triangle, the greater angle is opposite the greater side.

For a triangle  $ABC$ , if

$$\angle ACB < \angle ABC,$$

then

$$AB < AC.$$

The classical proof is indirect. One compares the two opposite sides  $AB$  and  $AC$ . Exactly one of the following three possibilities must occur:

$$AB \cong AC, \quad AB < AC, \quad AC < AB.$$

The first case contradicts Proposition I.5, because equal sides would imply congruent base angles. The third case contradicts Proposition I.18, because the greater side  $AB$  would force the opposite angle at  $C$  to be greater than the angle at  $B$ . Hence only

$$AB < AC$$

remains.

The formal reconstruction makes this trichotomy explicit through the new general theorem

$$\text{hilbert\_segment\_trichotomy.}$$

It also isolates a general irreflexivity theorem for strict angle comparison:

$$\text{hilbert\_angleLess\_irrefl.}$$

These two results are reusable structural components of the Hilbert layer rather than local devices specific to Proposition I.19.

## 19.2 The Classical Configuration

Let  $ABC$  be a nondegenerate triangle and suppose

$$\angle ACB < \angle ABC.$$

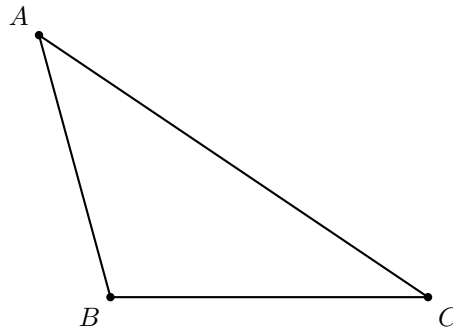


Figure 19.1: Classical configuration for Proposition I.19. The hypothesis is  $\angle ACB < \angle ABC$ ; the conclusion is  $AB < AC$ .

The theorem asserts that the side opposite the larger angle must be larger:

$$AB < AC.$$

No new auxiliary point is required. The proof is entirely comparative: it uses the trichotomy of the two sides and eliminates the two alternatives incompatible with the assumed angle inequality.

For this reason the single triangle is the right diagram. Adding separate pictures for the three side-comparison branches would falsely suggest three different geometric configurations. The branches are logical alternatives for the same fixed triangle.

## 19.3 Classical Proof

Assume

$$\angle ACB < \angle ABC.$$

Compare the two sides  $AB$  and  $AC$ .

If

$$AB = AC,$$

then triangle  $ABC$  is isosceles. Proposition I.5 gives

$$\angle ABC = \angle ACB,$$

contradicting the hypothesis that

$$\angle ACB < \angle ABC.$$

If instead

$$AC < AB,$$

then Proposition I.18, applied to triangle  $ACB$ , yields

$$\angle ABC < \angle ACB,$$

which again contradicts the original hypothesis.

The only remaining possibility is

$$AB < AC.$$

Thus the greater angle is subtended by the greater side.

The classical dependency pattern is therefore

$$\begin{array}{c}
 \angle ACB < \angle ABC \\
 \downarrow \\
 AB \cong AC \vee AB < AC \vee AC < AB \\
 \downarrow \\
 \begin{array}{ccc}
 AB \cong AC & AB < AC & AC < AB \\
 \downarrow & & \downarrow \\
 \text{I.5 contradiction} & \text{desired case} & \text{I.18 contradiction}
 \end{array}
 \end{array}$$

## 19.4 Segment Trichotomy

The key structural theorem introduced for the formal proof is

$$\text{hilbert\_segment\_trichotomy.}$$

Its conclusion has the form

$$AB \cong CD \vee AB < CD \vee CD < AB.$$

The proof does not introduce a new axiom. It is derived from segment construction and the order theory of collinear points.

A copy of  $AB$  is laid off on the ray  $CD$ , producing a point  $X$  such that

$$CX \cong AB.$$

If  $X = D$ , then

$$AB \cong CD.$$

Otherwise,  $C, D, X$  are distinct collinear points. The Hilbert betweenness trichotomy gives three possibilities. The case

$$D - C - X$$

is incompatible with the fact that  $X$  lies on the ray  $CD$ . The remaining two orders give respectively

$$CD < CX$$

and

$$CX < CD.$$

Transporting these inequalities through

$$CX \cong AB$$

yields the required segment trichotomy.

Thus the theorem packages a general completeness property of strict segment comparison.

This helper does have its own useful diagrammatic content, but it belongs to the Hilbert interface rather than to Proposition I.19 itself. A copy of one segment is laid off on the ray of the other, and the order of the copied endpoint relative to the original endpoint determines the three alternatives. For the I.19 documentation, the ASCII dependency diagram above is enough; a second geometric figure would mostly duplicate the meaning of `hilbert_segment_trichotomy`.

## 19.5 Irreflexivity of Strict Angle Comparison

The second structural theorem used in I.19 is

`hilbert_angleLess_irrefl.`

It states

$$\neg(\alpha < \alpha).$$

In the synthetic definition of `HilbertAngleLess`, the relation

$$\alpha < \beta$$

means that a congruent copy of  $\alpha$  occurs as a proper interior subangle of  $\beta$ .

Therefore

$$\alpha < \alpha$$

would imply that an angle is congruent to one of its own proper interior subangles. This is excluded by the previously proved theorem

`hilbert_interior_subangle_not_congruent_whole.`

The irreflexivity result provides exactly the contradiction mechanism needed in both non-target branches of Proposition I.19.

## 19.6 Proof Architecture of the Reconstruction

The Lean proof divides naturally into three stages.

```

given: ACB < ABC
      |
      v
segment trichotomy for AB and AC
      |
      +-----+-----+
      |       |       |
      v       v       v
AB~AC AB<AC AC<AB
      |       |       |
      |       |       +---> I.18

```

```

|      |      -> ABC < ACB
|      |      -> contradiction
|      |
|      +-----> goal
|
+--> I.5
      -> ABC ~= ACB
      -> contradiction

```

therefore  $AB < AC$

The geometry itself does not branch. What branches is the order relation between the two opposite sides.

### 19.6.1 Nondegeneracy of the Compared Side

From the noncollinearity hypothesis

$$\neg \text{Collinear}(A, B, C)$$

the proof first derives

$$A \neq C.$$

This is the nondegeneracy condition required to apply segment construction in `hilbert_segment_trichotomy`.

### 19.6.2 Apply Segment Trichotomy

The theorem

`hilbert_segment_trichotomy`

is applied to the segments  $AB$  and  $AC$ .

This produces three branches:

$$AB \cong AC, \quad AB < AC, \quad AC < AB.$$

The middle branch is exactly the desired conclusion.

### 19.6.3 Eliminate the Two Impossible Branches

In the congruent-side branch, Proposition I.5 gives

$$\angle ABC \cong \angle ACB.$$

The assumed strict inequality

$$\angle ACB < \angle ABC$$

is transported across that congruence, producing

$$\angle ACB < \angle ACB.$$

This contradicts `hilbert_angleLess_irrefl`.

In the branch

$$AC < AB,$$

Proposition I.18 applied to triangle  $ACB$  gives

$$\angle ABC < \angle ACB.$$

Combining this with the original inequality by transitivity yields

$$\angle ACB < \angle ACB,$$

again contradicting irreflexivity.

Only

$$AB < AC$$

remains.

## 19.7 Lean Formalization

The public theorem is:

```
theorem euclid_proposition_19
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not PrimCollinear Geo A B C)
  (hAngle :
    HilbertAngleLess Geo A C B A B C) :
  HilbertSegmentLess Geo A B A C := by
```

The proof first derives the nondegeneracy of  $AC$ :

```
have hAC : A != C :=
  hilbert_noncollinear_ne_first
    Geo A C B
  (by
    intro h
    exact hABC
      (PrimCollinearRotate Geo A C B h))
```

Then segment trichotomy is applied:

```
rcases
  hilbert_segment_trichotomy
    Geo
    A B
    A C
    hAC with
  hEq | hABltAC | hACltAB
```

The middle branch closes immediately:

```
$\cdot$ exact hABltAC
```

For the congruent-side branch, Proposition I.5 gives:

```
have hIso :
  Geo.AngleCongruent A B C A C B :=
  hilbert_isosceles_base_angles
  Geo
  A B C
  hABC
  hEq
```

The strict inequality is transported to the congruent target angle:

```
have hSame :
  HilbertAngleLess Geo A C B A C B :=
  hilbert_angleLess_transport_right
  Geo
  A C B
  A B C
  A C B
  hAngle
  hACB
  hIso
```

This contradicts irreflexivity:

```
exact
  False.elim
    ((hilbert_angleLess_irrefl
      Geo A C B)
      hSame)
```

In the final branch, Proposition I.18 gives the opposite strict inequality:

```
have hOpp :
  HilbertAngleLess Geo A B C A C B :=
  euclid_proposition_18
  Geo
  A C B
  hACB
  hACltAB
```

Transitivity then creates an impossible strict self-comparison:

```
have hCycle :
  HilbertAngleLess Geo A C B A C B :=
  hilbert_angleLess_trans
```

```

Geo
A C B
A B C
A C B
hAngle
hOpp

```

and finally:

```

exact
  False.elim
    ((hilbert_angleLess_irrefl
      Geo A C B)
     hCycle)

```

## 19.8 Relationship with Earlier Propositions

### 19.8.1 Proposition I.5

I.5 is used to eliminate the equality case. If

$$AB \cong AC,$$

then the base angles are congruent:

$$\angle ABC \cong \angle ACB.$$

This is incompatible with the assumed strict angle inequality.

### 19.8.2 Proposition I.18

I.18 is used to eliminate the reversed side inequality. If

$$AC < AB,$$

then applying I.18 to triangle  $ACB$  gives

$$\angle ABC < \angle ACB.$$

Together with the hypothesis

$$\angle ACB < \angle ABC,$$

this produces a strict cycle and therefore a contradiction.

### 19.8.3 Logical Relation between I.18 and I.19

The two propositions form a converse pair:

$$AB < AC \implies \angle ACB < \angle ABC$$



for I.18, and

$$\angle ACB < \angle ABC \implies AB < AC$$

for I.19.

Together they establish the equivalence between ordering of opposite sides and ordering of opposite angles in a nondegenerate triangle.

## 19.9 Hilbert and Book Zero Components Used

### 19.9.1 `hilbert_segment_trichotomy`

This theorem provides the complete three-way comparison of two segments and is the main new structural component required by I.19.

### 19.9.2 `hilbert_angleLess_irrefl`

This theorem excludes strict self-comparison of an angle. It is used in both contradictory branches of the proof.

### 19.9.3 `hilbert_isosceles_base_angles`

This is the formal version of Proposition I.5 needed to eliminate the equality case.

### 19.9.4 `hilbert_angleLess_transport_right`

This theorem transports a strict angle comparison across congruence of the containing angle. It converts the equality branch into an impossible self-comparison.

### 19.9.5 `hilbert_angleLess_trans`

Transitivity combines the original strict inequality with the opposite inequality supplied by I.18.

## 19.10 Formalization Notes

Proposition I.19 is formally much shorter than Proposition I.18, but its proof is logically revealing.

The argument does not construct a new geometric configuration. Instead, it performs a case analysis on the two sides whose order is to be determined. Segment trichotomy gives three possibilities. Two of them are incompatible with the assumed strict angle inequality.

The equality branch is eliminated by the isosceles-triangle theorem: equal sides force the corresponding angles to be congruent, contradicting strict angle comparison.

The reverse-inequality branch is eliminated by Proposition I.18. If the supposedly smaller side were actually larger, I.18 would force the opposite angle inequality. Together with the hypothesis this would produce a strict angle inequality from an angle to itself, contradicting irreflexivity.

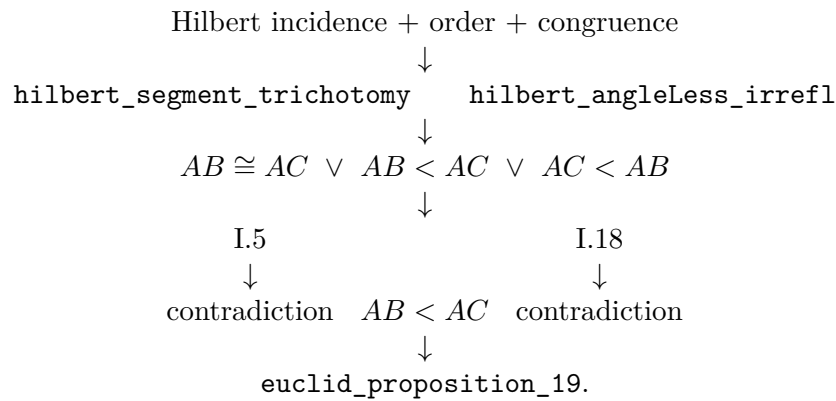
Thus the proof is a characteristic formal converse argument:

$$\text{trichotomy} + \text{direct theorem I.18} + \text{irreflexivity} \implies \text{I.19.}$$

This organization is useful beyond Proposition I.19. It shows that converse comparison theorems need not be proved by reproducing the geometric construction of the direct theorem. Once strict order has trichotomy and irreflexivity, the converse can often be obtained by eliminating the impossible alternatives.

## 19.11 Dependency Summary

The formal dependency structure is



No new geometric axiom is introduced.

## 19.12 Conclusion

Proposition I.19 is formally shorter than I.18, but it exposes an important structural issue: the proof requires a complete comparison principle for segments.

Once the theorem `hilbert_segment_trichotomy` is available, the Euclidean argument becomes exactly the expected three-case proof. Equal sides are ruled out by I.5, the reversed inequality is ruled out by I.18, and the remaining case is the desired conclusion.

The proof also clarifies the role of strict angle order. Instead of introducing an explicit asymmetry theorem, the development uses the more fundamental statement `hilbert_angleLess_irrefl` together with transitivity. This keeps the order theory modular and avoids unnecessary duplication.

Thus I.19 strengthens the reusable Hilbert layer in two directions: completeness of segment comparison and irreflexivity of strict angle comparison. The Euclidean theorem itself then becomes a concise application of those general principles.

The completed Lean development compiles cleanly and contains no `sorry`.

## Chapter 20

# Proposition I.20

### 20.1 Introduction

Euclid's Proposition I.20 is the triangle inequality in purely synthetic form. For a nondegenerate triangle  $ABC$ , Euclid proves that any two sides taken together are greater than the third. For one of the three cyclic cases the statement is

$$BA + AC > BC.$$

The Hilbert reconstruction does not introduce a numerical length function and does not define an arithmetic sum of real numbers. Instead, the sum  $BA + AC$  is represented geometrically. Extend  $BA$  beyond  $A$  to a point  $D$  such that

$$B - A - D \quad \text{and} \quad AD \cong AC.$$

Then the whole segment  $BD$  represents the concatenation of  $BA$  with a copy of  $AC$ . The required inequality becomes

$$BC < BD.$$

Thus the formal theorem asserts the existence of such a point  $D$  together with the strict segment comparison  $BC < BD$ .

The completed proof remains entirely inside neutral Hilbert geometry. It uses incidence, order, congruence, Proposition I.5, the synthetic angle-order relation, and Proposition I.19. No parallel postulate, numerical angle measure, or numerical segment length is used.

### 20.2 The Classical Configuration

Let  $ABC$  be a nondegenerate triangle. Extend  $BA$  beyond  $A$  to a point  $D$  and choose  $D$  so that

$$AD \cong AC.$$

The construction records

$$B - A - D, \quad AD \cong AC.$$

The target is

$$BC < BD.$$

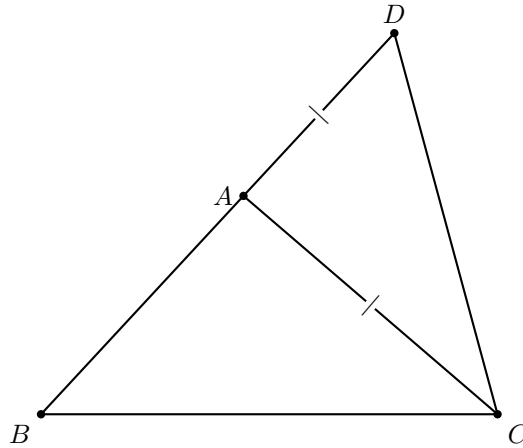


Figure 20.1: Classical configuration for Proposition I.20. The side  $BA$  is produced beyond  $A$  to  $D$ , with  $AD \cong AC$ .

Geometrically, this is exactly the inequality

$$BC < BA + AC.$$

### 20.3 Classical Proof

After extending  $BA$  to  $D$  so that  $AD = AC$ , join  $CD$ .

The auxiliary triangle  $ADC$  is isosceles. Proposition I.5 therefore gives

$$\angle ADC = \angle ACD.$$

Since  $A$  lies between  $B$  and  $D$ , the ray  $CA$  lies inside the angle  $BCD$ . Hence

$$\angle ACD < \angle BCD.$$

Combining this with the isosceles-triangle equality gives

$$\angle BDC < \angle BCD.$$

Now apply Proposition I.19 to triangle  $BCD$ . The greater angle is opposite the greater side, so

$$BC < BD.$$

Finally, because  $B - A - D$  and  $AD = AC$ , the segment  $BD$  is the geometric concatenation of  $BA$  and  $AC$ . Therefore

$$BA + AC > BC.$$

The classical dependency pattern is

$$\begin{array}{c}
 B - A - D, AD \cong AC \\
 \downarrow \\
 \angle ADC \cong \angle ACD \quad (\text{I.5}) \\
 \downarrow \\
 \angle BDC < \angle BCD \\
 \downarrow \quad (\text{I.19}) \\
 BC < BD.
 \end{array}$$

The single diagram above already contains every geometric ingredient needed for this proof. There is no case split to visualize. In particular, the important graphical information is not merely the shape of triangle  $BCD$ , but the collinear order  $B - A - D$  together with the marked equality  $AD \cong AC$ . These two facts explain why  $BD$  represents the synthetic sum  $BA + AC$ .

Wilkins also records an ancient alternative proof, attributed by Proclus to Heron or Porphyry, which bisects  $\angle BAC$  and uses I.16 and I.19. That proof has a genuinely different diagram, but it is not the proof formalized here, so it is not added to the main reconstruction.

## 20.4 Synthetic Form of the Triangle Inequality

The formal reconstruction deliberately avoids introducing an arithmetic addition operation on segments. The public result is expressed by an extension point  $D$ :

$$\exists D (B - A - D \wedge AD \cong AC \wedge BC < BD).$$

This is the natural Hilbert form of Euclid’s phrase “two sides taken together are greater than the remaining one.”

The representation has two advantages. First, it remains purely geometric. Second, it makes explicit which part of the statement belongs to order and congruence geometry: the “sum” is a concatenated segment, not an externally imposed numerical quantity.

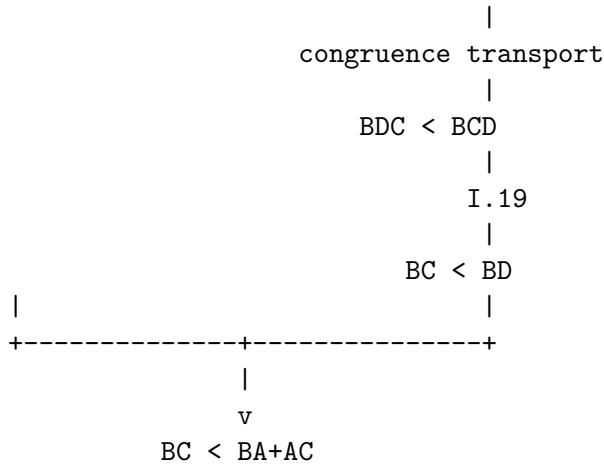
## 20.5 Proof Architecture of the Reconstruction

The formal proof follows Euclid’s geometric strategy closely, but it separates the argument into reusable logical stages. An extension point turns a sum of two sides into a single segment; an interior-angle comparison and an isosceles-triangle transport produce the angle inequality required by Proposition I.19; finally the resulting strict segment comparison is transported back to the original sides.

```

triangle ABC
|
extend BA beyond A
|
construct D with B-A-D and AD ~= AC
|
+-----+
|                                     |
v                                     v
BD represents BA+AC           triangle ACD is isosceles
|
I.5
|
ADC ~= ACD
|
CA lies inside angle DCB
|
ACD < DCB

```



Unlike I.21–I.24, no second geometric picture is needed here: the formal architecture is a refinement of the same Euclidean configuration rather than a different case structure or a stronger primitive construction.

### 20.5.1 Construction of the Extension Point

The point  $D$  is constructed in two stages.

First, Hilbert’s order axiom extends the line beyond  $A$ :

$$B - A - R.$$

In Lean this is obtained from

`HilbertOrder.between_extension.`

Second, a copy of  $AC$  is laid off from  $A$  on the ray  $AR$  using

`HilbertCongruence.segment_construction.`

This produces a point  $D$  satisfying

$$AD \cong AC$$

and placing  $D$  on the same ray from  $A$  as  $R$ .

The theorem

`hilbert_between_transport_sameRays`

then transports the original betweenness relation  $B - A - R$  to

$$B - A - D.$$

Thus the geometric “sum” segment  $BD$  is obtained entirely from existing order and congruence infrastructure.

### 20.5.2 The Interior-Angle Step

The key order-theoretic observation is that the ray  $CA$  meets the open segment  $DB$  at the point  $A$  itself. Since

$$D - A - B,$$

the witness for

HilbertRayMeetsSegment Geo C A D B

is simply  $A$ .

The strict angle comparison is then introduced directly with

hilbert\_angleLess\_intro.

The first angle is  $\angle ACD$ . The corresponding proper subangle of  $\angle DCB$  is represented by  $\angle DCA$ , which is the same unoriented angle as  $\angle ACD$ . The representation-level symmetry of angles is supplied by

angle\_congruent\_reverse\_second

together with reflexivity of angle congruence.

This yields

$$\angle ACD < \angle DCB.$$

No auxiliary half-plane or perpendicular construction is required. The entire strict comparison follows from the ray meeting the opposite segment.

### 20.5.3 Transport Through the Isosceles Triangle

From

$$AD \cong AC$$

and the noncollinearity of  $A, D, C$ , Proposition I.5 is available in the form

hilbert\_isosceles\_base\_angles.

It gives

$$\angle ADC \cong \angle ACD.$$

For Proposition I.20 we also introduced the reusable theorem

hilbert\_angleLess\_transport\_left.

It expresses invariance of strict angle comparison under replacing the smaller angle by a congruent angle.

Applied here, it transports

$$\angle ACD < \angle DCB$$

across

$$\angle ADC \cong \angle ACD,$$

producing

$$\angle ADC < \angle DCB.$$

Since  $B - A - D$ , the rays  $DA$  and  $DB$  coincide. The theorem

`hilbert_angle_eq_of_sameRay_first`

therefore identifies the angle  $\angle ADC$  with  $\angle BDC$ . Hence

$$\angle BDC < \angle DCB.$$

Finally, reversal of the two boundary rays at  $C$  identifies

$$\angle DCB \cong \angle BCD.$$

The existing theorem

`hilbert_angleLess_transport_right`

then gives

$$\angle BDC < \angle BCD.$$

#### 20.5.4 Application of Proposition I.19

At this point the proof has produced exactly the hypothesis needed for Proposition I.19 in triangle  $BCD$ :

$$\angle BDC < \angle BCD.$$

Applying

`euclid_proposition_19`

yields

$$BC < BD.$$

This is the synthetic triangle inequality for the chosen pair of sides. The other two cyclic cases follow by renaming the vertices.

## 20.6 Lean Formalization

The public theorem is:

```
theorem euclid_proposition_20
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not PrimCollinear Geo A B C) :
```



```

exists D : Geo.Point,
  Geo.Between B A D /\
  Geo.Congruent A D A C /\
  HilbertSegmentLess Geo B C B D := by

```

The first stage derives  $A \neq B$  and extends  $BA$  beyond  $A$ :

```

have hAB : A != B :=
  hilbert_noncollinear_ne_first
  Geo A B C hABC

have hBA : B != A :=
  hAB.symm

rcases
  HilbertOrder.between_extension
    B A hBA with
  <R, hBAR>

```

A copy of  $AC$  is then laid off on the ray  $AR$ :

```

have hAR : A != R :=
  (HilbertOrder.between_incidence
    B A R hBAR).2.1

rcases
  HilbertCongruence.segment_construction
    (Geo := Geo)
    A C
    A R
    hAR with
  <D, hRayARD, hAD_AC>

```

Betweenness is transported to the actual constructed point  $D$ :

```

have hRayABA :
  HilbertSameRay Geo A B B :=
  hilbert_sameRay_refl
  Geo A B hBA

have hBAD :
  Geo.Between B A D :=
  hilbert_between_transport_sameRays
  Geo
  B A R
  B D
  hBAR
  hRayABA
  hRayARD

```

The theorem then packages the construction and continues with the strict segment comparison:

```
refine <D, hBAD, hAD_AC, ?_>
```

The interior ray is represented directly by the point  $A$ :

```
have hDAB :
  Geo.Between D A B :=
  (HilbertOrder.between_incidence
   B A D hBAD).2.2.2.2

have hRayCAA :
  HilbertSameRay Geo C A A :=
  hilbert_sameRay_refl
  Geo C A hAC

have hInside :
  HilbertRayMeetsSegment Geo C A D B :=
  <A, hDAB, hRayCAA>
```

Angle reversal is obtained through the representation-level congruence laws:

```
have hAngleRefl :
  Geo.AngleCongruent A C D A C D :=
  Geometry.Geo.angle_congruent_reflexive
  Geo A C D

have hAngleSwap :
  Geo.AngleCongruent A C D D C A :=
  (Geo.angle_congruent_reverse_second
   A C D
   A C D).mp
  hAngleRefl
```

This gives the strict subangle relation:

```
have hLessACD_DCB :
  HilbertAngleLess Geo A C D D C B :=
  hilbert_angleLess_intro
  Geo
  A C D
  D C B
  A
  hACD
  hDCB
  hInside
  hAngleSwap
```

Proposition I.5 and left transport give the corresponding comparison with the base angle at  $D$ :

```
have hIso :
  Geo.AngleCongruent A D C A C D :=
  hilbert_isosceles_base_angles
  Geo
```

```

A D C
hADC
hAD_AC

have hLessADC_DCB :
  HilbertAngleLess Geo A D C D C B :=
  hilbert_angleLess_transport_left
  Geo
  A C D
  A D C
  D C B
  hLessACD_DCB
  hADC
  hIso

```

The ray  $DA$  is then replaced by the same ray  $DB$ :

```

have hRayDAB :
  HilbertSameRay Geo D A B :=
  hilbert_sameRay_of_between
  Geo D A B hDAB

have hRayDBA :
  HilbertSameRay Geo D B A :=
  hilbert_sameRay_symm
  Geo D A B hRayDAB

have hBDC_ACD :
  Geo.AngleCongruent B D C A C D := by
  unfold Geometry.Geo.AngleCongruent at hIso |-
  rw [hilbert_angle_eq_of_sameRay_first
    Geo D B A C hRayDBA]
  exact hIso

```

Using left transport again gives

```

have hLessBDC_DCB :
  HilbertAngleLess Geo B D C D C B :=
  hilbert_angleLess_transport_left
  Geo
  A C D
  B D C
  D C B
  hLessACD_DCB
  hBDC
  hBDC_ACD

```

Finally, the right-hand angle is reversed and Proposition I.19 closes the proof:

```

have hAngleRef1DCB :
  Geo.AngleCongruent D C B D C B :=
  Geometry.Geo.angle_congruent_reflexive
  Geo D C B

```

```

have hDCB_BCD :
  Geo.AngleCongruent D C B B C D :=
  (Geo.angle_congruent_reverse_second
   D C B
   D C B).mp
  hAngleRef1DCB

have hLessBDC_BCD :
  HilbertAngleLess Geo B D C B C D :=
  hilbert_angleLess_transport_right
    Geo
    B D C
    D C B
    B C D
    hLessBDC_DCB
    hBCD
    hDCB_BCD

exact
  euclid_proposition_19
    Geo
    B C D
    hBCD
    hLessBDC_BCD

```

## 20.7 Relationship with Earlier Propositions

### 20.7.1 Proposition I.5

The construction creates an isosceles triangle. Its base-angle congruence is the mechanism that transports the relevant angle from the auxiliary configuration back toward the original triangle.

### 20.7.2 Proposition I.16

The exterior-angle theorem supplies the strict angle comparison needed inside the auxiliary triangle. This is the order-theoretic part of the argument: the diagram alone is not used to infer that one angle is larger than another.

### 20.7.3 Proposition I.19

Proposition I.19 converts the strict angle comparison into a strict comparison of the opposite sides. In this sense I.20 is not an independent segment-order argument: its decisive inequality is obtained through the angle-order theory developed immediately before it.

### 20.7.4 Transition to Proposition I.21

Proposition I.20 proves the triangle inequality by representing the sum of two sides as a single constructed segment. Proposition I.21 develops this same synthetic segment-arithmetic viewpoint

further. Thus I.20 is the first clear bridge from the angle-comparison block I.16–I.19 to the more explicit segment arithmetic used in I.21 and I.22.

## 20.8 Hilbert and Book Zero Components Used

Proposition I.20 required one new general theorem in the Hilbert interface:

`hilbert_angleLess_transport_left.`

Its role is symmetric to the previously available

`hilbert_angleLess_transport_right.`

Together they express that strict synthetic angle comparison is invariant under congruent replacement of either compared angle. This is not specific to the triangle inequality and is expected to be reusable in later propositions.

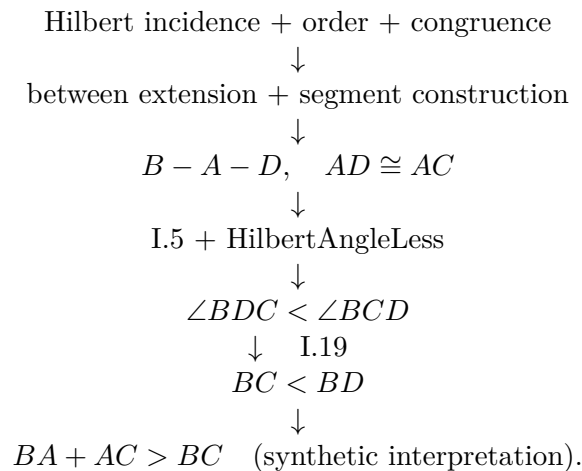
The proof also reuses the existing construction pattern

`between_extension`  $\longrightarrow$  `segment_construction`  $\longrightarrow$  `between_transport_sameRays`,

which already occurs elsewhere in the Hilbert development.

## 20.9 Dependency Summary

The formal dependency structure of Proposition I.20 can be summarized as



No Euclidean parallel axiom appears in this chain. Proposition I.20 is therefore a theorem of neutral geometry in the present Hilbert reconstruction.

## 20.10 Conclusion

Proposition I.20 is the first place in this sequence where Euclid's language of adding lengths is naturally reconstructed without introducing numerical length. The formal statement replaces

the expression  $BA + AC$  by an explicit extension segment  $BD$  satisfying

$$B - A - D, \quad AD \cong AC.$$

The rest of the argument is purely synthetic. Proposition I.5 converts the copied side into an angle congruence, the interior-ray definition produces a strict angle inequality, congruence transport normalizes the two angles, and Proposition I.19 converts the angle inequality back into the desired side inequality.

The resulting Lean proof therefore reproduces the classical geometry while exposing its exact logical structure. No new axiom is introduced, and the only new general infrastructure is the left-transport theorem for strict angle comparison.

## Chapter 21

# Proposition I.21

### 21.1 Introduction

Euclid's Proposition I.21 concerns two straight lines drawn from the ends of one side of a triangle to a point inside the triangle.

Let  $ABC$  be a nondegenerate triangle. Let  $D$  be an interior point, and let the line  $BD$  meet the side  $AC$  at  $E$ , so that

$$A - E - C \quad \text{and} \quad B - D - E.$$

Euclid asserts simultaneously that

$$BD + DC < BA + AC$$

and

$$\angle BAC < \angle BDC.$$

The proposition is therefore the first result in this part of Book I whose reconstruction naturally separates into two substantial comparison arguments: one for segments and one for angles.

The formal reconstruction preserves this separation. The segment inequality is obtained by combining Proposition I.20 with a synthetic “addition of unequal segments” argument. The angle inequality is proved independently, using Proposition I.16 and the strict angle comparison machinery developed in the preceding propositions.

A particularly important feature of the formalization is that no numerical length function is introduced. Expressions such as  $BA + AC$  and  $BD + DC$  are represented by ordinary constructed segments.

### 21.2 The Classical Configuration

Let  $ABC$  be a nondegenerate triangle. Choose a point  $D$  inside the triangle and let the line  $BD$  meet  $AC$  at  $E$ .

Thus

$$A - E - C, \quad B - D - E.$$

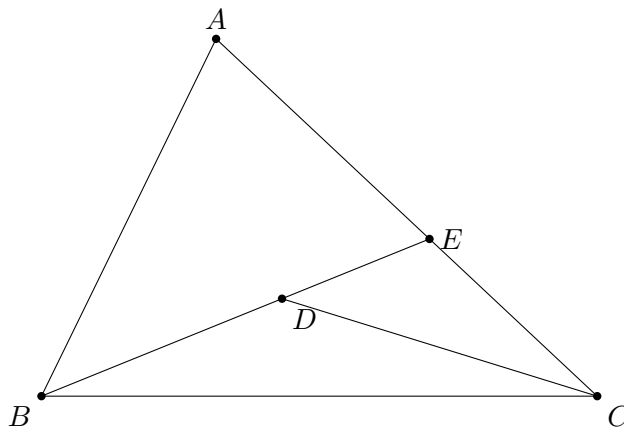


Figure 21.1: Classical configuration for Proposition I.21. The point  $D$  lies inside triangle  $ABC$ ; the line  $BD$  meets  $AC$  at  $E$ .

For the segment inequality, Euclid first compares the broken line  $BA + AE$  with  $BE$  by applying Proposition I.20 to triangle  $BAE$ . The remaining pieces  $EC$  and the corresponding copied segment are then added to obtain the comparison with  $BA + AC$ .

For the angle inequality, the point  $E$  allows the exterior-angle theorem to be applied to appropriate auxiliary triangles. The resulting strict angle comparisons are transported along the collinear rays determined by  $A, E, C$  and  $B, D, E$ .

The formal proof makes these two uses of the same configuration explicit.

Wilkins's presentation is especially useful here because the same diagram supports two logically independent chains. For segments, Proposition I.20 gives

$$DC < DE + EC, \quad DE < DA + AE,$$

and therefore

$$BD + DC < BD + DE + EC = BE + EC < BA + AE + EC = BA + AC.$$

For angles, the same point  $E$  exposes a clean two-step exterior-angle chain:

$$\angle BDC > \angle BEC > \angle BAC.$$

The single classical diagram above therefore serves both parts of the proof. There is no mathematical gain in repeating the same triangle a second time. I.21 is instead a useful example of one configuration supporting two distinct arguments: a segment-sum comparison and an angle-order comparison.

### 21.3 Classical Proof

Since  $D$  lies inside triangle  $ABC$ , the line  $BD$  meets  $AC$  at an interior point  $E$ .

Apply Proposition I.20 to triangle  $ABE$ . It gives

$$BE < BA + AE.$$



Add  $EC$  to the two sides in the geometric sense. Since

$$AE + EC = AC,$$

we obtain

$$BE + EC < BA + AC.$$

Now apply Proposition I.20 to triangle  $CDE$ . Since  $B - D - E$ ,

$$BD + DE = BE.$$

Proposition I.20 gives

$$DC + DE > CE,$$

or equivalently

$$CE < DC + DE.$$

Combining the appropriate segment comparisons in the classical configuration yields

$$BD + DC < BA + AC.$$

For the angle part, the exterior-angle theorem I.16 is applied to the triangles determined by the interior point and the two transversals. It yields strict comparisons which, after identifying the collinear rays, give

$$\angle BAC < \angle BDC.$$

Thus both assertions of I.21 follow:

$$BD + DC < BA + AC \quad \text{and} \quad \angle BAC < \angle BDC.$$

The classical proof is extremely compact because segment addition, subtraction, and replacement of collinear rays are read directly from the diagram. These are precisely the operations that must be made explicit in the formal reconstruction.

## 21.4 From Euclid's Diagram to the Formal Argument

The classical proof compresses several different kinds of reasoning into a single diagram. The formal reconstruction separates them.

First, statements such as “the whole is greater than the part” are represented by the strict synthetic relation `HilbertSegmentLess`. Second, sums and differences of segment lengths are not numerical expressions: they are realized by actual collinear configurations and transported by congruence. Third, the angle comparison needed in the proof is obtained from the already formalized exterior-angle machinery rather than inferred from the appearance of the figure.

Thus the reconstruction preserves Euclid's geometric strategy while making explicit the segment arithmetic and order-theoretic steps that the classical presentation leaves implicit.

## 21.5 Synthetic Segment Arithmetic

The segment part of I.21 exposes an important feature of the Hilbert reconstruction.

The formal development does not introduce a numerical length function, nor does it define an algebraic addition operation on segments. Instead, a sum is represented by a constructed segment.

For example, a point  $X$  satisfying

$$B - A - X \quad \text{and} \quad AX \cong AC$$

makes  $BX$  a representative of

$$BA + AC.$$

Similarly, a point  $Y$  satisfying

$$B - D - Y \quad \text{and} \quad DY \cong DC$$

makes  $BY$  a representative of

$$BD + DC.$$

Consequently the desired inequality is represented formally by

$$BY < BX.$$

The formal representatives can be pictured independently of the classical triangle:

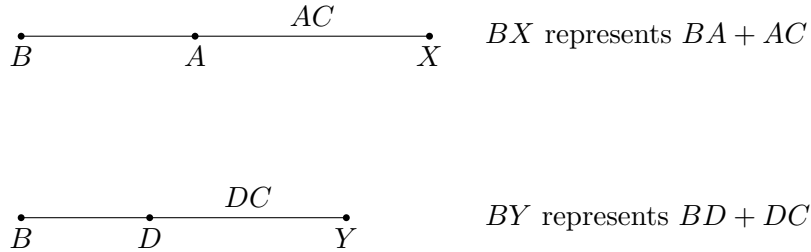


Figure 21.2: Synthetic representatives of the two segment sums in the Lean statement. The target inequality is the ordinary strict segment comparison  $BY < BX$ .

In Lean this is

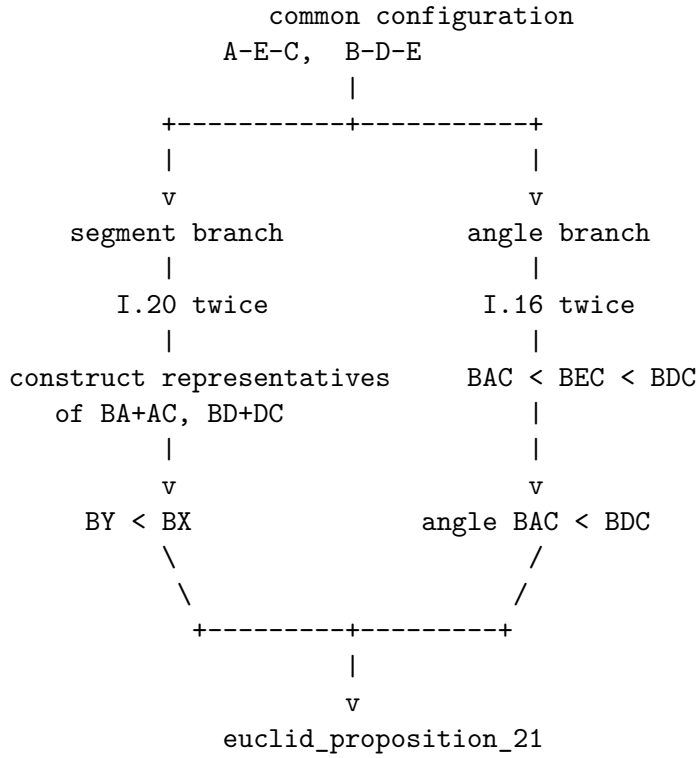
```
HilbertSegmentLess Geo B Y B X
```

This observation made an explicit predicate for segment sums unnecessary. The existing language of betweenness, congruence, and strict segment comparison is sufficient.

This is not merely a simplification of I.21. It is evidence that a useful fragment of segment arithmetic can be expressed internally in the synthetic language of Book Zero.

## 21.6 Proof Architecture of the Reconstruction

The Lean proof divides naturally into three major components.



The diagram is deliberately architectural rather than metric. In particular, the points  $X$  and  $Y$  in the formal theorem are not additional features of Euclid's printed diagram; they are witnesses that reify the two segment sums inside the synthetic language.

### 21.6.1 The Segment Helper

The first reusable component is

`euclid_proposition_21_segment_helper`

It begins with

$$A - E - C, \quad B - A - H,$$

together with

$$AH \cong AE$$

and

$$BE < BH.$$

A point  $F$  is constructed on the ray  $AH$  with

$$AF \cong AC.$$

Since  $AE < AC$ , congruence transport gives

$$AH < AF.$$

The Book Zero order machinery then yields

$$A - H - F.$$

Together with  $B - A - H$ , this produces

$$B - A - F \quad \text{and} \quad B - H - F.$$

Thus  $BF$  represents  $BA + AC$ .

Next a point  $G$  is constructed beyond  $E$  with

$$EG \cong EC.$$

Hence

$$B - E - G,$$

and  $BG$  represents  $BE + EC$ .

From

$$A - E - C, \quad A - H - F,$$

and

$$AE \cong AH, \quad AC \cong AF,$$

the theorem

`bookZero_differenceOfParts`

gives

$$EC \cong HF.$$

The inequality  $BE < BH$  can therefore be extended by congruent terminal parts. The theorem

`bookZero_53_lessThanAdditive`

yields

$$BG < BF.$$

This is the synthetic form of

$$BE + EC < BA + AC.$$

### 21.6.2 The Angle Branch

The angle comparison is isolated as

`euclid_proposition_21_angle`

with conclusion

`HilbertAngleLess Geo B A C B D C`

representing

$$\angle BAC < \angle BDC.$$

The proof first reconstructs the noncollinearity information required for the auxiliary triangles. This is a substantial formal task: Euclid simply reads these facts from the diagram.

Proposition I.16 then supplies the strict exterior-angle comparisons. The results are transported through collinear and same-ray configurations, and strict angle comparison is propagated to the required pair of angles.

The essential point is that no numerical angle measure is used. The proof remains entirely inside the synthetic relation `HilbertAngleLess`.

### 21.6.3 Final Segment Comparison

The main theorem first applies Proposition I.20 to triangle  $BAE$ . It obtains a point  $H$  with

$$B - A - H, \quad AH \cong AE,$$

and

$$BE < BH.$$

The segment helper extends this inequality to a comparison of constructed representatives.

The remaining part of the proof constructs a point  $Y$  satisfying

$$B - D - Y, \quad DY \cong DC.$$

The point  $X$  on the other side satisfies

$$B - A - X, \quad AX \cong AC.$$

After a sequence of congruence transports and applications of the Book Zero comparison lemmas, the proof reaches

`hBY_BX :`

`HilbertSegmentLess Geo B Y B X`

which is exactly

$$BD + DC < BA + AC.$$

## 21.7 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_21
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E : Geo.Point)
  (hABC : Not (PrimCollinear Geo A B C))
  (hAEC : Geo.Between A E C)
  (hBDE : Geo.Between B D E) :
  (exists X Y : Geo.Point,
    Geo.Between B A X /\
    Geo.Congruent A X A C /\
    Geo.Between B D Y /\
    Geo.Congruent D Y D C /\
    HilbertSegmentLess Geo B Y B X) /\
  HilbertAngleLess Geo B A C B D C := by
```

The existential part is the exact synthetic encoding of the segment inequality.

The conditions

`Geo.Between B A X`

`Geo.Congruent A X A C`

say that  $BX$  represents  $BA + AC$ .

Likewise

`Geo.Between B D Y`

`Geo.Congruent D Y D C`

say that  $BY$  represents  $BD + DC$ .

The comparison

`HilbertSegmentLess Geo B Y B X`

therefore expresses the first assertion of I.21.

The second assertion appears directly as

`HilbertAngleLess Geo B A C B D C`

The two branches are deliberately proved separately and assembled only at the end.

## 21.8 Relationship with Earlier Propositions

### 21.8.1 Proposition I.16

Proposition I.16 provides the exterior-angle inequalities needed for the angle branch.

This is the principal earlier proposition behind the proof of

$$\angle BAC < \angle BDC.$$

The formal reconstruction also reuses the same strict-angle language that was introduced and strengthened in the development surrounding I.16–I.18.

### 21.8.2 Proposition I.20

Proposition I.20 is fundamental to the segment branch.

Applied to triangle  $BAE$ , it supplies the initial inequality

$$BE < BA + AE.$$

In the formal proof the sum  $BA + AE$  is represented by a constructed point  $H$ :

$$B - A - H, \quad AH \cong AE,$$

so that

$$BE < BH.$$

This inequality is then enlarged synthetically by adding congruent parts.

### 21.8.3 The Cumulative Character of I.21

Proposition I.21 is markedly more cumulative than the immediately preceding propositions.

Its proof is no longer dominated by one construction. Instead it requires a coordinated use of order, congruence, segment comparison, ray transport, angle comparison, I.16, I.20, and numerous Book Zero lemmas.

For this reason I.21 is a useful test of whether the preceding formal development has produced a genuine working language rather than a collection of isolated theorem statements.

## 21.9 Hilbert and Book Zero Components Used

### 21.9.1 HilbertSegmentLess

Strict segment comparison supplies a synthetic replacement for numerical inequalities between lengths.

Its witness-based definition also makes it possible to recover interior points and to turn inequalities into betweenness configurations.

### 21.9.2 Congruence transport

The proof repeatedly changes representatives of the same segment. This requires strict inequalities to be transported through segment congruence.

Among the relevant Book Zero results are

`bookZero_30_lessThanCongruence`  
`bookZero_32_lessThanCongruence2`

These operations play the role that substitution of equal numerical lengths would play in an algebraic proof.

### 21.9.3 Difference of parts

The theorem

`bookZero_differenceOfParts`

expresses the synthetic subtraction principle.

From two collinear decompositions with congruent wholes and congruent initial parts, it identifies the remaining parts.

In I.21 it produces

$$EC \cong HF.$$

#### 21.9.4 Addition of inequalities

The theorem

`bookZero_53_lessThanAdditive`

is especially significant here.

It permits an existing strict segment inequality to be extended by congruent parts. In I.21 it turns

$$BE < BH$$

into the corresponding comparison after  $EC$  and  $HF$  have been added.

This is one of the clearest examples so far of Book Zero functioning as a synthetic arithmetic of segments.

#### 21.9.5 Segment-Inequality Transitivity

The theorem

`bookZero_52_lessThanTransitive`

allows the several intermediate comparisons to be composed into the final inequality.

#### 21.9.6 Sum of parts

The existing theorem

`bookZero_sumOfParts`

is used when two adjacent congruent pieces must be recognized as giving congruent whole segments.

Together with difference-of-parts and inequality transport, this provides much of the elementary segment calculus required by I.21.

### 21.10 What I.21 Reveals About Book Zero

The reconstruction of I.21 gives a useful answer to a structural question about the project.

It might appear natural to introduce explicit objects representing formal sums of segments. Indeed, an intermediate version of the development contained such a definition.

The completed proof shows that this is not presently necessary.



The combination of

betweenness + congruence + strict segment comparison

already permits sums to be constructed, compared, enlarged, reduced, and transported.

Thus the expressions

$$AB + CD, \quad AB < CD,$$

need not force the development prematurely into an algebra of lengths.

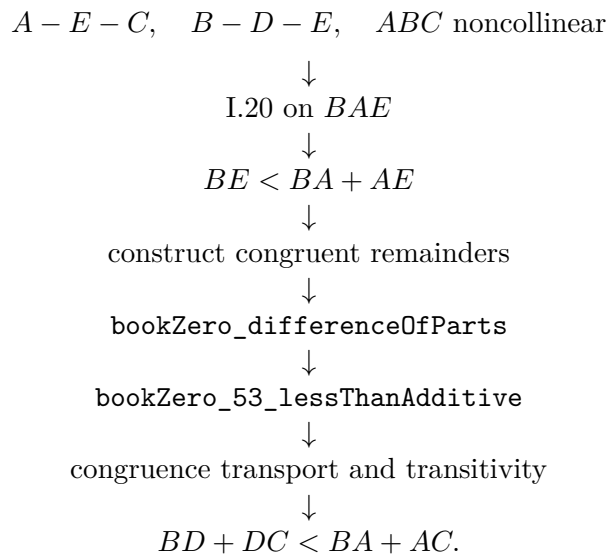
The geometric witnesses themselves carry the arithmetic.

This is important for the intended role of Book Zero. Its purpose is not merely to shorten proofs. It supplies the intermediate operations by which ordinary synthetic reasoning can be expressed without abandoning the primitive Hilbert geometry.

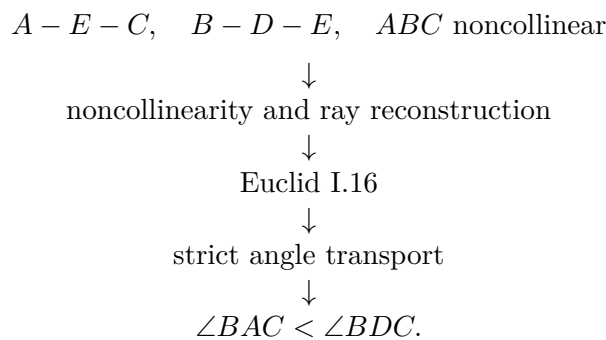
I.21 is the strongest example of this role encountered so far.

## 21.11 Dependency Summary

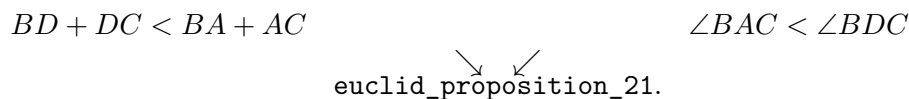
The segment branch can be summarized schematically as



The angle branch is



Finally,



No new geometric axiom is introduced.

## 21.12 Formalization Notes

Several points that are nearly invisible in the classical proof become substantial in Lean.

First, every assertion that a point lies on the same line, on the same ray, or on the opposite continuation of a ray must be derived explicitly.

Second, the notation  $AB + AC$  cannot silently denote a number. A concrete point must be constructed whose position and congruence properties make an ordinary segment represent that sum.

Third, inequalities cannot simply be rewritten by replacing equal lengths. Congruence transport requires explicit theorems.

Fourth, the two assertions of I.21 genuinely have different proof architectures. Keeping them separate in the formal development makes their dependencies visible and produces reusable intermediate results.

The resulting Lean proof is much longer than Euclid's proof, but the additional length is not caused by a different mathematical idea. It records the incidence, order, congruence, and transport operations which the classical diagram suppresses.

## 21.13 Conclusion

Proposition I.21 is an important transition point in the formalization of Book I.

The classical proposition combines a comparison of broken lines with a comparison of angles. Its proof assumes a mature ability to manipulate segment sums, strict inequalities, collinear configurations, and exterior angles.

The Lean reconstruction preserves this synthetic structure. It does not replace Euclid's argument by coordinates, numerical lengths, or angle measures.

More importantly, the reconstruction demonstrates that the Book Zero layer is beginning to perform its intended function. Results such as difference of parts, addition of inequalities, congruence transport, ray conversion, and strict-comparison transitivity combine into a usable language in which a substantially more complicated Euclidean argument can be expressed.

The fact that no separate `HilbertSegmentSum` object is needed is part of this conclusion. Synthetic segment arithmetic can, at least at this stage, remain geometry rather than become an external algebra of lengths.

The Hilbert reconstruction of Euclid I.21 is complete. It introduces no new geometric axiom, and the complete development through Proposition I.21 compiles successfully.

## Chapter 22

# Proposition I.22

### 22.1 Introduction

Euclid’s Proposition I.22 is the classical existence theorem for a triangle with three prescribed side lengths.

Three nondegenerate segments are given. Euclid assumes that any two of them, taken together, are greater than the remaining one. Under these hypotheses he constructs a triangle whose three sides are congruent to the three given segments.

In the present reconstruction the given segments are denoted

$$Aa, \quad Bb, \quad Cc.$$

The three hypotheses are the strict triangle inequalities

$$Aa < Bb + Cc, \quad Bb < Aa + Cc, \quad Cc < Aa + Bb.$$

The formal conclusion is the existence of a point  $K$  such that

$$BK \cong Aa \quad \text{and} \quad bK \cong Cc.$$

Since  $Bb$  is already the third prescribed side, the triangle  $BbK$  has precisely the required side lengths.

Proposition I.22 is especially important for the formal development because its proof exposes two pieces of infrastructure which classical notation tends to hide:

1. the phrase “two segments taken together are greater than a third” must receive a synthetic, witness-based meaning;
2. the existence of the vertex  $K$  is an application of circle-circle continuity.

No numerical length function is introduced. Segment sums, strict inequalities, and the circle intersection are all expressed in the synthetic language of betweenness and congruence.

The resulting formal architecture is therefore different from the visual shortcut suggested by the classical drawing:

```

triangle inequalities
|
+-- synthetic sum witnesses
|
+-- construct a point on the G-circle
|     that lies inside the F-circle
|
+-- construct a point on the G-circle
|     that lies outside the F-circle
|
v
circle-circle continuity
|
v
intersection point K
|
v
BK ~= Aa,  bK ~= Cc

```

This is the hidden logical content of the apparently simple instruction “draw two intersecting circles.”

## 22.2 The Classical Configuration

Euclid’s construction is best understood in the classical ray configuration used by Wilkins. Let the three prescribed lengths be represented by  $A$ ,  $B$ , and  $C$ . On one ray choose consecutive segments

$$DF = A, \quad FG = B, \quad GH = C.$$

Draw the circle centered at  $F$  with radius  $A$ , and the circle centered at  $G$  with radius  $C$ . Under the three strict triangle inequalities the circles intersect properly in two points  $K$  and  $L$ . Either triangle  $FGK$  or  $FGL$  then has side lengths  $A, B, C$ .

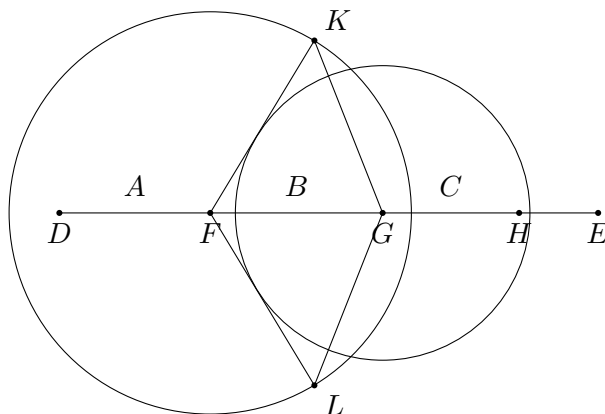


Figure 22.1: Wilkins-style configuration for Euclid I.22. The circles centered at  $F$  and  $G$  intersect in  $K$  and  $L$ , producing triangles with side lengths  $A, B, C$ .

The important point is that the three inequalities are not decorative side conditions. They control the relative position of the two circles. Wilkins makes this explicit by displaying the

six boundary and failure configurations obtained when one of the strict triangle inequalities is replaced by equality or reversed.

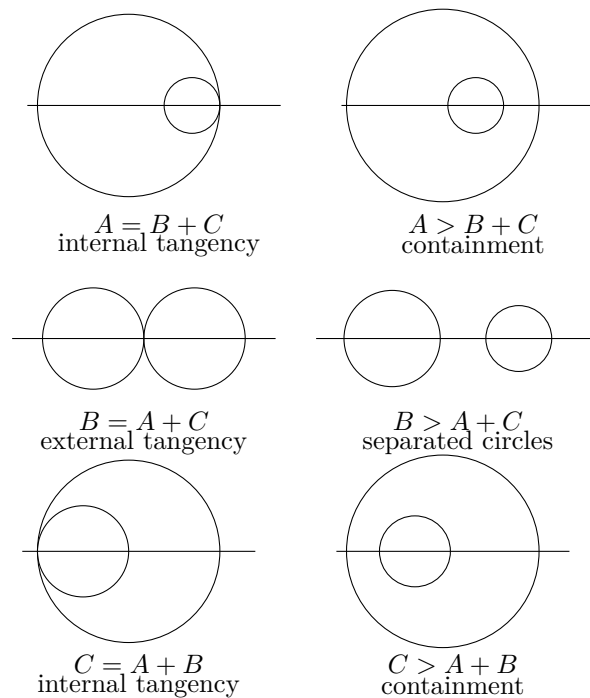


Figure 22.2: The six boundary and failure configurations corresponding to the three triangle inequalities. Equality gives tangency; reversing the inequality destroys proper intersection.

This makes the geometric meaning of the hypotheses visible:

$A < B + C$   
 $B < A + C$   
 $C < A + B$   
 |  
 v  
 two circles intersect properly  
 |  
 v  
 triangle with sides A, B, C

The formal proof does not use a numerical metric criterion for circle intersection. Instead, it constructs explicit witnesses on one circle that are respectively inside and outside the other circle. Circle-circle continuity then produces the common point.

## 22.3 Classical Proof

Euclid's proof is a direct ruler-and-compass construction.

Let the three prescribed segments be represented by

$$Aa, \quad Bb, \quad Cc,$$

and assume that every two taken together are greater than the remaining one:

$$Aa < Bb + Cc, \quad Bb < Aa + Cc, \quad Cc < Aa + Bb.$$

Choose the segment  $Bb$  as the base of the required triangle. Draw

- the circle centered at  $B$  with radius congruent to  $Aa$ ;
- the circle centered at  $b$  with radius congruent to  $Cc$ .

Let  $K$  be an intersection point of these two circles. By construction,

$$BK \cong Aa \quad \text{and} \quad bK \cong Cc.$$

Together with the base  $Bb$ , the triangle  $BbK$  therefore has the three prescribed side lengths.

The classical proof may be summarized as

$$\begin{array}{c}
 Aa < Bb + Cc, \quad Bb < Aa + Cc, \quad Cc < Aa + Bb \\
 \downarrow \\
 \text{draw two circles with centers } B \text{ and } b \\
 \downarrow \\
 K \text{ is their intersection} \\
 \downarrow \\
 BK \cong Aa, \quad bK \cong Cc \\
 \downarrow \\
 \text{triangle } BbK.
 \end{array}$$

The point at which the axiomatic reconstruction must say more is the existence of  $K$ . In the traditional diagram the two circles are simply drawn as intersecting. A formal proof must derive an appropriate intersection theorem from explicit hypotheses.

Thus Proposition I.22 is one of the places where the difference between a construction diagram and an axiomatic existence proof becomes especially visible.

## 22.4 Statement

The synthetic statement used in the final Lean theorem is:

```

theorem euclid_proposition_22
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A a B b C c : Geo.Point)
  (hAa : A != a)
  (hBb : B != b)
  (hCc : C != c)
  (hAa_lt_BbCc :
    HilbertSegmentSumGreater Geo B b C c A a)
  (hBb_lt_AaCc :
    HilbertSegmentSumGreater Geo A a C c B b)

```

```

(hCc_lt_AaBb :
  HilbertSegmentSumGreater Geo A a B b C c) :
exists K : Geo.Point,
  Geo.Congruent B K A a /\
  Geo.Congruent b K C c

```

The three assumptions involving `HilbertSegmentSumGreater` express

$$Bb + Cc > Aa, \quad Aa + Cc > Bb, \quad Aa + Bb > Cc.$$

The conclusion gives the third vertex  $K$  of the required triangle:

$$BK \cong Aa, \quad bK \cong Cc,$$

while the base itself is the given segment  $Bb$ .

## 22.5 A Synthetic Predicate for “Two Together Greater”

Proposition I.21 showed that segment sums need not in general be reified as numerical objects. A concrete collinear segment can serve as a representative of a sum.

Proposition I.22 introduces a slightly different requirement. The three triangle inequalities are not merely intermediate steps in the proof: they are part of the public hypotheses of the theorem. Consequently it is useful to package the geometric witness explicitly.

The Book Zero predicate is

```

def HilbertSegmentSumGreater
  [HilbertCongruence Geo]
  (A B C D E F : Geo.Point) : Prop :=
exists P : Geo.Point,
  Geo.Between A B P /\
  Geo.Congruent B P C D /\
  HilbertSegmentLess Geo E F A P

```

Geometrically, a witness  $P$  satisfies

$$A - B - P, \quad BP \cong CD.$$

Therefore  $AP$  represents the sum

$$AB + CD.$$

The final condition

$$EF < AP$$

says precisely

$$AB + CD > EF.$$

For example,

```
HilbertSegmentSumGreater Geo B b C c A a
```

means that there exists  $P$  with

$$B - b - P, \quad bP \cong Cc, \quad Aa < BP.$$

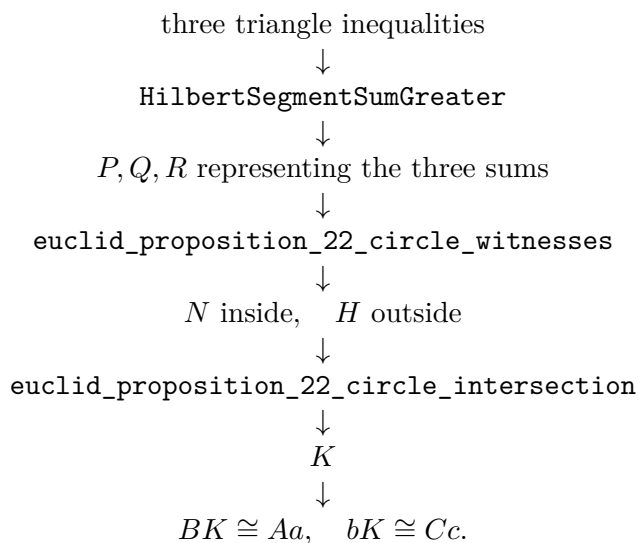
Thus  $BP$  is a synthetic representative of  $Bb + Cc$ .

This definition does not introduce an algebra of lengths. It merely packages an ordinary geometric construction together with the strict comparison already provided by `HilbertSegmentLess`.

## 22.6 Proof Architecture of the Reconstruction

The reconstruction is intentionally layered. The final theorem is short because the difficult circle geometry is isolated in reusable helpers.

The overall dependency is



The principal intermediate results are discussed below.

### 22.6.1 Transporting the outside inequality

The helper

`euclid_proposition_22_outside_helper`

handles the synthetic arithmetic behind the outside point.

Suppose  $BP$  represents

$$Bb + Cc,$$

and another collinear segment  $FH$  is assembled from pieces

$$FG \cong Bb, \quad GH \cong Cc.$$

Using `bookZero_sumOfParts`, the proof obtains

$$BP \cong FH.$$

Hence the triangle inequality

$$Aa < BP$$



can be transported to

$$Aa < FH$$

by `bookZero_30_lessThanCongruence`.

This is the formal replacement for the classical step “equal sums may be substituted in an inequality.”

### 22.6.2 Same-ray trichotomy

A point constructed on a ray can lie before, at, or beyond the reference point. The helper

`euclid_proposition_22_sameRay_trichotomy`

makes this order distinction explicit.

For points  $G, F, N$  on the same ray from  $G$ , it proves

$$G - N - F \quad \vee \quad N = F \quad \vee \quad G - F - N.$$

The proof uses strict segment trichotomy together with `bookZero_51_lessThanBetween` and `layoff uniqueness`.

This case distinction is essential when the point  $N$  on the second circle is compared with the first circle.

### 22.6.3 Cancellation for strict segment inequalities

The helper

`euclid_proposition_22_lessThan_cancel_right`

is a synthetic cancellation principle.

If

$$A - B - E, \quad C - D - F,$$

the terminal parts satisfy

$$BE \cong DF,$$

and

$$AE < CF,$$

then

$$AB < CD.$$

Classically this is an immediate cancellation of equal addends. Formally it requires a proof from the developed strict-order and congruence theory.

This helper is one of the clearest examples in I.22 of Book Zero providing arithmetic-like behavior without numerical lengths.

### 22.6.4 The inside point

The proof constructs a point  $N$  on the ray from  $b$  through  $B$  such that

$$bN \cong Cc.$$

The two possible order configurations are handled by

```
euclid_proposition_22_inside_left
euclid_proposition_22_inside_right
euclid_proposition_22_inside_helper
```

and the same-ray trichotomy theorem.

The goal is not merely to locate  $N$  on the second circle. One must also prove that it is inside, or on the center of, the first circle:

$$N = B \quad \vee \quad BN < Aa.$$

The relevant triangle inequalities are used to derive this comparison in each possible collinear configuration.

### 22.6.5 Constructing explicit sum representatives

The helper

```
euclid_proposition_22_extend_with_segment
```

constructs the ordinary geometric representative of a sum.

Starting from a nondegenerate segment  $AB$  and another segment  $CD$ , it extends the ray through  $B$  to a point  $P$  satisfying

$$A - B - P, \quad BP \cong CD.$$

Hence

$$AP$$

represents

$$AB + CD.$$

This is the basic geometric operation behind the witnesses  $P, Q, R$  used in the core construction.

### 22.6.6 Circle witnesses

The theorem

```
euclid_proposition_22_circle_witnesses
```

is the main preparatory result.

From the three represented triangle inequalities it constructs points  $H$  and  $N$  on the circle centered at  $b$  with radius congruent to  $Cc$ , with the following properties:

$$bH \cong Cc, \quad Aa < BH,$$

and

$$bN \cong Cc, \quad N = B \vee BN < Aa.$$

Thus  $H$  is outside the circle centered at  $B$  with radius  $Aa$ , while  $N$  is inside it or coincides with its center.

This is precisely the configuration required by circle-circle continuity.

## 22.7 Circle-Circle Continuity

The central existence step is isolated in

`euclid_proposition_22_circle_intersection`

The Hilbert circle-continuity theorem does not use the radius  $Aa$  directly as a numerical quantity. The proof first lays off a congruent segment  $BT$ :

$$BT \cong Aa.$$

The first circle is therefore represented by center  $B$  and radius segment  $BT$ .

The second circle has center  $b$  and radius  $bN$ , with

$$bN \cong Cc.$$

The already constructed points satisfy

$$bH \cong bN,$$

so  $H$  lies on the second circle, and

$$Aa < BH.$$

After transport through  $BT \cong Aa$ , this becomes

$$BT < BH,$$

showing that  $H$  is outside the first circle.

Similarly,

$$N = B \vee BN < Aa$$

is transported to

$$N = B \vee BN < BT,$$

showing that  $N$  is inside the first circle.

The theorem

`hilbert_circle_circle_intersection`

then yields a point  $K$  satisfying

$$BK \cong BT \quad \text{and} \quad bK \cong bN.$$

Finally congruence transitivity returns to the original given segments:

$$BK \cong Aa, \quad bK \cong Cc.$$

This is the required vertex.

## 22.8 The Core and the Public Theorem

Once the circle witnesses and the circle-intersection theorem are available, the remaining layers become very short.

The theorem

`euclid_proposition_22_core`

accepts explicit representatives  $P, Q, R$  of the three sums:

$$BP = Bb + Cc, \quad AQ = Aa + Cc, \quad AR = Aa + Bb,$$

together with

$$Aa < BP, \quad Bb < AQ, \quad Cc < AR.$$

It invokes the two successive helpers

`euclid_proposition_22_circle_witnesses`  
`euclid_proposition_22_circle_intersection`

The intermediate wrapper

`euclid_proposition_22_from_sum_witnesses`

packages the same construction when all three sum witnesses are supplied explicitly.

The public theorem finally removes  $P, Q, R$  from its interface. It opens the three witnesses stored in `HilbertSegmentSumGreater` and passes them to the internal construction.

Thus the public API states exactly the geometric theorem rather than the implementation details of its proof.

## 22.9 Lean Formalization

The public theorem deliberately exposes only the three given segments, their nondegeneracy, the three triangle inequalities, and the required vertex:

```
theorem euclid_proposition_22
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A a B b C c : Geo.Point)
  (hAa : A != a)
  (hBb : B != b)
  (hCc : C != c)
  (hAa_lt_BbCc :
    HilbertSegmentSumGreater Geo B b C c A a)
  (hBb_lt_AaCc :
    HilbertSegmentSumGreater Geo A a C c B b)
  (hCc_lt_AaBb :
    HilbertSegmentSumGreater Geo A a B b C c) :
  exists K : Geo.Point,
    Geo.Congruent B K A a /\
    Geo.Congruent b K C c
```

The public assumptions are therefore the exact synthetic counterparts of

$$Aa < Bb + Cc, \quad Bb < Aa + Cc, \quad Cc < Aa + Bb.$$

Internally, each occurrence of `HilbertSegmentSumGreater` is unpacked into a concrete witness. Schematically:

```
rcases hAa_lt_BbCc with
  <P, hBbP, hbP_Cc, hAa_BP>

rcases hBb_lt_AaCc with
  <Q, hAaQ, haQ_Cc, hBb_AQ>

rcases hCc_lt_AaBb with
  <R, hAaR, haR_Bb, hCc_AR>
```

The points  $P, Q, R$  are not abstract numbers representing sums. They are ordinary points producing collinear segments whose lengths realize the three sums geometrically.

The next layer constructs the circle witnesses. In schematic form:

```
have hWitnesses :=
  euclid_proposition_22_circle_witnesses
  ...

rcases hWitnesses with
  <H, N, hH_on_circle, hH_outside,
    hN_on_circle, hN_inside>
```

The important output is the inside/outside configuration

$$Aa < BH, \quad N = B \vee BN < Aa,$$

while both  $H$  and  $N$  lie on the circle centered at  $b$  with radius congruent to  $Cc$ .

The actual existence step is then concentrated in

`euclid_proposition_22_circle_intersection`

which invokes

`hilbert_circle_circle_intersection`

after transporting the radius and inequality data into the orientation expected by the continuity theorem.

Finally, congruence transitivity converts the constructed radius representatives back to the original segments and yields

$$BK \cong Aa, \quad bK \cong Cc.$$

The Lean proof is therefore longer than Euclid's paper proof not because it changes the construction, but because it makes three suppressed layers explicit:

segment arithmetic + ray/order geometry + circle continuity.

## 22.10 Relationship with Earlier Propositions

### 22.10.1 Proposition I.20

Proposition I.20 established the triangle inequality inside an already given triangle.

Proposition I.22 uses the converse direction as an existence hypothesis: three prescribed segment lengths satisfying the three triangle inequalities should determine a triangle.

The logical roles are therefore different. I.20 proves inequalities from a triangle; I.22 assumes the inequalities in order to construct a triangle.

### 22.10.2 Proposition I.21

Proposition I.21 demonstrated that segment sums can be handled synthetically by constructing ordinary segments representing sums of parts.

I.22 develops this idea further. Because the triangle inequalities are part of the public statement rather than temporary intermediate facts, the witness-bearing predicate `HilbertSegmentSumGreater` is introduced as an interface notion.

Thus I.22 turns the segment-arithmetic techniques of I.21 into reusable public infrastructure.

### 22.10.3 Proposition I.23

Proposition I.23 follows I.22 in Euclid's sequence and uses it essentially in Euclid's own construction of an angle congruent to a given angle.

In the Hilbert reconstruction, however, I.23 does not formally depend on I.22 because Hilbert III.4 already supplies angle construction.

This makes I.22 a useful reference point for distinguishing the historical dependency graph of the *Elements* from the dependency graph induced by the Hilbert foundation.

## 22.11 Hilbert and Book Zero Components Used

### 22.11.1 HilbertSegmentLess

The relation

`HilbertSegmentLess`

is the synthetic strict order on segments.

It supplies the comparisons needed to distinguish points lying inside and outside a circle and supports the transport and transitivity operations used throughout the proof.

### 22.11.2 bookZero\_49\_layoff

The theorem

`bookZero_49_layoff`

copies a given nondegenerate segment onto a chosen ray.

In I.22 it is used to represent the radius  $Aa$  by a segment based at the center  $B$ :

$$BT \cong Aa.$$

### 22.11.3 bookZero\_sumOfParts

The theorem

`bookZero_sumOfParts`

recognizes congruent wholes assembled from congruent adjacent parts.

It is used when a segment representing  $Bb + Cc$  must be identified with another segment built from a copy of  $Bb$  followed by a copy of  $Cc$ .

### 22.11.4 Congruence transport for inequalities

The theorems

`bookZero_30_lessThanCongruence`  
`bookZero_32_lessThanCongruence2`

transport strict segment comparisons through congruent representatives.

They replace the informal algebraic operation of substituting equal lengths into inequalities.

### 22.11.5 `bookZero_31_trichotomy1`

This theorem supplies the equality case when neither of two nondegenerate segments is strictly shorter than the other.

Together with layoff uniqueness it provides the middle case

$$N = F$$

in the same-ray trichotomy used by I.22.

### 22.11.6 `bookZero_51_lessThanBetween`

The theorem

`bookZero_51_lessThanBetween`

converts strict segment comparison on a common ray into strict betweenness.

It is the bridge between the order relation on segment lengths and the incidence-order geometry needed to locate the constructed circle points.

### 22.11.7 Circle-circle intersection

The theorem

`hilbert_circle_circle_intersection`

is the decisive existence principle.

Given a point of one circle lying inside a second circle and another point of the same first circle lying outside the second, it produces a common point of the two circles.

Proposition I.22 is therefore not merely an application of elementary segment arithmetic. It is the point where that arithmetic is used to establish the hypotheses of a genuine continuity theorem.



## 22.12 Comparison with Euclid's Original Organization

Euclid presents I.22 as an explicit ruler-and-compass construction. He places copies of the three given segments on a straight line and draws two circles whose intersection gives the required triangle.

The Hilbert reconstruction preserves the same geometric idea but changes its logical organization.

First, the base  $Bb$  is already available as one of the given segments, so there is no need to reproduce Euclid's preliminary line solely to manufacture a base of the correct length.

Second, the three assumptions "two together greater than the remaining one" are exposed through `HilbertSegmentSumGreater`. Their witnesses give concrete segments representing the required sums.

Third, the existence of the intersection is not treated as visually obvious. Explicit points  $N$  and  $H$  are constructed on one circle, and they are proved to lie respectively inside and outside the other. Only then is circle-circle continuity invoked.

Thus the formal proof separates three issues which Euclid's diagram compresses:

$$\text{segment arithmetic} \quad + \quad \text{order on rays} \quad + \quad \text{continuity}.$$

The resulting proof is longer, but the additional length identifies the exact logical content required for the construction.

## 22.13 What Proposition I.22 Adds to Book Zero

I.21 suggested that explicit segment-sum objects were unnecessary: inside a proof, sums could be represented directly by constructed segments.

I.22 refines that conclusion rather than contradicting it.

The new predicate

`HilbertSegmentSumGreater`

does not turn segment sums into algebraic objects. It is a witness-bearing proposition:

$$AB + CD > EF$$

means that a point exists which geometrically realizes  $AB + CD$ , and that the resulting whole segment is strictly greater than  $EF$ .

This distinction is important.

The project still has no numerical length map and no binary addition operation on lengths. What has been added is a convenient *interface proposition* for a classical phrase that occurs as a hypothesis of Euclid's theorem.

Proposition I.22 also shows that the Book Zero layer is now strong enough to prepare the exact inside/outside data required by a continuity theorem. The layer is therefore doing more than abbreviating elementary congruence arguments: it is connecting synthetic segment arithmetic with topological existence principles.

## 22.14 Dependency Summary

The public assumptions are

$$\begin{aligned} Aa \neq 0, \quad Bb \neq 0, \quad Cc \neq 0, \\ Aa < Bb + Cc, \\ Bb < Aa + Cc, \\ Cc < Aa + Bb. \end{aligned}$$

In Lean the three inequalities are encoded by

```
HilbertSegmentSumGreater Geo B b C c A a,
HilbertSegmentSumGreater Geo A a C c B b,
HilbertSegmentSumGreater Geo A a B b C c.
```

The complete proof architecture is

```
triangle inequalities
↓
P, Q, R sum witnesses
↓
segment transport + same-ray order
↓
N = B ∨ BN < Aa    Aa < BH
↓
bN ≅ Cc,    bH ≅ Cc
↓
hilbert_circle_circle_intersection
↓
BK ≅ Aa,    bK ≅ Cc
↓
euclid_proposition_22.
```

No new geometric axiom is introduced in Proposition I.22. The proof uses the already available circle-circle continuity theorem together with the developed Hilbert and Book Zero segment theory.

## 22.15 Formalization Notes

Several features of I.22 are easy to miss in an informal proof.

First, a circle radius is represented by an ordinary segment, not by a real number. Even when the intended radius is  $Aa$ , the circle-continuity theorem is applied using a constructed representative  $BT$  with

$$BT \cong Aa.$$

Second, “inside a circle” is expressed synthetically by a strict segment comparison such as

$$BN < BT,$$

with the center case  $N = B$  handled separately.

Third, “outside a circle” is likewise a strict segment comparison:

$$BT < BH.$$

Fourth, the fact that  $N$  lies on the ray from  $b$  through  $B$  is not enough to determine its order relative to  $B$ . The proof requires an explicit three-way analysis of the possible positions.

Fifth, cancellation of equal terminal pieces from strict inequalities is not automatic. It is isolated as a theorem and proved from the existing segment-order machinery.

Finally, the public theorem is deliberately cleaner than the internal construction. The implementation works with many auxiliary points, but the theorem exposes only the three original segments, their three triangle inequalities, and the required vertex.

## 22.16 Conclusion

Proposition I.22 is a major structural milestone in the reconstruction of Book I.

Its classical content is elementary: three admissible lengths determine a constructible triangle. Formalizing that statement reveals, however, a precise interaction between three layers of synthetic geometry.

The first layer is segment arithmetic. Betweenness, congruence, strict segment comparison, addition of adjacent parts, cancellation, and transport are used to express the three triangle inequalities and to manipulate their witnesses.

The second layer is ray and order geometry. Constructed points must be located explicitly on rays, and their possible orders must be distinguished.

The third layer is continuity. Once one point of the second circle is proved inside the first and another outside it, `hilbert_circle_circle_intersection` supplies the required common point.

The final theorem therefore has a particularly clear conceptual form: the triangle inequalities are exactly the hypotheses needed to build the inside/outside witnesses for two circles.

No coordinates, numerical lengths, or angle measures enter the proof. The reconstruction remains entirely synthetic, and the complete development through Euclid I.22 compiles successfully.

## Chapter 23

# Proposition I.23

### 23.1 Introduction

Euclid's Proposition I.23 asks for the construction, at a prescribed point on a prescribed straight line, of an angle congruent to a given rectilinear angle.

In Euclid's development this is a genuine construction theorem. The construction proceeds indirectly: Euclid chooses points on the arms of the given angle, obtains three segment lengths, uses Proposition I.22 to construct a triangle with those three sides, and then invokes Proposition I.8 (SSS) to identify the required angle.

In the Hilbert reconstruction used by CGJteamLab, the logical status of the proposition is different. Hilbert's congruence axiom III.4 already provides angle transport on a prescribed side of a prescribed ray, together with uniqueness of the resulting ray. Consequently the formal Proposition I.23 is a short wrapper over a stronger foundational principle.

This contrast is mathematically important. Proposition I.23 is an example in which the order of Euclid's propositions and the dependency order induced by the chosen axiomatic foundation no longer coincide.

### 23.2 The Classical Configuration

Euclid's construction is best understood in the original notation of the classical argument. A straight segment  $AB$  and a given angle at  $C$  are prescribed. The objective is to construct at  $A$ , on the given straight line, an angle equal to the angle at  $C$ .

Wilkins's presentation makes the constructional character of the proposition especially clear. The initial data are simply the line segment and the given angle:

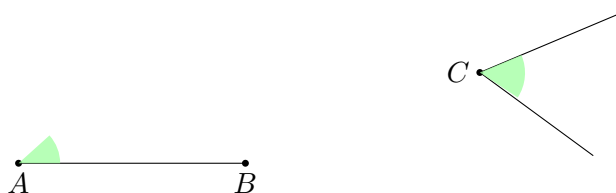


Figure 23.1: Initial data for Euclid I.23: a prescribed segment  $AB$  and a given rectilinear angle at  $C$ .

Euclid does not transport the angle directly. He first chooses points  $D$  and  $E$  on the two arms of the given angle and forms the triangle  $CDE$ . A second triangle  $AFG$  is then constructed with  $G$  lying on  $AB$ .

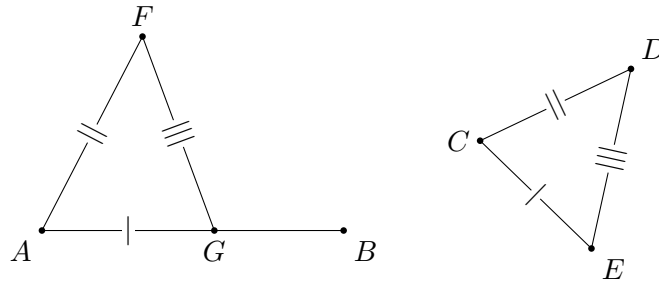


Figure 23.2: Euclid's completed auxiliary construction:  $AG \cong CE$ ,  $AF \cong CD$ , and  $FG \cong DE$ .

The three side congruences give

$$\triangle AFG \cong \triangle CDE$$

by Proposition I.8, and hence

$$\angle FAG \cong \angle DCE.$$

The drawings therefore encode the actual Euclidean construction rather than merely displaying two congruent angles.

### 23.3 Classical Proof

Euclid derives the construction from earlier propositions rather than treating angle transport as primitive.

Choose points  $A$  and  $C$  on the two arms of the given angle  $\angle ABC$ . The three segments

$$AB, \quad BC, \quad AC$$

determine the triangle  $ABC$ .

Now use Proposition I.22 to construct a triangle  $DEF$  whose corresponding sides satisfy

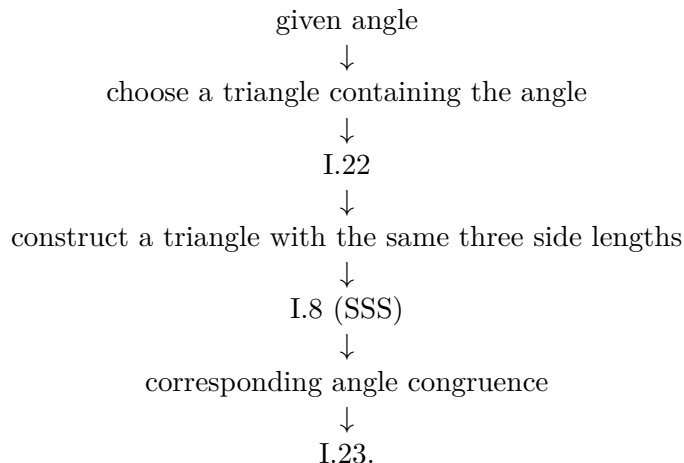
$$DE \cong AB, \quad DF \cong BC, \quad EF \cong AC.$$

By Proposition I.8, the two triangles are congruent by SSS. Hence their corresponding included angles are congruent, and in particular

$$\angle ABC \cong \angle EDF$$

after the appropriate correspondence of vertices.

Thus the classical dependency pattern is



In Euclid's route, angle transport is therefore reduced to segment construction and triangle congruence.

## 23.4 Hilbert III.4

Hilbert reverses the foundational emphasis.

Instead of deriving angle transport from triangle construction, Hilbert includes angle construction among the congruence axioms. In the present formal layer the relevant operation is

`HilbertCongruence.angle_construction`

Conceptually, it states that from

- a nondegenerate source angle;
- a prescribed initial ray;
- a choice of one side of the supporting line,

one obtains a ray on that side forming an angle congruent to the source angle.

The Hilbert axiom is stronger than Euclid I.23 because it also contains a uniqueness assertion: on the chosen side of the line, the constructed ray is unique.

This additional information is not needed in the public statement of Proposition I.23, but it remains available in the Hilbert interface.

## 23.5 Proof Architecture of the Reconstruction

The formal reconstruction separates naturally into four stages.

### 23.5.1 The Source Angle

The hypothesis

$$\neg \text{Collinear}(A, B, C)$$

ensures that  $\angle ABC$  is nondegenerate.

This is the source angle that will be transported.

### 23.5.2 The Prescribed Ray

The points  $D$  and  $E$  determine the initial ray of the new angle.

The hypotheses assert that

$$D \neq E$$

and that both points lie on a common supporting line. Thus the formal data specify not merely a vertex, but an oriented ray represented by two distinct points.

### 23.5.3 Selecting the Side

A point  $S$  not lying on the supporting line selects one of its two sides.

The constructed point  $F$  is required to satisfy

$$\text{SameSide}(F, S, \text{base}).$$

This datum has no explicit counterpart in Euclid's informal formulation, where the intended side is supplied by the diagram and by the instruction to construct the angle "on" a given straight line.

In the axiomatic reconstruction it must be stated.

### 23.5.4 Applying Angle Construction

The Hilbert angle-construction theorem supplies a point  $F$  satisfying the required same-side condition and

$$\angle ABC \cong \angle DEF.$$

Its uniqueness component is discarded because Proposition I.23 asks only for existence.

Thus the entire formal proof is an interface extraction from Hilbert III.4.

The Hilbert configuration is consequently much simpler than the Euclidean construction. The source angle, the prescribed ray  $DE$ , and the side selector  $S$  determine the required ray directly:

This diagram is intentionally not the Euclidean  $CDE/AFG$  construction. It represents the stronger primitive available in the Hilbert congruence layer.

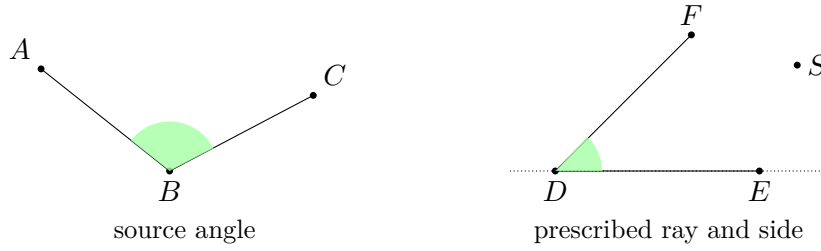


Figure 23.3: Hilbert III.4 configuration. The point  $S$  selects the side of the supporting line on which the second ray is constructed.

## 23.6 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_23
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E S : Geo.Point)
  (hABC : Not (PrimCollinear Geo A B C))
  (hDE : D != E)
  (base : Geo.Line)
  (hDbase : HilbertIncidence.OnLine D base)
  (hEbase : HilbertIncidence.OnLine E base)
  (hSbase : Not (HilbertIncidence.OnLine S base)) :
  Exists fun F : Geo.Point =>
    HilbertSameSide Geo F S base /\
    Geo.AngleCongruent A B C D E F := by
```

The mathematical meaning of the arguments is:

$$\angle ABC$$

is the source angle,

$$DE$$

is the prescribed initial ray, and  $S$  selects the side of the supporting line on which the second ray must be constructed.

The proof then invokes the Hilbert construction principle and extracts the components required by the Euclidean statement. Schematically:

```
rcases HilbertCongruence.angle_construction ... with
  <F, hSameSide, hAngle, hUnique>

exact <F, hSameSide, hAngle>
```

The uniqueness witness `hUnique` is deliberately not returned.



This is the essential feature of the Lean proof: its brevity reflects the strength of the Hilbert interface rather than an omitted geometric argument.

## 23.7 The Euclidean Route and the Hilbert Route

### 23.7.1 The Euclidean Route

Euclid reduces angle transport to triangle construction:

$$\text{I.22} + \text{I.8} \longrightarrow \text{I.23}.$$

Proposition I.22 constructs a triangle with prescribed side lengths, while Proposition I.8 identifies the corresponding angle by SSS.

The contrast is especially transparent in dependency form:

```

Euclid
-----
given angle at C
  |
choose D,E on its arms
  |
triangle CDE
  |
I.20 -> triangle inequalities
  |
I.22 -> construct AFG
  |
I.8 / SSS
  |
angle FAG ~= angle DCE

Hilbert / Lean
-----
source angle + prescribed ray + side selector
  |
Hilbert III.4
  |
constructed ray + angle congruence
  |
euclid_proposition_23

```

This route passes through segment arithmetic, circle intersection, continuity, and triangle congruence.

### 23.7.2 The Hilbert Route

In the Hilbert foundation the corresponding dependency is simply

III.4  $\longrightarrow$  I.23.

Angle transport has already been placed in the congruence layer.

Thus Proposition I.23 does not formally depend on Proposition I.22 or Proposition I.8.

### 23.7.3 Foundational Interpretation

The difference is not merely one of proof length.

Euclid treats angle transport as something to be constructed from more elementary operations. Hilbert treats the ability to copy an angle to a prescribed side of a ray as part of the primitive congruence structure.

The same geometric statement therefore occupies two different logical levels in the two systems.

This is precisely the kind of distinction that becomes visible when Book I is reconstructed over an explicit axiomatic interface.

## 23.8 Relationship with Earlier Propositions

### 23.8.1 Proposition I.8

In Euclid's proof, Proposition I.8 is the final congruence theorem. Once the auxiliary triangle has been constructed with the required three side lengths, SSS identifies the transported angle.

In the formal Hilbert proof I.8 is not used.

### 23.8.2 Proposition I.22

Euclid uses Proposition I.22 to construct the auxiliary triangle with prescribed sides.

The formal Hilbert proof again does not use it. Reintroducing I.22 would reconstruct Euclid's historical dependency chain on top of a foundation that already contains the stronger angle-construction principle.

This is especially noteworthy because I.22 required explicit continuity machinery in the reconstruction, whereas I.23 requires none.

### 23.8.3 Book I Order versus Formal Dependency

The file for Proposition I.23 naturally follows Proposition I.22 in the Book I sequence.

The logical dependency graph, however, does not contain an edge

I.22  $\longrightarrow$  I.23.

The textual order of the *Elements* and the dependency order induced by Hilbert geometry therefore diverge at this point.

## 23.9 Hilbert and Book Zero Components Used

### 23.9.1 HilbertCongruence.angle\_\_construction

This is the central ingredient of the proof.

It already contains the existence of the required point, the side condition, the angle congruence, and a uniqueness property stronger than the conclusion of I.23.

### 23.9.2 HilbertSameSide

The same-side relation makes explicit which of the two mirror-image constructions is intended.

The point  $S$  serves as a side selector, and the constructed point  $F$  is required to lie on the same side of the supporting line.

### 23.9.3 HilbertSameRay

The uniqueness clause of Hilbert III.4 is naturally expressed in terms of equality of rays. In the project, rays are represented through points together with the relation `HilbertSameRay` rather than as a separate primitive type.

### 23.9.4 No New Book Zero Lemma

Proposition I.23 exposes no missing low-level operation.

Unlike several preceding propositions, it does not require a new Book Zero theorem. The needed construction already exists at the Hilbert congruence level.

This negative result is structurally useful: a Book I proposition should not generate helper infrastructure merely because it appears as a separate proposition in Euclid.

## 23.10 Uniqueness and the Strength of the Hilbert Statement

Euclid asks for a construction.

Hilbert III.4 gives existence together with uniqueness on the selected side of the supporting line.

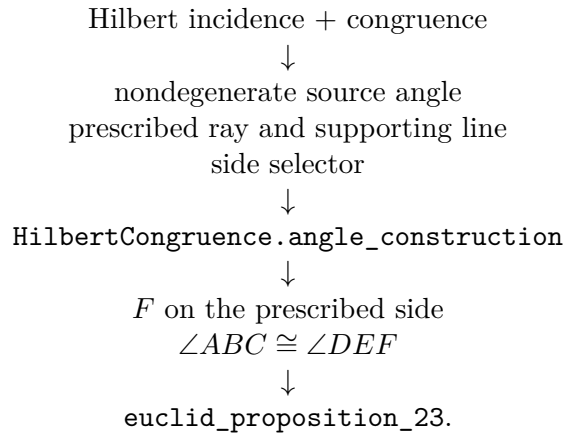
Suppose another point  $F'$  lies on the selected side and determines an angle congruent to the same source angle. The Hilbert uniqueness clause says that the ray from the new vertex through  $F'$  coincides with the ray through  $F$ .

Thus the formal public theorem intentionally forgets information.

This is a useful interface-design principle: the theorem representing Euclid I.23 should state exactly the Euclidean conclusion, while the stronger foundational theorem remains available separately.

## 23.11 Dependency Summary

The formal dependency structure is



No use is made of

I.22,      I.8,

or of a new continuity argument.

No new geometric axiom and no new Book Zero lemma are introduced.

## 23.12 Formalization Notes

Several features of the formal statement deserve emphasis.

First, the point  $S$  is additional explicit data. It selects one side of the line containing  $D$  and  $E$ . Without this information there are two mirror-image rays satisfying the angle-congruence requirement.

Second, the conditions that  $D$  and  $E$  lie on the base line and that  $D \neq E$  ensure that the prescribed initial ray is nondegenerate.

Third, the hypothesis that  $A, B, C$  are noncollinear ensures that the source angle is nondegenerate.

Fourth, the theorem returns a point  $F$ , not an abstract ray. This matches the point-based representation of rays used throughout the Hilbert layer.

Finally, the short proof should not be interpreted as evidence that Proposition I.23 is mathematically trivial. Rather, its mathematical content has already been absorbed into the axiomatic foundation.

## 23.13 Conclusion

Proposition I.23 provides a particularly clean example of the difference between historical construction order and formal axiomatic dependency.

Euclid constructs the transported angle indirectly. Proposition I.22 produces a triangle with prescribed side lengths, and Proposition I.8 identifies the desired angle by SSS.

Hilbert places the corresponding construction directly in the congruence axioms. The formal reconstruction can therefore obtain Euclid's conclusion immediately from III.4.

The important result is not merely that the Lean proof is short. The reconstruction identifies exactly *why* it is short: a theorem in Euclid has become part of the foundational interface in Hilbert geometry.

This also explains why Proposition I.23 introduces no new Book Zero machinery. The proposition does not reveal a missing elementary operation; instead, it reveals a change in foundational organization.

The completed formal theorem therefore preserves Euclid's visible Book I milestone while recording a substantially different dependency graph underneath it.

## Chapter 24

# Proposition I.24

### 24.1 Introduction

Euclid's Proposition I.24 is the classical hinge theorem. Two triangles have two corresponding pairs of congruent sides. If the included angle of the first triangle is greater than the included angle of the second, then the side opposite the greater angle is greater than the corresponding side of the second triangle.

In the notation of the formal development, the hypotheses are

$$AB \cong DE, \quad AC \cong DF,$$

and

$$\angle EDF < \angle BAC.$$

The conclusion is

$$EF < BC.$$

The reconstruction is entirely synthetic. No numerical length function, coordinate system, or numerical angle measure is introduced. Proposition I.24 is nevertheless substantially more delicate than its statement suggests. The classical diagram suppresses a genuine case distinction in the position of the auxiliary point. The Lean reconstruction makes this order-theoretic issue explicit and proves all possible configurations.

This is also a historically significant feature of I.24. The proof printed in the traditional text of Euclid treats only one of three possible configurations. Proclus supplied arguments for the omitted configurations, and later editors and commentators discussed alternative repairs. Thus the case split forced by the formal reconstruction is not an artifact of Lean: it reflects a classical issue in the textual and mathematical history of the proposition.

### 24.2 The Classical Configuration

Let  $ABC$  and  $DEF$  be nondegenerate triangles satisfying

$$AB \cong DE, \quad AC \cong DF, \quad \angle EDF < \angle BAC.$$

Euclid constructs at  $D$  an angle congruent to the larger angle at  $A$  and lays off a segment  $DG$  congruent to  $AC$  (and hence to  $DF$ ). Thus

$$\angle EDG \cong \angle BAC, \quad DG \cong AC \cong DF.$$

By SAS,

$$\triangle ABC \cong \triangle DEG,$$

so that

$$BC \cong EG.$$

It remains to prove

$$EF < EG.$$

The historical difficulty is best seen by displaying all three configurations distinguished in modern commentary. Following Wilkins, the distinction is made by the positions of  $D$  and  $E$  relative to the line through  $G$  and  $F$ .

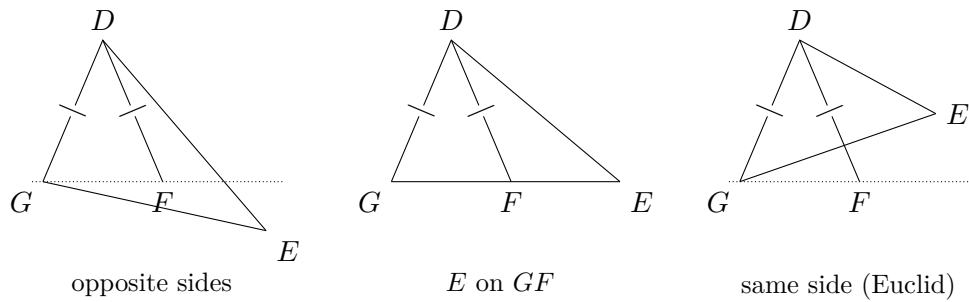


Figure 24.1: The three configurations distinguished in the classical discussion of Proposition I.24. Euclid's printed proof treats only the third.

The equal marks on  $DG$  and  $DF$  record the isosceles triangle used in the classical argument. The three diagrams are not three Lean branches; they belong to the historical analysis of Euclid's construction.

## 24.3 Classical Proof

Euclid's argument proceeds as follows. Since

$$AB \cong DE, \quad AC \cong DG, \quad \angle BAC \cong \angle EDG,$$

Proposition I.4 gives

$$BC \cong EG.$$

Also

$$DF \cong DG,$$

so triangle  $DFG$  is isosceles. Proposition I.5 therefore supplies the corresponding equality of its base angles. In the configuration displayed by Euclid, this angle equality, together with the position of  $E$ , yields a strict comparison between the two angles of triangle  $EFG$ . Proposition I.19 then gives

$$EF < EG.$$

Since  $EG \cong BC$ , one obtains

$$EF < BC.$$

The classical dependency pattern is therefore

$$\begin{array}{c}
 \angle EDF < \angle BAC \\
 \downarrow \\
 \text{construct } G : \quad DG \cong AC, \quad \angle EDG \cong \angle BAC \\
 \downarrow \text{ I.4} \\
 EG \cong BC \\
 \downarrow \\
 DF \cong DG \\
 \downarrow \text{ I.5} \\
 \text{angle comparison in } \triangle EFG \\
 \downarrow \text{ I.19} \\
 EF < EG \\
 \downarrow \\
 EF < BC.
 \end{array}$$

The problem is not the metric or congruence part of this argument. The problem is the unproved assumption about the relative position of the auxiliary points.

## 24.4 The Classical Configuration Gap

Modern commentary makes the difficulty precise. After the construction used in I.24, three configurations are possible. In Wilkins's formulation they are distinguished by the position of the relevant points with respect to the line through the two auxiliary points. The proof given by Euclid covers only one of these configurations; Proclus supplied proofs for the other two.

This point was also discussed by later editors. Simson modified the setup by adding a restriction on which of the two equal sides is chosen in the construction, with the aim of avoiding the three-case analysis. Heath's commentary records both the added condition and the underlying issue.

The historical point should therefore be stated carefully. Hilbert did not first discover the missing cases of I.24. They were already known in the ancient commentary tradition. What the Hilbert axiomatic framework provides is a language in which the relevant incidence, separation, and betweenness information must be stated and proved rather than inferred from a picture.

The Lean reconstruction follows this axiomatic discipline. It does not assume that the auxiliary point occupies the position suggested by one particular diagram.

For the third configuration, which is the one explicitly treated by Euclid, Wilkins makes the angle comparison visually explicit. Since

$$DG \cong DF,$$

the base angles of the isosceles triangle  $DGF$  are congruent, and the displayed configuration gives the chain

$$\angle EFG > \angle DFG = \angle DGF > \angle EGF.$$

Proposition I.19 then yields  $EG > EF$ .



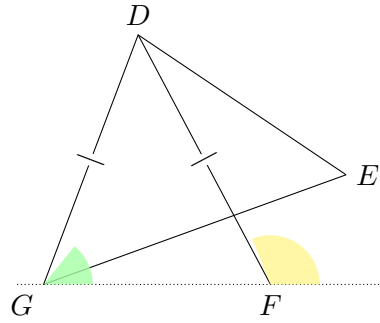


Figure 24.2: Euclid's third configuration, with the two angle-comparison regions highlighted schematically. The colours are explanatory only; the formal proof uses synthetic angle order, not numerical angle measure.

## 24.5 Strict Angle and Segment Comparison

Two synthetic order relations are central to the reconstruction.

The relation `HilbertAngleLess` expresses strict comparison of angles without introducing numerical angle measure. Informally,

$$\alpha < \beta$$

means that a congruent copy of  $\alpha$  occurs as a proper interior subangle of  $\beta$ .

Similarly, `HilbertSegmentLess` expresses strict comparison of segments through betweenness and congruence, without introducing a real-valued length function.

Proposition I.24 combines these two order theories. Its hypothesis is a strict angle inequality, while its conclusion is a strict segment inequality. The hinge construction is the bridge between them.

## 24.6 Proof Architecture of the Reconstruction

The formal proof separates naturally into seven stages.

### 24.6.1 The SAS Construction

The helper `euclid_proposition_24_sas_aux` starts from

$$AB \cong DE, \quad AC \cong DF, \quad \angle EDF < \angle BAC.$$

The strict angle comparison supplies an interior ray of  $\angle BAC$ . A point  $H$  is laid off on this ray so that

$$AH \cong AC.$$

Since  $AC \cong DF$ , congruence transport gives

$$AH \cong DF.$$

The auxiliary angle at  $A$  is congruent to  $\angle EDF$ , and SAS applied to  $ABH$  and  $DEF$  yields

$$BH \cong EF.$$

Thus the original goal  $EF < BC$  is reduced to the hinge comparison

$$BH < BC.$$

### 24.6.2 The Hinge Configuration

The interior ray through  $H$  meets the open segment  $BC$  at a point  $Y$ . The same-ray information does not determine the order of  $H$  and  $Y$  on the ray from  $A$ . The formal proof therefore uses same-ray trichotomy and obtains exactly three possibilities:

$$A - H - Y, \quad H = Y, \quad A - Y - H.$$

This is the central order-theoretic decomposition of the reconstruction.

It is important not to identify these three Lean cases with the three historical configurations above. The historical split concerns the position of  $D$  and  $E$  relative to the line  $GF$ . The Lean split concerns the order of  $H$  and  $Y$  on a single ray from  $A$  after the auxiliary hinge point has already been constructed.

The formal geometry can be displayed independently of the historical diagram:

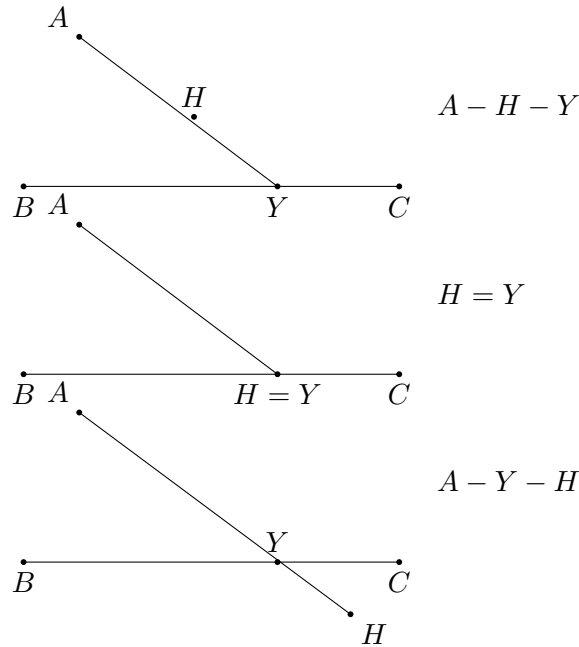


Figure 24.3: The three order configurations used by the Lean hinge helper. In all three,  $Y$  lies in the open segment  $BC$ ; the split concerns only the order of  $H$  and  $Y$  on their common ray from  $A$ .

#### 24.6.3 Case 1: $A - H - Y$

This is the longer nontrivial branch. After reconstructing the necessary noncollinearity facts, Proposition I.16 is used to obtain strict exterior-angle comparisons. The relation

$$AH \cong AC$$

makes triangle  $AHC$  isosceles, allowing the relevant angles at  $H$  and  $C$  to be transported across congruence.

The point  $Y$  lies between  $B$  and  $C$ , so the appropriate ray from  $H$  meets the opposite side internally. The theorem `hilbert_interior_angle_less` turns this incidence fact into a strict

angle comparison. After the necessary same-ray normalizations and transitivity steps, the proof obtains

$$\angle BCH < \angle BHC.$$

Proposition I.19 applied to triangle  $BHC$  now gives

$$BH < BC.$$

#### 24.6.4 Case 2: $H = Y$

If the auxiliary point is itself the intersection point with  $BC$ , then

$$B - H - C.$$

The definition of strict segment comparison immediately gives

$$BH < BC.$$

No angle comparison is required.

#### 24.6.5 Case 3: $A - Y - H$

This branch has a different geometry. At  $C$ , the ray through  $Y$  meets the open segment  $HA$ , so

$$\angle HCY < \angle HCA.$$

Because  $C - Y - B$ , this comparison can be transported from the ray  $CY$  to the ray  $CB$ .

Again  $AH \cong AC$  makes triangle  $AHC$  isosceles. The base-angle congruence transfers the comparison to the corresponding angle at  $H$ . At  $H$ , the ray through  $Y$  meets the open segment  $CB$ , giving a second proper interior-subangle comparison. Since  $H - Y - A$ , the ray  $HY$  agrees with the ray  $HA$ . Transitivity and angle-congruence transport therefore yield

$$\angle BCH < \angle BHC.$$

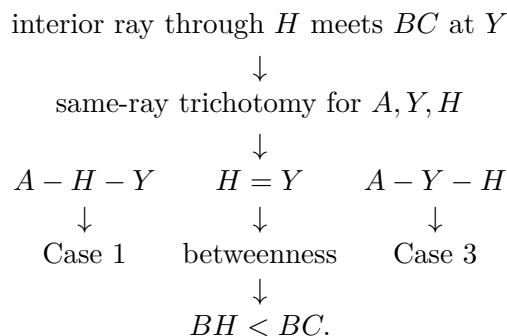
A final application of I.19 gives

$$BH < BC.$$

The two nontrivial cases therefore reach the same final comparison through different angle-order paths.

#### 24.6.6 The Hinge Helper

The theorem `euclid_proposition_24_hinge_aux` packages the complete case analysis:



No location of  $H$  is inferred from the diagram.

### 24.6.7 Completion of I.24

The SAS helper supplies

$$BH \cong EF,$$

and the hinge helper supplies

$$BH < BC.$$

Strict segment comparison is transported through congruence, giving

$$EF < BC.$$

This is exactly the conclusion of Proposition I.24.

## 24.7 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_24
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E F : Geo.Point)
  (hABC : Not (PrimCollinear Geo A B C))
  (hDEF : Not (PrimCollinear Geo D E F))
  (hAB_DE : Geo.Congruent A B D E)
  (hAC_DF : Geo.Congruent A C D F)
  (hAngle :
    HilbertAngleLess Geo E D F B A C) :
  HilbertSegmentLess Geo E F B C := by
  ...
```

The first major helper packages the SAS construction. Its output contains the auxiliary ray and point together with the two metric facts needed later:

$$AH \cong AC, \quad BH \cong EF.$$

Schematically, the public proof begins by unpacking this construction:

```
rcases
  euclid_proposition_24_sas_aux
    (Geo := Geo)
    A B C D E F
    hABC hDEF hAB_DE hAC_DF hAngle with
  <X, H, hMeet, hRayAXH, hAH_AC, hBH_EF>
```

The hinge comparison is then delegated to the second major helper:

```
have hBH_BC :
  HilbertSegmentLess Geo B H B C :=
  euclid_proposition_24_hinge_aux
    (Geo := Geo)
    A B C H X
    hABC
```

```

hMeet
hRayAXH
hAH_AC

```

Inside that helper the intersection witness is extracted and the point order is split exhaustively:

```

rcases hMeet with <Y, hBYC, hRayAXY>

rcases
  euclid_proposition_22_sameRay_trichotomy
    (Geo := Geo)
    A Y H hAY hAH hRayAYH with
  hAHY | hYH | hAYH

```

The three branches are then dispatched to Case 1, the immediate betweenness case, and Case 3. The public theorem contains none of this order analysis; it sees only the reusable result  $BH < BC$ .

Finally the segment inequality is transported from  $BH$  to the congruent segment  $EF$ . Thus the public proof is short because the difficult geometry has been isolated in named synthetic helpers.

## 24.8 Relationship with Earlier Propositions

### 24.8.1 Proposition I.5

The isosceles base-angle theorem is used repeatedly after

$$AH \cong AC.$$

It is the main congruence bridge between the angle at  $C$  and the angle at  $H$  in the hinge triangle.

### 24.8.2 Proposition I.16

Case 1 uses the exterior-angle theorem to obtain strict angle comparisons which are not available directly from the interior-ray configuration. This is the principal reason that Case 1 is longer than Case 3.

### 24.8.3 Proposition I.19

Both nontrivial branches end with the same angle inequality in triangle  $BHC$ :

$$\angle BCH < \angle BHC.$$

Proposition I.19 converts this angle comparison into the required side comparison

$$BH < BC.$$

Thus I.19 is the final metric step of the hinge argument.

### 24.8.4 Proposition I.23

The classical Euclidean proof uses I.23 to construct at  $D$  an angle equal to the given larger angle. In the Hilbert reconstruction the corresponding construction is expressed through the already available Hilbert angle construction machinery. Proposition I.23 is therefore part of the classical dependency picture, while the formal proof uses the more general interface theorem behind that construction.

## 24.9 Hilbert and Book Zero Components Used

### 24.9.1 HilbertAngleLess

Strict angle comparison is the input order relation of I.24. It also provides the interior-ray witness needed for the auxiliary construction.

### 24.9.2 HilbertSegmentLess

Strict segment comparison is the conclusion of I.24. Its witness-based form permits the middle case  $H = Y$  to close directly from betweenness.

### 24.9.3 HilbertSameRay

Same-ray machinery is used both to represent the auxiliary interior ray and to normalize angles when two syntactically different rays determine the same geometric direction.

### 24.9.4 Same-ray trichotomy

The trichotomy is the formal mechanism that exposes the hidden case split. It produces the exhaustive alternatives

$$A - H - Y, \quad H = Y, \quad A - Y - H.$$

This is the order-theoretic core of the reconstruction.

### 24.9.5 hilbert\_interior\_angle\_less

This theorem turns a ray meeting the opposite segment internally into a strict comparison between the resulting subangle and the whole angle. It is used repeatedly in the two nontrivial hinge cases.

### 24.9.6 Angle-congruence transport

The formal proof must explicitly transport strict inequalities across congruent angles and across coincident rays. These steps replace informal phrases such as “this is the same angle” which are normally read directly from a diagram.

## 24.10 Historical Note: Euclid, Proclus, Simson, and Hilbert

Proposition I.24 is a useful example of the distinction between a classical geometric argument and a complete axiomatic treatment of its diagram.

The traditional Euclidean proof presents one configuration. As emphasized in modern study notes by David R. Wilkins, three configurations can arise, and Euclid's printed argument covers only one of them. Proclus supplied proofs for the two omitted configurations, and later commentators reproduced or reorganized these supplements.

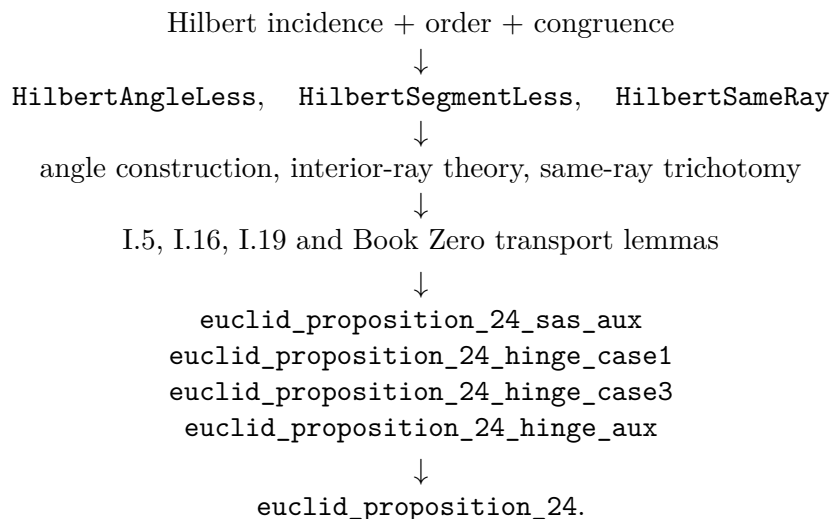
Robert Simson pursued a different repair. He added a condition governing which of the two equal sides is to be used in the construction, thereby seeking to force the auxiliary point into the configuration required by the standard proof. Heath records this modification in his commentary on I.24.

Hilbert's significance is different. His axiomatic treatment of order and incidence supplies a framework in which such positional assumptions cannot remain implicit. Betweenness, collinearity, ray identity, and separation must be justified formally. The Lean reconstruction inherits precisely this discipline.

Accordingly, the three-way split in the present proof should not be read as a peculiarity of proof-assistant programming. It is a machine-checked form of a genuine classical issue in Proposition I.24.

## 24.11 Dependency Summary

The formal dependency structure is



No numerical metric, numerical angle measure, circle continuity, or additional hinge axiom is introduced.

## 24.12 Conclusion

Proposition I.24 is one of the clearest examples so far of what formal reconstruction can reveal about a classical synthetic proof.

The metric core is simple. The auxiliary SAS construction gives

$$BH \cong EF,$$

and the final application of Proposition I.19 converts an angle comparison into

$$BH < BC.$$

The real difficulty is positional: the auxiliary point does not have a single order relation forced by the hypotheses.

The Lean proof therefore separates the possibilities

$$A - H - Y, \quad H = Y, \quad A - Y - H$$

and proves each one. This exhaustive split corresponds to a difficulty already recognized in the classical commentary tradition. The formal proof does not replace Euclid's synthetic geometry; it supplies the order and incidence information that the traditional diagram leaves implicit.

The completed development compiles cleanly and contains no `sorry`. Proposition I.24 introduces no new geometric axiom. It uses the Hilbert framework, the Book Zero proof language, and the synthetic results established earlier in Book I.



## Chapter 25

# Proposition I.25

### 25.1 Introduction

Euclid's Proposition I.25 is the converse comparison theorem to Proposition I.24.

Suppose two triangles have two corresponding sides congruent:

$$AB \cong DE, \quad AC \cong DF.$$

If the third side of the first triangle is greater than the third side of the second,

$$EF < BC,$$

then the included angle of the first triangle is greater:

$$\angle EDF < \angle BAC.$$

The formal reconstruction is strikingly short. Unlike Proposition I.24, no auxiliary geometric construction is needed inside the proposition itself. The proof is an elimination argument based on the previously established trichotomy for strict angle comparison.

The difficult geometric work needed for I.25 has already been absorbed into the Hilbert interface, in particular into `angle_trichotomy`. Once that theorem is available, Proposition I.25 becomes almost purely logical.

### 25.2 The Classical Configuration

Let  $\triangle ABC$  and  $\triangle DEF$  satisfy

$$AB \cong DE, \quad AC \cong DF,$$

and suppose

$$EF < BC.$$

The conclusion is

$$\angle EDF < \angle BAC.$$

No new point is constructed in Proposition I.25. The same two triangles are compared throughout the proof.

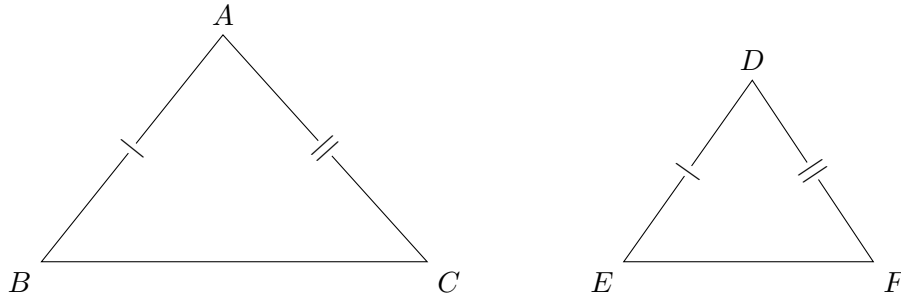


Figure 25.1: The configuration of Proposition I.25.

### 25.3 Classical Proof

Euclid excludes the two alternatives incompatible with the required conclusion.

First suppose

$$\angle BAC \cong \angle EDF.$$

Then Proposition I.4 (SAS), applied to the two pairs of congruent sides, gives

$$BC \cong EF,$$

contradicting  $EF < BC$ .

Next suppose

$$\angle BAC < \angle EDF.$$

Apply Proposition I.24 with the two triangles interchanged. Since the included angle of  $\triangle DEF$  is greater, I.24 gives

$$BC < EF,$$

again contradicting  $EF < BC$ .

Therefore the only remaining possibility is

$$\angle EDF < \angle BAC.$$

### 25.4 Angle Trichotomy

The decisive interface result is `angle_trichotomy`. For two nondegenerate angles it provides exactly the three alternatives

$$\angle BAC \cong \angle EDF, \quad \angle BAC < \angle EDF, \quad \angle EDF < \angle BAC.$$

This theorem was established in the Hilbert layer immediately before the formalization of I.25. Its proof uses Hilbert angle construction, same-side information, the order of rays in a half-plane, and the principle that two collinear points lying on the same side of a line through the vertex determine the same ray.

Thus angle trichotomy is not an additional axiom introduced for I.25. It is a derived structural theorem about the already defined relation `HilbertAngleLess`.

## 25.5 Proof Architecture of the Reconstruction

The formal proof is a three-branch elimination argument. Its complete architecture is short enough to display directly in ASCII:

```
angle_trichotomy
|
+-- angles congruent
|   -> SAS
|   -> BC ~= EF
|   -> contradiction with EF < BC
|
+-- BAC < EDF
|   -> Proposition I.24
|   -> BC < EF
|   -> contradiction with EF < BC
|
+-- EDF < BAC
    -> goal
```

This is not merely a summary of the Lean tactics. It is the actual mathematical dependency structure of the reconstructed proof.

The first branch is eliminated by triangle congruence. The second is eliminated by the direct hinge theorem I.24. The third branch is already the desired statement.

Consequently, after angle trichotomy has been established, I.25 contains no hidden construction and no additional geometric case analysis.

## 25.6 Hilbert Reconstruction

### 25.6.1 The Congruent-Angle Branch

Assume

$$\angle BAC \cong \angle EDF.$$

Together with

$$AB \cong DE, \quad AC \cong DF,$$

Hilbert SAS yields

$$BC \cong EF.$$

The interface theorem `hilbert_segmentLess_not_congruent` states that a strictly shorter segment cannot be congruent to the longer one. Hence this branch is impossible.

### 25.6.2 The Wrong Strict-Inequality Branch

Assume

$$\angle BAC < \angle EDF.$$

Interchange the roles of the triangles and reverse the side congruences:

$$DE \cong AB, \quad DF \cong AC.$$

Then Proposition I.24 gives

$$BC < EF.$$

But the original hypothesis is

$$EF < BC.$$

The asymmetry theorem `hilbert_segmentLess_asymm` excludes these simultaneous strict inequalities.

### 25.6.3 The Desired Branch

The remaining case is exactly

$$\angle EDF < \angle BAC,$$

so no further argument is needed.

## 25.7 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_25
  [HilbertCongruence Geo]
  (A B C D E F : Geo.Point)
  (hABC : not (PrimCollinear Geo A B C))
  (hDEF : not (PrimCollinear Geo D E F))
  (hAB_DE : Geo.Congruent A B D E)
  (hAC_DF : Geo.Congruent A C D F)
  (hEF_BC : HilbertSegmentLess Geo E F B C) :
  HilbertAngleLess Geo E D F B A C
```

The proof begins by invoking `angle_trichotomy` on the two included angles.

In the equality branch, `SAS` produces a `TriangleCongruenceResult`; its field `sideBC` gives  $BC \cong EF$ . After reversing the congruence, the contradiction is discharged by `hilbert_segmentLess_not_congruent`.

In the second branch, Proposition I.24 is called with the two triangles interchanged. This gives  $BC < EF$ , contradicting the hypothesis through `hilbert_segmentLess_asymm`.

The third branch is returned directly.

## 25.8 Relationship with Earlier Propositions

### 25.8.1 Proposition I.4

Proposition I.4 supplies the SAS criterion used to eliminate the equality branch:

$$\angle BAC \cong \angle EDF, \quad AB \cong DE, \quad AC \cong DF \implies BC \cong EF.$$

### 25.8.2 Proposition I.24

Proposition I.24 is the central direct dependency of I.25. I.24 states that, under two pairs of congruent sides, a greater included angle implies a greater third side. I.25 reverses the comparison: a greater third side implies a greater included angle.

The formal proof of I.25 uses I.24 exactly once, to eliminate the wrong direction of angle inequality.

Together they give, under

$$AB \cong DE, \quad AC \cong DF,$$

the order correspondence

$$\angle EDF < \angle BAC \iff EF < BC.$$

## 25.9 Hilbert and Book Zero Components Used

### 25.9.1 angle\_trichotomy

This is the principal structural result exposed by I.25. It converts the comparison of two nondegenerate angles into an exhaustive three-way split: congruent, smaller in one direction, or smaller in the other direction.

### 25.9.2 SAS

The existing SAS theorem converts equality of the included angles and the two side congruences into congruence of the third sides.

### 25.9.3 hilbert\_segmentLess\_not\_congruent

This theorem expresses incompatibility between strict segment order and segment congruence. It eliminates the first trichotomy branch.

### 25.9.4 hilbert\_segmentLess\_asymm

This theorem expresses asymmetry of strict segment comparison:

$$AB < CD \implies \neg(CD < AB).$$

It eliminates the second branch after I.24 produces the opposite inequality.

## 25.10 Formalization Notes

Proposition I.25 provides a useful contrast with I.24.

I.24 required substantial geometric work. Its formal reconstruction had to expose point-order cases hidden by Euclid's diagram and develop the corresponding hinge argument in each configuration.

I.25 does not repeat any of that geometry. Instead, the reconstruction first strengthens the reusable Hilbert interface by proving angle trichotomy. Once this has been done, I.25 is a direct logical consequence of

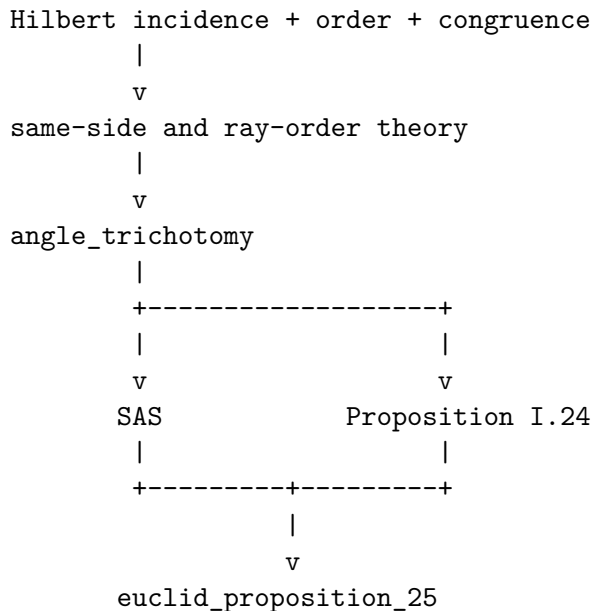
angle trichotomy,      SAS,      I.24.

This separation is important for the architecture of the project. A long foundational or order-theoretic argument should not be buried inside every later Euclidean proposition that needs it. Once identified, it becomes part of the intermediate synthetic language and later proofs become correspondingly transparent.

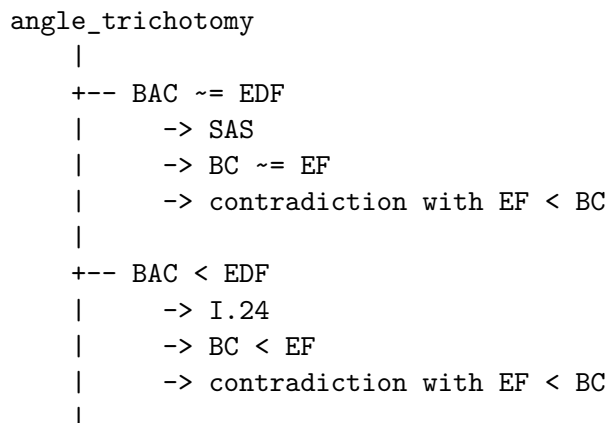
I.25 is therefore a particularly clean example of the role of the Hilbert interface: difficult geometric infrastructure is proved once, after which a classical theorem can recover its natural short proof.

## 25.11 Dependency Summary

The underlying dependency graph is



At the level of the final theorem:



```

+-- EDF < BAC
  -> exact

```

No new geometric axiom, continuity principle, coordinate system, or numerical angle measure is introduced.

## 25.12 Conclusion

Proposition I.25 is the converse comparison theorem to Proposition I.24.

Its formal reconstruction is exceptionally concise, but the concision is not accidental. The theorem becomes short only after the underlying angle-order theory has been completed.

The proof begins with angle trichotomy. Congruence of the included angles is excluded by SAS, because it would force congruence of the third sides. The wrong strict inequality is excluded by Proposition I.24, because it would force the opposite strict comparison of the third sides. The remaining branch is precisely the desired conclusion.

Thus I.25 gives a particularly clear instance of the architecture

foundational Hilbert theory  $\longrightarrow$  reusable synthetic interface  $\longrightarrow$  short Euclidean theorem.

## Chapter 26

# Proposition I.26

### 26.1 Introduction

Euclid's Proposition I.26 contains two triangle-congruence criteria.

For two nondegenerate triangles  $ABC$  and  $DEF$ , assume the two corresponding angles

$$\angle ABC \cong \angle DEF, \quad \angle BCA \cong \angle EFD.$$

Euclid then considers two possibilities for the given corresponding side.

1. In the ASA case,

$$BC \cong EF.$$

2. In the SAA case,

$$AB \cong DE.$$

In either case the conclusion is full triangle congruence:

$$AB \cong DE, \quad BC \cong EF, \quad AC \cong DF,$$

together with

$$\angle BAC \cong \angle EDF.$$

The two parts have very different formal behavior.

The ASA part is almost entirely absorbed by the existing Hilbert congruence theory. The SAA part is not. Its proof still requires the essential contradiction argument of Euclid: a hypothetical strict inequality of the corresponding sides is converted into an auxiliary SAS configuration and then contradicted by Proposition I.16.

Thus Proposition I.26 provides a useful boundary case. One half is already available as interface-level geometry; the other still exposes genuine Book I proof structure.

### 26.2 Classical Configurations

Wilkins emphasizes that Proposition I.26 establishes both the ASA and the SAA congruence rules. The same basic pair of triangles is used for both arguments, but the auxiliary point lies on a different side in the two cases.



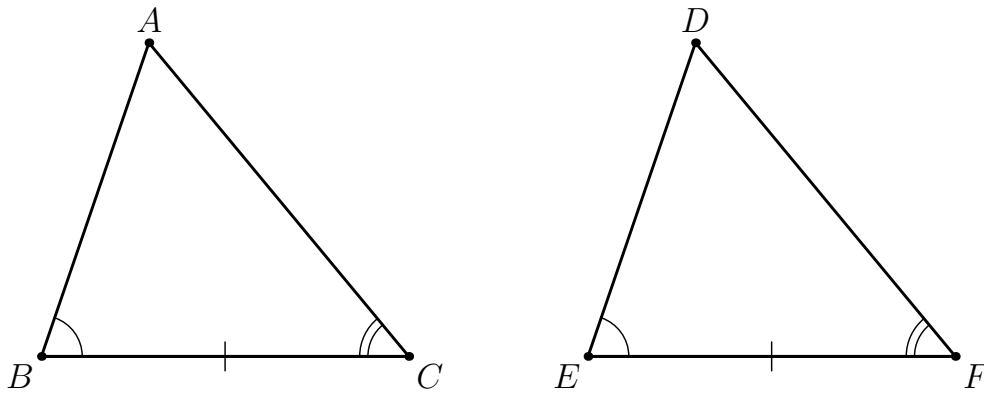


Figure 26.1: The common input configuration for Proposition I.26. The two pairs of corresponding angles are marked by one and two arcs respectively, and in the ASA case the corresponding side  $BC \cong EF$  is marked by a single tick.

### 26.2.1 ASA auxiliary configuration

Assume first

$$BC \cong EF.$$

Euclid supposes, for contradiction, that the corresponding sides  $AB$  and  $DE$  are unequal. If  $AB$  is the greater side, choose a point  $G$  between  $A$  and  $B$  such that

$$GB \cong DE,$$

and join  $G$  to  $C$ .

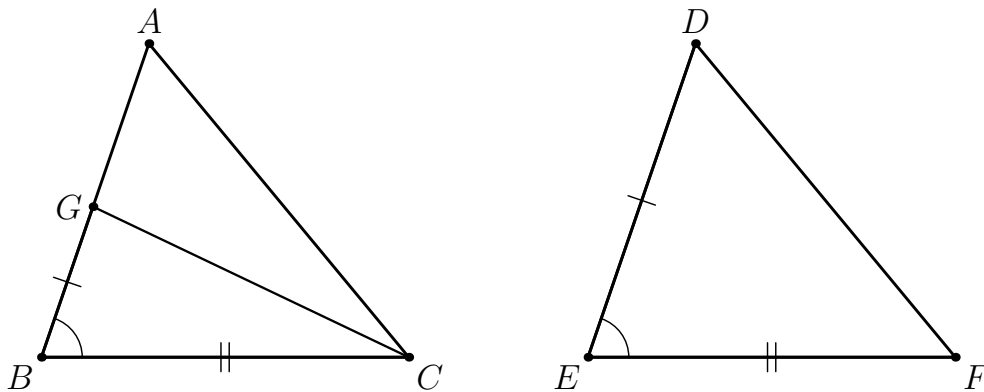


Figure 26.2: Euclid's ASA reductio configuration. The point  $G$  lies between  $A$  and  $B$ . The diagram marks exactly the local SAS data  $GB \cong DE$ ,  $BC \cong EF$ , and  $\angle GBC \cong \angle DEF$ .

SAS gives

$$\angle BCG \cong \angle EFD.$$

Together with the original hypothesis

$$\angle EFD \cong \angle BCA,$$

this would imply

$$\angle BCG \cong \angle BCA.$$

But  $G$  lies inside the side  $AB$ , so  $\angle BCG$  is a proper part of  $\angle BCA$ . This contradiction forces

$$AB \cong DE.$$

A final SAS application to the original triangles gives the remaining side and angle congruences.

### 26.2.2 SAA auxiliary configuration

Now assume instead

$$AB \cong DE.$$

Suppose, for contradiction, that  $EF < BC$ . Then there is a point  $H$  between  $B$  and  $C$  such that

$$BH \cong EF.$$

Join  $A$  to  $H$ .

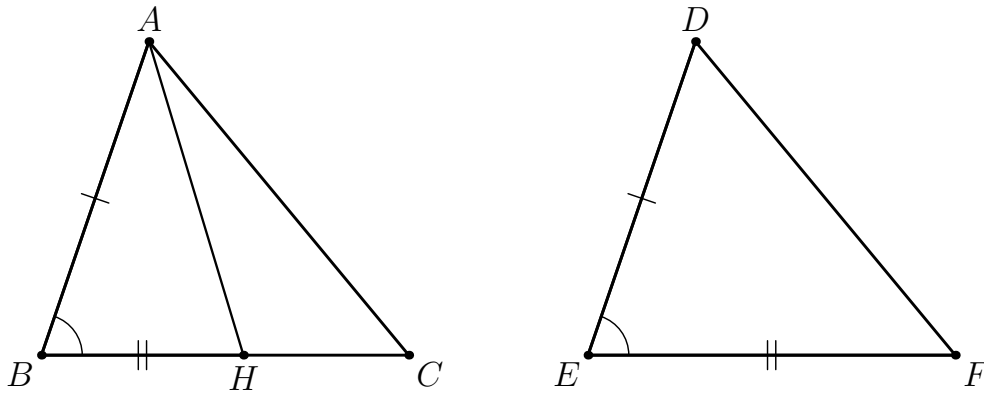


Figure 26.3: Euclid's SAA reductio configuration. The point  $H$  lies between  $B$  and  $C$ . The diagram marks exactly the local SAS data  $AB \cong DE$ ,  $BH \cong EF$ , and  $\angle ABH \cong \angle DEF$ . Proposition I.16 is then applied to  $\triangle AHC$ .

Since  $B, H, C$  are collinear on the same ray from  $B$ ,

$$\angle ABH \cong \angle ABC.$$

Hence SAS on  $\triangle BAH$  and  $\triangle EDF$  gives

$$\angle BHA \cong \angle EFD.$$

Using the original angle hypothesis,

$$\angle EFD \cong \angle BCA,$$

we obtain

$$\angle BHA \cong \angle BCA.$$

On the other hand,  $C - H - B$ , so  $\angle BHA$  is an exterior angle of triangle  $AHC$ . Proposition I.16 gives the strict inequality

$$\angle BCA < \angle BHA,$$

contradicting their congruence.

Thus

$$\neg(EF < BC).$$

Interchanging the two triangles and repeating the same argument gives

$$\neg(BC < EF).$$

Segment trichotomy therefore yields

$$BC \cong EF.$$

The SAA case has now been reduced to the ASA case.

## 26.3 Diagrammatic Audit

Three figures are retained because they have three genuinely different jobs.

Figure 26.1 records the common input configuration. Figure 26.2 isolates the auxiliary SAS configuration used in the ASA reductio. Figure 26.3 isolates the corresponding SAS configuration used in the SAA reductio.

The last two diagrams deliberately do not mark every hypothesis of Proposition I.26. They mark only the congruences used in the local SAS step. This follows the logic of Wilkins’s presentation and avoids overloading a single figure with relations that belong to different stages of the proof.

The auxiliary points are constructed geometrically:  $G$  lies genuinely on the segment  $AB$ , and  $H$  lies genuinely on the segment  $BC$ . Thus the incidence information used by the proof is represented directly rather than being approximated by hand-chosen coordinates.

These are not successive snapshots of one construction. They represent the two different reductio mechanisms contained in Proposition I.26.

## 26.4 Hilbert Reconstruction

The two halves of Proposition I.26 behave differently over the present Hilbert interface.

### 26.4.1 ASA

The theorem `hilbert_asa_sides` already contains the central mathematical content of the ASA case.

The given side in Euclid’s formulation is

$$BC \cong EF.$$

The theorem is applied after cyclically viewing the triangles as

$$\triangle BCA \quad \text{and} \quad \triangle EFD.$$

After the required angle and endpoint reorientations it gives

$$BA \cong ED, \quad CA \cong FD.$$

These are transported to

$$AB \cong DE, \quad AC \cong DF.$$

At that point all three pairs of sides are congruent. The existing `HilbertSSS` theorem supplies the remaining corresponding angle:

$$\angle BAC \cong \angle EDF.$$

Thus the formal ASA proof does not reconstruct Euclid’s reductio.

### 26.4.2 SAA

No permutation of `hilbert_asa_sides` directly solves the SAA problem. With the side  $AB \cong DE$  fixed, ASA would require the angles at  $A$  and  $B$ , while the hypotheses supply the angles at  $B$  and  $C$ .

The formal reconstruction therefore isolates Euclid's essential contradiction argument in the helper

```
euclid_proposition_26_saa_not_less
```

The helper proves

$$\neg(EF < BC).$$

Its architecture is

```
EF < BC
  |
witness H with B-H-C and BH ~= EF
  |
same-ray transport:
angle ABH ~= angle ABC
  |
SAS on BAH and EDF
  |
angle BHA ~= angle EFD ~= angle BCA
  |
I.16 on triangle ACH
  |
angle BCA < angle BHA
  |
contradiction
```

The same helper is then called again with the two triangles interchanged, producing

$$\neg(BC < EF).$$

Book Zero segment trichotomy yields

$$BC \cong EF,$$

and the public SAA theorem finishes by calling the already proved ASA case.

## 26.5 Proof Architecture of the Reconstruction

The two formal paths can be summarized separately.

I.26 ASA

-----

```

two angle congruences
+ BC ~= EF
  |
  v
hilbert_asa_sides
on BCA and EFD
  |
  v
AB ~= DE
AC ~= DF
  |
  v
HilbertSSS
  |
  v
angle BAC ~= angle EDF

```

## I.26 SAA

```

-----

```

```

two angle congruences
+ AB ~= DE
  |
  +-----+
  |               |
  v               v
not (EF < BC)    not (BC < EF)
SAS + I.16      same helper,
                triangles swapped
  |               |
  +-----+-----+
                |
                v
bookZero_31_trichotomy1
  |
  v
      BC ~= EF
  |
  v
      I.26 ASA case
  |
  v
AC ~= DF
angle BAC ~= angle EDF

```

The contrast with Euclid is precise.

For ASA, the Hilbert interface has already absorbed the reductio into a general theorem.

For SAA, Proposition I.16 remains a genuine formal dependency.

## 26.6 Lean Formalization

The complete implementation is available in the repository. Only the decisive interface steps are reproduced here.

For ASA, the first substantial call is

```
have hASA :=
  hilbert_asa_sides
    Geo
    B C A
    E F D
    hBCA
    hEFD
    hBC
    hAngleB'
    hAngleC
```

After endpoint reversal this gives the two missing sides. The final angle is then obtained from

```
have hSSS :=
  HilbertSSS
    Geo
    A B C
    D E F
    hABC
    hAB
    hBC
    hAC
```

For SAA, the public proof first excludes both strict segment inequalities:

```
have hNotEF_BC :=
  euclid_proposition_26_saa_not_less
    Geo A B C D E F
    hABC hDEF hAngleB hAngleC hAB
```

The same helper is applied with the two triangles interchanged. Segment trichotomy then produces the missing side congruence, and the theorem reduces to the already established ASA case.

This organization keeps the long exterior-angle argument inside one named helper rather than duplicating it in the public theorem.

## 26.7 Relationship with Earlier Propositions

### 26.7.1 Proposition I.4

Euclid uses SAS in both halves of I.26.

In the ASA proof, SAS is used after cutting off a segment congruent to the corresponding side of

the second triangle.

In the SAA proof, SAS is applied to the auxiliary triangle  $BAH$  and  $\triangle EDF$ .

The formal SAA helper preserves this use of SAS directly.

### 26.7.2 Proposition I.16

Proposition I.16 is the decisive difference between the ASA and SAA arguments.

The SAA helper requires the exterior-angle inequality

$$\angle BCA < \angle BHA.$$

This strict inequality contradicts the angle congruence obtained from SAS. Without I.16 the helper would not close.

### 26.7.3 Segment trichotomy

The formal SAA proof separates the two possible strict segment inequalities:

$$EF < BC \quad \text{and} \quad BC < EF.$$

After both are excluded, `bookZero_31_trichotomy1` returns the remaining possibility

$$BC \cong EF.$$

Thus Book Zero supplies the final order-theoretic bridge from the two contradiction arguments to the ASA case.

## 26.8 Hilbert and Book Zero Components Used

### 26.8.1 `hilbert_asa_sides`

This is the main interface theorem for the ASA half. It converts a common corresponding side together with the two adjacent angle congruences into the remaining two side congruences.

Its availability explains why the formal ASA proof is much shorter than Euclid's proof.

### 26.8.2 `HilbertSSS`

Once all three corresponding side congruences have been recovered, SSS provides the remaining angle congruence required by the public statement.

### 26.8.3 `euclid_proposition_26_saa_not_less`

This local helper is the formal core of the SAA proof. It packages the classical argument

strict side inequality  $\longrightarrow$  SAS  $\longrightarrow$  angle congruence  $\longrightarrow$  I.16 contradiction.

The helper is reusable under interchange of the two triangles, so the symmetric strict-inequality branch does not duplicate the proof.

#### 26.8.4 bookZero\_31\_trichotomy1

This theorem converts the exclusion of both strict comparisons into segment congruence.

It is the precise formal replacement for the classical conclusion that, if neither segment is greater than the other, the two segments are equal.

### 26.9 Formalization Notes

Proposition I.26 is a particularly useful diagnostic theorem.

The ASA part shows what happens when a classical Euclidean proof has already been absorbed into the lower Hilbert interface. The Book I wrapper is then short and need not reproduce the historical reductio.

The SAA part shows the opposite situation. The available ASA theorem cannot be reached merely by permuting the vertices, because the given side is not the side between the two supplied angles. A genuine argument is still required.

The formal proof therefore mirrors an important part of Euclid's historical structure: it constructs an interior point, applies SAS, and invokes the exterior-angle theorem I.16.

At the same time, the proof is reorganized to expose the reusable logical unit. Instead of proving two symmetric strict-inequality branches independently, one helper is applied twice.

This gives the architecture

classical geometric contradiction  $\longrightarrow$  named helper  $\longrightarrow$  segment trichotomy  $\longrightarrow$  existing ASA interface.

No new geometric axiom, numerical length, coordinate system, or numerical angle measure is introduced.

### 26.10 Dependency Summary

At the classical level:

ASA:

I.3-style segment cut

|

I.4 (SAS)

|

proper-angle contradiction

|



equal corresponding side

|

I.4

|

triangle congruence

SAA:

segment cut

|

I.4 (SAS)

|

I.16

|

contradiction

|

equal corresponding side

|

I.4

|

triangle congruence

At the formal level:

ASA:

hilbert\_asa\_sides

|

v

two missing sides

|

v

HilbertSSS

|

v

euclid\_proposition\_26\_asa

SAA:

SAS + Proposition I.16

|

v

euclid\_proposition\_26\_saa\_not\_less

|

+++ exclude  $EF < BC$

|

+++ exclude  $BC < EF$

|

v

bookZero\_31\_trichotomy1

|

v

```

BC ~= EF
  |
  v
euclid_proposition_26_asa
  |
  v
euclid_proposition_26_saa

```

Thus Proposition I.26 exhibits both modes of the current reconstruction in a single theorem: one half collapses to established interface geometry, while the other still requires a substantial Euclidean argument.

## 26.11 Conclusion

Proposition I.26 completes the elementary triangle-congruence theory of the first part of Book I by establishing the ASA and SAA criteria.

The formal reconstruction makes the distinction between the two cases sharper than the classical presentation.

ASA is already essentially available in the Hilbert interface. The Book I proof only has to orient the triangles correctly, recover the missing sides, and use SSS for the remaining angle.

SAA cannot be reduced by permutation alone. Its proof requires the exterior-angle theorem I.16 to eliminate a hypothetical strict inequality of the corresponding sides. Once both strict inequalities are excluded, Book Zero segment trichotomy reduces SAA to ASA.

The proposition therefore gives a compact example of the layered architecture of the project:

Hilbert interface  $\longrightarrow$  Book Zero order theory  $\longrightarrow$  Euclidean congruence theorem.

## Chapter 27

# Proposition I.27

### 27.1 Introduction

Proposition I.27 is the first proposition in Book I whose conclusion is parallelism. Its mathematical content is the converse direction of the familiar alternate-angle criterion:

$$\text{equal alternate interior angles} \implies \text{parallel straight lines.}$$

No form of Euclid's Fifth Postulate is required. The proposition belongs to neutral geometry: Euclid proves it by showing that any hypothetical intersection of the two lines would contradict Proposition I.16.

The present formalization is especially interesting because the Hilbert development already contains the alternate-angle implication as a general theorem. At the same time, translating the classical transversal diagram into the hypotheses of that theorem exposes additional incidence, order, ray-orientation, and side-of-line information that is suppressed in the traditional picture.

For this reason the Lean file deliberately retains two formulations of Proposition I.27. Their precise proof-theoretic relation is left visible rather than being eliminated by premature refactoring.

### 27.2 Classical Statement

Let a straight line meet the straight lines  $AB$  and  $CD$  at  $E$  and  $F$ , respectively. Suppose that the alternate interior angles

$$\angle AEF \quad \text{and} \quad \angle EFD$$

are equal. Then

$$AB \parallel CD.$$

The basic configuration is shown in Figure [27.1](#).

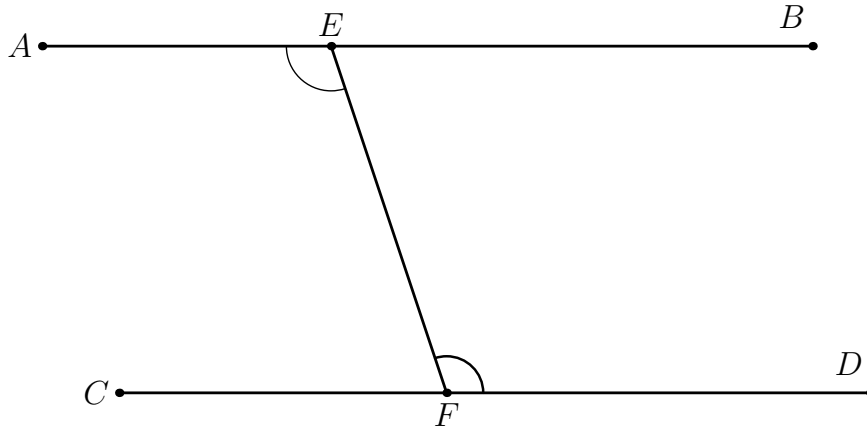


Figure 27.1: The transversal configuration of Proposition I.27. The alternate interior angles  $\angle AEF$  and  $\angle EFD$  are congruent.

### 27.3 Euclid's Proof

The proof is by contradiction.

Suppose first that the straight lines  $AB$  and  $CD$ , when produced beyond  $B$  and  $D$ , meet at a point  $G$ . Then  $\angle AEF$  is an exterior angle of the triangle  $EFG$ . By Proposition I.16,

$$\angle EFD < \angle AEF.$$

This contradicts the hypothesis

$$\angle AEF \cong \angle EFD.$$

The corresponding reductio configuration is shown in Figure 27.2.

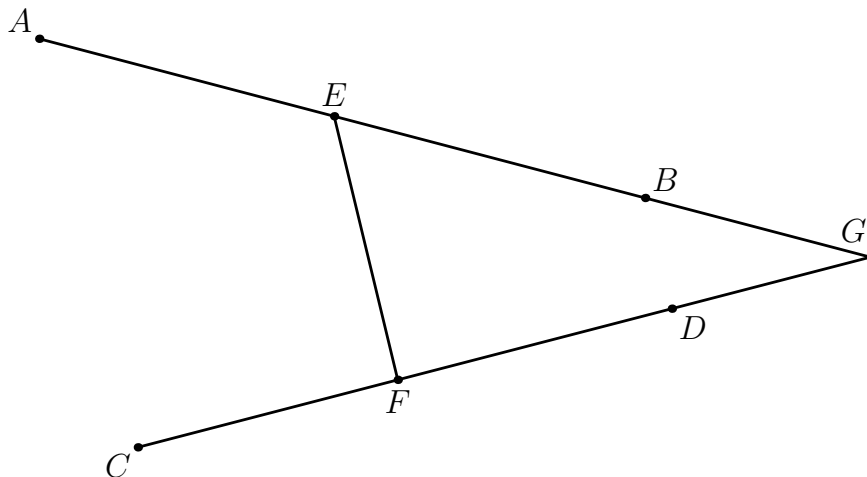


Figure 27.2: First reductio branch. If the two lines meet beyond  $B$  and  $D$  at  $G$ , Proposition I.16 makes  $\angle AEF$  strictly greater than  $\angle EFD$ .

There is a second possible direction of intersection. Suppose that the lines, when produced beyond  $A$  and  $C$ , meet at a point  $H$ . Then  $\angle EFD$  is an exterior angle of the triangle  $EFH$ . Again Proposition I.16 gives

$$\angle AEF < \angle EFD,$$

which contradicts the same angle-congruence hypothesis.

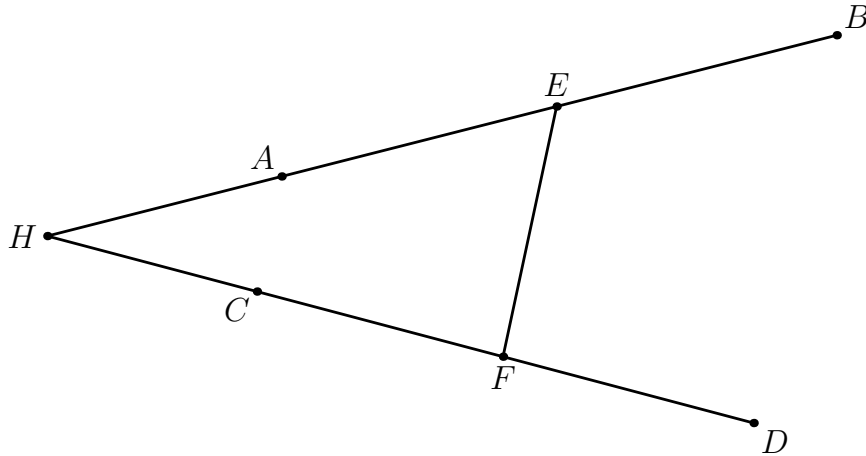


Figure 27.3: Second reductio branch. If the two lines meet beyond  $A$  and  $C$  at  $H$ , Proposition I.16 makes  $\angle EFD$  strictly greater than  $\angle AEF$ .

Both possible directions of intersection therefore lead to contradictions. Hence the two straight lines do not meet:

$$AB \parallel CD.$$

This is exactly the classical content of Proposition I.27.

## 27.4 Diagrammatic Audit

Three figures are retained because they represent three logically distinct configurations.

Figure 27.1 records only the hypothesis of the proposition: a transversal and a pair of congruent alternate interior angles.

Figure 27.2 represents the first hypothetical intersection, beyond  $B$  and  $D$ . Figure 27.3 represents the second hypothetical intersection, beyond  $A$  and  $C$ .

The last two figures are therefore not construction stages and are not simultaneously realizable with the original angle-congruence hypothesis. They are diagrams of the two reductio branches. Their purpose is to make the two applications of Proposition I.16 geometrically transparent.

As in Proposition I.26, the figures should encode the local geometric step being used rather than accumulate every relation occurring elsewhere in the proof.

## 27.5 Hilbert Reconstruction

The Hilbert development reaches the mathematical core of I.27 before the Euclidean sequence reaches Proposition I.27.

At the synthetic-interface level the implication

$$\text{equal alternate angles} \implies \text{parallel lines}$$

is already available through

```
parallel_from_equal_angles
```

The first Lean formulation therefore exposes I.27 directly at this level. Its proof is only an application of the existing theorem.

This is not an appeal to the Euclidean parallel postulate. The alternate-angle implication is a theorem of the neutral Hilbert development. The logical direction is important: I.27 proves a sufficient condition for parallelism; the converse direction used in Euclidean geometry appears later, in Proposition I.29, where the Fifth Postulate enters.

## 27.6 Two Formal Formulations

The current Lean file intentionally retains two formulations.

### 27.6.1 The interface-level formulation

The theorem

```
euclid_proposition_27
```

uses the ordered configuration already expected by `parallel_from_equal_angles`. Its proof is:

```
exact
  parallel_from_equal_angles
    Geo
    A C D
    B E F
    hADC
    hCEB
    hDEF
    hCED
    hAngle
```

At this level the central mathematical content of Proposition I.27 has already been packaged by the Hilbert theory. The Book I theorem merely exposes that result in the Euclidean sequence.

### 27.6.2 The explicit transversal formulation

The second theorem is

```
euclid_proposition_27_transversal
```

and begins with a configuration closer to the classical diagram:

$$A - E - B, \quad C - F - D,$$

with  $E$  and  $F$  on an explicitly supplied transversal *trans*. It also records that the selected interior arms  $EA$  and  $FD$  lie on opposite sides of the transversal and assumes

$$\angle AEF \cong \angle EFD.$$

This version reveals information that the traditional phrase “alternate interior angles” leaves implicit.

## 27.7 Normalization of the Transversal Configuration

The explicit proof first chooses a point  $M$  between  $E$  and  $F$ :

$$E - M - F.$$

This provides explicit representatives for the two directions of the transversal. Betweenness gives the corresponding same-ray relations. After reversing the first angle and transporting its transversal arm along these rays, the original congruence

$$\angle AEF \cong \angle EFD$$

is normalized to

$$\angle MEA \cong \angle MFD.$$

The theorem

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

then yields the local parallelism

$$EA \parallel FD.$$

The remainder of the proof is transport along the two original straight lines. Since

$$A - E - B$$

the first local line is extended from  $EA$  to  $AB$ . Since

$$C - F - D,$$

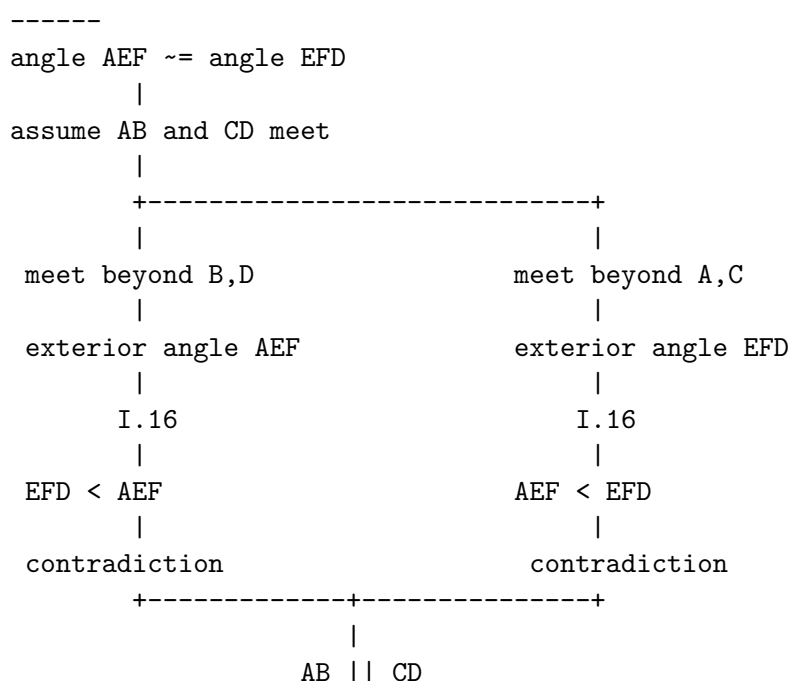
the second is transported from  $FD$  to  $CD$ . Thus

$$AB \parallel CD.$$

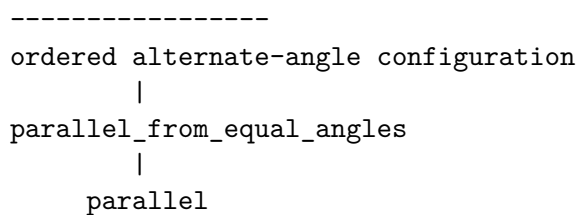
## 27.8 Proof Architecture

The classical and formal routes may be displayed schematically as follows.

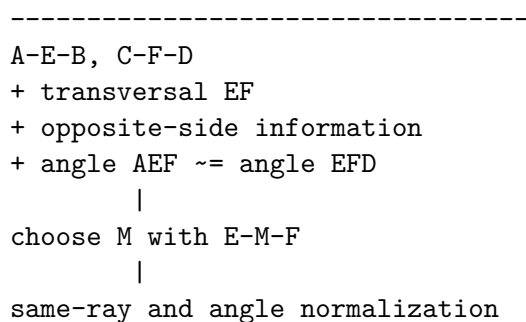
Euclid



Hilbert interface



Explicit transversal formalization





```

      |
angle MEA ~= angle MFD
      |
hilbert_parallel_of_alternate_angles_oppositeSide_lines
      |
EA || FD
      |
transport along A-E-B and C-F-D
      |
AB || CD

```

## 27.9 Proof-Theoretic Status

The coexistence of the two Lean formulations is deliberate.

The short formulation makes the mathematical core of I.27 unusually visible: the proposition coincides closely with an already established general theorem of the Hilbert layer.

The explicit transversal formulation answers a different question: what formal information is required to pass from the classical diagram to that general theorem? Its additional work concerns incidence, betweenness, same rays, side-of-line data, orientation of angles, and transport of parallelism.

At present we do not force a final interpretation of this distinction. It may eventually be preferable to regard the short theorem as the canonical Book I interface and the transversal theorem as a normalization result. It may instead prove useful to retain the explicit configuration as part of the public Euclidean layer.

Keeping both versions preserves this question for later comparison with the proof corpus and with the dependency graph of the Hilbert reconstruction.

### 27.10 Relationship with Proposition I.16

Euclid's classical proof depends directly on Proposition I.16. Each possible intersection converts one of the supposedly equal alternate angles into an exterior angle of a triangle, and I.16 makes that angle strictly greater than the remote interior angle.

The present Lean theorem does not invoke `euclid_proposition_16` directly at the Book I level because the corresponding alternate-angle theorem has already been established inside the Hilbert theory.

Thus the difference is one of proof organization:

$$\begin{array}{ll} \text{Euclid} & : \text{I.16} \longrightarrow \text{I.27}, \\ \text{Hilbert reconstruction} & : \text{order and angle theory} \longrightarrow \text{alternate-angle parallelism} \longrightarrow \text{I.27}. \end{array}$$

The Book I order and the formal dependency order therefore need not coincide.

## 27.11 Neutral Geometry and the Fifth Postulate

Proposition I.27 is a useful logical boundary marker.

It establishes

$$\text{equal alternate interior angles} \implies \text{parallel lines}$$

without using the Fifth Postulate. This direction is valid in neutral geometry.

The converse statement,

$$\text{parallel lines} \implies \text{equal alternate interior angles},$$

is part of Proposition I.29 and requires Euclidean parallelism.

Consequently I.27 should not be classified as a theorem depending on Hilbert’s Euclidean parallel axiom merely because its conclusion contains the word “parallel”. Its proof uses the definition of parallelism as nonintersection, together with the neutral theory developed before it.

## 27.12 Lean Formalization

The complete implementation is contained in `Proposition27.lean`. The public file currently records both the interface-level theorem and the explicit transversal theorem.

The latter proof has the following principal stages:

1. choose  $M$  with  $E - M - F$ ;
2. obtain the two same-ray relations along the transversal;
3. normalize the given angle congruence;
4. apply the Hilbert alternate-angle parallelism theorem;
5. transport the resulting local parallelism along the collinear triples  $A - E - B$  and  $C - F - D$ .

No Euclidean parallel axiom is used.

## 27.13 Conclusion

Proposition I.27 is formally small at the level of the developed Hilbert interface but conceptually revealing.

Euclid proves parallelism by excluding the two possible intersection directions through Proposition I.16. The Hilbert development has already abstracted this reasoning into a reusable alternate-angle theorem, so the core Book I formulation becomes a thin wrapper.

The explicit transversal theorem shows what that compression hides: the classical diagram silently carries order, orientation, side-of-line, and ray information that must be made explicit before the abstract theorem can be applied.

For the moment both formalizations are retained. Their coexistence is not duplication to be removed immediately, but a useful record of the difference between the mathematical content of I.27 and the formal normalization of its classical diagram.

## Chapter 28

# Proposition I.28

### 28.1 Introduction

Proposition I.28 gives two sufficient angle conditions for parallelism. A transversal meets two straight lines. Parallelism follows either from equality of a pair of corresponding angles or from the condition that the two interior angles on the same side are together equal to two right angles.

Both branches reduce to Proposition I.27. In each case the essential target is the same equality of alternate interior angles.

No form of the Fifth Postulate is used. Proposition I.28 belongs to neutral geometry.

### 28.2 Classical Configuration

Let the straight line  $EF$  meet  $AB$  at  $G$  and  $CD$  at  $H$ . The configuration is

$$A - G - B, \quad C - H - D, \quad E - G - H - F.$$

The proposition has two alternative hypotheses.

#### 28.2.1 Corresponding-angle criterion

Assume

$$\angle EGB \cong \angle GHD.$$

Since  $A - G - B$  and  $E - G - H$ , the angles

$$\angle EGB \quad \text{and} \quad \angle AGH$$

are vertical angles. By Proposition I.15,

$$\angle AGH \cong \angle EGB.$$

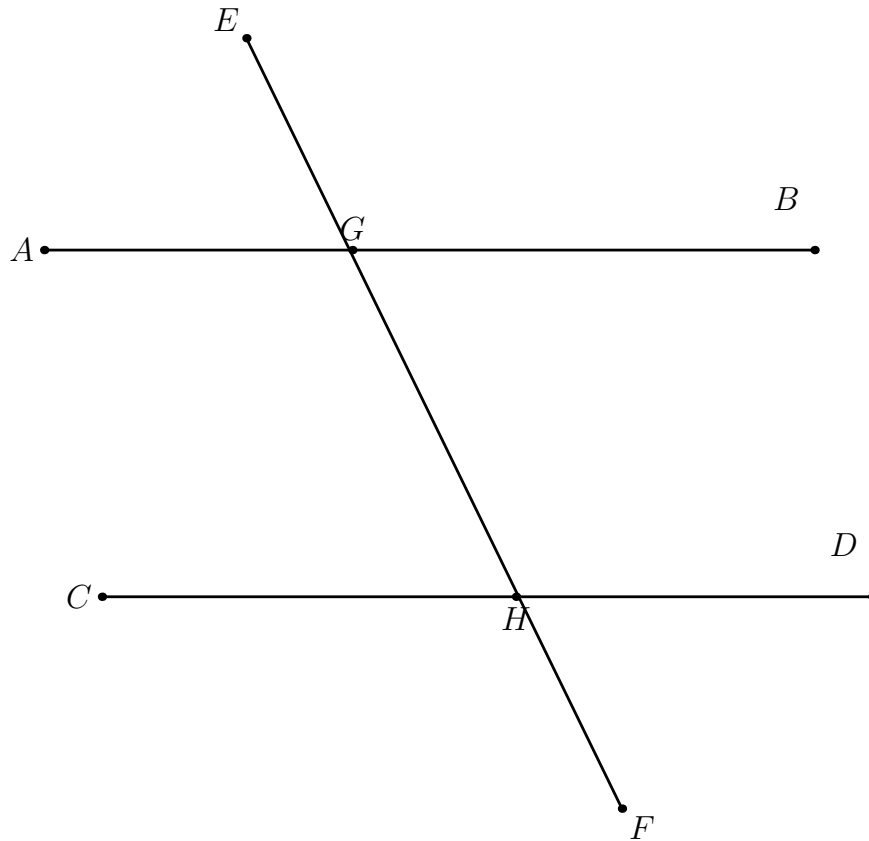


Figure 28.1: The common transversal configuration for Proposition I.28.

Hence

$$\angle AGH \cong \angle GHD.$$

These are alternate interior angles, so Proposition I.27 gives

$$AB \parallel CD.$$

### 28.2.2 Same-side interior-angle criterion

Assume that

$$\angle BGH \text{ and } \angle GHD$$

are together equal to two right angles.

Because  $A - G - B$ , the angles  $\angle AGH$  and  $\angle BGH$  form a supplementary pair. Proposition I.13 gives the second equality to two right angles. Subtracting the common angle  $\angle BGH$  leaves

$$\angle AGH \cong \angle GHD.$$

Again these are alternate interior angles, and Proposition I.27 gives

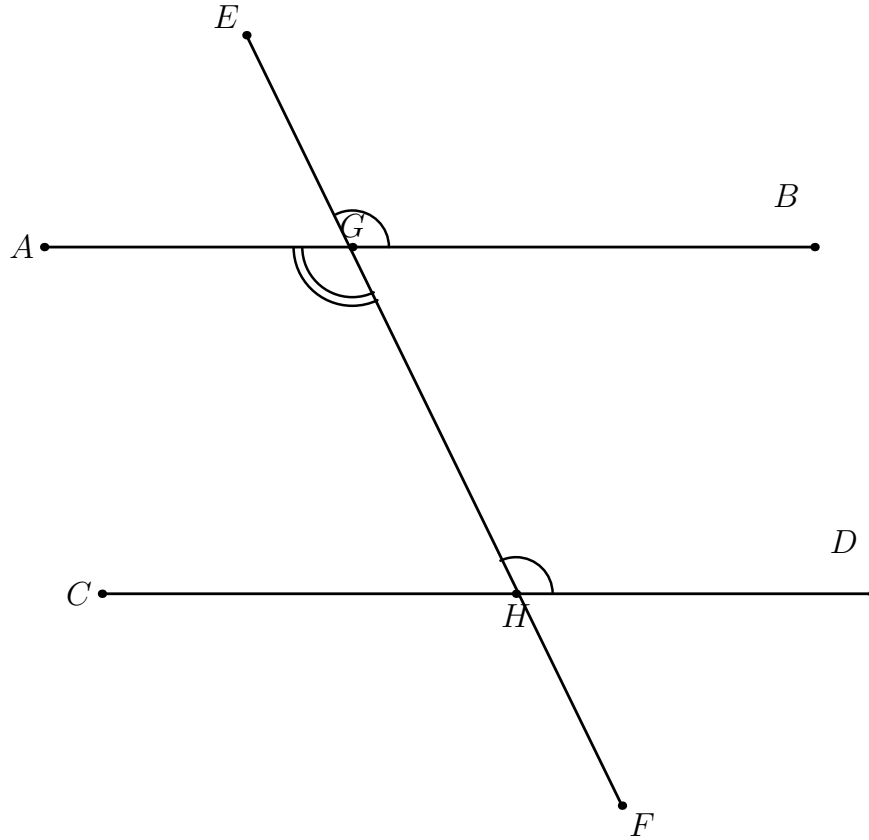


Figure 28.2: The first criterion. The corresponding angles  $\angle EGB$  and  $\angle GHD$  are congruent. Proposition I.15 identifies  $\angle EGB$  with the vertical angle  $\angle AGH$ , reducing the argument to Proposition I.27.

$$AB \parallel CD.$$

### 28.3 Hilbert Reconstruction

The formal reconstruction preserves the common geometric core of the two branches:

$$\angle AGH \cong \angle GHD \implies AB \parallel CD.$$

This final implication is exactly the explicit transversal form of Proposition I.27.

#### 28.3.1 First branch

The Lean theorem for the corresponding-angle criterion first applies **VerticalAngles**. From the two betweenness relations at  $G$  it obtains the vertical-angle congruence. A reversal of the second angle and transitivity with the given corresponding-angle congruence yield

$$\angle AGH \cong \angle GHD.$$

The proof then invokes

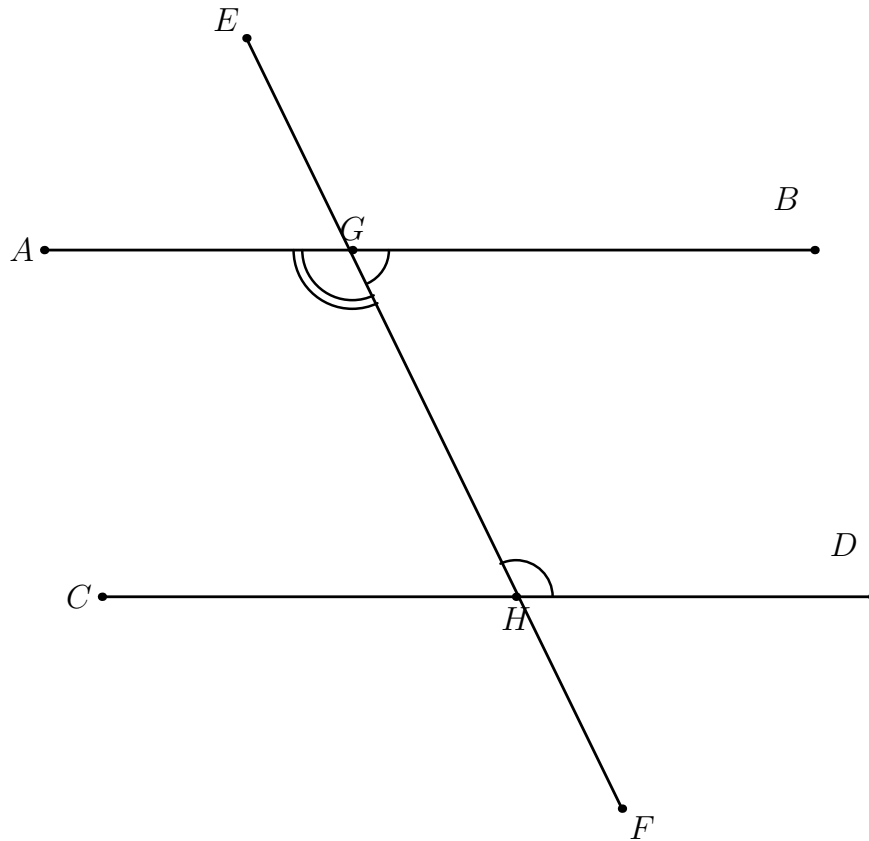


Figure 28.3: The second criterion. The interior angles  $\angle BGH$  and  $\angle GHD$  on the same side are together equal to two right angles. The supplementary angle to  $\angle BGH$  is  $\angle AGH$ , so the condition reduces to the alternate-angle hypothesis of Proposition I.27.

`euclid_proposition_27_transversal`

with  $G$  and  $H$  as the two intersection points of the transversal.

Schematically,

$$\begin{array}{c}
 \angle EGB \cong \angle GHD \\
 \angle AGH \cong \angle EGB \quad (\text{vertical angles}) \\
 \downarrow \\
 \angle AGH \cong \angle GHD \\
 \downarrow \\
 \text{euclid\_proposition\_27\_transversal} \\
 \downarrow \\
 AB \parallel CD.
 \end{array}$$

### 28.3.2 Second branch

The project does not introduce numerical angle measure or an operation of angle addition. The statement that two angles are together equal to two right angles is represented synthetically by an explicit supplementary ray.

The local predicate used in Proposition I.28 is

```
def HilbertAnglesEqualTwoRightAnglesWithSupplement
  (A O B C P D E : Geo.Point) : Prop :=
  Geo.Between D P E /\
  Geo.AngleCongruent A O B C P E
```

For the I.28 configuration it is instantiated as

```
HilbertAnglesEqualTwoRightAnglesWithSupplement
  Geo G H D H G B A
```

and therefore supplies

$$B - G - A$$

together with

$$\angle GHD \cong \angle HGA.$$

Betweenness symmetry recovers  $A - G - B$ . Symmetry and reversal of angle congruence then give

$$\angle AGH \cong \angle GHD.$$

The proof again ends with

```
euclid_proposition_27_transversal
```

and hence with  $AB \parallel CD$ .

## 28.4 Neutral Geometry

The logical direction in both parts of Proposition I.28 is

$$\text{angle condition} \implies \text{parallel lines.}$$

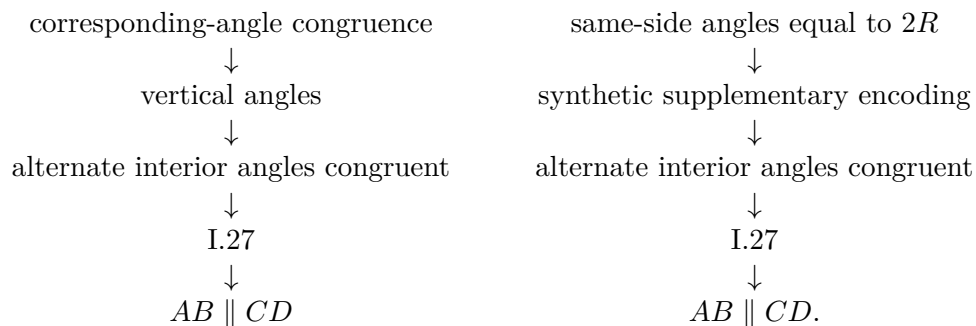
This direction does not require Hilbert's Group IV. The proof ultimately reduces to the neutral alternate-angle criterion already formalized in Proposition I.27.

Thus Propositions I.27 and I.28 belong to the neutral part of the theory. The reverse direction, from parallelism to the corresponding angle relations, appears in Proposition I.29 and is the point at which Euclidean parallelism enters.



## 28.5 Dependency Summary

The two formal branches are



No new parallel axiom is introduced.

## 28.6 Conclusion

Proposition I.28 adds two practical recognition criteria for parallel lines, but it does not introduce a new mechanism of parallelism. Both criteria are converted into the alternate-angle condition of Proposition I.27.

The first conversion uses vertical angles. The second uses the synthetic supplementary-angle language already developed in the reconstruction. Consequently the formal proof remains entirely within Hilbert incidence, order, congruence, and the neutral theory of parallelism.

## Chapter 29

# Proposition I.29

### 29.1 Introduction

Proposition I.29 is the first proposition in Book I whose proof genuinely uses the Euclidean parallel axiom.

The preceding Propositions I.27 and I.28 establish implications of the form

angle condition  $\longrightarrow$  parallel lines.

Those implications belong to neutral geometry. Proposition I.29 reverses the direction:

parallel lines  $\longrightarrow$  angle relations.

This is the point at which Hilbert's Group IV enters the reconstruction.

The proposition has three conclusions. If a transversal cuts two parallel straight lines, then:

$$\angle AGH \cong \angle GHD,$$

the alternate interior angles are congruent;

$$\angle EGB \cong \angle GHD,$$

the exterior angle is congruent to the interior and opposite angle on the same side; and the same-side interior angles

$$\angle BGH \quad \text{and} \quad \angle GHD$$

are together equal to two right angles.

Only the first conclusion contains the genuinely Euclidean step. The other two are consequences of the first by vertical-angle and supplementary-angle geometry.

## 29.2 Classical Configuration

Let the straight line  $EF$  meet the parallel straight lines  $AB$  and  $CD$  at  $G$  and  $H$ , respectively:

$$A - G - B, \quad C - H - D, \quad E - G - H - F.$$

Assume

$$AB \parallel CD.$$

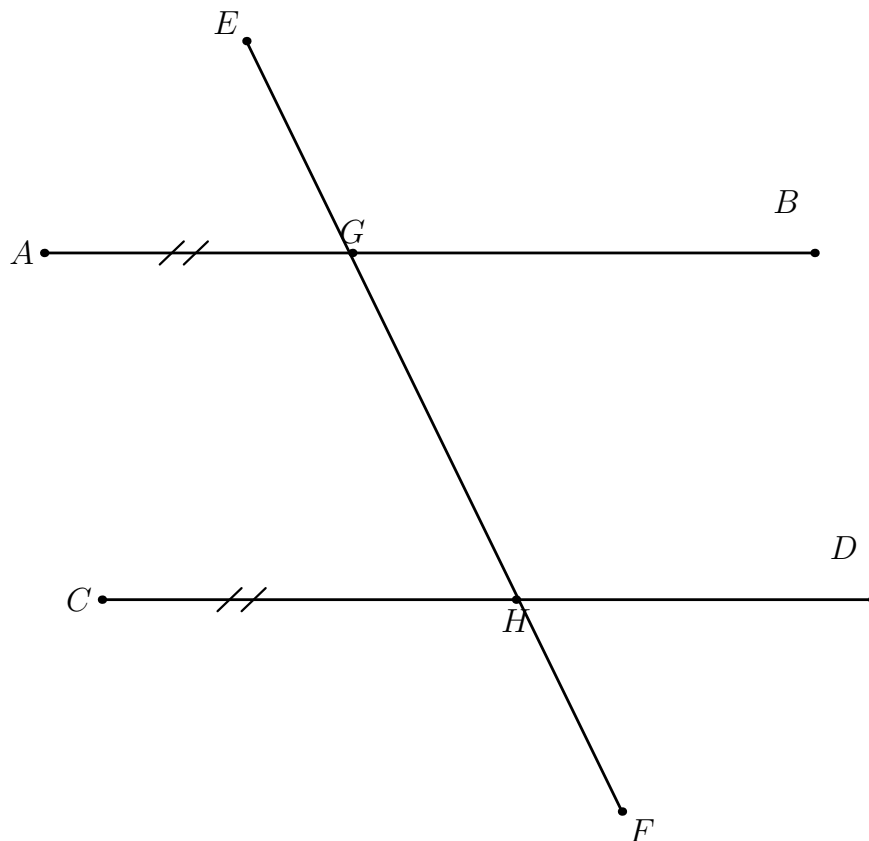


Figure 29.1: The transversal configuration of Proposition I.29. The straight lines  $AB$  and  $CD$  are parallel.

## 29.3 First Conclusion: Alternate Interior Angles

The principal assertion is

$$\angle AGH \cong \angle GHD.$$

This is the converse direction of Proposition I.27.

In the Hilbert reconstruction the corresponding general theorem is the Euclidean direction of Hilbert's Theorem 30. The proof constructs through the second intersection point a line making the required alternate angle. The neutral alternate-angle criterion shows that this constructed

line is parallel to the first given line. Hilbert's parallel axiom then identifies the constructed line with the already given parallel line.

Structurally, the first part is exactly the construction mechanism isolated later in Proposition I.31: copy an arbitrary angle at the external point and use I.27 to obtain a parallel. No Euclidean parallel axiom is involved in this existence step.

Schematically,

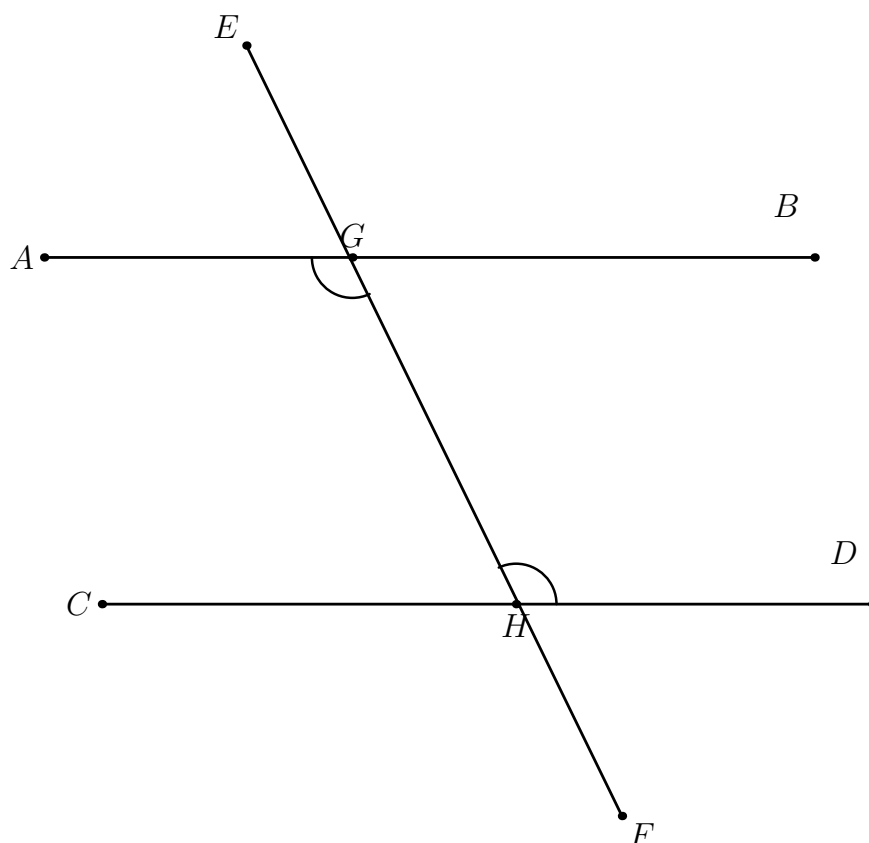
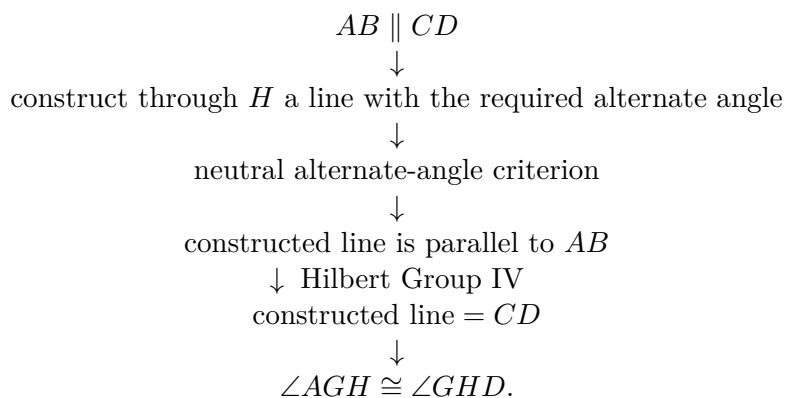


Figure 29.2: The Euclidean core of Proposition I.29: parallelism implies congruence of the alternate interior angles.

## 29.4 Second Conclusion: Corresponding Angles

Once

$$\angle AGH \cong \angle GHD$$

has been established, the second conclusion is neutral.

Because  $A - G - B$  and  $E - G - H$ , the angles

$$\angle AGH \quad \text{and} \quad \angle EGB$$

are vertical angles. Proposition I.15 therefore gives

$$\angle EGB \cong \angle AGH.$$

Combining this with the alternate-angle conclusion yields

$$\angle EGB \cong \angle GHD.$$

Thus the second branch has the dependency pattern

I.29 alternate  $\longrightarrow$  I.15 vertical angles  $\longrightarrow$  corresponding-angle congruence.

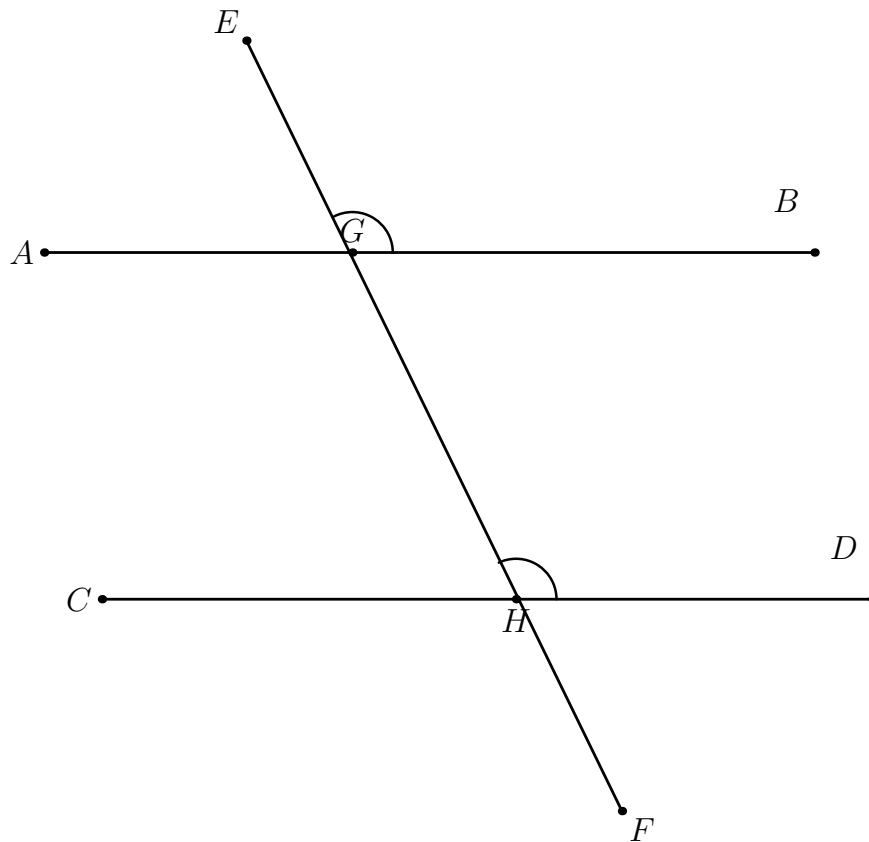


Figure 29.3: The corresponding-angle conclusion. The exterior angle  $\angle EGB$  is vertical to  $\angle AGH$ , which is already congruent to  $\angle GHD$ .

## 29.5 Third Conclusion: Same-Side Interior Angles

The third conclusion states that

$$\angle BGH \quad \text{and} \quad \angle GHD$$

are together equal to two right angles.

The reconstruction does not introduce numerical angle measure or a binary operation of angle addition. Instead it uses the synthetic predicate

`HilbertAnglesEqualTwoRightAnglesWithSupplement`

already placed in the Hilbert interface.

Because

$$A - G - B,$$

the rays  $GA$  and  $GB$  are opposite. Hence  $\angle AGH$  is the supplementary angle associated with  $\angle BGH$ .

From the first conclusion,

$$\angle AGH \cong \angle GHD.$$

Therefore the same-side pair satisfies the synthetic condition expressing that it is together equal to two right angles.

The dependency is

$$\begin{array}{c} \angle AGH \cong \angle GHD \\ A - G - B \\ \downarrow \\ \text{supplementary configuration} \\ \downarrow \\ \angle BGH + \angle GHD = 2R \quad (\text{synthetically}). \end{array}$$

## 29.6 Lean Formalization

The final Lean development contains five theorems.

The interface-level form is

`euclid_proposition_29`

and is a direct wrapper around

`equal_angles_from_parallel`

which is the Euclidean direction of Hilbert's Theorem 30.

The explicit transversal form is

`euclid_proposition_29_transversal`

with the conclusion

$$\angle AGH \cong \angle GHD.$$

Its proof first transports the global parallel relation

$$AB \parallel CD$$

to the local lines determined by the actual angle arms,

$$GA \parallel HD.$$

A point  $M$  is then chosen with

$$G - M - H.$$

The line-level Hilbert theorem gives

$$\angle MGA \cong \angle MHD.$$

Angle reversal and same-ray transport along the transversal convert this into

$$\angle AGH \cong \angle GHD.$$

The second theorem,

`euclid_proposition_29_corresponding`

combines the alternate-angle theorem with `VerticalAngles` to prove

$$\angle EGB \cong \angle GHD.$$

The third theorem,

`euclid_proposition_29_same_side`

combines the alternate-angle theorem with the supplementary relation  $A - G - B$  and returns

`HilbertAnglesEqualTwoRightAnglesWithSupplement`

for the same-side interior angles.

Finally,

`euclid_proposition_29_all`

packages the three conclusions in one theorem.

## 29.7 Proof Architecture

The formal dependency structure is

$$\begin{array}{c}
 AB \parallel CD \\
 \downarrow \\
 \text{hilbert\_alternate\_angles\_of\_parallel\_oppositeSide\_lines} \\
 \downarrow \text{ Hilbert Group IV} \\
 \angle AGH \cong \angle GHD
 \end{array}$$

and then

$$\begin{array}{ccc}
 & \angle AGH \cong \angle GHD & \\
 \swarrow & & \searrow \\
 \text{vertical angles} & & \text{supplementary configuration} \\
 \downarrow & & \downarrow \\
 \angle EGB \cong \angle GHD & & \angle BGH + \angle GHD = 2R.
 \end{array}$$

Thus the Euclidean axiom is needed only once. After the alternate-angle conclusion has been obtained, the remaining two conclusions are neutral.

## 29.8 The Logical Boundary at I.29

Propositions I.27–I.29 form a particularly clear logical block.

- I.27 : equal alternate angles  $\longrightarrow$  parallel lines,
- I.28 : other angle conditions  $\longrightarrow$  parallel lines,
- I.29 : parallel lines  $\longrightarrow$  angle relations.

The first two implications are neutral. The third is Euclidean.

This distinction is visible directly in the Lean typeclasses: I.27 and I.28 do not require the Euclidean plane structure, while I.29 uses

`[HilbertEuclideanPlane Geo]`

at the theorem that converts parallelism into alternate-angle congruence.



## 29.9 Existence and Uniqueness Behind I.29

The first conclusion of I.29 admits a useful structural decomposition.

The neutral part of the argument constructs through  $H$  a line having the required direction. Angle construction supplies the copied angle, and I.27 proves that the constructed line is parallel to  $AB$ . This is the same existence mechanism that appears explicitly in Proposition I.31.

At this stage neutral geometry gives only

there exists a line through  $H$  parallel to  $AB$ .

But the hypotheses of I.29 already contain another line through  $H$ , namely  $CD$ , with

$AB \parallel CD$ .

To conclude that the newly constructed line is actually  $CD$ , one needs uniqueness of the parallel through  $H$ . This is precisely the point at which Hilbert's Group IV, equivalently the Playfair-type uniqueness principle used in the reconstruction, enters.

Thus the logical core of the first conclusion can be displayed as

$$\begin{array}{c}
 \text{angle construction} \\
 \downarrow \\
 \text{equal alternate angles} \\
 \downarrow \\
 \text{I.27} \\
 \downarrow \\
 \text{a constructed parallel through } H \\
 \downarrow \text{ Group IV / parallel uniqueness} \\
 \text{the constructed line is } CD \\
 \downarrow \\
 \text{the required angle congruence of I.29.}
 \end{array}$$

This makes precise the observation that the Euclidean axiom is needed only once. It is not needed to construct a parallel direction; it is needed to identify the constructed parallel with the parallel line already present in the hypotheses.

Wilkins' discussion of the parallel propositions highlights this existence–uniqueness separation. Read retrospectively after I.31, the mechanism can be summarized schematically as

I.31-type construction + I.27 + Playfair uniqueness  $\implies$  the angle conclusion of I.29.

This is a structural analysis of I.29, not a claim that Euclid's original proof of I.29 depends on the later Proposition I.31.

## 29.10 Conclusion

Proposition I.29 is the first genuine Euclidean-parallelism theorem in the Book I reconstruction.

Its essential mathematical content is the single implication

$$AB \parallel CD \longrightarrow \angle AGH \cong \angle GHD.$$

That implication is Hilbert's Euclidean direction of the alternate-angle theorem and depends on Group IV.

The remaining statements of Euclid's proposition are consequences of this result. Corresponding angles are obtained through vertical-angle congruence, and the same-side two-right-angles condition is represented synthetically by a supplementary configuration.

The resulting formalization therefore isolates the exact point at which the parallel axiom enters and avoids attributing additional Euclidean strength to the two corollary branches.

## Chapter 30

# Proposition I.30

### 30.1 Introduction

Proposition I.30 establishes a structural property of Euclidean parallelism:

$$AB \parallel EF, \quad CD \parallel EF$$

imply

$$AB \parallel CD.$$

In Euclid's proof the conclusion is obtained by drawing a transversal through the three lines, applying Proposition I.29 twice, using transitivity of angle congruence, and then applying Proposition I.27.

In the Hilbert reconstruction the same theorem is more naturally organized through the uniqueness of a parallel line through a point. This is exactly the content of Hilbert's Group IV.

A formal subtlety must be made explicit. In the project, `Geo.Parallel` is a strict relation: parallel lines are distinct nondegenerate lines with disjoint point-line carriers. Therefore the statement must exclude the case in which the first and third lines are actually the same line.

The formal hypothesis is

$$\text{PointLine}(A, B) \neq \text{PointLine}(C, D).$$

This distinctness is implicit in Euclid's classical diagram, where the three straight lines are drawn as three separate lines.

### 30.2 Classical Configuration

Let  $AB$ ,  $CD$ , and  $EF$  be three distinct straight lines such that

$$AB \parallel EF$$

and

$$CD \parallel EF.$$

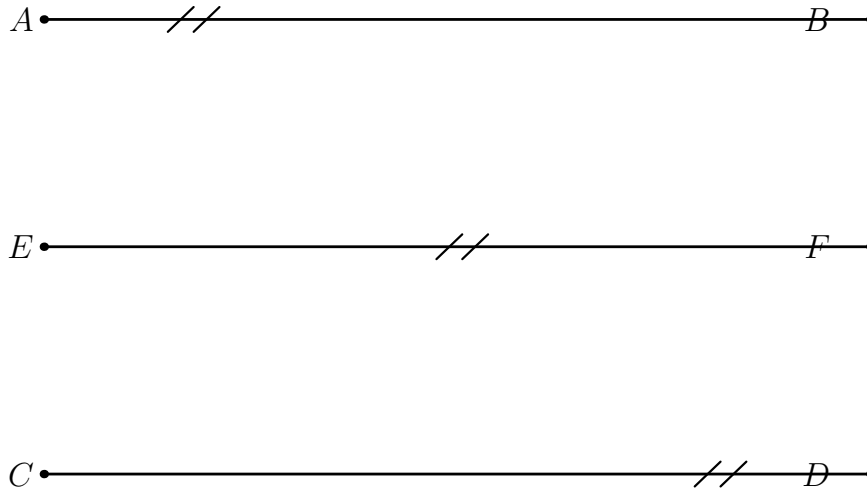


Figure 30.1: The configuration of Proposition I.30. The two outer lines are each parallel to the same middle line.

The conclusion is

$$AB \parallel CD.$$

### 30.3 Euclid's Proof Architecture

Euclid introduces a transversal meeting the three parallel lines.

The first application of Proposition I.29 gives an equality of alternate angles for the first pair of parallel lines. A second application of Proposition I.29 gives the corresponding equality for the second pair.

Thus one obtains a chain of angle congruences

$$\angle AGK \cong \angle GHF \cong \angle GKD.$$

By transitivity,

$$\angle AGK \cong \angle GKD.$$

These are alternate interior angles for the first and third lines. Proposition I.27 therefore gives

$$AB \parallel CD.$$

The classical dependency graph is therefore

$$\begin{array}{ccc}
AB \parallel EF & & CD \parallel EF \\
\downarrow \text{I.29} & & \downarrow \text{I.29} \\
\angle AGK \cong \angle GHF & & \angle GHF \cong \angle GKD \\
& \searrow & \downarrow & \swarrow \\
& & \angle AGK \cong \angle GKD & \\
& & \downarrow \text{I.27} & \\
& & AB \parallel CD. & 
\end{array}$$

## 30.4 Hilbert Reconstruction

The Hilbert reconstruction isolates the more general structural theorem

`hilbert_parallel_transitive_distinct`

with the mathematical content

$$AB \parallel EF, \quad CD \parallel EF, \quad \text{PointLine}(A, B) \neq \text{PointLine}(C, D)$$

imply

$$AB \parallel CD.$$

The proof does not pass through angle congruences.

Assume, for contradiction, that the first and third lines meet at a point  $P$ . Then both lines pass through  $P$ , and both are disjoint from  $EF$ .

Since  $P$  cannot lie on  $EF$ , there are two lines through the same point  $P$ , each parallel to  $EF$ .

Hilbert's Group IV says that through a point outside a line there is at most one line disjoint from the given line. Therefore the two lines through  $P$  must coincide.

Hence

$$\text{PointLine}(A, B) = \text{PointLine}(C, D),$$

contradicting the distinctness hypothesis.

Thus the first and third lines have no common point, and therefore

$$AB \parallel CD.$$

## 30.5 Lean Formalization

The reusable Hilbert-level theorem is

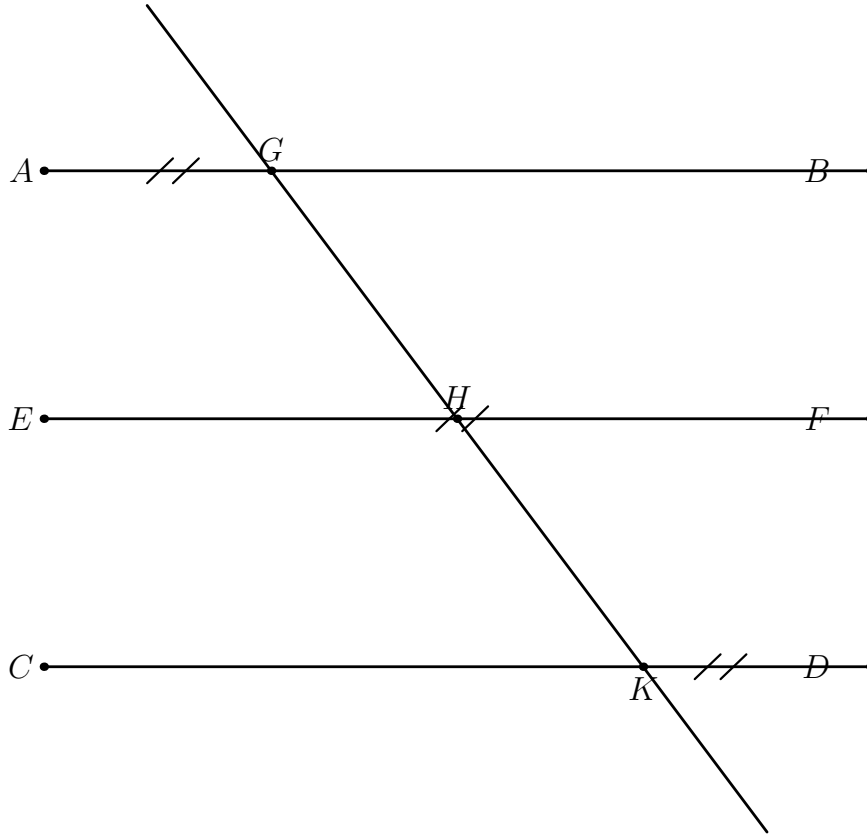


Figure 30.2: The Hilbert proof by contradiction. If the two outer lines met at  $P$ , they would be two distinct parallels through the same point to the middle line  $EF$ , contradicting Group IV.

```

theorem hilbert_parallel_transitive_distinct
  [HilbertEuclideanPlane Geo]
  (A B C D E F : Geo.Point)
  (hAB_EF : Geo.Parallel A B E F)
  (hCD_EF : Geo.Parallel C D E F)
  (hDistinct :
    Geo.PointLine A B != Geo.PointLine C D) :
  Geo.Parallel A B C D

```

The proof first obtains incidence lines through the three endpoint pairs:

$$AB, \quad CD, \quad EF.$$

The two hypotheses of parallelism are converted into disjointness of the corresponding incidence lines.

To prove  $AB \parallel CD$ , the proof assumes a common point  $P$  of the first and third point-line carriers. Membership in the carriers is transported to incidence on the chosen Hilbert lines.

The two parallel hypotheses imply that  $P$  is outside the line  $EF$ . The call

```
HilbertEuclideanPlane.parallel_unique
```

then identifies the two lines through  $P$ .

Finally,

`hilbert_pointLine_eq_of_points_on_line`

converts equality of the incidence carriers into equality of the project point-line carriers, contradicting `hDistinct`.

The Book I theorem

`euclid_proposition_30`

is consequently only a wrapper around this reusable Hilbert theorem.

## 30.6 Logical Position of Proposition I.30

The sequence I.27–I.30 now has a clear structure:

- I.27 : equal alternate angles  $\longrightarrow$  parallel,
- I.28 : other angle conditions  $\longrightarrow$  parallel,
- I.29 : parallel  $\longrightarrow$  angle relations,
- I.30 : parallel to the same line  $\longrightarrow$  parallel to one another.

The first two propositions are neutral.

Proposition I.29 introduces the Euclidean direction by using Hilbert's parallel axiom.

Proposition I.30 is then another direct manifestation of the same Euclidean principle: uniqueness of a parallel through a point.

Thus I.30 does not introduce a new independent parallel mechanism. It packages a structural consequence of Group IV.

## 30.7 Conclusion

Proposition I.30 becomes particularly transparent in the Hilbert reconstruction.

Euclid proves the theorem indirectly through angle relations:

$$\text{I.29} \longrightarrow \text{angle transitivity} \longrightarrow \text{I.27}.$$

The formal Hilbert development instead exposes the structural content:

two distinct lines parallel to the same line  $\longrightarrow$  they cannot meet,

because any intersection point would produce two different parallels through one point to the same given line.

The resulting theorem

`hilbert_parallel_transitive_distinct`

is reusable beyond Proposition I.30, while

`euclid_proposition_30`

records the corresponding result in the sequence of Book I.



## Chapter 31

# Proposition I.31

### 31.1 Introduction

Proposition I.31 is a construction theorem:

given a straight line  $AB$  and a point  $P \notin AB$ ,

construct a straight line through  $P$  parallel to  $AB$ .

At first sight the proposition belongs naturally to the Euclidean theory of parallel lines developed in I.29–I.30. Its logical status is different, however. Proposition I.31 does not require Hilbert's Group IV. It is an existence theorem in neutral geometry.

The construction uses an arbitrary transversal through the given point, copies an angle by Hilbert's angle-construction axiom, and then applies the neutral alternate-angle criterion corresponding to Euclid I.27.

Thus the essential mechanism is

angle construction  $\longrightarrow$  equal alternate angles  $\longrightarrow$  parallel lines.

No numerical angle measure and no construction of a right angle are needed.

### 31.2 Classical Configuration

Let  $AB$  be the given straight line and let  $P$  be a point outside it.

The goal is to construct a point  $Q \neq P$  such that

$$AB \parallel PQ.$$

Choose the line  $AP$  as a transversal. Then choose a point  $E$  with

$$A - E - P.$$

$P$   
•

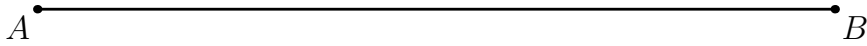


Figure 31.1: The initial data for Proposition I.31: the given straight line  $AB$  and the external point  $P$ .

The source angle is

$$\angle EAB.$$

At  $P$ , with  $PE$  as the first ray, construct a ray  $PQ$  on the appropriate side of the transversal such that

$$\angle EAB \cong \angle EPQ.$$

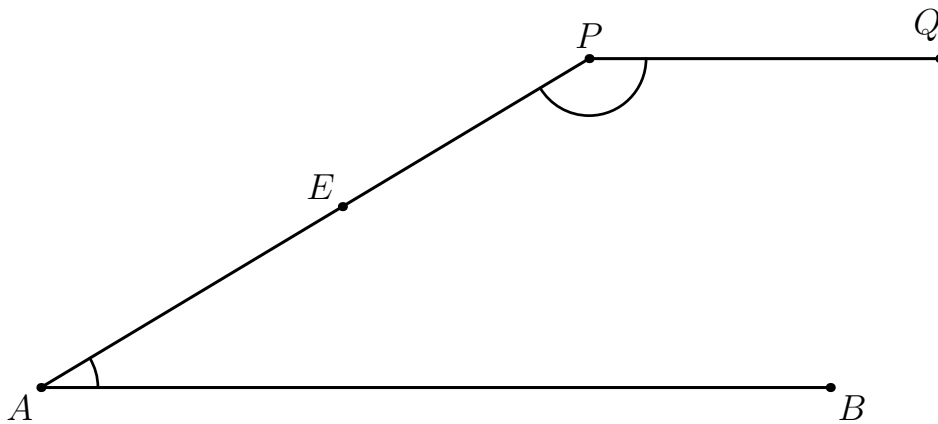


Figure 31.2: The transversal  $AP$ , the point  $E$  with  $A - E - P$ , and the copied angle at  $P$ .

The two congruent angles are alternate angles for the lines  $AB$  and  $PQ$ . Proposition I.27 therefore gives

$$AB \parallel PQ.$$

### 31.3 Why No Right Angle Is Needed

The construction does not proceed through perpendicular lines.

The angle

$$\angle EAB$$

is completely arbitrary. Its numerical size is irrelevant. Hilbert's angle-construction principle allows this angle to be copied at  $P$ , with the new ray placed on a prescribed side of the transversal.

Thus the direction of the new line is transferred synthetically by angle congruence:

$$\angle EAB \cong \angle EPQ.$$

The parallelism then follows from the converse alternate-angle theorem, that is, Proposition I.27.

This is conceptually important. The construction of a parallel does not require a prior theory of perpendiculars or a notion of  $90^\circ$ . It requires only the ability to reproduce a given angle and control its position relative to the transversal.

## 31.4 The Hidden Configuration

A classical diagram makes the side relation almost invisible. In the formal proof it must be explicit.

The equality

$$\angle EAB \cong \angle EPQ$$

alone is not sufficient. The point  $Q$  must be chosen so that  $B$  and  $Q$  lie on opposite sides of the transversal  $AP$ .

The formal construction therefore contains three distinct ingredients:

$$A - E - P,$$

which fixes the directions of the transversal rays;

$$B \notin AP,$$

which guarantees that the source angle is nondegenerate; and

$$B, Q \text{ on opposite sides of } AP,$$

which identifies the copied angles as genuine alternate angles.

This is exactly the type of configuration information that is compressed into an ordinary Euclidean picture but becomes explicit in a Hilbert-style formalization.

## 31.5 Hilbert Reconstruction

The substantial geometric content is isolated in the reusable theorem

`hilbert_parallel_through_point_exists`

with the formal statement

```
theorem hilbert_parallel_through_point_exists
  [HilbertCongruence Geo]
  (A B P : Geo.Point)
  (hAB : A != B)
  (hABP : not Collinear Geo A B P) :
  exists Q : Geo.Point,
    P != Q /\
    Geo.Parallel A B P Q
```

The proof first constructs the incidence line through  $A$  and  $B$ , and then the transversal through  $A$  and  $P$ .

Next it chooses  $E$  with

$$A - E - P.$$

The non-collinearity assumption implies that  $B$  lies off the transversal, so the angle  $\angle EAB$  is genuine.

A point  $S$  is then chosen on the side of the transversal opposite to  $B$ . Hilbert's angle-construction principle constructs  $Q$  on the side of  $S$  such that

$$\angle EAB \cong \angle EPQ.$$

The side relation is transported from  $S$  to  $Q$ , yielding the required opposite-side configuration for  $B$  and  $Q$ .

Finally,

`hilbert_parallel_of_alternate_angles_oppositeSide_lines`

applies the neutral alternate-angle criterion and returns

$$AB \parallel PQ.$$

## 31.6 Lean Formalization

The Book I theorem is

```
theorem euclid_proposition_31
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B P : Geo.Point)
```

```

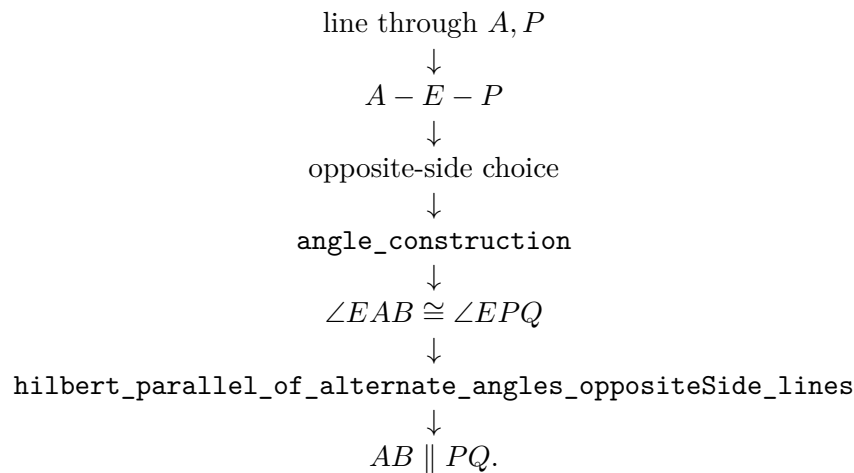
(hAB : A != B)
(hABP : not Collinear Geo A B P) :
exists Q : Geo.Point,
  P != Q /\
  Geo.Parallel A B P Q := by

exact
  hilbert_parallel_through_point_exists
    Geo
    A B P
    hAB
    hABP

```

Thus the Book I file is only a wrapper. The reusable geometric theorem belongs to the Hilbert interface, while the Proposition file records the corresponding result in Euclid's sequence.

The dependency graph of the actual Lean proof is



## 31.7 Neutral Existence Versus Euclidean Uniqueness

Proposition I.31 establishes the existence of a parallel through an external point.

It does not establish uniqueness.

This distinction separates I.31 from the immediately preceding Euclidean parallel propositions. In the Hilbert reconstruction, uniqueness is the additional content of Group IV:

through a point outside a line there is at most one parallel.

Hence

I.31: existence

belongs to neutral geometry, while

Group IV: uniqueness

is the Euclidean addition.

This also explains a useful structural relation with Proposition I.29. Once a parallel has been constructed by the mechanism of I.31, a uniqueness principle such as Playfair’s axiom can identify that constructed line with an already given parallel through the same point. This observation belongs to the logical analysis surrounding I.29; it is not an additional step in the proof of I.31 itself.

## 31.8 Final Configuration

The construction produces a point  $Q \neq P$  satisfying

$$AB \parallel PQ.$$

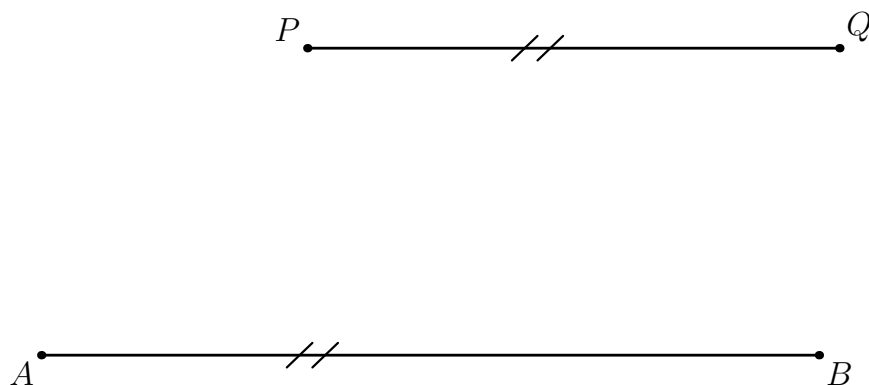


Figure 31.3: The final configuration of Proposition I.31: the constructed line through  $P$  is parallel to the given straight line  $AB$ .

## 31.9 Conclusion

Proposition I.31 is a particularly clear example of the difference between existence and uniqueness in the theory of parallels.

Its proof requires no Euclidean parallel axiom. A transversal is chosen, an arbitrary angle is copied at the external point, the side configuration is controlled explicitly, and Proposition I.27 proves that the resulting line is parallel to the given line.

The Hilbert reconstruction therefore isolates the reusable theorem

`hilbert_parallel_through_point_exists`

as a neutral construction result.

The Book I theorem

`euclid_proposition_31`

is then a thin wrapper around this more general piece of the developed Hilbert theory.

## Chapter 32

# Proposition I.32

### 32.1 Introduction

Proposition I.32 is the first proposition in Book I that applies the Euclidean parallel theory developed in I.29–I.31 directly to a triangle.

Let  $ABC$  be a nondegenerate triangle and extend  $BC$  through  $C$  to  $D$ . Euclid asserts first that the exterior angle at  $C$  is equal to the two remote interior angles:

$$\angle ACD = \angle BAC + \angle ABC.$$

He then concludes that the three interior angles of the triangle are together equal to two right angles.

In the present reconstruction no numerical angle measure is introduced. Thus the familiar modern formula

$$\angle BAC + \angle ABC + \angle ACB = 180^\circ$$

is not the literal Lean statement. Instead the proposition is represented synthetically: a ray inside the exterior angle decomposes it into two angles congruent to the two remote interior angles, while  $B - C - D$  records the straight-angle configuration at  $C$ .

This distinction is mathematically important. The formal proof establishes the synthetic content of the  $180^\circ$  theorem without introducing degrees or an arithmetic operation of angle addition.

### 32.2 Classical Statement

Let  $ABC$  be a triangle and let  $BC$  be produced to  $D$ . Proposition I.32 has two conclusions.

First,

$$\angle ACD = \angle BAC + \angle ABC.$$

Second,

$$\angle BAC + \angle ABC + \angle ACB = 2R,$$

where  $2R$  denotes two right angles.

The classical Euclidean proof constructs through  $C$  a line parallel to  $AB$ . Proposition I.29 then identifies the two parts of the exterior angle with the two remote interior angles.

### 32.3 The Formal Configuration

The Lean proof refines the classical diagram because the formal argument must record the orientation of rays and the incidence data needed by the parallel-angle theorem.

The principal points satisfy

$$A - M - C, \quad B - M - E, \quad A - R - D, \quad B - C - D.$$

The auxiliary construction gives

$$AB \parallel CE.$$

Moreover  $E$  and  $R$  determine the same ray from  $C$ . Thus the ray  $CR$  lies inside the exterior angle  $ACD$  and decomposes it into the two angles used in the formal statement.

For the explicit application of Proposition I.29 the proof also extends the two parallel lines:

$$A - B - X, \quad E - C - F.$$

Hence

$$AX \parallel EF.$$

### 32.4 Synthetic Representation of the Exterior-Angle Equality

The first conclusion is represented by the predicate

`HilbertExteriorAngleEqualsRemoteAngles`

defined by

```
def HilbertExteriorAngleEqualsRemoteAngles
  [HilbertIncidence Geo]
  (A B C D : Geo.Point) : Prop :=
  exists R : Geo.Point,
    Geo.Between A R D /\
    Geo.AngleCongruent B A C A C R /\
    Geo.AngleCongruent A B C R C D
```



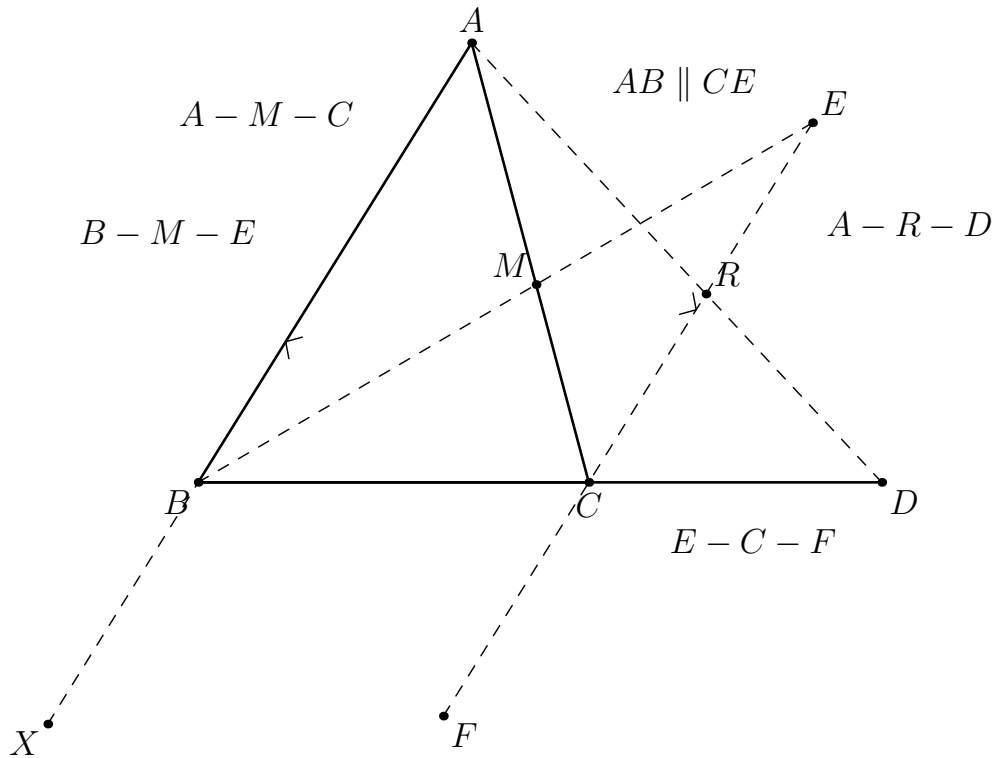


Figure 32.1: The formal configuration of Proposition I.32. The essential parallel relation is  $AB \parallel CE$ . The points  $X$  and  $F$  are technical extensions used to put the configuration into the explicit transversal form of Proposition I.29.

Thus the equation

$$\angle ACD = \angle BAC + \angle ABC$$

is encoded without angle addition. Instead there exists a point  $R$  with

$$A - R - D$$

such that

$$\angle BAC \cong \angle ACR$$

and

$$\angle ABC \cong \angle RCD.$$

The ray  $CR$  is therefore the synthetic “division point” of the exterior angle.

## 32.5 Construction of the Correctly Oriented Parallel Ray

A naive use of Proposition I.31 would produce a line through  $C$  parallel to  $AB$ , but a line alone does not determine which of its two rays lies inside the exterior angle  $ACD$ .

This orientation issue is genuine in the formal proof.

The reconstruction therefore uses the already established auxiliary configuration

`hilbert_exterior_angle_aux`

which supplies points  $M, E$  satisfying

$$A - M - C, \quad B - M - E,$$

together with

$$\angle BAC \cong \angle MCE.$$

The theorem

`hilbert_exterior_angle_aux_parallel`

then yields

$$AB \parallel CE.$$

The important point is that  $E$  is not an arbitrary point on the parallel through  $C$ . Its orientation is inherited from the auxiliary exterior-angle construction.

The helper

`hilbert_exterior_angle_inside`

uses this oriented configuration, together with  $B - C - D$ , to show that the ray  $CE$  meets the open segment  $AD$ . Hence there is a point  $R$  such that

$$A - R - D$$

and  $E, R$  lie on the same ray from  $C$ .

This is the formal crossbar step of the proof.

## 32.6 First Remote Interior Angle

From

$$A - M - C$$

the points  $M$  and  $A$  lie on the same ray from  $C$ . Therefore

$$\angle MCE = \angle ACE.$$

Since  $E$  and  $R$  lie on the same ray from  $C$ ,

$$\angle ACE = \angle ACR.$$

Combining these ray transports with the congruence supplied by the auxiliary construction gives

$$\boxed{\angle BAC \cong \angle ACR}.$$

In Lean this is the intermediate result

```
hFirst :
  Geo.AngleCongruent B A C A C R
```

and it establishes the first half of the exterior-angle decomposition.

## 32.7 Preparing Proposition I.29

The second congruence is obtained from the parallel relation, but the explicit theorem

```
euclid_proposition_29_transversal
```

expects the two intersections to lie between endpoints of the parallel lines.

The proof therefore extends  $AB$  beyond  $B$ :

$$A - B - X,$$

and extends  $EC$  beyond  $C$ :

$$E - C - F.$$

Parallelism is transported along the corresponding collinear triples to obtain

$$AX \parallel EF.$$

The transversal is the line  $BD$ , with intersection points  $B$  and  $C$ .

There remains an orientation condition. Since  $A - R - D$ , the points  $A$  and  $R$  lie on the same side of  $BD$ . From the same-ray relation between  $E$  and  $R$ , together with  $E - C - F$ , the proof obtains

$$R - C - F.$$

Thus  $R$  and  $F$  lie on opposite sides of  $BD$ . Transporting this relation through the same-side relation between  $A$  and  $R$  gives

$$A \text{ and } F \text{ on opposite sides of } BD.$$

At this point all hypotheses of the explicit transversal form of I.29 are available.

## 32.8 Second Remote Interior Angle

Proposition I.29 applied to

$$AX \parallel EF$$

with transversal  $BD$  gives

$$\angle ABC \cong \angle BCF.$$

At  $C$  the two straight lines are represented by

$$B - C - D \quad \text{and} \quad E - C - F.$$

Proposition I.15, through the theorem **VerticalAngles**, therefore gives

$$\angle BCF \cong \angle ECD.$$

By transitivity,

$$\angle ABC \cong \angle ECD.$$

Finally  $E$  and  $R$  lie on the same ray from  $C$ , so the first arm of the second angle may be transported from  $E$  to  $R$ :

$$\boxed{\angle ABC \cong \angle RCD}.$$

This is the Lean result

hSecond :

Geo.AngleCongruent A B C R C D

and completes the exterior-angle theorem.

## 32.9 Proof Architecture of the Exterior-Angle Part

The actual formal dependency structure is

$$\begin{array}{c}
 ABC \text{ noncollinear, } B - C - D \\
 \downarrow \\
 \text{hilbert\_exterior\_angle\_aux} \\
 \downarrow \\
 A - M - C, \quad B - M - E, \quad \angle BAC \cong \angle MCE \\
 \downarrow \\
 AB \parallel CE \\
 \downarrow \\
 \text{hilbert\_exterior\_angle\_inside} \\
 \downarrow \\
 A - R - D, \quad \text{SameRay}(C, E, R)
 \end{array}$$

and then the proof splits:

$$\begin{array}{ccc}
 & A - R - D, \quad AB \parallel CE & \\
 \swarrow & & \searrow \\
 \text{same-ray transport} & & \text{extend the two parallels} \\
 \downarrow & & \downarrow \\
 \angle BAC \cong \angle ACR & & AX \parallel EF \\
 & & \downarrow \\
 & & \text{I.29 + vertical angles} \\
 & & \downarrow \\
 & & \angle ABC \cong \angle RCD.
 \end{array}$$

The two branches are then packaged as

$$\exists R, \quad A - R - D, \quad \angle BAC \cong \angle ACR, \quad \angle ABC \cong \angle RCD.$$

## 32.10 Where the Euclidean Parallel Axiom Enters

The construction and order-theoretic parts of the proof use the Hilbert incidence, order, congruence, side-of-line and crossbar machinery.

The specifically Euclidean step is the use of Proposition I.29:

$$AX \parallel EF \longrightarrow \angle ABC \cong \angle BCF.$$

As established in the reconstruction of I.29, this direction depends on Hilbert's Group IV, or equivalently on uniqueness of the parallel through a point.

Thus I.32 inherits its Euclidean character from I.29.

This is structurally different from I.27 and I.28, where angle conditions imply parallelism in neutral geometry. In I.32 the proof needs the converse direction:

parallelism  $\longrightarrow$  angle congruence.

### 32.11 The Triangle-Angle-Sum Statement

The second formal predicate is

`HilbertTriangleAnglesEqualTwoRightAngles`

defined by

```
def HilbertTriangleAnglesEqualTwoRightAngles
  [HilbertIncidence Geo]
  (A B C : Geo.Point) : Prop :=
  exists D : Geo.Point,
    Geo.Between B C D /\
    HilbertExteriorAngleEqualsRemoteAngles
      A B C D
```

The theorem

`euclid_proposition_32`

is consequently short. From noncollinearity one obtains  $B \neq C$ , extends  $BC$  through  $C$  to a point  $D$ , and applies

`euclid_proposition_32_exterior`

to that extension.

The proof architecture is simply

$$\begin{array}{c}
 ABC \text{ noncollinear} \\
 \downarrow \\
 B \neq C \\
 \downarrow \\
 \text{extend } BC \text{ through } C \\
 \downarrow \\
 B - C - D \\
 \downarrow \\
 \text{euclid\_proposition\_32\_exterior} \\
 \downarrow \\
 \text{HilbertTriangleAnglesEqualTwoRightAngles.}
 \end{array}$$

## 32.12 What “Equal to Two Right Angles” Means Here

There is an important distinction between the classical mathematical consequence and the literal formal statement.

In ordinary angle-measure notation, I.32 yields

$$\angle BAC + \angle ABC + \angle ACB = 180^\circ.$$

The present Hilbert layer, however, does not assign numerical measures to angles. Nor does it define a general arithmetic operation

$$\alpha + \beta + \gamma.$$

Instead the formalization records the synthetic configuration from which the two-right-angles conclusion follows:

$$B - C - D$$

makes the rays  $CB$  and  $CD$  opposite, so  $\angle ACB$  and  $\angle ACD$  form a straight-angle configuration. The exterior angle is then decomposed by  $CR$  into angles congruent to  $\angle BAC$  and  $\angle ABC$ .

Consequently the formal theorem proves the geometric content

$$\angle BAC + \angle ABC + \angle ACB = 2R$$

without translating  $2R$  into the numerical value  $180^\circ$ .

Strictly speaking, the current predicate `HilbertTriangleAnglesEqualTwoRightAngles` stores the extension  $B - C - D$  and the exterior-angle decomposition; the supplementary interpretation of the straight line at  $C$  is implicit in this data rather than represented by a separate angle-addition object.

This is not merely a coding detail. It identifies exactly what must be formalized in a synthetic geometry before the familiar numerical  $180^\circ$  formulation can even be stated.

## 32.13 Lean Formalization

The principal theorem is

`euclid_proposition_32_exterior`

with hypotheses

```
(A B C D : Geo.Point)
(hABC : Not (PrimCollinear Geo A B C))
(hBCD : Geo.Between B C D)
```

and conclusion

`HilbertExteriorAngleEqualsRemoteAngles A B C D`

The proof uses the following main components:

```
hilbert_exterior_angle_aux
hilbert_exterior_angle_aux_parallel
hilbert_exterior_angle_inside
hilbert_angle_eq_of_sameRay_first
hilbert_angle_eq_of_sameRay_second
hilbert_between_points_sameSide_transversal
euclid_proposition_29_transversal
VerticalAngles
```

The second theorem,

`euclid_proposition_32`

is a corollary obtained by extending  $BC$  and invoking the exterior-angle theorem.

The typeclass

`[HilbertEuclideanPlane Geo]`

appears because the proof ultimately uses Proposition I.29 and hence the Euclidean parallel axiom.

## 32.14 Relation to the Classical Proof

At the level of a paper proof, I.32 is short:

construct a parallel through  $C \longrightarrow$  use I.29 twice  $\longrightarrow$  add the adjacent angles.

The Lean reconstruction exposes several pieces hidden by this description.

First, a parallel *line* through  $C$  is insufficient: the correct orientation of the ray must be established.

Second, the ray used to split the exterior angle must be shown to meet the segment  $AD$ . This is the crossbar component.

Third, the explicit I.29 theorem carries side-of-line hypotheses that must be proved from the betweenness configuration.

Fourth, the proof does not “add angles” numerically. It records a synthetic decomposition of the exterior angle.

Thus the expanded formal proof is not evidence that Euclid’s argument is mathematically complicated. It is an audit of the incidence, order and orientation information suppressed by the classical diagram.



## 32.15 Conclusion

Proposition I.32 combines the parallel theory of I.29–I.31 with the synthetic theory of triangles. Its essential Euclidean step is

$$AB \parallel CE \longrightarrow \text{the required angle congruence,}$$

which is supplied by I.29.

The formal proof then shows that the correctly oriented parallel ray through  $C$  cuts the exterior angle into two parts congruent to the remote interior angles:

$$\angle BAC \cong \angle ACR, \quad \angle ABC \cong \angle RCD.$$

Together with the straight-line relation  $B - C - D$ , this is the synthetic content of the classical theorem

$$\boxed{\angle BAC + \angle ABC + \angle ACB = 180^\circ}.$$

The reconstruction therefore reaches the familiar  $180^\circ$  theorem without introducing degrees: the numerical formulation appears only as the modern interpretation of a purely synthetic Hilbert-style configuration.

## Chapter 33

# Proposition I.33

### 33.1 Introduction

Euclid I.33 states that if two equal and parallel straight segments are joined towards the same parts, then the joining segments are themselves equal and parallel. In the notation used here, the hypotheses are

$$AB \parallel CD, \quad AB \cong CD,$$

and the conclusion is

$$AC \parallel BD, \quad AC \cong BD.$$

The proposition is Euclidean rather than neutral. The genuinely Euclidean step is the passage from parallel lines to congruent alternate angles, corresponding to Euclid I.29 and Hilbert's parallel axiom.

The formal reconstruction is particularly instructive because Euclid's phrase "towards the same parts" contains geometric orientation information which must be made explicit in Lean.

### 33.2 The Classical Configuration

For the explicit reconstruction we use the configuration occurring in Beeson's formal proof. The cross-joins  $AD$  and  $BC$  meet internally at a point  $M$ :

$$A - M - D, \quad B - M - C.$$

Thus  $M$  lies between  $A$  and  $D$ , and also between  $B$  and  $C$ . The two given segments satisfy

$$AB \parallel CD, \quad AB \cong CD.$$

The required conclusion is

$$AC \parallel BD, \quad AC \cong BD.$$

The betweenness assumptions are not merely a convenient way of drawing the figure. They encode the orientation information hidden in Euclid's phrase "towards the same parts."

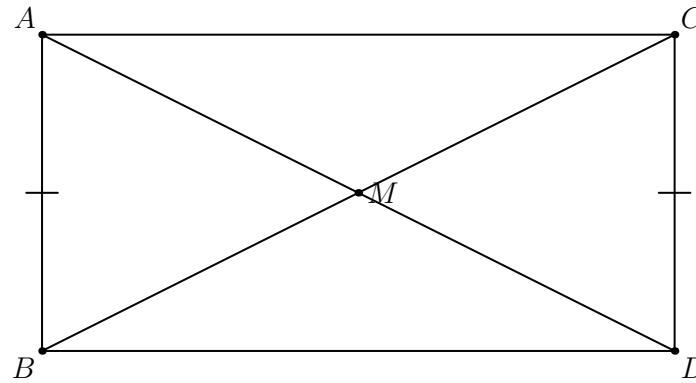


Figure 33.1: The configuration for Euclid I.33. The segments  $AB$  and  $CD$  are equal and parallel, while  $AD$  and  $BC$  meet internally at  $M$ .

### 33.3 Classical Proof

The classical dependency chain is short:

$$\text{I.29} \longrightarrow \text{SAS (I.4)} \longrightarrow \text{I.27.}$$

The line  $BC$  is a transversal of the parallel lines  $AB$  and  $CD$ . By I.29 the appropriate alternate angles are congruent.

Together with

$$AB \cong CD$$

and the common side  $BC$ , this gives the hypotheses of SAS for the two triangles determined by the configuration.

SAS gives, in particular,

$$AC \cong BD,$$

and also a second pair of congruent alternate angles.

The converse theorem I.27 can then be applied to this second pair of angles, yielding

$$AC \parallel BD.$$

Hence the joining segments are both equal and parallel.

### 33.4 The Hidden Orientation Data

The classical diagram makes several facts visually immediate. The formal proof cannot use them implicitly.

From

$$A - M - D, \quad B - M - C,$$

the Lean proof first obtains a line through  $B, M, C$ . It then proves that neither  $A$  nor  $D$  lies on this transversal.

Since  $M$  lies on the transversal and is between  $A$  and  $D$ , the open segment  $AD$  meets the transversal. Consequently  $A$  and  $D$  lie on opposite sides of the line  $BC$ .

This produces the formal separation datum

```
hOppositeAD :
HilbertOppositeSide Geo A D trans
```

which is required by the line-level alternate-angle theorem.

The proof also chooses an auxiliary point  $N$  between  $B$  and  $C$ . Thus

$$B - N - C$$

and, by symmetry of betweenness,

$$C - N - B.$$

These relations produce the corresponding same-ray data:

```
hRayBNC :
HilbertSameRay Geo B N C

hRayCNB :
HilbertSameRay Geo C N B
```

They are used to normalize the unoriented angles appearing in the Hilbert interface.

Thus the apparently simple phrase “towards the same parts” expands formally into incidence, betweenness, opposite-side, and same-ray information.

### 33.5 Proof Architecture of the Reconstruction

The Lean file contains two levels of I.33.

The first theorem is

```
euclid_proposition_33
```

It is a thin theorem at the level of the developed Hilbert API. Instead of starting from the explicit crossing configuration, it accepts an orientation witness suitable for the existing one-pair-parallel-and-congruent criterion.

The relevant reusable theorem is

```
OnePairParallelCongruentCriterion
```

which recognizes the corresponding quadrilateral as a parallelogram. Equality of the remaining opposite sides is then obtained from

```
ParallelogramOppositeSidesCongruent
```

The second theorem is

```
euclid_proposition_33_crossing
```

This is the configuration-faithful reconstruction. Its hypotheses contain the explicit point  $M$  and the two betweenness relations used by Beeson. The proof then reconstructs the classical I.29–SAS–I.27 argument.

The distinction between the two theorems is deliberate. The first shows that the mathematical content of I.33 is already available at a higher level in the Hilbert interface. The second shows how the classical Euclidean configuration is translated into the lower-level data required by that theory.

## 33.6 Lean Formalization

The explicit theorem has the following statement:

```
theorem euclid_proposition_33_crossing
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D M : Geo.Point)
  (hParallel : Geo.Parallel A B C D)
  (hCongruent : Geo.Congruent A B C D)
  (hAMD : Geo.Between A M D)
  (hBMC : Geo.Between B M C) :
  Geo.Parallel A C B D /\
  Geo.Congruent A C B D := by
```

The first stage extracts the incidence information associated with  $B - M - C$ , including a transversal through  $B, M, C$ .

Parallelism of  $AB$  and  $CD$  is then used to prove that  $A$  and  $D$  cannot lie on this transversal. Together with  $A - M - D$ , this gives

```
hOppositeAD :
HilbertOppositeSide Geo A D trans
```

An interior point  $N$  of  $BC$  is introduced to obtain the same-ray normalizations needed by the angle interface.

The Euclidean direction of the alternate-angle theorem is invoked through the following interface theorem:

```
hilbert_alternate_angles_of_parallel_
oppositeSide_lines
```

After normalization of the rays, this gives the included-angle congruence required for SAS.

The two triangles are then compared by

```
SAS
```

and the resulting congruence object is stored as

```
hTriangles :
TriangleCongruenceResult
Geo B A C C D B
```

One component gives the required segment congruence:

```
hAC_BD :
Geo.Congruent A C B D
```

Another component gives the angle congruence from which the second parallelism is obtained.

After same-ray normalization the relevant angle statement is

```
hAlternate :
Geo.AngleCongruent N C A N B D
```

The converse alternate-angle theorem is then applied:

```
hilbert_parallel_of_alternate_angles_
oppositeSide_lines
```

and yields the required parallelism of  $AC$  and  $BD$ .

### 33.7 Relationship with Earlier Propositions

Proposition I.33 combines two directions of the parallel-angle theory developed immediately before it.

The first direction is the Euclidean one:

parallel lines  $\implies$  congruent alternate angles.

This is the content represented by I.29. It is the point at which the Euclidean parallel axiom enters the proof.

The second direction is

congruent alternate angles  $\implies$  parallel lines.

This is the converse represented by I.27 and belongs to the neutral part of the theory.

I.33 therefore provides a compact test of the architecture developed for I.27–I.29. The proof begins with the Euclidean direction, passes through SAS, and ends with the neutral converse.

### 33.8 Hilbert and Book Zero Components Used

No new axiom is introduced for Proposition I.33.

The explicit proof uses existing infrastructure for:

- incidence and the line through two distinct points;
- betweenness and its incidence consequences;
- separation by a line;
- opposite-side relations;
- same-ray relations;
- segment-congruence symmetries;
- non-collinearity;
- SAS triangle congruence;
- the Euclidean alternate-angle theorem;
- the converse alternate-angle theorem.

The API-level theorem additionally uses the existing parallelogram machinery. In particular:

```
OnePairParallelCongruentCriterion
```

and

```
ParallelogramOppositeSidesCongruent
```

No helper specific to Proposition I.33 is added to `HilbertInterface` or to Book Zero.

## 33.9 Formalization Notes

The main formal issue in I.33 is not SAS itself. It is the orientation of the configuration.

In an ordinary diagram it is immediately apparent which sides of the transversal contain  $A$  and  $D$ , and which rays at  $B$  and  $C$  determine the relevant angles. None of this information is implicit in the formal language.

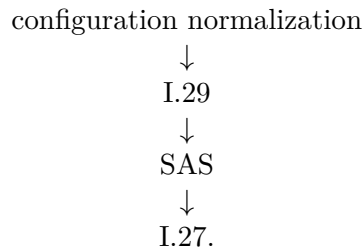
The crossing formulation makes the required information explicit through

$$A - M - D, \quad B - M - C.$$

From these two relations the proof reconstructs the separation and ray data required by the Hilbert interface.

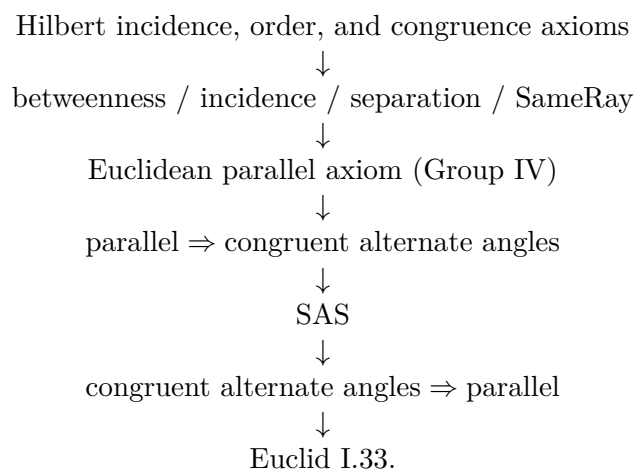
This also explains why the Lean proof is longer than the classical three-step argument. The additional code does not represent new Euclidean mathematics. It reconstructs configuration information which the classical proof obtains from the diagram and from the phrase “towards the same parts.”

The formal proof therefore separates two layers clearly:



### 33.10 Dependency Summary

The dependency structure of the explicit reconstruction is:



**New axiom:** no.

**New helper:** no.

**Build clean:** yes.

**Sorry:** no.

### 33.11 Conclusion

Euclid I.33 is mathematically short but formally revealing. Once the orientation hidden in the classical configuration is exposed, the original proof architecture survives:

$$\text{I.29} \longrightarrow \text{SAS} \longrightarrow \text{I.27.}$$

The explicit crossing theorem shows how the classical statement is recovered from betweenness and separation data. The shorter API-level theorem shows, independently, that the underlying parallelogram principle is already available in the developed Hilbert theory.

No new axiom and no Proposition-specific helper are required.



## Chapter 34

# Proposition I.34

### 34.1 Introduction

Euclid's Proposition I.34 collects the fundamental metric and angular properties of a parallelogram. For a parallelogram  $ABCD$ , it asserts that opposite sides are equal, opposite angles are equal, and a diagonal divides the parallelogram into two equal parts.

The present reconstruction separates these three assertions according to the language actually available in the CGJteamLab Hilbert development. The first two are represented directly by congruence of segments and angles. The final claim is represented by congruence of the two triangles cut off by the diagonal. No numerical area function is introduced.

This Proposition is therefore a useful example of a recurring distinction in the project: the classical theorem and the formal theorem have the same synthetic content, but the Lean proof uses a different proof architecture from Euclid's original argument.

### 34.2 The Classical Configuration

Let  $ABCD$  be a parallelogram, with

$$AB \parallel CD, \quad BC \parallel DA.$$

Draw the diagonal  $AC$ . The diagonal separates the parallelogram into the triangles  $ABC$  and  $CDA$ .

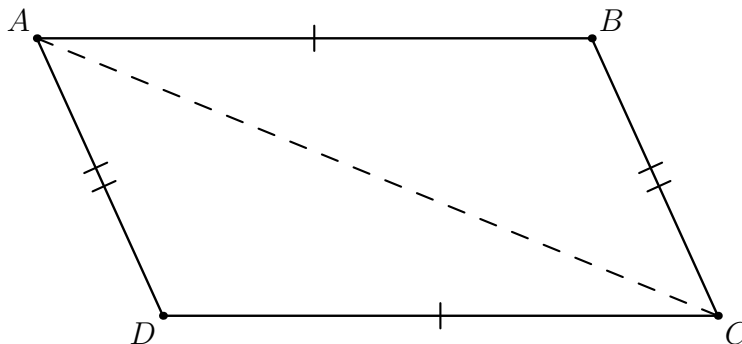


Figure 34.1: The configuration for Proposition I.34. The quadrilateral  $ABCD$  is a parallelogram and  $AC$  is the diagonal used in the reconstruction.

The required conclusions are

$$\begin{aligned} AB &\cong CD, & BC &\cong DA, \\ \angle DAB &\cong \angle BCD, & \angle ABC &\cong \angle CDA, \end{aligned}$$

and, in the language of the present formalization,

$$\triangle ABC \cong \triangle CDA.$$

### 34.3 Classical Proof

Euclid's proof uses the diagonal and the theory of parallel lines established immediately before Proposition I.34. Since  $AB \parallel CD$ , Proposition I.29 gives one pair of alternate angles. Since  $BC \parallel DA$ , a second application of Proposition I.29 gives another pair of alternate angles. The common diagonal then allows Proposition I.26 to identify the two triangles. The corresponding opposite sides and angles are therefore equal. The final statement that the diagonal bisects the parallelogram follows because the two triangles are congruent.

A compact classical dependency graph is

$$\text{I.29} + \text{I.29} \longrightarrow \text{I.26} \longrightarrow \begin{array}{l} \text{opposite sides equal,} \\ \text{opposite angles equal,} \\ \text{diagonal gives equal triangles.} \end{array}$$

The Beeson–Narboux–Wiedijk proof file makes explicit several configuration facts hidden by the classical diagram: non-collinearity of the relevant triangles, opposite-side data for the transversal configuration, and endpoint order needed to apply the formal versions of I.29 and I.26. These details are part of the classical reconstruction problem, even though they are not visible in Euclid's short paper proof.

### 34.4 The Main Formalization Issue: Reusing Parallelogram Theory

The current CGJteamLab proof does not replay the classical I.29–I.26 route. By the time Proposition I.34 is reached, the Hilbert interface already contains a reusable Euclidean theory of parallelograms. In particular, it contains a theorem stating that the opposite sides of a Euclidean parallelogram are congruent.

This changes the proof DAG substantially. Once

$$AB \cong CD \quad \text{and} \quad BC \cong DA$$

are available, the diagonal  $AC$  supplies the third side congruence

$$AC \cong CA.$$

The triangles  $ABC$  and  $CDA$  are therefore congruent by the already proved Hilbert SSS theorem. The required opposite-angle congruence is extracted from that triangle-congruence result.

Thus the formal proof uses the same diagonal as Euclid, but it enters the triangle argument through SSS rather than through two alternate-angle relations and ASA.

## 34.5 Proof Architecture of the Reconstruction

The reusable proof architecture is

$$\begin{array}{c}
 AB \parallel CD, \quad BC \parallel DA \\
 \Downarrow \\
 \text{IsParallelogram } ABCD \\
 \Downarrow \\
 AB \cong CD, \quad BC \cong DA \\
 AC \cong CA \\
 \Downarrow \quad \text{SSS} \\
 \triangle ABC \cong \triangle CDA \\
 \Downarrow \\
 \angle ABC \cong \angle CDA.
 \end{array}$$

The second pair of opposite angles is obtained by applying the same reusable angle theorem to the cyclically relabelled parallelogram  $BCDA$  and then using symmetry of angle congruence.

This produces two interface-level helpers:

```

ParallelogramOppositeAngleCongruent
ParallelogramOppositeAnglesCongruent

```

The first proves one pair of opposite angles. The second packages both pairs. They belong to `HilbertInterface` because they are general properties of every Euclidean parallelogram, not facts specific to Proposition I.34.

## 34.6 Lean Formalization

The main Proposition theorem is deliberately thin:

```

theorem euclid_proposition_34
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D : Geo.Point)
  (hParallelogram : IsParallelogram Geo A B C D) :
  And
    (OppositeSidesCongruent Geo A B C D)
    (OppositeAnglesCongruent Geo A B C D)

```

Its two branches are discharged by the established interface theorems

```

ParallelogramOppositeSidesCongruent
ParallelogramOppositeAnglesCongruent

```

The third part of Euclid’s statement is represented separately:

```
theorem euclid_proposition_34_diagonal
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D : Geo.Point)
  (hParallelogram : IsParallelogram Geo A B C D) :
  TriangleCongruenceResult Geo A B C C D A := by
```

The proof first derives non-collinearity of  $A, B, C$  from the disjointness of the parallel lines  $AB$  and  $CD$ . It then obtains the two pairs of opposite side congruences from the parallelogram theorem, uses reflexivity of congruence for the diagonal, and applies SSS:

```
have hSides :
  OppositeSidesCongruent Geo A B C D :=
  ParallelogramOppositeSidesCongruent
  Geo A B C D hParallelogram

have hAC_CA :
  Geo.Congruent A C C A :=
  CongruentSwapSecond
  Geo A C A C
  (hilbert_congruent_reflexive Geo A C)

have hSSS :=
  HilbertSSS
  Geo
  A B C
  C D A
  hABC
  hSides.1
  hSides.2
  hAC_CA

exact hSSS.2
```

No area object occurs in this theorem. The formal counterpart of Euclid’s “diameter bisects the areas” is the congruence of the two triangles cut off by the diagonal.

## 34.7 Relationship with Earlier Propositions

Classically, Proposition I.34 depends directly on I.29 and I.26. In the present Lean development these earlier Euclidean consequences have already been absorbed into the reusable parallelogram theory of `HilbertInterface`. Consequently the final Proposition file does not call `euclid_proposition_29` or `euclid_proposition_26` directly.

This is an intentional architectural compression. The classical dependency has not disappeared mathematically; it has been moved into general Hilbert-level theorems that are proved once and then reused.

The relation with I.33 is especially close. Proposition I.33 recognized a parallelogram from one pair of opposite sides that were both parallel and congruent, together with orientation data.

Proposition I.34 starts after the quadrilateral has already been recognized as a parallelogram and extracts its standard Euclidean consequences.

## 34.8 Hilbert and Book Zero Components Used

The formal Proposition uses the following established components:

- `IsParallelogram`, defined by two pairs of opposite parallel sides;
- `ParallelogramOppositeSidesCongruent`, the Euclidean theorem giving the two opposite-side congruences;
- `HilbertSSS`, the derived SSS congruence theorem;
- incidence uniqueness and the extensional representation of parallel lines, used to prove that  $A, B, C$  are non-collinear;
- reflexivity and endpoint symmetry of segment congruence;
- symmetry of angle congruence, used when packaging the second pair of opposite angles.

No new Book Zero theorem is required for Proposition I.34.

## 34.9 Formalization Notes

### 34.9.1 Axiomatic status

Proposition I.34 is Euclidean in the present formulation. The hypothesis is

```
[HilbertEuclideanPlane Geo]
```

The Euclidean strength is consumed in the general theorem asserting congruence of opposite sides of a parallelogram. This is not a neutral fact for a parallelogram defined only by parallel opposite sides: in hyperbolic geometry such a quadrilateral need not have congruent opposite sides.

After those side congruences are available, the local SSS step itself belongs to the congruence theory and does not require an additional use of the parallel axiom.

### 34.9.2 Hidden configuration data

The concise classical diagram suppresses a non-degeneracy issue. Lean must show explicitly that  $A, B, C$  are not collinear before applying SSS. The proof uses the fact that  $AB$  and  $CD$  are parallel and hence their extensional point-lines are disjoint. If  $C$  lay on  $AB$ , then  $C$  would belong to both parallel lines, a contradiction.

No new `SameSide`, `OppositeSide`, or `Between` witness is needed in the final Proposition I.34 proof. Those notions occur in the lower-level derivation of the reusable Euclidean parallelogram theory, but the API-level proof has already hidden them behind the established theorem.

### 34.9.3 Reusable helpers introduced

Two reusable parallelogram helpers were added while proving I.34:

```
ParallelogramOppositeAngleCongruent
ParallelogramOppositeAnglesCongruent
```

The first derives one pair of opposite angles from SSS on the two triangles cut by a diagonal. The second obtains both pairs by cyclic relabelling. These are not local Proposition helpers: they belong to `HilbertInterface` and are available to later propositions.

The packaging predicate

```
OppositeAnglesCongruent
```

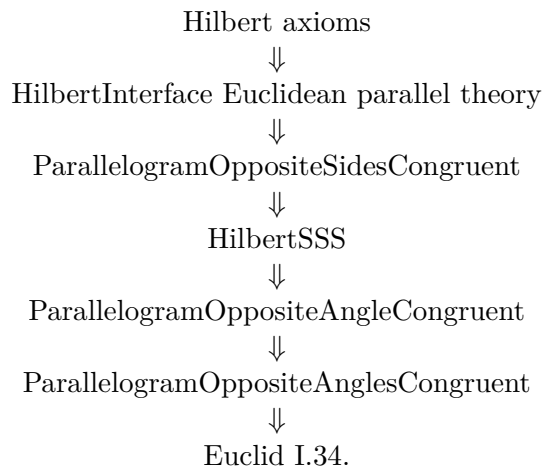
records the two required angle congruences.

## 34.10 Dependency Summary

The classical dependency is

$$\text{I.29} + \text{I.29} \longrightarrow \text{I.26} \longrightarrow \text{I.34}.$$

The actual Lean dependency is



For the diagonal theorem the last part is even shorter:

$$\text{opposite sides congruent} + AC \cong CA \longrightarrow \text{SSS} \longrightarrow \triangle ABC \cong \triangle CDA.$$

|                         |     |
|-------------------------|-----|
| New axiom:              | no  |
| New reusable helper:    | yes |
| Book Zero helper added: | no  |
| Build clean:            | yes |
| sorry:                  | no  |

## 34.11 Conclusion

Proposition I.34 is short at the Proposition-file level because the general Euclidean theory of parallelograms has already been established in the Hilbert interface. This is a deliberate gain from the layered architecture of the project.

Euclid obtains the theorem from parallel-angle relations and ASA. The present Lean reconstruction instead reuses the already derived congruence of opposite sides and applies SSS across the diagonal. The resulting proof is not a mechanical transcription of Euclid, but it preserves the same synthetic geometry while exposing where the Euclidean content actually resides.

The third classical assertion is represented without introducing area: the diagonal  $AC$  is proved to cut the parallelogram into the congruent triangles  $ABC$  and  $CDA$ .

## Chapter 35

# Proposition I.35

### 35.1 Introduction

Euclid’s Proposition I.35 is the first proposition in Book I in which the word “equal” no longer refers merely to congruent segments or congruent angles. The proposition states that two parallelograms on the same base and in the same parallels are equal to one another. The two parallelograms need not be congruent. What Euclid uses is a cut-and-paste comparison of rectilinear figures.

This makes I.35 a genuine transition point in the reconstruction. Up to I.34, the main formal objects are points, lines, betweenness, segment congruence, angle congruence, and parallelism. At I.35 one must decide what “equal figures” means before the classical argument can even be stated faithfully.

The present development does not introduce numerical area. Instead it introduces a small formal scissors calculus. Triangles are the basic pieces, finite formal sums of triangles represent decompositions, and a generated relation records the operations of splitting a triangle, replacing it by a congruent triangle, adding decompositions, and composing such steps.

The final Lean theorem therefore concludes not a numerical area equality but formal equicomplementability:

```
theorem euclid_proposition_35
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D E F : Geo.Point)
  (hABCD : IsParallelogram Geo A B C D)
  (hEBCF : IsParallelogram Geo E B C F)
  (hADF : Geo.Between A D F)
  (hAEF : Collinear Geo A E F) :
  HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo A B C D)
    (hilbertParallelogramTerm Geo E B C F)
```

The extra order and collinearity hypotheses are not cosmetic. They make explicit part of the configuration which is left implicit by the classical diagram and are essential to a complete formal case analysis.



## 35.2 The Classical Configuration

Let  $ABCD$  and  $EBCF$  be parallelograms on the common base  $BC$ . Their opposite sides  $AD$  and  $EF$  lie on one straight line which is parallel to  $BC$ . Euclid's statement is therefore

$$ABCD \text{ and } EBCF \text{ on the same base } BC \implies ABCD = EBCF$$

in Euclid's language of equality of rectilinear figures.

A complete synthetic proof has to distinguish three possible positions of the two upper sides. Wilkins isolates exactly these configurations:

1. the upper sides overlap, with order  $A, E, D, F$ ;
2. the upper sides meet at a common endpoint, so  $E = D$ ;
3. the upper sides are disjoint, with order  $A, D, E, F$ .

Only the third configuration is treated explicitly in Euclid's proof. The first two are not exceptional from the formal point of view: they are genuine cases which a theorem prover cannot recover from the picture.

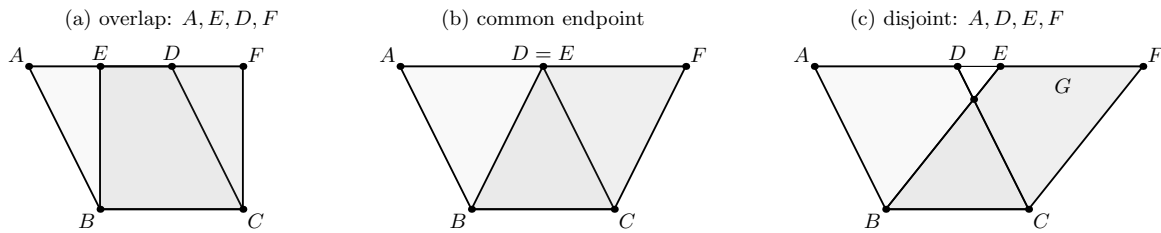


Figure 35.1: The three upper-side configurations relevant to Proposition I.35. The point  $G$  is required only in the disjoint case.

The current Lean theorem uses  $A - D - F$  as an outer container:

$$A - D - F, \quad A, E, F \text{ collinear.}$$

From the parallelogram hypotheses and this data, the reconstruction first proves  $A - E - F$ . The relative order of the two interior points  $D$  and  $E$  then yields precisely the three cases shown in Figure 35.1.

## 35.3 Classical Proof

Euclid's explicit proof is the disjoint case. Since  $ABCD$  and  $EBCF$  are parallelograms, Proposition I.34 gives

$$AD = BC = EF, \quad AB = DC.$$

Because  $DE$  is common to the two longer upper segments, Common Notions 1 and 2 yield

$$AE = DF.$$

The lines  $AB$  and  $DC$  are parallel, and the upper line  $AF$  is a transversal. Proposition I.29 therefore gives

$$\angle EAB = \angle FDC.$$

Euclid then uses Proposition I.4 to conclude that the outer triangles are equal:

$$\triangle EAB = \triangle FDC.$$

In the disjoint configuration the lines  $EB$  and  $DC$  meet at  $G$ . Euclid subtracts the common triangle  $DGE$  from the two equal outer triangles, obtaining equal trapezia, and then adds the common triangle  $GBC$ . This produces the two parallelograms.

The classical dependency graph is therefore

$$\begin{array}{c}
 \text{I.34} + \text{I.29} \\
 \Downarrow \\
 AE = DF, \quad AB = DC, \quad \angle EAB = \angle FDC \\
 \Downarrow \quad \text{I.4} \\
 \triangle EAB = \triangle FDC \\
 \Downarrow \quad \text{C.N.3} \\
 \text{equal remainders} \\
 \Downarrow \quad \text{C.N.2} \\
 ABCD = EBCF.
 \end{array}$$

The overlap and common-endpoint cases use the same outer-triangle comparison but a different addition pattern. Wilkins' three-case analysis is therefore a useful completion of Euclid's single drawn configuration.

## 35.4 Equidecomposition and Equicomplementability

The decisive modern control source for I.35 is Hilbert's theory of plane area. In Chapter IV, Section 18 of the *Foundations of Geometry*, Hilbert defines two polygons to be equidecomposable when they can be triangulated into finitely many pairwise congruent triangles. He then defines equicomplementability by allowing pairs of equidecomposable polygons to be adjoined before comparing the resulting figures by equidecomposition.

In Section 19 Hilbert states Theorem 44: two parallelograms with the same bases and the same altitudes are equicomplementable. This is the Hilbertian counterpart of the Euclidean area argument which begins at I.35.

The distinction between the two notions is essential. Equidecomposability is a direct finite scissors congruence. Equicomplementability is weaker: the figures may become equidecomposable only after suitable complements have been adjoined. Hilbert explicitly notes that in non-Archimedean geometry figures with equal bases and altitudes may be equicomplementable without being equidecomposable.

Hartshorne makes the same distinction under the name *equal content*. His definition requires figures  $P, Q$  and  $P', Q'$  to be nonoverlapping, requires  $Q$  and  $Q'$  to be equidecomposable, and requires the two enlarged figures to be equidecomposable. His Example 22.1.2 identifies Euclid I.35 as an example of equal content.

This matters for the present Lean development because the formal calculus is not yet a complete model of polygonal regions. It records formal sums of triangles, but it does not yet carry an interior or a nonoverlap predicate. Consequently

`HilbertScissorsEquicomplementable`

should be read exactly as its source comment says: a *term-level analogue* of Hilbert’s equicomplementability relation. The theorem proved in I.35 is therefore a faithful algebraic scissors certificate, but a later semantic layer would still be needed to identify every formal sum with an actual nonoverlapping polygonal union.

### 35.4.1 Why numerical area is deliberately avoided

A modern proof could assign an area to each parallelogram and write

$$\text{area} = \text{base} \cdot \text{height}.$$

That would be the wrong logical direction here. Numerical area is developed later than the elementary scissors theory in both Hilbert and Hartshorne. Using it at I.35 would replace Euclid’s synthetic argument by a stronger and logically downstream theorem.

The present reconstruction therefore stays at the level of finite cutting, congruence, and recombination.

## 35.5 The Hilbert Scissors Calculus

The file `HilbertScissors.lean` introduces two formal types:

```
abbrev HilbertTriangleTerm :=
  Multiset Geo.Point

abbrev HilbertScissorsTerm :=
  Multiset (HilbertTriangleTerm Geo)
```

A triangle is represented by an unordered multiset of its three vertices. A scissors term is an unordered multiset of triangle terms. This choice has two useful formal consequences:

- the order of the three names used to write a triangle is not geometric data;
- the order in which the triangular pieces are listed is not geometric data either.

Thus much of the bookkeeping which would otherwise be expressed by explicit list permutations disappears at the representation level.

The central generated relation is

```

inductive HilbertScissorsEq
  (Geo : Geometry.Geo) :
  HilbertScissorsTerm Geo ->
  HilbertScissorsTerm Geo ->
  Prop

| refl
  (P : HilbertScissorsTerm Geo) :
  HilbertScissorsEq Geo P P

| symm
  {P Q : HilbertScissorsTerm Geo}
  (h : HilbertScissorsEq Geo P Q) :
  HilbertScissorsEq Geo Q P

| trans
  {P Q R : HilbertScissorsTerm Geo}
  (hPQ : HilbertScissorsEq Geo P Q)
  (hQR : HilbertScissorsEq Geo Q R) :
  HilbertScissorsEq Geo P R

| add
  {P Q R S : HilbertScissorsTerm Geo}
  (hPQ : HilbertScissorsEq Geo P Q)
  (hRS : HilbertScissorsEq Geo R S) :
  HilbertScissorsEq Geo
    (P + R)
    (Q + S)

| split
  (A B C X : Geo.Point)
  (hBXC : Geo.Between B X C) :
  HilbertScissorsEq Geo
    (hilbertScissorsTriangle Geo A B C)
    (hilbertScissorsTriangle Geo A B X +
     hilbertScissorsTriangle Geo A X C)

| congruent
  (A B C D E F : Geo.Point)
  (hCong :
    TriangleCongruenceResult
      Geo A B C D E F) :
  HilbertScissorsEq Geo
    (hilbertScissorsTriangle Geo A B C)
    (hilbertScissorsTriangle Geo D E F)

```

The two geometric generators are therefore exactly the expected ones:

- **split**: cut a triangle at a point lying between two vertices;
- **congruent**: replace one triangle by a congruent triangle.

The remaining constructors make this relation an equivalence relation compatible with formal addition.

For formal equicomplementability, the implementation asks for formal complements  $R$  and  $S$  such that

$$R \sim S \quad \text{and} \quad P + R \sim Q + S,$$

where  $\sim$  denotes `HilbertScissorsEq`. This is the algebraic pattern of Hilbert's definition, while the geometric nonoverlap conditions remain outside the present term language.

### 35.5.1 Triangulation lemmas

Two reusable lemmas are particularly important:

```
parallelogram_two_triangles
crossing_quadrilateral_two_triangles
```

The first proves that the two diagonal triangulations of a parallelogram are scissors-equivalent. It uses the existing theorem that the diagonals of a Euclidean parallelogram meet internally.

The second is more general. If the diagonals  $AC$  and  $BD$  of a quadrilateral meet internally at  $M$ , then

$$ABC + ACD \sim ABD + BCD.$$

The scissors move is easier to read as a three-stage refinement.

$$ABC + ACD$$

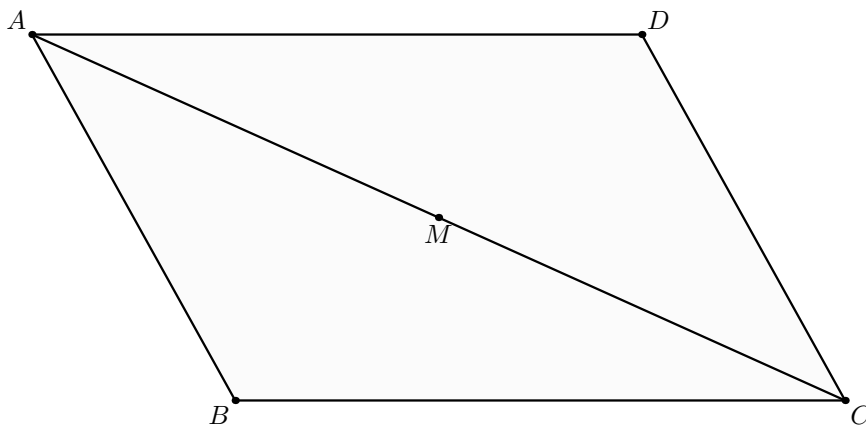


Figure 35.2: First triangulation of the quadrilateral, along the diagonal  $AC$ .

First, split the two triangles along the intersection point  $M$  of the diagonals. This refines the decomposition into the same four elementary triangles.

The four pieces can then be regrouped along the other diagonal  $BD$ .

Its proof is pure scissors algebra after the two betweenness witnesses are provided. This lemma is what makes the overlap case short: the common trapezoid  $EBCD$  can be flipped from one diagonal triangulation to the other without introducing a special area theorem.

$$ABM + BMC + CMD + DMA$$

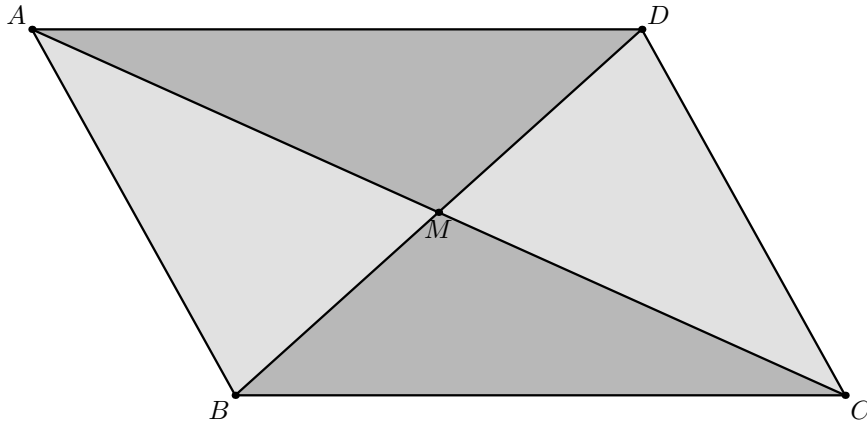


Figure 35.3: Common refinement at the diagonal intersection  $M$ .

$$ABD + BCD$$

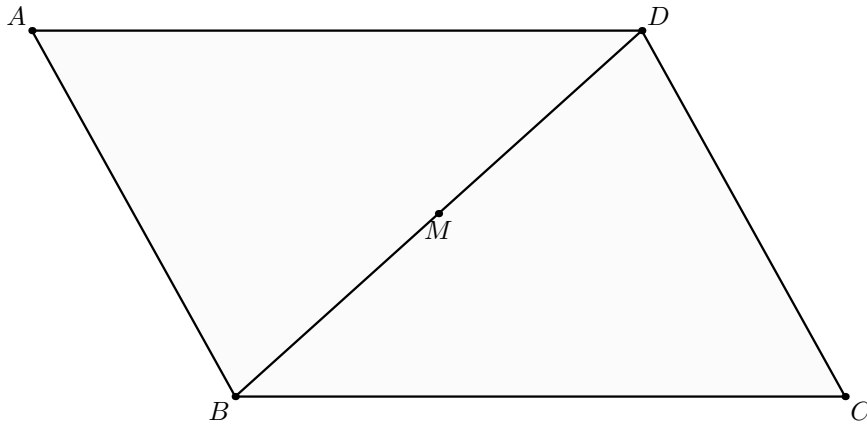


Figure 35.4: Second triangulation of the same quadrilateral, along the diagonal  $BD$ .

## 35.6 Proof Architecture of the Reconstruction

The actual Lean proof is not a mechanical transcription of Euclid's proof. It has two logically distinct stages.

### 35.6.1 Stage 1: recover the upper-line order

The final theorem is given

$$A - D - F \quad \text{and} \quad A, E, F \text{ collinear.}$$

The theorem `i35_upper_order` proves

$$A - E - F.$$

The proof uses opposite-side congruence for the two parallelograms to obtain  $AD \cong EF$ . Hilbert's betweenness trichotomy on  $A, E, F$  then gives three possible orders. The two bad orders are excluded separately.

The first exclusion,

`i35_not_EAF,`

is a segment-order argument. From  $A - D - F$  one gets  $AD < AF$ ; congruence transports this to  $EF < AF$ . The bad order  $E - A - F$  would imply the reverse strict inequality, contradicting asymmetry.

The second exclusion,

`i35_not_AFE,`

is genuinely planar. The helper `i35_cross_intersection` first constructs a point where  $BF$  crosses  $AC$ . The diagonal theorem for the second parallelogram gives another point where  $BF$  crosses  $EC$ . Under the bad order  $A - F - E$ , the same line would also meet the third side  $AE$  of the triangle  $AEC$ . The line-separation theorem `hilbert_line_avoids_third_triangle_side` forbids one line from meeting all three sides in the required interior configuration.

Once  $A - E - F$  is known, the pure order theorem `i35_right_cases_of_common_container` compares the two interior points  $D$  and  $E$  and produces exactly

$$A - E - D - F, \quad E = D, \quad \text{or} \quad A - D - E - F.$$

### 35.6.2 Stage 2: solve the three scissors cases

The three cases are dispatched by `i35_from_three_cases`.

#### Overlap case

The outer triangles  $EAB$  and  $FDC$  are shown congruent. The common trapezoid  $EBCD$  is then re-triangulated by `crossing_quadrilateral_two_triangulations`. No nonzero complement is needed. The two parallelogram terms are directly scissors-equivalent, so they are equicomplementable with zero complements.

#### Common-endpoint case

The two parallelograms share the upper endpoint. Again no auxiliary point  $G$  is required. The proof changes the triangulations and replaces the two outer triangles by congruent ones. The resulting parallelogram terms are directly scissors-equivalent, hence equicomplementable with zero complements.

#### Disjoint case

This is Euclid's explicit configuration. Here the point  $G$  is essential. The helper

## i35\_disjoint\_intersection

constructs  $G$  with

$$E - G - B, \quad D - G - C.$$

The existence proof is not read off from the diagram. It uses Hilbert order and Pasch-type intersection machinery, followed by parallel/order transport to place the intersection on both required segments.

Instead of literally subtracting  $DGE$  as Euclid does, the formal proof uses the equivalent completion pattern. It is useful to display the four stages separately.

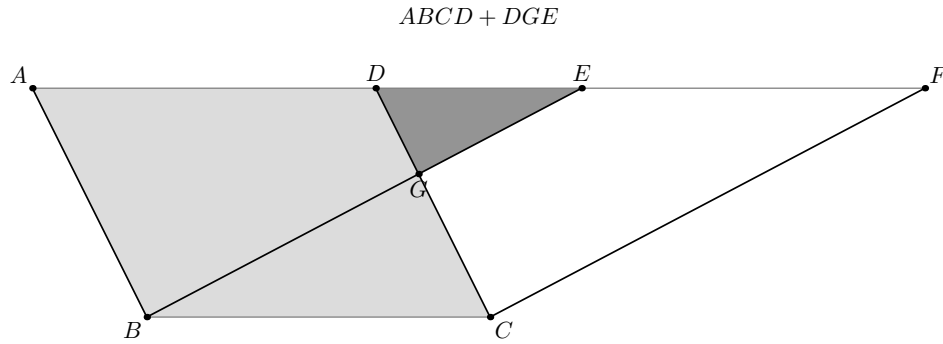


Figure 35.5: Step 1. Adjoin the triangle  $DGE$  to the parallelogram  $ABCD$ .

Using the betweenness relations  $E - G - B$  and  $D - G - C$ , split and regroup the augmented figure. The result is the outer triangle  $EAB$  together with the common triangle  $GBC$ .

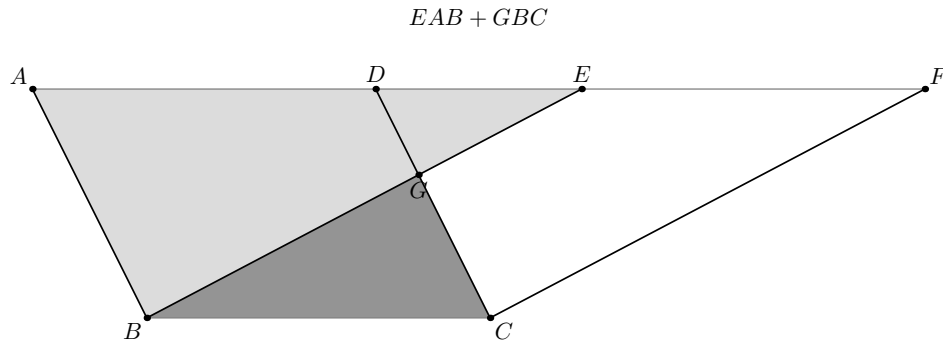


Figure 35.6: Step 2. The augmented left-hand figure is rearranged as  $EAB + GBC$ .

The outer triangles  $EAB$  and  $FDC$  are congruent. The scissors calculus therefore replaces the first outer triangle by the second while leaving the common triangle  $GBC$  unchanged.

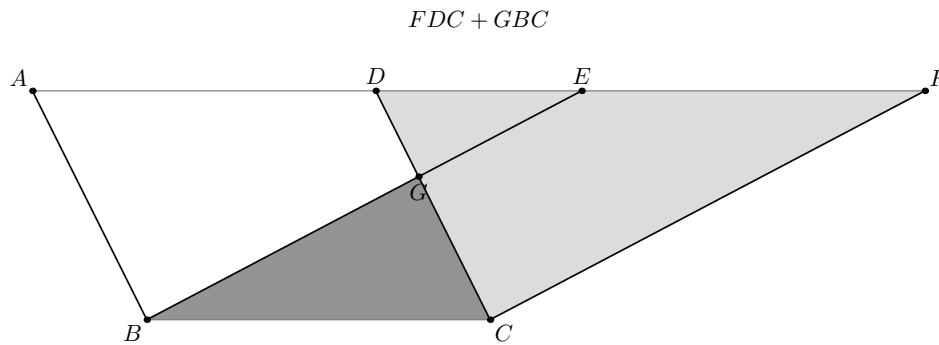
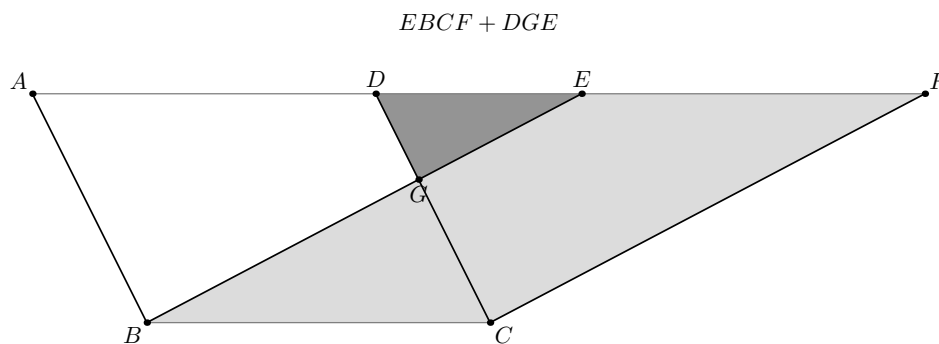
Finally, reverse the splitting and regrouping on the right-hand side.

Thus the complete formal chain is

$$\begin{aligned} ABCD + DGE &\sim EAB + GBC \\ &\sim FDC + GBC \\ &\sim EBCF + DGE. \end{aligned}$$

This is exactly the formal equicomplementability witness needed for I.35.



Figure 35.7: Step 3. Replace the congruent outer triangle:  $EAB + GBC \sim FDC + GBC$ .Figure 35.8: Step 4. Regroup the pieces as the second augmented parallelogram  $EBCF + DGE$ .

### 35.6.3 The complete Lean DAG

The reconstruction can be summarized as

$$\begin{array}{c}
 \text{two parallelograms on } BC + A - D - F + \text{Collinear}(A, E, F) \\
 \Downarrow \\
 AD \cong EF + \text{segment order} + \text{line separation} \\
 \Downarrow \\
 A - E - F \\
 \Downarrow \\
 \boxed{A - E - D - F} \quad \boxed{E = D} \quad \boxed{A - D - E - F} \\
 \Downarrow \\
 \text{overlap scissors proof} \quad \text{common-endpoint proof} \quad \text{disjoint proof with } G \\
 \Downarrow \\
 \text{HilbertScissorsEquicomplementable} \\
 \Downarrow \\
 \text{euclid\_proposition\_35.}
 \end{array}$$

## 35.7 Lean Formalization

The final Proposition file is intentionally short. It imports the separate scissors module and exposes only the Book I theorem:

```
import CGJteamLab.HilbertScissors
```

```

namespace Geometry

universe u

variable (Geo : Geometry.Geo)

theorem euclid_proposition_35
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D E F : Geo.Point)
  (hABCD : IsParallelogram Geo A B C D)
  (hEBCF : IsParallelogram Geo E B C F)
  (hADF : Geo.Between A D F)
  (hAEF : Collinear Geo A E F) :
  HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo A B C D)
    (hilbertParallelogramTerm Geo E B C F) := by
exact
  i35_complete
    Geo
    A B C D E F
    hABCD
    hEBCF
    hADF
    hAEF

end Geometry

```

All substantial work is therefore isolated in `HilbertScissors.lean`. This is deliberate. Proposition I.35 is the first client of the scissors language, but the formal content calculus is not part of the statement of Euclid I.35 itself.

The endpoint theorem `i35_complete` is also thin. Its role is to package the established ordered proof:

```

i35_upper_order
  ↓↓
i35_right_cases_of_common_container
  ↓↓
i35_from_three_cases
  ↓↓
i35_complete.

```

## 35.8 Relationship with Earlier Propositions

### 35.8.1 Proposition I.29

Euclid uses I.29 explicitly to identify the corresponding angles  $\angle EAB$  and  $\angle FDC$ . The current Lean proof does not replay this exact SAS route at the top level. Parallelism enters through the developed Hilbert theory and through the parallelogram infrastructure.

### 35.8.2 Proposition I.34

I.34 is the immediate metric input. Opposite sides of the two parallelograms are congruent, so

$$AD \cong BC \cong EF$$

and the second pair of opposite sides provides the remaining outer-triangle side congruence. The Lean helper `ParallelogramOppositeSidesCongruent` packages this information.

The current reconstruction then proves the outer triangles by the already available Hilbert SSS theorem rather than by Euclid’s SAS argument. This is another example of the project deliberately reusing a stronger developed Hilbert API rather than mechanically replaying the historical DAG.

### 35.8.3 Transition to Proposition I.36

I.36 is a direct generalization from the same base to equal bases. Euclid’s classical proof uses I.33 to construct an intermediate parallelogram and then applies I.35 twice. For the formal development this means that I.35 is not an isolated experiment: the scissors/equicomplementability language is now a real dependency of the next area propositions.

## 35.9 Hilbert and Book Zero Components Used

The I.35 reconstruction uses a much broader part of the developed Hilbert interface than the final 38-line Proposition file suggests.

- `IsParallelogram` and the Euclidean parallelogram theory;
- `ParallelogramOppositeSidesCongruent`;
- `ParallelogramDiagonalIntersectionExists`;
- the derived `HilbertSSS` triangle-congruence theorem;
- Hilbert betweenness trichotomy and inner/outer transitivity;
- synthetic segment comparison and transport under congruence;
- Pasch/intersection machinery for the disjoint case;
- line separation through `hilbert_line_avoids_third_triangle_side`;
- transport of betweenness from collinearity and parallelism;
- the newly introduced formal scissors layer in `HilbertScissors.lean`.

No new Book Zero theorem is introduced by the final Proposition file. The new reusable material is instead concentrated in the scissors module.

The source hierarchy also changes emphasis at this point. Borsuk remains a useful control source for order, congruence, and parallelogram geometry, but the new notion of equality of figures is controlled more directly by Hilbert’s Chapter IV and Hartshorne’s theory of content. The Beeson proof corpus remains useful primarily as an audit of hidden incidence and order conditions rather than as the architecture of the Lean proof.

## 35.10 Formalization Notes

### 35.10.1 Axiomatic status

The final theorem assumes

```
[HilbertEuclideanPlane Geo]
```

and is therefore Euclidean in the current project API. This is consistent with Hilbert's Chapter IV treatment, which assumes the parallel axiom in addition to incidence, order, and congruence. The present file does not claim a sharper minimal axiom boundary for every internal helper.

The Euclidean strength is already visible in the reusable parallelogram results. Opposite-side congruence and the developed intersection/parallel machinery are called through the Euclidean-plane instance used by the project. Once the relevant congruences and betweenness data are available, individual scissors rewrites themselves are purely formal.

### 35.10.2 Hidden configuration data

I.35 exposes more hidden diagrammatic information than the immediately preceding propositions. The formal proof must explicitly manage:

- the outer order  $A - D - F$ ;
- collinearity of  $A, E, F$ ;
- the derived order  $A - E - F$ ;
- the trichotomy between overlap, common endpoint, and disjoint upper sides;
- existence and exact betweenness position of the point  $G$  in the disjoint case;
- internal diagonal intersections used for re-triangulation;
- non-collinearity needed for Pasch and line-separation arguments.

The point  $G$  is especially instructive. It is not a universal witness for all forms of I.35. It appears only in the disjoint configuration. Trying to force one  $G$ -based proof onto every configuration would obscure the geometry and create unnecessary order obligations.

### 35.10.3 The scissors language is formal, not yet semantic

The nested `Multiset` representation is a formal algebra of triangle terms. It intentionally forgets vertex order and piece order, but it also forgets geometric interiors. Therefore formal addition is total even when two actual geometric pieces would overlap.

This is the main semantic limitation of the present module. A future polygonal semantics could add a nonoverlap predicate, partial composition, or a realization theorem connecting valid formal scissors derivations to nonoverlapping rectilinear figures. None of that is required to complete the current I.35 certificate, but the distinction is documented here so that the formal relation is not silently identified with a stronger geometric notion.

### 35.10.4 Reusable helpers introduced

The most obviously reusable scissors helpers are

```
triangle_swap12
triangle_swap23
triangle_cycle
scissors_triangle_swap12
scissors_triangle_swap23
scissors_triangle_cycle
parallelogram_two_triangulations
crossing_quadrilateral_two_triangulations
equicomplementable_refl
equicomplementable_symm
```

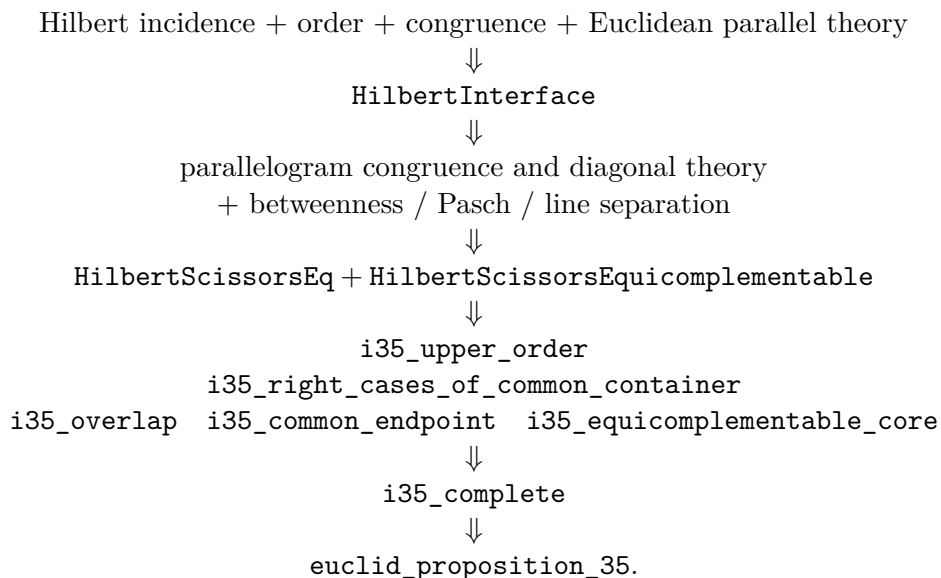
The remaining helpers whose names begin with `i35_` are deliberately left proposition-specific for now. Some may later reveal a more general Hilbert statement, but no architectural promotion is required for I.35.

## 35.11 Dependency Summary

The classical dependency is

$$\text{I.34} + \text{I.29} + \text{I.4} + \text{Common Notions 1-3} \longrightarrow \text{I.35.}$$

The actual Lean dependency is better represented as



|                       |                                         |
|-----------------------|-----------------------------------------|
| New axiom:            | no                                      |
| New formal framework: | yes — <code>HilbertScissors.lean</code> |
| New Book Zero helper: | no                                      |
| Build clean:          | yes                                     |
| sorry:                | no                                      |

## 35.12 Conclusion

Proposition I.35 is a structural turning point in the formal reconstruction of Book I. The geometry is still elementary, but the proposition introduces an equality of figures which cannot be represented faithfully by segment or angle congruence alone.

Hilbert's distinction between equidecomposability and equicomplementability provides the correct conceptual guide. The present Lean development follows that guide by introducing a formal scissors calculus rather than a numerical area function. The calculus is intentionally modest: it is a term-level algebra of triangular decompositions, not yet a full semantic theory of polygonal regions.

The proof itself also exposes a classical omission which is invisible on the usual diagram. There are three upper-side configurations, and the auxiliary intersection point  $G$  belongs only to the disjoint one. Once this case split is made at the correct level, the proof decomposes naturally into order geometry, triangle congruence, and formal scissors recombination.

The resulting Proposition file is short because the complexity has been placed in a reusable and explicitly named layer. This is exactly the kind of separation needed for the next area propositions, beginning with I.36.

## Chapter 36

# Proposition I.36

### 36.1 Introduction

Proposition I.36 is the first immediate reuse of the content machinery introduced at I.35. Euclid passes from parallelograms on the same base to parallelograms on equal bases. The classical proof constructs an intermediate parallelogram and then applies I.35 twice.

The present reconstruction keeps the same content architecture but changes the geometric route to the intermediate parallelogram. The final Lean theorem does not import Proposition I.33 and does not require an explicit crossing point. Instead it reconstructs the missing pair of parallel sides from the developed Hilbert theory.

The conclusion remains synthetic. No numerical area is introduced. The two parallelogram terms are proved formally equicomplementable in the scissors calculus already used for I.35.

### 36.2 The Classical Configuration

Let  $ABCD$  and  $EFGH$  be parallelograms on equal bases  $BC$  and  $FG$ , lying between the same two parallels. In the standard ordered configuration the points satisfy

$$A - D - E - H, \quad B - C - F - G.$$

The bases satisfy

$$BC \cong FG.$$

The current Lean theorem makes this order explicit through four strict betweenness hypotheses:

$$A - D - E, \quad D - E - H, \quad B - C - F, \quad C - F - G.$$

Thus the formal statement is configuration-faithful to the standard ordered diagram, but it has stronger positional hypotheses than Euclid's order-free wording.

### 36.3 Classical Proof

Euclid first uses Proposition I.34 on the parallelogram  $EFGH$ :

$$EH \cong FG.$$

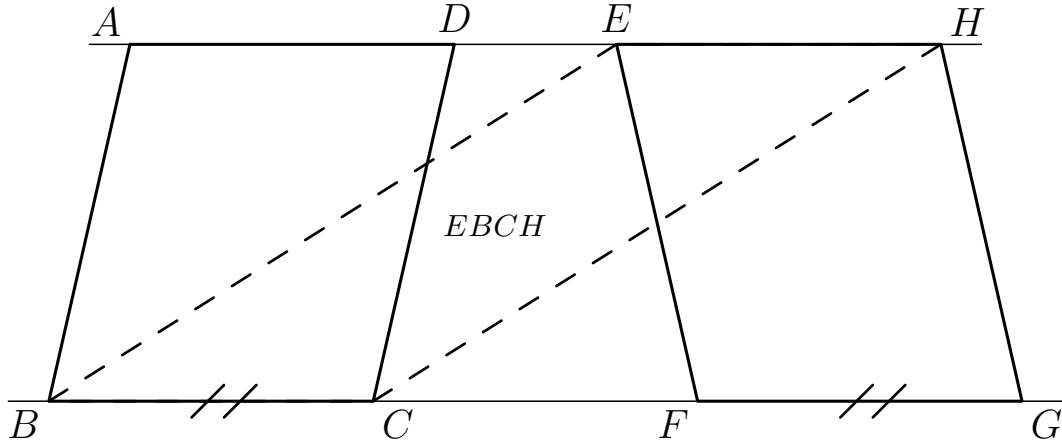


Figure 36.1: The ordered configuration used by the Lean theorem. The dashed quadrilateral  $EBCH$  is the intermediate parallelogram constructed internally by the reconstruction.

Since the bases are equal,

$$BC \cong FG,$$

Common Notion 1 gives

$$BC \cong EH.$$

The segments  $BC$  and  $EH$  are also parallel. Proposition I.33 is then used to conclude that the joining segments  $EB$  and  $HC$  are equal and parallel, hence  $EBCH$  is a parallelogram.

Now  $ABCD$  and  $EBCH$  are parallelograms on the same base  $BC$  and in the same parallels, so Proposition I.35 gives

$$ABCD = EBCH.$$

Similarly,  $EFGH$  and  $EBCH$  are equal by I.35. Common Notion 1 therefore yields

$$ABCD = EFGH.$$

The classical dependency graph is

$$\begin{array}{c}
 \text{I.34} + BC \cong FG \\
 \Downarrow \\
 BC \cong EH \\
 \Downarrow \quad \text{I.33} \\
 EBCH \text{ is a parallelogram} \\
 \Downarrow \quad \text{I.35 twice} \\
 ABCD = EBCH = EFGH \\
 \Downarrow \quad \text{C.N.1} \\
 ABCD = EFGH.
 \end{array}$$

This is exactly the role of I.36 in the classical sequence: it is a generalization of I.35 obtained by inserting one intermediate parallelogram.



## 36.4 The Main Issue: Constructing the Intermediate Parallelogram

The difficult part of the formal reconstruction is not the content transitivity. Once  $EBCH$  has been obtained, I.35 already supplies the required equality of figures. The real issue is geometric:

construct  $EBCH$  from the ordered I.36 data.

The final proof does this without Proposition I.33 and without introducing a crossing witness.

### 36.4.1 The upper and lower parallel sides

The helper

```
i36_equal_parallel_EH_BC
```

extracts two facts:

$$EH \parallel BC, \quad EH \cong BC.$$

Parallelism is transported from the side  $FG$  of  $EFGH$  to the collinear carrier  $BC$ . Congruence comes from opposite-side congruence in the parallelogram together with the hypothesis  $BC \cong FG$ .

### 36.4.2 The shifted bottom segments

From

$$B - C - F - G, \quad BC \cong FG,$$

the helper

```
i36_bottom_shift
```

derives

$$BF \cong CG$$

by segment additivity. This is the metric relation needed for the later SAS comparison.

### 36.4.3 The corresponding angle

The helper

```
i36_corresponding_angle
```

proves

$$\angle EFB \cong \angle HGC.$$

This is where the formal proof exposes the side-of-line data hidden by the ordinary diagram. A point  $X$  is constructed with

$$E - F - X.$$

The line through  $F$  and  $G$  is used as a carrier. The proof establishes the required SameSide and OppositeSide relations, transports  $EF \parallel GH$  to  $FX \parallel GH$ , and then applies the developed Hilbert theorem

```
hilbert_alternate_angles_of_parallel_
oppositeSide_lines
```

to obtain an alternate-angle congruence. Same-ray normalization and vertical angles convert that statement to the required corresponding-angle form.

#### 36.4.4 SAS across the two bases

The helper

```
i36_cross_triangles_congruent
```

compares the triangles

$$\triangle FEB \quad \text{and} \quad \triangle GHC.$$

The three inputs are

$$FE \cong GH, \quad FB \cong GC, \quad \angle EFB \cong \angle HGC.$$

SAS therefore produces a full TriangleCongruenceResult. In particular,

$$\angle FBE \cong \angle GCH.$$

#### 36.4.5 Recovering the second pair of parallel sides

The helper

```
i36_cross_parallel
```

uses the angle obtained from SAS to prove

$$EB \parallel HC.$$

Choose  $T$  with

$$B - T - C$$

and extend  $HC$  beyond  $C$  to a point  $Y$ :

$$H - C - Y.$$

The bottom carrier gives the required line-incidence data. Because  $E$  and  $H$  lie on the same side of that carrier, while  $H$  and  $Y$  lie on opposite sides, one obtains that  $E$  and  $Y$  are on opposite sides.

The angle from SAS is normalized to the rays through  $T$  and  $Y$ . Vertical angles at  $C$  give

$$\angle TBE \cong \angle TCY.$$

The neutral converse alternate-angle criterion

```
hilbert_parallel_of_alternate_angles_
oppositeSide_lines
```

then yields

$$BE \parallel CY.$$

Finally parallelism is transported along the collinear triple  $H, C, Y$ :

$$EB \parallel HC.$$

The helper

```
i36_intermediate_EBCH_no_crossing
```

combines

$$EH \parallel BC \quad \text{and} \quad EB \parallel HC$$

and returns

$$\text{IsParallelogram}(E, B, C, H).$$

## 36.5 Proof Architecture of the Reconstruction

After separating geometry from content, the final proof has a short top-level shape.

The geometric branch is

$$\begin{array}{c}
 BC \cong FG, \quad EFGH, \quad B - C - F - G \\
 \Downarrow \\
 EH \parallel BC, \quad EH \cong BC \\
 \Downarrow \\
 BF \cong CG \\
 \Downarrow \\
 \angle EFB \cong \angle HGC \\
 \Downarrow \text{SAS} \\
 \angle FBE \cong \angle GCH \\
 \Downarrow \\
 EB \parallel HC \\
 \Downarrow \\
 EBCH \text{ is a parallelogram.}
 \end{array}$$

The content branch is

$$\begin{array}{c}
 ABCD, EBCH, EFGH \\
 \Downarrow \text{I.35} \\
 ABCD \sim EBCH \\
 \Downarrow \text{I.35} \\
 EBCH \sim EFGH \\
 \Downarrow \text{transitivity} \\
 ABCD \sim EFGH.
 \end{array}$$

Here  $\sim$  denotes the formal `HilbertScissorsEquicomplementable` relation.

This separation is significant. The geometry of I.36 is new local work, but the equality-of-figures layer is inherited from I.35.

## 36.6 Lean Formalization

The current file imports only Proposition I.35:

```
import CGJteamLab.Proposition35
```

The public theorem is:

```
theorem euclid_proposition_36
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D E F G H : Geo.Point)
  (hABCD : IsParallelogram Geo A B C D)
  (hEFGH : IsParallelogram Geo E F G H)
  (hBCFG : Geo.Congruent B C F G)
  (hADE : Geo.Between A D E)
  (hDEH : Geo.Between D E H)
  (hBCF : Geo.Between B C F)
  (hCFG : Geo.Between C F G) :
  HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo A B C D)
    (hilbertParallelogramTerm Geo E F G H)
```

The local helper sequence is:

```
i36_equal_parallel_EH_BC
i36_bottom_shift
i36_corresponding_angle
i36_cross_triangles_congruent
i36_cross_parallel
i36_intermediate_EBCH_no_crossing
i36_from_intermediate
euclid_proposition_36
```

The final theorem first derives the outer order information

$$A - E - H, \quad A - D - H,$$

and

$$B - F - G, \quad B - C - G$$

from the four explicit betweenness hypotheses. It then constructs  $EBCH$  through the no-crossing helper and sends the result to the content helper.

The helper

```
i36_from_intermediate
```

is the exact content bridge. It applies I.35 once to  $ABCD$  and  $EBCH$ . For the second application it reverses the parallelogram representations so that  $EBCH$  and  $EFGH$  are presented on the common base  $EH$ . The formal reversal is converted into scissors equivalence and then into equicomplementability. Two transitivity steps finish the proof.

## 36.7 Relationship with Earlier Propositions

### 36.7.1 Proposition I.33

I.33 is part of Euclid’s classical DAG. After obtaining  $BC \cong EH$  and  $BC \parallel EH$ , Euclid uses I.33 to conclude that the joining segments form the intermediate parallelogram  $EBCH$ .

The final Lean proof does not import Proposition I.33. Instead it proves  $EB \parallel HC$  directly by the SAS and alternate-angle route described above. Thus I.33 remains historically relevant but is not a formal dependency of the final file.

### 36.7.2 Proposition I.34

Classically, I.34 supplies

$$EH \cong FG.$$

The current Lean proof accesses this content through the developed parallelogram API, in particular:

```
ParallelogramOppositeSidesCongruent
```

Thus the mathematical input of I.34 is present even though the Book I theorem is not called by name.

### 36.7.3 Proposition I.35

I.35 is the decisive content dependency. Once  $EBCH$  has been constructed, the final result is obtained by two applications of I.35 plus transitivity of equicomplementability.

This agrees with the classical organization and with modern content-based treatments of I.36.

## 36.8 Hilbert and Book Zero Components Used

The proof uses three layers of developed infrastructure.

First, the Hilbert geometry layer supplies:

- parallelogram opposite-side congruence;
- transport of parallelism along collinear carriers;
- betweenness extension and inner/outer transitivity;
- SameRay normalization of angles;

- SameSide and OppositeSide transport;
- vertical angles;
- SAS triangle congruence;
- the Euclidean theorem giving equal alternate angles from parallel lines;
- the neutral converse criterion giving parallel lines from equal alternate angles.

Second, the Book Zero layer supplies elementary order and congruence infrastructure used by the developed Hilbert interface. Proposition I.36 does not introduce a new Book Zero theorem.

Third, the scissors layer inherited from I.35 supplies:

- parallelogram terms;
- representation reversal;
- conversion from scissors equivalence to equicomplementability;
- transitivity of equicomplementability.

No polygonal realization layer is introduced. The existing `HilbertScissorsTerm` representation is sufficient.

## 36.9 Formalization Notes

### 36.9.1 Geometry, content, and representation

I.36 is a useful example of the three-way separation introduced after I.35.

**Geometry.** The new work is the intrinsic construction of  $EBCH$  from equal bases, parallel carriers, and the ordered configuration.

**Content.** No new equality-of-figures principle is required. I.35 and transitivity are sufficient.

**Representation.** The only extra bookkeeping is reversal of parallelogram terms before the second application of I.35. This is handled by existing scissors-equivalence helpers.

### 36.9.2 Hidden configuration data

The formal theorem exposes several facts left implicit by the classical diagram:

- the upper order  $A - D - E - H$ ;
- the lower order  $B - C - F - G$ ;
- the carrier line through  $F$  and  $G$ ;
- SameSide information for  $E$  and  $H$  relative to that carrier;

- the extension point  $X$  with  $E - F - X$ ;
- the interior point  $T$  with  $B - T - C$ ;
- the extension point  $Y$  with  $H - C - Y$ ;
- the required non-collinearity conditions for genuine angles;
- the OppositeSide condition used by the alternate-angle criterion.

These data are derived or constructed internally except for the four betweenness hypotheses in the theorem statement.

### 36.9.3 Axiomatic status

The public theorem assumes

```
[HilbertEuclideanPlane Geo]
```

and is therefore Euclidean in the current project API.

The local geometric route is informative about where the parallel axiom is used. The helper proving the corresponding angle invokes

```
hilbert_alternate_angles_of_parallel_
oppositeSide_lines
```

which is the Euclidean direction

parallel lines  $\implies$  equal alternate angles.

By contrast, the final step that recovers  $EB \parallel HC$  invokes

```
hilbert_parallel_of_alternate_angles_
oppositeSide_lines
```

which is the neutral converse direction.

Hilbert's Chapter IV theory of plane area works over Groups I–IV and does not require the continuity group. The formal reconstruction likewise introduces no continuity axiom at I.36.

### 36.9.4 Axiom audit

The final audit is:

```
'Geometry.euclid_proposition_36' depends on axioms:
[proptext, Classical.choice, bookZero_nullSegment1, Quot.sound]
```

The geometric helpers have the smaller audit:

```
i36_cross_parallel:
[propext, Classical.choice, Quot.sound]

i36_intermediate_EBCH_no_crossing:
[propext, Classical.choice, Quot.sound]
```

The local project axiom

```
bookZero_nullSegment1
```

enters only through Proposition I.35 and hence through the content helper. The new I.36 geometry introduces no additional local axiom.

### 36.9.5 Source audit

The source audit gives a consistent picture.

Euclid/Joyce and Wilkins give the classical DAG

$$\text{I.34} \rightarrow \text{I.33} \rightarrow \text{I.35 twice} \rightarrow \text{C.N.1.}$$

Hilbert’s Chapter IV, Section 19, Theorem 44 identifies equicomplementability of parallelograms with equal bases and equal altitudes as the correct synthetic area-theoretic target. It also warns against silently strengthening the result to unconditional equidecomposability in non-Archimedean settings.

Hartshorne’s Section 22 states that I.36 follows from transitivity of equal content. The additional property commonly denoted  $(Z)$  is not required for I.36; it first becomes relevant in the discussion of I.37.

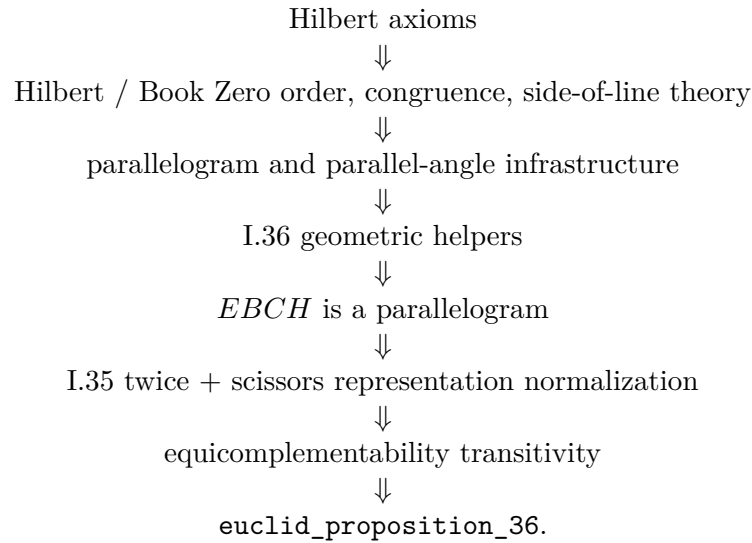
The Beeson proof corpus was used as an audit of hidden orientation and crossing data. The final Lean proof deliberately does not reproduce its case split: the intermediate parallelogram is derived intrinsically instead.

Borsuk–Szmielew and Greenberg remain control sources for order, parallelograms, neutral geometry, and the neutral/Euclidean boundary, but they do not determine the content architecture of the final proof.



## 36.10 Dependency Summary

The complete formal dependency chain is



The short comparison is

Euclid : I.34  $\rightarrow$  I.33  $\rightarrow$  I.35 twice,

Lean : equal-base geometry  $\rightarrow$  SAS  $\rightarrow$  alternate-angle parallelism  $\rightarrow$  *EBCH*  $\rightarrow$  I.35 twice.

|                                     |     |
|-------------------------------------|-----|
| New axiom:                          | no  |
| New proposition-local helpers:      | yes |
| New Book Zero theorem:              | no  |
| New polygon/content representation: | no  |
| Build clean:                        | yes |
| sorry:                              | no  |

## 36.11 Conclusion

Proposition I.36 confirms that the scissors layer introduced at I.35 is already reusable. No new notion of area, content, or polygon is needed. The content argument is exactly the expected transitivity pattern through an intermediate parallelogram.

The principal formal contribution lies instead in the geometry of that intermediate figure. The final reconstruction removes both an explicit crossing witness and the Book I dependency on I.33. Equal-base information, parallel-line angle theory, SAS, side-of-line control, and a neutral alternate-angle criterion are sufficient to recover *EBCH* directly.

Thus I.36 separates cleanly into a new geometric normalization stage followed by an already established I.35 content stage.

## Chapter 37

# Proposition I.37

### 37.1 Introduction

Proposition I.37 is the first triangle theorem in Euclid’s area block. Two triangles have the same base  $BC$ , while their third vertices  $A$  and  $D$  lie on one line parallel to  $BC$ . Euclid concludes that the two triangles are “equal.” At this point of Book I, as already in I.35 and I.36, this is an equality of content rather than congruence of triangles.

The formal reconstruction keeps that conclusion synthetic. It introduces no numerical area and does not strengthen the result to unconditional finite equidecomposability. The public Lean theorem concludes `HilbertScissorsEquicomplementable`.

The principal change from Euclid’s proof is the treatment of the final “halving” step. Euclid surrounds each triangle by a parallelogram, proves the two parallelograms equal by I.35, and then appeals to the fact that halves of equals are equal. The present reconstruction does not add a general cancellation or division-by-two principle. Instead it follows Hilbert’s Chapter IV organization: each triangle is first converted, by the midpoint construction of Hilbert’s Theorem 45, into an equidecomposable parallelogram; those two parallelograms are then compared by the already formalized I.35 scissors machinery.

This makes I.37 a particularly clear example of the three layers that now have to be kept distinct:

$$\text{geometry} \longrightarrow \text{representation} \longrightarrow \text{content}.$$

### 37.2 The Classical Configuration

Let  $ABC$  and  $DBC$  be triangles on the same base  $BC$ . Their top vertices  $A$  and  $D$  lie on a line parallel to  $BC$ :

$$AD \parallel BC.$$

Euclid says that the two triangles are in the same parallels  $AD$  and  $BC$ .

The current Lean theorem encodes the configuration with the single strict parallelism hypothesis

$$\text{Parallel}(A, D; B, C).$$

Because the project’s parallel relation is nondegenerate, this already gives  $A \neq D$  and  $B \neq C$ , and the disjointness of the two carrier lines forces both  $A$  and  $D$  off the base line. Thus the two triangles are internally proved noncollinear; nondegeneracy is not an extra theorem hypothesis.

The formal proof then constructs four midpoint witnesses:

$$\begin{array}{ll} M \text{ midpoint of } BA, & N \text{ midpoint of } CA, \\ P \text{ midpoint of } BD, & Q \text{ midpoint of } CD. \end{array}$$

Hilbert's Theorem 45 is applied to the two triangles and produces points  $F$  and  $G$  such that

$$MBCF \quad \text{and} \quad PBCG$$

are parallelograms, while the original triangles are scissors-equivalent to these parallelogram terms.

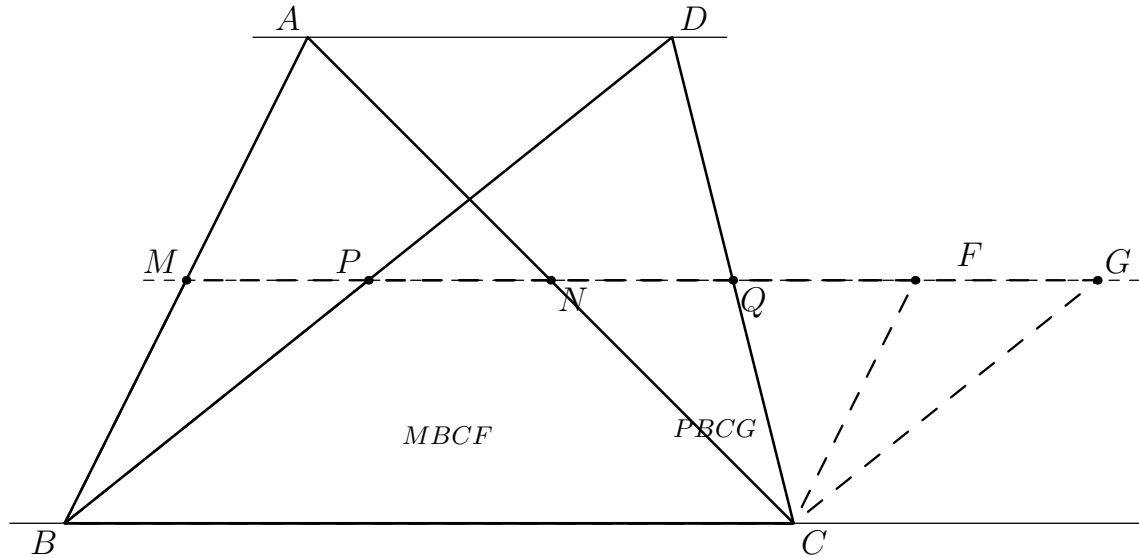


Figure 37.1: The geometric reduction used by the Lean proof. The solid triangles  $ABC$  and  $DBC$  are the figures in the public theorem. The points  $M, N, P, Q$  are the four midpoint witnesses used by two applications of Hilbert's Theorem 45; the dashed figures  $MBCF$  and  $PBCG$  are the resulting parallelograms. The displayed order  $M - P - F - G$  is one representative strict case only. The proof derives the common upper carrier and handles all possible orders rather than assuming the order shown in the picture.

### 37.3 Classical Proof

Euclid extends the top line  $AD$  in both directions to points  $E$  and  $F$ . Through  $B$  he draws a line parallel to  $CA$ , and through  $C$  a line parallel to  $BD$ , using I.31. This produces two parallelograms

$$EBCA \quad \text{and} \quad DBCF.$$

They have the same base  $BC$  and lie in the same parallels, so I.35 gives

$$EBCA = DBCF.$$

By I.34 the diagonal  $AB$  bisects the first parallelogram into two congruent triangles, and the diagonal  $DC$  bisects the second. Hence

$$\triangle ABC \text{ is half of } EBCA, \quad \triangle DBC \text{ is half of } DBCF.$$

Euclid then concludes that the two halves are equal.

The classical dependency graph is therefore

$$\begin{array}{c}
 AD \parallel BC \\
 \Downarrow \text{ I.31 twice} \\
 EBCA, DBCF \text{ parallelograms} \\
 \Downarrow \text{ I.35} \\
 EBCA = DBCF \\
 \Downarrow \text{ I.34 twice} \\
 \triangle ABC, \triangle DBC \text{ are corresponding halves} \\
 \Downarrow \text{ "halves of equals are equal"} \\
 \triangle ABC = \triangle DBC.
 \end{array}$$

The final step is the important foundational point. In Heath’s presentation the principle “halves of equal things are equal” is bracketed: it is not one of Euclid’s explicitly stated Common Notions. A formal development must therefore either justify a suitable cancellation principle or avoid the step. The present reconstruction takes the second route.

### 37.4 Hilbert’s Theorems 44–46 and the Halving Issue

Hilbert’s Chapter IV gives a direct synthetic organization of exactly the material needed here.

**Theorem 44.** Parallelograms with equal bases and equal altitudes are *equicomplementable*.

**Theorem 45.** Every triangle is *equidecomposable* with a parallelogram having an equal base and half the altitude. Hilbert’s construction bisects two sides of the triangle and extends the segment joining the midpoints by an equal amount. The two small triangles created by the construction are congruent, so the triangle can be rearranged into the parallelogram.

**Theorem 46.** From Theorems 44 and 45, together with transitivity of the content relation, Hilbert obtains that triangles with equal bases and equal altitudes are equicomplementable.

This is the correct control theorem for I.37. Hilbert also explicitly warns that the stronger assertion of finite equidecomposability is not available in full generality without an Archimedean hypothesis: in non-Archimedean geometries triangles of equal base and altitude can be equicomplementable without being equidecomposable.

Hartshorne makes the same logical issue visible in the language of equal content. In Section 22 he observes that Euclid’s original proof of I.37 uses “halves of equals are equal,” a property depending on his additional property (Z). He therefore gives a midpoint/parallelogram proof of I.37 which avoids (Z).

The Lean reconstruction follows the same conceptual escape from cancellation, but its exact proof graph is Hilbertian: Theorem 45 is formalized directly in the scissors calculus, and I.35 supplies the parallelogram content comparison. No general theorem of the form

$$2P \sim 2Q \implies P \sim Q$$

is introduced.

## 37.5 Proof Architecture of the Reconstruction

### 37.5.1 Step 1: nondegeneracy from the parallel hypothesis

The helper

```
i37_nondegenerate_triangles
```

derives

$$A, B, C \text{ noncollinear}, \quad D, B, C \text{ noncollinear}, \quad B, A, D \text{ noncollinear}.$$

The proof uses the disjoint carriers contained in the strict parallelism hypothesis. Thus the ordinary diagrammatic statement that the top vertices are off the base is converted into explicit Hilbert incidence information.

### 37.5.2 Step 2: four midpoint witnesses and Hilbert 45 twice

The helper

```
i37_half_parallelograms
```

constructs  $M, N, P, Q$  as the four midpoints shown in Figure 37.1. It then applies

```
hilbert_triangle_to_parallelogram
```

twice.

For triangle  $ABC$  the internal call is made in the cyclic presentation  $BCA$  and returns a point  $F$  with

$$BCFM \text{ a parallelogram}$$

and a formal scissors equality between the triangle and that parallelogram. For triangle  $DBC$  the analogous construction returns  $G$  and the parallelogram  $BCGP$ .

This theorem is the formal version of Hilbert's Theorem 45. No area value is computed. The result is a finite triangle-rearrangement certificate in `HilbertScissorsEq`.

### 37.5.3 Step 3: the common upper carrier

The midpoint theorem is applied in triangle  $BAD$ . Since  $M$  is the midpoint of  $BA$  and  $P$  is the midpoint of  $BD$ ,

$$MP \parallel AD.$$

The public hypothesis gives  $AD \parallel BC$ , and the developed Euclidean parallel theory yields

$$MP \parallel BC.$$

The two Hilbert-45 parallelograms also give

$$FM \parallel BC, \quad GP \parallel BC.$$

Uniqueness/transitivity of parallels then identifies their carriers with the carrier  $MP$ . Hence

$$M, F, P \text{ are collinear,} \quad M, P, G \text{ are collinear.}$$

This is packaged by

```
i37_upper_line
```

and is one of the most important hidden configuration facts of the proof.

#### 37.5.4 Step 4: equal upper sides

Opposite-side congruence in the two parallelograms gives

$$MF \cong BC, \quad PG \cong BC.$$

Therefore

$$MF \cong PG.$$

The helper

```
i37_upper_sides_congruent
```

packages this short metric step.

#### 37.5.5 Step 5: representation normalization

Hilbert 45 naturally returns parallelograms in the forms  $BCFM$  and  $BCGP$ . Proposition I.35 is most conveniently applied to the cyclic presentations  $MBCF$  and  $PBCG$ , in which  $BC$  is the common base and the upper sides are  $MF$  and  $PG$ .

The helper

```
i37_parallelogram_reduction
```

performs precisely this normalization. The geometric parallelograms are rotated by

```
ParallelogramRotateOne
```

and the corresponding formal scissors terms are transported by

```
parallelogram_term_rotateOne
```

This step changes no geometry; it only changes the representation expected by the downstream API.

At the end of the reduction we have points  $M, P, F, G$  satisfying

$$\begin{aligned} &MBCF, PBCG \text{ parallelograms,} \\ &MP \parallel BC, \\ &M, F, P \text{ collinear, } M, P, G \text{ collinear,} \\ &MF \cong PG, \end{aligned}$$

together with scissors equivalences

$$\triangle ABC \sim MBCF, \quad \triangle DBC \sim PBCG.$$

Here  $\sim$  in this display denotes `HilbertScissorsEq`. The stronger content relation used later is

```
HilbertScissorsEquicomplementable
```

### 37.5.6 Step 6: order normalization on the common upper line

Proposition I.35 is sensitive to the actual order of the four upper points. The formal proof therefore does not read that order from Figure 37.1. It first handles two boundary configurations:

$$M = P, \quad F = P.$$

If  $M = P$ , uniqueness of the fourth vertex of a parallelogram gives  $F = G$ , so the two reduced parallelograms coincide. This is handled by

```
i37_finish_same_left_vertex
```

using the reusable theorem

```
ParallelogramFourthVertexUnique
```

If  $F = P$ , the two parallelograms have a common upper endpoint, and the existing I.35 common-endpoint helper closes the content comparison. This is

```
i37_finish_common_upper_endpoint
```

In the remaining case, Hilbert's betweenness trichotomy gives exactly three strict orders of  $M, F, P$ :

$$M - F - P, \quad F - M - P, \quad M - P - F.$$

The helper

```
i37_upper_trichotomy
```

packages this split.

### 37.5.7 Step 7: the orientation lemma for the strict cases

The case  $M - F - P$  contains the deepest proposition-local order argument. The helper

```
i37_order_MFP_implies_FPG
```

proves

$$M - F - P \implies F - P - G.$$

The proof is genuinely synthetic. In triangle  $PMB$ , the line  $CF$  enters through the side  $PM$  at  $F$ . Since  $CF \parallel MB$ , it cannot leave through  $MB$ , so Pasch forces an intersection with  $PB$ . The existing helper `hilbert_collinear_between_of_parallel` locates this crossing on  $FC$ . The second parallelogram supplies SameSide information for  $C$  and  $G$  relative to the line  $PB$ . Hence  $F$  and  $G$  are on opposite sides of  $PB$ . Their segment must cross  $PB$ , and incidence uniqueness identifies that crossing point with  $P$ . Thus  $F - P - G$ .

This is exactly the sort of positional statement that is obvious in a well-drawn affine picture but cannot be taken from the picture in a Hilbert formalization.

The case  $F - M - P$  is closed by reversing the two parallelogram presentations and applying I.35 in the reversed configuration. The helper is

```
i37_finish_order_FMP
```

For the case  $M - P - F$ , the helper

```
i37_order_MPF_implies_PFG
```

uses the congruence  $MF \cong PG$ , strict segment comparison, and a reuse of the previous orientation lemma to eliminate the two impossible orders of  $P, F, G$ . The surviving order is

$$P - F - G.$$

The final content step for this branch is packaged in

```
i37_finish_order_MPF
```

### 37.5.8 Step 8: Proposition I.35 and transport back to the triangles

Once the correct upper order is known, Proposition I.35 compares the two reduced parallelograms:

$$MBCF \quad \text{and} \quad PBCG.$$

The resulting relation is `HilbertScissorsEquicomplementable`.

Finally

```
equicomplementable_transport
```



uses the two Hilbert-45 scissors equivalences to move the conclusion from the parallelogram representatives back to the original triangles. This is the content stage of the proof; the preceding midpoint, parallel, Pasch, and order arguments belong to the geometry stage.

## 37.6 Lean Formalization

The current file imports

```
import CGJteamLab.Proposition35
import CGJteamLab.MidsegmentParallel
```

Thus I.35 is the direct Book I content dependency, while the midpoint theorem is imported as developed Hilbert infrastructure. Proposition I.36 is not a formal dependency of I.37.

The public theorem is

```
theorem euclid_proposition_37
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D : Geo.Point)
  (hAD_BC : Geo.Parallel A D B C) :
  HilbertScissorsEquicomplementable Geo
    (hilbertScissorsTriangle Geo A B C)
    (hilbertScissorsTriangle Geo D B C)
```

The proposition-local proof path is represented by the following declarations:

```
i37_nondegenerate_triangles
i37_half_parallelograms
i37_upper_line
i37_upper_sides_congruent
i37_parallelogram_reduction
i37_finish_from_right_cases
i37_finish_same_left_vertex
i37_finish_common_upper_endpoint
i37_upper_trichotomy
i37_finish_order_FMP
i37_finish_order_MFP
i37_order_MFP_implies_FPG
i37_order_MPF_implies_PFG
i37_finish_order_MPF
euclid_proposition_37
```

At the top level the final theorem is intentionally short. It first calls

```
i37_parallelogram_reduction
```

then separates the two boundary cases, uses

```
i37_upper_trichotomy
```

for the three strict cases, and dispatches each case to its corresponding finishing helper.

The crucial point is that the top-level theorem contains no hidden arithmetic area operation and no cancellation axiom. The only equality-of-figures relations in the statement and final transport are the formal scissors relations already introduced at I.35.

## 37.7 Relationship with Earlier Propositions

### 37.7.1 Proposition I.31

I.31 belongs to Euclid's literal construction. Euclid draws two parallels in order to surround the two triangles by parallelograms. The final Lean proof does not call Proposition I.31 by name. Its construction is replaced by Hilbert's midpoint theorem and Hilbert 45.

### 37.7.2 Proposition I.34

Classically I.34 supplies the fact that a diagonal bisects a parallelogram and therefore that each original triangle is one half of its surrounding parallelogram. This is precisely the source of the final halving issue.

The Lean reconstruction does not use I.34 as a Book I theorem to divide equal parallelograms by two. Its relevant parallelogram congruence content is already available in the developed interface, while Hilbert 45 provides a direct scissors equivalence between each triangle and a smaller parallelogram.

### 37.7.3 Proposition I.35

I.35 is the decisive earlier proposition in the actual Lean dependency graph. After both triangles have been converted to parallelogram terms and their upper-order configuration has been normalized, I.35 proves those parallelograms equicomplementable. The result is then transported back to the triangles.

Thus the common core of Euclid and Lean is still I.35, but the bridge from triangles to parallelograms is different.

### 37.7.4 Proposition I.36

I.36 is historically adjacent but is not used by the final formal proof. It compares parallelograms on equal bases, whereas I.37 has a common base from the start. The two propositions nevertheless confirm the same architectural lesson: the content machinery belongs in a reusable scissors layer, while the new work in each proposition is geometric normalization.

## 37.8 Hilbert and Book Zero Components Used

The proof uses a broad but well-separated collection of reusable results.

### 37.8.1 Midpoint and midsegment infrastructure

The midpoint layer supplies

- midpoint existence;
- midpoint symmetry;
- conversion of midpoint data into strict betweenness and congruence;
- `hilbert_midpoints_noncollinear`;
- `MidsegmentTheorem`.

These results produce the four midpoint witnesses and the parallel line  $MP$ .

### 37.8.2 Hilbert Theorem 45 in the scissors layer

The reusable theorem

```
hilbert_triangle_to_parallelogram
```

is the central new content bridge. Its geometric construction is provided by

```
hilbert_triangle_parallelogram_data
```

and its scissors conclusion states that the triangle term is `HilbertScissorsEq` to the corresponding parallelogram term.

This theorem was introduced while reconstructing I.37 but belongs in the reusable scissors/interface layer rather than in the proposition-local file.

### 37.8.3 Parallel and parallelogram infrastructure

The proof uses

- transport of parallelism along collinear carriers;
- Euclidean transitivity/uniqueness of parallels;
- `ParallelogramOppositeSidesCongruent`;
- `ParallelogramFourthVertexUnique`;
- `ParallelogramRotateOne`;
- representation-level parallelogram rotations in the scissors layer.

### 37.8.4 Order, Pasch, and segment comparison

The strict-case normalization uses

- Hilbert betweenness trichotomy;
- inner and outer betweenness transitivity;
- Pasch’s axiom through the derived intersection machinery;
- SameSide and OppositeSide transport;
- incidence uniqueness of the intersection of two distinct lines;
- strict segment comparison and its transport under congruence.

No new Book Zero theorem is introduced by the final Proposition file. The proof instead exercises the existing Book Zero/order language and adds several reusable helpers to the Interface and Scissors layers.

## 37.9 Formalization Notes

### 37.9.1 Geometry, representation, and content

I.37 makes the three-layer architecture especially explicit.

**Geometry.** The geometric work constructs midpoint data, obtains two parallelograms, proves that their upper sides lie on one common carrier, and determines the possible point orders on that carrier. Pasch and side-of-line arguments occur only here.

**Representation.** The same geometric parallelogram can appear as *BCFM*, *MBCF*, *FCBM*, and other cyclic or reversed presentations. Lean cannot silently identify these argument lists. Helpers such as `ParallelogramRotateOne`, `parallelogram_term_rotateOne`, and the reversal lemmas normalize the presentation without adding geometric content.

**Content.** Once the representatives are aligned, I.35 and `equicomplementable_transport` are sufficient. No new area function, polygon semantics, or cancellation rule is introduced.

### 37.9.2 Hidden configuration data

The proof explicitly derives or manages facts that the classical diagram suppresses:

- noncollinearity of both original triangles;
- four strict midpoint betweenness relations;
- the midpoint parallel  $MP \parallel AD$ ;
- the derived common upper carrier  $MP \parallel BC$ ;

- collinearity of  $M, F, P, G$  on that carrier;
- congruence  $MF \cong PG$ ;
- the boundary cases  $M = P$  and  $F = P$ ;
- the three strict orders  $M - F - P$ ,  $F - M - P$ ,  $M - P - F$ ;
- Pasch data in triangle  $PMB$ ;
- SameSide/OppositeSide information relative to  $PB$ ;
- the exact crossing point used to prove  $F - P - G$ ;
- strict segment comparisons used to eliminate impossible orders in the  $M - P - F$  branch.

None of these order relations is inferred merely from the chosen drawing.

### 37.9.3 Axiomatic status

The public theorem assumes

```
[HilbertEuclideanPlane Geo]
```

and is therefore Euclidean in the current project API.

The Euclidean parallel axiom enters through already developed infrastructure, not through a new I.37-specific axiom. In particular, the proof uses the midsegment module and Euclidean transitivity/uniqueness of parallels, and it calls Proposition I.35, whose parallelogram/content infrastructure is also exposed under the Euclidean-plane instance.

This documentation does not claim that every internal sublemma has the minimal possible axiom signature. The relevant project-level statement is that the completed I.37 theorem is derived from the existing `HilbertEuclideanPlane` hierarchy and introduces no new geometric axiom. No continuity axiom is added at I.37.

### 37.9.4 Why the conclusion is equicomplementability

The conclusion is deliberately

```
HilbertScissorsEquicomplementable
```

rather than `HilbertScissorsEq`. This matches Hilbert's Theorem 46 and avoids an invalid strengthening in non-Archimedean settings. The individual Hilbert-45 reductions are scissors equivalences, but the comparison of the two resulting parallelograms comes from I.35 and therefore has the weaker, correct content relation.

### 37.9.5 Reusable helpers introduced around I.37

The formalization produced reusable material beyond the proposition-local case analysis.

In `HilbertInterface`:

```

hilbert_triangle_parallelagram_data
hilbert_midpoints_noncollinear
ParallelogramFourthVertexUnique
ParallelogramRotateOne

```

In HilbertScissors:

```

hilbert_triangle_to_parallelagram
equicomplementable_transport
parallelagram_term_rotateOne

```

The declarations beginning with `i37_` remain proposition-local because they package the particular reduction and order analysis required by I.37.

### 37.9.6 Source audit

The source audit separates four functions of the literature.

Euclid, Heath, and Wilkins control the historical statement and classical DAG: I.31 constructs the two parallelograms, I.35 compares them, I.34 identifies the triangles as halves, and the proof ends with the unstated halving principle.

Hilbert, Chapter IV, Sections 18–19, controls the semantics and the preferred formal architecture. Theorems 44–46 distinguish equidecomposability from equicomplementability and supply the midpoint route from triangles to parallelograms.

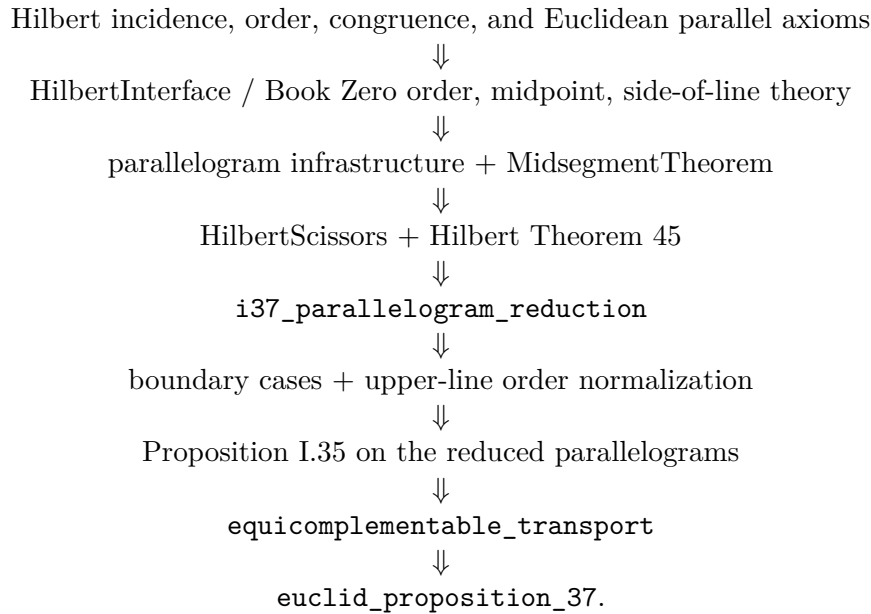
Hartshorne, Section 22, independently identifies the halving step as depending on property  $(Z)$  and gives a proof of I.37 that avoids  $(Z)$  by using midpoints and parallelograms. His treatment also confirms that equal content must not be silently replaced by numerical area or by unconditional finite equidecomposability.

Borsuk–Szmielw and Greenberg remain control sources for the underlying order, midpoint, parallelogram, and Euclidean parallel infrastructure. Neither is the primary source for the equal-content semantics at this stage: that role is filled more directly by Hilbert and Hartshorne.

The Beeson proof corpus remains useful as an audit of hidden configuration conditions and Euclidean gaps, but the final I.37 architecture is not organized around Tarski-style tactics or a literal reproduction of Beeson’s proof.

## 37.10 Dependency Summary

The actual formal dependency chain is



The historical and formal DAGs can be compared succinctly as

Euclid : I.31 twice  $\rightarrow$  I.35  $\rightarrow$  I.34 twice  $\rightarrow$  halving,  
 Lean : midpoints  $\rightarrow$  Hilbert 45 twice  $\rightarrow$  order geometry  $\rightarrow$  I.35  $\rightarrow$  transport.

|                                                    |     |
|----------------------------------------------------|-----|
| New geometric axiom:                               | no  |
| New proposition-local helpers:                     | yes |
| New reusable Interface/Scissors helpers:           | yes |
| New Book Zero theorem:                             | no  |
| New numerical area or polygon semantics:           | no  |
| Public theorem compiles:                           | yes |
| Full lake build re-run in this documentation pass: | no  |
| sorry in Proposition37.lean:                       | no  |

## 37.11 Conclusion

Proposition I.37 is not merely I.35 with triangles substituted for parallelograms. It is the point where the classical phrase “half of an equal figure” becomes a genuine foundational obligation.

The reconstruction resolves that obligation without adding a division or cancellation axiom. Hilbert’s Theorem 45 converts each triangle into an appropriate parallelogram by a finite congruent-piece rearrangement. The midsegment theorem and Euclidean parallel theory place the two resulting parallelograms on one common upper carrier. A substantial but localized order analysis then supplies the precise configuration needed by I.35. Finally the existing scissors transport returns the equicomplementability result to the original triangles.

The proof therefore makes the architecture of the area block more precise: geometry determines where the pieces and carriers lie, representation helpers put the same figures into the argument order required by the API, and the scissors layer alone carries equality of content. That separation is likely to be the reusable pattern for the propositions immediately following I.37.

## Chapter 38

# Proposition I.38

### 38.1 Introduction

Proposition I.38 continues the triangle part of Euclid’s area block. The two triangles no longer share a base:  $ABC$  is on  $BC$  and  $DEF$  is on  $EF$ . The bases are congruent, the four base points are collinear, and the two top vertices lie on a line parallel to the common base carrier. Euclid concludes that the triangles are “equal.” As in I.35–I.37, the relevant equality is synthetic equality of content, not congruence and not a numerical area value.

The formal statement therefore concludes `HilbertScissorsEquicomplementable`. It does not introduce an area function and it does not strengthen the conclusion to unconditional finite scissors equivalence. Hilbert’s Chapter IV is the controlling source for this choice: his Theorem 46 states that triangles with equal bases and equal altitudes are equicomplementable, while he explicitly warns that the stronger equidecomposability assertion cannot be obtained in full generality without an Archimedean hypothesis.

The proof architecture, however, is not a literal formalization of Euclid’s proof. The classical route constructs two surrounding parallelograms, invokes I.36, uses I.34 to identify the triangles as halves, and then cancels the common factor two. The Lean proof deliberately avoids this final cancellation. Instead it constructs a triangle  $GEF$  congruent to  $ABC$ , proves that  $G$  lies on the same upper parallel as  $A$  and  $D$ , applies I.37 to  $GEF$  and  $DEF$ , and finishes by transitivity of equicomplementability.

Thus I.38 is a particularly clean realization of Hartshorne’s observation that, once the equal-content theory and I.37 are available, I.38 follows by transitivity. The work hidden by that sentence is geometric: one must build the correct congruent copy and prove that its third vertex has the required position.

### 38.2 The Classical Configuration

Let  $ABC$  and  $DEF$  be triangles on equal bases  $BC$  and  $EF$ . In the configuration used by the formal theorem the base points occur in the strict order

$$B - C - E - F,$$

encoded by

$$\text{Between}(B, C, E), \quad \text{Between}(C, E, F).$$



The equality of bases is represented by segment congruence

$$BC \cong EF.$$

The top vertices  $A$  and  $D$  determine a line parallel to the base carrier:

$$AD \parallel BC.$$

These hypotheses are stronger than the informal phrase “on equal bases in the same parallels” because they expose the point order needed by the later synthetic argument. In particular, the Lean proof does not read the order of  $B, C, E, F$  from a drawing.

Figure 38.1 displays four stages of the proof. Panel (a) is the public configuration. Panel (b) shows the congruent-copy construction. Panel (c) displays the extension points and parallelogram that are hidden in the formal proof of the copy’s position. Panel (d) is the nondegenerate branch in which I.37 is applied. The separate boundary case  $G = D$  is handled directly in Lean and is not represented as a second picture.

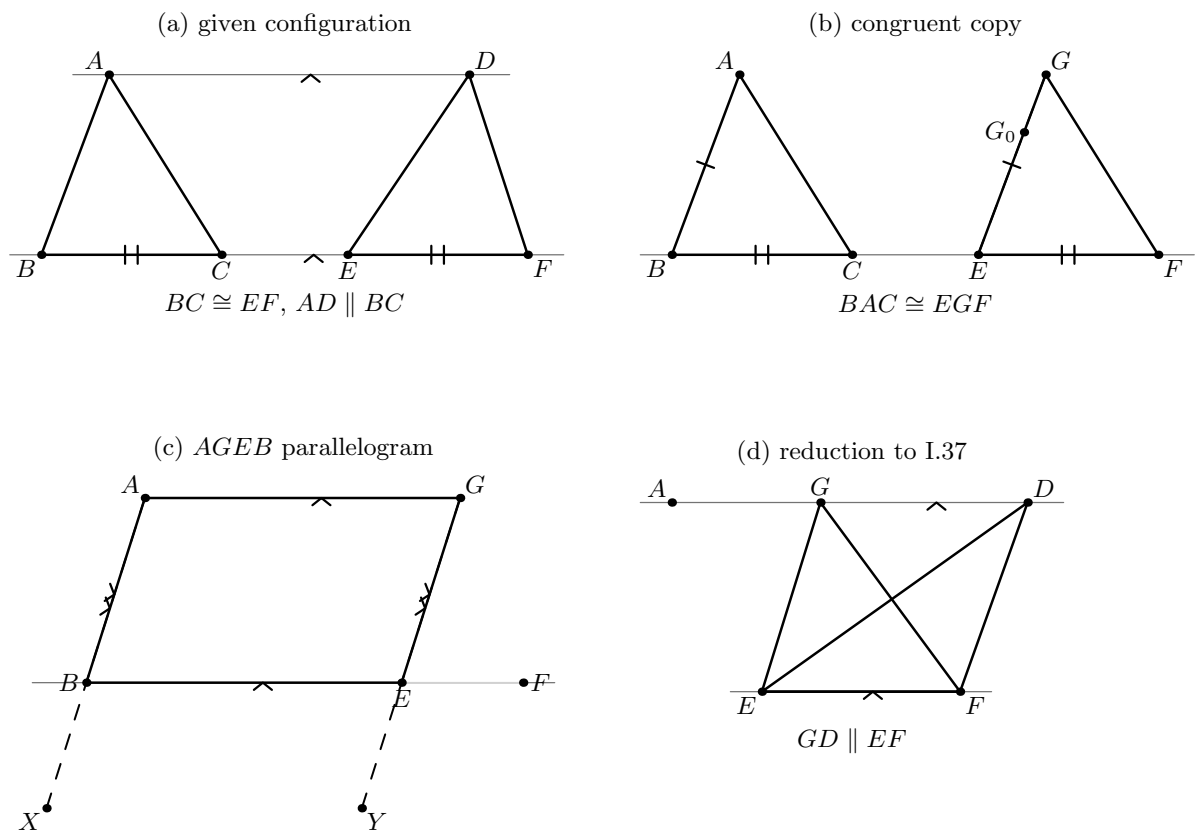


Figure 38.1: The principal geometric stages of the Lean reconstruction of I.38. (a) The given triangles have ordered collinear bases  $BC$  and  $EF$ , with  $BC \cong EF$  and the top carrier parallel to the base carrier. (b) The angle at  $B$  is copied at  $E$  on the side containing  $A$ , and  $EG \cong AB$  is laid off on the constructed ray; SAS gives  $BAC \cong EGF$ . The temporary point  $G_0$  names the ray used for the layoff. (c) Extensions  $A - B - X$  and  $G - E - Y$  provide the oriented transversal configuration used with I.28; the resulting parallel and congruence data give the parallelogram  $AGEB$ . (d) The top carrier is identified with  $AD$ ; in the branch  $G \neq D$  one obtains  $GD \parallel EF$ , so I.37 compares  $GEF$  and  $DEF$ .

### 38.3 Classical Proof

The classical proof follows the same pattern as I.37 but first uses I.36 to compare parallelograms on equal bases. In the standard configuration one constructs parallelograms around the two triangles so that a diagonal of each parallelogram cuts it into two congruent triangles. Wilkins describes the parallelograms as  $GBCA$  and  $DEFH$ .

Because their bases  $BC$  and  $EF$  are equal and they lie in the same parallels, I.36 gives equality of the two parallelograms. Proposition I.34 then says that each diagonal divides its parallelogram into two congruent triangles. Thus each of the given triangles is one half of one of the equal parallelograms, and the classical conclusion is that the two halves are equal.

The dependency graph is therefore

$$\begin{array}{c}
 BC \cong EF, \quad \text{same parallels} \\
 \Downarrow \quad \text{construct two parallelograms} \\
 GBCA, DEFH \\
 \Downarrow \quad \text{I.36} \\
 GBCA = DEFH \\
 \Downarrow \quad \text{I.34 twice} \\
 \triangle ABC, \triangle DEF \text{ are corresponding halves} \\
 \Downarrow \quad \text{“halves of equals are equal”} \\
 \triangle ABC = \triangle DEF.
 \end{array}$$

This is exactly the route visible in the Beeson proof corpus: the trace for `Prop38.prf` calls Proposition 36, Proposition 34 twice, converts triangle congruence to equality of figures, and ends with an explicit `halvesofequals` axiom. That trace is useful as a dependency audit, but it is not adopted as the architecture of the present proof.

### 38.4 Equal Bases by Congruent Copy and Transitivity

Hartshorne’s Section 22 gives the key conceptual simplification. After proving a version of I.37 that avoids the problematic halving principle, he states that I.38 follows by transitivity. This is possible because equal content is an equivalence relation and because equidecomposable figures have equal content.

In an informal affine picture, one can see the transitivity route immediately: translate a congruent copy of  $ABC$  onto the equal base  $EF$ . Its third vertex must lie on the same upper parallel, so I.37 compares that copied triangle with  $DEF$ . A Hilbert formalization cannot simply say “translate the triangle,” because translation is not a primitive operation in the project. The copy has to be reconstructed from angle layoff, segment layoff, side-of-line data, SAS, and parallel uniqueness.

The Lean DAG is therefore

$$\begin{array}{c}
BC \cong EF, B - C - E - F, AD \parallel BC \\
\Downarrow \\
\text{construct } G \text{ with } BAC \cong EGF \\
\Downarrow \\
AGEB \text{ is a parallelogram} \\
\Downarrow \\
AG \parallel BC \text{ and } AG \text{ has the same carrier as } AD \\
\Downarrow \\
G = D \quad \text{or} \quad GD \parallel EF \\
\Downarrow \\
\triangle ABC \sim \triangle GEF \\
\triangle GEF \approx \triangle DEF \quad \text{by I.37} \\
\Downarrow \quad \text{transitivity} \\
\triangle ABC \approx \triangle DEF.
\end{array}$$

Here  $\sim$  denotes the stronger relation `HilbertScissorsEq` produced by triangle congruence, while  $\approx$  denotes `HilbertScissorsEquicomplementable`. The distinction is important: the first step is a genuine congruent replacement, while I.37 supplies only the weaker content relation that is correct in the non-Archimedean setting.

Hilbert’s Theorem 46 supplies a broader control theorem: equal-base, equal-altitude triangles are equicomplementable. I.38 is exactly a Euclidean same-parallel instance of that geometry. The current library does not invoke Theorem 46 as a monolithic theorem; it realizes the equal-base case through the already completed I.37 and a congruent-copy reduction.

## 38.5 Proof Architecture of the Reconstruction

### 38.5.1 Step 1: recovering one common base carrier

The helper

`i38_base_carrier`

starts from  $AD \parallel BC$  and uses `parallel_endpoints_sameSide` to obtain an actual Hilbert line `base` carrying  $B$  and  $C$ , together with `SameSide` information for  $A$  and  $D$ . The two strict betweenness hypotheses then place  $E$  and  $F$  on that same carrier by incidence transport.

Mathematically, the result packages the statement that the apparently separate segments  $BC$  and  $EF$  belong to one lower line, while both top vertices are off that line and lie on the same side of it. This is the first place where the formal proof exposes data that the classical phrase “the same parallels” suppresses.

The helper remains proposition-local. It is a useful normalization for this ordered six-point configuration, but it is not promoted to the general interface because its statement is tailored to  $B - C - E - F$ .

### 38.5.2 Step 2: nondegeneracy is derived, not assumed

The helper

`i38_nondegenerate`

proves

$$A, B, C \text{ noncollinear}, \quad D, E, F \text{ noncollinear}.$$

Since  $B, C, E, F$  lie on the base carrier and SameSide data place  $A, D$  off that carrier, both conclusions are incidence consequences of the configuration.

This helper is logically small but important for the next stage. Hilbert's angle construction and SAS require explicit noncollinearity witnesses. The classical diagram treats these as evident; Lean requires them as proof terms. No new geometric hypothesis is added to the public theorem.

### 38.5.3 Step 3: constructing the congruent copy on the equal base

The central construction is

`i38_equal_base_copy`.

It first uses Hilbert's angle-construction axiom to copy the angle  $ABC$  at  $E$ , with the fixed ray directed through  $F$  and with the new ray chosen on the side of the base containing  $A$ . This creates a temporary point  $G_0$  and an angle congruence of the form

$$\angle ABC \cong \angle FEG_0.$$

Next, Hilbert's segment construction lays off on the ray  $EG_0$  a point  $G$  satisfying

$$EG \cong AB.$$

The relation `HilbertSameRay` between  $G_0$  and  $G$  is then used twice: first to preserve the side-of-line choice, and second to replace  $G_0$  by  $G$  in the copied angle. Thus the construction retains the orientation selected by the angle axiom instead of merely obtaining an unoriented congruent angle.

With the public hypothesis  $BC \cong EF$ , SAS is applied to the triangles in the formal orientation

$$BAC \quad \text{and} \quad EGF.$$

The result is

$$\text{TriangleCongruenceResult}(BAC, EGF).$$

Panel (b) of Figure 38.1 records this stage. The point  $G_0$  is not part of the final geometry; it exists solely to name the constructed ray on which the final segment is laid off.

This helper is proposition-local, but it reveals a reusable proof pattern: angle layoff + segment layoff + SameRay transport + SAS gives a controlled oriented copy of a triangle on a prescribed congruent base.

### 38.5.4 Step 4: proving that the copy is a parallelogram

The helper

`i38_copy_parallelogram`

contains the most substantial hidden positional argument of I.38. The desired conclusion is

$AGEB$  is a parallelogram.

At an affine level this is the statement that the congruent copy on  $EF$  is a translation of the original triangle. The project does not use translation as a primitive notion, so the claim is rebuilt synthetically.

First  $GE$  is extended through  $E$  to a point  $Y$ , giving  $G - E - Y$ , and  $AB$  is extended through  $B$  to a point  $X$ , giving  $A - B - X$ . From the common base carrier and the SameSide choice of  $G$ , the proof derives the required SameSide/OppositeSide relations for the transversal configuration. The lower order  $B - C - E - F$  also yields the derived order  $F - E - B$ .

The copied angle is normalized from the ray  $BC$  to the ray  $BE$  using `HilbertSameRay`; then the unordered angle representation is reoriented to exactly the corresponding-angle shape expected by I.28. Proposition I.28 is applied in its neutral corresponding-angle form to obtain parallelism of the extended side lines. Collinearity transport then gives

$$AB \parallel GE.$$

SAS had already supplied the corresponding side congruence

$$AB \cong GE.$$

Together with the fact that  $A$  and  $G$  lie on the same side of the lower carrier, these data satisfy `OnePairParallelCongruent`. The reusable criterion `OnePairParallelCongruentCriterion` therefore yields

$AGEB$  parallelogram.

Panel (c) of Figure 38.1 shows the extension witnesses  $X, Y$  that make this argument formally oriented.

This helper is local to I.38. The underlying one-pair parallelogram criterion is reusable Interface infrastructure and corresponds closely to the stronger parallelogram theorem isolated in the Borsuk–Szmielw development.

### 38.5.5 Step 5: identifying the upper carrier

The helper

`i38_copy_upper_position`

turns the parallelogram conclusion into the exact hypothesis needed by I.37. From  $AGEB$  it obtains

$$AG \parallel EB.$$

Because  $E, B, C$  are collinear, parallel transport gives

$$AG \parallel BC.$$

But the public hypothesis already says

$$AD \parallel BC.$$

The crucial point is now uniqueness of the Euclidean parallel. If the extensional point-lines  $AG$  and  $AD$  were distinct, the developed transitivity result `hilbert_parallel_transitive_distinct` would make them parallel to one another. This is impossible because they both contain  $A$ . Hence the two point-line carriers are equal, and  $G$  lies on the upper carrier through  $A, D$ .

The proof then separates a boundary case that a drawing normally ignores:

$$G = D$$

or

$$G \neq D.$$

If  $G = D$ , the copied triangle already has the desired top vertex. Otherwise, parallelism is transported along the common upper and lower carriers to obtain

$$GD \parallel EF.$$

Panel (d) of Figure 38.1 depicts this nondegenerate branch.

This step is the clearest point at which the Euclidean uniqueness theory of parallels becomes indispensable. The earlier angle construction, segment construction, side-of-line transport, SAS, and I.28 criterion are neutral.

### 38.5.6 Step 6: the content proof is now short

The final theorem converts the triangle congruence certificate into `HilbertScissorsEq` using `scissors_congruent`. Because triangle terms are represented as unordered multisets of vertices, two applications of `scissors_triangle_swap12` normalize

$$BAC \sim EGF$$

to

$$ABC \sim GEF.$$

This is a representation step only; it changes no geometry.

The stronger scissors equality is embedded into equicomplementability by the existing helper

$$\text{equicomplementable\_of\_scissorsEq}.$$

The two position cases then close as follows.

If  $G = D$ , the copied triangle is already  $DEF$ , so the congruent replacement finishes the proof. If  $G \neq D$ , the parallelism  $GD \parallel EF$  is exactly the hypothesis of Proposition I.37 for the two triangles  $GEF$  and  $DEF$ . Thus

$$GEF \approx DEF.$$

Finally `equicomplementable_trans` composes

$$ABC \approx GEF, \quad GEF \approx DEF$$

to obtain the public conclusion.

This is the content stage of the proof. All of the preceding line, ray, SameSide, extension, and parallel arguments exist solely to put the copied triangle into the configuration in which I.37 can be called.

## 38.6 Lean Formalization

The current Proposition file imports

```
import CGJteamLab.Proposition28
import CGJteamLab.Proposition37
```

Thus I.28 is the direct neutral parallel criterion used inside the geometric copy argument, while I.37 is the direct content theorem used at the end.

The public theorem is

```
theorem euclid_proposition_38
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D E F : Geo.Point)
  (hAD_BC : Geo.Parallel A D B C)
  (hBCE : Geo.Between B C E)
  (hCEF : Geo.Between C E F)
  (hBC_EF : Geo.Congruent B C E F) :
  HilbertScissorsEquicomplementable Geo
    (hilbertScissorsTriangle Geo A B C)
    (hilbertScissorsTriangle Geo D E F) := by
  ...
```

The ellipsis here is documentation notation only; the repository theorem is fully proved and contains no `sorry`.

The proposition-local proof path is represented by

```
i38_base_carrier
i38_nondegenerate
i38_equal_base_copy
i38_copy_parallelogram
i38_copy_upper_position
euclid_proposition_38
```

The relatively short list should not be mistaken for a trivial proof. The middle three helpers package the substantial geometry: orientation of the copied triangle, the I.28 extension configuration, construction of the parallelogram, and identification of the upper carrier.

At the top level the final theorem is intentionally small. It consumes the geometric package returned by `i38_copy_upper_position`, converts the congruent copy to a scissors equivalence, and dispatches the two position cases to either reflexive transport or I.37 plus transitivity.

## 38.7 Relationship with Earlier Propositions

### 38.7.1 Proposition I.28

I.28 is a direct formal dependency. The helper `i38_copy_parallelogram` uses the theorem

### euclid\_proposition\_28\_corresponding

after explicitly constructing the extension points and the opposite-side configuration required by the transversal theorem. This is a neutral use: equal corresponding angles imply parallelism without invoking the Euclidean uniqueness axiom.

#### 38.7.2 Proposition I.33

The final I.38 file does not call `euclid_proposition_33` by name, but its mathematical role is present in reusable form. The Interface theorem `OnePairParallelCongruentCriterion` says that one oriented pair of parallel and congruent opposite sides determines a parallelogram. This is the structural content needed to turn  $AB \parallel GE$  and  $AB \cong GE$  into the parallelogram  $AGEB$ .

#### 38.7.3 Proposition I.34

I.34 belongs to Euclid’s classical proof: it supplies the diagonal congruence showing that each surrounding parallelogram is double its triangle. The Lean proof does not call I.34. Avoiding that step is precisely what removes the need for a general “halves of equals” principle.

#### 38.7.4 Proposition I.36

I.36 is the decisive earlier proposition in the classical DAG. It compares the two constructed parallelograms on the equal bases  $BC$  and  $EF$ . Nevertheless, it is not an actual dependency of the final Lean proof. The formal route replaces the I.36-plus-halving argument by a congruent-copy reduction to I.37.

This difference is structurally important. It confirms that the formal Book I dependency graph is not forced to reproduce Euclid’s proposition order even when the public statement remains faithful to Euclid.

#### 38.7.5 Proposition I.37

I.37 is the decisive content dependency in the actual Lean proof. Once the copy point  $G$  has been placed on the upper carrier and the nondegenerate branch gives  $GD \parallel EF$ , I.37 applies immediately to the triangles  $GEF$  and  $DEF$  on the common base  $EF$ .

Thus the formal relationship is

$$\text{congruent replacement} \quad + \quad \text{I.37} \quad + \quad \text{transitivity}.$$

This is the exact formal meaning of Hartshorne’s compressed statement that I.38 follows by transitivity.

## 38.8 Hilbert and Book Zero Components Used

### 38.8.1 Incidence, order, and side-of-line infrastructure

The proof uses the developed Hilbert language for



- strict betweenness and its incidence consequences;
- propagation of collinearity to a fixed carrier line;
- noncollinearity from a point lying off a line;
- line existence and uniqueness through two distinct points;
- SameRay transport;
- SameSide and OppositeSide symmetry and transitivity;
- extension of a segment beyond an endpoint.

These mechanisms are responsible for most of the positional detail absent from the classical diagram.

### 38.8.2 Congruence construction and SAS

Hilbert’s congruence layer supplies two genuine constructions: `angle_construction` creates the oriented ray at  $E$ , and `segment_construction` lays off  $EG \cong AB$  on that ray. The theorem `hilbert_sameRay_points_sameSide` ensures that replacing  $G_0$  by the final layoff point  $G$  preserves the chosen half-plane.

SAS then gives the full triangle-congruence package

$$BAC \cong EGF.$$

No circle construction or continuity axiom is used in this stage.

### 38.8.3 Parallel and parallelogram infrastructure

The geometric reduction consumes

- I.28 corresponding-angle parallelism;
- parallel transport along collinear carriers;
- `OnePairParallelCongruentCriterion`;
- Euclidean transitivity and uniqueness of parallels;
- the extensional `PointLine` representation used to identify the upper carrier.

The distinction between neutral existence/angle criteria and Euclidean uniqueness is visible here rather than hidden in a generic “parallel” tactic.

### 38.8.4 Scissors and content infrastructure

No new scissors constructor is needed for I.38. The proof uses only established operations:

- `scissors_congruent`;

- triangle-term permutation lemmas;
- `equicomplementable_of_scissorsEq`;
- `equicomplementable_trans`;
- Proposition I.37.

This is a strong confirmation that the term-level content interface introduced at I.35 is adequate for I.38.

## 38.9 Formalization Notes

### 38.9.1 Geometry, representation, and content

I.38 again separates cleanly into three layers.

**Geometry.** The proof constructs the copied ray and segment, proves the copied triangle congruent, derives the parallelogram  $AGEB$ , identifies the common upper carrier, and transports the lower carrier from  $BC$  to  $EF$ .

**Representation.** Angles have to be reversed or transported along `SameRay` relations to match the argument order required by I.28 and SAS. At the end, the congruent triangles appear as  $BAC$  and  $EGF$ , while the public scissors terms are  $ABC$  and  $GEF$ . The triangle swap lemmas normalize this representation without adding geometric content.

**Content.** Only after the geometry is complete does the proof enter the scissors layer. Congruence gives a strong `HilbertScissorsEq`; I.37 gives `HilbertScissorsEquicomplementable`; transitivity composes them. No subtraction, cancellation, numerical area, or new polygon semantics is needed.

### 38.9.2 Hidden configuration data

The formal proof explicitly establishes the following facts instead of reading them from Figure 38.1:

- $B, C, E, F$  lie on one Hilbert line;
- $A, D$  lie off that line and on the same side of it;
- both original triangles are noncollinear;
- the copied point  $G_0$  is on the chosen side of the base;
- $G$  is on the same ray from  $E$  as  $G_0$ ;
- moving from  $G_0$  to  $G$  preserves `SameSide` data and the copied angle;
- $G - E - Y$  and  $A - B - X$  are strict extensions;
- the extension point  $Y$  is on the opposite side of the base from  $G$  and therefore from  $A$ ;

- the order  $B - C - E - F$  implies the transversal order  $F - E - B$ ;
- the exact angle orientation required by I.28;
- equality of the extensional carriers  $AG$  and  $AD$ ;
- the boundary case  $G = D$  separately from the strict case  $G \neq D$ .

The last two items are particularly significant: a standard affine drawing makes  $G$  look automatically located on the upper line, but in Hilbert geometry that position is a theorem obtained from Euclidean parallel uniqueness.

### 38.9.3 Axiomatic status

The public theorem assumes

[HilbertEuclideanPlane Geo]

and is therefore Euclidean in the current project API.

The proof nevertheless contains a substantial neutral prefix. Incidence, strict order, SameSide/OppositeSide reasoning, angle construction, segment construction, SameRay transport, SAS, and the corresponding-angle direction of I.28 do not require the Euclidean uniqueness of parallels.

Group IV enters when the copied side data are converted into the full parallelogram structure and, most visibly, when the proof identifies the two upper carriers through Euclidean transitivity/uniqueness of parallels. The final call to I.37 also inherits the Euclidean-plane instance.

No new parallel axiom is introduced in I.38. No continuity or Archimedean axiom is added. The construction of  $G$  is an existence argument inside the neutral congruence/order layer; the specifically Euclidean force lies in the uniqueness and structural consequences of parallelism.

### 38.9.4 Why the conclusion remains equicomplementability

Hilbert's Theorem 46 is the decisive semantic control. Equal-base, equal-altitude triangles are equicomplementable. Hilbert explicitly notes that without the Archimedean axiom one cannot in general strengthen this to finite equidecomposability.

The Lean proof reflects that distinction exactly. The replacement  $ABC \rightsquigarrow GEF$  is a direct scissors equality because the two triangles are congruent. The subsequent comparison  $GEF \rightsquigarrow DEF$  comes from I.37 and is only equicomplementability. Therefore the final theorem correctly returns the weaker relation.

### 38.9.5 Reusable helpers and proposition-local helpers

I.38 introduces no new reusable theorem in `HilbertInterface`, `Book Zero`, or `HilbertScissors`. Its new declarations are all prefixed `i38_` and remain proposition-local.

Their mathematical roles are nevertheless worth recording:

- `i38_base_carrier`: ordered lower-line normalization;
- `i38_nondegenerate`: noncollinearity consequences;
- `i38_equal_base_copy`: oriented congruent-copy construction;
- `i38_copy_parallelogram`: corresponding-angle and one-pair parallelogram argument;
- `i38_copy_upper_position`: Euclidean carrier identification and reduction to the I.37 configuration.

If the congruent-copy pattern recurs in later propositions, the third and fourth helpers are candidates for refactoring. At I.38 itself there is not yet enough evidence to enlarge the general Interface merely to shorten one proposition.

### 38.9.6 Source audit

The source audit assigns distinct roles to the standard references.

Euclid’s classical proof, as recorded by Wilkins, uses two constructed parallelograms, I.36, I.34 twice, and the concluding halving principle. The Beeson proof corpus confirms the same dependency at a formal trace level and makes the final `halvesofequals` assumption explicit.

Hilbert, Chapter IV, Section 19, supplies the semantic control. Theorems 44–46 distinguish equidecomposability from equicomplementability and culminate in the theorem that equal-base, equal-altitude triangles are equicomplementable.

Hartshorne, Section 22, is the most direct architectural source for the chosen proof. After replacing Euclid’s I.37 halving argument by a proof valid for equal content without property (*Z*), he states that I.38 follows by transitivity. The present Lean proof makes the geometric content of that transitivity step explicit by constructing *GEF*.

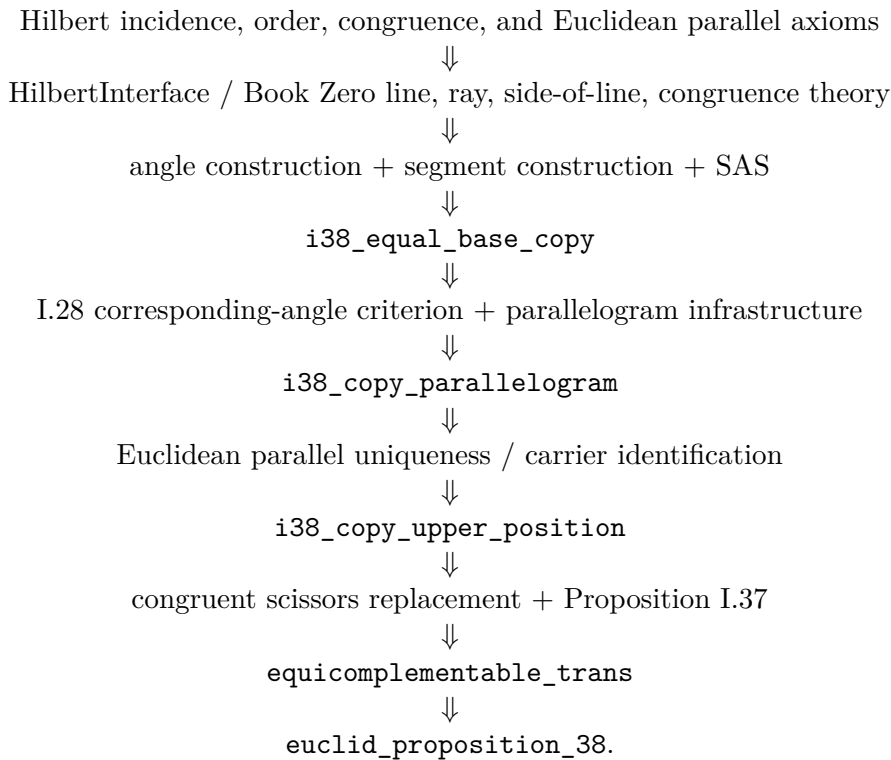
Borsuk–Szmielew are not the source for the equal-content semantics at this stage, but their Euclidean parallelogram theory is directly relevant to the one-pair parallel-plus-congruence criterion consumed by the proof. Their organization supports keeping this criterion in reusable interface infrastructure.

Greenberg remains a control source for the semantic boundary. He distinguishes numerical area from equal content and points to Hartshorne for the full synthetic theory rather than developing it himself. This supports the decision not to replace the scissors relation by a numerical area function.

The Euclid/Joyce edition and the project Book I source compilation remain statement and historical cross-checks; neither is used to force the Lean proof into the classical proposition DAG.

## 38.10 Dependency Summary

The actual formal dependency chain is



The historical and formal DAGs are therefore sharply different:

Euclid: construct parallelograms  $\rightarrow$  I.36  $\rightarrow$  I.34 twice  $\rightarrow$  halving,  
Lean: congruent copy  $\rightarrow$  upper-carrier proof  $\rightarrow$  I.37  $\rightarrow$  transitivity.

|                                          |       |
|------------------------------------------|-------|
| New geometric axiom:                     | no    |
| New proposition-local helpers:           | yes   |
| New reusable Interface/Scissors helper:  | no    |
| New Book Zero theorem:                   | no    |
| New numerical area or polygon semantics: | no    |
| Public theorem compiles:                 | yes   |
| Full lake build:                         | clean |
| sorry in Proposition38.lean:             | no    |

## 38.11 Conclusion

Proposition I.38 is classically presented as the equal-base analogue of I.37, but its formal reconstruction exposes a more useful structure. The content part of the theorem is almost entirely inherited from I.37 and transitivity. The new work is geometric: construct an oriented congruent copy on the second base, prove that the copy is related to the original triangle by a parallelogram, and use Euclidean parallel uniqueness to place its top vertex on the required upper carrier.

This explains why a short classical proposition can generate a substantial Lean development. The extra code is not a hidden area calculation. It is the formal cost of orientation, side-of-line data,

ray replacement, extension points, carrier transport, and uniqueness of parallels – all information that a conventional diagram suppresses.

At the same time, I.38 is evidence that the content architecture introduced at I.35 has stabilized. No new scissors operation, polygon semantics, numerical area, or cancellation principle is required. Congruence supplies the strong local replacement  $ABC \sim GEF$ ; I.37 supplies the equal-content comparison  $GEF \approx DEF$ ; and transitivity closes the theorem. In that precise sense, the final proof realizes Hartshorne's concise diagnosis: I.38 follows by transitivity, once the geometry needed to make the intermediate triangle explicit has been proved.

## Chapter 39

# Proposition I.39

### 39.1 Introduction

Proposition I.39 is the first genuine converse theorem in Euclid’s area block. Propositions I.35–I.38 proceed in the forward direction: geometric information about bases and parallels is converted into equality of content. Proposition I.39 reverses that direction. Two triangles on the same base are assumed to be equal, and from this equality one must recover a geometric conclusion about the position of their third vertices.

In the present formalization, Euclid’s equality of the two triangles is represented by the relation

```
HilbertScissorsEquicomplementable
```

Thus the theorem starts from a synthetic content relation rather than from numerical areas. The conclusion is also geometric rather than numerical: the two top vertices lie on a common Hilbert incidence line disjoint from the base line.

This converse direction exposes a new foundational issue. The formal scissors calculus developed for I.35–I.38 supports congruent replacement, splitting, addition, and equicomplementability, but it does not by itself express the strict statement that a whole figure cannot have the same content as one of its proper parts. Hartshorne isolates exactly this issue as property (Z), and he notes that Euclid’s proof of I.39 uses the principle that the whole is greater than the part.

The reconstruction therefore does not import a numerical area function and does not assume Hilbert’s Theorem 48 wholesale. Instead it isolates a configuration-free positivity principle for the scissors calculus: adjoining a nondegenerate triangle to an arbitrary scissors term cannot leave the content unchanged. A triangle split along an interior point of one side then derives, as a theorem, the specialized De Zolt statement actually used by I.39.

The result is a useful architectural checkpoint. The geometric part of the proof is derived from the existing Hilbert theory and Proposition I.37. The specialized proper-part statement is no longer an axiom. The remaining new foundational debt is the single positivity assumption `hilbert_scissors_triangle_positive`.

## 39.2 The Classical Configuration

Let  $ABC$  and  $DBC$  be triangles on the common base  $BC$ . The top vertices  $A$  and  $D$  lie on the same side of the line carrying  $BC$ , and the two triangles are assumed equal in content. The conclusion is that  $A$  and  $D$  lie on a line parallel to the base.

The classical *reductio* draws through  $A$  a line parallel to  $BC$ . If  $AD$  is not already this parallel, the new line meets the carrier of  $BD$  in a point  $E$ . There are two strict configurations compatible with  $A$  and  $D$  lying on the same side of the base:

$$B - E - D, \quad \text{or} \quad B - D - E.$$

In the first case  $EBC$  is a proper subtriangle of  $DBC$ . In the second case  $DBC$  is a proper subtriangle of  $EBC$ . The third collinear order

$$E - B - D$$

would place  $E$  and  $D$  on opposite sides of the base line, because the open segment  $ED$  would meet the base at  $B$ . The formal proof therefore excludes this third order from SameSide/OppositeSide theory rather than from the drawing.

Figure 39.1 displays exactly these three stages. Panel (a) is the public configuration. Panels (b) and (c) are the two proper-part alternatives that remain after the auxiliary parallel through  $A$  has been constructed.

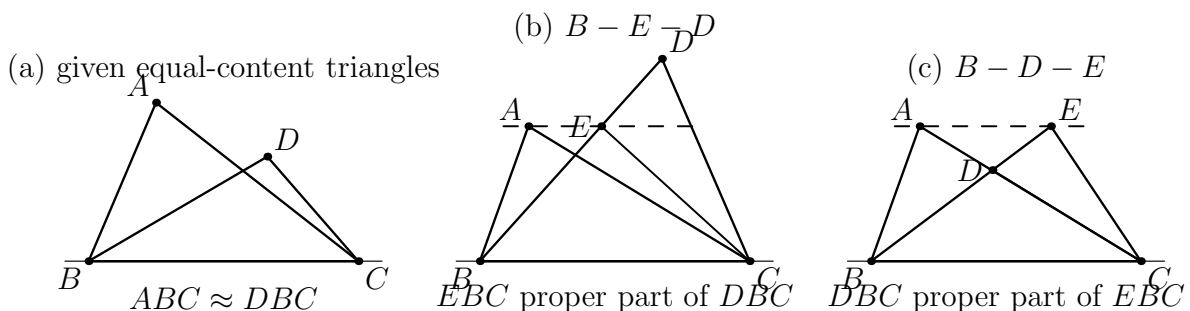


Figure 39.1: The geometric configuration used in the reconstruction of I.39. (a) The given triangles  $ABC$  and  $DBC$  share the base  $BC$ , lie on the same side of its carrier, and are assumed equicomplementable. (b) The parallel through  $A$  meets  $BD$  at  $E$  with  $B - E - D$ ; Proposition I.37 gives equal content for  $ABC$  and  $EBC$ , while  $EBC$  is a proper subtriangle of  $DBC$ . (c) The alternative order is  $B - D - E$ ; now  $DBC$  is a proper subtriangle of  $EBC$ . The order  $E - B - D$  is excluded formally because it would put  $E$  and  $D$  on opposite sides of the base.

## 39.3 Classical Proof

Wilkins emphasizes that a complete proof must treat two positional configurations, although Euclid explicitly discusses only one of them. In each configuration the same auxiliary idea is used.

Draw through  $A$  a straight line parallel to the common base  $BC$ . If this line already contains  $D$ , the conclusion is immediate. Otherwise let  $E$  be its intersection with the straight line  $BD$ , or with the continuation of  $BD$  beyond  $D$ .



The auxiliary triangle  $EBC$  has the same base  $BC$  as  $ABC$ , and its third vertex  $E$  lies on the parallel through  $A$ . Hence an earlier equal-content result shows that

$$ABC = EBC$$

in Euclid's area language. Wilkins' Study Note labels this step as Proposition I.36. The standard Book I dependency, and the Euclid/Joyce dependency table for I.39, identify the relevant triangle theorem as I.37: triangles on the same base and in the same parallels are equal. The present Lean development uses Proposition I.37 directly.

Now compare  $EBC$  with  $DBC$ . If

$$B - E - D,$$

then  $EBC$  is properly contained in  $DBC$ . If

$$B - D - E,$$

then  $DBC$  is properly contained in  $EBC$ . In either case the hypothesis that the original triangles have equal content, together with the equality  $ABC = EBC$ , would force a whole triangle to be equal to one of its proper parts.

This contradicts Euclid's Common Notion 5, that the whole is greater than the part. Therefore the assumption that  $D$  does not lie on the parallel through  $A$  is impossible. Hence the top vertices  $A$  and  $D$  lie on one straight line parallel to the common base.

The classical dependency is therefore schematically

$$\begin{array}{c} \text{construct the parallel through } A \\ \Downarrow \\ E \in BD \text{ or its extension} \\ \Downarrow \\ ABC = EBC \text{ by an earlier equal-content theorem} \\ \Downarrow \\ EBC \text{ is a proper part of } DBC \text{ or conversely} \\ \Downarrow \text{ C.N.5} \\ \text{contradiction} \\ \Downarrow \\ AD \parallel BC. \end{array}$$

The point of the reconstruction is not to suppress this argument but to make its hidden logical requirement explicit.

## 39.4 Property (Z), De Zolt, and the Converse Content Problem

### 39.4.1 Why I.39 is different from I.38

Hartshorne's analysis of Euclid's area theory makes the boundary especially clear. Once an adequate equal-content relation is available, I.38 follows by transitivity. I.39 is its converse, and the converse uses the claim that a proper part cannot have the same content as the whole. Hartshorne calls the relevant strictness condition property (Z).

Thus the implication

$$\text{same parallels} \longrightarrow \text{equal content}$$

belongs to the forward scissors theory, while

$$\text{equal content} \longrightarrow \text{same parallels}$$

requires an additional strictness principle.

This is exactly what the current formalization detects. The operations already present in `HilbertScissors` do not yield a contradiction merely from the fact that one triangle term arises from a geometric proper subtriangle of another. The term calculus intentionally has no primitive notion of geometric interior or nonoverlap.

### 39.4.2 Hilbert's Theorem 48

Hilbert reaches the converse by a different route. In Chapter IV he first develops segment arithmetic and an area function. In Section 21, equicomplementable polygons are shown to have equal areas. For two equicomplementable triangles with equal base  $g$  and altitudes  $h, h'$ , Hilbert writes the equality

$$\frac{1}{2}gh = \frac{1}{2}gh'$$

and cancels the common nonzero factor to obtain

$$h = h'.$$

This is Theorem 48.

That theorem would certainly be strong enough for I.39. It is not, however, the theorem formalized here. Importing it directly would hide the exact foundational obstacle encountered by the present scissors language and would force the development prematurely into multiplication of segment classes and the later area theory.

### 39.4.3 Positivity and the derived De Zolt principle

Hilbert later states a De Zolt theorem asserting, in effect, that after a rectilinear figure has been decomposed into triangular pieces, omitting one piece cannot leave enough material to reconstruct the whole. The current formalization isolates the strictness behind that phenomenon at a more general algebraic boundary.

The one new temporary content assumption is:

```
axiom hilbert_scissors_triangle_positive
  [HilbertIncidence Geo]
  (P : HilbertScissorsTerm Geo)
  (A B C : Geo.Point)
  (hNoncol : Not (Collinear Geo A B C)) :
  Not (HilbertScissorsEquicomplementable Geo
    (P + hilbertScissorsTriangle Geo A B C)
    P)
```

This statement contains no betweenness, parallelism, or proposition-specific configuration. It says only that adjoining a nondegenerate triangular piece strictly changes content.

The geometric proper-part statement needed in I.39 is then derived. First,

```
scissors_split_side
```

uses the primitive scissors split to prove that, when  $B - X - A$ ,

$$ABC \sim XBC + AXC,$$

where  $\sim$  denotes `HilbertScissorsEq`. The proof also establishes that the cut-off triangle  $AXC$  is nondegenerate. If one assumed

$$ABC \approx XBC,$$

then transport across the split would give

$$XBC + AXC \approx XBC,$$

contradicting positivity.

Consequently the former I.39 axiom is now a theorem:

```
theorem hilbert_scissors_triangle_proper_part
  [HilbertIncidence Geo]
  [HilbertOrder Geo]
  (A X B C : Geo.Point)
  (hBXA : Geo.Between B X A)
  (hBC : Ne B C)
  (hNoncol : Not (Collinear Geo B C A)) :
  Not (HilbertScissorsEquicomplementable Geo
    (hilbertScissorsTriangle Geo A B C)
    (hilbertScissorsTriangle Geo X B C))
```

The argument should not be read as a proof that positivity is equivalent, in the present Lean API, to Hilbert's Theorem 48. What has been established formally is narrower and cleaner: positivity implies the specialized proper-part theorem, and that theorem is sufficient for I.39. A future concrete additive valuation, nonzero on nondegenerate triangles, will discharge the positivity axiom directly.

## 39.5 Proof Architecture of the Reconstruction

The Lean proof separates the geometry from the one missing content principle.

### 39.5.1 Step 1: the coincident-vertex boundary case

If

$$A = D,$$

there is no need for a reductio. The point  $A$  lies off the base because the `SameSide` hypothesis contains this information. The existing parallel-through- point theorem constructs a line through  $A$  disjoint from the base, and the same incidence fact is used for both top vertices.

This boundary case is invisible in the usual diagram but must be handled explicitly because the strict point-line parallel relation requires two distinct points.

### 39.5.2 Step 2: the already-parallel branch

Assume now that  $A \neq D$ . The proof tests whether

$$AD \parallel BC.$$

If so, the helper

```
proposition39_parallel_points_to_top_line
```

converts point-line parallelism into the incidence-line form used by the public conclusion. It constructs the line through  $A, D$  and proves that it is disjoint from the given incidence line **base** through  $B, C$ .

### 39.5.3 Step 3: construction of the auxiliary point

The only nontrivial branch is

$$A \neq D, \quad AD \nparallel BC.$$

The helper

```
proposition39_auxiliary_point
```

draws the parallel through  $A$  to  $BC$  and proves that the carrier of  $BD$  must meet it. If the two incidence lines were disjoint, Euclidean uniqueness of the parallel through  $B$  would force the carrier  $BD$  to coincide with the base, contradicting the fact that  $D$  lies off the base.

The resulting point  $E$  satisfies

$$\begin{aligned} &B, E, D \text{ collinear,} \\ &E \neq B, \quad E \neq D, \\ &E \notin \text{base,} \\ &\text{SameSide}(A, E; \text{base}), \end{aligned}$$

and either

$$E = A$$

or

$$AE \parallel BC.$$

The last disjunction is exactly what is needed for Proposition I.37. If  $E = A$ , the content comparison is reflexive; otherwise the nondegenerate parallel  $AE \parallel BC$  gives it directly.

### 39.5.4 Step 4: normalization of the order on BD

The helper

```
proposition39_auxiliary_order
```

takes the collinearity of  $B, E, D$  and proves

$$B - E - D \quad \text{or} \quad B - D - E.$$

The raw betweenness trichotomy has a third branch,  $E - B - D$ . The proof excludes it by side-of-line theory. Because  $B$  lies on the base,  $E - B - D$  would make  $E$  and  $D$  opposite sides of the base. But transitivity and symmetry of SameSide, using the common point  $A$ , show that  $E$  and  $D$  are on the same side. The two statements are incompatible.

This is one of the main pieces of hidden configuration data revealed by the formalization.

### 39.5.5 Step 5: the I.37 content comparison

The auxiliary parallel gives

$$ABC \approx EBC,$$

where  $\approx$  denotes HilbertScissorsEquicomplementable. The original hypothesis gives

$$ABC \approx DBC.$$

Symmetry and transitivity therefore yield

$$DBC \approx EBC.$$

No subtraction, cancellation, numerical area, or new scissors constructor is used here. This part is entirely inside the content interface already established before I.39.

### 39.5.6 Step 6: the proper-part contradiction

There are now exactly two order branches.

If

$$B - E - D,$$

then  $EBC$  is a proper subtriangle of the nondegenerate triangle  $DBC$ . The derived proper-part theorem says

$$DBC \not\approx EBC,$$

contradicting the comparison obtained above.

If

$$B - D - E,$$

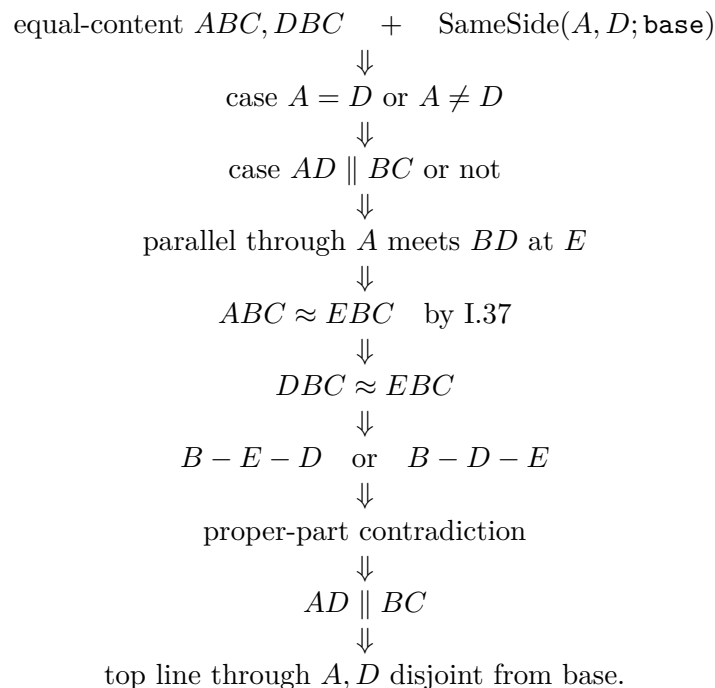
then  $DBC$  is a proper subtriangle of the nondegenerate triangle  $EBC$ . After symmetry of equicomplementability, the same derived theorem gives the opposite contradiction.

Thus the assumption

$$AD \parallel BC$$

has no possible realization. The remaining branch is  $AD \parallel BC$ , which is converted into the required incidence line through  $A$  and  $D$  disjoint from the base.

The formal DAG can be summarized as



## 39.6 Lean Formalization

The current file imports

```
import CGJteamLab.Proposition37
import CGJteamLab.HilbertScissorsPositivity
```

The positivity module in turn imports `HilbertScissorsInvariant`, which contains the generic valuation interface intended for the eventual discharge of the positivity assumption.

The public theorem is:

```
theorem euclid_proposition_39
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D : Geo.Point)
  (base : Geo.Line)
  (hBC : Ne B C)
  (hBbase : HilbertIncidence.OnLine B base)
  (hCbase : HilbertIncidence.OnLine C base)
  (hSame : HilbertSameSide Geo A D base)
  (hEqual :
    HilbertScissorsEquicomplementtable
      Geo
      (hilbertScissorsTriangle Geo A B C)
      (hilbertScissorsTriangle Geo D B C)) :
  Exists fun top : Geo.Line =>
    HilbertIncidence.OnLine A top /\
    HilbertIncidence.OnLine D top /\
    HilbertLinesDisjoint Geo top base := by
  ...
```

The ellipsis is documentation notation only. The repository theorem is fully proved and contains no `sorry`. Unicode logical symbols from the source signature have been normalized above to ASCII forms for the documentation listing; declaration names and argument structure are unchanged.

The mathematically significant proposition-local declarations are

```
proposition39_auxiliary_point
proposition39_auxiliary_order
proposition39_parallel_points_to_top_line
```

The content module contributes the chain

```
scissors_split_side
hilbert_scissors_triangle_positive
hilbert_scissors_triangle_proper_part
```

where the middle declaration is the axiom and the last declaration is a theorem derived from the first two.

The final axiom audit reported by Lean is:

```
'Geometry.euclid_proposition_39' depends on axioms:
[propext,
 Classical.choice,
 bookZero_nullSegment1,
 hilbert_scissors_triangle_positive,
 Quot.sound]
```

Here `propext`, `Classical.choice`, and `Quot.sound` are logical foundations inherited by the Lean development. The pre-existing Book Zero debt shown in the audit is inherited transitively. The only new I.39-specific content assumption is the positivity declaration shown there.

The full project command

```
lake build
```

is clean with the current `Proposition39.lean`.

## 39.7 Relationship with Earlier Propositions

### 39.7.1 Proposition I.31

Classically, the proof requires drawing through  $A$  a line parallel to the base. Euclid would cite the parallel construction of I.31. The Lean proof does not call I.31 by name. The construction is already available as

```
hilbert_parallel_through_point_exists
```

in the developed Hilbert interface.

This is another example of the distinction between the historical proposition DAG and the actual dependency graph of the reconstruction.

### 39.7.2 Proposition I.37

I.37 is the decisive earlier proposition in the actual Lean proof. Once  $E$  has been constructed on the parallel through  $A$ , the two triangles  $ABC$  and  $EBC$  have the same base and lie between the same parallels. Therefore

$$ABC \approx EBC.$$

This use of I.37 is particularly important because the project’s proof of I.37 was deliberately constructed without a general “halves of equals” or property- $(Z)$  principle. Consequently there is no circularity: I.37 provides the forward equal-content theorem, while I.39 is exactly the later converse that first introduces a strict proper-part assumption.

### 39.7.3 Proposition I.38

I.38 is the immediate conceptual contrast. Its content conclusion is still forward and can be obtained from a congruent-copy construction, I.37, and transitivity. No strict comparison of contents is needed.

I.39 starts with equal content and concludes a geometric relation. This is why the existing I.35–I.38 scissors interface is almost sufficient but not quite: the missing ingredient is not another equality operation but a strictness principle.

### 39.7.4 Proposition I.40

The clean project build confirms that the public theorem name `euclid_proposition_39` is now available to Proposition I.40. Thus I.39 is not an isolated experimental theorem; it is part of the continuing Book I dependency chain.

## 39.8 Hilbert and Book Zero Components Used

### 39.8.1 Incidence and order

The auxiliary construction uses the Hilbert incidence API for:

- the line through two distinct points;
- uniqueness of a line through two distinct points;
- conversion between point-line carriers and incidence lines;
- collinearity induced by a common incidence line;



- betweenness trichotomy for three distinct collinear points;
- transport of betweenness to incidence information.

These tools are responsible for proving that the constructed point  $E$  is actually on the carrier  $BD$  and for reducing its position to the two strict orders needed by the reductio.

### 39.8.2 SameSide and OppositeSide

The public hypothesis

$$\text{SameSide}(A, D; \text{base})$$

contains more information than a conventional drawing usually records. It implies that both top vertices lie off the base. After the auxiliary point is constructed, the proof establishes

$$\text{SameSide}(A, E; \text{base})$$

and hence, by symmetry and transitivity,

$$\text{SameSide}(E, D; \text{base}).$$

This is used against

```
hilbert_oppositeSide_not_sameSide
```

to eliminate the order  $E - B - D$ .

### 39.8.3 Euclidean parallelism

The reconstruction is Euclidean in the current project API. Parallelism is used in two logically distinct ways.

First, the proof constructs a parallel through  $A$  to the base. Second, Euclidean uniqueness is used in the contradiction showing that the carrier  $BD$  cannot be disjoint from this new parallel unless it coincides with the base, which would place  $D$  on the base.

The helper

```
collinear_parallel_trans
```

then transports the auxiliary parallel along a collinear carrier so that, when  $E \neq A$ , the exact hypothesis

$$AE \parallel BC$$

required by I.37 is available.

### 39.8.4 Scissors content

The content stage uses only the already established equivalence operations:

- `equicomplementable_refl`;
- `equicomplementable_symm`;
- `equicomplementable_trans`;
- Proposition I.37.

No new scissors constructor is introduced. The existing split constructor is packaged by `scissors_split_side`. The specialized proper-part principle is derived from that split and the single new positivity axiom.

### 39.8.5 Book Zero

No new Book Zero theorem is introduced for I.39. The axiom audit nevertheless shows the inherited dependency

```
bookZero_nullSegment1
```

through earlier project infrastructure. This dependency predates I.39 and is not part of the new converse-content argument.

## 39.9 Formalization Notes

### 39.9.1 Geometry, representation, and content

From I.35 onward it is important to keep three layers separate.

**Geometry.** The proof constructs the parallel through  $A$ , proves that it meets the carrier  $BD$ , obtains the point  $E$ , establishes its side-of-line position, classifies the order of  $B, E, D$ , and finally proves the required parallelism.

**Representation.** The public theorem speaks about an explicit incidence line `base` and returns an explicit incidence line `top`. Proposition I.37, in contrast, uses point-line parallelism. The dedicated top-line packaging helper bridges these two representations. Its exact declaration name is

```
proposition39_parallel_points_to_top_line
```

This is a representation issue, not an additional geometric axiom.

At the triangle-term level no special permutation machinery is central to I.39: all three triangles are written on the common ordered base  $BC$ .

**Content.** The equality information is entirely synthetic. The formal theorem never assigns a number to the area of a triangle. It derives

$$DBC \approx EBC$$

by symmetry and transitivity of equicomplementability and then invokes one strict-content principle to contradict the proper inclusion.

Thus the proof does not identify `HilbertScissorsEquicomplementable` with numerical equality of area. It uses only the operations explicitly available in the formal content layer.

### 39.9.2 Hidden configuration data

The Lean proof establishes explicitly:

- $A$  and  $D$  are off the base line;
- $B \neq D$  because  $B$  is on the base and  $D$  is off it;
- the parallel through  $A$  is disjoint from the base;
- the carrier  $BD$  must meet that parallel in the nonparallel branch;
- the intersection point  $E$  is off the base;
- $E \neq B$  and  $E \neq D$ ;
- $B, E, D$  are collinear;
- $A$  and  $E$  are on the same side of the base;
- therefore  $E$  and  $D$  are on the same side of the base;
- the order  $E - B - D$  is impossible;
- in the remaining two orders the larger triangle is nondegenerate;
- the proper-subtriangle relation is therefore strict rather than degenerate.

None of these facts is read from Figure 39.1. The diagram is a summary of conclusions proved inside the geometry layer.

### 39.9.3 Axiomatic status

The public theorem assumes

`[HilbertEuclideanPlane Geo].`

The proof uses incidence, order, congruence-derived side-of-line theory, and Euclidean parallelism. It introduces no continuity axiom and no Archimedean axiom.

There is, however, one new non-geometric assumption:

```
hilbert_scissors_triangle_positive
```

This should be read as an explicit positivity axiom for content, not as part of Hilbert Group IV. Its role is analogous to the strictness isolated by Hartshorne's property ( $Z$ ) and to the De Zolt phenomenon in the classical theory of area. The proposition-specific proper-part declaration is now a theorem:

```
hilbert_scissors_triangle_proper_part
```

The current theorem is therefore conditional in a very precise sense: all geometry of I.39 is proved from the established Hilbert/Euclidean infrastructure, while the missing semantic strength of the term-level scissors calculus is exposed as one configuration-free positivity assumption.

### 39.9.4 Why Hilbert Theorem 48 is not an imported axiom

An earlier experimental route packaged the converse as an equal-altitude statement close to Hilbert Theorem 48. That route was abandoned for the final I.39 module because it concealed too much of the real dependency.

Hilbert’s own proof of Theorem 48 is downstream from segment arithmetic and the area function. Reproducing it faithfully would require substantially more infrastructure than I.39 itself needs. The present route instead follows the classical *reductio* and isolates the exact strict-content principle at the point where the current formal scissors calculus stops.

This makes the remaining debt cleaner and easier to study independently. The module `HilbertScissorsInvariant` already provides the generic valuation interface and proves that equicomplementable terms have equal valuation. What is still missing is a concrete additive valuation whose value on every nondegenerate triangle is nonzero. Such a valuation immediately discharges `hilbert_scissors_triangle_positive`, after which the derived proper-part theorem and I.39 require no change.

### 39.9.5 Source audit

The sources play different roles in the reconstruction.

Wilkins supplies the clearest complete audit of the classical configuration. His Study Note explicitly distinguishes the two positions of the intersection point  $E$  and notes that Euclid discusses only one of them. This two-case analysis matches the two strict betweenness branches retained by the Lean proof.

Hartshorne supplies the semantic diagnosis. Section 22 distinguishes forward equal-content arguments from converse arguments and explicitly states that I.39 uses “the whole is greater than the part,” hence property (Z). This is the principal source for interpreting the new formal positivity requirement.

Hilbert supplies the broader theory of plane area. Theorems 44–46 justify equicomplementability as the correct synthetic target for the forward equal-base/equal-altitude theorems. Section 21 proves Theorem 48 only after the development of area, while Theorem 52 records a De Zolt-type impossibility of reconstructing a whole after omitting a triangular part. These results explain both why the converse requires strictness and why it is reasonable to isolate a positivity principle from which the proper-part statement can be derived.

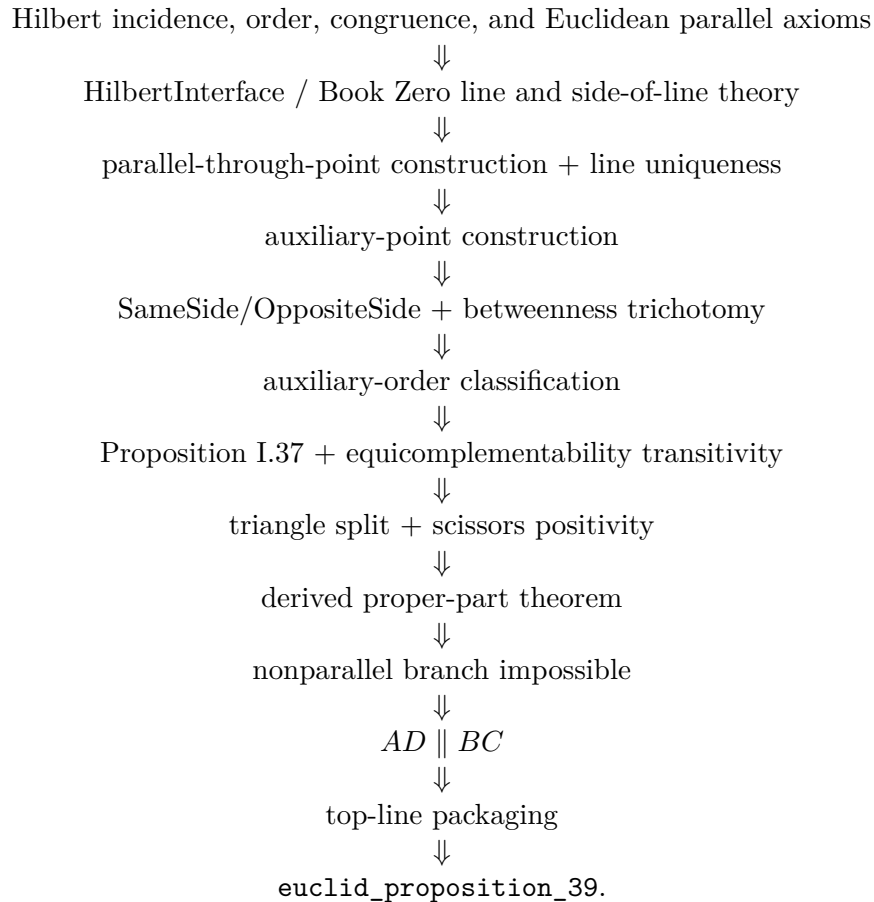
The Beeson proof corpus remains a control source for hidden positional and order obligations, but it is not used as the architecture of the final Lean proof. In particular, the project does not import a Tarski-style proof trace or reorganize the Hilbert interface around Beeson tactics.

Borsuk–Szmieliew and Greenberg remain control sources for incidence, order, parallelism, and the neutral/Euclidean boundary. They are not used here to replace the content relation by a numerical area theory.

The Euclid/Joyce edition controls the classical statement and Common Notion 5. The project Book I compilation is used as a consistency check against the documentation of I.35–I.38. Neither source overrides the current Lean module, which is authoritative for the formal dependency graph.

## 39.10 Dependency Summary

The actual formal dependency chain is



The historical and formal DAGs are therefore related but not identical:

|                 |                                                                                                |
|-----------------|------------------------------------------------------------------------------------------------|
| Euclid/Wilkins: | parallel through $A \rightarrow E \rightarrow$ earlier equal-area theorem $\rightarrow$ C.N.5, |
| Lean:           | parallel through $A \rightarrow E \rightarrow$ I.37 $\rightarrow$ derived proper-part theorem. |

|                                          |                                  |
|------------------------------------------|----------------------------------|
| New geometric axiom:                     | no                               |
| New content axiom:                       | yes — scissors positivity        |
| New proposition-local helpers:           | yes                              |
| New reusable Scissors theorems:          | yes — split-side and proper-part |
| New Book Zero theorem:                   | no                               |
| New numerical area or polygon semantics: | no                               |
| Public theorem compiles:                 | yes                              |
| Full lake build:                         | clean                            |
| sorry in Proposition39.lean:             | no                               |

### 39.11 Conclusion

Proposition I.39 marks a real change in the logical direction of Euclid’s area theory. The preceding propositions show that appropriate geometric configurations imply equality of content. I.39 asks equal content to force a geometric conclusion. That reversal is exactly where a strict content principle becomes unavoidable.

The formal reconstruction makes the boundary unusually transparent. All of the geometry needed for the classical reductio – the parallel through  $A$ , its intersection with  $BD$ , the SameSide analysis, the collinear order trichotomy, and the reduction to Proposition I.37 – is derived inside the existing Hilbert/Euclidean infrastructure. No new geometric axiom is needed.

What remains is one explicit content assumption: adjoining a nondegenerate triangle to an arbitrary scissors term cannot leave the content unchanged. Rather than encoding the I.39 configuration itself as an axiom, the development records this configuration-free positivity statement in the declaration

```
hilbert_scissors_triangle_positive
```

The triangle split then proves the proposition-specific proper-part statement:

```
hilbert_scissors_triangle_proper_part
```

The axiom audit confirms that positivity, not the proper-part statement, is the only new I.39-specific foundational debt.

This is the main structural result of the proposition. I.39 itself is now stable. A future positive additive valuation can discharge the remaining positivity assumption without reopening either the proper-part derivation or the geometric proof of I.39.

## Chapter 40

# Proposition I.40

### 40.1 Introduction

Proposition I.40 is the equal-base companion to Proposition I.39. The four triangle propositions at the end of the first area block therefore form a natural pattern:

- I.37 : same base + same parallels  $\longrightarrow$  equal content,
- I.38 : equal bases + same parallels  $\longrightarrow$  equal content,
- I.39 : same base + equal content  $\longrightarrow$  same parallels,
- I.40 : equal bases + equal content  $\longrightarrow$  same parallels.

The present Lean theorem keeps the same synthetic content language used since I.35. Equality of triangles is represented by

```
HilbertScissorsEquicomplementable
```

rather than by a numerical area function. Its conclusion is geometric: the two top vertices lie on a Hilbert incidence line disjoint from the common base carrier.

The main architectural fact is that I.40 does not need a second converse-area argument. The difficult same-base converse has already been isolated in I.39. The only additional task is to reduce the equal-base configuration to a same-base one. For that reduction the formal proof reuses the congruent-copy construction developed for I.38.

This reuse was not merely cosmetic. The first draft of I.40 repeated the angle construction, segment layoff, SAS argument, oriented parallel argument, and parallelogram criterion already present in I.38. Inspection of I.40 showed that two I.38 helpers had been stated with unnecessary proposition-specific data. Their signatures were generalized, the full project was rebuilt successfully, and I.40 now calls that common construction directly. Thus the proposition records a genuine API improvement discovered by a later proof.

No new content principle is introduced in I.40. The theorem depends on I.39, and therefore inherits the positivity assumption already isolated there, but I.40 itself adds no new axiom, no numerical area, and no new polygon semantics.

## 40.2 The Classical Configuration

Euclid’s statement concerns two equal triangles on equal bases lying in one straight line and on the same side of that line. In the current formal statement the triangles are

$$ABC \quad \text{and} \quad DEF,$$

with equal bases

$$BC \cong EF.$$

The four base points are placed in the strict order

$$B - C - E - F,$$

encoded by

$$\text{Between}(B, C, E), \quad \text{Between}(C, E, F).$$

The points  $B$  and  $C$  are given on an incidence line **base**. The betweenness hypotheses then force  $E$  and  $F$  onto that same carrier. The top vertices  $A$  and  $D$  satisfy

$$\text{SameSide}(A, D; \text{base}).$$

This is a deliberate specialization of the fully general Euclidean statement. Wilkins’ Study Note points out that a complete treatment of arbitrary equal bases on one line requires six positional configurations. These arise from the possible relative positions of the two base segments together with the position of the second top vertex relative to the auxiliary parallel through the first. Euclid explicitly discusses only one of those cases.

The present Lean theorem does *not* claim to formalize all six configurations. It works with the ordered separated-base arrangement  $B - C - E - F$ , the same carrier normalization already used by Proposition I.38. The advantage is that the equal-base geometry can be reused without reopening a large order case split. A future fully order-free wrapper could normalize other base arrangements before calling this theorem; that normalization is not part of the current proposition.

Figure 40.1 shows the actual formal configuration and the two conceptual reductions used in Lean.

## 40.3 Classical Proof

In Euclid’s displayed configuration the two equal bases share an endpoint: the triangles are  $ABC$  and  $CDE$ , with bases  $BC$  and  $CE$ . Join the two top vertices  $A$  and  $D$ . The aim is to prove that  $AD$  is parallel to the base line.

Assume that  $AD$  is not the required parallel. Through  $A$  draw another line parallel to the base. Let it meet the side of the second triangle, or its extension, at an auxiliary point  $F$ , and join  $F$  to the appropriate base endpoint. Proposition I.38 applies to the original triangle and the auxiliary triangle because their bases are equal and they lie in the same parallels. Hence the auxiliary triangle is equal to the first triangle. By hypothesis the first triangle is equal to the second, so transitivity of equality forces the auxiliary triangle to be equal to the second triangle. But one of these is geometrically a proper part of the other. This is the “greater equals the less” contradiction used by Euclid.



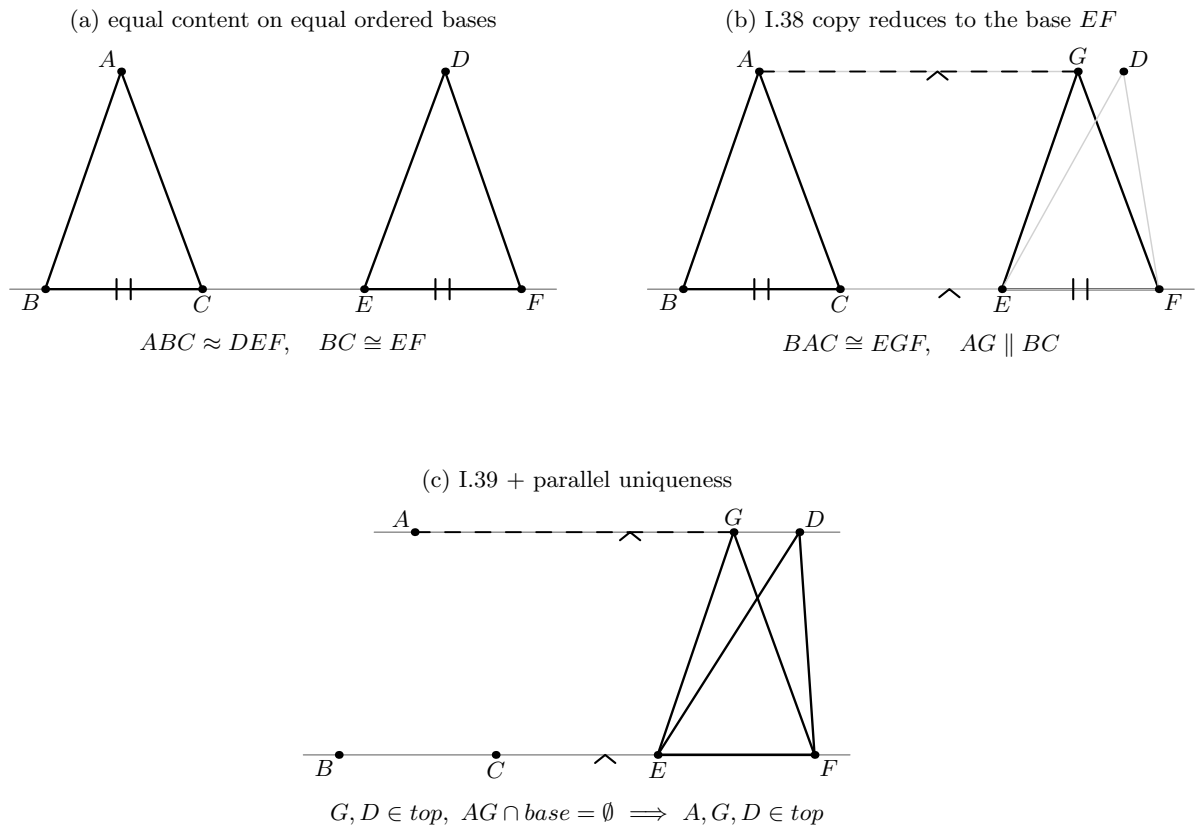


Figure 40.1: The formal reconstruction of I.40. (a) The given triangles  $ABC$  and  $DEF$  have equal content, equal ordered bases  $BC$  and  $EF$ , and top vertices on the same side of the common base carrier. (b) The reusable I.38 construction produces a point  $G$  such that  $BAC \cong EGF$  and  $AG \parallel BC$ ; internally this comes from the parallelogram  $AGEB$ . Hence  $ABC$  and  $GEF$  are related by the stronger scissors equality, and transitivity gives equal content for  $GEF$  and  $DEF$ . (c) Proposition I.39 applied on the common base  $EF$  places  $G$  and  $D$  on a line  $\text{top}$  disjoint from  $\text{base}$ . The carrier of  $AG$  is also disjoint from  $\text{base}$ ; since both candidate top lines pass through  $G$ , Euclidean parallel uniqueness identifies them and places  $A$  on  $\text{top}$  as well.

The classical dependency in the displayed case is therefore

equal triangles on equal bases  
 $\Downarrow$   
 draw a parallel through the first top vertex  
 $\Downarrow$   
 auxiliary triangle in the same parallels  
 $\Downarrow$  I.38  
 auxiliary triangle = first triangle  
 $\Downarrow$  C.N.1  
 auxiliary triangle = second triangle  
 $\Downarrow$   
 proper-part contradiction  
 $\Downarrow$   
 the top vertices lie on the required parallel.

The historical proof and the current Lean proof therefore share the idea that I.40 should be reduced to an earlier equal-base theorem, but they reduce in opposite directions. Euclid uses the forward theorem I.38 inside a reductio. The Lean development first uses the I.38 *construction* to

move to a common base and then invokes the already completed converse I.39.

Wilkins' expanded analysis is important here. Once the two bases are allowed to be adjacent, separated, or interlocking, and once both possible positions of the second top vertex relative to the auxiliary parallel are considered, six cases must be ruled out. The present formal theorem avoids claiming this full classical case analysis: its explicit  $B - C - E - F$  hypotheses select one ordered base arrangement from the outset.

## 40.4 Equal Bases by Reduction to the Same-Base Converse

The central structural observation is that the passage from I.39 to I.40 is the converse analogue of the passage from I.37 to I.38.

For the forward direction, I.38 constructs a congruent copy of one triangle on the other equal base and then compares the copied triangle with the target by I.37. For the converse direction, I.40 uses the same copy construction, but now the copied triangle and the target triangle are equal in content and share *the same base*. Proposition I.39 can therefore be applied directly.

Schematically,

$$\begin{array}{ccc}
 \text{I.37} & \xrightarrow{\text{equal-base copy}} & \text{I.38} \\
 \Updownarrow & & \Updownarrow \\
 \text{I.39} & \xrightarrow{\text{same equal-base copy}} & \text{I.40}
 \end{array}$$

where the vertical symbols indicate only the forward/converse relationship, not a formal biconditional theorem in Lean.

This observation led to a small but significant refactor of `Proposition38.lean`. The helpers

```
i38_equal_base_copy
i38_copy_parallelogram
```

had originally accepted the second top vertex  $D$  and a hypothesis of the form  $AD \parallel BC$ . Inspection showed that these data were not intrinsic to the copy itself. What the construction actually needs is:

- one explicit base carrier;
- $B, C, E, F$  on that carrier;
- the ordered-base data  $B - C - E - F$ ;
- $BC \cong EF$ ;
- the point  $A$  off the base.

After the signatures were generalized, I.38 continued to build cleanly and I.40 could call the construction without repeating its proof.

The resulting Lean DAG is

$$\begin{array}{c}
ABC \approx DEF, \quad BC \cong EF, \quad B - C - E - F, \quad \text{SameSide}(A, D; \mathbf{base}) \\
\Downarrow \\
E, F \in \mathbf{base} \\
\Downarrow \quad \text{i38\_copy\_parallelogram} \\
\exists G : \quad BAC \cong EGF, \quad \text{SameSide}(A, G; \mathbf{base}), \quad AG \parallel BC \\
\Downarrow \\
ABC \sim GEF \\
\Downarrow \quad \text{content symmetry/transitivity} \\
GEF \approx DEF \\
\Downarrow \quad \text{I.39 on base } EF \\
G, D \in \mathbf{top}, \quad \mathbf{top} \cap \mathbf{base} = \emptyset \\
\Downarrow \quad AG \parallel BC \\
\text{carrier}(AG) \cap \mathbf{base} = \emptyset \\
\Downarrow \quad \text{Euclidean parallel uniqueness through } G \\
A, G, D \in \mathbf{top} \\
\Downarrow \\
\text{euclid\_proposition\_40.}
\end{array}$$

Here  $\sim$  denotes the stronger relation

```
HilbertScissorsEq
```

while  $\approx$  denotes

```
HilbertScissorsEquicomplementtable
```

## 40.5 Proof Architecture of the Reconstruction

### 40.5.1 Step 1: putting the second base on the common carrier

The first proposition-local helper is

```
i40_base_endpoints
```

It starts from  $B, C \in \mathbf{base}$  together with

$$B - C - E, \quad C - E - F,$$

and proves

$$E, F \in \mathbf{base}.$$

The proof is pure incidence/order transport. Betweenness supplies both collinearity and endpoint distinctness, and line uniqueness transports the new points to the already chosen carrier.

This helper is intentionally local. It packages the particular ordered four-point configuration used by I.40; it does not introduce a new general notion of collinear base segments.

### 40.5.2 Step 2: reusing the I.38 congruent-copy construction

The second helper is

```
i40_copy_parallel
```

Its output is exactly the geometric bridge needed by the converse proof:

$$\exists G : \text{SameSide}(A, G; \mathbf{base}), \quad BAC \cong EGF, \quad AG \parallel BC.$$

The helper does not rebuild this geometry. It calls

```
i38_copy_parallelogram
```

which returns the stronger package

$$\text{SameSide}(A, G; \mathbf{base}), \quad AGE B \text{ is a parallelogram}, \quad BAC \cong EGF.$$

From the parallelogram one gets  $AG \parallel EB$ . Since  $B, C, E$  are collinear, parallelism is transported from the carrier  $EB$  to  $BC$ . The result is  $AG \parallel BC$ .

This is the point at which the I.38 refactor pays off. The substantial angle, ray, SAS, extension, and I.28 machinery remains proved once, in I.38. I.40 uses only its mathematical output.

### 40.5.3 Step 3: transporting equal content to the copied triangle

Triangle congruence gives the strong scissors relation

$$BAC \sim EGF.$$

The scissors representation of a triangle is insensitive to exchanging its first two vertices, so the proof rewrites this as

$$ABC \sim GEF.$$

The relation `HilbertScissorsEq` implies `HilbertScissorsEquicomplementable`, hence

$$ABC \approx GEF.$$

By symmetry and transitivity with the hypothesis

$$ABC \approx DEF,$$

we obtain

$$GEF \approx DEF.$$

No subtraction, cancellation, strict content comparison, or numerical area is used in this stage. It is ordinary equivalence transport in the existing scissors layer.

#### 40.5.4 Step 4: reducing to I.39 on the common base $EF$

The copy construction puts  $G$  on the same side of **base** as  $A$ , while the public hypothesis puts  $D$  on that side as well. Symmetry and transitivity of **HilbertSameSide** therefore yield

$$\text{SameSide}(G, D; \text{base}).$$

The strict betweenness  $C - E - F$  gives  $E \neq F$ . All hypotheses of I.39 are now available for the triangles  $GEF$  and  $DEF$  on the same base  $EF$ .

Proposition I.39 returns a line **top** such that

$$G, D \in \text{top}, \quad \text{HilbertLinesDisjoint}(\text{top}, \text{base}).$$

This is the only converse-content step in I.40. Consequently all strictness needed to turn equal content back into geometry is inherited from I.39 rather than reintroduced locally.

#### 40.5.5 Step 5: identifying the final **top** line

It remains to place  $A$  on the line returned by I.39. The helper `i40_copy_parallel` already supplied  $AG \parallel BC$ . Let **topAG** be the incidence line through  $A$  and  $G$ . Since  $BC$  lies on **base**, point-line parallelism implies that **topAG** and **base** are disjoint.

Now both **topAG** and **top**

- pass through  $G$ , and
- are disjoint from **base**.

The Euclidean uniqueness of a parallel through an external point therefore forces

$$\text{topAG} = \text{top}.$$

Since  $A \in \text{topAG}$ , it follows that  $A \in \text{top}$ . Together with  $D \in \text{top}$  and disjointness from the base, this is exactly the public conclusion.

## 40.6 Lean Formalization

The current file imports the two propositions that carry its geometry and its converse-content step:

```
import CGJteamLab.Proposition38
import CGJteamLab.Proposition39
```

The public theorem is:

```
theorem euclid_proposition_40
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D E F : Geo.Point)
  (base : Geo.Line)
```

```

(hBCE : Geo.Between B C E)
(hCEF : Geo.Between C E F)
(hBbase : HilbertIncidence.OnLine B base)
(hCbase : HilbertIncidence.OnLine C base)
(hBC_EF : Geo.Congruent B C E F)
(hSame : HilbertSameSide Geo A D base)
(hEqual :
  HilbertScissorsEquicomplementable
  Geo
  (hilbertScissorsTriangle Geo A B C)
  (hilbertScissorsTriangle Geo D E F)) :
Exists fun top : Geo.Line =>
  HilbertIncidence.OnLine A top /\
  HilbertIncidence.OnLine D top /\
  HilbertLinesDisjoint Geo top base := by
...

```

The ellipsis is documentation notation only. The repository theorem is fully proved. Unicode logical symbols from the Lean source have been normalized to ASCII forms in the listing above.

The two proposition-local helpers are:

```

i40_base_endpoints
i40_copy_parallel

```

The second of these calls the generalized I.38 helper

```

i38_copy_parallelogram

```

rather than repeating its proof.

The scissors part of the final theorem uses the following existing operations:

```

scissors_congruent
scissors_triangle_swap12
equicomplementable_of_scissorsEq
equicomplementable_symm
equicomplementable_trans

```

The line-placement part then uses:

```

euclid_proposition_39
HilbertPlaneIncidence.line_through
hilbert_mem_pointLine_iff_onLine
HilbertEuclideanPlane.parallel_unique

```

## 40.7 Relationship with Earlier Propositions

### 40.7.1 Proposition I.38

I.38 supplies the reusable equal-base geometry. Its original formal proof had to construct  $G$  by angle layoff and segment layoff, establish triangle congruence by SAS, recover the appropriate orientation through extensions and I.28, and package the result as the parallelogram  $AGEB$ .

I.40 revealed that this construction was more general than its original signature. The helpers `i38_equal_base_copy` and `i38_copy_parallelogram` were therefore generalized to accept the base carrier and off-line data directly. The public I.38 theorem continues to use them, and I.40 now reuses the same package. This is a concrete example of a later proposition improving the reusable boundary of an earlier proof.

### 40.7.2 Proposition I.39

I.39 is the decisive converse theorem. It says that equal-content triangles on one common base and on the same side lie between the same parallels. Its proof is where the strict content issue appears: the proper-part contradiction ultimately depends on the positivity principle isolated in the scissors layer.

I.40 does not reproduce that argument. Once the copied triangle  $GEF$  has been made equal in content to  $DEF$ , I.39 is applied directly on the base  $EF$ . Thus the foundational debt of I.40 is inherited from I.39 rather than multiplied.

### 40.7.3 Proposition I.37

I.37 is not called directly by `Proposition40.lean`, but it remains in the transitive dependency graph through I.39. It is the forward same-base theorem used inside the I.39 reductio. The four propositions I.37–I.40 therefore form a coherent block rather than four independent area arguments.

## 40.8 Hilbert and Book Zero Components Used

The proposition sits on several already established layers.

### 40.8.1 Incidence and order

Strict betweenness provides endpoint distinctness and collinearity. The helper

```
hilbert_collinear_on_line
```

transports  $E$  and  $F$  onto the chosen base carrier. The theorem also uses incidence lines through distinct points and the equivalence between membership in a point-line carrier and incidence on a chosen Hilbert line.

### 40.8.2 Side-of-line geometry

The public statement exposes

```
HilbertSameSide Geo A D base
```

explicitly. The copied point  $G$  comes with a second `SameSide` relation from I.38. Symmetry and transitivity then produce the exact `SameSide` hypothesis required by I.39. This information is invisible in the short classical phrase “on the same side” but is a first-class proof object in Lean.

### 40.8.3 Parallelism and parallelograms

The I.38 construction returns the parallelogram  $AGEB$ . The existing parallel transport lemmas convert  $AG \parallel EB$  into  $AG \parallel BC$  using the collinearity of  $B, C, E$ . At the final stage the Euclidean uniqueness of the parallel through  $G$  identifies the carrier through  $A, G$  with the line through  $G, D$  returned by I.39.

Thus Group IV enters through the Euclidean parallel layer. No continuity axiom is used by the proposition.

### 40.8.4 Scissors content

The content calculation is intentionally small. Triangle congruence gives a strong scissors equality; representation transport reorders the triangle vertices; and the existing equivalence operations convert and compose the relations. No new scissors constructor or theorem is introduced in I.40.

The strict content principle needed by the converse direction is already encapsulated upstream in I.39 through

```
hilbert_scissors_triangle_positive
```

I.40 introduces no second positivity or cancellation assumption.

## 40.9 Source Audit

### 40.9.1 Euclid and Joyce

The classical Proposition I.40 is explicitly the equal-base converse of I.38. In Euclid’s displayed proof, a parallel is drawn through one top vertex; I.38 makes the corresponding auxiliary triangle equal to the first triangle, and transitivity of equality then produces the impossible equality of a greater triangle with a lesser one. Joyce also notes that the transmitted statement is less precise than it should be: the bases must lie on one straight line, and the proof should not depend on their sharing an endpoint.



### 40.9.2 Wilkins

Wilkins makes the hidden configuration problem explicit. A complete proof for two arbitrary equal bases on one line must exclude six configurations; only one appears explicitly in Euclid's proof. This is an important control on the formal statement. The current Lean theorem does not silently claim those six cases: it assumes the concrete order  $B - C - E - F$ .

### 40.9.3 Hilbert and Hartshorne

Hilbert's area theory and Hartshorne's Section 22 remain the controlling sources for the meaning of equal content. The present reconstruction keeps `HilbertScissorsEquicomplementable` distinct from numerical area and from unconditional finite equidecomposability.

For the converse direction, I.39 has already isolated the proper-part strictness issue. I.40 therefore needs no new area semantics. Its genuinely new work is only the geometric reduction from equal bases to one common base.

### 40.9.4 Borsuk–Szmielw and Greenberg

Borsuk–Szmielw remain useful for the incidence, order, side-of-line, congruence, and parallelogram infrastructure, but they are not used as the primary source for equal-content semantics. Greenberg similarly provides a control on the Euclidean/neutral boundary and distinguishes equal content from numerical area, while referring the detailed content theory elsewhere.

### 40.9.5 Beeson

The Beeson proof corpus is retained as a configuration audit rather than as the architectural source of the present proof. No Tarski-style proof trace is copied into I.40. In particular, the final Lean architecture is determined by the already developed Hilbert/I.38/I.39 interfaces.

## 40.10 Formalization Notes

### 40.10.1 Geometry

The geometric data actually used by the theorem are:

- the strict order  $B - C - E - F$ ;
- one common base carrier;
- equal base segments  $BC \cong EF$ ;
- SameSide data for  $A$  and  $D$ ;
- the reusable congruent-copy/parallelogram package from I.38;
- Euclidean uniqueness of the parallel through a point.

The theorem does not infer all possible relative positions of equal base segments. Those alternatives are outside its current statement.

### 40.10.2 Content

The public content hypothesis is exactly `HilbertScissorsEquicomplementable`. The local copy step uses the stronger `HilbertScissorsEq` because congruent triangles can be replaced directly. The proof then weakens this relation and uses only symmetry and transitivity.

There is no local subtraction, cancellation, comparison relation, numerical area, or whole-greater-than-part argument in I.40. The only strict content principle in the dependency graph is the positivity axiom already used to prove I.39.

### 40.10.3 Representation

The congruence theorem returned by I.38 is oriented as

$$BAC \cong EGF,$$

while the content theorem is stated for  $ABC$  and  $GEF$ . The formal proof therefore uses

```
scissors_triangle_swap12
```

twice. This is representational transport, not additional geometry.

Likewise, the conclusion of I.39 is expressed by an actual incidence line `top`, whereas  $AG \parallel BC$  is initially expressed by point-line carriers. The final proof explicitly bridges these two representations before applying parallel uniqueness.

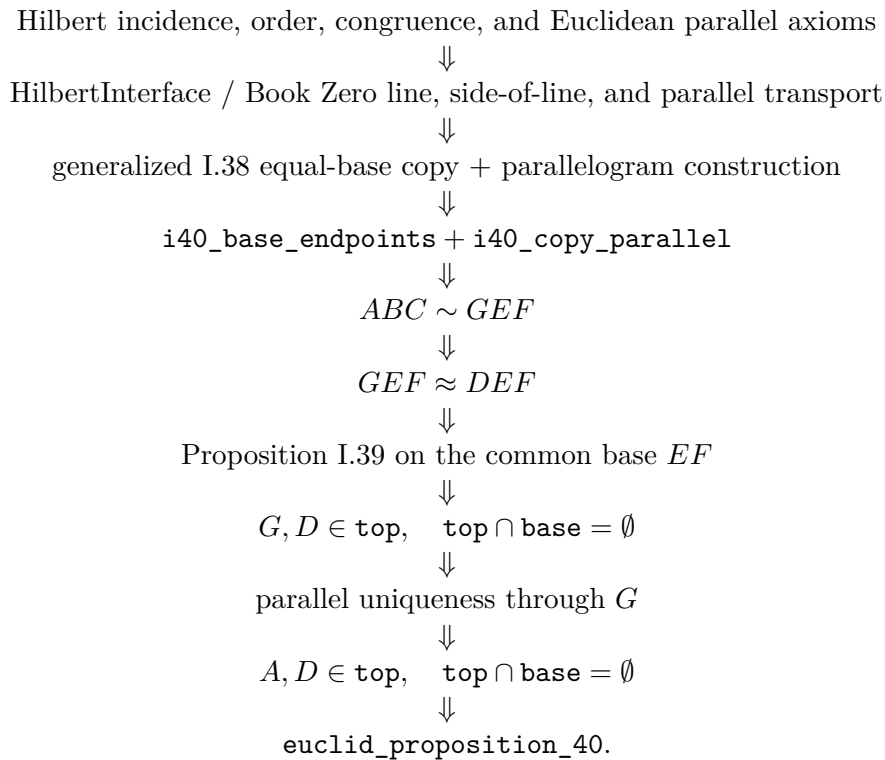
### 40.10.4 Axiomatic status

The public theorem assumes `HilbertEuclideanPlane`; it is therefore an Euclidean theorem in the current hierarchy. The final carrier identification uses the uniqueness of the parallel through  $G$ , and the reused I.38 construction also lives in the Euclidean layer. No continuity assumption is introduced.

On the content side, I.40 adds no axiom. It imports I.39 and therefore inherits the previously isolated scissors positivity assumption. This inheritance should not be confused with a new I.40-specific foundational principle.

## 40.11 Dependency Summary

The actual formal dependency chain is



The historical and formal DAGs are therefore related but distinctly organized:

Euclid : I.31 → I.38 → proper-part contradiction → same parallels,  
 Lean : I.38 copy geometry → content transport → I.39 → parallel uniqueness.

|                                          |                                        |
|------------------------------------------|----------------------------------------|
| New geometric axiom in I.40:             | no                                     |
| New content axiom in I.40:               | no                                     |
| Inherited content axiom:                 | yes — scissors positivity through I.39 |
| New proposition-local helpers:           | yes — two                              |
| New reusable declaration in I.40:        | no                                     |
| Reusable refactor required:              | yes — generalized I.38 copy helpers    |
| New Interface/Scissors theorem:          | no                                     |
| New Book Zero theorem:                   | no                                     |
| New numerical area or polygon semantics: | no                                     |
| Public theorem compiles:                 | yes                                    |
| Full lake build:                         | clean                                  |
| sorry in Proposition40.lean:             | no                                     |

## 40.12 Conclusion

Proposition I.40 is a useful test of whether the preceding area block has produced reusable mathematics rather than proposition-specific code. Once I.39 supplies the same-base converse and I.38 supplies an equal-base congruent copy, the new theorem does not need another proper-part

proof. It transports the original equal-content hypothesis to a copied triangle on the base  $EF$ , applies I.39, and finishes by uniqueness of the parallel through the copied top vertex.

The most important development result occurred just before the final proof was written. I.40 exposed that the equal-base copy construction in I.38 had been stated too narrowly. Generalizing the existing I.38 helpers eliminated a large duplicate proof while leaving I.38 itself unchanged. The clean rebuild of the entire project confirmed that the refactor preserved the earlier proposition and produced a genuinely reusable boundary for I.40.

The content architecture also remains stable. No numerical area, new scissors operation, or new strictness axiom is introduced. I.40 inherits the single positivity debt already made explicit by I.39 and otherwise consists of geometry, representation transport, equivalence transport, and Euclidean parallel uniqueness.

Finally, the formal statement remains intentionally narrower than the most general classical proposition: it treats the explicit ordered carrier  $B - C - E - F$ . Wilkins' six-configuration analysis therefore remains a useful record of what a future order-normalizing wrapper would have to cover. The current theorem is complete for its stated configuration, builds cleanly, and fits the architecture established by I.35–I.39.

# Chapter 41

## Proposition I.41

### 41.1 Introduction

Proposition I.41 begins a new use of the area block rather than a new area foundation. The preceding propositions I.37–I.40 compare triangles. I.41 combines the same-base theorem I.37 with the diagonal theorem I.34 and concludes that a parallelogram is “double” a triangle on the same base and between the same parallels.

The present formalization keeps this statement entirely synthetic. It does not introduce a numerical area function and it does not interpret “double” as multiplication by the real number 2. Instead the conclusion is expressed in the existing scissors language: the formal parallelogram term is equicomplementable with the formal sum of two copies of the triangle term. Thus the relevant target has the form

$$P \approx (T + T),$$

where  $\approx$  abbreviates the project’s term-level

```
HilbertScissorsEquicomplementable
```

relation.

I.41 is therefore a forward content theorem. No content relation is inverted, no proper-part comparison is required, and no positivity, cancellation, or whole-greater-than-part principle is introduced. This sharply separates I.41 from the converse propositions I.39 and I.40.

The reconstruction also exposed one missing reusable law of the scissors calculus. Equicomplementability must be compatible with formal addition. The initial proposition-local “doubling” helper was consequently replaced by the general theorem

```
equicomplementable_add
```

in `HilbertScissors.lean`. The full project rebuild remained clean, and the same reusable theorem now also replaces the former I.41-specific helper in Proposition I.47.

## 41.2 The Classical Configuration

Let  $ABCD$  be a parallelogram and let  $EBC$  be a triangle with the same base  $BC$ . Both figures lie between the same two parallel carriers. In Euclid's notation the upper carrier passes through  $A$  and  $E$ , so

$$AE \parallel BC.$$

The parallelogram has the same lower base  $BC$ , and its upper side  $AD$  lies on the same upper carrier in the classical picture.

The public Lean theorem encodes this configuration economically by the two hypotheses

```
(hParallelogram : IsParallelogram Geo A B C D)
(hAE_BC : Geo.Parallel A E B C)
```

There is no additional hypothesis asserting explicitly that  $A, D, E$  are collinear. The proof does not need such a witness. The parallelogram structure is used by I.34, while the parallel pair  $AE, BC$  is used independently by I.37. This is an example where the formal statement contains exactly the data needed by the proof rather than reproducing every incidence relation visible in the classical diagram.

Figure 41.1 shows both the classical configuration and the two content comparisons used by the Lean proof.

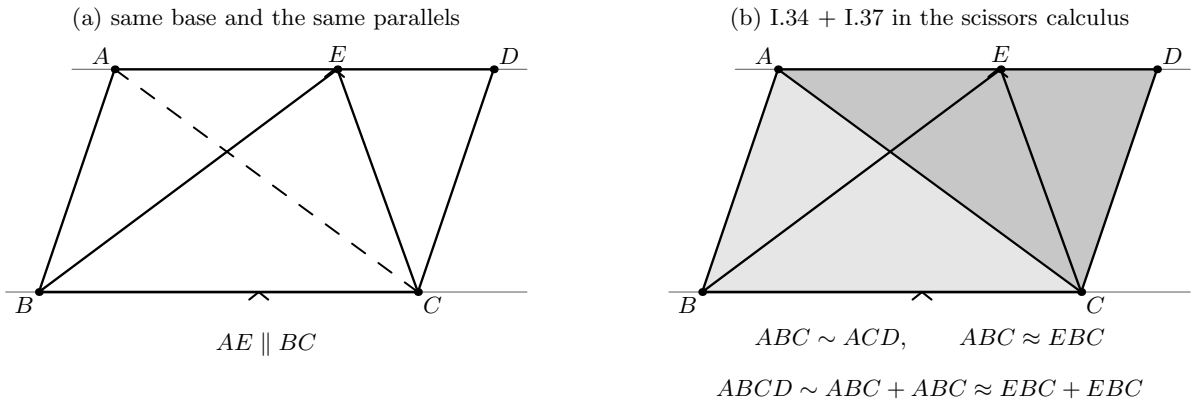


Figure 41.1: The configuration and proof structure of I.41. (a) The parallelogram  $ABCD$  and triangle  $EBC$  share the base  $BC$  and lie between the parallel carriers through  $AE$  and  $BC$ ; the diagonal  $AC$  is the diagonal used by I.34. (b) I.34 makes the two diagonal triangles scissors-equivalent, so the parallelogram term can be represented as two copies of  $ABC$ . I.37 gives equal content for  $ABC$  and  $EBC$ . The reusable addition theorem then transports this relation to the two formal sums.

## 41.3 Classical Proof

Euclid's proof is one of the shortest in the area block.

Join  $A$  to  $C$ . Proposition I.37 applies to the triangles  $ABC$  and  $EBC$ : they stand on the same base  $BC$  and lie in the same parallels  $AE$  and  $BC$ . Hence the two triangles are equal in Euclid's content sense.

Next apply Proposition I.34 to the parallelogram. Its diagonal  $AC$  bisects the parallelogram, so the parallelogram is double the triangle  $ABC$ . Since  $ABC$  is equal to  $EBC$ , the parallelogram is also double the triangle  $EBC$ .

The classical dependency graph is therefore

$$\begin{array}{ccc}
 ABCD \text{ parallelogram,} & AE \parallel BC & \\
 \Downarrow & & \\
 \triangle ABC = \triangle EBC & \text{by I.37} & \\
 \Downarrow & & \\
 ABCD = 2\triangle ABC & \text{by I.34} & \\
 \Downarrow & & \\
 ABCD = 2\triangle EBC. & & 
 \end{array}$$

Joyce’s text makes exactly this sequence explicit: join  $AC$ , use I.37 for the same-base triangles, and use I.34 for the diagonal bisection of the parallelogram. Unlike I.39 and I.40, there is no *reductio* and no proper-part argument.

## 41.4 Doubling in the Scissors Calculus

The word “double” needs a precise synthetic interpretation in the current library. The type

```
HilbertScissorsTerm
```

is a formal multiset sum of triangle terms. Thus the term-level double of a triangle  $T$  is simply

$$T + T.$$

No scalar multiplication is introduced.

The crucial reusable statement is

```

theorem equicomplementable_add
  {P1 Q1 P2 Q2 : HilbertScissorsTerm Geo}
  (h1 : HilbertScissorsEquicomplementable Geo P1 Q1)
  (h2 : HilbertScissorsEquicomplementable Geo P2 Q2) :
  HilbertScissorsEquicomplementable Geo
    (P1 + P2) (Q1 + Q2)

```

Conceptually, if  $P_1 \approx Q_1$  and  $P_2 \approx Q_2$ , then

$$P_1 + P_2 \approx Q_1 + Q_2.$$

I.41 uses the same relation twice, obtaining

$$ABC + ABC \approx EBC + EBC.$$

This theorem belongs to `HilbertScissors.lean`, not to `Proposition41.lean`. Its proof is pure algebra of the existing scissors relation: the complements for the two hypotheses are added, the two whole-term relations are combined with `HilbertScissorsEq.add`, and associativity and commutativity of multiset addition rearrange the resulting terms.

This is stronger architecturally than an I.41-specific “double” lemma. The law is closed under arbitrary pairs of equicomplementable terms and can be reused whenever a later proposition adds equal-content components.

## 41.5 Proof Architecture of the Reconstruction

The Lean proof is short enough to be displayed as four conceptual stages.

### 41.5.1 Step 1: replace the second diagonal triangle by the first

Proposition I.34 supplies

```
euclid_proposition_34_diagonal
```

with result

$$ABC \cong CDA.$$

The theorem

```
scissors_congruent
```

turns this triangle congruence into a scissors equality. A cyclic permutation of the triangle representation identifies  $CDA$  with  $ACD$ , giving

$$ABC \sim ACD,$$

where  $\sim$  denotes `HilbertScissorsEq`.

The formal parallelogram term is defined by the diagonal  $AC$ :

$$\text{Par}(ABCD) = ABC + ACD.$$

Adding reflexivity on the first triangle to the reverse of the relation above yields

$$\text{Par}(ABCD) \sim ABC + ABC.$$

This is the formal counterpart of the classical statement that the diagonal bisects the parallelogram.

### 41.5.2 Step 2: apply Proposition I.37 on the common base

The hypothesis

$$AE \parallel BC$$

is already in exactly the form expected by Proposition I.37. The call

```
euclid_proposition_37 Geo A B C E hAE_BC
```



returns

$$ABC \approx EBC.$$

No new carrier construction, SameSide witness, midpoint construction, or order case split appears in I.41 itself. All geometric normalization required by the same-base theorem is encapsulated in the already proved I.37.

### 41.5.3 Step 3: add the equal-content relation to itself

The new reusable theorem is applied twice to the same hypothesis:

```
equicomplementable_add
  Geo
  hABC_EBC
  hABC_EBC
```

This gives

$$ABC + ABC \approx EBC + EBC.$$

This is the precise term-level operation corresponding to Euclid’s use of “double.”

### 41.5.4 Step 4: transport from the doubled triangle to the parallelogram

The final call

```
equicomplementable_transport
```

uses the scissors equality

$$\text{Par}(ABCD) \sim ABC + ABC$$

on the left and reflexivity on  $EBC + EBC$  on the right. It therefore transports the doubled I.37 conclusion to the public target

$$\text{Par}(ABCD) \approx EBC + EBC.$$

The complete Lean DAG is

$$\begin{array}{c}
 \text{IsParallelogram}(A, B, C, D) \quad + \quad AE \parallel BC \\
 \Downarrow \text{I.34} \\
 ABC \cong CDA \\
 \Downarrow \text{scissors\_congruent} \\
 \text{Par}(ABCD) \sim ABC + ABC \\
 AE \parallel BC \\
 \Downarrow \text{I.37} \\
 ABC \approx EBC \\
 \Downarrow \text{equicomplementable\_add} \\
 ABC + ABC \approx EBC + EBC \\
 \Downarrow \text{equicomplementable\_transport} \\
 \text{Par}(ABCD) \approx EBC + EBC \\
 \Downarrow \\
 \text{euclid\_proposition\_41.}
 \end{array}$$

## 41.6 Lean Formalization

The current proposition file imports exactly the two earlier propositions used by the mathematical proof:

```
import CGJteamLab.Proposition34
import CGJteamLab.Proposition37
```

The public theorem is:

```
theorem euclid_proposition_41
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D E : Geo.Point)
  (hParallelogram : IsParallelogram Geo A B C D)
  (hAE_BC : Geo.Parallel A E B C) :
  HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo A B C D)
    (hilbertScissorsTriangle Geo E B C +
     hilbertScissorsTriangle Geo E B C) := by
  ...
```

The ellipsis is documentation notation only. The repository theorem is fully proved and the full project build is clean.

The mathematically significant declarations called directly by the proof are:

```
euclid_proposition_34_diagonal
scissors_congruent
scissors_triangle_cycle
euclid_proposition_37
equicomplementable_add
equicomplementable_transport
```

The remaining operations are reflexivity, symmetry, and formal addition in `HilbertScissorsEq`. There are no proposition-local helper theorems in the final I.41 file.

## 41.7 Relationship with Earlier Propositions

### 41.7.1 Proposition I.34

I.34 supplies the diagonal congruence of the parallelogram. In the classical proof this is stated as bisection of the parallelogram by its diagonal. In the formal proof the stronger triangle-congruence result is converted directly into scissors equality and then used to rewrite

$$ABC + ACD$$

as

$$ABC + ABC.$$

Thus I.34 is the representation bridge from the geometric parallelogram to the formal doubled triangle.

### 41.7.2 Proposition I.37

I.37 is the direct content theorem. It states that triangles on the same base and between the same parallels are equicomplementable. With the substitution  $D := E$ , its public hypothesis is exactly the I.41 hypothesis  $AE \parallel BC$ . The result is therefore available without any new geometric construction.

I.37 is also the reason I.41 needs no halving or cancellation theorem. The formal proof proceeds in the forward direction

$$ABC \approx EBC \implies ABC + ABC \approx EBC + EBC,$$

not in the converse direction from doubled figures back to their halves.

### 41.7.3 Propositions I.35–I.40

I.35–I.38 established the forward equal-content machinery; I.39 and I.40 then showed where converse reasoning requires strict content information. I.41 returns to the forward direction. Consequently it uses only equivalence transport and addition in the scissors calculus. The positivity issue isolated for I.39 is not needed here.

This position in the Book I sequence is important: I.41 is not another test of proper-part semantics. It is a test of whether the already established content relation behaves compositionally under addition.

## 41.8 Hilbert and Book Zero Components Used

### 41.8.1 Triangle congruence and parallelogram infrastructure

The direct geometric input from I.34 ultimately rests on the established Hilbert congruence and parallelogram theory. I.41 itself does not reopen SSS, angle transport, or diagonal incidence. It consumes the packaged theorem

```
euclid_proposition_34_diagonal
```

as a black box.

### 41.8.2 Same-base triangle content

All of the substantial geometry behind equal-content triangles on a common base is encapsulated by I.37. Its Hilbertian reconstruction uses the midpoint and parallelogram machinery developed earlier in the project, but none of those witnesses is reconstructed in I.41.

### 41.8.3 Scissors representation

Three levels must remain distinct:

- `hilbertParallelogramTerm` is the chosen formal triangulation of a parallelogram along  $AC$ ;
- `HilbertScissorsEq` is the stronger scissors-equivalence relation;
- `HilbertScissorsEquicomplementable` is the equal-content relation used in the public theorem.

I.34 first produces the stronger relation; I.37 produces the weaker content relation; I.41 combines them by addition and transport.

No new Book Zero theorem is introduced. The only new reusable declaration in the scissors layer is

```
equicomplementable_add
```

## 41.9 Source Audit

### 41.9.1 Euclid and Joyce

The classical source gives exactly the short proof reflected in the present architecture: join  $AC$ , apply I.37 to  $ABC$  and  $EBC$ , then use I.34 to say that the parallelogram is double the diagonal triangle. Joyce also notes that I.41 generalizes the diagonal statement I.34 and that an equal-base version would be slightly more general.

### 41.9.2 Hartshorne

Hartshorne’s Section 22 is the main semantic control for the area block. His classification explicitly places I.41–I.45 among the remaining Book I results that are valid for equal content. This is consistent with the present proof: I.41 never uses equal content as a hypothesis to recover metric or incidence information and therefore needs no new property (Z)-style strictness principle.

### 41.9.3 Hilbert, Borsuk–Szmielw, Greenberg, and Beeson

For I.41 these sources play their established control roles rather than supplying a new proposition-specific theorem. Hilbert remains the architectural source for the separation between congruence, equidecomposition, and equicomplementability. Borsuk–Szmielw and Greenberg control the developed incidence/congruence/parallel background. The Beeson corpus remains a check on formal side conditions, but no Tarski-style local construction is imported into the final Lean proof.

The current Lean source is authoritative for the exact formal claim. In particular, the documentation does not strengthen its conclusion to numerical area equality or to an unconditional geometric dissection statement.

## 41.10 Formalization Notes

### 41.10.1 Geometry

The public theorem exposes only two geometric inputs:

- the parallelogram structure of  $ABCD$ ;
- the strict parallel relation  $AE \parallel BC$ .

There are no explicit Between, SameRay, SameSide, OppositeSide, Pasch, or extension hypotheses in I.41. Any lower-level nondegeneracy needed by I.34 or I.37 is handled inside those earlier theorems.

The absence of an explicit  $A, D, E$  collinearity hypothesis is intentional. It is not needed by the actual proof DAG: I.34 consumes the parallelogram, and I.37 consumes  $AE \parallel BC$ .

### 41.10.2 Content

The content direction is purely forward:

$$ABC \approx EBC \implies ABC + ABC \approx EBC + EBC.$$

Accordingly there is no subtraction, cancellation, positivity, strict content order, or whole-greater-than-part step. The only new content fact is the closure of equicomplementability under formal addition.

### 41.10.3 Representation

The diagonal theorem returns the triangle orientation  $ABC \cong CDA$ , while the parallelogram term is written as  $ABC + ACD$ . Two applications of

```
scissors_triangle_cycle
```

identify the cyclic presentations. This is representation transport only; it contains no new geometric argument.

The final statement should therefore be read literally at the term level:

$$\text{parallelogram term} \approx \text{triangle term} + \text{triangle term}.$$

It is the current scissors representation of Euclid’s “double,” not a claim about a real-valued area function.

### 41.10.4 Axiomatic status

The public theorem assumes

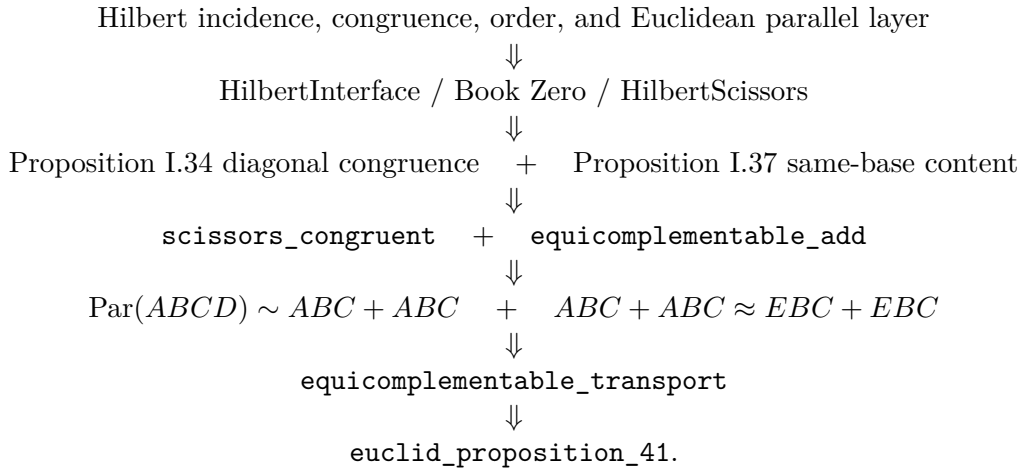
```
HilbertEuclideanPlane Geo
```

and is therefore Euclidean in the current hierarchy. I.41 introduces no new parallel construction and does not call parallel uniqueness directly; its Euclidean dependence is inherited through the already proved I.34/I.37 layer, most importantly the same-parallel content theorem I.37.

No continuity axiom is introduced. No new geometric or content axiom is added by Proposition I.41.

## 41.11 Dependency Summary

The actual formal dependency chain is



The historical and formal DAGs are unusually close:

$$\begin{array}{ll}
 \text{Euclid} & : \quad \text{I.37} + \text{I.34} \longrightarrow \text{I.41}, \\
 \text{Lean} & : \quad \text{I.37} + \text{I.34} + \text{scissors addition/transport} \longrightarrow \text{I.41}.
 \end{array}$$

The extra Lean layer is representational rather than geometric.

|                                            |                                                    |
|--------------------------------------------|----------------------------------------------------|
| New geometric axiom in I.41:               | no                                                 |
| New content axiom in I.41:                 | no                                                 |
| Inherited strict-content/positivity axiom: | no                                                 |
| New proposition-local helper:              | no                                                 |
| New reusable declaration:                  | yes — <code>equicomplementtable_add</code>         |
| Reusable refactor required:                | yes — helper moved to <code>HilbertScissors</code> |
| New Interface/Scissors theorem:            | yes                                                |
| New Book Zero theorem:                     | no                                                 |
| New numerical area or polygon semantics:   | no                                                 |
| Public theorem compiles:                   | yes                                                |
| Full <code>lake build</code> :             | clean                                              |
| sorry in <code>Proposition41.lean</code> : | no                                                 |

## 41.12 Conclusion

Proposition I.41 is a compact confirmation that the area architecture developed since I.35 is compositional. Its geometry is already contained in two earlier results: I.34 identifies the two diagonal halves of a parallelogram, and I.37 compares triangles on the same base and between

the same parallels. The new formal ingredient is not another geometric construction but the algebraic fact that equicomplementability is preserved by formal addition.

That observation led directly to the reusable theorem

```
equicomplementable_add
```

in `HilbertScissors.lean`. Once this theorem is available, the I.41 proof becomes almost the same DAG as Euclid's: replace the second diagonal triangle by the first, apply I.37, double the content relation by addition, and transport back to the parallelogram term.

The proposition therefore introduces no new axiom, no numerical area, no strict content order, and no new polygon semantics. The full project rebuild is clean, and the general addition theorem is already reusable beyond I.41.

## Chapter 42

# Proposition I.42

### 42.1 Introduction

Proposition I.42 is the first construction in Book I whose target is not merely a line, angle, triangle, or parallel relation, but a new figure with prescribed *content*. Given a triangle and a rectilineal angle, Euclid constructs a parallelogram equal to the triangle and having an angle equal to the prescribed one. In the traditional sequence this is the opening move of the “application of areas” block that continues through I.44 and I.45.

The present reconstruction preserves that synthetic character. There is no real-valued area function. Equality of content is represented by

```
HilbertScissorsEquicomplementable
```

and the constructed figure is represented by

```
hilbertParallelogramTerm
```

The final theorem therefore asserts the existence of points  $E, F, G$  such that  $FECG$  is a parallelogram, its angle  $FEC$  is congruent to the prescribed angle, and the triangle term  $ABC$  is equicomplementable with the parallelogram term  $FECG$ .

I.42 is also a useful test of the separation between three layers that has become essential since I.35:

- **geometry**: midpoint construction, angle layoff, side-of-line information, intersection of a ray carrier with a parallel, and completion of a parallelogram;
- **content**: equal-content comparison by I.38 and the “double” relation supplied by I.41;
- **representation**: carrier identification, ray orientation, and the use of a collinearity statement rather than an invalid strict parallel statement in the boundary case  $F = A$ .

The classical proof is short. The formal proof is longer because one configuration fact suppressed by the classical diagram becomes decisive: the copied angle ray must actually meet the parallel through  $A$ , and the meeting point is allowed to be  $A$  itself. The final Lean file proves this rather than assuming it.



## 42.2 The Classical Configuration

Let  $ABC$  be the given triangle and let the prescribed angle be represented formally by the noncollinear triple  $XYZ$ . Euclid bisects  $BC$  at  $E$ , so  $BE = EC$ . He then constructs at  $E$  an angle  $CEF$  equal to the prescribed angle, choosing the new ray on the side of the line  $BC$  that contains  $A$ . Through  $A$  he draws a line parallel to  $EC$ , meeting the constructed ray at  $F$ . Through  $C$  a line parallel to  $EF$  completes the parallelogram  $FECG$ .

The intended final configuration is therefore

$$B - E - C, \quad F, A, G \text{ on one upper carrier}, \quad FG \parallel EC,$$

with  $FECG$  a parallelogram and

$$\angle FEC \cong \angle XYZ.$$

The point  $A$  is not required to be strictly between  $F$  and  $G$ , and the formal construction must not exclude the legitimate case  $F = A$ .

Figure 42.1 separates the three geometric configurations that matter in the reconstruction.

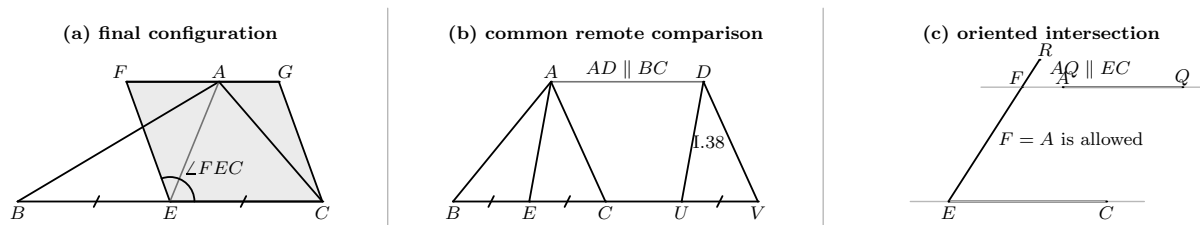


Figure 42.1: Three geometric stages of I.42. (a) The final configuration:  $E$  is the midpoint of  $BC$ , the upper carrier contains  $F, A, G$ , and  $FECG$  is the required parallelogram. (b) The formal proof that the median halves have equal content does not appeal to numerical altitude: both  $ABE$  and  $AEC$  are compared by I.38 with one remote triangle  $DUV$ , where  $BE \cong EC \cong UV$  and the upper carrier through  $A, D$  is parallel to the common base carrier. (c) The copied angle ray  $ER$  is proved to meet the parallel  $AQ$  at  $F$ ; SameSide information then selects the correct ray. The special case  $F = A$  is valid and is handled explicitly.

## 42.3 Classical Proof

Euclid's dependency graph is compact.

First bisect  $BC$  at  $E$  by I.10 and join  $AE$ . Since  $BE = EC$ , Proposition I.38 gives

$$\triangle ABE = \triangle AEC$$

in Euclid's equality-of-figures sense: the triangles stand on the equal bases  $BE$  and  $EC$  and between the same parallels. Hence

$$\triangle ABC = 2\triangle AEC.$$

Next construct at  $E$  the angle  $CEF$  equal to the given angle by I.23. Draw through  $A$  the line parallel to  $EC$  by I.31 and complete the parallelogram  $FECG$ . Proposition I.41 then gives

$$FECG = 2\triangle AEC.$$

Thus both the original triangle and the constructed parallelogram are double the same triangle, so Common Notion 1 yields

$$\triangle ABC = FECG.$$

The angle  $FEC$  has the prescribed value by construction.

The classical DAG can be displayed as

$$\begin{array}{c}
 E \text{ midpoint of } BC \quad (\text{I.10}) \\
 \Downarrow \\
 ABE = AEC \quad (\text{I.38}) \\
 \Downarrow \\
 ABC = 2AEC \\
 \angle FEC \cong \angle XYZ \quad (\text{I.23}), \quad AG \parallel EC \quad (\text{I.31}) \\
 \Downarrow \\
 FECG \text{ parallelogram} \\
 \Downarrow \\
 FECG = 2AEC \quad (\text{I.41}) \\
 \Downarrow \\
 ABC = FECG.
 \end{array}$$

The historical proof is therefore a forward construction. Equal content is never assumed in order to recover a metric or incidence conclusion. This will matter for the axiomatic status below.

## 42.4 Application of Areas and the Hidden Intersection

I.42 marks a change in the use of the area block. Propositions I.35–I.41 mainly compare already given figures. I.42 uses those comparison theorems as an engine for *constructing* a new figure with prescribed content and a prescribed angle. This is the first of the Book I “application of areas” constructions that will feed I.44 and I.45.

There is, however, a geometric step hidden by the ordinary diagram. After constructing the ray at  $E$  and the line through  $A$  parallel to  $EC$ , Euclid names their intersection  $F$  without separately proving that the two carriers meet. Wilkins makes the intended orientation explicit by directing the ray into the side of  $BC$  containing  $A$ . The Beeson proof trace goes further: it explicitly stops to prove that the angle ray meets the upper parallel. Its case analysis also contains the equality branch in which the intersection is  $A$  itself.

That observation changes the correct formal API. An earlier draft attempted to encode the upper-carrier condition as

```
Geo.Parallel F A E C
```

This is too strong because the project’s parallel relation is strict. If  $F = A$ , the expression would ask the degenerate carrier  $AA$  to be parallel to  $EC$ . The final theorem instead returns

```
Collinear Geo F A G
```

for the completed parallelogram  $FECG$ . This says exactly what the geometry needs:  $A$  lies on the upper carrier  $FG$ , including the boundary case  $F = A$ .

This is not a cosmetic change of representation. It is an example of a hidden configuration condition changing the mathematically correct formal statement of an intermediate construction.

## 42.5 Proof Architecture of the Reconstruction

The final Lean proof is organized into seven conceptual stages. The local helpers are proposition-specific because they package the exact geometry of I.42 rather than a new general synthetic theory.

### 42.5.1 Step 1: midpoint and equal-content halves

The public theorem first obtains a strict midpoint  $E$  of  $BC$ . The helper

```
i42_median_bisects_area
```

proves

$$ABE \approx AEC,$$

where  $\approx$  denotes `HilbertScissorsEquicomplementable`.

The proof is deliberately not a numerical same-altitude argument. It creates one remote comparison triangle  $DUV$ . A line through  $A$  and  $D$  is parallel to the base carrier. The base is extended so that the ordered points contain  $B - E - C - U - V$ , and a segment  $UV$  is laid off with

$$UV \cong BE \cong EC.$$

Proposition I.38 is then applied twice:

$$ABE \approx DUV, \quad AEC \approx DUV.$$

Transitivity gives the desired midpoint-halves relation.

This reduction is a consequence of the current I.38 API: I.38 is expressed using two separated equal bases, so a common remote triangle supplies a single comparison target for both halves.

### 42.5.2 Step 2: copy the angle on the correct side and draw the parallel

The helper

```
i42_angle_parallel_setup
```

constructs the carrier  $EC$ , uses Hilbert's angle-construction axiom/theorem to copy  $XYZ$  at  $E$ , and selects the side containing  $A$ . It returns a point  $R$  satisfying

$$R \text{ and } A \text{ are on the same side of } EC, \quad \angle CER \cong \angle XYZ.$$

It also constructs a point  $Q$  such that

$$EC \parallel AQ.$$

Thus the two objects that must meet – the ray carrier  $ER$  and the upper parallel carrier  $AQ$  – are both explicit Lean data.

### 42.5.3 Step 3: prove that the two carriers meet

The helper

```
i42_angle_parallel_intersection
```

is the central Euclidean existence argument of the proposition. Suppose the lines  $ER$  and  $AQ$  were disjoint. The bridge

```
intersection_test_parallel_of_lines_disjoint
```

would convert this incidence-level disjointness into strict parallelism

$$ER \parallel AQ.$$

But  $EC \parallel AQ$  as well. If the carriers  $ER$  and  $EC$  were distinct, Euclidean transitivity of parallels would imply

$$ER \parallel EC,$$

which is impossible because both contain  $E$ . If the two carriers were the same, then  $R$  would lie on the base  $EC$ , contradicting the SameSide output of angle construction. Therefore  $ER$  and  $AQ$  meet, and their intersection is named  $F$ .

This is exactly the step that the classical text suppresses.

### 42.5.4 Step 4: orient the intersection onto the correct ray

An intersection of the two *lines* is not yet enough. I.42 needs  $F$  on the chosen ray from  $E$ , not on the opposite ray. The helper

```
i42_angle_parallel_oriented_intersection
```

proves this using side-of-line data.

First,  $A$  and  $F$  lie on the same side of the base. If  $F = A$ , this is simply SameSide reflexivity. Otherwise the carrier  $AF$  is the carrier  $AQ$ , hence is parallel to  $EC$ , and its endpoints lie on the same side of  $EC$ . Since  $R$  and  $A$  were already constructed on the same side, transitivity gives that  $R$  and  $F$  are on the same side of  $EC$ .

Now  $E, R, F$  are collinear. They cannot satisfy  $R - E - F$ , because then the base line through  $E$  would separate  $R$  and  $F$ . Hence

```
HilbertSameRay Geo E R F
```

and the angle can be transported from the auxiliary ray  $ER$  to the actual intersection ray  $EF$ . The result is

$$\angle FEC \cong \angle XYZ.$$

### 42.5.5 Step 5: complete the parallelogram and identify its upper carrier

The helper

```
i42_construct_parallelogram
```

first proves that  $F$  is off the base  $EC$ : it lies on  $AQ$ , and  $AQ$  is parallel to  $EC$ . Thus  $E, F, C$  are noncollinear. The reusable theorem

```
hilbert_parallelogram_fourth_vertex_exists
```

then completes  $EFC$  to a parallelogram  $EFGC$ , which is reversed to the orientation  $FE CG$  used by I.42.

The upper side  $GF$  and the constructed carrier  $AQ$  are both parallel to  $EC$  and share  $F$ . Euclidean parallel uniqueness/transitivity identifies their carriers. Therefore  $G$  also lies on  $AQ$ , and the helper returns

```
Collinear Geo F A G
```

rather than the invalid strict relation  $FA \parallel EC$ .

### 42.5.6 Step 6: use I.41 on a genuine side and transport along the carrier

The carrier formulation requires a small wrapper around I.41:

```
i42_parallelogram_double_triangle_on_carrier
```

The current public I.41 expects a strict parallel pair. Instead of using the possibly degenerate pair  $FA$ , the wrapper uses the genuine parallelogram side  $FG$  and applies I.41 with apex  $G$ :

$$\text{Par}(FE CG) \approx GEC + GEC.$$

It then compares  $GEC$  with  $AEC$ . If  $G = A$ , this is reflexivity. Otherwise  $G, A, F$  are collinear, so the carrier  $GA$  is the same upper carrier as  $GF$ ; therefore  $GA \parallel EC$ , and I.37 gives

$$GEC \approx AEC.$$

Using `equicomplementable_add` twice yields

$$GEC + GEC \approx AEC + AEC,$$

and transitivity gives

$$\text{Par}(FE CG) \approx AEC + AEC.$$

### 42.5.7 Step 7: assemble the content equality

The top-level theorem uses the scissors splitting identity

$$ABC \sim ABE + AEC,$$

where  $\sim$  denotes `HilbertScissorsEq`. From Step 1 and reflexivity, `equicomplementable_add` gives

$$ABE + AEC \approx AEC + AEC.$$

Transport across the split yields

$$ABC \approx AEC + AEC.$$

Step 6 gives the same doubled triangle as the content of the constructed parallelogram. A final symmetry and transitivity step closes

$$ABC \approx \text{Par}(FECG).$$

The full formal DAG is therefore

$$\begin{array}{c}
 E \text{ midpoint of } BC \\
 \Downarrow \\
 \text{I.38 twice via } DUV \\
 \Downarrow \\
 ABE \approx AEC \\
 \Downarrow \\
 ABC \approx AEC + AEC \\
 \text{angle construction} + EC \parallel AQ \\
 \Downarrow \\
 ER \cap AQ = \{F\} \quad \text{and} \quad E, R, F \text{ same ray} \\
 \Downarrow \\
 FECG \text{ parallelogram, } F, A, G \text{ collinear} \\
 \Downarrow \\
 \text{I.41 at } G \quad + \quad \text{I.37 along the upper carrier} \\
 \Downarrow \\
 \text{Par}(FECG) \approx AEC + AEC \\
 \Downarrow \\
 ABC \approx \text{Par}(FECG).
 \end{array}$$

## 42.6 Lean Formalization

The current file imports the two earlier content propositions used directly by the reconstruction:

```
import CGJteamLab.Proposition38
import CGJteamLab.Proposition41
```

The public theorem is the following. The snippet uses the project's preferred ASCII notation for documentation while preserving the exact theorem data.

```
theorem euclid_proposition_42
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C X Y Z : Geo.Point)
  (hABC : Not (Collinear Geo A B C))
  (hXYZ : Not (Collinear Geo X Y Z)) :
  exists E F G : Geo.Point,
    IsParallelogram Geo F E C G /\
```

```

Geo.AngleCongruent F E C X Y Z /\
HilbertScissorsEquicomplementable Geo
  (hilbertScissorsTriangle Geo A B C)
  (hilbertParallelogramTerm Geo F E C G) := by
-- proof in Proposition42.lean

```

The six mathematically significant proposition-local helpers are:

```

i42_median_bisects_area
i42_angle_parallel_setup
i42_angle_parallel_intersection
i42_angle_parallel_oriented_intersection
i42_construct_parallelogram
i42_parallelogram_double_triangle_on_carrier

```

They are not six unrelated technical conveniences. They compress the proof into the six geometric/content transitions described above.

### 42.6.1 The midpoint-content helper

Its public shape is

```

theorem i42_median_bisects_area
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C E : Geo.Point)
  (hABC : Not (Collinear Geo A B C))
  (hE : HilbertIsMidpoint Geo E B C) :
  HilbertScissorsEquicomplementable Geo
    (hilbertScissorsTriangle Geo A B E)
    (hilbertScissorsTriangle Geo A E C) := by
-- I.38 twice through one remote triangle DUV

```

The helper is proposition-local because its remote-triangle construction is a way of adapting the current I.38 API to the midpoint geometry of I.42. A future general same-altitude theorem might subsume it, but the present file does not introduce such a new interface theorem.

### 42.6.2 The construction helper

The decisive output is

```

theorem i42_construct_parallelogram
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A E C X Y Z : Geo.Point)
  (hAEC : Not (Collinear Geo A E C))
  (hXYZ : Not (Collinear Geo X Y Z)) :
  exists F G : Geo.Point,
    IsParallelogram Geo F E C G /\
    Geo.AngleCongruent F E C X Y Z /\

```

```
Collinear Geo F A G := by
-- angle ray, intersection, orientation, and fourth vertex
```

The third conjunct is the important final API decision. It records the common upper carrier without imposing strict distinctness on  $F$  and  $A$ .

### 42.6.3 The carrier form of I.41

The last local helper has the shape

```
theorem i42_parallelagram_double_triangle_on_carrier
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (F E C G A : Geo.Point)
  (hParallelogram : IsParallelogram Geo F E C G)
  (hFAG : Collinear Geo F A G) :
  HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo F E C G)
    (hilbertScissorsTriangle Geo A E C +
      hilbertScissorsTriangle Geo A E C) := by
-- I.41 at G, then I.37 from GEC to AEC
```

This wrapper is not a new area principle. It is a geometry/API adapter that keeps the strictness of I.41 on the genuine side  $FG$  and uses I.37 to move the triangle apex along the same upper carrier.

## 42.7 Relationship with Earlier Propositions

### 42.7.1 Proposition I.10

Classically, I.10 supplies the midpoint  $E$  of  $BC$ . The Lean theorem does not call I.10 by proposition name. It uses the developed Hilbert midpoint existence theorem directly. This is a typical Hilbert reorganization: the construction is reusable foundational infrastructure rather than a historical Book I dependency.

### 42.7.2 Propositions I.23 and I.31

Euclid uses I.23 to reproduce the given angle and I.31 to draw the parallel through  $A$ . The reconstruction again consumes the corresponding Hilbert operations directly:

```
HilbertCongruence.angle_construction
hilbert_parallel_through_point_exists
```

The formal proof then adds information absent from the classical invocation: the chosen side of the base, the line carriers, and the proof that the two constructed carriers meet.



### 42.7.3 Proposition I.37

I.37 enters only after the parallelogram has already been constructed. The carrier wrapper for I.41 first obtains the double of  $GEC$ , then uses the fact that  $G$  and  $A$  lie on the same upper carrier to obtain

$$GEC \approx AEC.$$

This is a forward use of same-base equal content.

### 42.7.4 Proposition I.38

I.38 supplies the equal-content halves cut by the median. The current I.42 proof applies I.38 twice through the auxiliary triangle  $DUV$  rather than restating a new theorem about equal altitudes. This is the main content input before the final application of I.41.

### 42.7.5 Proposition I.41

I.41 states that a parallelogram on the same base and in the same parallels is double the corresponding triangle. In I.42 the strict public API is applied with  $G$  as the apex because  $FG$  is a genuine side of the parallelogram. I.37 then transports from  $GEC$  to  $AEC$ . This avoids any need to assume  $F \neq A$ .

### 42.7.6 Propositions I.44 and I.45

Historically, I.42 is immediately reusable: I.44 starts by constructing a parallelogram equal to a given triangle in a prescribed angle, and I.45 builds a parallelogram equal to a general rectilinear figure by repeated application of the triangle construction. The current project follows the same broad dependency direction: Proposition I.45 imports Proposition I.42.

Cleaning the final I.42 file also exposed one dependency-hygiene issue in I.45. I.45 had been using an old proposition-local helper for adding an unchanged scissors term to an equicomplementability relation. That call was replaced by the already reusable

```
equicomplementable_add
```

from the scissors layer. No helper had to be restored to I.42, and the full project build remained clean.

## 42.8 Hilbert and Book Zero Components Used

### 42.8.1 Midpoint, betweenness, and segment construction

The midpoint data are strict Hilbert data:  $B - E - C$  together with  $BE \cong EC$ . The proof uses betweenness incidence facts, base extensions, segment construction on a ray, and transitivity/symmetry of segment congruence to build the auxiliary equal base  $UV$ .

These operations belong to the existing Hilbert/Book Zero language. I.42 introduces no new midpoint or segment-arithmetic theorem.

### 42.8.2 Angle construction, SameSide, and SameRay

Hilbert angle construction is stronger as an API than the classical phrase “construct an equal angle”: it also records the chosen side of a carrier. That SameSide witness becomes essential later. It is combined with the parallel carrier  $AQ$  to prove that the line intersection  $F$  is on the correct ray, expressed by

```
HilbertSameRay Geo E R F
```

The angle is then transported along that ray.

This is a good example of hidden order data becoming a first-class part of the formal proof rather than a diagrammatic convention.

### 42.8.3 Euclidean parallel and intersection infrastructure

The central intersection proof uses the established bridge from disjoint incidence lines to strict point-line parallelism, together with Euclidean transitivity of distinct parallels and the fact that parallel carriers cannot share a point. Carrier identification later uses the same Euclidean uniqueness principle.

Thus the parallel axiom enters I.42 not merely because a parallel line is drawn, but because the proof must control uniqueness and intersection of the constructed carriers.

### 42.8.4 Parallelogram infrastructure

Once  $E, F, C$  are shown noncollinear, the theorem

```
hilbert_parallelogram_fourth_vertex_exists
```

constructs the fourth vertex. The representation is normalized by

```
ParallelogramReverse
```

and the upper side is identified with the already constructed carrier through  $A$ . No new parallelogram axiom is introduced.

### 42.8.5 Scissors content infrastructure

The content layer uses only previously established operations:

```
equicomplementable_add
equicomplementable_refl
equicomplementable_symm
equicomplementable_trans
equicomplementable_transport
```

The triangle split itself is a `HilbertScissorsEq` identity. At no point is a real-valued area, scalar multiplication, subtraction, or content order introduced.

## 42.9 Source Audit

### 42.9.1 Euclid, Heath, and Joyce

The classical text gives the exact short DAG used as the historical reference: bisect  $BC$ , construct the prescribed angle at  $E$ , draw the parallel through  $A$ , complete the parallelogram, use I.38 for the two half-triangles, and use I.41 for the parallelogram. Joyce also identifies I.42 as the beginning of the application-of-areas phase of Book I.

This source controls the statement and the classical dependency graph. It does not by itself resolve the hidden existence proof for the intersection point  $F$ .

### 42.9.2 Wilkins

Wilkins makes one orientation choice explicit that is important for the formal reconstruction: the newly constructed ray from  $E$  is directed into the side of the line  $BC$  containing  $A$ . His account then describes the parallel through  $A$  as meeting that ray at  $F$  and completes the parallelogram. This is the correct classical configuration control for the `SameSide/SameRay` argument in Lean.

### 42.9.3 Beeson

The Beeson formal proof corpus is especially useful as a gap audit here. Its I.42 trace explicitly interrupts the Euclidean construction to prove that the angle ray meets the upper parallel, noting that this meeting is omitted in Euclid's prose. More importantly, the trace includes a branch in which the meeting point is  $A$  itself.

The final Lean architecture is not copied from the Tarski-style Beeson proof, but these observations directly validate two design decisions: intersection must be proved, and the intermediate API must allow  $F = A$ .

### 42.9.4 Hilbert

Hilbert's Chapter IV remains the semantic control for the content block. The distinction between congruent-piece equivalence and equicomplementability is maintained throughout the project. In I.42 the final proof does not call Hilbert's Theorems 45–46 directly; instead their content architecture is already embodied in the previously proved I.37–I.41 layer.

For the geometric construction, the Hilbert organization is even more direct: angle layoff, side-of-line data, line incidence, and the Euclidean uniqueness of parallels are reusable operations rather than historical Book I propositions.

### 42.9.5 Hartshorne

Hartshorne's Section 22 is the main semantic check on the forward content argument. He distinguishes the results that require the extra strictness property ( $Z$ ) from those that remain

valid for equal content. The remaining Book I results I.41–I.45 are in the latter class.

That classification matches the actual Lean DAG. I.42 never starts from an equal-content hypothesis and concludes a metric or incidence fact. It only constructs geometry and then combines forward equal-content statements. Accordingly no cancellation, positivity, whole-greater-than-part principle, or property (Z) is needed.

Hartshorne also places I.42–I.45 in the “algebra of areas” or application of areas: synthetic content is manipulated constructively without assigning a numerical area value to each figure.

### 42.9.6 Borsuk–Szmielew and Greenberg

Borsuk–Szmielew are strongest here as a control source for the underlying synthetic API. Their organization emphasizes angle layoff on a chosen side, midpoint data with positional information, and – after the Euclidean axiom – parallel uniqueness, parallel transitivity, and parallelogram theory. These are precisely the kinds of components used by the geometric half of I.42.

They do not supply the equal-content semantics of the I.35–I.45 block. As in the preceding documentation, that role remains with Hilbert and Hartshorne. Greenberg likewise serves as a control on the Euclidean parallel boundary and on the distinction between synthetic content and numerical area rather than as the source of an I.42-specific proof.

## 42.10 Formalization Notes

### 42.10.1 Geometry

The formal geometry contains substantially more data than the classical picture displays:

- strict noncollinearity for the input triangle and prescribed angle;
- strict midpoint betweenness  $B - E - C$ ;
- a concrete carrier line for  $EC$ ;
- the fact that the copied angle point  $R$  and  $A$  lie on the same side of that carrier;
- a concrete carrier  $AQ$  parallel to  $EC$ ;
- a proof that the ray carrier  $ER$  meets  $AQ$ ;
- a proof that the resulting intersection  $F$  lies on the selected ray;
- noncollinearity of  $E, F, C$  before the fourth vertex can be constructed;
- identification of the completed upper side  $FG$  with the carrier through  $A$ .

No Pasch step is introduced locally in the top-level theorem, but the proof uses the developed side-of-line and intersection theory whose lower layers are built from Hilbert order/separation infrastructure.

### 42.10.2 Content

The content argument is entirely forward:

$$ABE \approx AEC \implies ABC \approx AEC + AEC,$$

and

$$\text{Par}(FECG) \approx GEC + GEC \implies \text{Par}(FECG) \approx AEC + AEC.$$

Transitivity then compares the original triangle with the constructed parallelogram.

There is no inverse “halving” theorem. There is no cancellation of a common summand and no deduction from equal content to segment or angle congruence. Thus I.42 stays on the safe forward side of the content semantics established since I.35.

### 42.10.3 Representation

Three representational choices deserve explicit mention.

First, the angle-construction helper initially returns the angle in the order  $CER$  and the final theorem wants  $FEC$ . SameRay transport and the symmetry of angle representation normalize this without changing the geometry.

Second, the parallelogram fourth-vertex theorem naturally returns  $EFGC$ , while the Euclid statement is expressed as  $FECG$ . The theorem `ParallelogramReverse` performs this normalization.

Third, and most importantly, the upper-carrier datum is

```
Collinear Geo F A G
```

rather than a strict parallel using the pair  $FA$ . This is the correct representation because the pair may degenerate when  $F = A$  even though the upper carrier itself is perfectly nondegenerate through  $F$  and  $G$ .

### 42.10.4 Axiomatic status

The public theorem assumes

```
HilbertEuclideanPlane Geo
```

and is therefore Euclidean in the current hierarchy.

The Euclidean parallel axiom is used essentially in the developed theory of parallel uniqueness/transitivity. The most visible direct use in I.42 is the proof that the angle-ray carrier and the parallel through  $A$  cannot remain disjoint: two distinct carriers parallel to the same carrier are forced into the Euclidean parallel relation, which contradicts their shared point when appropriate. The same uniqueness theory identifies the final upper carrier.

Angle construction and segment construction are congruence/order operations; they do not require a continuity axiom in this proof. No circle intersection or completeness principle is introduced. Proposition I.42 adds no new geometric axiom and no new content axiom.

## 42.11 Dependency Summary

The actual formal dependency chain is

```

Hilbert incidence, order, congruence, and Euclidean parallel axioms
  ↓
HilbertInterface / Book Zero line, ray, side-of-line, midpoint theory
  ↓
angle construction + parallel-through-point + intersection theory
  ↓
Proposition I.38 twice
  ↓
i42_median_bisects_area
  ↓
 $ABC \approx AEC + AEC$ 
  ↓
i42_angle_parallel_setup
  ↓
i42_angle_parallel_intersection
  ↓
i42_angle_parallel_oriented_intersection
  ↓
i42_construct_parallelogram
  ↓
I.41 + I.37 + scissors addition
  ↓
i42_parallelogram_double_triangle_on_carrier
  ↓
euclid_proposition_42.

```

The historical and formal DAGs have the same content skeleton but different geometric packaging:

Euclid : I.10 → I.38 → I.23/I.31 → I.41 → I.42,

Lean : Hilbert midpoint → I.38 twice → angle/parallel intersection → I.41+I.37 → I.42.

|                                          |           |
|------------------------------------------|-----------|
| New geometric axiom in I.42:             | no        |
| New content axiom in I.42:               | no        |
| New proposition-local helpers:           | yes — six |
| New reusable Interface/Scissors theorem: | no        |
| New Book Zero theorem:                   | no        |
| Property (Z) / content cancellation:     | no        |
| New numerical area semantics:            | no        |
| Continuity axiom introduced:             | no        |
| Boundary case $F = A$ supported:         | yes       |
| Public theorem compiles:                 | yes       |
| Full lake build:                         | clean     |
| sorry in Proposition42.lean:             | no        |

## 42.12 Conclusion

Proposition I.42 is a construction theorem built on the content theory of I.35–I.41, but the main formal difficulty is geometric rather than algebraic. The equal-content calculation itself is simple

once the right configuration has been produced: the midpoint divides the original triangle into two equal content pieces, I.41 makes the new parallelogram double a triangle on its base, and I.37 moves that triangle along the common upper carrier.

What the formal reconstruction adds is the configuration data that the classical diagram suppresses. The copied angle ray is placed on the correct side, its carrier is proved to meet the upper parallel, the intersection is proved to lie on the correct ray, and the completed upper side is identified with the carrier through  $A$ . The boundary case  $F = A$  is not discarded; it is the reason the final construction returns collinearity on the upper carrier instead of a strict parallel relation involving  $FA$ .

The proposition therefore illustrates the mature architecture of the Book I reconstruction. Geometry constructs and orients the carriers; the scissors layer carries equal content; representation lemmas normalize the same figure without changing its meaning. No new axiom, numerical area, cancellation principle, or continuity assumption is needed. The final full project build is clean, and I.42 is available as the construction input required by the later application-of-areas propositions.

## Chapter 43

# Proposition I.43

### 43.1 Introduction

Proposition I.43 returns from the construction problem of I.42 to a purely relational statement about an already given parallelogram. A diagonal of the large parallelogram is divided at a point  $K$ . Two smaller parallelograms are placed “about the diameter”, one on each part of the diagonal. The two remaining quadrilateral regions are the classical *complements*, and Euclid proves that they are equal.

The formal reconstruction keeps the argument entirely synthetic. It does not introduce numerical area, scalar multiplication, a content order, or a cancellation axiom. Instead, it turns Euclid’s Common Notion 3 step into a direct witness for the already defined relation `HilbertScissorsEquicomplementable`.

This makes I.43 a particularly clean test of the scissors language introduced at I.35. The proof is short enough that the historical and formal dependency graphs remain visibly related, but the formal representation makes explicit which triangular pieces are removed and which remain.

### 43.2 The Classical Configuration

Let  $ABCD$  be a parallelogram with diagonal  $AC$ . Let  $K$  lie strictly on the diagonal, and choose points

$$A - G - B, \quad A - E - D, \quad B - F - C, \quad D - H - C,$$

so that  $AGKE$  and  $KFCH$  are parallelograms. The two quadrilateral regions left over are

$$GBFK \quad \text{and} \quad EKHD.$$

These are the two complements in the statement of I.43.

The Lean theorem records the strict order data explicitly:

$$A - K - C, \quad A - G - B, \quad A - E - D, \quad B - F - C, \quad D - H - C.$$

The two small parallelograms are also hypotheses rather than constructions. Thus I.43 is not a construction theorem in the current development; it is a content theorem for a given configuration.

Figure 43.1 shows exactly this formal configuration. The dashed segments  $BK$  and  $KD$  are not extra hypotheses. They are the diagonals used to represent the two complement regions as sums of two triangles.



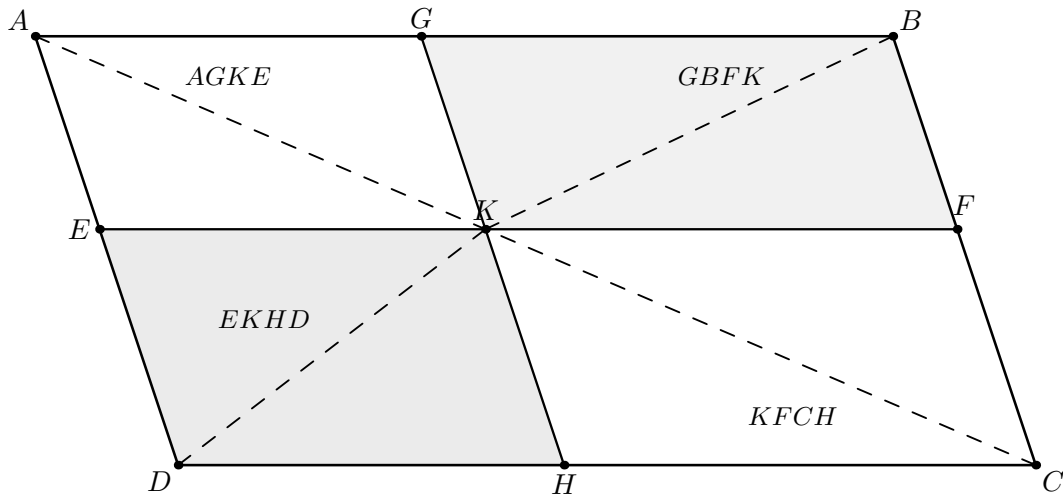
triangulation diagonals  $BK$ ,  $KD$ 

Figure 43.1: The configuration of Proposition I.43. The large parallelogram is  $ABCD$ , with  $A - K - C$ . The parallelograms  $AGKE$  and  $KFCH$  lie about the two parts of the diagonal. The remaining regions are the complements  $GBFK$  and  $EKHD$ . In the Lean representation the first complement is triangulated by  $BK$  as  $GBK + KBF$ , while the second is triangulated by  $KD$  as  $EDK + KDH$ .

### 43.3 Classical Proof

Euclid's proof uses Proposition I.34 three times. Since the diagonal of a parallelogram bisects it into equal triangles,

$$\triangle ABC = \triangle ACD.$$

The same theorem applied to the two smaller parallelograms gives

$$\triangle AGK = \triangle AEK, \quad \triangle KFC = \triangle KHC.$$

Equivalently, in Euclid's usual lettering, these are the two equal pairs of triangles belonging to the parallelograms about the diameter.

Common Notion 2 combines the two small equalities. The corresponding pairs of corner triangles therefore have equal total content. Since the two large diagonal triangles are equal, Common Notion 3 removes equal content from equal content and leaves the two complements equal.

The classical dependency graph is therefore

$$\begin{array}{c}
 ABCD, AGKE, KFCH \text{ parallelograms} \\
 \Downarrow \text{I.34 three times} \\
 ABC = ACD, \quad AGK = AEK, \quad KFC = KHC \\
 \Downarrow \text{C.N.2} \\
 AGK + KFC = AEK + KHC \\
 \Downarrow \text{C.N.3} \\
 GBFK = EKHD.
 \end{array}$$

The historical proof is therefore a forward content argument. No equal-content hypothesis is used to recover a metric, incidence, or parallelism conclusion. This places I.43 on the opposite side of the logical boundary exposed by I.39–I.40.

## 43.4 Complements and Equicomplementability

The central formal point is that Euclid’s phrase “subtract equals from equals” need not be encoded as a general cancellation rule

$$P + R \sim Q + R \implies P \sim Q.$$

No such theorem is required here.

The existing scissors relation defines formal equicomplementability by the existence of two equidecomposable complements  $R, S$  such that

$$R \sim S, \quad P + R \sim Q + S.$$

Thus I.43 can be proved by exhibiting the pieces removed by Euclid as the witnesses  $R$  and  $S$ .

For the two complement terms put

$$\begin{aligned} P &= GBK + KBF, \\ Q &= EDK + KDH, \end{aligned}$$

and choose

$$\begin{aligned} R &= AGK + KFC, \\ S &= AEK + KHC. \end{aligned}$$

Proposition I.34 gives  $R \sim S$ . The two full-triangle decompositions and the large diagonal equality give

$$P + R \sim Q + S.$$

These two facts are already exactly the data required by `HilbertScissorsEquicomplementable`.

This is the key difference between an abstract cancellation theorem and the present proof. The formalization does not infer equality after deleting an unrepresented common summand. It records the deleted pieces explicitly as part of the certificate.

## 43.5 Proof Architecture of the Reconstruction

The working theorem `i43_raw_complements` has five mathematical stages.

### 43.5.1 Step 1: decompose the first large diagonal triangle

Using  $A - K - C$ , the triangle  $ABC$  is split at  $K$ . The resulting  $AK$ -piece is split at  $G$  using  $A - G - B$ , and the  $KC$ -piece is split at  $F$  using  $B - F - C$ . The result is

$$ABC \sim (AGK + GBK) + (KBF + KFC).$$

In the Lean file this is the local identity `hT1`.

### 43.5.2 Step 2: decompose the second large diagonal triangle

The mirror construction uses  $A - E - D$  and  $D - H - C$ . After orienting the first betweenness relation as  $D - E - A$ , the triangle  $ACD$  is decomposed as

$$ACD \sim (EDK + AEK) + (KDH + KHC).$$

This is `hT2`.

### 43.5.3 Step 3: obtain the three I.34 equalities

The big parallelogram gives

$$ABC \sim ACD.$$

The two parallelograms about the diagonal give

$$AGK \sim AEK, \quad KFC \sim KHC.$$

These are the local facts `hT3`, `hT4`, and `hT5`. The implementation obtains all three through the already proved theorem `euclid_proposition_34_diagonal` and the congruent-triangle generator of the scissors calculus.

### 43.5.4 Step 4: assemble and regroup the full decompositions

Transitivity gives one four-piece relation

$$\begin{aligned} & (AGK + GBK) + (KBF + KFC) \\ & \sim (EDK + AEK) + (KDH + KHC). \end{aligned}$$

The code then uses only associativity and commutativity of multiset addition to regroup this as

$$(GBK + KBF) + (AGK + KFC) \sim (EDK + KDH) + (AEK + KHC).$$

No geometric theorem is hidden in this regrouping.

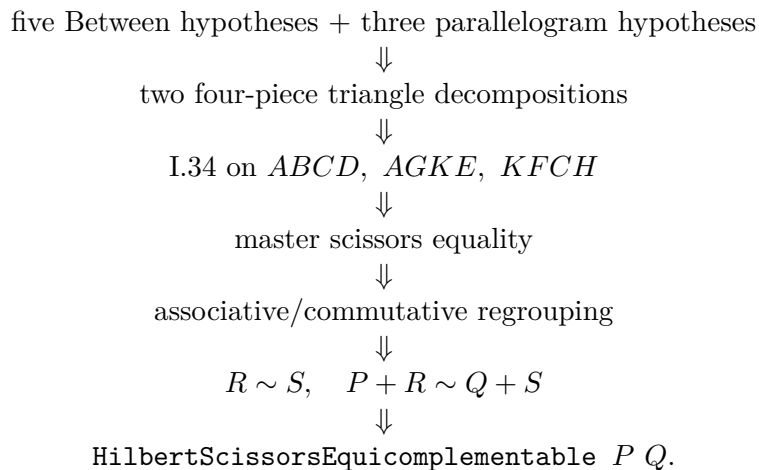
### 43.5.5 Step 5: package the equicomplementability witness

Finally `hT4` and `hT5` are combined with `HilbertScissorsEq.add`, giving

$$AGK + KFC \sim AEK + KHC.$$

Together with the regrouped master relation, these are exactly the two fields required by the definition of equicomplementability.

The complete Lean DAG is therefore



## 43.6 Lean Formalization

The file imports only the already established diagonal theorem for parallelograms and the scissors calculus:

```
import CGJteamLab.Proposition34
import CGJteamLab.HilbertScissors
```

The substantial proof is isolated in the proposition-local helper `i43_raw_complements`. Its public shape is the following.

```
theorem i43_raw_complements
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D K G E F H : Geo.Point)
  (hParallelogram : IsParallelogram Geo A B C D)
  (hAKC : Geo.Between A K C)
  (hAGB : Geo.Between A G B)
  (hAED : Geo.Between A E D)
  (hBFC : Geo.Between B F C)
  (hDHC : Geo.Between D H C)
  (hPar1 : IsParallelogram Geo A G K E)
  (hPar2 : IsParallelogram Geo K F C H) :
  HilbertScissorsEquicomplementable Geo
    (hilbertScissorsTriangle Geo G B K +
     hilbertScissorsTriangle Geo K B F)
    (hilbertScissorsTriangle Geo E D K +
     hilbertScissorsTriangle Geo K D H) := by
  -- proof in Proposition43.lean
```

The final Book I endpoint is deliberately thin:

```
theorem euclid_proposition_43
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D K G E F H : Geo.Point)
  (hParallelogram : IsParallelogram Geo A B C D)
  (hAKC : Geo.Between A K C)
  (hAGB : Geo.Between A G B)
  (hAED : Geo.Between A E D)
  (hBFC : Geo.Between B F C)
  (hDHC : Geo.Between D H C)
  (hPar1 : IsParallelogram Geo A G K E)
  (hPar2 : IsParallelogram Geo K F C H) :
  HilbertScissorsEquicomplementable Geo
    (hilbertScissorsTriangle Geo G B K +
     hilbertScissorsTriangle Geo K B F)
    (hilbertScissorsTriangle Geo E D K +
     hilbertScissorsTriangle Geo K D H) := by
exact
  i43_raw_complements
    Geo
    A B C D K G E F H
    hParallelogram
```

```

hAKC
hAGB
hAED
hBFC
hDHC
hPar1
hPar2

```

The final certificate returned by the raw helper uses

```

hilbertScissorsTriangle Geo A G K +
hilbertScissorsTriangle Geo K F C

```

and

```

hilbertScissorsTriangle Geo A E K +
hilbertScissorsTriangle Geo K H C

```

as the two equidecomposable complements.

## 43.7 Relationship with Earlier Propositions

### 43.7.1 Proposition I.34

I.34 is the only earlier Book I proposition used directly by I.43. It enters three times through `euclid_proposition_34_diagonal`: once for the large parallelogram and once for each parallelogram about the diagonal. The current proof does not replay the parallel-angle or congruence argument behind I.34; it consumes the already established diagonal triangle congruence.

### 43.7.2 Propositions I.35–I.41

These propositions established and exercised the synthetic content language, but I.43 does not call them by proposition name. Their architectural legacy is **HilbertScissors**: finite triangle sums, splitting, congruent replacement, addition, transitivity, and equicomplementability.

In particular, I.43 is not a new converse theorem of the I.39 type. It needs no proper-part contradiction and no positivity principle.

### 43.7.3 Proposition I.42

I.42 is a construction theorem: it creates a parallelogram with prescribed content and angle. I.43 is complementary in character. The entire geometric configuration is supplied as input, and the theorem proves a relation between the two remaining regions.

### 43.7.4 Proposition I.44

The next proposition uses I.43 immediately. Historically, I.44 first creates a parallelogram of the required content and angle and then uses the complement theorem to transfer that content to a parallelogram applied to the prescribed segment. Thus I.43 is not an isolated scissors identity; it is the bridge from I.42 to the application-of-areas construction in I.44.

## 43.8 Hilbert and Book Zero Components Used

### 43.8.1 Betweenness and incidence

The five strict betweenness hypotheses are the hidden configuration data needed for the formal triangle splits. They guarantee that  $K, G, E, F, H$  lie on the intended segments, not merely on the corresponding carrier lines. Apart from reversing  $A - E - D$  to  $D - E - A$  for one split, no new order theorem is introduced.

### 43.8.2 Parallelogram diagonal congruence

The metric input is entirely packaged by I.34. The current proof therefore uses the developed Euclidean parallelogram layer rather than reconstructing the three congruence proofs locally.

### 43.8.3 Scissors content infrastructure

The proof uses only existing operations:

```
HilbertScissorsEq.split
HilbertScissorsEq.congruent
HilbertScissorsEq.add
HilbertScissorsEq.symm
HilbertScissorsEq.trans
scissors_congruent
scissors_triangle_swap12
scissors_triangle_swap23
scissors_triangle_cycle
```

Together with associativity and commutativity of multiset addition, these are sufficient to reconstruct Common Notions 2 and 3 in the present configuration.

No new theorem is added to `HilbertInterface`, `HilbertBookZero`, or `HilbertScissors`.

## 43.9 Source Audit

### 43.9.1 Euclid, Heath, and Joyce

The classical sources agree on the short proof: I.34 gives equality of the two large diagonal triangles and of the two pairs of small diagonal triangles; C.N.2 combines the small equalities; C.N.3 leaves the two complements equal. Joyce also emphasizes the operational meaning of

“parallelograms about the diameter” and notes that the next proposition uses I.43 to change the shape of a parallelogram without changing its content.

### 43.9.2 Wilkins

Wilkins is used here as a control on the classical dependency graph and the configuration around the diagonal. No additional hidden case split is needed in the current Lean theorem because all five segment-order relations are already explicit hypotheses.

### 43.9.3 Hilbert

The closest Hilbertian source is Chapter IV, Section 18. Immediately before Hilbert’s Theorem 43, the general theory states that if equidecomposable polygons are removed from equidecomposable polygons, the remaining polygons are equicomplementable. Hilbert’s Theorem 43 then supplies the corresponding transitivity-style properties for equidecomposability and equicomplementability.

The numerical coincidence is worth avoiding: Hilbert’s Theorem 43 is not Euclid’s Proposition I.43. The connection is structural, not numerical. The present Lean proof specializes the general removal pattern directly at the term level by exhibiting the removed pieces as the two complements in the definition of `HilbertScissorsEquicomplementable`.

### 43.9.4 Hartshorne

Hartshorne’s area theory is a semantic control on the direction of the argument. I.41–I.45 belong to the forward equal-content part of Book I; the extra strictness property used by the converse propositions I.39–I.40 is not needed. This supports the present choice to remain entirely inside synthetic cut-and-paste content rather than numerical area.

### 43.9.5 Beeson, Borsuk–Szmielw, and Greenberg

These sources add no separate I.43-specific theorem to the current architecture. Beeson remains a control on omitted formal conditions, while Borsuk–Szmielw and Greenberg control the underlying Euclidean parallel and parallelogram infrastructure. The actual proof is already expressed more directly by I.34 plus the existing scissors calculus.

## 43.10 Formalization Notes

### 43.10.1 Geometry

The formal theorem assumes more explicit positional data than Euclid writes in the proposition statement:

- $K$  is strictly between  $A$  and  $C$ ;
- $G$  is strictly between  $A$  and  $B$ ;
- $E$  is strictly between  $A$  and  $D$ ;

- $F$  is strictly between  $B$  and  $C$ ;
- $H$  is strictly between  $D$  and  $C$ ;
- the large and two small quadrilaterals are explicitly `IsParallelogram` hypotheses.

These hypotheses are not auxiliary constructions introduced by the proof. They are the formal unpacking of the classical diagram required by the splitting constructor.

### 43.10.2 Content

The content argument is entirely forward. There is no appeal to `HilbertScissorsPositivity`, no whole-greater-than-part theorem, and no cancellation principle. The two deleted collections of corner triangles are retained as explicit witnesses of equicomplementability.

This is exactly why I.43 remains inexpensive after I.39: Common Notion 3 is not being used to infer geometry from equality of content. It is being realized as a constructive scissors certificate.

### 43.10.3 Representation

The public theorem does not return the two complements as `hilbertParallelogramTerm`. Instead it returns the exact triangulated terms used by the proof:

$$GBK + KBF \quad \text{and} \quad EDK + KDH.$$

This is faithful to the current term-level semantics of `HilbertScissors`, which represents finite sums of triangles rather than polygonal interiors. The theorem therefore establishes the equal-content relation needed for I.43 without claiming a stronger polygon-semantics result that the library does not yet encode.

No nonoverlap predicate, rectilinear-region type, or numerical area function is introduced.

### 43.10.4 Axiomatic status

The public theorem is stated under

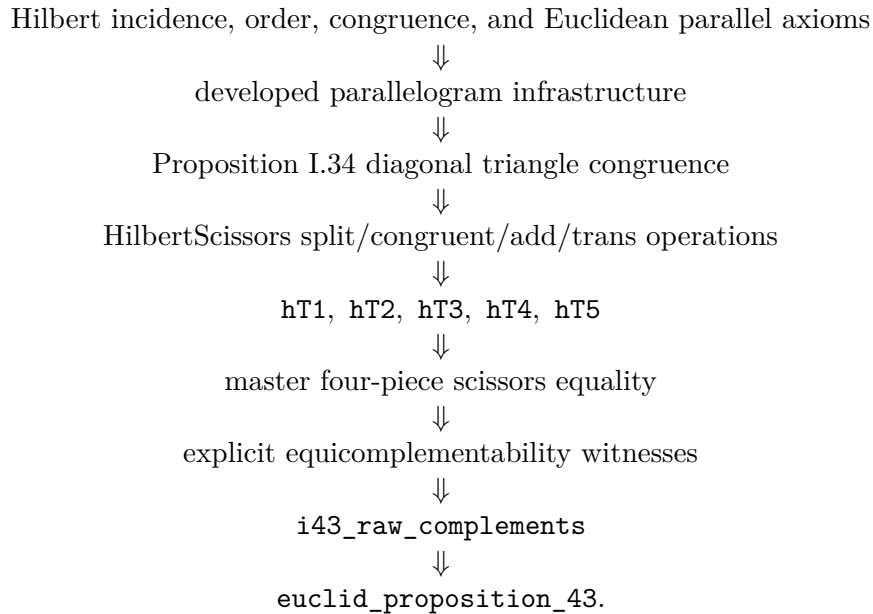
```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

because its I.34 dependency belongs to the developed Euclidean parallelogram layer. No continuity axiom is used locally, no Archimedean principle is used, and I.43 adds no new geometric or content axiom.



## 43.11 Dependency Summary

The actual formal dependency chain is



|                                 |                           |
|---------------------------------|---------------------------|
| New geometric axiom:            | no                        |
| New content axiom:              | no                        |
| New proposition-local helper:   | yes — i43_raw_complements |
| New reusable helper:            | no                        |
| New Book Zero theorem:          | no                        |
| New Interface/Scissors theorem: | no                        |
| Numerical area introduced:      | no                        |
| Polygon semantics introduced:   | no                        |
| Full lake build:                | clean                     |
| sorry in Proposition43.lean:    | no                        |

## 43.12 Conclusion

Proposition I.43 is a compact confirmation that the scissors architecture built from I.35 onward has reached the right level of abstraction. Euclid’s proof uses I.34 together with addition and subtraction of equal figures. The Lean proof preserves the same mathematical structure without numerical area and without introducing a general cancellation theorem.

The decisive formal move is to keep the pieces that Euclid “subtracts” as explicit data. The two collections of corner triangles are proved scissors-equivalent, and the two complements become equidecomposable after those collections are adjoined. That is precisely the existing definition of equicomplementability.

Thus I.43 introduces no new foundational debt. Its one local helper packages a configuration-specific decomposition, while the reusable mathematics remains where it already belongs: I.34 for the diagonal congruences and `HilbertScissors` for finite cutting, congruent replacement, addition, and transitivity. This also leaves the project in the natural position for I.44, where the complement theorem becomes an active construction tool.

## Chapter 44

# Proposition I.44

### 44.1 Introduction

Proposition I.44 is the first Book I theorem that takes the synthetic content machinery developed in I.35–I.43 and applies it to a prescribed segment. The input consists of a nondegenerate segment  $AB$ , a nondegenerate triangle  $PQR$ , and a nondegenerate angle  $XYZ$ . The output is a parallelogram on the given side  $AB$ , with the prescribed angle at  $B$ , and with the same content as the given triangle.

The current Lean statement is deliberately synthetic. It does not introduce a numerical area function. The target parallelogram is represented by `hilbertParallelogramTerm`, and equality of content is represented by `HilbertScissorsEquicomplementable`. Thus the theorem constructs points  $M, L$  such that

$$ABML \text{ is a parallelogram,} \quad \angle ABM \cong \angle XYZ,$$

and

$$\triangle PQR \approx ABML,$$

where  $\approx$  denotes the project's formal equicomplementability relation.

I.44 is also the first proposition in the application-of-areas block where a small phrase in the classical proof hides a substantial formal obligation. Euclid first invokes I.42 to obtain a parallelogram of the correct content and angle, and then says that one side of this parallelogram is to be placed in a straight line with the given segment. The final Lean proof does not treat this rigid placement as a primitive operation. It is reconstructed from segment construction, angle construction, SAS, SSS, and completion of a parallelogram.

A second hidden obligation occurs later. In the classical diagram two produced lines meet at a point  $K$ . Euclid justifies this by an angle argument using I.29 and Postulate 5. The Lean proof reaches the same intersection by a different Euclidean route: if the two carriers were disjoint, they would be parallel; uniqueness and transitivity of parallels would then force two lines through a common point to be parallel, which is impossible. The conclusion is the same, but the formal DAG is not a mechanical transcription of the classical angle-sum proof.

The proof therefore has three sharply separated layers.

- **Geometry:** rigid placement, extensions, intersections, parallel carriers, SameSide/OppositeSide arguments, strict order, and completion of several parallelograms.

- **Content:** a congruent copy preserves scissors content; I.43 identifies the two complements; transitivity then connects the I.42 parallelogram to the final one.
- **Representation:** the I.43 complement terms are sums of triangles, while I.44 exposes whole parallelogram terms. The proof therefore uses alternate triangulations and triangle permutations to bridge the two representations.

The main architectural result of the reconstruction is that I.44 closes with no new geometric axiom and no new content axiom. In particular, the former provisional transport axiom has been eliminated. The public theorem now rests on the previously developed Hilbert and scissors infrastructure together with proposition-local lemmas that expose the geometry suppressed by the classical diagram.

## 44.2 The Classical Configuration

Let  $AB$  be the given straight segment, let  $PQR$  be the given triangle, and let  $XYZ$  represent the prescribed rectilinear angle. Euclid first uses I.42 to construct a parallelogram of the required content and angle. In the lettering used by the reconstruction, after rigid placement this auxiliary parallelogram is  $BEFG$ , with

$$A - B - E \quad \text{and} \quad \angle EBG \cong \angle XYZ.$$

The side  $FG$  is produced beyond  $G$  to  $H$ . Through  $A$  a line is drawn parallel to  $GB$ . The formal order relation is

$$H - G - F, \quad AH \parallel GB.$$

The carrier  $HB$  then meets the carrier  $FE$  at  $K$ , and the proof recovers the strict relations

$$H - B - K, \quad F - E - K.$$

Through  $K$  a line is drawn parallel to  $BE$ . Its intersections with the produced carriers  $HA$  and  $GB$  are denoted by  $L$  and  $M$ . The final order data are

$$H - A - L, \quad G - B - M, \quad L - M - K.$$

These relations are not decorative labels. They are the exact strict-order facts used by the formal proof.

The resulting large figure is the parallelogram  $HLKF$ , with diameter  $HK$ . The two parallelograms about the diameter are

$$HABG \quad \text{and} \quad BMKE.$$

The two complements are precisely the old placed parallelogram  $BEFG$  and the target parallelogram  $ABML$ . This is the configuration to which Proposition I.43 applies.

Figure 44.1 is an affine realization of all the strict betweenness and parallelogram hypotheses used by the current Lean proof. Unlike a purely illustrative Euclidean sketch, it has been chosen so that the relations  $H - A - L$ ,  $H - G - F$ ,  $H - B - K$ ,  $G - B - M$ , and  $F - E - K$  hold simultaneously and the four relevant quadrilaterals are genuine parallelograms.

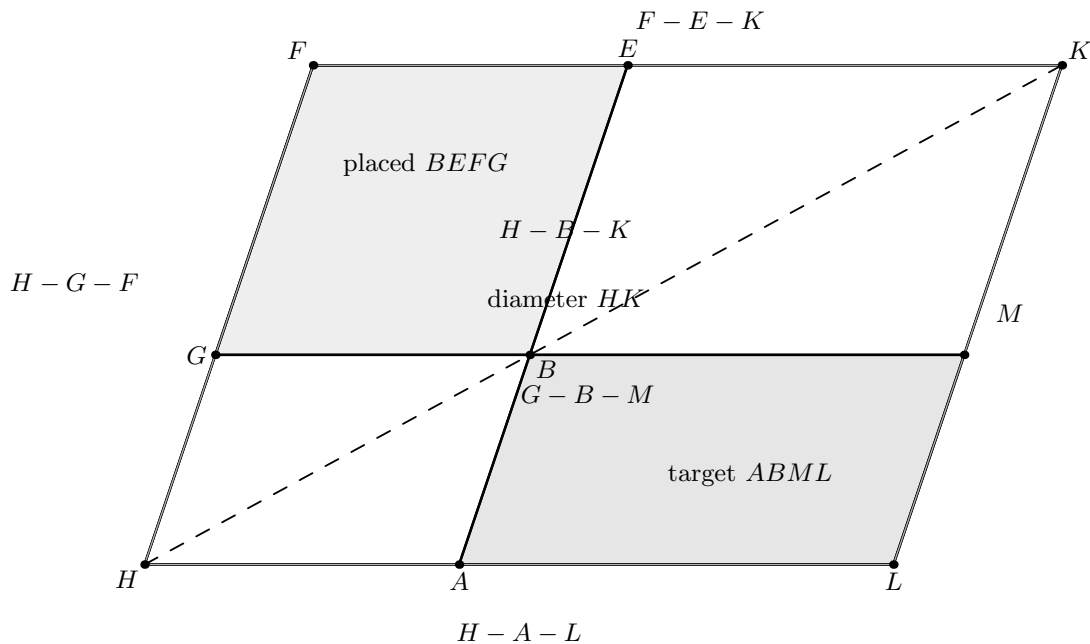


Figure 44.1: The central configuration of Proposition I.44. The large parallelogram is  $HLKF$  with diameter  $HK$ . The two parallelograms about the diameter are  $HABG$  and  $BMKE$ . The shaded complements are the placed auxiliary parallelogram  $BEFG$  and the target parallelogram  $ABML$ . Proposition I.43 identifies their synthetic content. The strict order relations displayed in the figure are exactly the relations recovered by the Lean helpers before I.43 is invoked.

### 44.3 Classical Proof

The classical proof has a compact dependency graph but several diagrammatic steps.

First apply Proposition I.42 to construct a parallelogram  $BEFG$  equal to the given triangle and having

$$\angle EBG = \angle XYZ.$$

Place it so that  $BE$  is in the same straight line as the prescribed segment  $AB$ , with  $B$  between  $A$  and  $E$ . Euclid then produces  $FG$  to  $H$  and draws  $AH$  parallel to  $BG$ .

Because  $AH \parallel EF$ , the transversal  $HF$  creates the supplementary angle configuration used in Euclid's text. With I.29 and the angle facts already available from the parallelogram, the interior angles made by the produced lines  $HB$  and  $FE$  are less than two right angles. Postulate 5 therefore forces the two lines to meet; call their intersection  $K$ .

Through  $K$  draw a line parallel to  $EA$  and produce  $HA$  and  $GB$  until they meet this new line at  $L$  and  $M$ . The quadrilateral  $HLKF$  is a parallelogram,  $HK$  is its diameter, and  $HABG$  and  $BMKE$  are the two parallelograms about the diameter. Proposition I.43 now gives equality of the complements:

$$ABML = BEFG$$

in Euclid's equality-of-figures sense.

By I.42,  $BEFG$  is equal to the given triangle. Common Notion 1 therefore gives the required equality between  $ABML$  and the triangle. Finally,  $A - B - E$  and  $G - B - M$  make  $\angle ABM$  and  $\angle EBG$  vertical angles, so I.15 gives

$$\angle ABM = \angle EBG = \angle XYZ.$$

Thus the required parallelogram has been applied to  $AB$  in the prescribed angle.

The historical dependency graph is therefore

$$\begin{array}{c}
 \triangle PQR, AB, \angle XYZ \\
 \Downarrow \text{I.42} \\
 BEFG \text{ with the required content and angle} \\
 \Downarrow \text{place } BE \text{ on the produced carrier of } AB \\
 A - B - E \\
 \Downarrow \text{I.31, I.29, Postulate 5} \\
 H - G - F, H - B - K, F - E - K \\
 \Downarrow \text{parallel through } K \\
 HLKF, HABG, BMKE \\
 \Downarrow \text{I.43} \\
 ABML = BEFG \\
 \Downarrow \text{C.N.1 and I.15} \\
 ABML = \triangle PQR, \quad \angle ABM = \angle XYZ.
 \end{array}$$

Two points in this graph are especially important for formalization. The phrase “place  $BE$  in a straight line with  $AB$ ” is not a theorem reference in the classical text, and the meeting of  $HB$  and  $FE$  is justified through a specifically Euclidean angle argument. Both steps are reconstructed explicitly in Lean.

## 44.4 Application of Areas and Rigid Placement

The proposition-specific novelty of I.44 is not the existence of a parallelogram with prescribed content and angle; I.42 already supplies that. The new problem is to put such a parallelogram on a *specified segment* without changing either its shape or its content.

The helper

```
i44_place_parallelogram_shape
```

solves the geometric part of this problem. Suppose  $STUV$  is a given parallelogram and  $A \neq B$ . The proof first extends  $AB$  beyond  $B$ . On the resulting ray it lays off  $BE$  congruent to  $TS$ . It then copies the angle  $STU$  at  $B$  with first arm  $BE$ , lays off  $BG$  congruent to  $TU$  on the constructed ray, and completes  $BEG$  to the parallelogram  $BEFG$ . The output contains

$$A - B - E, \quad BE \cong TS, \quad BG \cong TU, \quad \angle EBG \cong \angle STU.$$

Figure 44.2 shows only this rigid-placement stage. It is separate from Figure 44.1 because the two diagrams solve mathematically different problems: the first transports the shape to the prescribed base, while the second performs Euclid’s I.43 complement construction.

The companion helper

```
i44_placed_parallelogram_content
```

proves that this rigid copy preserves the scissors content of the original parallelogram. The argument is stronger than a bare appeal to an undefined notion of “same area”. It triangulates the two parallelograms and proves congruence of corresponding triangles.

$$BE \cong TS, \quad BG \cong TU$$

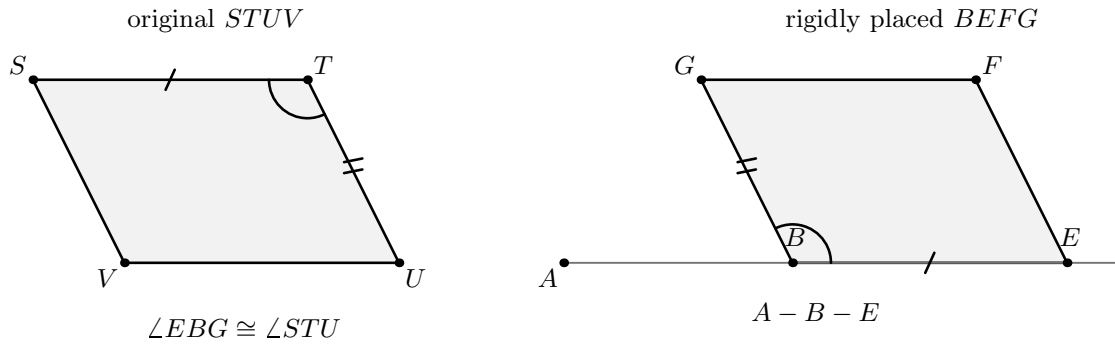


Figure 44.2: The rigid-placement helper. The original parallelogram  $STUV$  is copied to  $BEFG$  with  $A - B - E$ ,  $BE \cong TS$ ,  $BG \cong TU$ , and  $\angle EBG \cong \angle STU$ . The construction uses segment layoff, angle construction on a chosen side, ray orientation, and completion of a parallelogram. No content theorem is used in the geometric placement itself.

The triangle  $BEG$  is congruent to  $TSU$  by SAS, since the two sides and their included angle were copied explicitly. This also gives the diagonal congruence

$$EG \cong SU.$$

Opposite-side congruence in the two parallelograms supplies the remaining side relations needed for SSS, yielding

$$\triangle EFG \cong \triangle SVU.$$

The two congruences are converted into `HilbertScissorsEq` relations. The existing helper

```
parallelogram_two_triangulations
```

then identifies their sums with the whole parallelogram terms. Hence the original and placed parallelograms are scissors-equivalent, and therefore equicomplementable.

This separation between *shape placement* and *content preservation* is important. The first helper belongs to synthetic geometry; the second is a certificate that the rigid copy has not changed the formal content. Neither helper introduces numerical area.

## 44.5 Proof Architecture of the Reconstruction

The final Lean proof can be compressed into seven mathematical stages. The source file contains more helper declarations, but many of them are local normalization lemmas for order or carrier orientation. The stages below are the mathematically significant transitions.

### 44.5.1 Step 1: obtain a content-and-angle parallelogram from I.42

The public theorem begins by calling

```
euclid_proposition_42
```

on the triangle  $PQR$  and the angle  $XYZ$ . It obtains points  $E, F, G$  such that  $FERG$  is a parallelogram,

$$\angle FER \cong \angle XYZ,$$

and

$$\triangle PQR \approx FERG.$$

At this point the parallelogram has the correct content and angle but is not located on the prescribed segment  $AB$ .

#### 44.5.2 Step 2: rigidly place the I.42 parallelogram beyond B

The theorem `i44_transport_onto_line` treats the I.42 output as a general parallelogram  $STUV$ . The helper `i44_place_parallelogram_shape` constructs the placed copy  $BEFG$  with  $A - B - E$  and with two adjacent sides and the included angle copied from  $STUV$ .

This is the formal replacement for the classical phrase that the side of the I.42 parallelogram is to be placed in the same straight line as the given segment. In an earlier provisional architecture this step was packaged as a local transport axiom. The current file derives it completely.

#### 44.5.3 Step 3: certify the content of the rigid copy

The helper `i44_placed_parallelogram_content` proves

$$STUV \approx BEFG.$$

Its proof uses SAS, opposite-side congruence of parallelograms, SSS, and the two triangulations of a parallelogram. Thus the content transport is reduced to finite congruent-piece geometry rather than postulated as an area law.

#### 44.5.4 Step 4: construct the positioned Euclid configuration

Starting with  $A - B - E$  and the placed parallelogram  $BEFG$ , the helper

```
i44_positioned_content
```

constructs the full I.43 configuration. The first key result is

```
i44_construct_H_ordered
```

which returns  $H$  with

$$H - G - F, \quad AH \parallel GB.$$

The incidence part first constructs the parallel through  $A$  and proves that it meets the carrier  $FG$ . The order part then uses line separation to show that the intersection lies beyond  $G$  rather than in another collinear position.

The next key result is

```
i44_construct_K_ordered
```

which returns  $K$  with

$$H - B - K, \quad F - E - K.$$

This is where the formal proof departs most visibly from Euclid's literal argument. Instead of reproducing the sum-of-interior-angles calculation, it supposes that the carriers  $HB$  and  $EF$  are disjoint. Disjointness gives  $HB \parallel EF$ . But the old parallelogram already gives  $GB \parallel EF$ . Euclidean transitivity/uniqueness of parallels would then force  $HB$  and  $GB$  to be parallel, impossible because they share  $B$ . Hence the carriers meet. SameSide/OppositeSide arguments then identify the correct strict order on both lines.

Finally `i44_construct_LM` draws the carrier through  $K$  parallel to  $BE$  and finds its intersections with the carriers  $HA$  and  $GB$ . The order helpers recover

$$H - A - L, \quad L - M - K, \quad G - B - M.$$

The last relation is obtained by a separate line-separation argument in `i44_order_GBM` and is later essential for the vertical-angle step.

#### 44.5.5 Step 5: recognize the four parallelograms needed by I.43

The construction is then repackaged into the four parallelogram statements that express the classical diagram:

$HABG$  first parallelogram about the diameter,  
 $BMKE$  second parallelogram about the diameter,  
 $HLKF$  large parallelogram,  
 $ABML$  target parallelogram.

These conclusions are supplied by the helpers

```
i44_first_diagonal_parallelogram
i44_second_diagonal_parallelogram
i44_big_parallelogram
i44_target_parallelogram
```

The proofs are mainly parallel-carrier transport and recognition of the `IsParallelogram` definition.

#### 44.5.6 Step 6: use I.43 and transfer the angle

The helper

```
i44_i43_content_bridge
```

feeds the constructed data to `euclid_proposition_43`. The point mapping is

| I.43 name  | $A$ | $B$ | $C$ | $D$ | $K$ | $G$ | $E$ | $F$ | $H$ |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| I.44 point | $H$ | $L$ | $K$ | $F$ | $B$ | $A$ | $G$ | $M$ | $E$ |

Thus the I.43 large parallelogram is  $HLKF$ , its diameter point is  $B$ , and the two parallelograms about the diameter are  $HABG$  and  $BMKE$ .



The public I.43 theorem returns its complements as sums of triangle terms, whereas I.44 wants whole parallelogram terms. The bridge uses

```
parallelogram_two_triangulations
```

together with triangle permutation identities to prove

$$ABML \approx BEFG.$$

This is a representation bridge, not a new content axiom.

For the angle,  $A - B - E$  and  $G - B - M$  expose vertical angles. The helper

```
i44_angle_transfer
```

first proves that  $ABM$  is nondegenerate from the target parallelogram, then invokes the existing vertical-angle theorem and angle transitivity to obtain

$$\angle ABM \cong \angle EBG \cong \angle XYZ.$$

#### 44.5.7 Step 7: chain the two content transports

The derived transport theorem now has

$$STUV \approx BEFG \quad \text{and} \quad BEFG \approx ABML.$$

Transitivity gives

$$STUV \approx ABML.$$

At the top level, I.42 already supplied

$$\triangle PQR \approx STUV,$$

so one last transitivity step proves

$$\triangle PQR \approx ABML.$$

The actual formal DAG is therefore

$$\begin{array}{c}
 PQR, AB, XYZ \\
 \Downarrow \text{I.42} \\
 STUV \text{ with prescribed content and angle} \\
 \Downarrow \text{segment and angle construction} \\
 A - B - E, BEFG \text{ rigid copy of } STUV \\
 \Downarrow \text{SAS + SSS + parallelogram triangulations} \\
 STUV \approx BEFG \\
 \Downarrow \text{parallel/intersection/order construction} \\
 H - G - F, H - B - K, F - E - K, H - A - L, L - M - K, G - B - M \\
 \Downarrow \\
 HLKF, HABG, BMKE, ABML \text{ parallelograms} \\
 \Downarrow \text{I.43 + representation bridge} \\
 ABML \approx BEFG \\
 \Downarrow \text{I.15 / VerticalAngles} \\
 \angle ABM \cong \angle XYZ \\
 \Downarrow \text{content transitivity} \\
 \triangle PQR \approx ABML.
 \end{array}$$

## 44.6 Lean Formalization

The current file imports exactly the two immediately preceding construction and complement results used by the public theorem:

```
import CGJteamLab.Proposition42
import CGJteamLab.Proposition43
```

The public theorem, written in the project’s documentation-friendly ASCII notation, is

```
theorem euclid_proposition_44
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B P Q R X Y Z : Geo.Point)
  (hAB : Not (A = B))
  (hPQR : Not (Collinear Geo P Q R))
  (hXYZ : Not (Collinear Geo X Y Z)) :
  exists M L : Geo.Point,
    IsParallelogram Geo A B M L /\
    Geo.AngleCongruent A B M X Y Z /\
    HilbertScissorsEquicomplementable Geo
      (hilbertScissorsTriangle Geo P Q R)
      (hilbertParallelogramTerm Geo A B M L) := by
  -- proof in Proposition44.lean
```

The decisive derived transport theorem has the following shape:

```
theorem i44_transport_onto_line
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B S T U V X Y Z : Geo.Point)
  (hAB : Not (A = B))
  (hParallelogram : IsParallelogram Geo S T U V)
  (hAngle : Geo.AngleCongruent S T U X Y Z) :
  exists M L : Geo.Point,
    IsParallelogram Geo A B M L /\
    Geo.AngleCongruent A B M X Y Z /\
    HilbertScissorsEquicomplementable Geo
      (hilbertParallelogramTerm Geo S T U V)
      (hilbertParallelogramTerm Geo A B M L) := by
  -- rigid placement + positioned I.43 construction
```

This theorem is the formal content of the phrase “apply the parallelogram to the given line”. It is no longer an axiom.

The two helpers that isolate rigid placement and its content certificate are

```
theorem i44_place_parallelogram_shape
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B S T U V : Geo.Point)
  (hAB : Not (A = B))
```

```

(hParallelogram : IsParallelogram Geo S T U V) :
exists E F G : Geo.Point,
  Geo.Between A B E /\
  IsParallelogram Geo B E F G /\
  Geo.AngleCongruent E B G S T U /\
  Geo.Congruent B E T S /\
  Geo.Congruent B G T U := by
-- segment layoff, angle copy, ray control, fourth vertex

```

and

```

theorem i44_placed_parallelogram_content
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (B E F G S T U V : Geo.Point)
  (hOriginal : IsParallelogram Geo S T U V)
  (hPlaced : IsParallelogram Geo B E F G)
  (hAngle : Geo.AngleCongruent E B G S T U)
  (hBE_TS : Geo.Congruent B E T S)
  (hBG_TU : Geo.Congruent B G T U) :
  HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo S T U V)
    (hilbertParallelogramTerm Geo B E F G) := by
-- SAS + SSS + the two parallelogram triangulations

```

The positioned core is

```

theorem i44_positioned_content
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B E F G : Geo.Point)
  (hABE : Geo.Between A B E)
  (hPar : IsParallelogram Geo B E F G) :
exists M L : Geo.Point,
  IsParallelogram Geo A B M L /\
  Geo.Between G B M /\
  HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo A B M L)
    (hilbertParallelogramTerm Geo B E F G) := by
-- H, K, L, M construction + I.43

```

The mathematically significant local declarations can be grouped by role.

- `i44_place_parallelogram_shape`: rigid placement on the prescribed base;
- `i44_placed_parallelogram_content`: congruent-piece certificate for the rigid copy;
- `i44_construct_H_ordered`: construct  $H - G - F$  with  $AH \parallel GB$ ;
- `i44_construct_K_ordered`: construct  $H - B - K$  and  $F - E - K$ ;
- `i44_construct_LM`: construct the parallel through  $K$  and its two intersections;
- the three order helpers for  $H - A - L$ ,  $L - M - K$ , and  $G - B - M$ : recover strict order from carrier data;

- `i44_i43_content_bridge`: adapt I.43 to whole parallelogram terms;
- `i44_positioned_content`: assemble the full positioned construction;
- `i44_angle_transfer`: transfer the prescribed angle by vertical angles;
- `i44_transport_onto_line`: package the reusable high-level transport theorem.

The remaining proposition-local helpers mostly recognize parallelograms, transport parallelism along collinear carriers, or normalize line orientation. They are essential to compilation but do not represent independent new mathematical principles.

## 44.7 Relationship with Earlier Propositions

### 44.7.1 Proposition I.15

The last angle step is exactly the vertical-angle theorem. Once the order lemmas have established

$$A - B - E \quad \text{and} \quad G - B - M,$$

I.15 identifies  $\angle ABM$  with  $\angle EBG$ . In Lean this appears through the already exposed helper `VerticalAngles`; no new angle theorem is proved in I.44.

### 44.7.2 Proposition I.31 and the Euclidean parallel layer

Classically, I.31 is invoked to draw the parallels through  $A$  and  $K$ . The Lean file does not call the historical proposition by name. It uses the Hilbert-level construction

```
hilbert_parallel_through_point_exists
```

and the developed parallel-carrier infrastructure.

More importantly, the existence of the required intersections is proved by Euclidean uniqueness/transitivity of parallels. Thus the formal proof has a clear Euclidean dependency even where it does not replay the literal I.29 plus Postulate-5 angle argument.

### 44.7.3 Proposition I.42

I.42 is the first direct Book I dependency of the final theorem. It converts the given triangle and angle into a parallelogram with the desired content and angle. I.44 adds the missing location constraint: the parallelogram must be applied to a specified segment.

The final architecture therefore treats I.42 as a content-and-shape producer, not as a theorem that already knows where the resulting parallelogram is to be placed.

### 44.7.4 Proposition I.43

I.43 is the central content theorem inside the positioned construction. Once  $HLKF$ ,  $HABG$ , and  $BMKE$  have been recognized, I.43 states that the two remaining complements have equal

synthetic content. The local bridge `i44_i43_content_bridge` does not strengthen I.43. It only converts its triangle-sum representation of the complements into the parallelogram terms  $ABML$  and  $BEFG$  used by I.44.

#### 44.7.5 Propositions I.34–I.41

These propositions are not replayed directly by the public I.44 theorem, but they remain present through its two major dependencies. I.42 already rests on the forward content theory of I.37–I.41, and I.43 rests on the diagonal parallelogram theorem I.34 together with the scissors calculus. I.44 is therefore the first point where those two branches are combined in an active construction.

#### 44.7.6 Proposition I.45

I.45 generalizes the application-of-areas idea from a triangle to an arbitrary rectilineal figure. The transport theorem isolated in I.44 is therefore a natural reusable unit for the next stage: once a suitable parallelogram of the required total content is available, it can be applied to a prescribed segment with a prescribed angle.

### 44.8 Hilbert and Book Zero Components Used

#### 44.8.1 Betweenness, extensions, and carrier incidence

The formal proof uses strict betweenness much more extensively than the classical text writes. The core order relations are

$$A - B - E, \quad H - G - F, \quad H - B - K, \quad F - E - K, \quad H - A - L, \quad L - M - K, \quad G - B - M.$$

They are obtained by segment extension, line intersection, and plane-separation arguments. Carrier identification repeatedly uses incidence uniqueness for two distinct points.

The proof also makes heavy use of the bridge between a Hilbert incidence line and the extensional `Geo.PointLine` representation. This is what lets the code move safely between line-uniqueness arguments and the strict `Geo.Parallel` relation.

#### 44.8.2 SameSide and OppositeSide

SameSide/OppositeSide data are the main device for turning a bare line intersection into the correct order statement. For example, `i44_construct_H_ordered` uses the line  $GB$  as a separator to prove that the intersection on  $FG$  lies beyond  $G$ . The  $K$ ,  $L$ , and  $M$  order helpers use analogous arguments.

The proof of  $G - B - M$  is particularly revealing. It shows that  $H$  and  $G$  are on the same side of  $BE$ , while  $K$  and  $M$  are on the same side of  $BE$ . Since  $H - B - K$ , the points  $H$  and  $K$  lie on opposite sides. Transporting through the two SameSide relations gives  $G$  and  $M$  on opposite sides, and uniqueness of the intersection with  $BE$  identifies the crossing point with  $B$ .

These side-of-line facts are invisible in the short Euclidean proof but are central to the formal reconstruction.

### 44.8.3 Segment construction, SameRay, and angle construction

Rigid placement uses Hilbert segment construction twice: once to copy the first side of the original parallelogram onto the extension of  $AB$ , and once to copy the second side onto the constructed angle ray. The first copy must remain on the ray beyond  $B$ , and the second must remain on the copied angle ray. SameRay machinery records these orientation facts.

Angle construction is used with an explicit carrier and a chosen side. This produces more information than the classical phrase “construct an equal angle”, and that information is exactly what is needed to prove that the new point  $G$  is off the line  $BE$  and that  $BEG$  is a genuine triangle.

### 44.8.4 Euclidean parallelism

The theorem is genuinely Euclidean. The class assumption is

```
[HilbertEuclideanPlane Geo]
```

and the Euclidean input is used in several places:

- construction and uniqueness of a parallel through a point;
- transitivity of distinct parallel carriers;
- proving that the constructed parallel through  $A$  meets  $FG$ ;
- proving that  $HB$  meets  $EF$ ;
- identifying carriers that are both parallel to the same line and share a point;
- completing and recognizing the several parallelograms in the I.43 configuration.

This is the formal counterpart of Euclid’s use of I.29 and Postulate 5. In neutral geometry the relevant uniqueness/intersection conclusions are not available in this form.

### 44.8.5 Congruence and parallelogram infrastructure

The rigid-copy content proof uses SAS and SSS. It also uses

```
ParallelogramOppositeSidesCongruent  
hilbert_parallelogram_fourth_vertex_exists
```

Together they show that a shape copied by two adjacent sides and the included angle is congruent piecewise after triangulation, and that the copied three vertices determine a genuine parallelogram.

No new congruence axiom or parallelogram axiom is introduced in I.44.

### 44.8.6 Scissors content infrastructure

The content layer uses only the existing scissors operations, principally

```

scissors_congruent
HilbertScissorsEq.add
HilbertScissorsEq.trans
equicomplementable_of_scissorsEq
equicomplementable_symm
equicomplementable_trans
equicomplementable_transport
parallelogram_two_triangulations

```

I.44 neither defines numerical area nor introduces subtraction or a content order. The only equality-of-figures conclusion is the already established synthetic relation of equicomplementability.

## 44.9 Source Audit

### 44.9.1 Euclid, Heath, and Joyce

The classical sources agree on the main construction. I.42 supplies a parallelogram equal to the given triangle in the prescribed angle. One side is placed on the produced carrier of the given segment. The side  $FG$  is produced, a parallel through  $A$  is drawn, and the two produced lines are forced to meet by the Fifth Postulate. A parallel through the intersection creates the large parallelogram and the two parallelograms about its diameter. I.43 identifies the complements, and I.15 supplies the final vertical-angle identity.

These sources control the historical statement and the classical DAG. They also expose the two formal pressure points: the unreferenced placement step and the specifically Euclidean meeting step.

### 44.9.2 Wilkins

Wilkins gives the same dependency pattern with explicit attention to the produced lines and the I.43 configuration. His account is useful as a control on point order and on which quadrilaterals are intended to be the two complements. The current Lean diagram follows those order constraints rather than relying on a visually plausible but logically weaker collinearity sketch.

### 44.9.3 Hilbert and Borsuk–Szmielew

Hilbert’s Chapter IV and the Borsuk–Szmielew treatment control the semantics of content. Their plane-area theory distinguishes equidecomposability from equicomplementability and develops the forward parallelogram/triangle content theorems without requiring the continuity group. In particular, the non-Archimedean distinction warns against replacing the current target by an unjustified global equidecomposability statement.

The same sources reorganize the geometry away from Euclid’s proposition numbers. Segment layoff, angle construction, line order, and parallel uniqueness are reusable parts of the axiomatic geometry. This is why the Lean proof can derive the placement and intersection steps without introducing a new I.44-specific axiom.

#### 44.9.4 Hartshorne

Hartshorne places I.35–I.48 in the area part of Book I and describes I.44 as the construction of a parallelogram with a given side and angle equal to a given triangle. His discussion of “equal content” is also the correct semantic control for the present documentation: Euclid has not introduced a real-valued area function, so equality of figures in this block should not be silently rewritten as a numerical formula.

Hartshorne’s analysis of the parallel postulate also supports the formal classification. The Fifth Postulate and Playfair uniqueness are equivalent over the developed neutral theory. The Lean proof chooses the uniqueness form of the Euclidean input when proving the hidden intersections.

#### 44.9.5 Greenberg

Greenberg supplies an independent modern control on the same Euclidean boundary: Euclid’s Fifth Postulate and Hilbert/Playfair parallel uniqueness are equivalent in the neutral setting. This explains why the formal replacement of Euclid’s less-than-two-right-angles argument by a uniqueness-of-parallels contradiction does not change the axiomatic strength of the proposition.

#### 44.9.6 Beeson proof corpus

The Beeson/Tarski corpus remains useful in the project as a gap audit for classical constructions, especially where a traditional diagram names an intersection without isolating its existence or orientation. The final I.44 proof, however, is not organized around Tarski tactics. Its actual DAG is the Hilbert-style one documented above: explicit line carriers, SameSide/OppositeSide order recovery, Euclidean parallel uniqueness, and the project’s own scissors representation.

### 44.10 Formalization Notes

#### 44.10.1 Geometry

The geometric cost of I.44 is substantially higher than the length of the classical text suggests. The formal proof must establish all of the following, rather than reading them from a picture:

- the first copied side lies beyond  $B$ , giving  $A - B - E$ ;
- the copied angle ray produces a point genuinely off the carrier  $BE$ ;
- the fourth vertex completing the placed parallelogram exists;
- the parallel through  $A$  actually meets the produced carrier  $FG$ ;
- that intersection is ordered as  $H - G - F$ ;
- the carriers  $HB$  and  $EF$  actually meet;
- their intersection satisfies both  $H - B - K$  and  $F - E - K$ ;
- the parallel through  $K$  meets the two required carriers;
- those intersections satisfy  $H - A - L$ ,  $L - M - K$ , and  $G - B - M$ ;



- the four quadrilaterals used by I.43 satisfy the strict `IsParallelogram` relation.

No continuity axiom is used to obtain these intersections. Their existence comes from the developed Euclidean parallel theory plus incidence and plane-separation arguments.

### 44.10.2 Content

There are two content transports and they have different proofs.

First, the arbitrary I.42 parallelogram is rigidly copied to  $BEFG$ . Here the proof actually produces a direct scissors equality by congruent triangulation: SAS for one triangle, SSS for the other, and addition of the two congruent pieces.

Second,  $BEFG$  is transformed into the target  $ABML$  by the I.43 complement construction. Here the conclusion is equicomplementability, exactly matching the public content strength of I.43. The final theorem composes these relations by transitivity.

At no stage is equal content used to infer an incidence, order, or metric fact. All geometry is constructed first; content is then transported forward. Consequently I.44 needs neither the positivity principle used at the converse boundary I.39 nor any numerical area theorem.

### 44.10.3 Representation

The public orientation of the target is

```
IsParallelogram Geo A B M L
Geo.AngleCongruent A B M X Y Z
```

This orientation matters. The prescribed angle is the angle at  $B$  between the actual adjacent sides  $BA$  and  $BM$ . It should not be represented using a diagonal arm.

The I.43 bridge has a second representation issue. I.43 returns complements as explicit sums of two triangle terms, while I.44 wants a whole parallelogram term. The helper

```
i44_i43_content_bridge
```

uses alternate triangulations and triangle permutation identities to prove that these are the same formal scissors objects up to `HilbertScissorsEq`. No new polygon-region semantics is introduced.

The formal figures therefore represent finite triangle sums and parallelogram terms, not geometric interiors with a measure attached to them.

### 44.10.4 Axiomatic status

Proposition I.44 is Euclidean, not neutral. The explicit type-class context is

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

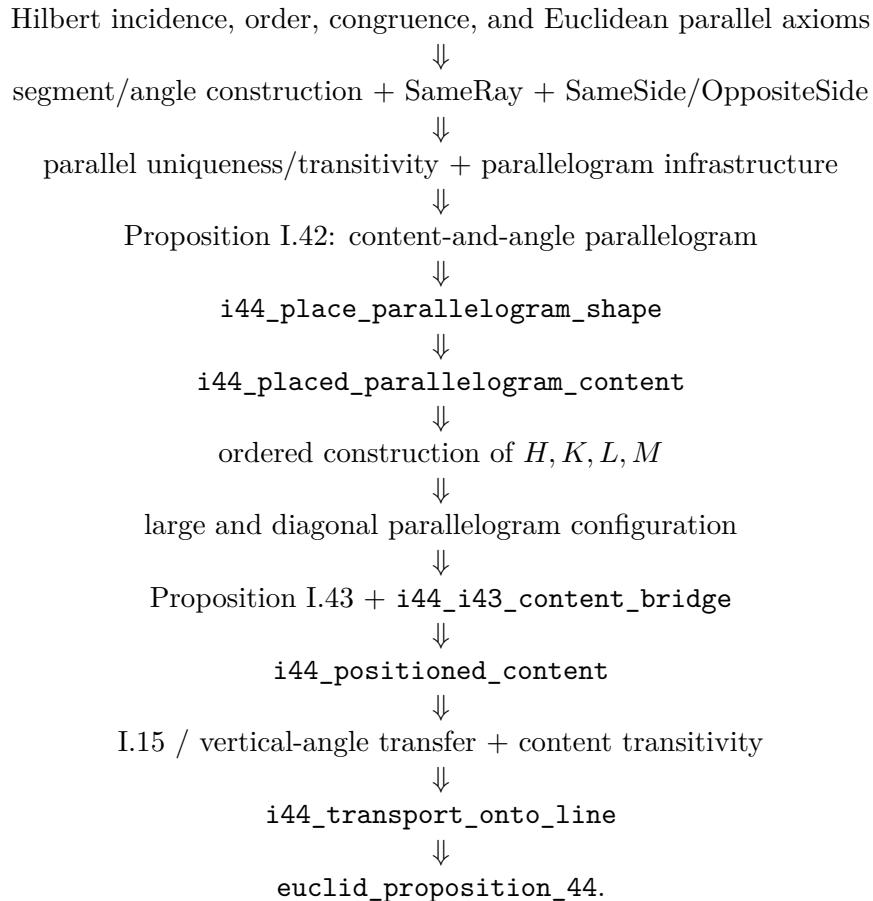
Hilbert Group IV (the Euclidean parallel axiom) is therefore used. Its formal role is uniqueness/transitivity of parallels and the intersection arguments described above. Parallel existence is provided by the developed construction layer; the specifically Euclidean strength is needed when the proof excludes a second distinct parallel through the same point. This has the same axiomatic strength as the classical use of Postulate 5, although the formal proof uses a different equivalent route.

Group III congruence construction is also essential: segment layoff, angle construction, SAS, SSS, and opposite-side congruence of parallelograms are used in the rigid-placement certificate. No Group V continuity principle is used.

The current proposition introduces no active local axiom. In particular, the provisional `i44_transport_onto_line` axiom has been replaced by the derived theorem of the same mathematical purpose. No new theorem is added to `HilbertInterface`, `HilbertBookZero`, or `HilbertScissors` by I.44 itself.

## 44.11 Dependency Summary

The formal dependency chain is



The classical and formal DAGs may be summarized as

Euclid: I.42  $\rightarrow$  placement  $\rightarrow$  I.31/I.29/Post.5  
 $\rightarrow$  I.43  $\rightarrow$  I.15,  
 Lean: I.42  $\rightarrow$  derived rigid placement  
 $\rightarrow$  Euclidean parallel/intersection order  
 $\rightarrow$  I.43 bridge  $\rightarrow$  VerticalAngles.

|                                                                    |               |
|--------------------------------------------------------------------|---------------|
| New geometric axiom:                                               | no            |
| New content axiom:                                                 | no            |
| Provisional I.44 transport axiom retained as active axiom:         | no            |
| New proposition-local helpers:                                     | yes           |
| New reusable high-level transport helper:                          | yes           |
| New Book Zero theorem:                                             | no            |
| New Interface/Scissors theorem:                                    | no            |
| Numerical area introduced:                                         | no            |
| Hilbert Group IV used:                                             | yes           |
| Continuity used:                                                   | no            |
| Public <code>Proposition44.lean</code> compiles:                   | yes           |
| Full <code>lake build</code> re-run in this documentation pass:    | no            |
| GitHub CI status checks on the committed code:                     | none reported |
| <code>sorry</code> introduced in <code>Proposition44.lean</code> : | no            |

## 44.12 Conclusion

Proposition I.44 is a major architectural checkpoint in the Book I reconstruction. The classical proof combines I.42 and I.43 with a short parallel construction and a vertical-angle observation. The formal proof shows that the apparent simplicity hides two genuine tasks: a rigid placement of the I.42 parallelogram on a prescribed segment, and a large amount of order information needed to instantiate I.43 correctly.

Both tasks are now derived rather than assumed. The rigid placement is built from segment and angle construction and certified by SAS/SSS triangulation. The positioned construction derives  $H, K, L, M$ , recovers the strict order relations by plane separation, recognizes the four required parallelograms, and then invokes I.43 through a representation bridge. Vertical angles close the prescribed-angle condition, and equicomplementability transitivity closes the content condition.

The proposition therefore introduces no new foundational debt. It remains fully synthetic: no numerical area, no polygon measure, no continuity axiom, and no application-of-area transport axiom. At the same time it produces a useful high-level theorem, `i44_transport_onto_line`, which packages a substantial geometric construction into a form suitable for later Book I work. This is precisely the kind of compression the Hilbert/Book Zero architecture is intended to provide: the final theorem is short because the hidden geometry has been made explicit and proved once.

## Chapter 45

# Proposition I.45

### 45.1 Introduction

Proposition I.45 is the culmination of the application-of-areas sequence I.42–I.45. Proposition I.42 constructs a parallelogram equal in content to one triangle and having a prescribed angle. Proposition I.44 adds the location constraint that such a parallelogram be applied to a prescribed segment. Proposition I.45 passes from one triangle to a rectilineal figure. Classically, the new idea is finite composition: triangulate the figure, apply the preceding construction to its triangular pieces, and combine the resulting parallelograms into one parallelogram with the prescribed angle.

The current Lean reconstruction follows that compositional idea, but makes its data flow explicit. A triangulated figure is represented by `HilbertTriangulatedFigure Geo`, and its executable scissors content is the sum of the corresponding triangle terms. For a single triangle the proof calls I.42 directly. For a longer list, the induction hypothesis produces a parallelogram for the tail, the next triangle is converted into an oriented adjacent parallelogram, and the two parallelograms are pasted into a larger one. The paste must preserve both the synthetic content and the prescribed angle.

The final theorem remains entirely synthetic. It does not attach a number to a figure and does not use a numerical area formula. Equality of content is represented by

```
HilbertScissorsEquicomplementable
```

If  $L$  is the input list of triangles, the theorem constructs a parallelogram  $STUV$  satisfying

$$\angle STU \cong \angle XYZ$$

and

$$\text{Term}(L) \approx \text{Par}(STUV),$$

where  $\approx$  denotes the project's equicomplementability relation.

I.45 also resolves an architectural question left deliberately open earlier in the area block. The existing Hilbert layer already contains `HilbertTriangle`, `HilbertTriangulatedFigure`, and relations between triangulations. For the constructive content theorem of I.45, this list-of-triangles representation is sufficient: the proof needs only finite addition of triangle content and an induction step that adjoins one triangle. No new polygon datatype and no numerical area layer are required.

There is, however, an important representation boundary. In the current library,

```
abbrev HilbertTriangulatedFigure
  (Geo : Geometry.Geo) :=
  List (HilbertTriangle Geo)
```

is only a finite list of triangles. The statement of `euclid_proposition_45` assumes that the list is nonempty and that each triangle is nondegenerate. It does not additionally assert that the triangles have disjoint interiors, fit along common edges, or form the interior of one simple polygon. Thus the formal theorem is a valid and in one sense stronger additive statement about any finite nonempty list of nondegenerate triangles. To read such a list as one geometric rectilinear figure requires a representation invariant supplied outside this theorem. The documentation therefore distinguishes the proved scissors statement from a full internal semantics of polygonal regions.

As in the preceding content propositions, the proof has three distinct layers.

- **Geometry:** extensions, SameRay orientation, SameSide/OppositeSide separation, parallel uniqueness, Pasch, strict betweenness, and recognition of the pasted outer parallelogram.
- **Content:** scissors refinement, equicomplementability of the new triangular piece with an adjacent parallelogram, addition, and transitivity.
- **Representation:** a rectilinear input is represented by a triangle list and a multiset term; the I.44 construction is locally strengthened to retain one strict order fact that its public wrapper intentionally hides.

The most important foundational result is negative: I.45 introduces no new geometric axiom, no new content axiom, no continuity principle, and no positivity or cancellation principle. The temporary `i45_extend_parallelogram` assumption used during development has been completely eliminated and replaced by a theorem.

## 45.2 The Classical Configuration

Euclid states the proposition for a given rectilinear figure and a given rectilinear angle. In the first nontrivial case, let the figure be a quadrilateral  $ABCD$  divided by the diagonal  $BD$  into the two triangles  $ABD$  and  $DBC$ .

First construct, by I.42, a parallelogram equal to triangle  $ABD$  in the given angle. Then apply, by I.44, a second parallelogram equal to triangle  $DBC$  to a side of the first parallelogram. The second parallelogram is placed so that the two pieces are adjacent rather than overlapping. Euclid then uses parallel-line angle relations and straight-line criteria to show that the outer boundary is again a parallelogram. Since the two inner parallelograms are respectively equal to the two triangles, their union is equal to the whole rectilinear figure.

For a rectilinear figure with more triangular pieces, the same operation is repeated. One may therefore read the classical construction recursively: start with a parallelogram for one triangle and append one triangle at a time by applying an adjacent parallelogram to the current boundary.

The current Lean proof adopts exactly this recursive shape, but the local configuration is normalized to the notation

$$STUV, \quad TUML, \quad SLMV.$$

Here  $STUV$  is the already constructed parallelogram,  $TUML$  is the parallelogram representing the next triangle, and  $SLMV$  is the pasted outer parallelogram. The strict order conditions are

$$S - T - L, \quad V - U - M.$$

These two relations are the formal content of saying that the second parallelogram is attached externally along the side  $TU$ .

Figure 45.1 is an affine realization of this exact Lean configuration. It also displays the point  $X$  used by the scissors proof, with

$$S - X - M, \quad T - X - U.$$

The point is not needed merely to recognize the outer parallelogram. It is needed to refine the two triangulations into the same finite collection of triangle pieces.

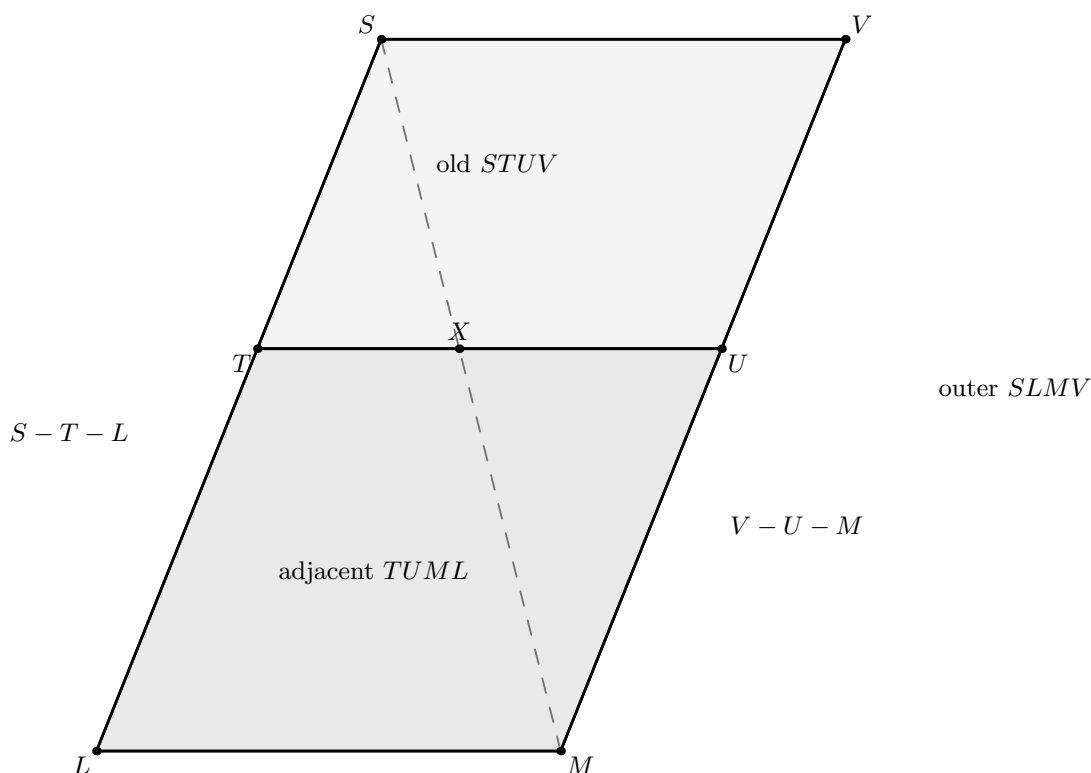


Figure 45.1: Two adjacent parallelograms in the exact order configuration used by the Lean proof. The old parallelogram is  $STUV$ , the new one is  $TUML$ , and the strict relations  $S - T - L$  and  $V - U - M$  put the new piece outside the old one. Their outer boundary is the parallelogram  $SLMV$ . The dashed segment  $SM$  meets the common side  $TU$  at  $X$ , giving  $S - X - M$  and  $T - X - U$  for the scissors refinement.

### 45.3 Classical Proof

The classical proof is a finite composition of I.42 and I.44.

For a quadrilateral  $ABCD$ , draw the diagonal  $BD$ . Proposition I.42 gives a parallelogram equal to triangle  $ABD$  in the prescribed angle. Proposition I.44 then applies to one side of that parallelogram a second parallelogram equal to triangle  $DBC$ , in the supplementary orientation required for the two pieces to form a straight outer boundary.

The angle equalities supplied by the two constructions, together with I.29, show that the relevant sides of the two parallelograms are parallel. Euclid uses I.14 to recognize the straight-line relations created at the common side. The outer opposite sides are then shown parallel using the established parallel theory, including I.30 and the parallelogram criterion behind I.33. Thus the two applied pieces form one larger parallelogram.

The content conclusion is purely additive. One inner parallelogram is equal to triangle  $ABD$ , the other to triangle  $DBC$ , and the two triangles are the whole rectilinear figure. Therefore the outer parallelogram is equal to the given rectilinear figure. The prescribed outer angle is inherited from the first construction.

The two-triangle historical DAG is therefore

$$\begin{array}{c}
 ABCD \text{ divided by } BD \\
 \Downarrow \\
 \triangle ABD + \triangle DBC \\
 \Downarrow \text{ I.42 and I.44} \\
 \text{two adjacent parallelograms} \\
 \Downarrow \text{ I.29, I.14, I.30, I.33} \\
 \text{one outer parallelogram} \\
 \Downarrow \text{ additivity of equality of figures} \\
 \text{outer parallelogram} = ABCD.
 \end{array}$$

The general finite case repeats the same step for each triangular component. This is the part of the classical argument that the final Lean induction captures most directly.

## 45.4 Triangulated Figures and the Representation Boundary

The proposition is the first point in Book I where the input is not naturally one triangle or one quadrilateral but an arbitrary rectilinear figure. This forces a distinction between a geometric region and a formal decomposition of that region.

The Hilbert layer already contains

```

structure HilbertTriangle
  (Geo : Geometry.Geo) where
  A : Geo.Point
  B : Geo.Point
  C : Geo.Point

abbrev HilbertTriangulatedFigure
  (Geo : Geometry.Geo) :=
  List (HilbertTriangle Geo)

```

and a relation `HilbertSameFigure` generated by triangle splitting and triangle permutations. This is genuine infrastructure for reasoning about triangulations. It is intentionally more structural than a coordinate polygon representation.

Proposition I.45 introduces the executable term

```

def rectilinearTerm
  (L : HilbertTriangulatedFigure Geo) :

```

```
HilbertScissorsTerm Geo :=
(L.map
 (fun t =>
   hilbertScissorsTriangle Geo t.A t.B t.C)).sum
```

which forgets all incidence between different list entries and retains exactly what the proof needs for content: a finite formal sum of triangle terms.

This choice has two consequences.

First, it is sufficient for the theorem actually proved. The induction step uses only the identity

$$\text{Term}(T :: L) = \text{Tri}(T) + \text{Term}(L),$$

followed by addition and transitivity of equicomplementability. No operation in the proof needs to inspect a polygon boundary.

Second, the theorem should not be described as proving from first principles that every such list is a simple rectilinear region. A geometric rectilinear figure requires a compatible finite triangulation of one region, in particular the appropriate nonoverlap and coverage conditions. The current Lean theorem does not carry those conditions as hypotheses. When  $L$  comes from a genuine triangulation, the theorem formalizes Euclid I.45. Abstractly, it proves a slightly more general finite-additive statement.

This is an important architectural checkpoint. The existing `HilbertTriangulatedFigure` plus scissors terms are sufficient for the constructive content calculus of I.45. A stronger polygon-region type may still be useful later for theorems whose hypotheses or conclusions depend on the boundary, interior, overlap, containment, or equality of geometric regions. I.45 itself does not force that extension.

## 45.5 Proof Architecture of the Reconstruction

The 2300-line Lean file is best understood as eight conceptual stages rather than as a sequence of low-level incidence manipulations.

### 45.5.1 Step 1: turn a triangle list into a scissors term

The small definitions

```
rectilinearTerm
rectilinearTerm_cons
rectilinearTerm_nil
```

form the additive interface between the list representation and `HilbertScissorsTerm`. They isolate the only list algebra used by the final induction.

This stage is purely representational. It introduces no new geometric relation and no content axiom.



### 45.5.2 Step 2: prove the geometry of pasting adjacent parallelograms

Suppose  $ABCD$  and  $BCML$  are parallelograms and the strict orders are

$$D - C - M, \quad A - B - L.$$

The helpers

```
i45_outer_first_parallel
i45_outer_second_parallel
i45_paste_parallelograms_geometry
```

prove that the outer quadrilateral  $ALMD$  is a parallelogram.

The first outer parallel is obtained mostly by transporting parallelism along the two collinear produced sides. The second is subtler: the proof has two lines parallel to the common carrier and must show that the two candidate outer carriers are distinct before invoking Euclidean transitivity/uniqueness of parallels. This is one of the places where the formal proof uses Group IV in a way hidden by the ordinary picture.

### 45.5.3 Step 3: refine the paste into a scissors identity

The geometric union of two adjacent parallelograms is not automatically an identity of the project's multiset terms. The proof therefore constructs an internal crossing point.

The helper

```
i45_paste_crossing
```

returns a point  $X$  satisfying

$$A - X - M, \quad B - X - C.$$

Its existence is obtained by side-of-line information and a forced Pasch argument. Parallel/order transport is then used to identify the second betweenness relation.

With this witness,

```
i45_paste_parallelograms_content_of_crossing
```

splits the relevant triangles and uses the generic diagonal-flip theorem

```
crossing_quadrilateral_two_triangulations
```

Explicit multiset associativity and commutativity then normalize the pieces. The packaged result

```
i45_paste_parallelograms_content
```

is the formal identity

$$\text{Par}(ALMD) \sim \text{Par}(ABCD) + \text{Par}(BCML).$$

The companion

```
i45_paste_equicomplementable
```

combines that scissors equality with an equicomplementability certificate for the right-hand piece. This is the additive content engine used by the induction step.

#### 45.5.4 Step 4: preserve the prescribed angle

The helper

```
i45_paste_angle
```

proves that pasting does not change the prescribed angle carried by the first parallelogram.

For the outer parallelogram  $ALMD$ , opposite angles give

$$\angle ALM \cong \angle MDA.$$

The order  $D - C - M$  identifies the rays  $DC$  and  $DM$ , so

$$\angle MDA = \angle CDA.$$

The old parallelogram  $ABCD$  gives

$$\angle ABC \cong \angle CDA.$$

Transitivity therefore yields

$$\angle ALM \cong \angle ABC.$$

This argument uses only the developed parallelogram-angle theorem and SameRay angle transport; it introduces no new angle principle.

#### 45.5.5 Step 5: reduce the orientation problem to one strict order

The main obstruction to using public I.44 directly is orientation. Public I.44 returns a parallelogram on a prescribed segment, but it does not expose which side of the segment contains the new parallelogram. I.45 needs the new piece to be external to the old one.

The first reduction is

```
i45_other_side_order
```

which shows that for parallelograms  $STUV$  and  $TUML$ ,

$$V - U - M \implies S - T - L.$$

The proof uses  $TU$  as a separator. Parallel endpoints give SameSide data for  $V, S$  and for  $M, L$ . Since  $V - U - M$ , points  $V$  and  $M$  are on opposite sides of the carrier  $TU$ . OppositeSide transport therefore puts  $S$  and  $L$  on opposite sides. Euclidean uniqueness of the parallel identifies the carrier  $SL$  through  $T$ , and incidence uniqueness identifies the crossing point with  $T$ . Thus only one strict order must be controlled by the construction.

The second reduction is

```
i45_extension_angle
```

If  $STUV$  is a parallelogram and  $T - U - E$ , then

$$\angle EUV \cong \angle STU.$$

The proof explicitly constructs the side-of-line configuration for the transversal, invokes the developed alternate-angle theorem for parallel lines, and replaces interior points by the corresponding SameRay endpoints. This lemma makes it possible to orient the copied I.44 shape using the old ray  $UV$  itself.

### 45.5.6 Step 6: construct the new triangle piece on the correct side

The central new helper is

```
i45_oriented_triangle_placement
```

Starting from  $STUV$  and a nondegenerate triangle  $PQR$ , it constructs points  $E, F, G$  such that

$$T - U - E, \quad UEFG \text{ is a parallelogram,} \quad G \text{ lies on the ray } UV,$$

and

$$\triangle PQR \approx \triangle UFG.$$

The proof first uses I.42 to obtain a source parallelogram with the correct triangle content and the angle  $STU$ . It then extends  $TU$  beyond  $U$  and lays off one copied side along that extension. The other copied side is laid off directly on the old ray  $UV$ . The lemma `i45_extension_angle` identifies the included angle, and the existing I.44 helper `i44_placed_parallelogram_content` certifies the piecewise scissors content.

Figure 45.2 shows this stage. The important feature is not merely that  $UEFG$  has the right shape, but that its point  $G$  is forced onto the chosen ray  $UV$ .

The next helper

```
i45_positioned_content_ordered
```

is a deliberately stronger local wrapper over the internal I.44 construction. The public helper `i44_positioned_content` returns the target parallelogram and its content equivalence but discards the intermediate order fact  $G - B - M$ . I.45 needs that fact, so it reuses the same I.44 construction of  $H, K, L, M$  and retains

$$G - B - M$$

before hiding the internal witnesses.

This is a representation/API issue, not a new geometric principle. No part of I.44 is assumed again. The local theorem replays already proved I.44 helpers and exposes one conclusion needed by the stronger downstream orientation invariant.

Combining these two helpers gives

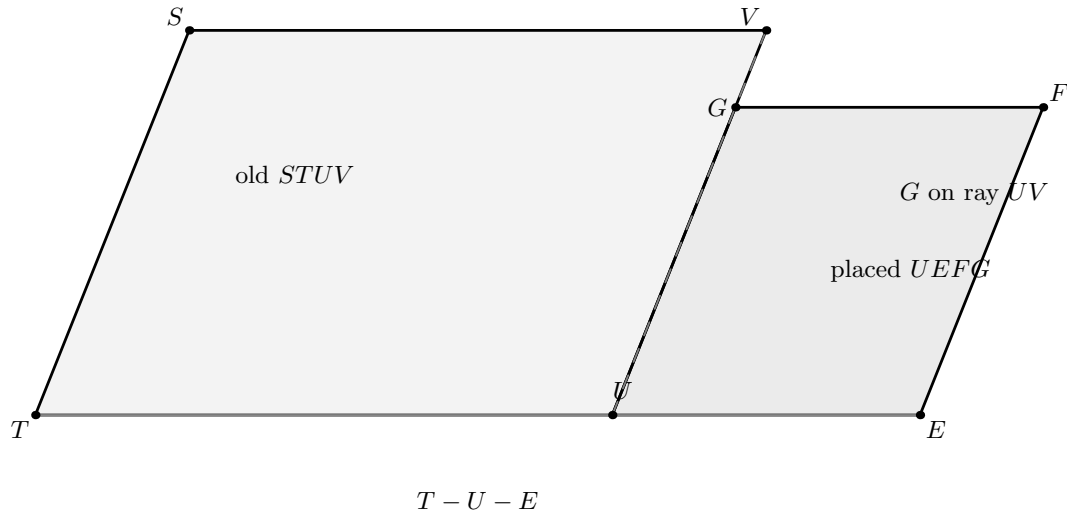


Figure 45.2: The orientation step used before the I.44 positioned construction. The old parallelogram is  $STUV$ . The base carrier is produced to  $E$ , giving  $T - U - E$ , while  $G$  is laid off on the old ray  $UV$ . Completing  $U, E, G$  gives the auxiliary parallelogram  $UEFG$ , which has the content of the new triangle. The explicit ray choice is the information not retained by the public I.44 wrapper.

```
i45_adjacent_triangle_parallelogram
```

with the clean output

$$TUML \text{ parallelogram, } \quad V - U - M, \quad \triangle PQR \approx TUML.$$

This is the exact object required by the paste lemmas.

### 45.5.7 Step 7: package one induction extension

The theorem

```
i45_extend_parallelogram
```

is the mathematical induction step.

Given an old parallelogram  $STUV$  whose angle  $STU$  is congruent to the prescribed angle and a new triangle  $PQR$ , it first constructs the adjacent parallelogram  $TUML$ . The order theorem supplies  $S - T - L$ . The paste theorem produces the outer parallelogram  $SLMV$ . The angle helper preserves the prescribed angle, while the content helper proves

$$\text{Par}(SLMV) \approx \triangle PQR + \text{Par}(STUV).$$

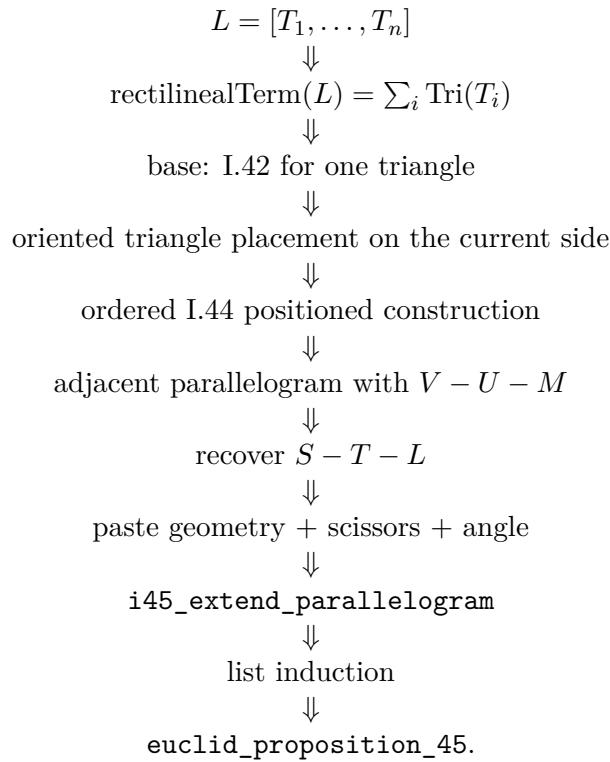
This theorem was initially isolated as a provisional local axiom while the required orientation and paste geometry were being developed. In the final file it is a derived theorem. Eliminating that provisional assumption is the main foundational closure of I.45.

### 45.5.8 Step 8: induction on the triangle list

The public proof finally becomes short.

If the list contains one triangle, I.42 gives the required parallelogram directly. If the list contains a head triangle and a nonempty tail, the induction hypothesis gives a parallelogram for the tail and `i45_extend_parallelogram` adjoins the head triangle. The only remaining content step is `equicomplementable_add` followed by transitivity.

The actual Lean DAG is therefore



## 45.6 Lean Formalization

The current file imports only the immediately preceding proposition:

```
import CGJteamLab.Proposition44
```

All earlier dependencies therefore enter through the established module DAG.

### 45.6.1 The rectilinear term

The executable content of a triangulated input is

```
def rectilinearTerm
  (L : HilbertTriangulatedFigure Geo) :
  HilbertScissorsTerm Geo :=
  (L.map
    (fun t =>
```

```
hilbertScissorsTriangle Geo t.A t.B t.C)).sum
```

The two normalization lemmas for cons and nil are used directly by the final induction.

### 45.6.2 The adjacent-piece theorem

The decisive geometric/content constructor has the following shape, written in the documentation's ASCII convention:

```
theorem i45_adjacent_triangle_parallelogram
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (S T U V P Q R : Geo.Point)
  (hLeft : IsParallelogram Geo S T U V)
  (hPQR : Not (Collinear Geo P Q R)) :
  exists M L : Geo.Point,
    IsParallelogram Geo T U M L /\
    Geo.Between V U M /\
    HilbertScissorsEquicomplementable Geo
      (hilbertScissorsTriangle Geo P Q R)
      (hilbertParallelogramTerm Geo T U M L)
```

The order clause is part of the public output of this helper because the paste step genuinely depends on orientation.

### 45.6.3 The induction-step theorem

The formerly provisional construction is now a theorem:

```
theorem i45_extend_parallelogram
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (S T U V P Q R X Y Z : Geo.Point)
  (hParallelogram : IsParallelogram Geo S T U V)
  (hAngle : Geo.AngleCongruent S T U X Y Z)
  (hPQR : Not (Collinear Geo P Q R)) :
  exists T' U' V' : Geo.Point,
    IsParallelogram Geo S T' U' V' /\
    Geo.AngleCongruent S T' U' X Y Z /\
    HilbertScissorsEquicomplementable Geo
      (hilbertParallelogramTerm Geo S T' U' V')
      (hilbertScissorsTriangle Geo P Q R +
       hilbertParallelogramTerm Geo S T U V)
```

This is the clean mathematical interface between the large local geometry and the final list induction.

### 45.6.4 The public proposition

The public theorem is

```

theorem euclid_proposition_45
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (X Y Z : Geo.Point)
  (hXYZ : Not (Collinear Geo X Y Z))
  (L : HilbertTriangulatedFigure Geo)
  (hNonempty : L != [])
  (hTriangles :
    forall t in L,
      Not (Collinear Geo t.A t.B t.C)) :
  exists S T U V : Geo.Point,
    IsParallelogram Geo S T U V /\
    Geo.AngleCongruent S T U X Y Z /\
    HilbertScissorsEquicomplementable Geo
      (rectilinearTerm Geo L)
      (hilbertParallelogramTerm Geo S T U V)

```

The displayed code is documentation-friendly ASCII. The compiled source uses the corresponding Lean quantifier and membership notation.

The theorem requires no compatibility condition between distinct triangles in  $L$ . This is intentional in the current statement and is the reason the representation discussion above is part of the formal result rather than a purely historical aside.

## 45.7 Relationship with Earlier Propositions

### 45.7.1 Proposition I.42

I.42 is the base case of the induction and also the shape/content source inside

```
i45_oriented_triangle_placement
```

It already knows how to turn one nondegenerate triangle into a parallelogram with a prescribed angle. I.45 does not reprove that content theorem.

The new task is to place the I.42 shape in a controlled orientation relative to an already constructed parallelogram.

### 45.7.2 Proposition I.44

I.44 is the principal direct dependency of the induction step. Its internal rigid-placement certificate

```
i44_placed_parallelogram_content
```

is used to certify the oriented auxiliary copy. Its ordered construction of  $H, K, L, M$  is reused by

`i45_positioned_content_ordered` so that I.45 can retain the strict order  $G - B - M$ .

This is a useful API lesson. Public I.44 intentionally exposes only the parallelogram, its prescribed angle, and its content. I.45 needs a stronger orientation invariant for adjacency. Rather than add an axiom or alter the meaning of public I.44, the local wrapper reuses the proved internal construction and exposes the one extra order fact downstream.

### 45.7.3 Proposition I.43

I.43 does not appear explicitly in the final induction theorem, but it remains inside the positioned I.44 construction. The content equivalence between the placed auxiliary parallelogram and the target parallelogram on the prescribed side is still the I.43 complement theorem, transported through the I.44 representation bridge.

Thus the content DAG is genuinely cumulative:

$$\text{I.42} \longrightarrow \text{I.44 (using I.43)} \longrightarrow \text{I.45}.$$

### 45.7.4 The Euclidean parallel block

Classically I.45 invokes I.29, I.30, and I.33 to recognize the final parallelogram. The Lean proof consumes the corresponding Hilbert-level parallel infrastructure rather than replaying those proposition numbers literally.

The important operations are transport of parallelism along collinear carriers, transitivity of parallels, and uniqueness of a parallel through a point. These are exactly the operations used in the paste geometry and in the proof that the second strict order follows from the first.

### 45.7.5 The content block I.35–I.41

These propositions are present transitively through I.42, I.43, and I.44. I.45 itself does not use a converse-content theorem and never infers geometry from equality of content. It only composes forward content certificates. Consequently it does not require the proper-part/positivity machinery that marks the difficult converse boundary at I.39.

## 45.8 Hilbert and Book Zero Components Used

### 45.8.1 Betweenness and SameRay

Strict betweenness is essential to adjacency. The two decisive relations for one paste are

$$S - T - L, \quad V - U - M.$$

They say more than collinearity: they determine which side of the common segment contains the new parallelogram.

SameRay transport is used at two levels. First, segment construction places copied side lengths on chosen rays. Second, angle identities replace an auxiliary point by another point on the same ray without changing the angle. The oriented placement depends critically on the fact that  $G$  is not merely collinear with  $U, V$ , but lies on the same ray from  $U$  as  $V$ .



### 45.8.2 SameSide, OppositeSide, and plane separation

The theorem `i45_other_side_order` is a concentrated use of Hilbert plane separation. Parallel endpoints supply SameSide facts, one strict betweenness supplies an OppositeSide witness, and OppositeSide transport propagates the orientation to the opposite boundary.

The crossing theorem also uses side-of-line data. The common side carrier is used as a separator, and the proof rules out one segment-line intersection before applying a forced Pasch argument. These are exactly the kinds of order facts that are visible in Euclid's diagram but must be named in Lean.

### 45.8.3 Pasch and crossing

The scissors proof requires an actual point  $X$  lying on both the outer diagonal-like segment and the common side. The helper `i45_paste_crossing` obtains it from the order/separation machinery rather than assuming the obvious visual crossing.

The conclusion is stronger than bare line intersection:

$$A - X - M, \quad B - X - C.$$

These strict relations are what allow the triangle-splitting constructors in `HilbertScissorsEq` to be invoked.

### 45.8.4 Euclidean uniqueness of parallels

The proposition is Euclidean. The type-class context

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

appears throughout the major local helpers.

Group IV enters when the proof identifies carriers by uniqueness/transitivity of parallels. It is used in the outer-parallelogram geometry, in the orientation theorem that recovers  $S - T - L$ , and inside the I.44 machinery used for positioned content. This is the formal counterpart of the specifically Euclidean parallel reasoning in the classical I.45 construction.

### 45.8.5 Congruence and angle construction

The oriented copy uses segment construction on prescribed rays and the existing congruence theory of parallelograms. The initial source parallelogram comes from I.42. No new SAS, SSS, or angle-construction theorem is introduced in I.45; those results enter through the established interface and through I.44.

### 45.8.6 Scissors operations

The content layer uses the existing operations

```

HilbertScissorsEq.split
HilbertScissorsEq.add
HilbertScissorsEq.trans
equicomplementable_of_scissorsEq
equicomplementable_refl
equicomplementable_symm
equicomplementable_trans
equicomplementable_add
crossing_quadrilateral_two_triangulations

```

No subtraction, cancellation, positivity, or scalar area is required. The entire direction is constructive and additive.

## 45.9 Formalization Notes

### 45.9.1 Geometry

The geometric induction step requires substantially more than the phrase “apply the next parallelogram adjacent to the first.” The formal proof must establish:

- a produced point  $E$  with  $T - U - E$ ;
- a copied point  $G$  on the selected ray  $UV$ ;
- noncollinearity needed to complete the auxiliary parallelogram;
- the ordered I.44 output  $G - U - M$  and its transport to  $V - U - M$ ;
- the opposite boundary order  $S - T - L$ ;
- both outer pairs of opposite sides parallel;
- an actual crossing point with two strict betweenness relations for the scissors refinement;
- preservation of the prescribed angle after the paste.

None of these facts is inferred from a drawing. They are proved by the existing Hilbert order, separation, incidence, congruence, and Euclidean parallel theory.

### 45.9.2 Content

The content proof is forward and additive.

The new triangle is first shown equicomplementable with the oriented auxiliary parallelogram. The ordered I.44 construction transfers that content to the adjacent parallelogram on the common side. The paste theorem gives a direct scissors equality between the outer parallelogram and the formal sum of the two inner parallelograms. Finally `equicomplementable_add` and transitivity combine the previous parallelogram with the new triangle.

No equal-content hypothesis is ever used to deduce a line, angle, side, or order relation. Therefore no de Zolt/proper-part principle is needed. I.45 also does not use numerical area, multiplication of lengths, or cancellation of an area factor.

### 45.9.3 Representation

There are two representation issues.

The first is the triangle-list input. `rectilinealTerm` is a sum of triangle terms, and the public theorem only requires each listed triangle to be nondegenerate. It does not prove that the list is a valid nonoverlapping triangulation of one polygonal region. This should remain explicit in any reuse of I.45.

The second is the orientation information hidden by public I.44. The theorem

```
i45_positioned_content_ordered
```

duplicates no axiom; it simply replays the proved I.44 construction while retaining the intermediate order  $G - B - M$ . This is a local API strengthening needed to compose constructions. A future refactoring could expose such oriented placement as a shared theorem, but the present I.45 proof does not require changing the public I.44 statement.

### 45.9.4 Axiomatic status

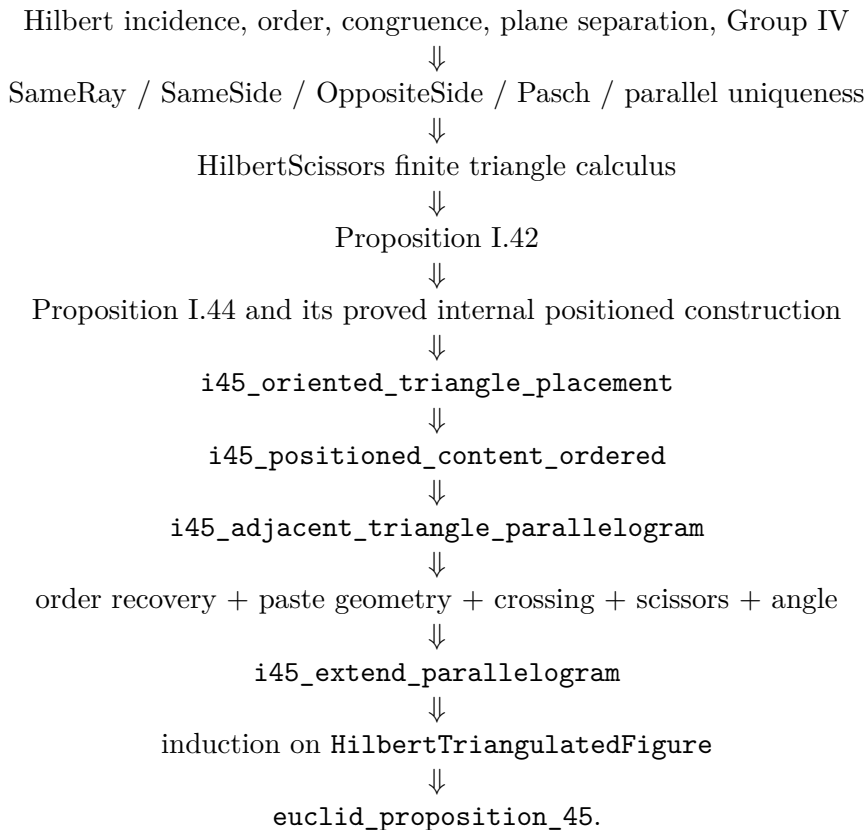
Proposition I.45 is Euclidean. Hilbert Group IV is used through the developed uniqueness/transitivity theory of parallels. The proposition also uses the incidence, order, plane-separation, congruence, and angle-construction layers inherited from the Hilbert interface and from I.44.

No continuity axiom is used by I.45. No Archimedean axiom is used. No new content axiom is introduced. The conclusion is equicomplementability, not a stronger universal equidecomposability claim.

The current source contains no active `axiom`, `sorry`, or `admit` declaration for I.45. The former provisional `i45_extend_parallelogram` assumption has been replaced by the derived theorem documented above.

## 45.10 Dependency Summary

The formal dependency chain is



The classical and formal DAGs can be compared succinctly as

Euclid: triangulate  $\rightarrow$  I.42/I.44 pieces  $\rightarrow$  parallel paste  $\rightarrow$  add equality,  
 Lean: triangle list  $\rightarrow$  oriented I.44 piece  $\rightarrow$  strict-order recovery  
 $\rightarrow$  scissors paste  $\rightarrow$  induction.

|                                                     |                 |
|-----------------------------------------------------|-----------------|
| New geometric axiom:                                | no              |
| New content axiom:                                  | no              |
| New proposition-local helpers:                      | yes             |
| New reusable helper promoted to Interface/Scissors: | no              |
| New Book Zero theorem:                              | no              |
| New Interface/Scissors theorem:                     | no              |
| New numerical area semantics:                       | no              |
| New full polygon-region semantics:                  | no              |
| Continuity used:                                    | no              |
| Archimedean axiom used:                             | no              |
| Hilbert Group IV used:                              | yes             |
| Public theorem compiles:                            | yes             |
| Full lake build:                                    | yes, 1458 jobs  |
| Pre-existing unrelated linter warning:              | yes, unused hBC |
| sorry in Proposition45.lean:                        | no              |

## 45.11 Conclusion

Proposition I.45 closes the application-of-areas construction by showing that the triangle-level machinery can be iterated without introducing a new area axiom. The final Lean proof is not a literal transcription of Euclid's quadrilateral diagram. Its central operation is a derived extension theorem: construct the next triangular content as an oriented adjacent parallelogram, recover the strict boundary order, paste the geometry, refine the paste in the scissors calculus, and preserve the prescribed angle.

The most delicate issue is orientation. Public I.44 intentionally hides the side on which its target parallelogram lies. I.45 solves this synthetically, not axiomatically: one copied side is placed on the produced base ray, the other on the old parallelogram ray, the ordered internal I.44 construction is reused, and plane separation converts one retained order relation into the second. Once this is done, the rest of the proof becomes a clean additive induction.

The proposition also answers the earlier representation checkpoint in a qualified way. The existing `HilbertTriangulatedFigure` and `HilbertScissorsTerm` layers are sufficient for the constructive content statement of I.45. No stronger polygon datatype is forced by this proof. At the same time, the public theorem treats a triangulated figure as a finite list of nondegenerate triangles and does not internalize nonoverlap, coverage, boundary, or simple-polygon semantics. Those notions remain a separate possible extension of the figure layer rather than a hidden claim of I.45.

With the provisional extension axiom eliminated and the full project build successful, I.45 is therefore closed as a theorem of the existing Hilbert plus scissors architecture.

# Appendix A

## Greenberg: A Hilbert-Style Reconstruction

### A.1 Purpose and scope

This appendix records a source audit of Euclid Book I, Propositions I.1–I.45, against Marvin Jay Greenberg’s *Euclidean and Non-Euclidean Geometries: Development and History*, fourth edition (2008) [4].

The purpose of the audit is deliberately stricter than identifying mathematically related results. For each proposition we ask:

- whether Greenberg explicitly identifies the Euclidean proposition;
- whether he supplies a complete proof, a proof skeleton, a hint, an exercise, an axiom, or only related infrastructure;
- whether the argument is genuinely synthetic or is presented using numerical measurement;
- what additional foundational issue Greenberg explicitly records, such as continuity, betweenness, separation, or the status of SAS;
- how the result relates to the present Lean reconstruction.

No observation from the source audit is intentionally discarded merely because it is not needed for the present proof. Such remarks may become important later when Book I is extended beyond the present checkpoint or when the Hilbert and Book Zero interfaces are reorganized.

**Edition-control note for the I.36–I.42 extension.** The existing I.1–I.35 audit was prepared against the fourth edition. The copy currently retained in the project Library for re-checking the extension is the third edition. The earlier fourth-edition observations are therefore left unchanged. New rows I.36–I.42 are phrased only at points where the available text gives edition-stable structural evidence: Greenberg’s Euclidean parallel and parallelogram infrastructure, his construction primitives, and his later numerical discussion of area. Where no proposition-level counterpart was located, the table says so explicitly rather than silently promoting a later area formula into a synthetic equal-content proof.

**Edition-control note for the I.43–I.45 extension.** The same conservative policy is retained for the final application-of-areas rows. No proposition-level Greenberg proof is asserted where none was located. The new entries record only the geometry/content separation that is supported by Greenberg’s Hilbert-style construction infrastructure and his explicit decision not to develop the full synthetic equal-content calculus.

## A.2 Why the fourth edition matters

The fourth edition is not a bibliographical reprint of the third. Greenberg explicitly reports substantial changes to Chapters 3 and 4. In particular, the proof of the crossbar theorem was moved from the exercises into the text; the terminology of Hilbert planes was aligned with Hartshorne; Dedekind’s axiom was no longer assumed as part of the basic Euclidean setup; the existence proof for midpoints was moved from the exercises into the text; and the discussion of measurement was rewritten to stress that real-number measurement is convenient rather than mathematically necessary.

These changes directly affect the status of several propositions in I.1–I.45, especially I.9, I.10, I.16, I.17, I.20, I.34, I.35, and the content block I.36–I.42.

## A.3 Comparison table

Table A.1: Euclid I.1–I.45 in Greenberg’s fourth edition

| Euclid | Greenberg counter-part                                                             | Status in Greenberg                                                                                                                                 | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                                           |
|--------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.1    | Construction of an equilateral triangle on a given segment.                        | Greenberg explicitly treats Euclid’s first proposition as containing a continuity gap. The gap is filled by the circle-circle continuity principle. | This is a foundational warning, not merely a proof detail. Greenberg emphasizes that a weak quadratic continuity principle is needed here. Thus I.1 has a different status from the later neutral results derived from incidence, betweenness, and congruence alone. This should be retained when the formal dependency graph of Book I is documented. |
| I.2    | Euclid’s second postulate; extension of a segment and laying off a congruent copy. | Greenberg asks the reader explicitly to prove Euclid’s second postulate and later states that it follows easily from Axioms C-1 and B-2.            | This is the primitive mechanism behind geometric segment addition. Greenberg later uses exactly this observation in his discussion of I.20. The point is important for Book Zero: “addition” of segments need not be numerical; it is implemented synthetically by extension plus segment construction.                                                |

Continued on next page

Table A.1 – continued from previous page

| Euclid | Greenberg counterpart                                                              | Status in Greenberg                                                                                                                                                                                                                          | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                                                    |
|--------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.3    | Cutting off from a greater segment a segment congruent to a given smaller segment. | No separate Greenberg proposition labelled as Euclid I.3 was located in the fourth-edition text. Its content is absorbed into segment construction, segment ordering, and betweenness.                                                       | The absence of a named counterpart is itself informative. Modern Hilbert-style organization decomposes I.3 into lower-level operations rather than treating it as an independent theorem. The Lean proof should therefore be compared with C-1, Proposition 3.12, and Proposition 3.13 rather than with a single Greenberg theorem.                             |
| I.4    | SAS triangle congruence.                                                           | SAS is Congruence Axiom C-6. Greenberg explicitly rejects Euclid's superposition argument as a proof in Euclid's stated system.                                                                                                              | This proposition becomes foundational in the Hilbert-style system. Greenberg stresses that SAS cannot be proved from the preceding incidence, betweenness, and congruence axioms without adding an appropriate congruence axiom. This remains an important distinction between Euclid's proposition order and the dependency order of the formal Hilbert layer. |
| I.5    | Proposition 3.10: base angles of an isosceles triangle.                            | A complete synthetic proof is printed, using SAS.                                                                                                                                                                                            | This is a direct Hilbert-style reconstruction of the mathematical content of I.5. It is one of the cleanest reference proofs for the formal development.                                                                                                                                                                                                        |
| I.6    | Proposition 3.18: converse of Proposition 3.10.                                    | The proposition is stated, but the proof is assigned as Exercise 27.                                                                                                                                                                         | Greenberg confirms the theorem and its dependency location but does not supply the completed derivation in the main text. The formal Lean proof therefore fills a genuine proof-level omission in the textbook presentation, without claiming a new mathematical theorem.                                                                                       |
| I.7    | Uniqueness configuration for triangles on the same base and on the same side.      | No explicit fourth-edition counterpart labelled as Euclid I.7 was located in the source audit. Related uniqueness is distributed through triangle laying-off, congruence axioms, side-of-line machinery, and subsequent congruence criteria. | This row should not be simplified to "absent." The relevant mathematics is present as infrastructure, but Greenberg does not appear to preserve I.7 as a named theorem. For future work it is worth recording exactly which formal uniqueness lemma replaces I.7 in the Hilbert interface.                                                                      |
| I.8    | Proposition 3.22: SSS criterion for congruence.                                    | SSS is stated as a proposition. Its proof is left as Exercise 32, with a substantial hint reducing the argument by the corollary to SAS and a same/opposite-side configuration.                                                              | Greenberg supplies the mathematical route but not a completed printed proof. This is especially relevant to the formal project because SSS was previously a foundational debt in the Hilbert interface. The exercise shows that Greenberg regards SSS as derivable from the Hilbert congruence machinery rather than as an additional axiom.                    |

Continued on next page



Table A.1 – continued from previous page

| Euclid | Greenberg counter-part                                                  | Status in Greenberg                                                                                                                                                                                                                        | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                                                                                                      |
|--------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.9    | Proposition 4.4(a): every angle has a unique bisector.                  | The proposition is stated in the text; Exercise 6(b) asks for the proof of existence and uniqueness. Greenberg says the angle-bisector construction follows easily from the midpoint construction.                                         | Greenberg explicitly comments on Euclid's own I.9: Euclid uses I.1 and also relies tacitly on a betweenness fact visible in the diagram, namely that a constructed point lies on the required opposite side. This note should be retained because it identifies the hidden order/separation content that a Lean proof must expose.                                                                                |
| I.10   | Proposition 4.3: every segment has a unique midpoint.                   | A major change from the third edition: the existence proof is now in the text as an explicit 18-step proof skeleton. The reader is asked to justify the steps; uniqueness remains Exercise 6(a).                                           | Greenberg explicitly says that Euclid's proof of I.10 tacitly requires the crossbar theorem. His replacement construction avoids using an equilateral triangle and therefore avoids importing the continuity issue from I.1. This distinction is valuable for future optimization of the formal dependency graph.                                                                                                 |
| I.11   | Perpendicular erected at a point on a line; part of Proposition 3.16.   | Proposition 3.16 gives a complete proof of existence of a perpendicular through an arbitrary point, with a separate case when the point lies on the line.                                                                                  | The modern theorem merges the content of I.11 and I.12. When the point lies on the line, Greenberg uses the existence of a right angle and angle laying-off. The result is obtained without invoking Euclid's circle construction.                                                                                                                                                                                |
| I.12   | Perpendicular dropped from a point to a line; part of Proposition 3.16. | A complete proof is printed for the case in which the point is not on the line.                                                                                                                                                            | Greenberg explicitly warns that Euclid's original construction of I.12 tacitly assumes line-circle continuity. His proof of Proposition 3.16 avoids that route; indeed, he notes that using line-circle continuity there would be circular because his later derivation of that continuity principle uses Proposition 3.16. This is a particularly important dependency note for any future foundational cleanup. |
| I.13   | Supplementary-angle configuration when a line stands on another line.   | No direct proposition labelled I.13 was located. The relevant theory is distributed among the definition of supplementary angles, betweenness, Proposition 3.14 on supplements of congruent angles, and angle addition (Proposition 3.19). | The source audit therefore treats I.13 as an example of reorganization: Greenberg proves and uses more general structural facts rather than retaining the Euclidean proposition as a named unit. In Lean, the corresponding proof obligations naturally live in the angle and ray interface.                                                                                                                      |
| I.14   | Converse straight-line criterion from adjacent supplementary angles.    | No direct proposition labelled I.14 was located in the fourth-edition text. Its ingredients occur in angle laying-off uniqueness, supplementary-angle theory, ray order, and angle addition/subtraction.                                   | As with I.13, the mathematical content is absorbed into a stronger modern infrastructure. This should be preserved as a source-audit observation rather than recorded simply as "not found."                                                                                                                                                                                                                      |

Continued on next page

Table A.1 – continued from previous page

| Euclid    | Greenberg counterpart                                                                                                               | Status in Greenberg                                                                                                                                                                                                                         | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                                                                                                              |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.15      | Proposition 3.15(a): vertical angles are congruent.                                                                                 | The theorem is stated in the text. Exercise 25 asks the reader to deduce it from Proposition 3.14.                                                                                                                                          | Thus the result is present but the proof is not fully written out in the main text. The dependency on supplements of congruent angles matches the synthetic route used in formal reconstruction.                                                                                                                                                                                                                          |
| I.16      | Exterior Angle Theorem 4.2.                                                                                                         | A complete proof is printed. Greenberg then separately analyzes Euclid’s original proof and fills its diagrammatic gap with betweenness and plane separation.                                                                               | This is one of the richest entries in the audit. Greenberg stresses that the missing assumption is not that “lines are infinite in extent.” What is needed is the betweenness theory, especially plane separation B-4 and its consequences. He also notes that the theorem fails in elliptic geometry because the relevant separation/parallel structure fails. These remarks are potentially important well beyond I.16. |
| I.17      | Corollary 2 to the Exterior Angle Theorem: the sum of the degree measures of any two angles of a triangle is less than 180 degrees. | A complete short proof is printed, but it is stated in the language of numerical angle measure.                                                                                                                                             | This is not the purely synthetic presentation used in the Lean reconstruction. Greenberg is explicit, however, that the introduction of numbers is a convenience: Hartshorne’s treatment by congruence classes is mathematically preferable and valid in all Hilbert planes. Thus the numerical wording should not be mistaken for a foundational need for real numbers.                                                  |
| I.18–I.19 | Proposition 4.5: in a triangle, the greater side lies opposite the greater angle and conversely.                                    | Greenberg combines I.18 and I.19 into one proposition. The proof is Exercise 8, with a specific synthetic hint: choose a point on a side, use an isosceles triangle, apply the exterior angle theorem, and use trichotomy for the converse. | This is strong confirmation of the dependency structure reconstructed in Lean. The textbook does not provide the full formal derivation, so the Lean development must supply all order, noncollinearity, ray, and angle-transport details suppressed by the hint.                                                                                                                                                         |

Continued on next page

Table A.1 – continued from previous page

| Euclid | Greenberg counterpart                                                                                                                                   | Status in Greenberg                                                                                                                                                                                                                                                    | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.20   | Triangle inequality.                                                                                                                                    | A complete seven-step proof is printed. It is written using segment length notation and addition, but Greenberg immediately explains that the numerical language is only a convenience.                                                                                | This row changes significantly relative to a superficial reading. Greenberg says that Euclid’s proof is the same proof, with “two sides taken together” interpreted as geometric addition of segment congruence classes. He explains that Euclid II is obtained from C-1 and B-2 and that C-3 makes segment-class addition well-defined. Hence Archimedes’ axiom and the full measurement theorem are not required. This is very close in mathematical structure to the present synthetic Lean proof. Greenberg also records Heron’s alternative proof as an exercise: bisect the angle, use the crossbar theorem, the exterior angle theorem, and Proposition 4.5. |
| I.21   | Interior-point comparison in a triangle; related to the exterior-angle theorem, Proposition 4.5, segment order, and triangle inequality infrastructure. | No separate fourth-edition proposition explicitly identified as Euclid I.21 was located in the present audit. Its ingredients belong to the neutral comparison theory developed around the exterior-angle theorem, side-angle comparison, and the triangle inequality. | This is another case in which Greenberg’s organization is more structural than Euclid’s proposition sequence. The Lean reconstruction is especially informative here because it exposes the exact betweenness and angle-order transports needed to turn the interior-point configuration into the two strict segment comparisons.                                                                                                                                                                                                                                                                                                                                   |
| I.22   | Converse to the triangle inequality: construction of a triangle from three segments satisfying the pairwise triangle inequalities.                      | Greenberg explicitly ties the converse to the triangle inequality to circle-circle continuity. In his foundational organization this is a continuity result rather than a consequence of neutral incidence, betweenness, and congruence alone.                         | This is the principal new foundational observation beyond I.20. The Lean reconstruction likewise isolates the continuity-dependent existence step from the synthetic segment arithmetic expressing that each pair of sides taken together is greater than the third. The comparison confirms that the continuity assumption is genuine and should remain visible in the dependency graph.                                                                                                                                                                                                                                                                           |
| I.23   | Construction of an angle congruent to a given angle on a prescribed side of a given ray; Greenberg’s angle-laying-off infrastructure.                   | The mathematical content is already part of the Hilbert-style angle construction machinery used by Greenberg. It is therefore not naturally presented as a derived Euclidean construction depending on I.22.                                                           | This sharply illustrates the difference between Euclid’s historical dependency order and the Hilbert foundation. Euclid derives I.23 using earlier constructions, whereas the present Lean theorem is an interface wrapper around the stronger angle-construction principle already available in the congruence layer.                                                                                                                                                                                                                                                                                                                                              |

Continued on next page

Table A.1 – continued from previous page

| Euclid | Greenberg counter-part                                                                                                                                                                    | Status in Greenberg                                                                                                                                                                                                                                                                                                                            | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                                                                                                                    |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.24   | Comparison of third sides from two equal-side configurations with an inequality between the included angles; related to angle laying-off, SAS, Proposition 4.5, and neutral order theory. | No separate fourth-edition proposition explicitly identified as Euclid I.24 was located in the present audit. The required mathematical ingredients occur in Greenberg’s general triangle-congruence and side-angle comparison infrastructure.                                                                                                 | The formal reconstruction adds an important point not captured by a mere theorem-name correspondence: Euclid’s displayed construction has a configuration gap. The Lean proof must analyze the possible relative positions of the auxiliary points rather than reading one configuration from the diagram. Thus I.24 is a particularly clear example of the project’s role in exposing hidden order information.                |
| I.25   | Converse comparison for two triangles with two corresponding sides congruent; Greenberg Proposition 4.6.                                                                                  | Proposition 4.6 states, for two triangles with two corresponding sides congruent, that inequality of the included angles is equivalent to inequality of the third sides. Its proof is assigned as Exercise 22, with a detailed hint using the crossbar theorem, Proposition 4.5, isosceles-triangle arguments, and the Exterior Angle Theorem. | Greenberg packages the mathematical content of Euclid I.24 and I.25 into a single biconditional comparison theorem. Thus I.25 is not a separate stage in his organization: it is the converse direction of the same neutral comparison result. This strongly confirms the formal project’s decision to treat the two propositions as neighboring directions of one comparison mechanism rather than as unrelated constructions. |
| I.26   | ASA and SAA triangle congruence; Proposition 3.17 and Proposition 4.1.                                                                                                                    | Proposition 3.17 is the ASA criterion. Its proof is supplied as Exercise 26 in Chapter 3. Proposition 4.1 gives the SAA criterion after the Exterior Angle Theorem; Greenberg leaves its proof as an exercise, and later Exercise 10 gives a correct proof skeleton.                                                                           | Greenberg separates the two cases that Euclid combines in I.26. The ASA case belongs directly to the Hilbert congruence development, whereas the SAA case is placed after the neutral Exterior Angle Theorem. This is useful for the Lean dependency graph: the two formal branches need not have identical internal proofs even though Euclid states them together.                                                            |
| I.27   | Theorem 4.1, Alternate Interior Angle Theorem.                                                                                                                                            | A complete proof is printed: congruent alternate interior angles imply that the two lines are parallel. Greenberg proves this in neutral geometry, before assuming Hilbert’s parallel postulate.                                                                                                                                               | This is the precise modern Hilbert-style counterpart of the essential content of I.27. Greenberg explicitly warns immediately afterward not to use its converse. Hence the Lean reconstruction’s placement of I.27 on the neutral side of the parallel-postulate boundary is confirmed directly by the source.                                                                                                                  |

Continued on next page

Table A.1 – continued from previous page

| Euclid | Greenberg counter-part                                                                                       | Status in Greenberg                                                                                                                                                                                                                                                                                        | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                                                                                                                   |
|--------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.28   | Theorem 4.1 together with the corresponding-angle reduction recorded in Chapter 4, Exercise 32.              | The alternate-interior criterion is Theorem 4.1. Exercise 32 asks the reader to prove that corresponding angles are congruent if and only if alternate interior angles are congruent. The supplementary-angle machinery supplies the same-side-interior formulation.                                       | Greenberg does not preserve Euclid I.28 as one named theorem, but its two classical criteria reduce to the neutral Alternate Interior Angle Theorem and elementary angle relations. This agrees with the Lean reconstruction: I.28 remains neutral and should not be confused with the converse parallel-to-angle direction used in I.29.                                                                                      |
| I.29   | Proposition 4.8: the converse to Theorem 4.1.                                                                | Greenberg states that the converse Alternate Interior Angle Theorem is logically equivalent to Hilbert’s parallel postulate. Proposition 4.8 is assigned as Exercise 5; the hint proves both directions of the equivalence, using uniqueness of parallels and angle laying-off.                            | This is the exact foundational boundary reached by Euclid at I.29. Greenberg also remarks earlier that Euclid postponed use of the fifth postulate until his twenty-ninth proposition. The Lean theorem therefore correctly requires the Euclidean-plane structure precisely when parallelism is used to recover angle congruence.                                                                                             |
| I.30   | Transitivity of parallelism; consequence/equivalent form of the parallel postulate in Greenberg’s Chapter 4. | Immediately after Theorem 4.5 Greenberg lists equivalent forms of the parallel postulate. Proposition 4.7 concerns a transversal meeting both of two parallel lines, and its exercise explicitly asks the reader to deduce that transitivity of parallelism is equivalent to Hilbert’s parallel postulate. | I.30 is therefore genuinely Euclidean in Greenberg’s logical analysis, not a neutral theorem about the definition of parallel lines. This matches the formal reconstruction, where the theorem is downstream of the Euclidean uniqueness principle rather than merely of neutral existence of parallels.                                                                                                                       |
| I.31   | Corollary 2 to Theorem 4.1: existence of a parallel through a point not on a given line.                     | Greenberg proves existence in neutral geometry by dropping a perpendicular from the point to the given line and then erecting a second perpendicular. He emphasizes that existence is available without the parallel postulate; what cannot be proved neutrally is uniqueness.                             | This distinction is especially important for I.31. Euclid’s construction asks only for a parallel through the point, so its existence content is neutral in Greenberg’s Hilbert-style theory. Hilbert’s Group IV becomes relevant only when one asks that this parallel be the unique parallel. The source therefore confirms the existence-versus-uniqueness distinction that must remain explicit in the Lean documentation. |

Continued on next page

Table A.1 – continued from previous page

| Euclid | Greenberg counterpart                                                                                                                                                                | Status in Greenberg                                                                                                                                                                                                                                                                                                                                                                                                            | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.32   | Proposition 4.11: Hilbert’s parallel postulate implies that every triangle has angle sum $180^\circ$ ; together with Proposition 4.8, the converse Alternate Interior Angle Theorem. | Greenberg states Proposition 4.11 as a consequence of Hilbert’s parallel postulate and assigns its proof as Exercise 8. He then defines triangle defect and stresses that exact angle sum $180^\circ$ cannot be obtained in neutral geometry merely from the preceding theory. Elsewhere he identifies the converse Alternate Interior Angle Theorem as equivalent to the parallel postulate.                                  | This is the closest Greenberg counterpart to the second half of Euclid I.32 and confirms its genuinely Euclidean status. Greenberg formulates the result numerically using degree measure, whereas the Lean reconstruction deliberately keeps the synthetic content: extend a side, split the exterior angle by a correctly oriented parallel ray, and represent “two right angles” without introducing degree arithmetic. Thus the source and the formalization agree on the dependency while using different languages for angle addition. |
| I.33   | Equal and parallel segments joined in the same direction; the joining segments are themselves equal and parallel.                                                                    | No separate fourth-edition proposition explicitly identified as Euclid I.33 was located in the present audit. Its mathematical content belongs to Greenberg’s Euclidean parallel and parallelogram infrastructure rather than appearing as a preserved Euclidean proposition with its own proof.                                                                                                                               | This is a useful negative result for the source audit. Greenberg should not be cited as supplying a one-for-one proof of I.33. For the Lean reconstruction the relevant comparison is structural: once the Euclidean parallel theory is available, the configuration can be packaged as parallelogram data and congruence consequences.                                                                                                                                                                                                      |
| I.34   | Opposite sides and opposite angles of a parallelogram are congruent.                                                                                                                 | Greenberg explicitly lists I.34 among Euclid’s results using the parallel postulate. In the Chapter 4 exercises he also stresses the foundational point: in a general Hilbert plane opposite sides of a parallelogram need not be congruent, whereas in a plane satisfying the Euclidean parallel postulate opposite sides and opposite angles are congruent.                                                                  | This is a particularly strong source match for the Lean reconstruction. I.34 is not merely convenient parallelogram algebra; Greenberg uses it to exhibit the exact Euclidean boundary. The theorem therefore belongs in the reusable Euclidean parallelogram interface and supplies the congruence data needed immediately in I.35.                                                                                                                                                                                                         |
| I.35   | Parallelograms on the same base and in the same parallels have equal content.                                                                                                        | No dedicated fourth-edition proof of Euclid I.35 was located in the present audit. Greenberg does, however, explain the relevant semantic notion. He distinguishes numerical area from <i>equal content</i> , described informally by dissecting polygonal figures into pieces and reassembling them. He then explicitly says that he will not develop this theory in the book and refers the reader to Hartshorne, Chapter 5. | For formalization this absence is highly informative. Greenberg supplies the conceptual warning that Euclid’s equality of figures should not be silently replaced by numerical area, but he is not the source for the full synthetic calculus required by I.35. The present Lean scissors layer therefore agrees with Greenberg’s interpretation of content while its rigorous proof architecture must be controlled by sources that actually develop the theory, especially Hilbert and Hartshorne.                                         |

Continued on next page

Table A.1 – continued from previous page

| Euclid | Greenberg counterpart                                                       | Status in Greenberg                                                                                                                                                                                                                                                                                                                                                                                                                                  | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                                                                                                                |
|--------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.36   | Parallelograms on equal bases and in the same parallels have equal content. | No dedicated Greenberg proof of Euclid I.36 was located in the available text. The Euclidean geometry needed to formulate the theorem is present: Greenberg develops the ordinary consequences of the parallel postulate and treats the familiar congruence properties of Euclidean parallelograms. What is still missing is the synthetic equal-content calculus already noted at I.35.                                                             | This is exactly the distinction needed by the formal reconstruction. The geometric shell of I.36 belongs to Greenberg’s Euclidean parallel layer, but the conclusion belongs to the scissors/equicomplementability layer. A later numerical area theory could verify the equality after introducing measures, but that would be a different proof language and should not be retrofitted as the source of the Lean theorem. |
| I.37   | Triangles on the same base and in the same parallels have equal content.    | No proposition-level synthetic counterpart was located. In Greenberg’s later discussion of numerical area, an arbitrary Euclidean triangle has area one half of base times altitude, and area is required to be invariant under congruence and additive under a split into triangles. These facts make the numerical analogue of I.37 immediate, but they occur in a later measurement framework rather than in a developed theory of equal content. | The distinction is foundationally important. The Lean theorem concludes <code>HilbertScissorsEquicomplementable</code> , not equality of real numbers. The formal proof therefore follows Hilbert/Hartshorne-style content architecture rather than Greenberg’s later formula $K = \frac{1}{2}bh$ . Greenberg remains a semantic cross-check, not the proof source.                                                         |
| I.38   | Triangles on equal bases and in the same parallels have equal content.      | Again, no dedicated synthetic counterpart was located. Greenberg’s later triangle-area formula also makes the numerical statement immediate when the bases and altitudes agree. The source does not, however, provide the proposition-level congruent-copy and equicomplementability argument required by the current formalization.                                                                                                                 | The Lean reconstruction is deliberately more conservative semantically: it reduces the equal-base case to I.37 by an oriented congruent copy and transitivity of equal content. Greenberg supports the expected Euclidean truth of the result but does not determine that DAG.                                                                                                                                              |

Continued on next page

Table A.1 – continued from previous page

| Euclid | Greenberg counterpart                                                                                             | Status in Greenberg                                                                                                                                                                                                                                                                                                                                                                                                                              | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.39   | Equal-content triangles on the same base and on the same side are between the same parallels.                     | No Greenberg theorem corresponding to this converse equal-content statement was located in the available text. His later numerical area function is positive and additive, so a metric/area treatment can distinguish a proper subtriangle from the whole, but the book does not develop the synthetic content-order principle needed to reproduce Euclid’s converse argument in the same language as I.35–I.38.                                 | This is where Greenberg ceases to be an adequate semantic source for the formal proposition. The Lean development must expose a strict content principle explicitly. The current reconstruction isolates this as scissors positivity rather than silently importing a numerical area or a whole-greater-than-part axiom from the diagram.                                                                                                                                |
| I.40   | Equal-content triangles on equal bases and on the same side are between the same parallels.                       | No separate counterpart was located. Structurally this is the equal-base companion to I.39. Greenberg supplies the Euclidean congruence and parallel infrastructure needed to move between equal-base configurations, but not a synthetic converse theory of equal content.                                                                                                                                                                      | The present Lean proof therefore reuses the geometric copy construction from I.38 and the strict-content converse isolated at I.39. Greenberg is useful for the geometry of the reduction, but he does not remove or replace the content-positivity obligation inherited from I.39.                                                                                                                                                                                      |
| I.41   | A parallelogram on the same base and in the same parallels is double the triangle.                                | No dedicated Greenberg reconstruction of I.41 was located. His later numerical discussion gives the compatible measurement picture: a diagonal of a rectangle produces congruent halves, and a general triangle has area one half base times altitude. This confirms the familiar numerical meaning of “double,” but it is not a proof in Greenberg’s undeveloped equal-content language.                                                        | The formal theorem interprets “double” compositionally as a sum of two triangle terms. I.34 supplies the geometric diagonal congruence, I.37 supplies equal content on the common base, and the scissors addition theorem transports the equality. Greenberg’s text is therefore a consistency check, not the architectural source.                                                                                                                                      |
| I.42   | To construct a parallelogram equal in content to a given triangle and having an angle congruent to a given angle. | No proposition-level counterpart was located. The geometric ingredients are nevertheless all characteristic of Greenberg’s Hilbert-style organization: midpoint construction, laying off a congruent angle on a chosen side, drawing a parallel through a point, and Euclidean parallelogram theory. The missing piece is again the synthetic content calculus that turns the completed parallelogram into a figure equal to the given triangle. | This makes I.42 a particularly clean boundary case for the source audit. Greenberg can control the construction layer but not the final content identity in the language used by Lean. The formal proof therefore combines Greenberg-compatible geometric infrastructure with the already established I.38/I.41 scissors results. The hidden intersection and orientation data are formal obligations of the construction, not consequences of a numerical area formula. |

Continued on next page



Table A.1 – continued from previous page

| Euclid | Greenberg counterpart                                                                           | Status in Greenberg                                                                                                                                                                                                                                                                            | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                               |
|--------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.43   | Complements of parallelograms about a diagonal are equal in content.                            | No proposition-level Greenberg counterpart in a developed synthetic equal-content calculus was located. His Euclidean parallelogram theory controls the geometry, while his later area discussion is numerical.                                                                                | The formal content step should therefore remain in the scissors layer. The source is useful for the parallelogram and diagonal geometry but does not supply a new cancellation or complement axiom.                                                        |
| I.44   | Apply a parallelogram of prescribed content and angle to a specified segment.                   | No direct Greenberg reconstruction was located. The geometric operations are all native to his Hilbert-style setup: angle construction, segment order, parallel through a point, and Euclidean parallelogram theory. The equal-content conclusion is outside the synthetic theory he develops. | This is another clean geometry/content separation. Greenberg can control the placement and the Euclidean parallel strength; the content equality is supplied by the established I.42/I.43 scissors machinery.                                              |
| I.45   | Construct a parallelogram in a prescribed angle equal in content to a given rectilinear figure. | No proposition-level counterpart in Greenberg’s synthetic development was located. His text distinguishes equal content from numerical area but refers the complete synthetic treatment to Hartshorne; the geometric construction operations remain available.                                 | For Lean the source is therefore a control on incidence, order, angle construction, and Euclidean parallelism, not on the semantics of the final content relation. The finite-triangle induction belongs to the scissors and figure-representation layers. |

## A.4 Structural observations from the audit

The fourth edition makes several general points that are directly relevant to the formal reconstruction.

First, Greenberg states that the main purpose of Chapter 3 is to fill gaps in Euclid’s presentation, while also acknowledging that not every gap is filled there. He emphasizes that almost all elementary synthetic Euclidean geometry can nevertheless be developed from his axiomatic system.

Second, Greenberg separates three kinds of foundational phenomena that are easy to blur in an informal reading of Euclid:

1. **incidence, betweenness, and congruence**, which define a Hilbert plane and support a large neutral fragment;
2. **quadratic continuity**, needed for constructions such as Euclid I.1 and, later, the converse to the triangle inequality;
3. **measurement by real numbers**, which is convenient but not required for the elementary synthetic content under study.

Third, the fourth edition repeatedly points out that statements which look obvious in diagrams

encode nontrivial order information. The crossbar theorem is the most visible example. Its proof was promoted from an exercise to the main text in this edition, which makes the book especially useful as a reference for the formal project.

Fourth, Greenberg’s organization confirms that proposition numbering is not a dependency structure. Several Euclidean propositions disappear as named theorems because their content is absorbed into more general machinery; conversely, some single Greenberg propositions combine two Euclidean propositions.

Fifth, the extension through I.42 reveals a second change of language. Greenberg’s treatment of I.34 remains fully inside the Euclidean theory of parallels and parallelograms. At I.35 Euclid begins to compare plane figures by content. Greenberg explicitly distinguishes this from numerical area and describes the cut-and-paste interpretation, but he does not develop the resulting theory. Consequently I.35–I.45 cannot be recovered from Greenberg as a continuous synthetic proof block.

Sixth, Greenberg’s later numerical theory of area is useful as a semantic cross-check but not as an architectural substitute. There he requires area to be invariant under congruence and additive under subdivision, and derives for a Euclidean triangle the familiar value

$$K = \frac{1}{2}bh.$$

This makes the forward equal-area content of I.37, I.38, and I.41 transparent numerically. It does not by itself supply the finite-scissors relation used by the formal project, nor should it be used to erase the converse-content issue at I.39–I.40.

Seventh, I.42 separates the two roles of Greenberg especially clearly. Its geometry is naturally expressed in his Hilbert-style primitives: midpoint, angle layoff on a chosen side, parallel construction, and parallelogram theory. Its conclusion, however, belongs to the equal-content language that Greenberg does not develop. Thus the source remains highly useful for the construction API even after it ceases to be the proof source for the content calculus.

## A.5 Notes that should be retained for future work

The following source observations should be retained because they may matter in later Book I work.

- Greenberg defines a Hilbert plane as a model of the thirteen incidence, betweenness, and congruence axioms.
- His Euclidean setup adds Hilbert’s Euclidean parallel axiom and circle-circle continuity; Dedekind’s axiom is deliberately not assumed in the fourth edition.
- He explicitly distinguishes circle-circle continuity from Archimedes’ axiom and from Dedekind completeness.
- The converse to the triangle inequality, Euclid I.22, is tied directly to circle-circle continuity. This is now part of the completed I.1–I.45 block and provides the clearest new continuity issue after I.1.
- Greenberg repeatedly refers to Hartshorne for the technically complete treatment of elementary geometry without real numbers, especially for congruence classes of angles and their addition.

- The fourth edition states that numerical measurement is introduced in Chapter 4 to shorten the exposition, not because the geometry requires real numbers.
- Greenberg explicitly rejects the widespread claim that Euclid I.16 needs an additional assumption that lines are “infinite in extent.” His diagnosis is that the missing machinery is betweenness and plane separation.
- The existence proof for a midpoint in Proposition 4.3 is not Euclid’s own construction. Greenberg deliberately gives a route which avoids dependence on equilateral triangles and therefore avoids importing the continuity requirement of I.1.
- Greenberg gives Heron’s alternative proof of the triangle inequality as an exercise. This provides a second synthetic route to I.20 that may later be useful for comparing proof graphs or designing tactics.
- Greenberg explicitly treats I.34 as Euclidean: in a general Hilbert plane a parallelogram need not have congruent opposite sides, while the Euclidean parallel postulate restores the familiar opposite-side and opposite-angle congruences.
- Greenberg distinguishes numerical area from equal content and explains the latter by finite dissection and reassembly of polygonal figures.
- He explicitly declines to develop the theory of equal content and refers the reader to Hartshorne, Chapter 5. Consequently Greenberg should not be used as if he supplied a complete source proof of I.35.
- Greenberg later mentions the Wallace–Bolyai–Gerwien equidecomposability theorem, but that later area theory should not be retrofitted as the immediate semantic basis of Euclid I.35 in this audit.
- For I.36–I.38 the absence of a developed equal-content calculus persists. Greenberg’s Euclidean parallel and parallelogram theory controls the geometry, while the Lean scissors layer controls the conclusion.
- Greenberg’s later numerical area discussion assumes invariance under congruence and additivity and derives the Euclidean triangle formula  $K = \frac{1}{2}bh$ . This is a useful semantic check on I.37–I.41 but not a replacement for synthetic equicomplementability.
- No direct Greenberg counterpart for the converse content propositions I.39–I.40 was located. The formal positivity issue isolated there must therefore be documented from other sources rather than attributed to Greenberg.
- I.42 uses geometric operations that fit Greenberg’s Hilbert-style API, but its equal-content conclusion still lies outside the theory he develops. This construction/content split should be preserved in later source audits of I.44–I.45.

## A.6 I.43–I.45: the same construction/content split at larger scale

Nothing in I.43–I.45 changes Greenberg’s role in the source hierarchy. His Hilbert-style development supplies the geometric operations needed to recognize complements, place parallelograms, and control the Euclidean parallel constructions. It still does not provide the complete synthetic calculus of equal content used by the Lean files.

The growth from I.42 to I.45 is therefore not a reason to import Greenberg’s later numerical area formulas into the proof. The correct formal division remains:

Greenberg-compatible geometry + Hilbert/Hartshorne scissors content + triangle-list representation.

At I.45 the last term becomes visible as an independent architectural issue: a finite list suffices for the additive theorem, while a full rectilinear-region semantics would require additional representation invariants.

## A.7 Result of the fourth-edition query

The fourth edition does not provide a continuous sequence of forty-five fully written synthetic proofs corresponding one-for-one to Euclid I.1–I.45. It provides something more useful for foundational analysis: a reorganized Hilbert-style theory in which the Euclidean results have different statuses.

The extension from I.25 to I.32 makes the logical boundary at I.29 particularly sharp. I.25 is absorbed together with I.24 into Greenberg’s biconditional triangle-comparison Proposition 4.6. I.26 is split between the ASA criterion of Proposition 3.17 and the SAA criterion of Proposition 4.1. I.27 and I.28 remain on the neutral side: the Alternate Interior Angle Theorem proves angle congruence implies parallelism, and the corresponding-angle formulation is reduced to it.

At I.29 the direction reverses. Greenberg explicitly identifies the converse Alternate Interior Angle Theorem as logically equivalent to Hilbert’s parallel postulate. Transitivity of parallelism, corresponding to I.30, is likewise listed as an equivalent form. By contrast, the existence statement in I.31 is already available in neutral geometry; only uniqueness of the parallel requires Hilbert’s parallel postulate. Proposition 4.11 then states that Hilbert’s parallel postulate implies that every triangle has angle sum  $180^\circ$ , giving Greenberg’s numerical counterpart to the synthetic I.32 reconstruction.

The extension to I.33–I.34 remains within this Euclidean parallel block. No separate I.33 theorem was located, but I.34 is explicit and foundationally sharp. Greenberg notes that opposite sides of a parallelogram need not be congruent in a general Hilbert plane and asks the reader to prove the usual opposite-side and opposite-angle congruences under the Euclidean parallel postulate. Thus the familiar parallelogram theorem is genuinely Euclidean in his organization.

At I.35 the subject changes. Greenberg identifies the conceptual difference between numerical area and equality of figures by dissection, but does not develop a proposition-by-proposition synthetic content theory. That remains the decisive fact through I.42. The new propositions do not reveal seven missing Greenberg theorems waiting to be transcribed; instead they divide naturally into three classes.

For I.36–I.38 Greenberg supplies most of the geometric environment but not the formal equal-content relation. His later numerical area discussion confirms the familiar base-altitude consequences, yet those formulas belong to a measurement framework. The Lean proofs therefore use Hilbert/Hartshorne content semantics and Greenberg-compatible geometric infrastructure.

For I.39–I.40 the mismatch is sharper. These propositions reverse the content implication and attempt to recover geometric information from equality of figures. No direct Greenberg theorem in the available text supplies the strict content principle needed for that converse. The project’s explicit scissors positivity assumption must consequently be audited elsewhere rather than hidden under a numerical area calculation.

I.41 returns to the forward direction and is numerically compatible with Greenberg’s later area

formulas. I.42 then starts the application-of-areas construction block. Its midpoint, angle-copy, parallel, and parallelogram operations fit Greenberg’s Hilbert-style organization very well, while the statement that the resulting parallelogram has the same content as the given triangle still depends on the external scissors theory.

For the present project Greenberg should therefore be used as a foundational and dependency reference, not as a source from which I.35–I.45 can simply be transcribed. Up to I.34 he is especially valuable for locating the neutral/Euclidean boundary. From I.35 onward his role bifurcates: he remains a strong source for incidence, order, congruence, parallelism, and construction, but the synthetic semantics of content must be supplied by Hilbert, Hartshorne, and the formal scissors layer.

## A.8 Current checkpoint

The present source audit now concerns Euclid Book I, Propositions I.1–I.45.

The structural picture has three major transitions. Propositions I.25–I.28 remain within neutral comparison and angle-to-parallel theory. Proposition I.29 is the exact point at which the converse direction, parallel-to-angle, introduces the Euclidean parallel principle. I.30 inherits the same Euclidean status through transitivity of parallelism, while I.31 separates neutral existence from Euclidean uniqueness. I.32 gives the Euclidean angle-sum theorem.

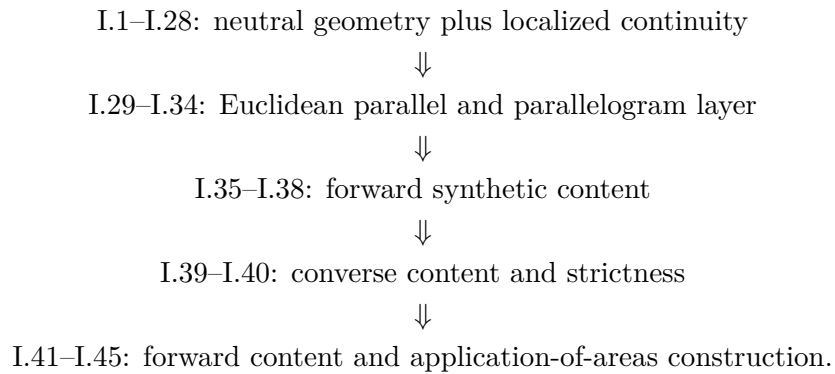
I.33 and I.34 remain applications of that same Euclidean parallel layer. Greenberg does not preserve I.33 as a separately identified theorem in the material located by this audit, whereas I.34 is explicit and is used to illustrate that Euclidean parallelogram congruence is not available in an arbitrary Hilbert plane.

I.35 marks the second boundary: Euclid starts comparing plane figures by content. Greenberg distinguishes the intended cut-and-paste idea from numerical measurement but does not develop the required synthetic calculus. The extension I.36–I.38 therefore continues to use Greenberg chiefly for geometry, not for the content relation itself. His later formula  $K = \frac{1}{2}bh$  confirms the numerical meaning of equal-base/equal-altitude comparisons without replacing the formal scissors semantics.

I.39–I.40 mark a third boundary inside the area block. They use equal content as a hypothesis and recover a geometric conclusion. The present Greenberg audit supplies no synthetic strict-content theorem that would justify this reversal. This agrees with the need to make positivity explicit in the Lean reconstruction rather than treating “the whole is greater than the part” as an automatic property of the chosen content relation.

I.41 returns to forward content composition. I.42 then changes from comparison to construction: a new parallelogram must be built with prescribed angle and prescribed content. Greenberg’s Hilbert-style geometry is highly relevant to the midpoint, angle-layoff, parallel, and parallelogram stages, while the final content equality remains a client of the separate scissors layer.

The durable architectural summary is therefore



This appendix should be treated as a durable research record for the I.1–I.45 checkpoint. Later Greenberg observations should be appended rather than replacing or silently deleting the earlier source notes, and any claim about a proposition-level Greenberg counterpart should remain separate from later numerical area arguments.

## Appendix B

# Hartshorne: Euclid in a Hilbert Plane

### B.1 Purpose and scope

This appendix records a source audit of Euclid Book I, Propositions I.1–I.45, against Robin Hartshorne’s *Geometry: Euclid and Beyond* [5].

The audit has a specific purpose. Hartshorne begins with Euclid’s own proofs and constructions, analyzes where their logical structure is incomplete, and then reconstructs the relevant part of Book I inside a Hilbert-plane framework. For each proposition I.1–I.45, we record:

- whether Hartshorne regards Euclid’s proof as valid as it stands;
- whether the proposition is replaced by an axiom or by a more general Hilbert-plane theorem;
- whether Hartshorne supplies a complete proof, a proof sketch, or an exercise;
- which hidden order, betweenness, separation, congruence, or existence issue he explicitly identifies;
- which points are especially relevant to a fully formal synthetic reconstruction.

The purpose is not to claim novelty for the classical mathematics. Rather, the audit identifies the exact level at which a conventional synthetic proof still leaves obligations that must be made explicit in a formal proof assistant.

### B.2 Hartshorne’s organization of the problem

Hartshorne separates the discussion into two stages. Chapter 1 studies Euclid’s own geometry and repeatedly asks whether a proof depends on a diagram, an unstated positional assumption, or a construction whose existence has not yet been justified. Chapter 2 then introduces Hilbert’s axioms of incidence, betweenness, and congruence and reconstructs the early propositions inside that framework.

The key structural statement appears in the section on Hilbert planes: Theorem 10.4 states that all of Euclid I.1–I.28, except I.1 and I.22, can be proved in an arbitrary Hilbert plane. The first exception is the circle-intersection problem in I.1; the second is the prescribed-side triangle construction I.22. Beginning with I.29 Hartshorne moves to a Hilbert plane with the parallel axiom (P), and Theorem 12.1 recovers Euclid’s theory of parallels I.29–I.34.

At I.35 the subject changes again. Hartshorne explicitly treats I.35–I.48 as the beginning of Euclid’s theory of area. Since Euclid does not attach numerical areas to figures, Hartshorne interprets the new equality by the synthetic relation of equal content. Section 22 later defines equal content and returns to I.35 as Example 22.1.2. Thus the present checkpoint contains not only the neutral-to-Euclidean parallel boundary at I.29 and the parallel-to-content boundary at I.35, but also the internal logical boundary inside the content theory itself. Hartshorne distinguishes forward equal-content results from converses that require the additional property (Z). By I.42 the emphasis changes again from comparison of given figures to Euclid’s “application of areas,” where equal content becomes an engine for construction.

### B.3 Comparison table

Table B.1: Euclid I.1–I.45 in Hartshorne’s Hilbert-plane development

| Euclid | Hartshorne counterpart                                                         | Status in Hartshorne                                                                                                                                                                                          | Source-audit remarks and relevance to formalization                                                                                                                                                                                                                                                                 |
|--------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.1    | Construction of an equilateral triangle on a given segment.                    | Hartshorne treats I.1 as the first genuine exception to the Hilbert-plane recovery of Book I. The intersection of the two circles used by Euclid cannot be derived from the basic Hilbert-plane axioms alone. | Chapter 1 already isolates the hidden existence problem: the diagram shows two circles meeting, but Euclid’s stated postulates do not guarantee that they do. Hartshorne therefore requires an additional intersection or continuity principle. This is a foundational dependency, not a minor gap in presentation. |
| I.2    | Transporting a given segment to a specified point.                             | Hartshorne states that I.2 is effectively replaced by the segment congruence construction axiom C1.                                                                                                           | The Euclidean proposition becomes primitive infrastructure in the Hilbert system. This is important for later segment arithmetic and for the synthetic interpretation of “adding” segments by extending a ray and laying off a congruent copy.                                                                      |
| I.3    | Cutting off from a greater segment a segment equal to a given smaller segment. | Hartshorne groups I.3 with I.2 as effectively replaced by C1 together with the order theory of segments.                                                                                                      | The proposition is no longer structurally fundamental as a named theorem. Its mathematical content is decomposed into segment construction, uniqueness, betweenness, and segment ordering.                                                                                                                          |
| I.4    | SAS triangle congruence.                                                       | SAS is taken as axiom C6. Hartshorne analyzes Euclid’s superposition argument and concludes that it relies on an unstated theory of rigid motions.                                                            | This is one of the sharpest differences between Euclid’s proposition order and a Hilbert-style dependency order. The proof by superposition is not accepted as a derivation from Euclid’s explicit postulates and common notions.                                                                                   |
| I.5    | Base angles of an isosceles triangle.                                          | Hartshorne explicitly reinterprets Euclid’s proof in the Hilbert system and gives the proof in detail using segment construction and SAS.                                                                     | This is a direct example of the program of recovering Euclid inside a Hilbert plane. Hartshorne also points out the hidden betweenness facts needed to justify some of Euclid’s positional claims.                                                                                                                  |

Continued on next page



Table B.1 – continued from previous page

| Euclid | Hartshorne counterpart                                                        | Status in Hartshorne                                                                                                                                                                                    | Source-audit remarks and relevance to formalization                                                                                                                                                                                                       |
|--------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.6    | Converse of the isosceles base-angle theorem.                                 | Hartshorne says that Euclid’s argument is essentially valid, but one point must be rewritten using the formally defined order relation on angles.                                                       | The phrase “less than” cannot be left intuitive. The proof depends on having an explicit angle-order relation and on using its asymmetry and trichotomy correctly. This is precisely the kind of obligation that becomes visible in Lean.                 |
| I.7    | Uniqueness configuration for triangles on the same base and on the same side. | Hartshorne identifies a hidden dependence on the relative position of the lines and points. The missing betweenness justification is assigned for completion in the Hilbert-plane exercises.            | This proposition is a canonical example of a proof whose diagram suppresses case distinctions. Hartshorne stresses that one must know where the lines and intersection points lie before the angle inequalities used in the proof are legitimate.         |
| I.8    | SSS triangle congruence.                                                      | Hartshorne replaces Euclid’s superposition proof with a new theorem, Proposition 10.1 (SSS), proved from the Hilbert congruence axioms.                                                                 | A complete Hilbert-style proof is supplied. The construction creates a copy of one triangle on the opposite side of a side of the other and uses angle addition and SAS. The positional cases are explicitly noted.                                       |
| I.9    | Construction of an angle bisector.                                            | Hartshorne reconstructs I.9 using Hilbert tools. Since I.1 is not available in an arbitrary Hilbert plane, he replaces Euclid’s equilateral-triangle step by the construction of an isosceles triangle. | He explicitly notes that it is not enough to perform the construction: one must prove that the constructed ray really lies in the interior of the original angle. That requires the order/separation machinery.                                           |
| I.10   | Construction of the midpoint of a segment.                                    | Hartshorne reconstructs the proposition by the same strategy: replace Euclid’s use of I.1 by an isosceles triangle. The corresponding Hilbert-tools construction is left as Exercise 10.2.              | The proposition is therefore recoverable in a Hilbert plane without using the circle-intersection principle required by I.1. This is an important alternative dependency route.                                                                           |
| I.11   | Erecting a perpendicular at a point of a line.                                | Hartshorne explains that the construction can be recovered using Hilbert tools and assigns the explicit construction as Exercise 10.3.                                                                  | The reconstruction also establishes the existence of right angles, which is not taken for granted beforehand. The proof is therefore part of the basic congruence-and-angle infrastructure rather than a compass construction in Euclid’s original sense. |
| I.12   | Dropping a perpendicular from a point to a line.                              | Hartshorne explicitly says that Euclid’s original compass construction does not work in an arbitrary Hilbert plane. A new existence proof is required; the explicit construction is Exercise 10.4.      | This is a significant foundational note. The failure is not that the theorem is false, but that Euclid’s chosen construction imports an intersection principle unavailable in the bare Hilbert system.                                                    |

Continued on next page

Table B.1 – continued from previous page

| Euclid | Hartshorne counterpart                                                | Status in Hartshorne                                                                                                                                                                                                            | Source-audit remarks and relevance to formalization                                                                                                                                                                                                                              |
|--------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.13   | Angles formed when a straight line stands on another straight line.   | Hartshorne replaces the Euclidean proposition by Proposition 9.2 on supplements of congruent angles.                                                                                                                            | He remarks explicitly that Proposition 9.2 may be viewed as the proper Hilbert-style replacement for I.13. The modern statement is more general and fits directly into the angle-congruence infrastructure.                                                                      |
| I.14   | Converse straight-line criterion from adjacent supplementary angles.  | Hartshorne treats I.14 as an easy consequence of the preceding supplement theory and asks for a careful Hilbert-plane formulation in Exercise 10.7.                                                                             | The theorem is mathematically routine only after the modern notions of rays, supplementary angles, and angle congruence have been made precise. The formal version therefore belongs naturally to the low-level angle interface.                                                 |
| I.15   | Vertical angles are equal.                                            | Corollary 9.3: vertical angles are congruent. A complete short proof is given from the theory of supplementary angles.                                                                                                          | This is a clean example in which Hartshorne replaces Euclid's common notion argument by an explicitly stated congruence theorem.                                                                                                                                                 |
| I.16   | Exterior Angle Theorem.                                               | Proposition 10.3 gives a full Hilbert-plane proof.                                                                                                                                                                              | Hartshorne returns to the exact point he had criticized in Chapter 1: the constructed ray must really lie inside the relevant angle. He closes this gap using betweenness and plane separation. Thus the proof makes explicit the hidden positional content of Euclid's diagram. |
| I.17   | Any two angles of a triangle are together less than two right angles. | Hartshorne says that I.17–I.21 are valid in a Hilbert plane after reinterpretation. For I.17 he avoids treating “the sum of two angles” as a primitive object and reformulates the statement using supplements and angle order. | This is especially relevant to formalization: the familiar informal statement must be replaced by a proposition expressible in the available angle language.                                                                                                                     |
| I.18   | The greater side of a triangle subtends the greater opposite angle.   | Hartshorne regards the Euclidean argument as recoverable but assigns a careful Hilbert-plane proof as Exercise 10.6.                                                                                                            | The exercise explicitly calls attention to betweenness and inequalities. These are exactly the two classes of hidden obligations that dominate a formal reconstruction of I.18.                                                                                                  |
| I.19   | The greater angle of a triangle subtends the greater opposite side.   | Included in Hartshorne's statement that I.17–I.21 are valid in a Hilbert plane after the preceding reconstruction of order and congruence.                                                                                      | Hartshorne does not single out a new main-text proof comparable to Proposition 10.3. Its validity rests on the now-formalized order relations for sides and angles and on the preceding comparison theorem.                                                                      |

Continued on next page

Table B.1 – continued from previous page

| Euclid | Hartshorne counterpart                                                                                              | Status in Hartshorne                                                                                                                                                                                                                                                            | Source-audit remarks and relevance to formalization                                                                                                                                                                                                                                     |
|--------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.20   | Triangle inequality.                                                                                                | Hartshorne says that I.17–I.21 are valid in a Hilbert plane and assigns a careful proof of I.20 as Exercise 10.8.                                                                                                                                                               | The theorem therefore belongs to neutral Hilbert geometry. Hartshorne’s exercise status is important: the mathematics is declared recoverable, but the complete chain of synthetic justifications is left to the reader. A formal proof must expose all those omitted steps explicitly. |
| I.21   | Comparison theorem for two triangles on a common base, with one vertex inside the other triangle.                   | Hartshorne includes I.21 in his statement that I.17–I.21 are valid in an arbitrary Hilbert plane.                                                                                                                                                                               | The proposition remains neutral. Its two conclusions depend on the already developed order theory for segments and angles; no new continuity or parallel principle is introduced.                                                                                                       |
| I.22   | Construction of a triangle from three prescribed side lengths satisfying the strict triangle inequalities.          | This is Hartshorne’s second explicit exception. In a bare Hilbert plane the required two-circle intersection need not exist, even when the side inequalities hold. Section 11 restores the construction by adding the intersection axiom (E); Theorem 11.7 then validates I.22. | This is not a defect in the inequality theory but a genuine existence gap. It sharply separates neutral order/congruence results from circle-intersection continuity.                                                                                                                   |
| I.23   | Transport of a given angle to a prescribed point and side.                                                          | Hartshorne does not reconstruct Euclid’s I.22-based proof in a bare Hilbert plane; instead I.23 is replaced directly by congruence axiom (C4), the angle transporter.                                                                                                           | A long Euclidean construction becomes primitive Hilbert infrastructure. This is a major dependency compression and explains why later neutral results can proceed even though I.22 itself fails without (E).                                                                            |
| I.24   | Two triangles with two corresponding sides congruent: the greater included angle has the greater third side.        | Hartshorne states that the remaining results before the parallel postulate, including I.24, are valid as they stand in a Hilbert plane.                                                                                                                                         | The proposition is a neutral comparison theorem. Its formal burden is therefore configuration and strict order, not Euclidean parallelism.                                                                                                                                              |
| I.25   | Converse comparison: with two corresponding sides congruent, the greater third side has the greater included angle. | Included among the remaining pre-parallel results valid in a Hilbert plane.                                                                                                                                                                                                     | Together I.24 and I.25 form a neutral comparison pair. The source supports treating them as neighboring directions of one order-theoretic mechanism.                                                                                                                                    |
| I.26   | ASA and AAS triangle congruence.                                                                                    | Hartshorne explicitly identifies I.26 with ASA and AAS and includes it among the results valid in an arbitrary Hilbert plane.                                                                                                                                                   | The theorem is neutral. For formalization the two cases may still be separated internally because their proof obligations need not be identical.                                                                                                                                        |

Continued on next page

Table B.1 – continued from previous page

| Euclid | Hartshorne counterpart                                                                                                               | Status in Hartshorne                                                                                                                                                                                                                                                                                                          | Source-audit remarks and relevance to formalization                                                                                                                                                                      |
|--------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.27   | Equal alternate interior angles imply that the two lines are parallel.                                                               | Hartshorne treats I.27 as a neutral theorem and uses it repeatedly before introducing the parallel axiom. In Chapter 1 he derives it from I.16 by excluding an intersection on either side.                                                                                                                                   | This direction, angle relation to parallelism, is available without Playfair. It is the neutral half of the later converse pair with I.29.                                                                               |
| I.28   | Corresponding-angle or supplementary-interior-angle criteria imply parallelism.                                                      | Hartshorne says that I.28 follows from I.27 using vertical angles I.15 or supplementary angles I.13, and Theorem 10.4 includes it in the Hilbert-plane fragment.                                                                                                                                                              | I.28 completes the neutral parallel criteria. Its conclusion is parallelism, but no uniqueness-of-parallels principle is needed.                                                                                         |
| I.29   | A transversal of two parallel lines gives equal alternate interior angles and the associated angle relations.                        | This is the first proposition where Hartshorne requires the parallel axiom. With Playfair axiom (P), construct through the relevant point a line making the required angle by (C4); I.27 proves that the constructed line is parallel to the second given line; uniqueness from (P) identifies it with the original parallel. | Hartshorne isolates the Euclidean step exactly: construction and I.27 are neutral, while identification of two parallels through the same point is the non-neutral uniqueness step.                                      |
| I.30   | Lines parallel to the same line are parallel.                                                                                        | Hartshorne states that I.30 is essentially equivalent to the parallel axiom (P).                                                                                                                                                                                                                                              | Transitivity of parallelism is therefore not merely a formal property of the definition. In Hartshorne's organization it is an equivalent interface form of Euclidean parallel uniqueness.                               |
| I.31   | Construction of a parallel through a given external point.                                                                           | Hartshorne explicitly notes that existence already follows in a Hilbert plane from the angle transporter (C4) and I.27, without (P). After (P) is assumed, I.31 can be strengthened to existence and uniqueness.                                                                                                              | This is the clearest existence-versus-uniqueness split in the source. The proposition's construction is neutral; only the stronger unique parallel statement is Euclidean.                                               |
| I.32   | Exterior angle equals the two remote interior angles; equivalently, the angles of a triangle are together equal to two right angles. | Hartshorne places I.32 among the remaining propositions following from (P). He also warns that literal angle addition is not primitive in his Hilbert system and proposes the scrupulous reformulation: the sum of any two angles is supplementary to the third.                                                              | The theorem is genuinely Euclidean through I.29. Hartshorne's reformulation is especially relevant to Lean, where the synthetic supplementary-angle statement can be preferable to numerical or additive angle language. |

Continued on next page

Table B.1 – continued from previous page

| Euclid | Hartshorne counterpart                                                                                                            | Status in Hartshorne                                                                                                                                                                                                                                                                                 | Source-audit remarks and relevance to formalization                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.33   | Lines joining the corresponding endpoints of equal and parallel segments in the same direction are themselves equal and parallel. | Hartshorne includes I.33 in Theorem 12.1, which states that Euclid’s theory of parallels I.29–I.34 holds in every Hilbert plane with the parallel axiom (P). He does not single I.33 out for a new foundational argument; it belongs to the remaining consequences of the Euclidean parallel theory. | For formalization, I.33 is best viewed as a parallelogram-producing bridge: parallelism plus segment congruence yields the second pair of equal and parallel sides. Its position after I.29 is logically significant, since the converse angle theory needed to control the parallel configuration is already Euclidean.                                                                                                                                                                                                      |
| I.34   | Opposite sides and opposite angles of a parallelogram are equal, and a diagonal divides it into two congruent triangles.          | Hartshorne includes I.34 in Theorem 12.1. Elsewhere in the book he uses I.34 as standard parallelogram infrastructure, in particular the equality of opposite sides. The theorem therefore closes the I.29–I.34 block of Euclid’s theory of parallels.                                               | This proposition is the immediate structural predecessor of I.35. It converts a parallel configuration into congruence data: opposite sides, opposite angles, and the diagonal triangle congruence. In a formal reconstruction it is natural for these consequences to become reusable interface lemmas rather than to be reproved inside the area argument.                                                                                                                                                                  |
| I.35   | Parallelograms on the same base and lying in the same parallels have equal content.                                               | I.35 is the first proposition of Hartshorne’s area block. In Chapter 1 he analyzes Euclid’s cutting-and-pasting proof. In Section 22, Example 22.1.2 states the result precisely using the defined relation of equal content: the two parallelograms have equal content.                             | This is a genuine change of formal language. Congruence of segments, angles, and triangles no longer suffices to state the conclusion. Hartshorne’s equal content is a synthetic cut-and-paste relation, not a numerical area formula. Moreover, equal content must not be silently strengthened to equidecomposability: Hartshorne gives a non-Archimedean example where the I.35 parallelograms have equal content but are not equidecomposable. This distinction is directly relevant to the formal scissors architecture. |
| I.36   | Parallelograms on equal bases and lying in the same parallels have equal content.                                                 | In Section 22 Hartshorne states that I.36 follows from I.35 by transitivity of equal content. No new area principle is introduced.                                                                                                                                                                   | This is the first confirmation that the equal-content relation is meant to be used compositionally. The formal burden is therefore geometric normalization to an intermediate common-base configuration plus transitivity, not numerical area and not cancellation.                                                                                                                                                                                                                                                           |
| I.37   | Triangles on the same base, with their top vertices on one line parallel to the base, have equal content.                         | Hartshorne isolates a defect in Euclid’s proof: “halves of equals are equal” depends on property (Z). He therefore gives a replacement, Proposition 22.6, proving I.37 in a Euclidean plane without using (Z).                                                                                       | The replacement proof uses midpoints and parallelograms, reduces both triangles to equal-content parallelograms, applies I.35, and concludes by transitivity. This is the key source control for a formal proof that must avoid hidden division by two or cancellation of doubled content.                                                                                                                                                                                                                                    |

Continued on next page

Table B.1 – continued from previous page

| Euclid | Hartshorne counterpart                                                                                                      | Status in Hartshorne                                                                                                                                                                        | Source-audit remarks and relevance to formalization                                                                                                                                                                                                                                                                                 |
|--------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.38   | Triangles on equal bases and in the same parallels have equal content.                                                      | Immediately after Proposition 22.6, Hartshorne states that I.38 follows by transitivity.                                                                                                    | The source therefore treats I.38 as a forward equal-content corollary, not as a new foundational area theorem. In a formal system the main work may still be geometric: one must construct or identify an intermediate same-base triangle before transitivity can be applied.                                                       |
| I.39   | Equal-content triangles on the same base and on the same side lie in the same parallels.                                    | Hartshorne identifies I.39 as a converse-type result. Euclid's proof uses "the whole is greater than the part," and Hartshorne states that this depends on property (Z).                    | This is the first sharp reversal of direction in the area block: equal content is now a hypothesis from which geometric position is recovered. Hartshorne explicitly says he knows no purely geometric proof of (Z) from the definition of content; Section 23 later derives (Z) using a measure-of-area/coordinate argument.       |
| I.40   | Equal-content triangles on equal bases and on the same side lie in the same parallels.                                      | Hartshorne places I.40 with I.39 on the converse side of the theory. He states that I.40 also depends on (Z) because equal content is being used to conclude a geometric relation.          | No second independent strictness principle is suggested. Once the same-base converse and the equal-base reduction are available, the foundational issue is the same property (Z) already exposed at I.39. This is precisely the boundary between forward content transport and content-to-geometry reasoning.                       |
| I.41   | A parallelogram is double a triangle on the same base and in the same parallels, interpreted as an equal-content statement. | Hartshorne groups I.41–I.45 among the remaining Book I results that are valid for equal content. Unlike I.39–I.40, no appeal to (Z) is required.                                            | The logical direction is again forward. Congruent pieces and addition of equal-content figures suffice; there is no inference from equality of content back to lengths, angles, or incidence. For formalization this supports a purely additive scissors interpretation of "double."                                                |
| I.42   | Construct a parallelogram equal in content to a given triangle and having a prescribed angle.                               | Hartshorne includes I.42 among the results valid for equal content and, in Chapter 3, singles out I.42–I.45 as Euclid's "application of areas."                                             | This proposition marks a new use of the content theory: comparison results become construction tools. Hartshorne describes I.42–I.45 as part of an "algebra of areas," in which rectangles and parallelograms play the role of products of segments. The result remains on the forward side of the theory and does not require (Z). |
| I.43   | The complements of the parallelograms about a diagonal have equal content.                                                  | Hartshorne does not single out I.43 for a separate foundational repair; he groups I.41–I.45 among the forward Book I results that remain valid for the synthetic relation of equal content. | The proposition stays entirely on the forward side of the theory. Its subtraction language should be implemented through the definition and closure properties of equal content, not through property (Z) or numerical area.                                                                                                        |

Continued on next page

Table B.1 – continued from previous page

| Euclid | Hartshorne counterpart                                                                                    | Status in Hartshorne                                                                                                                                                                                             | Source-audit remarks and relevance to formalization                                                                                                                                                                                                                                                                                                                        |
|--------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.44   | Apply a parallelogram equal in content to a given triangle to a prescribed segment in a prescribed angle. | Hartshorne treats I.42–I.45 as the application-of-areas block. In Exercise 22.7 he specifically invokes I.44 to show that any figure can be replaced by a rectangle of equal content having a fixed side.        | This confirms that I.44 is a reusable placement operation, not a theorem about one accidental diagram. It also supports keeping the target at equal content rather than strengthening it to unconditional equidecomposability.                                                                                                                                             |
| I.45   | Construct, in a prescribed angle, a parallelogram having equal content with a given rectilineal figure.   | Hartshorne defines a rectilineal figure as a finite nonoverlapping union of triangles. He states that I.41–I.45 are valid for equal content and groups I.42–I.45 as the application-of-areas construction block. | This supplies the sharp representation control for the Lean statement. The current <code>HilbertTriangulatedFigure</code> is a finite triangle list, whereas Hartshorne’s figure includes nonoverlap and region semantics. The Lean induction proves the additive theorem; reading the list as one rectilineal figure requires the corresponding representation invariant. |

## B.4 Foundational observations retained from Hartshorne

Several remarks in Hartshorne are important beyond the immediate I.1–I.35 audit and should be preserved for later work.

- Hartshorne repeatedly uses a proof from a diagram as a warning sign. When a step depends on the relative position of points, lines, or rays, that position must be derived from the axioms rather than read from the figure.
- Betweenness and plane separation are not secondary bookkeeping devices. They are the mechanism that repairs several of the early Euclidean proofs, especially I.7, I.9, and I.16.
- Hartshorne’s Hilbert plane is built from incidence, betweenness, and congruence. The parallel postulate is kept separate so that one can identify which propositions belong to neutral geometry.
- The first major exception is I.1. The basic Hilbert-plane axioms do not guarantee the intersection of the two circles used by Euclid. Hence an additional intersection/continuity principle is required.
- Euclid’s constructions and Hilbert’s existence axioms should not be conflated. A Euclidean compass construction may fail as a proof in an arbitrary Hilbert plane even when the corresponding geometric object can still be proved to exist by another route.
- Hartshorne replaces Euclid’s superposition method by axiomatic SAS. He interprets superposition as requiring a theory of rigid motions, which is not supplied by Euclid’s stated postulates.
- Segment addition is formalized synthetically. Hartshorne defines the sum of two segments by laying off a congruent copy on a ray and proves that this construction is well-defined up to congruence.

- Segment order is then defined using betweenness and congruence, and Hartshorne proves its compatibility with congruence and its trichotomy properties.
- The angle theory is developed in parallel: supplementary angles, vertical angles, angle addition, and angle order are all made explicit before the later Euclidean propositions are recovered.
- I.17 illustrates an important language issue. An informal sum of angles need not correspond directly to a primitive formal object, so the theorem is reformulated using inequalities and supplements.
- Hartshorne explicitly identifies I.22 as the second exception in Theorem 10.4. The continuity/intersection issue encountered at I.1 therefore returns inside the present checkpoint.
- I.23 is replaced by axiom (C4), the angle transporter. Thus the Hilbert reconstruction can recover later neutral propositions without reconstructing Euclid’s I.22-dependent angle-copy construction.
- Theorem 10.4 ends at I.28. The endpoint is structural: I.27–I.28 derive parallelism from angle relations in neutral geometry, whereas I.29 reverses the implication and requires uniqueness of parallels.
- Hartshorne states that I.30 is essentially equivalent to (P), making transitivity of parallelism an equivalent Euclidean interface form.
- I.31 separates existence from uniqueness. Existence of a parallel is neutral, from (C4) and I.27; uniqueness is supplied only after (P).
- For I.32 Hartshorne explicitly recommends replacing informal angle addition by a supplementary-angle formulation. This is directly relevant to a synthetic Lean statement without numerical angle measure.
- Theorem 12.1 closes the parallel block at I.34. Thus I.33 and I.34 do not open a new axiomatic layer; they are applications of the Euclidean parallel theory already entered at I.29.
- I.34 is especially important as the last congruence theorem before the area block. Its parallelogram consequences become input data for I.35 rather than part of the definition of equality of figures.
- I.35 is a new semantic boundary. Hartshorne treats I.35–I.48 as Euclid’s theory of area and interprets equality of figures by equal content rather than by numerical measure.
- Hartshorne’s Section 22 shows that equal content and equidecomposability are not interchangeable in arbitrary Euclidean planes. In particular, the I.35 conclusion is equal content; a stronger finite-dissection conclusion requires additional hypotheses in general.
- I.36 introduces no new content principle: Hartshorne derives it by transitivity of equal content.
- I.37 is the first place where Hartshorne explicitly rejects an apparently harmless Euclidean magnitude rule inside the new content semantics. “Halves of equals are equal” depends on (Z), so Proposition 22.6 replaces Euclid’s proof by a midpoint/parallelogram argument that avoids (Z).
- I.38 then follows by transitivity. Thus I.36–I.38 remain on the forward side of the content theory even though Euclid’s literal proof of I.37 does not.



- I.39 and I.40 reverse the logical direction. Equal content becomes a hypothesis from which a geometric conclusion is recovered, and Hartshorne states that these propositions depend on  $(Z)$ .
- Hartshorne does not treat  $(Z)$  as an unexplained permanent axiom. Section 23 derives it from the existence of a measure-of-area function after coordinates have been introduced. What matters for the source audit is that  $(Z)$  is not available from the bare definition of equal content alone.
- I.41–I.45 return to forward equal-content statements and are valid for content without  $(Z)$ . This sharply separates them from the converse pair I.39–I.40.
- Hartshorne identifies I.42–I.45 as the “application of areas” and describes the resulting technique as a kind of “algebra of areas.” At I.42 the area block stops being only comparative and becomes a construction technology.

## B.5 I.43–I.45: completion of the application-of-areas block

Hartshorne’s treatment is the cleanest semantic control for the end of the application-of-areas sequence. He explicitly classifies I.41–I.45 as valid for equal content, so I.43–I.45 do not cross the property  $(Z)$  boundary that appears at I.39–I.40. Their direction is geometric construction followed by an equal-content conclusion.

Two details are especially useful for the formal project. First, Exercise 22.7 uses I.44 as a genuine normalization operation: a figure can be replaced by an equal-content rectangle with a prescribed fixed side. Second, Hartshorne’s definition of a rectilineal figure requires a finite nonoverlapping union of triangles. The latter is stronger as a representation invariant than the present Lean abbreviation `HilbertTriangulatedFigure = List HilbertTriangle`.

Thus I.45 confirms that the current scissors term is sufficient for the constructive additive theorem, while also locating exactly what a future internal polygon semantics would have to add: nonoverlap and the assertion that the triangles describe one geometric region.

## B.6 Result of the Hartshorne audit

Hartshorne reorganizes Euclid I.1–I.45 according to several successive logical layers rather than treating the propositions as one homogeneous sequence.

The first layer is neutral geometry. Theorem 10.4 states precisely that all propositions I.1–I.28 except I.1 and I.22 can be recovered in an arbitrary Hilbert plane. The two exceptions are existence problems caused by missing circle-intersection continuity. Hartshorne later adds axiom (E), and Theorem 11.7 restores exactly the Euclidean constructions I.1 and I.22.

Within the neutral layer Hartshorne also changes the dependency graph. I.2 and I.3 are absorbed into segment transporter axiom (C1), I.4 becomes SAS axiom (C6), and I.23 is replaced by the angle transporter axiom (C4). Several Euclidean constructions are therefore compressed into primitive Hilbert infrastructure. Other propositions, especially I.5, I.8, I.15, and I.16, receive explicit Hilbert-style reconstructions in which betweenness, plane separation, angle interiors, and strict order are made visible.

The second layer begins at I.29. Hartshorne explicitly identifies this as the first use of the parallel axiom. His Hilbert-style proof of I.29 is particularly informative: construct a candidate line with

the desired angle by (C4), use neutral I.27 to show that it is parallel, and then use Playfair uniqueness (P) to identify it with the given parallel. Thus the non-neutral step is isolated exactly at uniqueness.

This clarifies the remainder of the parallel block. I.30 is essentially equivalent to (P); I.31 separates neutral existence from Euclidean uniqueness; I.32 recovers the angle-sum theory through I.29; and I.33–I.34 supply the standard consequences for equal parallel segments and parallelograms. Theorem 12.1 packages the entire block I.29–I.34 as Euclid’s theory of parallels in a Hilbert plane with (P).

The third layer begins sharply at I.35. Hartshorne identifies I.35–I.48 as Euclid’s theory of area. The relevant equality is not segment congruence, angle congruence, triangle congruence, or a numerical area function. Section 22 introduces the synthetic relation of equal content and Example 22.1.2 shows that Euclid’s proof of I.35 establishes equal content for parallelograms on the same base and in the same parallels. I.36 then follows by transitivity.

At I.37 Hartshorne exposes a second, internal boundary. Euclid’s phrase “halves of equals are equal” is not licensed by the basic equal-content relation: Hartshorne says it depends on property (Z). Instead of adding (Z) merely to save Euclid’s proof, Proposition 22.6 gives a new proof of I.37 using midpoints, parallelograms, I.35, and transitivity. I.38 then follows by transitivity. Thus the forward content theory survives without (Z), provided the proof architecture is changed.

I.39 and I.40 reverse the direction. They assume equal content and infer a geometric relation, namely equality of altitude expressed by common parallels. Hartshorne states that these converses depend on (Z) and remarks that he knows no purely geometric proof of (Z) from the given definition of content. Section 23 later proves (Z) through the coordinate/measure-of-area theory. For a formal reconstruction this is a precise warning: some positivity or strictness principle is genuinely needed at the converse boundary, even if it is eventually discharged by a stronger semantic construction.

I.41 changes direction once more. Hartshorne states that I.41–I.45 and I.47 are valid for equal content. These are forward statements: geometry and construction produce equal-content conclusions, so no recovery of metric data from content equality is required. In particular I.41 can be read additively as the parallelogram having the content of two copies of the triangle on the same base and between the same parallels.

At I.42 the area theory becomes constructive. Hartshorne groups I.42–I.45 under Euclid’s “application of areas” and describes the resulting technique as a kind of “algebra of areas.” Equal-content comparison is now used to construct a parallelogram with prescribed content and angle. This is a different role from I.35–I.41: the content calculus has become an interface for geometric construction rather than only a relation used to compare figures.

For formal work, Hartshorne’s central contribution is therefore not merely a classification into neutral and Euclidean theorems. He repeatedly identifies the exact hidden obligation suppressed by a classical diagram: intersection existence, betweenness, side-of-line data, interior-of-angle claims, order, uniqueness, the semantics of cutting and pasting figures, and finally the strictness needed when content equality is used in the converse direction. These are precisely the obligations that become explicit proof terms and interface layers in Lean.

## B.7 Current checkpoint

This appendix now concerns Euclid Book I, Propositions I.1–I.45.

The structural picture at this checkpoint is:

|                                |                                                         |
|--------------------------------|---------------------------------------------------------|
| Hilbert plane:                 | <i>I.1–I.28</i> , except <i>I.1</i> and <i>I.22</i> ;   |
| Hilbert plane + (E):           | restores the constructions <i>I.1</i> and <i>I.22</i> ; |
| Hilbert plane + (P):           | <i>I.29–I.34</i> (the theory of parallels);             |
| equal content, forward:        | <i>I.35–I.38</i> ;                                      |
| equal content + (Z), converse: | <i>I.39–I.40</i> ;                                      |
| equal content, forward again:  | <i>I.41–I.45</i> ;                                      |
| application of areas:          | <i>I.42–I.45</i> .                                      |

Within that classification, *I.31* still deserves separate emphasis: its existence statement belongs to the neutral layer even though its position in Euclid comes after *I.29*. Only uniqueness belongs to the Euclidean layer.

The content block now has its own internal logical structure. *I.35–I.38* show how geometry implies equal content. Hartshorne’s alternative proof of *I.37* is decisive because it removes the illicit “halves of equals” step rather than assuming the strictness principle (Z). By contrast, *I.39–I.40* run from equal content back to geometry and therefore cross the (Z) boundary.

*I.41–I.45* return to the forward direction. Hartshorne explicitly classifies this whole block as valid for equal content, and *I.42* starts the “application of areas.” At this point the synthetic content relation has become stable enough to support constructions with prescribed content and prescribed geometric data, without replacing the theory by a numerical area function.

This appendix should be treated as the durable Hartshorne source audit for the *I.1–I.45* checkpoint. Later propositions should extend the record rather than silently revise the distinctions established here.

## Appendix C

# Wilkins: The Euclidean Dependency Graph

### C.1 Purpose and scope

This appendix records a source audit of David R. Wilkins’s *Study Notes on Euclid’s Elements*, Book I, Propositions I.1–I.45.

Unlike the Greenberg and Hartshorne appendices, the primary object here is not the reorganization of Euclid inside a modern axiomatic system. Wilkins stays close to the proposition sequence and is therefore especially useful for reconstructing the internal Euclidean dependency graph. For each proposition the audit records:

- the earlier Euclidean propositions, postulates, and Common Notions used in the proof;
- alternative proofs and historically reported variants;
- converse statements and logical equivalences;
- configuration cases that Euclid’s surviving written proof omits or compresses;
- hidden order, side-of-line, intersection, or continuity assumptions made explicit in the Study Note;
- the significance of these observations for a fully formal synthetic reconstruction.

The entries below are source-audit summaries, not replacements for the Study Notes themselves. In particular, when Wilkins identifies several geometric configurations, the audit retains that case structure rather than collapsing it into a single diagrammatic argument.

### C.2 Proposition-by-proposition audit

Table C.1: Euclid I.1–I.45 in Wilkins’s Study Notes

| Prop. | Euclidean dependency DAG                                                                                                                                                                                                                                     | Wilkins additions                                                                                                                                                                                                                                                                                                                                         | Significance for formalization                                                                                                                                                                                                                                                                                                                           |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.1   | Postulates 1 and 3 construct the joining segments and the two circles. The circle definitions give $AB = AC$ and $AB = BC$ ; Common Notion 1 then gives $AC = BC$ .                                                                                          | Wilkins emphasizes that Euclid tacitly assumes that the two circles intersect. He formulates a topological crossing principle: a straight segment or circular arc with one endpoint inside a circle and the other outside must meet the circle. He also stresses that Euclidean construction concerns exact mathematical objects, not physical drawings.  | I.1 is not merely a short congruence argument. The formal burden is the existence of the intersection point $C$ . A proof assistant must represent the circle-intersection principle explicitly or replace Euclid’s construction by stronger primitive existence infrastructure. The Study Note therefore marks the first continuity boundary in Book I. |
| I.2   | I.1 constructs the auxiliary equilateral triangle. Postulates 2 and 3 extend rays and draw the two circles. The construction gives $BC = BG$ , $DG = DL$ , and $DB = DA$ ; Common Notion 3 gives $BG = AL$ , and Common Notion 1 gives $BC = AL$ .           | Wilkins examines the strength actually required from Postulate 2. The ray must be extendable far enough to exceed any prescribed segment, so that it meets the relevant circle. He again invokes a tacit topological intersection principle to justify the circle-line intersection.                                                                      | Segment transport is simple as a finished theorem but expensive as Euclid constructs it. The audit separates the algebraic cancellation step from the existence problem. In a Hilbert-style interface this explains why a segment-layoff axiom can absorb much of I.2 while the classical proof still records a nontrivial construction DAG.             |
| I.3   | I.2 transports the shorter segment to obtain $AD = C$ . A circle centered at $A$ meets the greater segment $AB$ at $E$ , giving $AE = AD$ ; Common Notion 1 yields $AE = C$ .                                                                                | Wilkins presents I.3 as the completion of the construction strategy developed through I.1–I.3. The key positional fact is that the circle meets $AB$ at a point $E$ between $A$ and $B$ .                                                                                                                                                                 | The proposition already mixes metric transport with order. A formal version must not infer the betweenness of $E$ from the picture. If segment layoff and uniqueness are primitive, I.3 may collapse to a small order-and-congruence corollary even though its Euclidean construction is longer.                                                         |
| I.4   | The proposition is the SAS Congruence Rule. Euclid’s proof uses superposition and Common Notion 4: after applying one triangle to the other, corresponding vertices and sides coincide, giving the remaining side and angle equalities and equality in area. | Wilkins makes explicit an orientation condition suppressed by Euclid: after matching the first side, the third vertex must be placed on the same side of the matched side as the third vertex of the target triangle. He then analyzes the result in modern language using Euclidean motions and notes the homogeneity and isotropy implicit in the rule. | This is a foundational rather than merely local issue. The superposition argument hides a theory of motions and side choice. Hilbert-style formalization normally takes SAS as congruence infrastructure and avoids reconstructing global rigid motions. The Study Note explains exactly what Euclid’s diagrammatic application is assuming.             |

Continued on next page

Table C.1 – continued from previous page

| Prop. | Euclidean dependency DAG                                                                                                                                                                                                                                                                                              | Wilkins additions                                                                                                                                                                                                                                                                                                                                                                                          | Significance for formalization                                                                                                                                                                                                                                                                                             |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.5   | I.3 chooses $G$ so that $BF = CG$ . I.4 is applied first to the large triangles $FAC$ and $GAB$ , then to $CFB$ and $BGC$ . Common Notion 3 subtracts equal angle parts to obtain the equality of the base angles.                                                                                                    | Wilkins retains both conclusions of the Pons Asinorum: equality of the base angles and equality of the angles under the base after the equal sides are extended. He notes that commentators such as Proclus and Heath need the second conclusion when completing omitted cases in I.7.                                                                                                                     | The second, exterior-angle conclusion is not decorative. It becomes reusable infrastructure for later configuration repairs. The formal library should therefore not reduce I.5 to only the familiar base-angle statement if later Euclidean proofs need the extended-side form.                                           |
| I.6   | Assume unequal opposite sides. I.3 cuts off a segment making a smaller triangle; I.4 then yields a contradiction because a proper part would be equal to the whole. Wilkins also gives an alternative angle-based contradiction using SAS and the principle that a proper part of an angle is smaller than the whole. | The Study Note separates two reductio routes: Euclid’s area-based argument and an alternative that avoids area by comparing an angle with a proper subangle. In the latter discussion Wilkins refers to the relevant “whole greater than part” principle as a Common Notion.                                                                                                                               | The alternative route is important before the area layer begins. It shows that the converse isosceles theorem can be kept within congruence and strict angle order. In formal work the trichotomy of segment lengths and the strictness of “part of an angle” must be explicit proof objects.                              |
| I.7   | The principal geometric engine is I.5. Depending on the position of the second vertex $D$ , the proof uses the base-angle or under-base angle conclusions of the isosceles theorem together with strict containment of angles.                                                                                        | Wilkins gives an extensive configuration audit. He states the required order and plane-separation principles, divides the relevant half-plane into four regions $K, L, M, N$ , and notes that Euclid explicitly treats only region $L$ . Region $K$ uses extensions and the under-base part of I.5; region $M$ is reduced by interchanging the roles of the two vertices; region $N$ is analogous to $L$ . | I.7 is one of the strongest warnings against proving from the diagram. Same-side, opposite-side, interior-of-angle, interior-of-triangle, and line-order data are all logically active. A complete formal proof must either reproduce the regional case split or establish a stronger uniqueness theorem that subsumes it. |
| I.8   | Euclid’s SSS proof uses superposition together with I.7 to force the third vertex to coincide once the three corresponding sides have been matched.                                                                                                                                                                   | Wilkins also presents the proof attributed by Proclus to Philo of Byzantium. That proof produces five intersection configurations. The first three are proved directly using I.5, I.4, and Common Notions 1, 2, or 3 as appropriate; the last two follow by relabelling from the first and third cases.                                                                                                    | SSS is not configuration-free in its classical derivation. Both Euclid’s and Philo’s routes hide motion/orientation or substantial case analysis. This is directly relevant when deciding whether SSS belongs as an axiom, an interface theorem, or a theorem reconstructed from lower Hilbert infrastructure.             |

Continued on next page

Table C.1 – continued from previous page

| Prop. | Euclidean dependency DAG                                                                                                                                                                                                                               | Wilkins additions                                                                                                                                                                                                                                                                        | Significance for formalization                                                                                                                                                                                                                                                                                                            |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.9   | I.3 chooses equal points $D, E$ on the two sides of the angle. I.1 constructs the equilateral triangle $DEF$ . I.8 then applies SSS to $DAF$ and $EAF$ , yielding equality of the two angles at $A$ .                                                  | Wilkins explicitly notes that I.9 cannot occur earlier in Euclid's sequence because it depends on I.1, I.3, and I.8. The auxiliary vertex $F$ is chosen on the opposite side of $DE$ from $A$ .                                                                                          | The angle-bisector theorem contains hidden side-of-line data in addition to congruence. If a Hilbert system has an angle-construction or angle-bisector theorem already available, the Euclidean construction DAG can be compressed, but the classical dependency record should be retained separately.                                   |
| I.10  | I.1 constructs an equilateral triangle on $AB$ . I.9 bisects the apex angle, and I.4 applies SAS to the two resulting triangles to obtain equality of the two halves of $AB$ .                                                                         | Wilkins also gives the self-contained construction attributed by Proclus to Apollonius: two equal-radius circles meet at two points on opposite sides of $AB$ ; I.8 and I.4 then prove that their joining line bisects $AB$ .                                                            | The alternative proof exposes a stronger circle-intersection obligation than the main Euclidean DAG. It is useful when comparing which existence principles a formalization chooses to make primitive. The mathematical theorem is midpoint existence; the cost of the construction depends heavily on the chosen foundational interface. |
| I.11  | Choose $D, E$ equidistant from the given point $C$ , construct an equilateral triangle $DEF$ by I.1, and apply I.8 to triangles $DCF$ and $ECF$ . Equality of the adjacent angles at $C$ gives a right angle by definition.                            | Wilkins describes the construction as straightforward and keeps the right-angle conclusion tied to the definition: equal adjacent angles on a straight line are right angles.                                                                                                            | The proposition is a good example of a Euclidean construction that can become a general existence theorem in a Hilbert interface. The formal burden moves from the visible equilateral triangle to precise definitions of opposite rays, adjacent angles, and right angle.                                                                |
| I.12  | A point $D$ is chosen on the opposite side of the given line from $C$ . A circle centered at $C$ meets the line at $E, G$ ; I.10 bisects $EG$ at $H$ ; I.8 gives congruence of $CEH$ and $CGH$ , so the equal adjacent angles at $H$ are right angles. | Wilkins explicitly states that Euclid tacitly assumes a line has two sides and a topological crossing principle for arcs or segments whose endpoints lie on opposite sides of a line. He explains how the two distinct intersections $E, G$ can be justified from the two circular arcs. | Dropping a perpendicular is therefore not just an SSS application. Existence of opposite-side points and two line-circle intersections is foundational. These are exactly the obligations that disappear if a Hilbert interface already provides perpendicular existence.                                                                 |
| I.13  | I.12 erects a perpendicular through the vertex. Postulate 4 makes all right angles equal, and repeated Common Notion 2 additions show that the two adjacent angles on the straight line sum to two right angles.                                       | Wilkins writes the proof for the positional case in which the original point $A$ lies in the interior of the relevant right angle, so the decomposition into three adjacent angle pieces is explicit.                                                                                    | The phrase “sum of angles” hides ray-order and interior-of-angle information. A synthetic formalization may prefer a supplementary-angle relation or an explicit decomposition theorem rather than numerical or untyped angle addition.                                                                                                   |

Continued on next page

Table C.1 – continued from previous page

| Prop. | Euclidean dependency DAG                                                                                                                                                                                                                                                                                      | Wilkins additions                                                                                                                                                                                                                                                                         | Significance for formalization                                                                                                                                                                                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.14  | Extend $CB$ through $B$ to $F$ . I.13 gives the supplementary relation for $ABC$ and $ABF$ . The assumed supplementary relation for $ABC$ and $ABD$ , followed by Common Notions 1 and 3, yields $\angle ABF = \angle ABD$ . Uniqueness of the straight line through $B, F$ gives collinearity of $C, B, D$ . | Wilkins says explicitly that his logical structure departs somewhat from Euclid's presentation. He also uses the fact that $D$ and $F$ lie on the same side of $AB$ when converting equality of the two angles into coincidence or collinearity.                                          | This is a converse straight-line criterion whose formal content is not merely angle cancellation. Same-side information and uniqueness of an incident line are needed to recover collinearity from an angle identity.                                                              |
| I.15  | Two applications of I.13 show that each vertically opposite angle, when added to the same adjacent angle, gives two right angles. Common Notions 1 and 3 cancel the common adjacent angle and yield equality of the vertical angles.                                                                          | Wilkins gives a precise verbal characterization of vertically opposite angles before running the short Euclidean cancellation proof.                                                                                                                                                      | Once supplementary-angle theory is available, I.15 is a compact interface lemma. The formal work lies mainly in recognizing the opposite-ray configuration and normalizing the order of the angle arguments.                                                                       |
| I.16  | I.10 bisects $AC$ ; the line $BE$ is extended so that $BE = EF$ ; I.15 gives the vertical angle equality at $E$ ; I.4 gives SAS congruence of $ABE$ and $CFE$ . The exterior-angle inequality then follows after locating $F$ inside the relevant exterior angle.                                             | Wilkins supplies a detailed plane-separation argument proving the diagrammatic claim that $F$ lies inside angle $ACD$ . He uses same-side and opposite-side information with respect to $BC$ and $AC$ , and then repeats the argument symmetrically for the second remote interior angle. | The central congruence calculation is short; the hidden positional proof is the hard part. I.16 is therefore a model case for why between/side-of-line infrastructure belongs in a formal Hilbert interface rather than being treated as diagrammatic bookkeeping.                 |
| I.17  | I.16 says the exterior angle $ACD$ is greater than the remote interior angle $ABC$ . I.13 says $ACD + ACB$ is two right angles. Therefore $ABC + ACB$ is less than two right angles.                                                                                                                          | Wilkins presents the proof as a direct comparison between one exterior-angle inequality and one supplementary-angle identity.                                                                                                                                                             | The theorem is structurally small once I.13 and I.16 exist. In formal language, however, it may be preferable to express the result through angle order and supplements rather than an undeveloped operation of adding angles.                                                     |
| I.18  | I.3 cuts off $AD = AB$ on the longer side. I.5 makes $ADB$ isosceles, I.16 compares the exterior angle at $D$ with the remote angle at $C$ , and strict inclusion of $ABD$ in $ABC$ completes the chain of angle inequalities.                                                                                | Wilkins writes the inequality chain explicitly: $\angle ABC > \angle ABD = \angle ADB > \angle ACB$ .                                                                                                                                                                                     | The proof uses three distinct kinds of order: segment order to choose $D$ , angle equality from isosceles congruence, and strict angle containment. Formalization should preserve these as separate interface operations rather than flattening them into numerical angle measure. |

Continued on next page



Table C.1 – continued from previous page

| Prop. | Euclidean dependency DAG                                                                                                                                                                                                                        | Wilkins additions                                                                                                                                                                                                                                                  | Significance for formalization                                                                                                                                                                                                                                                                                    |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.19  | If the opposite sides were equal, I.5 would force the angles equal. If the side opposite the smaller angle were longer, I.18 would reverse the assumed angle inequality. The only remaining trichotomy case gives the required side inequality. | Wilkins presents I.19 as a clean elimination of the two impossible segment-order cases rather than constructing a new auxiliary figure.                                                                                                                            | This is a useful example of a theorem whose Euclidean proof is mostly logical once segment and angle trichotomy are formalized. The formal DAG should expose the trichotomy step explicitly.                                                                                                                      |
| I.20  | Extend $BA$ beyond $A$ to $D$ with $DA = AC$ . I.5 makes $ACD$ isosceles; angle inclusion gives $\angle BCD > \angle BDC$ ; I.19 then gives $BD > BC$ , and segment addition yields $BA + AC > BC$ .                                            | Wilkins also records the ancient alternative attributed by Proclus to Heron or Porphyry: I.9 bisects the angle, I.16 gives two exterior-angle inequalities, and I.19 converts them into the two side inequalities whose sum is the Triangle Inequality.            | The alternative path is valuable for proof-graph comparison: the same theorem can be reached through an isosceles-extension route or through angle bisection and exterior-angle theory. Such alternate DAGs are potential inputs for tactic design and helper selection.                                          |
| I.21  | Two applications of I.20 give the side-sum inequality for the inner triangle. Two applications of I.16 give $\angle BDC > \angle BEC > \angle BAC$ .                                                                                            | Wilkins keeps the intermediate point $E$ visible in both the segment and angle arguments, showing how the two inequalities are assembled from repeated earlier propositions.                                                                                       | The hypothesis that $D$ lies inside triangle $ABC$ is substantial. A formal theorem needs the interior configuration or enough side-of-line data to justify all intermediate betweenness relations used in the two chains.                                                                                        |
| I.22  | I.20 supplies the necessary triangle inequalities. The three given segments are laid off successively on a ray, circles of the required radii are drawn, and their intersection gives the third vertex of the triangle.                         | Wilkins analyzes all boundary and failure cases. He shows that the three strict triangle inequalities are not only necessary but sufficient for the circle-circle construction; equality gives tangency and reversed inequality gives separated or nested circles. | I.22 is the second major circle-intersection checkpoint after I.1. The exact relation between segment inequalities and circle intersection is part of the theorem's existence content. A formalization cannot replace this by a diagrammatic assumption without recording a continuity or intersection principle. |
| I.23  | I.20 verifies the triangle inequalities for a small triangle cut out of the given angle. I.22 constructs a congruent triangle on the given ray, and I.8 (SSS) transfers the target angle to the specified point.                                | Wilkins attributes the construction, through Eudemus, to Oenopides. He chooses the sample points on the given angle so that one side is shorter than the available base segment before invoking I.22.                                                              | In a Hilbert development, angle transport may be primitive. The Study Note explains why Euclid's construction is much later: it factors through triangle existence and SSS. This distinction is important when comparing Euclid's linear order with a Hilbert API.                                                |

Continued on next page

Table C.1 – continued from previous page

| Prop. | Euclidean dependency DAG                                                                                                                                                                                                                                                         | Wilkins additions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Significance for formalization                                                                                                                                                                                                                                                                 |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.24  | I.23 constructs the comparison triangle and I.4 identifies it with the first given triangle. The remaining inequality $EG > EF$ is then proved according to configuration, using I.21 in one case and I.5 plus I.19 in Euclid's case.                                            | Wilkins identifies three configurations according to the positions of $D, E$ relative to line $GF$ , and states that Euclid explicitly covers only the third. He also supplies alternative proofs that exchange the strategies used in the first and third cases. In the final alternative paragraph the Study Note cites Proposition I.22 while calling it the Triangle Inequality; since the same note and the rest of Book I identify I.20 as the Triangle Inequality, this appears to be a citation slip in the source rather than a new dependency. | I.24 is another major case-split theorem. The formal burden is not the comparison theorem alone but classification of the relative position of $D, E, G, F$ . The apparent I.22/I.20 citation inconsistency should be recorded rather than silently normalized when tracing the classical DAG. |
| I.25  | If the included angles were equal, I.4 would force the third sides equal. If the second angle were larger, I.24 would force the second third side larger. Both alternatives contradict the hypotheses, leaving the desired angle inequality.                                     | Wilkins presents I.25 as the converse comparison theorem obtained by eliminating the equality and reversed-inequality cases.                                                                                                                                                                                                                                                                                                                                                                                                                             | As with I.19, the formal engine is trichotomy plus previously established comparison theorems. No new construction is needed once strict segment and angle order are available.                                                                                                                |
| I.26  | ASA: assume the corresponding included side differs, cut off an equal part, use I.4, and contradict strict angle inclusion. SAA: use the analogous reduction, I.4, and I.16 to contradict the Exterior Angle Theorem. A final application of I.4 gives full triangle congruence. | Wilkins explicitly separates ASA and SAA even though Euclid uses the same diagram. The SAA branch requires I.16 whereas the ASA branch does not.                                                                                                                                                                                                                                                                                                                                                                                                         | A formal congruence interface should not conflate the two rules' dependencies. ASA can be packaged from SAS plus order; SAA additionally uses the exterior-angle inequality in Euclid's route. This matters when minimizing assumptions for downstream propositions such as I.34.              |
| I.27  | I.16 $\rightarrow$ I.27. If the two candidate lines met on either side of the transversal, one of the equal alternate angles would be an exterior angle of a triangle and hence greater than the remote interior angle.                                                          | Wilkins explicitly treats both possible sides of intersection and identifies I.27 as the converse of the Alternate Angles Theorem obtained later from I.29.                                                                                                                                                                                                                                                                                                                                                                                              | The proof DAG is exceptionally small. Parallelism is obtained from the neutral exterior-angle inequality; the formal burden is chiefly the two possible intersection orientations. This is the key neutral angle-to-parallel criterion.                                                        |

Continued on next page

Table C.1 – continued from previous page

| Prop. | Euclidean dependency DAG                                                                                                                                                                                                                                       | Wilkins additions                                                                                                                                                                                                                  | Significance for formalization                                                                                                                                                                                                                                                                       |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.28  | First criterion: I.15 converts corresponding-angle equality to alternate-angle equality, then I.27 gives parallelism. Second criterion: I.13 plus Common Notion 3 converts same-side supplementarity to alternate-angle equality, then I.27 gives parallelism. | Wilkins treats the two criteria separately and identifies the first as the converse of the Corresponding Angles Theorem later contained in I.29.                                                                                   | I.28 naturally splits into wrappers over I.27 after different angle normalizations. The geometric engine remains the neutral theorem I.27; parallelism in the conclusion does not by itself make the proposition Euclidean.                                                                          |
| I.29  | Fifth Postulate plus I.13 rules out either same-side interior-angle sum being less or greater than two right angles. From equality to two right angles, cancellation yields the alternate-angle and corresponding-angle equalities.                            | Wilkins explicitly states that I.29 is the first proposition in Book I whose proof requires the Fifth Postulate. He identifies the alternate and corresponding angle theorems and points back to their converses in I.27 and I.28. | This is the exact logical reversal of I.27–I.28: angle relations imply parallelism neutrally, whereas parallelism implies the angle relations only after the Fifth Postulate enters. This is the central axiomatic boundary in the Book I DAG.                                                       |
| I.30  | Euclid’s route uses I.29 twice to transport angle equalities and I.27 to conclude that the two outer lines are parallel. A more detailed variant inserts I.15; another variant uses corresponding angles and I.28.                                             | Wilkins proves that I.30 is logically equivalent to Playfair’s Axiom. He also derives I.29 from Playfair’s Axiom together with I.27.                                                                                               | I.30 is therefore not merely transitivity bookkeeping. It is an equivalent interface form of Euclidean parallel uniqueness. The Study Note gives a direct bridge between Euclid’s proposition DAG and a modern axiomatic formulation of the parallel postulate.                                      |
| I.31  | I.23 constructs at the external point an angle equal to the required alternate angle; I.27 proves that the resulting line is parallel to the given line.                                                                                                       | Wilkins separately notes that I.29 gives uniqueness of the infinite parallel through the external point.                                                                                                                           | The proposition cleanly separates existence from uniqueness. Existence uses only angle construction plus the neutral criterion I.27; the Euclidean input I.29 is needed only for uniqueness. This distinction must remain visible in a Hilbert-style interface.                                      |
| I.32  | I.31 constructs the auxiliary parallel through a vertex. I.29 is applied twice to identify the two remote angles with pieces of the exterior angle. I.13 then closes the two-right-angles conclusion.                                                          | Wilkins also presents the proof attributed by Eudemus to the Pythagoreans and explains how it is related to Euclid’s proof by extension, reorientation, relabelling, and I.15.                                                     | The invariant proof mechanism is clearer than either particular diagram: construct a parallel, transport two angles by I.29, then close a straight-angle relation. Equivalent classical configurations can have substantially different formal costs depending on orientation and ray normalization. |

Continued on next page

Table C.1 – continued from previous page

| Prop. | Euclidean dependency DAG                                                                                                                                                                                                                                                                                                                                                                                             | Wilkins additions                                                                                                                                                                                                                                                                               | Significance for formalization                                                                                                                                                                                                                                                                                              |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.33  | I.29 gives equality of the alternate angles determined by the given equal parallel sides. I.4 applies SAS to the two triangles, yielding $AC = BD$ and a second angle equality. I.27 converts that equality back into $AC \parallel BD$ .                                                                                                                                                                            | Wilkins makes the closed loop explicit: parallelism $\rightarrow$ angle equality $\rightarrow$ triangle congruence $\rightarrow$ angle equality $\rightarrow$ parallelism.                                                                                                                      | I.33 is a compact bridge between the parallel and congruence interfaces. In Lean it should remain small if I.29, SAS, and I.27 are already packaged; a long proof would indicate missing reusable transport lemmas rather than new geometry.                                                                                |
| I.34  | Two applications of I.29 give the alternate-angle equalities cut by the diagonal. I.26 (ASA) on the common diagonal gives congruence of the two triangles and therefore equality of opposite sides and one pair of opposite angles. The remaining opposite-angle equality is obtained by comparing the angle sums at the two common vertices; I.4 then immediately gives equality in area of the diagonal triangles. | Wilkins formulates the proposition first as a complete congruence package for the two diagonal triangles and only then notes the area consequence.                                                                                                                                              | This separation is ideal for formalization. The reusable content of I.34 is parallelogram congruence data; the area statement should not force a numerical area theory. I.34 becomes the immediate metric and angular interface consumed by I.35.                                                                           |
| I.35  | In every branch I.34 supplies opposite-side equalities, I.29 supplies the corresponding-angle equality, and I.4 gives congruence of the outer triangles. In the overlap and common-endpoint branches Common Notion 2 adds a common trapezium or triangle. In the disjoint branch Common Notion 3 subtracts $DGE$ , then Common Notion 2 adds $GBC$ .                                                                 | Wilkins insists on three configurations for a complete proof: overlap with upper-point order $A, E, D, F$ ; common endpoint with order $A, D, F$ ; and disjoint sides with order $A, D, E, F$ . He states explicitly that Euclid's surviving written proof covers only the third configuration. | I.35 is the strongest configuration audit in the present block. A formal theorem must derive which upper-point order holds and dispatch the corresponding cut-and-paste argument. The Study Note predicts precisely why a single diagram is insufficient and why a scissors or equal-content layer is needed at this point. |
| I.36  | I.34 gives $EH = FG$ ; with the hypothesis $BC = FG$ , Common Notion 1 gives $BC = EH$ . I.33 then makes $EBCH$ an intermediate parallelogram. I.35 is applied twice, and Common Notion 1 closes the chain<br><br>$ABCD = EBCH = EFGH$ .                                                                                                                                                                             | Wilkins presents the intermediate parallelogram $EBCH$ explicitly and keeps the proof very close to Euclid's local proposition DAG. The argument is entirely a reuse of I.33–I.35 plus equality to a common figure; no new area operation is introduced in the Study Note.                      | For formalization, I.36 should be expected to be a thin client of the I.35 content layer once the equal-base parallelogram geometry has been packaged. The source separates the geometric construction of $EBCH$ from the final transitivity of equality of area.                                                           |

Continued on next page

Table C.1 – continued from previous page

| Prop. | Euclidean dependency DAG                                                                                                                                                                                                                                                                                                                                             | Wilkins additions                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Significance for formalization                                                                                                                                                                                                                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.37  | Construct parallelograms $EBCA$ and $DBCF$ on the common base $BC$ , with $AB$ and $DC$ as diagonals. I.35 makes the two parallelograms equal in area; I.34 says each is double its corresponding triangle. Therefore $ABC$ and $DBC$ are equal in area.                                                                                                             | Wilkins gives the classical “double both triangles” proof without a separate configuration discussion. The note treats the inference from equal doubles to equal halves as immediate.                                                                                                                                                                                                                                                                                 | The hidden operation is halving/cancellation at the level of content. A formal reconstruction need not follow this route literally; if it does, the principle cannot be smuggled in as ordinary arithmetic. This is exactly where a synthetic equal-content semantics must specify what operations are available.                                   |
| I.38  | Construct parallelograms $GBCA$ and $DEFH$ so that $AB$ and $DF$ bisect them. I.36 gives equality of the parallelograms because their bases are equal and they lie in the same parallels; I.34 makes each parallelogram double its corresponding triangle. Hence $ABC = DEF$ in area.                                                                                | The Study Note makes the ordered collinearity of $B, C, E, F$ and the parallel upper carrier through $A, D$ explicit in its statement of the configuration. As in I.37, the final passage from equal doubles to equal triangles is left implicit.                                                                                                                                                                                                                     | I.38 is the equal-base analogue of I.37. The classical DAG predicts that a modern formal proof can profitably reduce it to an already proved same-base content theorem plus congruent-copy geometry rather than introduce another halving principle.                                                                                                |
| I.39  | Draw through $A$ the line parallel to the common base $BC$ . If the other apex $D$ is not on this parallel, an intersection point $E$ is obtained on the line $BD$ or on its extension. A triangle with apex $E$ has the same area as $ABC$ ; depending on the order, $DBC$ is then strictly larger or strictly smaller than $ABC$ , contradicting equality of area. | Wilkins explicitly supplies the configuration completion omitted by Euclid: there are two cases, according as $B$ and $D$ lie on opposite sides or on the same side of the parallel through $A$ . The Study Note cites Proposition I.36 for the equal-area comparison of the same-base triangles $ABC$ and $EBC$ . This is an apparent local citation inconsistency: I.36 concerns parallelograms, while I.37 is the same-base triangle theorem used by the argument. | I.39 reverses the direction of the preceding content theorems: equal area is now used to force parallelism. A formal proof therefore needs a strict proper-part/positivity principle in addition to the forward equal-content relation. Wilkins’s two cases also show that the order classification cannot be read from a single Euclidean diagram. |

Continued on next page

Table C.1 – continued from previous page

| Prop. | Euclidean dependency DAG                                                                                                                                                                                                                                                                                                                                                             | Wilkins additions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Significance for formalization                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.40  | The strategy is the equal-base analogue of I.39. Draw through one apex a line parallel to the common base carrier. If the second apex is not on that parallel, construct a comparison triangle on the second equal base; it has the same area as the first triangle, while the given second triangle is strictly larger or smaller according to the point order.                     | Wilkins states that a complete proof has six configurations, only one of which is explicitly treated by Euclid. The six arise from adjacent, separate, and interlocking equal bases, each combined with the two possible positions of the second apex relative to the auxiliary parallel. The first configuration also contains an apparent local label slip, writing that $C$ lies between $A$ and $E$ with $AC = CE$ , whereas the figure and the equal-base setup require $C$ between $B$ and $E$ with $BC = CE$ . As in the I.39 note, the text cites Proposition I.36 when comparing equal-area triangles; in context this is an apparent citation error for the equal-base triangle theorem I.38. | I.40 is a major configuration warning. The logical content is not a new converse-area principle beyond I.39, but the reduction to the same-base/equal-base comparison requires a complete six-way positional audit. A formal theorem must either encode these cases or hide them behind a stronger reusable carrier-and-order lemma.                                                                                                                             |
| I.41  | I.34 says that the diagonal $AC$ bisects the parallelogram $ABCD$ , so the parallelogram is double the triangle $ABC$ . I.37 says $ABC = EBC$ in area because the triangles have the same base and lie in the same parallels. Therefore $ABCD$ is double $EBC$ .                                                                                                                     | Wilkins gives a one-page proof with no new geometric construction. The entire argument is a composition of the diagonal-bisection result and the already established same-base triangle equality.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | This is a clean test of addition/doubling in the content calculus. No converse or cancellation is required if the proof is organized in the forward direction: replace one triangle by an equal-content triangle inside a formal double.                                                                                                                                                                                                                         |
| I.42  | Bisect $BC$ at $E$ ; reproduce the prescribed angle at $E$ by I.23, choosing the ray on the side of $BC$ containing $A$ ; draw through $A$ the parallel to $BC$ by I.31 and let it meet the ray at $F$ ; complete the parallelogram $FECG$ . I.38 gives $ABE = AEC$ , while I.41 makes $FECG$ double $AEC$ . Thus the triangle $ABC$ and the parallelogram $FECG$ are equal in area. | Wilkins explicitly records the orientation of the copied-angle ray: it is directed into the side of $BC$ containing $A$ . The note then treats the existence of the point $F$ , where the parallel through $A$ meets that ray, as part of the construction rather than proving a separate intersection lemma. There is also a local naming inconsistency: the construction first says “parallelogram $AECG$ ” but the area argument and figure use $FECG$ .                                                                                                                                                                                                                                             | For formalization I.42 is simultaneously a construction theorem and a content theorem. The side choice and the existence of the ray-parallel intersection are genuine geometric obligations; after they are discharged, the content calculation is only I.38, I.41, and forward addition/transitivity. The intermediate API must also avoid encoding a degenerate segment as a strict parallel when the intersection happens to coincide with an existing point. |

Continued on next page

Table C.1 – continued from previous page

| Prop. | Euclidean dependency DAG                                                                                                                                                                                                                                                                                                                                    | Wilkins additions                                                                                                                                                                                                                                                                                                          | Significance for formalization                                                                                                                                                                                                                                                                                                                                                               |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.43  | I.34 is applied to the large parallelogram and to the two smaller parallelograms about the diagonal. Common Notion 2 combines the equal corner triangles, and Common Notion 3 leaves the two complements equal in area.                                                                                                                                     | Wilkins fixes the standard configuration explicitly: a point on the diagonal, one line through it parallel to one pair of sides, and another line through it parallel to the other pair. The two remaining regions are identified as the complements to be compared.                                                       | For formalization the important point is that “subtract equals from equals” is a statement about a specific finite decomposition. The Lean proof can keep the removed pieces explicit and prove equicomplementability without adding a general cancellation theorem.                                                                                                                         |
| I.44  | I.42 first constructs a parallelogram equal to the given triangle in the given angle. The construction is then placed on the prescribed segment; parallel constructions and the Fifth Postulate produce the large parallelogram, I.43 identifies the complements, and I.15 supplies the final vertical-angle equality.                                      | Wilkins states the location requirement from the beginning: the required parallelogram must have the given segment as one of its sides. His diagram and proof retain the produced lines and the complement configuration used by I.43.                                                                                     | This is a placement theorem, not merely a content theorem. A formal proof must recover the order of the produced points and prove the required intersections instead of reading them from the diagram. Those obligations are exactly what the I.44 order helps expose.                                                                                                                       |
| I.45  | Divide the rectilinear figure into finitely many triangles $T_1, \dots, T_k$ . Use I.44 to associate to each triangle a parallelogram with one fixed side $A$ , the prescribed angle $\theta$ , and a second side $B_i$ . Concatenate the segments $B_i$ to obtain one side $B$ ; the resulting parallelogram is equal in area to the sum of the triangles. | Wilkins makes the general finite case completely explicit with modern subscript notation. He assumes the triangular interiors are disjoint and states that the rectilinear figure is their finite union. His figure shows the final parallelogram subdivided into the adjacent parallelograms corresponding to the $B_i$ . | This is the closest classical control for the final Lean induction. The recursive operation “adjoin one triangle” matches the theorem <code>i45_extend_parallelogram</code> . It also makes the representation boundary precise: Wilkins’s geometric figure includes nonoverlap, whereas <code>HilbertTriangulatedFigure</code> is only a list unless that invariant is supplied separately. |

## C.3 Major dependency blocks

The complete I.1–I.45 audit reveals several overlapping dependency blocks rather than a single linear chain.

### C.3.1 Construction and continuity: I.1–I.3, I.9–I.12, I.22–I.23

The first three propositions build Euclid’s segment-construction machinery. Wilkins repeatedly distinguishes the formal equality argument from the existence of the constructed intersection points. I.1 requires the two circles to intersect. I.2 requires the relevant ray to be extendable far enough to meet a circle. I.3 then combines segment transport with a betweenness conclusion.

The same issue returns later. I.9 and I.10 use the early construction machinery for angle and segment bisection. I.12 explicitly requires a two-sided line and two line-circle intersections. I.22 is the second major circle-circle existence theorem: the triangle inequalities are shown to be exactly the conditions under which the prescribed-side triangle construction succeeds. I.23 then

uses I.22 and SSS to transport an angle.

For the formal project, this block records a recurring distinction:

construction DAG  $\neq$  minimal modern existence interface.

A Hilbert axiom or interface theorem can compress several Euclidean construction steps without changing the classical source audit.

### C.3.2 Congruence and triangle comparison: I.4–I.8, I.18–I.26

I.4 introduces SAS by superposition. Wilkins’s discussion makes the hidden same-side placement explicit and analyzes the modern motion interpretation. I.5 then supplies both the base-angle and under-base angle conclusions for an isosceles triangle. I.7 shows how much topological and order information is hidden by a seemingly elementary uniqueness statement, and I.8 builds SSS from I.7 while also recording Philo’s five-case alternative proof.

The comparison block I.18–I.26 repeatedly converts between strict segment order and strict angle order. I.18 and I.19 form a converse pair, as do I.24 and I.25. I.20 supplies the Triangle Inequality and I.21 combines it with the Exterior Angle Theorem. I.26 closes the triangle-congruence block by deriving ASA and SAA from SAS, with the SAA route requiring I.16.

This block is particularly useful for testing whether a formal development has a sufficiently strong order API. Many classical proofs become short only after the following operations are reusable: trichotomy, transport of order under congruence, strict part-whole inequalities, and orientation of the relevant rays or sides.

### C.3.3 The neutral-to-Euclidean parallel boundary: I.27–I.34

Wilkins makes the logical boundary at I.29 exceptionally explicit. I.27 and I.28 are neutral converses:

angle condition  $\longrightarrow$  parallelism.

At I.29 the direction reverses:

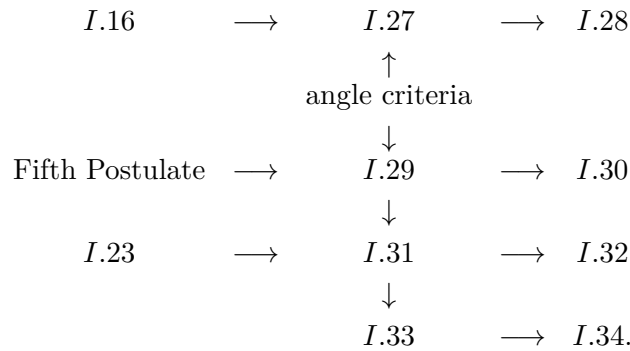
parallelism  $\longrightarrow$  angle condition,

and the Fifth Postulate is used for the first time in Book I.

I.30 then reveals the equivalence structure surrounding this boundary: I.30 is equivalent to Playfair’s Axiom, and Playfair together with I.27 implies I.29. I.31 separates neutral existence of a parallel from Euclidean uniqueness. I.32 uses the new parallel-to-angle direction to obtain the angle-sum theorem. I.33 and I.34 convert the Euclidean parallel theory into reusable parallelogram congruence data.



A schematic DAG is



The diagram is intentionally schematic: I.31 does not depend on I.29 for existence, while its uniqueness remark does; I.33 uses both I.29 and I.27; and I.34 uses I.29 and I.26.

### C.3.4 The area and application block: I.35–I.45

I.35 changes the kind of proof obligation. The outer-triangle congruence still belongs to the established congruence and parallel theory:

$$I.34 + I.29 + I.4.$$

But the conclusion compares whole parallelograms by cutting, subtracting, and adding figures. Wilkins's three configurations show that the cut-and-paste pattern itself depends on the order of the upper endpoints.

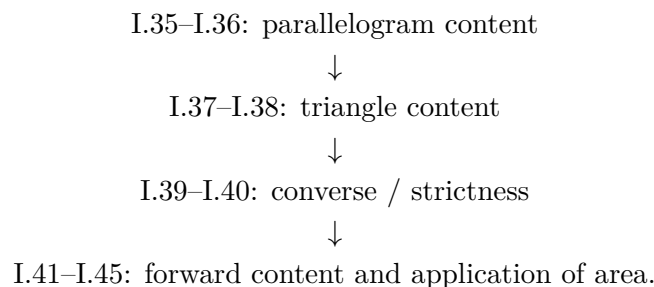
I.36 then transports this same-area relation from common bases to equal bases through an intermediate parallelogram. I.37 and I.38 pass from parallelograms to triangles by doubling each triangle to a parallelogram. Their classical proofs therefore contain the familiar but logically nontrivial move

$$2T_1 = 2T_2 \implies T_1 = T_2.$$

I.39 and I.40 reverse the implication. Equal area is assumed, and the conclusion is geometric parallelism. Wilkins's configuration audit becomes substantial again: two cases for I.39 and six for I.40. The contradictions are proper-part arguments: if the second apex misses the required parallel, one triangle must be strictly larger or smaller in area than the other.

I.41 returns to a forward content calculation: a parallelogram is double a triangle on the same base and in the same parallels. I.42 then uses I.38 and I.41 as a construction engine, producing a parallelogram of prescribed angle and equal area to a given triangle.

Thus the classical DAG reaches a stable new layer:



Wilkins does not supply a modern axiomatic theory of equal content; that is the role of other sources. His contribution is the exact Euclidean proposition DAG, the Common-Notion algebra,

and – especially in I.39, I.40, and I.42 – the configuration and construction data suppressed by a single textbook diagram.

## C.4 Configuration completeness

One of the most important results of the complete Wilkins audit is that several propositions require substantially more configuration analysis than Euclid’s surviving text displays.

| Prop. | Configuration issue recorded by Wilkins                                                                                                                                                                                                                                  |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.7   | The relevant half-plane is divided into four regions $K, L, M, N$ . Euclid explicitly treats only region $L$ ; the other cases require additional side-of-line and angle-interior arguments.                                                                             |
| I.8   | Philo’s alternative SSS proof produces five cases according to where the joining line meets the line containing the common side; three are proved directly and two follow by symmetry/relabelling.                                                                       |
| I.24  | Three configurations arise according to the positions of $D, E$ relative to $GF$ . Euclid explicitly treats only the third. Wilkins also gives alternative proofs that exchange the principal strategies used in the first and third configurations.                     |
| I.35  | Three configurations arise for the upper sides of the two parallelograms: overlap, common endpoint, and disjoint. Euclid explicitly treats only the disjoint case. The corresponding upper-point orders are $A, E, D, F$ , $A, D, F$ , and $A, D, E, F$ .                |
| I.39  | Two configurations must be excluded if the two same-base equal-area triangles are not in the same parallels: the second apex lies on the opposite side or on the same side of the auxiliary parallel through the first apex. Euclid explicitly discusses only the first. |
| I.40  | Six configurations must be excluded: adjacent, separate, or interlocking equal bases, each combined with the two possible positions of the second apex relative to the auxiliary parallel. Euclid explicitly discusses only one of the six.                              |

These examples justify treating “configuration audit” as a separate source function. A complete formal theorem cannot appeal to the same picture in all cases. Either the proof performs the corresponding case split, or an earlier stronger theorem must already have packaged the missing positional information.

## C.5 Converses, equivalences, and alternative proof paths

The Study Notes expose a network of logical relations that is almost invisible if Book I is read only as a linear sequence.

- I.6 is the converse of the base-angle conclusion of I.5.
- I.18 and I.19 convert strict side order to strict opposite-angle order and conversely.
- I.24 and I.25 form the analogous comparison pair for two triangles with two corresponding sides fixed.
- I.27 is the converse of the Alternate Angles Theorem contained in I.29.

- The corresponding-angle criterion in I.28 is the converse of the Corresponding Angles Theorem contained in I.29.
- I.30 is logically equivalent to Playfair's Axiom.
- Playfair's Axiom together with I.27 implies I.29.
- I.31 proves existence using I.23 and I.27, whereas I.29 supplies uniqueness.
- I.39 is the converse-area counterpart of I.37: equal-area same-base triangles on the same side force the same parallels.
- I.40 is the equal-base converse counterpart of I.38 and expands the configuration audit to six cases.
- I.41 is a forward composition result: I.34 plus I.37 turns a parallelogram into twice any triangle on the same base and in the same parallels.
- I.42 is the first construction in this block that uses equal-area results as an engine: midpoint + I.23 + I.31 + I.38 + I.41.
- I.32 has both Euclid's proof and the proof attributed by Eudemus to the Pythagoreans; Wilkins explains how the two configurations are related.
- I.20 has both Euclid's isosceles-extension proof and the ancient angle-bisector proof attributed by Proclus to Heron or Porphyry.
- I.24 admits alternative configuration-wise proofs in addition to the main Euclidean/Proclean treatment.

The formal significance is that there is rarely a unique mathematically natural proof path. A reconstruction can therefore be evaluated not only for correctness but also for how well its helper library exposes the short routes already visible in the classical dependency network.

## C.6 Common Notions as proof operations

Wilkins's Study Notes are also useful because they name the Common Notion operations instead of treating them as invisible algebra.

Repeated patterns include:

- Common Notion 1: transitivity through equality to a common magnitude, used already in I.1 and repeatedly thereafter;
- Common Notion 2: addition of equal magnitudes, used in angle decompositions and explicitly in the overlap and common-endpoint branches of I.35;
- Common Notion 3: subtraction or cancellation of equal magnitudes, prominent in I.2, I.5, I.15, I.28, and the disjoint branch of I.35, and again as the final transitivity step in I.36;
- the part-whole principle, used in comparison and reductio arguments such as I.6 and later order proofs, and in area form in the reductio arguments of I.39–I.40;
- Common Notion 4 / superposition in I.4, whose geometric meaning is considerably stronger than a bare logical equality rule.

For formalization these operations should not automatically be converted into numerical arithmetic. Their synthetic counterparts are segment addition/subtraction, angle addition or supplement relations, strict part-whole order, congruence transport, and finite cut-and-paste operations on figures.

## C.7 Source-audit observations retained for the formal project

The complete I.1–I.45 audit supports the following durable observations.

- Wilkins is strongest as a source for the classical Euclidean dependency DAG, not for reorganizing Euclid into Hilbert’s axiomatic order.
- Construction propositions must be audited separately for existence. Circle-circle and line-circle intersections are explicit issues in I.1, I.2, I.12, and I.22.
- Euclid’s diagram often suppresses order and plane separation. I.7 and I.16 make this especially visible, while I.24 and I.35 show that omitted positional cases can survive far into Book I.
- Congruence rules have different classical costs. SAS is I.4; SSS is I.8 and depends on I.7 in Euclid’s route; ASA and SAA appear only at I.26, with SAA additionally using I.16.
- Proposition numbering is not the same thing as logical strength. I.27 and I.28 are neutral despite concluding parallelism, while I.29 is the first proposition whose proof uses the Fifth Postulate.
- I.30 should be remembered as an equivalence result, not merely as transitivity of parallelism: Wilkins proves its equivalence with Playfair’s Axiom.
- I.31 supplies the cleanest existence-versus-uniqueness split in the classical notes: I.23 plus I.27 gives existence; I.29 gives uniqueness.
- I.35 is the first proposition in the present checkpoint where the proof architecture itself becomes a finite cut-and-paste calculus. Wilkins’s three branches provide the configuration specification that the formal scissors proof must cover.
- I.36–I.38 stabilize the forward area block: I.36 reuses I.35, while I.37 and I.38 obtain triangle equality through doubled parallelograms. The Study Notes leave the “halves of equals” step implicit.
- I.39 and I.40 mark a converse boundary. Wilkins’s two-case and six-case analyses show that a proof from equal area to parallelism requires both strict part-whole content and explicit positional classification.
- The Study Notes for I.39 and I.40 contain an apparent repeated local citation error: they name I.36 when invoking equality of areas of comparison triangles. The surrounding geometry points to I.37 in I.39 and I.38 in I.40. The audit records the source wording instead of silently normalizing it.
- I.41 returns to a purely forward content calculation and is correspondingly short.
- I.42 records a construction detail essential for formal work: the copied-angle ray is chosen on the side of the base containing the original apex. It does not separately prove that this ray meets the parallel through the apex. The note also contains the local label inconsistency *AECG/FECE*, which the audit retains.

- The Study Note for I.24 contains an apparent local citation inconsistency in one alternative proof: it names Proposition I.22 while calling it the Triangle Inequality. Since Wilkins explicitly identifies I.20 as the Triangle Inequality elsewhere, the audit records the inconsistency instead of silently rewriting the source.

## C.8 I.43–I.45: complements and finite application of areas

The last three propositions make the role of the Study Notes especially clear. I.43 is still a comparison theorem: the diagram determines a finite subtraction pattern inside one parallelogram. I.44 turns that content calculus into a rigid placement construction on a prescribed segment. I.45 then iterates the placement operation over a finite triangulation.

For I.45 Wilkins goes beyond Euclid’s first quadrilateral example and writes the general case directly as a finite family of triangles  $T_1, \dots, T_k$  with disjoint interiors. The corresponding applied parallelograms share one fixed side and one fixed angle, while their other sides are concatenated. This is almost exactly the outer architecture of the Lean proof: a one-triangle base case followed by repeated extension of the current parallelogram.

The difference is representational rather than mathematical. Wilkins starts with one rectilinear region and a genuine triangulation of it. The current Lean theorem starts with a nonempty list of nondegenerate triangles and proves an additive statement for that list. The nonoverlap/one-region invariant is therefore external to `euclid_proposition_45`, even though the inductive content calculation is the same.

## C.9 Conclusion

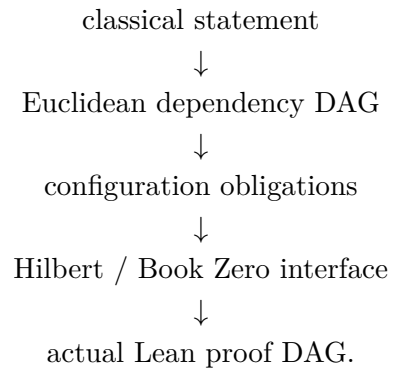
The full Wilkins audit I.1–I.45 confirms that these Study Notes deserve a permanent place in the Book I source procedure.

Greenberg and Hartshorne are especially valuable for identifying the axiomatic layer in which a proposition belongs. Wilkins serves a different function. He reconstructs the local Euclidean proof graph: which earlier propositions are actually invoked, which Common Notions perform the algebraic work, which converse or equivalent theorem is nearby, and which positional cases must be added to obtain a complete proof.

Four structural transitions are particularly clear in this checkpoint.

First, the early construction propositions repeatedly expose hidden intersection and continuity assumptions. Second, I.27–I.30 isolate the neutral-to-Euclidean reversal in the theory of parallels, culminating in the equivalence of I.30 with Playfair’s Axiom. Third, I.35 moves from parallelogram congruence to finite decomposition and recombination of figures, and I.36–I.38 stabilize the resulting forward area calculus. Fourth, I.39–I.40 expose the converse boundary through two-case and six-case positional audits, after which I.41–I.45 return to forward composition and use area equality constructively.

For the Lean reconstruction, this is precisely the information needed before implementation begins. The source audit can distinguish:



The present appendix should therefore be treated as the durable Wilkins record for Euclid Book I, Propositions I.1–I.45. Later propositions should extend this record rather than replacing the observations established here.

## Appendix D

# Hilbert: The *Grundlagen* Architecture

### D.1 Purpose and scope

This appendix records a source audit of Euclid Book I, Propositions I.1–I.45, against David Hilbert’s *Foundations of Geometry* (*Grundlagen der Geometrie*), in the second English edition translated from the tenth German edition [2].

The purpose of this appendix is different from the Wilkins audit. Wilkins is especially useful for the internal Euclidean dependency DAG. Hilbert reorganizes the subject itself. He replaces Euclid’s linear sequence by five axiom groups and then derives a relatively small number of structural theorems from them.

For each Euclidean proposition the audit therefore asks:

- whether Hilbert preserves an essentially identical theorem;
- whether the Euclidean result has become an axiom, a numbered theorem, an immediate corollary, or merely part of the developed infrastructure;
- which axiom groups are active at the point where the relevant Hilbert theorem is proved;
- whether the proposition lies before or after the introduction of Group IV, the axiom of parallels;
- whether continuity is used;
- how Hilbert treats the change from congruence to plane area at Euclid I.35;
- what this organization suggests for the Hilbert/Book Zero/Lean architecture of the present project.

A warning about terminology is important. Hilbert does not normally label his theorems by Euclid’s proposition numbers. Consequently the phrase “Hilbert counterpart” below does not mean that Hilbert himself explicitly wrote “Euclid I.n” unless this is stated in the audit. It means that the theorem, axiom, or developed piece of infrastructure performs the corresponding mathematical role.

## D.2 Hilbert’s organization of the elementary theory

Chapter I divides the axioms into five groups: incidence, order, congruence, parallels, and continuity. For plane geometry the fundamental layers relevant here are

- Group I: incidence,
- Group II: order and plane separation,
- Group III: segment and angle congruence,
- Group IV: uniqueness of parallels,
- Group V: Archimedean and completeness axioms.

The first major part of Euclid Book I is absorbed into Groups I–III and the theorems derived from them. In particular, Hilbert’s Chapter I, Section 6 contains the base-angle theorem, three triangle congruence theorems, quantitative comparison of angles, the theorem that all right angles are congruent, the Exterior Angle Theorem, midpoint existence, and angle-bisection existence.

A second boundary occurs in Section 7. Before Group IV is stated, Hilbert uses the Exterior Angle Theorem to show that through a point outside a line one can draw a line that does not meet the given line. Group IV then asserts that there is *at most one* such line. Thus existence and uniqueness of parallels are separated at the axiomatic level.

A third boundary occurs at Chapter IV. Hilbert introduces a new language for plane figures: equidecomposability and equicomplementability. This chapter assumes the line and plane axioms of Groups I–IV but explicitly omits the continuity axioms. Theorems 44–46 organize the forward content theory behind Euclid I.35–I.38, while Theorem 48 isolates the converse difficulty that appears at I.39.

## D.3 Comparison table

Table D.1: Euclid I.1–I.45 in Hilbert’s *Grundlagen der Geometrie*

| Euclid | Hilbert counterpart                                                                                                                                                                                                                          | Status in the Grundlagen                                                                                                                                                                                                                                    | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                                                                                                                        |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.1    | No separately numbered Chapter I theorem reproducing Euclid’s circle–circle construction was located. Relevant primitive infrastructure is supplied instead by incidence and congruence, while circles are introduced only after Theorem 31. | The proposition is not preserved one-for-one in Hilbert’s initial theorem sequence. Hilbert’s aim is to axiomatize congruence and construction directly rather than to rebuild the early Euclidean straightedge-and-compass DAG proposition by proposition. | The absence of a literal I.1 counterpart is structurally important. The present Lean reconstruction likewise need not imitate Euclid’s two circles if an equidistant-point existence theorem is already present in the Hilbert interface. The source audit should therefore distinguish the Euclidean construction from the Hilbert theorem architecture rather than forcing a circle-intersection proof into the Grundlagen route. |

Continued on next page



Table D.1 – continued from previous page

| Euclid | Hilbert counterpart                                                                                                                                                                                                | Status in the Grundlagen                                                                                                                                                                                                                             | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                 |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.2    | Axiom III,1: a segment congruent to a given segment can be laid off from a prescribed point on a prescribed side of a line.                                                                                        | The central content of Euclid I.2 has become primitive congruence infrastructure. Hilbert explicitly says that III,1 requires the possibility of constructing segments; uniqueness is proved later.                                                  | This is a paradigmatic compression of Euclid by Hilbert. Segment transport should therefore live low in the formal interface. A Book I proof that reconstructs Euclid I.2 from circles would be historically faithful to Euclid but architecturally contrary to the Grundlagen organization used by the project.             |
| I.3    | Axiom III,1 together with order on a line and the later uniqueness of segment construction.                                                                                                                        | No separately numbered theorem corresponding exactly to I.3 is needed. Once a congruent segment can be laid off on a chosen ray, the operation of cutting off a prescribed length from a longer segment is a basic order-and-congruence consequence. | Hilbert separates the metric operation from Euclid's circle construction. In Lean this is exactly the sort of statement that should normally be a Book Zero or interface lemma over segment layoff, betweenness, and uniqueness rather than a new foundational construction.                                                 |
| I.4    | Axiom III,5 and Theorem 12, the first congruence theorem for triangles (SAS).                                                                                                                                      | The core of Euclid I.4 is axiomatic in Hilbert's system. Axiom III,5 gives the angle consequences from two corresponding sides and their included angle; Theorem 12 packages the full triangle congruence, including the third side.                 | This is one of the clearest examples of Hilbert changing the dependency graph. SAS is not a Book I theorem to be reconstructed from superposition; it belongs to Group III. The Lean interface should therefore expose SAS as foundational congruence machinery and use I.4 only as the Euclidean-facing wrapper if desired. |
| I.5    | Theorem 11: the base angles of an isosceles triangle are congruent.                                                                                                                                                | This is a direct mathematical counterpart of the principal isosceles-triangle conclusion of Euclid I.5. In the main text it follows from the broad triangle congruence axiom III,5 and the last part of III,4.                                       | Theorem 11 belongs entirely to the congruence layer of Chapter I. A later appendix of the Grundlagen studies how sensitive this theorem is to weakening III,5, but the main development treats it as an immediate Group III consequence. This supports keeping the isosceles theorem in the reusable Hilbert interface.      |
| I.6    | Theorem 24: a triangle with two equal angles is isosceles.                                                                                                                                                         | Hilbert states the converse of Theorem 11 explicitly and derives it immediately from Theorem 23.                                                                                                                                                     | Euclid's reductio construction disappears. Once strict comparison of angles and sides has been developed, I.6 becomes a short comparison corollary. This is a strong precedent for preferring the developed Hilbert order API over replaying Euclid's longer proof.                                                          |
| I.7    | No one-to-one numbered counterpart was located. The relevant uniqueness content is distributed through plane separation, angle construction uniqueness, Theorem 17, and the SSS development leading to Theorem 18. | Hilbert does not preserve Euclid I.7 as a named step in the route to SSS. Instead he develops general same-side/different-side and congruence machinery and proves the third triangle congruence theorem directly.                                   | This is precisely the sort of proposition that disappears when the axiomatic interface is strengthened. Formal work should record which general uniqueness/side-of-line lemmas replace I.7, but should not invent a Hilbert theorem number that the source does not provide.                                                 |

Continued on next page

Table D.1 – continued from previous page

| Euclid | Hilbert counterpart                                                                                                                                                                                                              | Status in the Grundlagen                                                                                                                                                                                                                                                                  | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                      |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.8    | Theorem 18, the third congruence theorem for triangles (SSS).                                                                                                                                                                    | Hilbert proves SSS from the preceding congruence machinery. The proof uses Theorem 17, Theorem 12, segment congruence, and uniqueness of angle construction.                                                                                                                              | This is a central source result for the project: in the main Grundlagen development SSS is a theorem, not an extra Euclidean axiom. Any temporary <code>HilbertSSS</code> assumption in Lean should therefore be regarded as foundational debt unless the chosen Hilbert interface deliberately takes a stronger primitive basis. |
| I.9    | Immediately after Theorem 26 Hilbert states that every angle can be bisected, as an immediate consequence of Theorems 11 and 26.                                                                                                 | Angle bisection is not assigned a new theorem number. It is a corollary of the isosceles theorem and midpoint existence.                                                                                                                                                                  | This is a useful architectural signal. Angle-bisector existence belongs in the reusable Hilbert layer; Euclid I.9 need not carry a separate equilateral-triangle construction once Theorems 11 and 26 are available.                                                                                                              |
| I.10   | Theorem 26: every segment can be bisected.                                                                                                                                                                                       | This is a direct counterpart of Euclid I.10, but Hilbert uses a different construction. Equal angles are constructed at the endpoints, equal segments are laid off, and Theorem 25 is applied after an order argument establishes that the intersection lies inside the original segment. | Hilbert’s proof is especially relevant to Book Zero because it makes the positional work explicit: the candidate midpoint must first be shown to lie between the endpoints. The theorem is a natural reusable interface result rather than a construction that later propositions should re-prove.                                |
| I.11   | Existence of right angles follows after Theorem 14; Axiom III,4 then permits a congruent angle to be constructed on a prescribed side of a prescribed ray.                                                                       | Hilbert does not label a separate theorem as Euclid I.11. The needed operation is absorbed into right-angle existence plus the angle construction axiom.                                                                                                                                  | The formal analogue should likewise be a general <code>right_angle_exists</code> /perpendicular construction theorem. Euclid I.11 is then naturally a thin wrapper over the developed Hilbert infrastructure.                                                                                                                     |
| I.12   | No separately numbered Chapter I counterpart was located. Perpendicular constructions are used throughout the later text, and Chapter VII treats “construct a perpendicular to a given line” as a standard construction problem. | Hilbert does not retain Euclid I.12 as an isolated early proposition in the main theorem list. The construction belongs to the developed incidence/congruence construction theory rather than to a Euclidean proposition sequence.                                                        | This supports the current project architecture in which perpendicular-from-point existence is extracted as a reusable Hilbert theorem. The appendix deliberately does not claim a theorem number that the Grundlagen does not assign.                                                                                             |
| I.13   | Definitions of supplementary and right angles; Theorems 14, 15, and 21 provide supplement transport, angle composition, and congruence of all right angles.                                                                      | No theorem is labelled as the literal Euclid I.13 statement. Hilbert replaces the informal arithmetic of adjacent angles by a developed synthetic supplement and angle-composition theory.                                                                                                | For Lean this is a direct warning against introducing degree arithmetic merely to express “two right angles.” Supplementary-angle congruence and ordered angle composition are the Hilbert-native language.                                                                                                                       |

Continued on next page

Table D.1 – continued from previous page

| Euclid | Hilbert counterpart                                                                                                                                    | Status in the Grundlagen                                                                                                                                                                                                                                                             | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                           |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.14   | Axiom III,4 (unique angle construction), together with the supplementary-angle theory of Theorem 14.                                                   | The converse straight-line criterion is not isolated as a numbered theorem. Its logical content is absorbed by uniqueness of constructing an angle on a given side of a ray and by supplement transport.                                                                             | The Hilbert route suggests treating I.14 as a collinearity/angle uniqueness lemma rather than as a new major Euclidean construction.                                                                                                                                                                                                   |
| I.15   | Immediate corollary of Theorem 14: vertical angles are congruent.                                                                                      | Hilbert says explicitly that the congruence theorem for vertical angles is an immediate corollary of the supplementary-angle theorem.                                                                                                                                                | Vertical-angle congruence is therefore low-level reusable infrastructure. It should not be reproved inside later propositions such as I.16, I.26, or the midpoint constructions.                                                                                                                                                       |
| I.16   | Theorem 22, the Exterior Angle Theorem.                                                                                                                | This is a direct structural counterpart: an exterior angle of a triangle is greater than either nonadjacent interior angle. It is proved inside Section 6 from Groups I–III, before the axiom of parallels is introduced.                                                            | This is one of the most important neutral theorems in the whole route. It powers Hilbert’s pre-Group-IV construction of a nonintersecting line and therefore explains why Euclid I.27 is neutral in the project.                                                                                                                       |
| I.17   | Immediate consequence of Theorem 22 together with the supplementary-angle theory.                                                                      | Hilbert does not assign a separate theorem number to the Euclidean statement that any two angles of a triangle sum to less than two right angles. The required inequality is already contained in the Exterior Angle Theorem plus supplements.                                       | This distinction matters later in Chapter II: Legendre’s Theorem 35 is the stronger statement about the sum of all three angles and invokes Archimedes’ axiom, whereas the Euclid I.17 two-angle inequality is an earlier neutral consequence and should not be conflated with it.                                                     |
| I.18   | A direct comparison corollary of Hilbert’s quantitative angle comparison together with Theorems 11, 23, and 24.                                        | Hilbert explicitly numbers the converse direction as Theorem 23 (greater angle opposite greater side). The side-to-angle direction is not separately numbered in the main Chapter I list.                                                                                            | The formal result should be placed in the common comparison API. Once side and angle trichotomy are available, Euclid’s construction can be replaced by the short Hilbert comparison argument.                                                                                                                                         |
| I.19   | Theorem 23: in every triangle the greater angle lies opposite the greater side.                                                                        | This is a direct counterpart of Euclid I.19.                                                                                                                                                                                                                                         | Hilbert’s quantitative comparison of angles and segments makes the theorem a basic order result. It is also used later when the triangle inequality is expressed in the segment arithmetic of Chapter III.                                                                                                                             |
| I.20   | Triangle inequality, explicitly said later to follow from Theorem 23 once the sum of segments is interpreted by the segment arithmetic of Chapter III. | Hilbert does not make I.20 a numbered theorem in the Chapter I sequence. In Chapter IV he states that from Theorem 23 it follows that the sum of two sides of a triangle is greater than the third, where the sum is understood in the segment arithmetic introduced in Chapter III. | This is a striking reorganization. The geometric comparison theorem is established early; the literal additive statement is postponed until segment addition has a formal meaning. The present Lean project can instead use its synthetic segment-addition/less-than interface, but the same conceptual separation should be retained. |

Continued on next page

Table D.1 – continued from previous page

| Euclid | Hilbert counterpart                                                                                                                                                          | Status in the Grundlagen                                                                                                                                                                                          | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                    |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.21   | No one-to-one numbered counterpart was located. Its two ingredients are the triangle inequality just described and the Exterior Angle Theorem 22.                            | The proposition is naturally a compound corollary in Hilbert’s developed comparison theory rather than a new foundational theorem.                                                                                | For formalization the important data are the interior-point configuration and two repeated comparison chains. No new Hilbert axiom should be introduced merely because Euclid gives the result its own proposition number.                                                                                      |
| I.22   | No direct numbered Chapter I counterpart was located for Euclid’s three-prescribed-sides construction.                                                                       | Hilbert has segment layoff (III,1), SSS (Theorem 18), and later a general chapter on ruler-and-compass constructions, but the proposition is not preserved as a named theorem in the elementary Chapter I list.   | The audit should therefore keep two questions separate: uniqueness/congruence of a triangle with three sides, which is Theorem 18, and existence of such a triangle under triangle inequalities, which is a construction problem. This is exactly the distinction that becomes important in a formal interface. |
| I.23   | Axiom III,4: every angle can be constructed uniquely on a prescribed side of a prescribed ray.                                                                               | The construction content of Euclid I.23 has become primitive angle congruence infrastructure.                                                                                                                     | This is the angular analogue of I.2/III,1. It explains why later Hilbert proofs can copy angles directly without replaying Euclid’s I.20–I.22 dependency chain. In Lean, angle transport belongs in the interface, not in each downstream proposition.                                                          |
| I.24   | No separately numbered counterpart. The relevant machinery is Theorem 20 (quantitative comparison of angles), Theorem 12 (SAS), and Theorem 23 (angle/side comparison).      | Hilbert develops a general order of angles before stating the fundamental exterior-angle and triangle-comparison theorems, rather than preserving Euclid’s two-triangle comparison proposition as a named result. | This is best formalized as a reusable SAS monotonicity/comparison lemma. The source supports viewing I.24 as derived comparison infrastructure, not a new axiomatic layer.                                                                                                                                      |
| I.25   | Converse comparison obtained from the same quantitative comparison machinery and the congruence theorems.                                                                    | No one-to-one theorem number was located. Hilbert’s general trichotomy and transitivity for angle and segment comparison absorb the logical structure of Euclid I.25.                                             | The project can profit from the same compression: once the forward comparison and trichotomy are in the interface, the converse should be a small logical theorem rather than a second large construction.                                                                                                      |
| I.26   | Theorem 13 (second triangle congruence theorem) and Theorem 25 (extension of it): ASA and the second two-angle/one-side congruence configuration.                            | Hilbert splits Euclid I.26 across two numbered triangle congruence theorems. Theorem 13 treats the side between the two given angles; Theorem 25 extends the result using the Exterior Angle Theorem.             | This is a useful dependency refinement. The two forms should not be conflated if the formal project is tracking minimal hypotheses. Theorem 25 explicitly depends on the neutral Exterior Angle Theorem 22.                                                                                                     |
| I.27   | The paragraph immediately preceding Group IV: equal intersection angles constructed through an external point give a line that cannot meet the original line, by Theorem 22. | Hilbert proves existence of a nonintersecting line before stating the parallel axiom. Parallelism is then defined as coplanarity plus nonintersection.                                                            | This is the exact Hilbert source for the project’s treatment of Euclid I.27 as neutral. The result uses Groups I–III and the Exterior Angle Theorem, not Group IV. It is the crucial separation between angle-to-parallel and parallel-to-angle reasoning.                                                      |

Continued on next page

Table D.1 – continued from previous page

| Euclid | Hilbert counterpart                                                                                                                                                                               | Status in the Grundlagen                                                                                                                                                                                                                          | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                              |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.28   | The neutral angle-to-nonintersection direction is contained in the same pre-Group-IV mechanism as I.27; the full corresponding/alternate angle equivalence is later bundled into Theorem 30.      | Hilbert does not isolate Euclid's two I.28 criteria as a separately numbered theorem. The source first establishes nonintersection from an appropriate equality of intersection angles, then after Group IV states Theorem 30 in both directions. | For the formal audit the direction matters more than the theorem number. The implication from an angle criterion to parallelism belongs on the neutral side; the converse direction is the Euclidean content introduced with Group IV.                                                    |
| I.29   | Theorem 30, after adjoining the axiom of parallels: a transversal of two parallels gives congruent corresponding and alternate angles.                                                            | This is Hilbert's principal counterpart of Euclid I.29. It appears only after Group IV has supplied uniqueness of the parallel through an external point.                                                                                         | The source confirms the exact axiomatic boundary used in the project. The non-neutral ingredient is not existence of a nonintersecting line; that was already available. It is the uniqueness principle that allows one to identify the constructed candidate with the given parallel.    |
| I.30   | Equivalent formulation of Group IV: if two lines in a plane do not meet a third line, then they do not meet each other.                                                                           | Hilbert states and proves the equivalence immediately after the parallel axiom. This is the transitivity-of-parallelism content of Euclid I.30.                                                                                                   | I.30 is therefore an interface form of the Euclidean parallel axiom, not merely an innocent derived convenience. Any use of general parallel transitivity in Lean should be recognized as consuming the Group IV layer.                                                                   |
| I.31   | Existence is established before Group IV by constructing an equal angle and using Theorem 22; Group IV then gives exactly one parallel through the external point.                                | Hilbert explicitly separates the two claims: a nonintersecting line can always be drawn before the axiom of parallels; the axiom says there is at most one, and hence exactly one.                                                                | This is the strongest source precedent for keeping existence-versus-uniqueness explicit. The Euclidean proposition asks for a construction, but the Hilbert theory reveals that construction existence is neutral while uniqueness is Euclidean.                                          |
| I.32   | Theorem 31: the angles of a triangle add up to two right angles.                                                                                                                                  | The theorem is stated immediately after Theorem 30 in the Group IV section. Hilbert explicitly places the triangle angle-sum theorem in the Euclidean parallel layer.                                                                             | This provides the direct Grundlagen counterpart to the project's synthetic I.32 reconstruction. The Lean statement can avoid numerical degrees, but its dependency should still visibly pass through the Euclidean parallel interface.                                                    |
| I.33   | No separately numbered theorem was located. The result is a standard corollary of Theorem 30 together with the triangle congruence theorems and the neutral converse angle-to-parallel criterion. | Hilbert's main theorem list moves from Theorem 30 and Theorem 31 to circle theory rather than preserving Euclid I.33 as a named proposition.                                                                                                      | The proposition belongs to the already developed Euclidean parallel-plus-congruence interface. A formal proof should therefore be short once I.29/Theorem-30 transport and triangle congruence are reusable; a long proof would signal missing interface lemmas rather than new geometry. |

Continued on next page

Table D.1 – continued from previous page

| Euclid | Hilbert counterpart                                                                                                                                                                                                                 | Status in the Grundlagen                                                                                                                                                                                                                                           | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                                                                                                                                           |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.34   | No separately numbered Chapter I theorem, but the Grundlagen later uses “the theorem on the equality of opposite sides in a parallelogram.” The full Euclidean result follows from Theorem 30 and the triangle congruence theorems. | Hilbert treats parallelogram congruence as familiar derived infrastructure rather than as a numbered theorem in the initial list. The equality of opposite sides is used later in the theory of proportion.                                                        | This supports placing the I.34 consequences in the reusable Euclidean parallelogram interface. For the current project I.34 is precisely the last congruence package before the language changes at I.35.                                                                                                                                                                                                                                              |
| I.35   | Chapter IV, Theorem 44: two parallelograms with the same bases and the same altitudes are equicomplementable.                                                                                                                       | Hilbert explicitly says that Euclid’s well-known argument gives Theorem 44. Chapter IV first defines finite decomposition into congruent triangles, equidecomposability, and equicomplementability. The chapter assumes Groups I–IV but not the continuity axioms. | This is the decisive Grundlagen source for the I.35 formalization. The correct Hilbert-strength conclusion is equicomplementability, not numerical area and not automatically the stronger claim of equidecomposability. Hilbert immediately warns that equal-base, equal-altitude figures can be equicomplementable without being equidecomposable in non-Archimedean geometry. The formal scissors layer should therefore preserve this distinction. |
| I.36   | Theorem 44 again: parallelograms with equal bases and equal altitudes are equicomplementable.                                                                                                                                       | Hilbert’s formulation is already at the general equal-base/equal-altitude level. Thus no second area theorem is needed when Euclid passes from the common-base case I.35 to the equal-base case I.36.                                                              | This confirms that I.36 should introduce no new content principle. The extra work in a Euclid-facing or Lean proof is geometric normalization of the equal-base configuration; once equal altitude is available, Theorem 44 already supplies the correct synthetic content relation.                                                                                                                                                                   |
| I.37   | Theorem 46: two triangles with equal bases and altitudes are equicomplementable.                                                                                                                                                    | Hilbert derives Theorem 46 immediately from Theorems 44 and 45 together with Theorem 43. Theorem 45 first makes every triangle equidecomposable with a parallelogram of equal base and half the altitude.                                                          | This is the exact Hilbert-strength target for I.37. In particular, Hilbert warns that the stronger general claim of equidecomposability cannot be proved without the Archimedean axiom. The project’s use of <code>HilbertScissorsEquicomplementable</code> therefore matches the source rather than weakening a numerical-area statement.                                                                                                             |
| I.38   | Theorem 46 again, now used at its full equal-base generality.                                                                                                                                                                       | I.38 is not assigned a separate Hilbert theorem number. Equal bases and the same parallel carriers give equal altitudes, so the proposition is an immediate geometric instance of Theorem 46.                                                                      | The formal project need not expose Theorem 46 as one monolithic call. Reducing I.38 to I.37 by an oriented congruent copy is compatible with the Grundlagen architecture: the new work is the geometry of the copy, while the content conclusion remains exactly Theorem-46 strength.                                                                                                                                                                  |

Continued on next page

Table D.1 – continued from previous page

| Euclid | Hilbert counterpart                                                                                                                                                                                                                                                                                                                         | Status in the Grundlagen                                                                                                                                                                                                                                                                                                                                                                                                                                          | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                            |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.39   | Theorem 48: if two equicomplementable triangles have the same base, then they have the same altitude.                                                                                                                                                                                                                                       | Hilbert explicitly identifies this as the fundamental theorem occurring in Euclid Book I as Theorem 39. It is the converse of Theorem 46 and is not treated as a routine consequence. Hilbert notes that Euclid’s appeal to “the whole is greater than any of its parts” amounts to an additional principle for equicomplementability, and then develops area in Sections 20–21 in order to prove Theorem 48 from the plane theory without the Archimedean axiom. | This is the decisive converse-content boundary. A formal reconstruction should isolate Theorem-48 strength explicitly rather than hide it inside an I.39-shaped construction axiom. The missing ingredient is not more line incidence; it is the strictness/cancellation content needed to recover altitude from equicomplementability. |
| I.40   | No separately numbered counterpart was located. The converse core is Theorem 48, combined with an equal-base-to-common-base reduction and the transitivity machinery of Theorem 43.                                                                                                                                                         | Hilbert names Euclid I.39 when stating Theorem 48 and does not add a second theorem for the equal-base companion I.40. The developed congruence and area theory is already sufficient to transport one triangle to a common base and reuse the converse.                                                                                                                                                                                                          | This is exactly the natural formal architecture: reuse the congruent-copy machinery of I.38 and the hard same-base converse of I.39. I.40 should not create a second independent converse-area principle.                                                                                                                               |
| I.41   | No one-to-one numbered theorem was located. In the Chapter IV framework, its content can be recovered from Theorem 45 together with the composition and comparison machinery of Theorems 43–44: a triangle can be replaced by a half-altitude parallelogram and the resulting doubled content can be compared with the given parallelogram. | Hilbert does not preserve Euclid’s word “double” as a separate theorem. Instead the conclusion is expressed through finite composition, equidecomposability, and equicomplementability.                                                                                                                                                                                                                                                                           | This supports the project’s term-level reading of “double” as a formal sum of two triangle terms. No scalar-valued area and no cancellation principle is required; I.41 is a forward content theorem.                                                                                                                                   |

Continued on next page

Table D.1 – continued from previous page

| Euclid | Hilbert counterpart                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Status in the Grundlagen                                                                                                                                                                                                                                                                                                                                     | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                                                                                                                                                                                           |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.42   | No separately numbered prescribed-angle counterpart was located. The necessary ingredients are already distributed through Hilbert’s system: Theorem 26 for the midpoint, Axiom III,4 for angle construction on a chosen side, the developed Group-IV parallel theory for the upper carrier, and the Chapter IV content theorems. The later Theorem 47 gives a broader but different canonical area construction, sending every triangle or polygon to an equicomplementable right triangle with a unit leg. | Hilbert therefore does not retain Euclid I.42 as a named stage. The exact prescribed-angle parallelogram construction is reconstructed from general operations rather than from a dedicated area theorem. Theorem 47 should not be misidentified as I.42: it demonstrates the strength of the developed area-construction theory but has a different target. | This is closely aligned with the Lean reconstruction. Midpoint, angle lay-off, line incidence, side-of-line information, and Euclidean parallel uniqueness are consumed as reusable Hilbert operations, while the content comparison is delegated to the already established I.38/I.41 layer. The formal proof must additionally expose the intersection and ray-orientation data suppressed by the classical diagram. |
| I.43   | No one-to-one numbered counterpart is needed. Section 18 already defines equidecomposability and equicomplementability, and Theorem 43 supplies their transitivity through a common comparison figure. The same section observes that removing equidecomposable pieces from equidecomposable polygons leaves equicomplementable remainders.                                                                                                                                                                  | Hilbert has already abstracted the common-notion calculation behind Euclid’s complements into the general content relations. The proposition therefore adds no new area principle in the <i>Grundlagen</i> .                                                                                                                                                 | This is exactly the useful reading for Lean I.43: keep the finite pieces explicit and use the scissors/equicomplementability definitions rather than postulating a free-standing cancellation law.                                                                                                                                                                                                                     |
| I.44   | No dedicated numbered prescribed-side application theorem was located. Its ingredients are already distributed through angle construction, segment/ray construction, Euclidean parallel uniqueness, parallelogram geometry, and the Chapter IV content calculus.                                                                                                                                                                                                                                             | Hilbert’s organization absorbs the Euclidean construction into reusable operations rather than preserving I.44 as a separate foundational stage. The content conclusion is forward; the difficult part is geometric placement.                                                                                                                               | The Lean reconstruction follows this organization: I.42 supplies content and shape, while the I.44 helps reconstruct the position and order data and use I.43 only as the content bridge. No new axiom is justified.                                                                                                                                                                                                   |

Continued on next page



Table D.1 – continued from previous page

| Euclid | Hilbert counterpart                                                                                                                                                                                                                                                   | Status in the Grundlagen                                                                                                                                                                                                                                                      | Source-audit remarks and relation to the Lean reconstruction                                                                                                                                                                                     |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.45   | Theorem 47 is the nearest structural analogue: every triangle, and hence every simple polygon, can be replaced by an equicomplementable right triangle with a unit leg. For polygons Hilbert explicitly triangulates the polygon and combines the triangular results. | The target is not Euclid’s prescribed-angle parallelogram, so Theorem 47 is not a literal counterpart of I.45. It nevertheless confirms that finite triangulation plus additive content composition is the intended general mechanism for passing from triangles to polygons. | This is the architectural control for the Lean induction. The theorem needs no numerical area, no continuity axiom, and no Archimedean axiom: the I.45 construction stays on the forward scissors/equicomplementability side of the area theory. |

## D.4 Hilbert counterparts and structural reductions in the I.1–I.45 block

The strongest one-to-one correspondences and the most important structural reductions are worth collecting separately:

|                  |                    |                                               |
|------------------|--------------------|-----------------------------------------------|
| Euclid I.2       | $\rightsquigarrow$ | Axiom III,1 (segment layoff),                 |
| Euclid I.4       | $\rightsquigarrow$ | Axiom III,5 / Theorem 12 (SAS),               |
| Euclid I.5       | $\rightsquigarrow$ | Theorem 11 (isosceles base angles),           |
| Euclid I.6       | $\rightsquigarrow$ | Theorem 24,                                   |
| Euclid I.8       | $\rightsquigarrow$ | Theorem 18 (SSS),                             |
| Euclid I.10      | $\rightsquigarrow$ | Theorem 26 (midpoint),                        |
| Euclid I.15      | $\rightsquigarrow$ | corollary of Theorem 14,                      |
| Euclid I.16      | $\rightsquigarrow$ | Theorem 22 (Exterior Angle),                  |
| Euclid I.19      | $\rightsquigarrow$ | Theorem 23,                                   |
| Euclid I.23      | $\rightsquigarrow$ | Axiom III,4 (angle construction),             |
| Euclid I.26      | $\rightsquigarrow$ | Theorems 13 and 25,                           |
| Euclid I.29      | $\rightsquigarrow$ | Theorem 30,                                   |
| Euclid I.32      | $\rightsquigarrow$ | Theorem 31,                                   |
| Euclid I.35–I.36 | $\rightsquigarrow$ | Theorem 44,                                   |
| Euclid I.37–I.38 | $\rightsquigarrow$ | Theorem 46,                                   |
| Euclid I.39      | $\rightsquigarrow$ | Theorem 48 (explicitly identified),           |
| Euclid I.40–I.42 | $\rightsquigarrow$ | derived Chapter IV constructions/corollaries. |

This display also shows why a formal reconstruction should not be judged by whether its local proposition files reproduce Euclid’s proof lengths. Several Euclidean propositions have become axioms or immediate corollaries, while I.40–I.42 are absorbed into the already developed Chapter IV construction and content machinery.

## D.5 The neutral theory before Group IV

The most important structural feature of Chapter I is the amount of geometry already available before the axiom of parallels.

From Groups I and II Hilbert obtains:

- ordering of arbitrary finite sets of collinear points;
- line separation of a plane into two sides;
- rays, polygonal segments, and interiors/exterior of polygons.

From Group III and its consequences he obtains:

- segment layoff and segment additivity;
- unique angle construction;
- SAS, ASA-type criteria, and SSS;
- supplementary and vertical angle transport;
- existence and congruence of right angles;
- quantitative comparison of segments and angles;
- the Exterior Angle Theorem;
- comparison of sides and angles in triangles;
- midpoint existence and angle bisection.

This is the true Hilbert background of Euclid I.1–I.28. The theorem numbering is different because the elementary propositions have been reorganized around primitive relations and reusable structural results.

The most consequential result is Theorem 22. It is strong enough to show, before Group IV, that through a point outside a line one can construct a line that does not intersect the given line. Hilbert then defines two coplanar nonintersecting lines to be parallel.

Thus the existence of a parallel is not the Euclidean step.

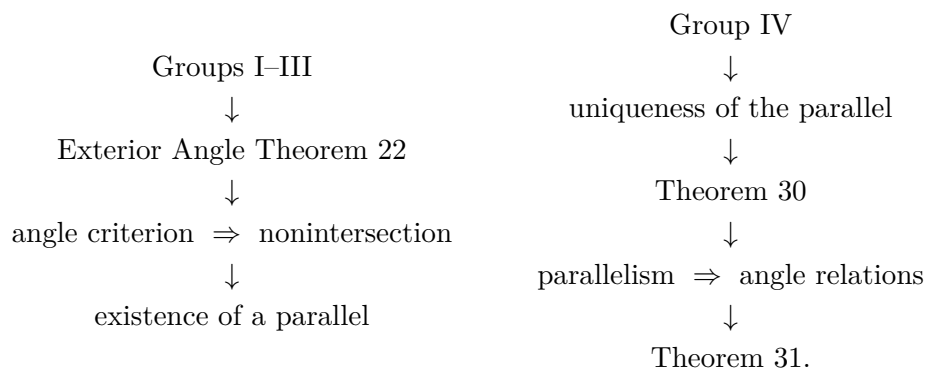
## D.6 The exact role of Group IV

Hilbert's Group IV reads, in substance:

through a point outside a line there is at most one line in the relevant plane that passes through the point and does not intersect the given line.

Since existence was already proved, Group IV converts “at least one” into “exactly one.”

This single observation organizes Euclid I.27–I.32:



The same section states an equivalent form of Group IV: if two lines do not meet a third line, they do not meet each other. This is the transitivity pattern corresponding to Euclid I.30.

For the present Lean development the consequences are concrete:

- I.27 must remain on the neutral side;
- the existence part of I.31 must remain on the neutral side;
- I.29 is the first theorem in this block that genuinely consumes the Euclidean uniqueness principle;
- general parallel transitivity should be classified as Euclidean;
- I.32 is Euclidean even if stated synthetically without degrees.

## D.7 Congruence architecture and the status of SSS

The Grundlagen is especially informative for the congruence layer. Axiom III,5 is the triangle-congruence axiom in the broad form. From it Hilbert obtains Theorem 11, the isosceles base-angle theorem, and Theorem 12, the first triangle congruence theorem. Theorem 18 then derives SSS.

Thus in the main text the hierarchy is schematically

$$\text{III,4} + \text{III,5} \longrightarrow \text{Theorems 11–17} \longrightarrow \text{Theorem 18 (SSS)}.$$

This matters directly to the formal project. SSS is stronger than the primitive SAS layer but is still intended as a theorem of the developed Hilbert theory.

The later Appendix II of the Grundlagen studies what happens if the triangle congruence axiom III,5 is weakened to the narrower III,5\*. There Hilbert/Bernays emphasize that the base-angle theorem is no longer automatic and analyze additional assumptions needed to recover it. The present audit follows the *main Chapter I system*, with the broader III,5. The Appendix II independence questions should therefore not be silently imported into the classification of the main Book I route, but they remain important when discussing possible reductions of the primitive congruence basis.

## D.8 Continuity and the I.1–I.45 checkpoint

Group V contains Archimedes’ axiom and the line-completeness axiom. They are not uniformly part of the elementary development relevant to this checkpoint.

In particular, Chapter IV begins by stating explicitly that the theory of plane area assumes the line and plane axioms of Groups I–IV and *excludes the continuity axiom*. Hilbert’s equicomplementability theory and Theorems 44–48 therefore do not require Group V. In particular, when introducing Theorem 48 Hilbert states that the converse can be established with the plane axioms and without the Archimedean axiom.

This should be distinguished from two different facts:

1. Euclid’s own early circle constructions contain intersection existence assumptions that modern sources may formulate as continuity principles.
2. Hilbert’s later independence investigations use Archimedes’ axiom for questions such as Legendre’s theorem and for comparing equicomplementability with stronger dissection properties.

Neither fact licenses the conclusion that the Hilbert Chapter IV proofs of the I.35–I.42 content block use continuity. The source explicitly says the opposite.

## D.9 The transition at I.35: from congruence to plane area

Chapter IV opens a genuinely new language.

Hilbert defines two simple polygons to be equidecomposable when they can be decomposed into finitely many triangles congruent in pairs. He then defines equicomplementability by allowing finitely many pairs of equidecomposable polygons to be adjoined so that the enlarged figures become equidecomposable.

Theorem 43 supplies the transitivity-style refinement properties needed for these relations. Section 19 then begins:

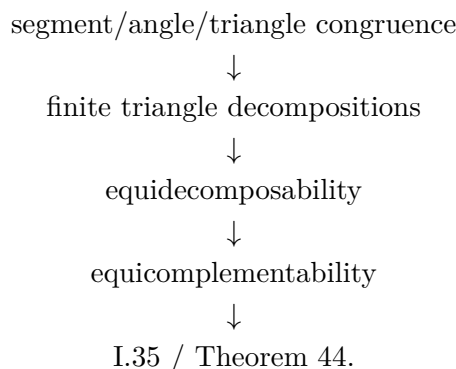
Euclid’s well-known argument ... furnishes the following theorem.

The resulting Theorem 44 states that two parallelograms with the same bases and the same altitudes are equicomplementable.

For Euclid I.35, “same base and between the same parallels” is exactly the classical configuration that produces the same altitude. Thus Theorem 44 is the correct Hilbert area-theoretic target.

The strength of the conclusion is crucial. Hilbert immediately distinguishes equicomplementability from equidecomposability. He notes that although equal-base/equal-altitude parallelograms and triangles are easily seen to be equidecomposable in the ordinary setting, a proof of that stronger statement without the Archimedean axiom is impossible in general: non-Archimedean examples can be equicomplementable without being equidecomposable.

Accordingly the formal hierarchy should be read as



This is the strongest direct source justification for the decision not to identify Proposition I.35 with a numerical area formula and not to strengthen its target indiscriminately to equidecomposability.

## D.10 The I.36–I.42 extension: forward content, converse, and construction

The continuation from I.35 to I.42 makes the internal organization of Hilbert’s Chapter IV much clearer. The seven Euclidean propositions do not correspond to seven new Hilbert theorems. Instead they fall into three structural groups.

### D.10.1 I.36–I.38: the forward content theorems

Theorems 44–46 already state the forward equal-content theory at a more general level than Euclid’s individual propositions.

- Theorem 44 : equal-base, equal-altitude parallelograms are equicomplementable,
- Theorem 45 : a triangle is equidecomposable with a half-altitude parallelogram,
- Theorem 46 : equal-base, equal-altitude triangles are equicomplementable.

Consequently I.36 is already contained in the generality of Theorem 44, while I.37 and I.38 are instances of Theorem 46. The distinction between “same base” and “equal bases” that drives Euclid’s proposition sequence is therefore not a new content distinction for Hilbert. It is a matter of geometric placement before the general equal-base/equal-altitude theorem is applied.

Theorem 45 is also architecturally important. It explains why the project can move between triangles and parallelograms without introducing numerical area. The individual conversion is stronger – equidecomposability – while the comparison theorem is deliberately stated at the weaker Equicomplementability level appropriate without the Archimedean axiom.

### D.10.2 I.39–I.40: the converse boundary

The direction changes at I.39. Hilbert says explicitly that one now needs the converse of Theorem 46 and states Theorem 48:

If two equicomplementable triangles have the same bases then they also have the same altitudes.

He immediately identifies this as the fundamental theorem appearing in Euclid Book I as Theorem 39. The source is unusually explicit about the logical cost. Euclid's use of "the whole is greater than any of its parts" is not treated as a harmless Common Notion inside the new content language; Hilbert says that this amounts to introducing a new geometric principle of equicomplementability.

Hilbert's response is not to add such a principle. Sections 20–21 develop area and use it to prove Theorem 48 from the plane theory, without the Archimedean axiom. Thus I.39 is the exact point where the forward scissors calculus ceases to be sufficient by equivalence laws alone.

I.40 adds no new converse theorem in the Grundlagen. Its natural Hilbertian reading is: reduce the equal-base configuration to a common-base one and then reuse Theorem 48. This is also the preferred library architecture, because it prevents the equal-base companion from duplicating the foundational debt already isolated at I.39.

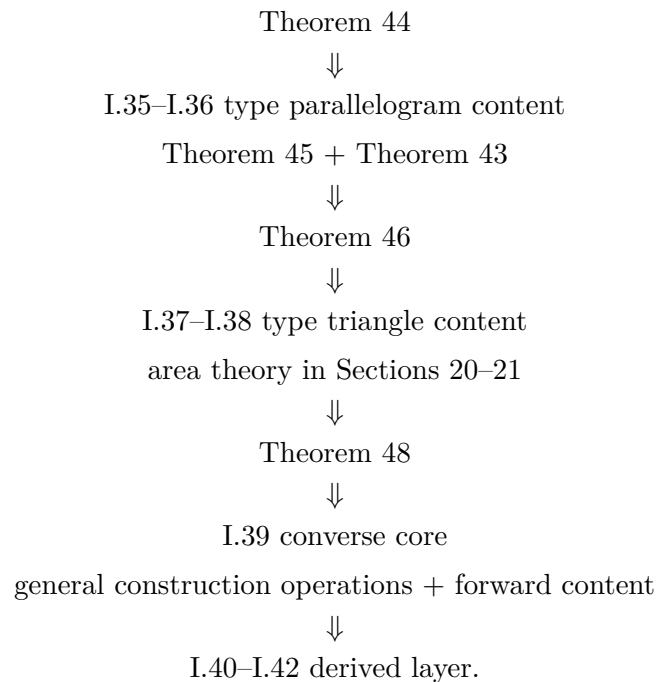
### D.10.3 I.41–I.42: composition and application of areas

After the converse pair, the direction becomes forward again. I.41 does not need Theorem-48-style cancellation. It composes the already developed triangle/parallelogram content theory: Theorem 45 provides the half-altitude parallelogram representation and Theorem 43 provides the composition and transitivity behavior of the content relations.

I.42 then turns equal content into a construction problem. Hilbert does not state a numbered theorem saying "construct a parallelogram equal to a given triangle in a prescribed angle." The geometric ingredients have already been made primitive or reusable: midpoint construction, angle layoff on a chosen side, line incidence, and the Euclidean uniqueness of parallels. The Chapter IV results supply the content comparison after the configuration has been constructed.

Theorem 47 is worth recording at this point, but only as a broader structural analogue. Hilbert proves that every triangle, and hence every simple polygon, can be transformed into an equicomplementable right triangle having a unit leg. This shows that Chapter IV has already become a genuine construction calculus for content. It is not, however, a literal counterpart of I.42, because the prescribed angle of Euclid's construction is replaced by a canonical right angle and unit leg.

The extension therefore has the schematic form



This is a stronger architectural description than a proposition-by-proposition parallel with Euclid. Hilbert compresses the forward content theorems into 44–46, isolates the converse as the genuinely difficult Theorem 48, and leaves the application-of-area constructions to the already developed general geometry and content machinery.

#### D.10.4 I.43–I.45: from complement calculus to finite polygon construction

The continuation through I.45 does not open a new Hilbertian layer. I.43 is already absorbed by the definitions and closure properties of Section 18. I.44 is a geometric placement problem assembled from the existing construction and parallel infrastructure. I.45 is then the finite composition step.

Theorem 47 is the decisive structural control. Hilbert proves its polygon case by decomposing a simple polygon into triangles, constructing the required canonical representative for each triangle, and combining the results. The target differs from Euclid I.45, but the proof architecture is the same one that matters for the library:

triangulation  $\longrightarrow$  triangle-level content constructions  $\longrightarrow$  finite additive composition.

Thus the completed I.45 proof confirms, rather than extends, the Chapter IV architecture already present in the project.

### D.11 Result of the Grundlagen audit

The audit shows that Hilbert’s *Grundlagen* is not a parallel edition of Euclid Book I. It is a re-factoring of the theory.

The early Euclidean constructions separate into three types.

First, some become axioms: segment transport (I.2) becomes III,1, angle transport (I.23) becomes III,4, and the core of SAS (I.4) becomes III,5.

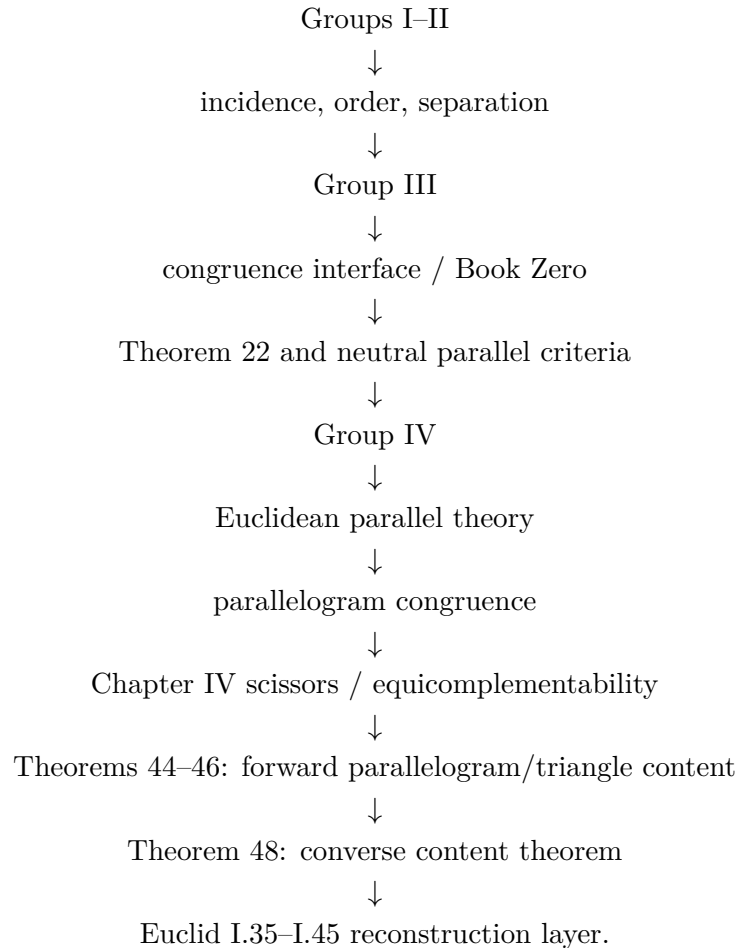
Second, some become strong reusable theorems: I.5 becomes Theorem 11, SSS/I.8 becomes Theorem 18, midpoint existence/I.10 becomes Theorem 26, the Exterior Angle Theorem/I.16 becomes Theorem 22, and the relevant comparison theorems become Theorems 23 and 24.

Third, several Euclidean propositions cease to be named at all because their content is absorbed into order, side-of-line, angle-composition, or construction infrastructure.

The parallel block then has a particularly clean logical shape. Theorem 22 already proves enough for the neutral existence of a parallel. Group IV contributes uniqueness. Theorem 30 is the full Euclidean parallel-angle theorem, and Theorem 31 is the triangle angle-sum theorem.

Finally I.35 does not merely add another consequence of parallelism. It changes the language of equality of figures. Chapter IV introduces equidecomposability and equicomplementability, explicitly without continuity. Theorem 44 controls I.35–I.36, Theorem 45 supplies the triangle-to-parallelogram conversion, Theorem 46 controls I.37–I.38, and Theorem 48 isolates the converse difficulty visible at I.39. I.40–I.45 then reuse this developed content theory together with the general Hilbert construction operations rather than opening new axiomatic layers.

For the current formal project this yields the architectural picture



## D.12 Current checkpoint

This appendix concerns Euclid Book I, Propositions I.1–I.45.

At this checkpoint the most important source conclusions are:



- Hilbert’s main ordering is by axiom group, not by Euclid number.
- Many Euclidean propositions are intentionally compressed into primitive segment or angle construction.
- SSS is Theorem 18 in the main development and is therefore a derived Hilbert theorem.
- The Exterior Angle Theorem 22 is established before Group IV.
- Existence of a parallel is established before Group IV.
- Group IV contributes uniqueness of the parallel.
- Theorem 30 is the Euclidean parallel-angle theorem.
- Theorem 31 is the Euclidean angle-sum theorem.
- I.33 and I.34 belong to the developed Euclidean parallel/congruence layer rather than opening a new foundation.
- Chapter IV introduces the new plane-area language.
- Theorem 44 controls the parallelogram content of I.35–I.36.
- Theorem 45 gives the reusable triangle-to-half-altitude- parallelogram conversion.
- Theorem 46 is the direct Hilbert target for I.37–I.38.
- Theorem 48 is explicitly identified by Hilbert with Euclid I.39 and marks the converse-content boundary.
- I.40 adds no separate Hilbert converse theorem; its natural role is reduction to the common-base Theorem 48 case.
- I.41 returns to forward composition of content and needs no Theorem-48-style cancellation principle.
- I.42 has no separately numbered prescribed-angle counterpart; Hilbert’s general midpoint, angle-construction, parallel, and Chapter IV machinery already supplies its components.
- Theorem 47 is a broader area-construction theorem – every triangle or simple polygon can be replaced by an equicomplementable right triangle with a unit leg – but it is not a literal counterpart of I.42.
- I.43 is absorbed by the general Section 18 calculus of equidecomposability and equicomplementability.
- I.44 introduces no new Hilbertian content principle; it is a placement construction built from existing angle, order, parallel, parallelogram, and Chapter IV machinery.
- I.45 has Theorem 47 as its closest structural analogue: both pass from triangles to a general polygonal figure by triangulation and finite composition of content.
- Chapter IV explicitly omits the continuity axioms, and Hilbert states that Theorem 48 can be established without the Archimedean axiom.
- Equicomplementability must not be confused with the stronger equidecomposability statement in non-Archimedean geometry.

The appendix should therefore be treated as the durable *Grundlagen* source audit for the I.1–I.45 checkpoint. Later Book I propositions can be appended while preserving the distinction between Euclid’s linear proposition DAG and Hilbert’s layered axiomatic organization.

## Appendix E

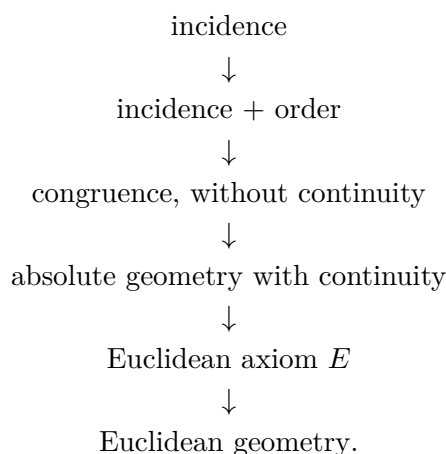
# Borsuk–Szmielw: The Axiomatic Layers

### E.1 Why this source matters

Karol Borsuk and Wanda Szmielw organize elementary geometry in a way that is unusually close to the architectural questions arising in a formal reconstruction.

Their primitive geometric language consists of points, lines, planes, betweenness, and equidistance. The axioms are divided into five groups: incidence, order, congruence, continuity, and finally either the Euclidean axiom  $E$  or the Bolyai–Lobachevskian axiom. The first four groups constitute absolute geometry.

More important for the present purpose, the authors explicitly state that one should study not only which theorems are true, but also which means are necessary to prove them. Their exposition is therefore deliberately layered:



This makes the book especially useful for deciding where a theorem belongs in a formal library. A result corresponding to a Euclidean proposition may turn out to be:

- part of the incidence/order language;
- a consequence of congruence alone;
- available before continuity;

- available only after continuity;
- genuinely Euclidean, requiring  $E$ ;
- or absent as a named theorem because Borsuk–Szmielw replace it by a stronger structural result.

The last column of the table below therefore asks a practical question: what does the Borsuk–Szmielw placement suggest for the Hilbert interface and the Book Zero layer?

## E.2 Layer notation

For compactness the table uses the following descriptions.

| Layer         | Meaning                                                                                                                                                                                        |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inc/Ord       | Chapter I: incidence and order, including betweenness, half-lines, half-planes, angles as ordered ray configurations, triangles, quadrangles, Pasch-type results, polygons, and triangulation. |
| Congr-no-Co   | Chapter II: congruence added, but the Axiom of Continuity is not used.                                                                                                                         |
| Absolute      | Chapter III: the Axiom of Continuity is added; the full first four groups of axioms are now available.                                                                                         |
| Euclidean $E$ | Chapter V: absolute geometry plus the Axiom of Euclid $E$ , equivalent in the book to Euclid’s Fifth Postulate.                                                                                |

## E.3 Euclid in the Borsuk–Szmielw development

Table E.1: Euclidean propositions in the axiomatic layers of Borsuk–Szmielw

| Euclid | Borsuk–Szmielw counterpart                                                                                                                                                                    | Layer                 | Structural significance for the reconstruction                                                                                                                                                                                                                                                                         |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.1    | No one-to-one construction theorem was located. Equilateral triangles are defined in Chapter II, but the audited text does not reproduce Euclid’s two-circle construction as a named theorem. | No direct counterpart | This is a useful absence. Borsuk–Szmielw do not organize the congruence theory by reconstructing Euclid’s first circle construction. The formal project should therefore keep the classical construction DAG separate from the axiomatic layer in which an equilateral-point existence theorem is eventually supplied. |
| I.2    | Axiom C5: on every half-line with origin $a$ , for every segment $pq$ , there exists exactly one point $b$ such that $ab = pq$ .                                                              | Congr-no-Co           | Euclid’s segment transport has become primitive and is strengthened to existence plus uniqueness on a prescribed half-line. This is precisely interface-level segment layoff, not a proposition that later proofs should reconstruct from circles.                                                                     |
| I.3    | Axiom C5 together with segment order and betweenness.                                                                                                                                         | Congr-no-Co           | Cutting a prescribed length from a longer segment is absorbed into ray layoff plus order. The natural formal location is a small Book Zero lemma over segment layoff and betweenness.                                                                                                                                  |

Continued on next page

Table E.1 – continued from previous page

| Euclid | Borsuk–Szmielew counterpart                                                                                                                                                                              | Layer       | Structural significance for the reconstruction                                                                                                                                                                                                                            |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.4    | Theorem 22(I): two sides and the included angle determine a congruent triangle. The text immediately records this as the SAS congruence criterion.                                                       | Congr-no-Co | SAS is developed before continuity. This is strong evidence for keeping it in the neutral congruence interface. Euclid’s superposition argument is not part of the Borsuk–Szmielew proof architecture.                                                                    |
| I.5    | Theorem 23: in a triangle, congruent sides are opposite congruent angles; in particular, the base angles of an isosceles triangle are congruent.                                                         | Congr-no-Co | The isosceles theorem is packaged together with its converse in one biconditional result. The reusable API should reflect this stronger relationship instead of treating I.5 and I.6 as unrelated endpoints.                                                              |
| I.6    | The converse direction of Theorem 23: congruent opposite angles imply congruent opposite sides.                                                                                                          | Congr-no-Co | Borsuk–Szmielew eliminate Euclid’s separate reductio construction. Once segment and angle congruence are organized properly, I.6 is simply the reverse implication of the same structural theorem as I.5.                                                                 |
| I.7    | No separate theorem corresponding exactly to Euclid I.7 was located. Its uniqueness content is absorbed into the order theory, unique angle construction (Theorem 13), and the strong congruence axioms. | Congr-no-Co | This is an example of a Euclidean proposition disappearing under a stronger interface. The formal question is not how to imitate I.7, but which general uniqueness and side-of-line lemmas make its role unnecessary.                                                     |
| I.8    | Theorem 21: if the three corresponding sides of two triangles are congruent, then the three corresponding angles are congruent.                                                                          | Congr-no-Co | This is the SSS content in the Borsuk–Szmielew organization, and it is available without continuity and without the Euclidean axiom. For the formal project this strongly supports treating SSS as derived neutral congruence infrastructure rather than Euclidean input. |
| I.9    | Theorems 33–34: a bisector lies between the sides of the angle, and every angle has exactly one bisector.                                                                                                | Congr-no-Co | The source gives a stronger result than Euclid’s bare construction: existence and uniqueness are packaged together. Angle-bisector existence therefore belongs naturally in the reusable interface.                                                                       |
| I.10   | Theorems 31–32: an equidistant collinear point is between the endpoints, and every segment has exactly one midpoint.                                                                                     | Congr-no-Co | This is especially valuable formally. Borsuk–Szmielew do not take the diagrammatic position of the midpoint for granted; they first prove the betweenness theorem and only then existence and uniqueness. This is exactly the separation needed in Lean.                  |
| I.11   | Theorem 76: through a point lying on a line there exists exactly one line in the plane perpendicular to the given line.                                                                                  | Congr-no-Co | Perpendicular erection is already a neutral interface theorem. Later propositions should consume it directly rather than replay an auxiliary Euclidean triangle construction.                                                                                             |
| I.12   | Theorem 77: through a point outside a given line there exists exactly one perpendicular to that line.                                                                                                    | Congr-no-Co | Dropping a perpendicular is likewise available before continuity. The theorem is stronger and cleaner than Euclid’s construction-oriented formulation and belongs naturally in the Hilbert interface.                                                                     |

Continued on next page

Table E.1 – continued from previous page

| Euclid | Borsuk–Szmielew counterpart                                                                                                                                                                       | Layer       | Structural significance for the reconstruction                                                                                                                                                                                                     |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.13   | Sections 21–24 of Chapter II: addition of free angles, adjacent angles, right angles, and the fact that an angle adjacent to a right angle is right (Theorems 63, 69–71).                         | Congr-no-Co | Borsuk–Szmielew replace informal angle arithmetic by an explicit calculus of free angles. This is a strong source precedent for using synthetic supplement/addition relations rather than numerical degrees in the formalization.                  |
| I.14   | Theorem 13 gives unique construction of a congruent angle on a specified side; Theorems 14–15 control adjacent angles and their uniqueness.                                                       | Congr-no-Co | The converse straight-line criterion is absorbed into a more general uniqueness theory for angle placement. The formal analogue should be a small collinearity/adjacency consequence of that interface.                                            |
| I.15   | Theorem 16: vertical angles are congruent.                                                                                                                                                        | Congr-no-Co | A direct reusable theorem. Its low position in the congruence chapter confirms that vertical-angle congruence should be generic infrastructure for later triangle and parallel arguments.                                                          |
| I.16   | Theorem 35: an exterior angle of a triangle is greater than each nonadjacent interior angle.                                                                                                      | Congr-no-Co | This is one of the central neutral theorems. The proof explicitly uses midpoint, vertical-angle congruence, SAS, and angle order. It is also the engine for constructing nonintersecting lines before the Euclidean axiom enters.                  |
| I.17   | A short consequence of Theorem 35 together with the developed theory of adjacent and right angles. It is not necessary to use the later Saccheri–Legendre theorem for this Euclidean proposition. | Congr-no-Co | This distinction prevents an important misclassification. Borsuk–Szmielew later prove stronger angle-sum results after continuity, but Euclid I.17 itself already belongs to the earlier neutral congruence layer.                                 |
| I.18   | Theorem 40, one direction: in a triangle the greater side lies opposite the greater angle.                                                                                                        | Congr-no-Co | I.18 and I.19 are unified. Borsuk–Szmielew package strict side and angle comparison into a biconditional theorem, exactly the sort of strong reusable comparison lemma useful in Book Zero.                                                        |
| I.19   | Theorem 40, converse direction: the greater angle lies opposite the greater side.                                                                                                                 | Congr-no-Co | The converse is not a separate construction. It is the reverse direction of the same strict comparison equivalence used for I.18.                                                                                                                  |
| I.20   | Theorem 50: for distinct points $a, b, c$ , $[ab] + [bc] > [ac]$ , with equality exactly when $B(a, b, c)$ .                                                                                      | Congr-no-Co | This is substantially stronger than the triangle-only formulation. It combines triangle inequality with the collinear equality case and uses the explicit algebra of free segments. It is an excellent model for a robust segment-order interface. |
| I.21   | Theorem 52 generalizes the side-sum part to a convex polygonal line inside a triangle; the angle part is controlled by the Exterior Angle Theorem 35 and the order theory of rays.                | Congr-no-Co | The source replaces the one-off proposition by a more general polygonal inequality. For formal work the useful target is therefore the reusable generalized lemma rather than the exact Euclidean diagram.                                         |

Continued on next page

Table E.1 – continued from previous page

| Euclid | Borsuk–Szmielew counterpart                                                                                                                                                                                                                                    | Layer                 | Structural significance for the reconstruction                                                                                                                                                                                                                         |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.22   | No one-to-one prescribed-three-sides construction theorem was located in the audited elementary sequence. The book has the triangle inequality and a strong congruence construction axiom, but does not replay Euclid’s circle-intersection construction here. | No direct counterpart | This should be kept as an explicit gap in the source map. Existence of a triangle with three prescribed side lengths is logically different from SSS uniqueness/congruence and should not be silently identified with Theorem 21.                                      |
| I.23   | Theorem 13: for a given ray, chosen half-plane, and given angle, there exists exactly one ray on that side making a congruent angle.                                                                                                                           | Congr-no-Co           | Euclid’s angle-copy construction becomes a primitive-style transport theorem with uniqueness. This is one of the strongest direct precedents for the angle-construction API already used in the Hilbert route.                                                         |
| I.24   | Theorem 41: with two corresponding sides fixed, comparison of the included angles is equivalent to comparison of the third sides.                                                                                                                              | Congr-no-Co           | Borsuk–Szmielew combine the Euclidean comparison theorem and its converse into one biconditional. This is a natural interface theorem for replacing configuration-heavy local proofs.                                                                                  |
| I.25   | The converse direction of Theorem 41.                                                                                                                                                                                                                          | Congr-no-Co           | As with I.18/I.19, the source suggests exposing one symmetric comparison theorem rather than maintaining two large proposition-sized proofs.                                                                                                                           |
| I.26   | Theorem 22(II): one side and the two angles adjacent to it determine a congruent triangle (ASA).                                                                                                                                                               | Congr-no-Co           | The ASA part of I.26 is explicit and neutral. The audited text does not package Euclid’s second AAS configuration under a separate theorem number here, so the appendix does not silently identify the two formulations.                                               |
| I.27   | Theorems 37–39. Equal appropriate transversal angles give nonintersection; Theorem 39 concludes that through an external point there is at least one line not intersecting the given line.                                                                     | Congr-no-Co           | This is one of the most important source placements in the book. The angle-to-parallel direction and existence of a nonintersecting line are established before continuity and before the Euclidean axiom.                                                             |
| I.28   | A consequence of the same Theorems 37–38 together with the earlier adjacent/vertical-angle normalization theorems.                                                                                                                                             | Congr-no-Co           | The two Euclidean angle criteria are wrappers over the neutral nonintersection theorem. The conclusion “parallel” does not by itself make the result Euclidean.                                                                                                        |
| I.29   | After $E$ , Chapter V develops the Euclidean characterization of parallel lines and Theorem 10: angles whose corresponding sides are parallel are congruent.                                                                                                   | Euclidean $E$         | This is the reversal of I.27. Before $E$ , appropriate angle data produce nonintersection; after $E$ , parallelism can be converted back into angle congruence. The source therefore marks the same neutral-to-Euclidean boundary that the formal reconstruction uses. |
| I.30   | Chapter V, Theorem 7: parallelism of lines is transitive.                                                                                                                                                                                                      | Euclidean $E$         | Parallel transitivity is explicitly downstream of the Euclidean axiom. It should therefore be classified as Euclidean interface infrastructure, not as a neutral convenience lemma.                                                                                    |

Continued on next page

Table E.1 – continued from previous page

| Euclid | Borsuk–Szmielew counterpart                                                                                                                                                                                                                                    | Layer                                                                                 | Structural significance for the reconstruction                                                                                                                                                                                                                                                                                                                |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.31   | Chapter II, Theorem 39 gives existence of a non-intersecting line. Axiom $E$ plus that theorem gives Chapter V, Theorem 1, uniqueness; Theorem 6 states the corresponding unique parallel through a point.                                                     | Existence:<br>Congr-no-Co;<br>uniqueness:<br>Euclidean $E$                            | Borsuk–Szmielew make the existence/uniqueness split exceptionally clean. This is a direct architectural model for keeping the two layers separate in Lean.                                                                                                                                                                                                    |
| I.32   | Chapter V, Theorem 2: the sum of the angles of every triangle is $\pi$ .                                                                                                                                                                                       | Euclidean $E$                                                                         | The equality form of the triangle angle-sum theorem is placed immediately after the Euclidean axiom. A synthetic Lean statement need not use numerical measure, but its dependency must remain Euclidean.                                                                                                                                                     |
| I.33   | Chapter V, Theorem 12, sufficiency direction: a plane quadrangle is a parallelogram when one pair of corresponding directed sides is parallel and the corresponding segments are congruent.                                                                    | Euclidean $E$                                                                         | This theorem is stronger and more useful than a literal copy of I.33. It characterizes a parallelogram from exactly the sort of parallel-plus-congruence data that arise in I.33. It is a strong candidate for reusable interface infrastructure.                                                                                                             |
| I.34   | Chapter V, Theorem 12, necessity direction and proof: in a parallelogram the diagonal triangles are congruent, yielding equality of opposite sides and the corresponding angle data.                                                                           | Euclidean $E$                                                                         | The proof architecture is almost tailor-made for the formal I.34 interface: parallel half-lines give angle congruences, then triangle congruence supplies opposite-side data. I.34 should expose this package for immediate downstream use.                                                                                                                   |
| I.35   | No Euclidean equal-content or equicomplementability theorem corresponding directly to I.35 was located. Chapter I already has a substantial theory of polygons and triangulation, while Chapter V has the parallelogram theory needed immediately before I.35. | No direct counterpart                                                                 | The absence is informative. Borsuk–Szmielew supply rigorous figure/triangulation infrastructure and Euclidean parallelogram infrastructure, but they do not insert a Hilbert/Hartshorne-style equal-content layer at this point. Thus they are highly useful for the components feeding I.35, but not the primary source for the semantics of its conclusion. |
| I.36   | No direct equal-area theorem was located. Chapter V, Theorem 12 supplies the parallelogram characterization used by the classical construction, while Chapter I, Theorem 100 supplies common refinements of polygon triangulations.                            | Geometry:<br>Euclidean $E$ ;<br>representation:<br>Inc/Ord;<br>no content counterpart | The two ingredients needed around I.36 are present on opposite sides of the library: rigorous figure decomposition below, and Euclidean parallelogram geometry above. What is absent is the relation that says two such parallelograms have equal content.                                                                                                    |

Continued on next page

Table E.1 – continued from previous page

| Euclid | Borsuk–Szmielew counterpart                                                                                                                                                                                                                                                                             | Layer                                            | Structural significance for the reconstruction                                                                                                                                                                                                                                                                      |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.37   | No direct equal-content theorem was located. In the necessity proof of Chapter V, Theorem 12, a parallelogram is cut by a diagonal into two congruent triangles.                                                                                                                                        | Geometry: Euclidean $E$ ; no content counterpart | Borsuk–Szmielew contain the geometric half of Euclid’s “halves” argument: each individual parallelogram is decomposed into two congruent triangles. They do not provide the additional additive content principle needed to compare halves of different equal-content parallelograms.                               |
| I.38   | No direct equal-area counterpart was located. Chapter V, Theorem 13 organizes parallel projectivity between lines as a similitude, and Theorem 12 supplies the parallelogram geometry, but neither theorem introduces equality of content.                                                              | Euclidean $E$ , with no content counterpart      | The source gives strong control of equal-base and parallel projection geometry, but the conclusion of I.38 belongs to a semantic layer that Borsuk–Szmielew do not develop. The formal proof should therefore use their geometry only through reusable interface lemmas and obtain content from the scissors layer. |
| I.39   | No direct counterpart in the Euclidean development. The book has no synthetic equal-content relation from which equality of triangles could force a parallelism conclusion. In the later Bolyai–Lobachevskian chapter the authors explicitly state that they will not develop a general theory of area. | No direct content counterpart                    | This is a sharp source boundary. Borsuk–Szmielew remain useful for the parallel-through-point, incidence, order, and polygon geometry entering a reductio, but they do not supply the positivity or proper-part principle needed by the converse content argument.                                                  |
| I.40   | No direct counterpart. As for I.39, the Euclidean line and parallel infrastructure is available, but there is no equal-content relation whose hypothesis can be converted back into geometry.                                                                                                           | No direct content counterpart                    | The equal-base converse has the same architectural status as I.39. The source can control the geometric case analysis, but the strict content step must come from Hilbert/Hartshorne-style area theory or an explicit positive additive invariant.                                                                  |
| I.41   | Chapter V, Theorem 12 proves that the two triangles cut from a parallelogram by a diagonal are congruent. No theorem stating that a parallelogram is double a triangle in content was located.                                                                                                          | Geometry: Euclidean $E$ ; no content counterpart | The geometry needed for “double” is present exactly: the parallelogram is the union of two congruent diagonal triangles. The missing step is not geometric but additive – turning that decomposition into an equality of content with two copies of a triangle.                                                     |

Continued on next page



Table E.1 – continued from previous page

| Euclid | Borsuk–Szmielw counterpart                                                                                                                                                                                                                                                                                                                                                                           | Layer                                                                         | Structural significance for the reconstruction                                                                                                                                                                                                                                                                                          |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.42   | No named application-of-area theorem was located. Its construction is, however, assembled from infrastructure already present in the book: midpoint existence and uniqueness (Chapter II, Theorems 31–32), unique angle construction on a chosen side (Chapter II, Theorem 13), the unique parallel through a point (Chapter V, Theorem 6), and the parallelogram criterion (Chapter V, Theorem 12). | Construction: Congr-no-Co + Euclidean $E$ ; no content counterpart            | This is the clearest separation in the block. Borsuk–Szmielw almost completely supply the geometric constructor for I.42 as reusable API. What they do not supply is the I.38/I.41 content calculus proving that the resulting parallelogram equals the given triangle.                                                                 |
| I.43   | No direct equal-content counterpart was located. The book supplies the parallelogram geometry and the polygon/triangulation language needed to describe the configuration, but not a Hilbert-style relation of equal content.                                                                                                                                                                        | Geometry/representation available; no content counterpart                     | This source can control the finite decomposition of the large and small parallelograms. The conclusion that the two complements have equal content must still come from the Hilbert/scissors layer.                                                                                                                                     |
| I.44   | No named application-of-area theorem was located. The geometric constructor can be assembled from unique angle layoff, segment/ray order, the unique parallel through a point, and Chapter V parallelogram criteria.                                                                                                                                                                                 | Construction: Congr-no-Co + Euclidean $E$ ; no content counterpart            | This reinforces the geometry/content split already visible at I.42. The book supplies strong placement infrastructure, but the statement that the placed parallelogram has the required content lies outside its developed Euclidean theory.                                                                                            |
| I.45   | Chapter I supplies a rigorous theory of polygons and triangulation, including common refinements of triangulations. No theorem applying a parallelogram of given angle to an arbitrary figure with equal content was located.                                                                                                                                                                        | Representation: Inc/Ord; construction: Euclidean $E$ ; no content counterpart | For I.45 this source is strongest exactly where the Lean statement is weakest as a region representation: it knows what a polygonal figure and a triangulation are. Conversely it does not supply the synthetic equal-content semantics, so the formal theorem must continue to obtain that conclusion from the Hilbert/scissors layer. |

## E.4 The strongest architectural correspondences

The most useful feature of the Borsuk–Szmielw source is that several Euclidean propositions are replaced by stronger interface theorems.

### E.4.1 Transport is primitive-strength infrastructure

Axiom C5 is stronger than the bare construction in I.2: it gives exactly one point on a chosen half-line at a prescribed distance. Theorem 13 is the angular analogue: on a chosen side of a ray, an angle congruent to a given angle can be constructed uniquely.

Thus two operations that appear relatively late or require elaborate construction in Euclid become basic transport mechanisms:

$$\text{segment layoff} \quad \text{and} \quad \text{angle layoff}.$$

This is directly consonant with the Hilbert-style formal interface.

### E.4.2 Pairs of Euclidean propositions become biconditionals

Theorem 23 unifies I.5 and I.6:

$$\text{equal sides} \iff \text{equal opposite angles}.$$

Theorem 40 unifies I.18 and I.19:

$$\text{greater side} \iff \text{greater opposite angle}.$$

Theorem 41 similarly unifies I.24 and I.25 when two corresponding sides are fixed:

$$\text{greater included angle} \iff \text{greater third side}.$$

For library design, these are more informative than the proposition numbers. They identify the natural reusable theorem boundaries.

### E.4.3 Midpoints and bisectors include positional information

The midpoint discussion is especially close to the needs of formal geometry. Borsuk–Szmielew first prove that a collinear point equidistant from the endpoints of a segment must lie between them. Only then do they prove existence and uniqueness of the midpoint.

The angle-bisector discussion has the same pattern: first the bisector is shown to lie between the sides of the angle, then existence and uniqueness are established.

The lesson for formalization is precise. “Midpoint” and “bisector” should not be treated merely as metric equalities; their order/ray placement is part of the reusable theorem package.

## E.5 The neutral parallel theory is already large

The source is particularly valuable around Euclid I.27–I.31.

From the Exterior Angle Theorem and the congruence/order theory, Borsuk–Szmielew prove that an appropriate equality of transversal angles forces two lines to be disjoint. They then obtain Theorem 39:

through an external point there exists at least one nonintersecting line.

This theorem belongs to Chapter II, before continuity and before the Euclidean axiom.

Only much later, in Chapter V, is Axiom *E* added:

through an external point there is at most one nonintersecting line.

Together the two statements give uniqueness. Chapter V then derives the Euclidean parallel theory: characterization of disjoint coplanar lines, uniqueness of the parallel, transitivity, angles with parallel sides, and parallelograms.

This produces a clean formal boundary:

|                                               |                   |
|-----------------------------------------------|-------------------|
| angle criterion $\Rightarrow$ nonintersection | before <i>E</i> , |
| existence of a nonintersecting line           |                   |

while

|                                            |                  |
|--------------------------------------------|------------------|
| uniqueness of the parallel                 | after <i>E</i> . |
| parallelism $\Rightarrow$ angle congruence |                  |
| parallel transitivity                      |                  |
| triangle angle sum                         |                  |

For the present reconstruction this is one of the strongest independent checks on the placement of I.27, I.29, I.30, I.31, and I.32.

## E.6 Theorem 12 of Chapter V as an interface theorem

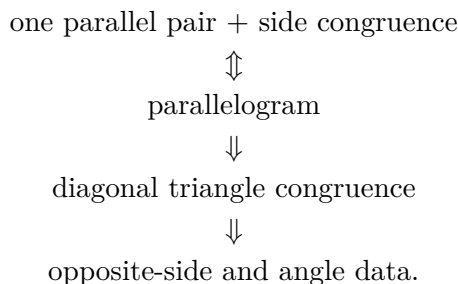
The parallelogram theorem deserves special attention.

Borsuk–Szmielw define a parallelogram in the expected way, by two pairs of opposite sides lying on parallel lines. Their Theorem 12 then proves a necessary and sufficient condition using only one pair of directed parallel sides and equality of the corresponding segments.

In the necessity direction, the proof starts from a parallelogram, transports angles using parallel half-lines, and obtains congruence of the two triangles cut by a diagonal. Opposite-side congruence follows.

In the sufficiency direction, one parallel pair together with one segment congruence forces the remaining pair to be parallel.

This theorem simultaneously explains why I.33 and I.34 should sit close to the reusable Euclidean parallelogram interface:



It is therefore not merely a historical counterpart. It is a plausible API boundary for subsequent formal work.

## E.7 What the book contributes at I.35–I.45

The transition at I.35 is revealing because the source provides two sides of the problem but not the final semantic relation, and this same pattern persists through I.45.

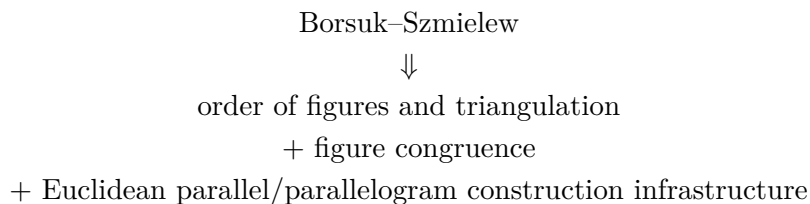
Chapter I develops polygons, intersections of polygons with half-planes, and triangulation. In particular, Theorem 100 gives a common refinement of any two triangulations of the same polygon. This is a rigorous combinatorial theory of plane figures built very low in the axiomatic hierarchy.

Chapter II also defines congruence of arbitrary figures by an isometry-like one-to-one correspondence preserving all segment congruences. This relation is reflexive, symmetric, and transitive. It is important not to confuse this strong notion of congruence with Euclid’s later “equality” of noncongruent figures.

Chapter V develops the Euclidean parallelogram theory and supplies the congruence infrastructure used throughout I.35–I.45. Theorem 12 is particularly close to I.41: its proof shows that a diagonal divides a parallelogram into two congruent triangles. Theorem 6 supplies the unique parallel through a point, so the main Euclidean construction used in I.42 is also already interface-level infrastructure.

What is not developed there is a Euclidean theory of equal content or equicomplementability comparable to Hilbert Chapter IV or Hartshorne Section 22. Later, in the Bolyai–Lobachevskian chapter, the authors explicitly note the relation between defect and area and then state that they will not develop a general theory of area. Their defect of polygons is therefore a separate invariant, not a substitute for the Euclidean content relation needed in I.35–I.45.

For the formal reconstruction the source therefore has a sharply defined role:



but

Hilbert/Hartshorne remain the control sources for equicomplementability/equal content.

Thus I.36–I.38 and I.41–I.45 are not absent geometrically; rather, their geometric DAGs are largely available while their content semantics are not. I.39–I.40 make the boundary even sharper because their hypotheses themselves are content statements. This is not a weakness of the audit. It tells us exactly where one source stops and another must take over.

## E.8 I.42 as a geometry/content separation test

Proposition I.42 is especially informative in the Borsuk–Szmielw organization because nearly every geometric construction appearing in the classical proof has already been promoted to reusable infrastructure.

The base can be bisected by the midpoint theorems. A prescribed angle can be laid off uniquely on a chosen side. After Axiom *E*, the unique parallel through a point is Chapter V, Theorem 6. Finally, Chapter V, Theorem 12 supplies a compact criterion for completing and recognizing a parallelogram.

Schematically, the geometric part of I.42 can therefore be read as

$$\begin{array}{c} \text{midpoint} + \text{angle layoff} + \text{unique parallel} \\ \Downarrow \\ \text{parallelogram construction.} \end{array}$$

What is not present is the second half of the proof:

$$\triangle ABE \approx \triangle AEC, \quad \text{parallelogram } FECG \approx 2 \triangle AEC.$$

Those are equal-content assertions, not missing incidence or congruence facts. This makes I.42 a particularly clean example of why the formal library should keep geometric constructors and the scissors/content layer separate.

## E.9 I.43–I.45: representation and construction without equal-content semantics

The final application-of-areas propositions sharpen the division already visible at I.42. Borsuk–Szmielw supply two of the three layers needed by the formal reconstruction. Chapter I has genuine polygon and triangulation machinery, and Chapter V supplies the Euclidean parallel/parallelogram constructor. What remains absent is a synthetic Euclidean relation playing the role of Hilbert/Hartshorne equal content.

This is particularly informative at I.45. The source shows what it would mean to strengthen `HilbertTriangulatedFigure` from an arbitrary triangle list to a representation of one polygonal region: the pieces must participate in a genuine triangulation, not merely in a formal sum. At the same time, nothing in the Borsuk–Szmielw audit suggests replacing the current scissors semantics by numerical area. Representation and content remain separate library concerns.

## E.10 Consequences for the formal library

The Borsuk–Szmielew audit suggests a practical classification of reusable results.

### E.10.1 Hilbert interface candidates

The following belong naturally to a general synthetic interface rather than to individual Euclidean proposition files:

- unique segment layoff on a ray;
- unique angle layoff on a chosen side;
- SAS and SSS-style triangle congruence;
- isosceles theorem and converse;
- midpoint existence, uniqueness, and betweenness;
- angle-bisector existence, uniqueness, and ray order;
- vertical angles;
- Exterior Angle Theorem;
- strict side/angle comparison;
- perpendicular existence and uniqueness;
- neutral angle criteria for nonintersection.

### E.10.2 Book Zero candidates

The developed calculus of free segments and free angles also points to a natural macro-language:

- segment less-than and trichotomy;
- transport of order under congruence;
- segment addition and subtraction;
- triangle inequality with the collinear equality case;
- angle less-than and trichotomy;
- angle addition and adjacency;
- supplement/right-angle normalization.

These are exactly the sorts of operations that prevent later proposition files from expanding into repeated low-level incidence and order proofs.

### E.10.3 Euclidean interface candidates

After Axiom *E*, the natural reusable layer contains:

- unique parallel through a point;
- characterization of Euclidean parallels;
- parallel transitivity;
- congruence of angles with respectively parallel sides;
- Euclidean triangle angle sum;
- the Chapter V parallelogram characterization.

The source thus gives not just another list of theorem counterparts, but a plausible decomposition of the formal library itself.

## E.11 Conclusion

Borsuk and Szmielew are particularly valuable for the reconstruction because their book asks almost the same structural question as the formal project: not only whether a theorem is true, but which axiomatic resources are actually needed to prove it.

Their treatment makes several boundaries unusually visible. Much of the elementary order and congruence theory, including the Exterior Angle Theorem, midpoints, bisectors, perpendiculars, triangle comparison, and existence of nonintersecting lines, is developed before continuity. The Axiom of Euclid then contributes the uniqueness needed to obtain the full Euclidean theory of parallels. Their parallelogram theorem packages the material around I.33–I.34 in a form that is especially attractive as formal interface infrastructure.

At I.35–I.45 the source becomes useful in a different way. Its early theory of polygons and triangulation, its general notion of figure congruence, and its later Euclidean parallel/parallelogram theory provide most of the geometric and representational infrastructure. The semantic theory of equal content is left to other sources. I.42 makes this division particularly visible: its constructor is almost entirely available, while the proof that the constructed parallelogram has the required content is not.

The resulting appendix should therefore be read as an axiomatic-layer map and an API-design aid, not merely as another historical comparison with Euclid.

# Bibliography

- [1] Euclid, *The Thirteen Books of Euclid's Elements*, edited and translated by Sir Thomas L. Heath, 3 vols., Dover Publications, New York, 1956. (Reprint of the second edition, Cambridge University Press, 1926.)
- [2] David Hilbert, *Grundlagen der Geometrie*, B. G. Teubner, Leipzig, 1899.
- [3] Karol Borsuk and Wanda Szmielew, *Foundations of Geometry: Euclidean and Bolyai–Lobachevskian Geometry, Projective Geometry*, North-Holland Publishing Company, Amsterdam, 1960.
- [4] Marvin Jay Greenberg, *Euclidean and Non-Euclidean Geometries: Development and History*, 4th ed., W. H. Freeman and Company, New York, 2008.
- [5] Robin Hartshorne, *Geometry: Euclid and Beyond*, Springer, New York, 2000.
- [6] Michael Beeson, *A Constructive Version of Tarski's Geometry*, *Annals of Pure and Applied Logic*, Vol. 166, No. 11, pp. 1199–1273, 2015.
- [7] Michael Beeson, Julien Narboux, and Freek Wiedijk, “Proof-checking Euclid,” *Annals of Mathematics and Artificial Intelligence*, Vol. 85, pp. 213–257, 2019.
- [8] Michael Beeson and Larry Wos, “Axiomatizing Geometry,” *Annals of Mathematics and Artificial Intelligence*, 2019.
- [9] David R. Wilkins, *Study Notes on Euclid's Elements, Book I*, Trinity College Dublin.
- [10] Michael Beeson, *Formal Proof Archive for Euclidean Geometry*, proof scripts accompanying *A Constructive Version of Tarski's Geometry*, 2019.
- [11] Euklides, *Elementy*, opracowanie i tłumaczenie: Piotr Błaszczyk, Kazimierz Mrowka, Copernicus Center Press, Krakow, 2014.